

INTERNATIONAL STANDARD

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**Adjustable speed electrical power drive systems –
Part 7-202: Generic interface and use of profiles for power drive systems –
Profile type 2 specification**

**Entraînements électriques de puissance à vitesse variable –
Partie 7-202: Interface générique et utilisation de profils pour les entraînements
électriques de puissance – Spécification de profil de type 2**



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IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00
info@iec.ch
www.iec.ch

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CONTENTS

FOREWORD	10
INTRODUCTION	12
0.1 General	12
0.2 Patent declaration	15
1 Scope	17
2 Normative references	17
3 Terms, definitions and abbreviated terms	17
3.1 Terms and definitions	17
3.2 Abbreviated terms	25
4 Overview	25
4.1 General	25
4.2 Control modes	26
4.2.1 General	26
4.2.2 Control methods	26
4.2.3 Control nomenclature	27
4.2.4 Position control	27
4.2.5 Velocity control	28
4.2.6 Acceleration control	30
4.2.7 Torque control	30
4.2.8 No Control	31
5 Data types	32
5.1 Data type overview	32
5.2 Conventions	32
6 CIP Motion drive profile	32
6.1 Object model	32
6.1.1 Object overview	32
6.1.2 Object description	33
6.2 How objects affect behavior	34
6.3 Defining object interfaces	34
6.4 I/O connection messages	35
6.4.1 General	35
6.4.2 CIP Motion I/O Connection	35
6.4.3 Controller-to-Device Connection	39
6.4.4 Device-to-Controller Connection	59
6.4.5 Fixed Motion I/O connection format	68
6.4.6 CIP Motion I/O Connection timing model	69
6.5 Device startup procedure	85
6.5.1 General	85
6.5.2 Motion I/O Connection creation	85
6.5.3 Motion Device Axis Object configuration	88
6.5.4 Time Synchronization	90
6.6 Device visualisation	92
6.7 Ethernet/IP Quality of Service (QoS)	93
7 Motion Device Axis Object	93
7.1 General considerations	93

7.1.1	General	93
7.1.2	Revision history	93
7.1.3	Object overview	93
7.1.4	Motion Device Axis Object abstraction	94
7.1.5	Motion Control Axis Object	95
7.1.6	Device control classification.....	95
7.1.7	Required vs. Optional in implementation.....	96
7.2	Class attributes	107
7.2.1	General	107
7.2.2	Semantics.....	111
7.3	Instance attributes	114
7.3.1	General	114
7.3.2	Motion Control configuration attributes	116
7.3.3	Motion Scaling attributes	117
7.3.4	Connection Data attributes	123
7.3.5	Motor attributes	129
7.3.6	Feedback attributes	140
7.3.7	Event Capture attributes	149
7.3.8	Command reference generation attributes	155
7.3.9	Control mode attributes	158
7.3.10	Stopping & Braking attributes	176
7.3.11	DC Bus Control attributes	186
7.3.12	Power and thermal management attributes	190
7.3.13	Axis Status attributes.....	193
7.3.14	Exception, fault, and alarm attributes.....	199
7.3.15	Fault and alarm Log attributes	205
7.3.16	Exception limit attributes.....	210
7.3.17	Axis exception action configuration attribute	213
7.3.18	Initialization fault attributes	215
7.3.19	Start inhibit attributes	216
7.3.20	APR fault attributes	217
7.3.21	Axis statistical attributes	219
7.3.22	Axis info attributes	219
7.3.23	General purpose I/O attributes.....	220
7.3.24	Local Mode attributes	222
7.3.25	Axis Safety attributes.....	222
7.4	Common services	223
7.4.1	Supported services	223
7.4.2	Service specific data.....	224
7.5	Object specific services	226
7.5.1	Supported services	226
7.5.2	Service specific data.....	226
7.6	Behavior	241
7.6.1	State model	241
7.6.2	State behavior	252
7.6.3	Fault and alarm behavior	259
7.6.4	Start Inhibit behavior	261
7.6.5	Visualization behavior.....	261
7.6.6	Command generation behavior	266

7.6.7	Feedback interface behavior.....	271
7.6.8	Event Capture Behavior.....	273
7.6.9	Control Mode behavior.....	274
Bibliography.....		288
Figure 1	– Structure of IEC 61800-7.....	15
Figure 2	– Open loop position control.....	27
Figure 3	– Closed loop position control	28
Figure 4	– Open loop velocity control	29
Figure 5	– Closed loop velocity control	29
Figure 6	– Acceleration control	30
Figure 7	– Torque control.....	31
Figure 8	– No Control (Feedback Only).....	31
Figure 9	– Object Model for a CIP Motion device	33
Figure 10	– CIP Motion I/O Connection model	35
Figure 11	– CIP Motion I/O Connection channels	36
Figure 12	– Controller-to-Device Connection format (Connection Point 2)	37
Figure 13	– Device-to-Controller Connection format (Connection Point 2)	38
Figure 14	– CIP Motion Controller-to-Device Connection format.....	39
Figure 15	– Connection Header	39
Figure 16	– Connection Format.....	39
Figure 17	– Connection Header	40
Figure 18	– Instance Data Block	43
Figure 19	– Instance Data Header	43
Figure 20	– Cyclic Data Block	44
Figure 21	– Control Mode	44
Figure 22	– Feedback Mode.....	44
Figure 23	– Cyclic Write Data Block	50
Figure 24	– Cyclic Write Data Block example	50
Figure 25	– Event Data Block.....	51
Figure 26	– Service Data Block.....	58
Figure 27	– CIP Motion Device-to-Controller Connection format.....	59
Figure 28	– Connection Header	59
Figure 29	– Connection Header	60
Figure 30	– Node Fault/Alarm	61
Figure 31	– Adjustment of actual position data based on Device Time Stamp	62
Figure 32	– Instance Data Block	63
Figure 33	– Instance Data Header	63
Figure 34	– Cyclic Data Block	63
Figure 35	– Cyclic Read Data Block	65
Figure 36	– Cyclic Read Data Block example	65
Figure 37	– Event Data Block.....	66
Figure 38	– Service Data Block.....	68

Figure 39 – Fixed Controller-to-Device Connection format (fixed size = 16 bytes)	69
Figure 40 – Fixed Device-to-Controller Connection format (fixed size = 16 bytes)	69
Figure 41 – CIP Motion 1-Cycle timing model.....	70
Figure 42 – CIP Motion 2-Cycle timing model.....	72
Figure 43 – CIP Motion 3-Cycle timing model.....	73
Figure 44 – Controller-to-Device Connection timing with fine interpolation	74
Figure 45 – Controller-to-Device Connection timing with extrapolation	76
Figure 46 – Use of Time Stamp to adjust actual position to the controller's timebase	77
Figure 47 – Coordination of two drives with different Update Periods	79
Figure 48 – Coordination of multiple drive axes in case of delayed Controller-to-Device Connection packets.....	80
Figure 49 – Propagation of a step change in time	81
Figure 50 – Configuration Block Format Revision 1 (Connection Point 81)	86
Figure 51 – Configuration Block Format Revision 2 (Connection Point 82)	87
Figure 52 – Typical initial C-to-D connection data block	88
Figure 53 – Typical initial D-to-C connection data block	88
Figure 54 – Typical contents of first C-to-D class attribute configuration packet	88
Figure 55 – Typical response to first C-to-D class configuration packet.....	89
Figure 56 – Typical contents of first C-to-D axis instance configuration packet.....	89
Figure 57 – Typical response to first C-to-D axis configuration packet	90
Figure 58 – Typical contents of C-to-D Time Sync service request packet.....	90
Figure 59 – Group Sync of CIP Motion devices	91
Figure 60 – Object components for CIP Motion control architecture	94
Figure 61 – Command Control Word field.....	127
Figure 62 – IEEE Std 112 per phase motor model.....	130
Figure 63 – Event Checking Control Word field	152
Figure 64 – Event Checking Status word field	153
Figure 65 – Brake Control Sequence (Category 0 Stop).....	182
Figure 66 – Brake Control Sequence (Category 1 Stop).....	183
Figure 67 – Brake Control Sequence (Category 2 Stop).....	184
Figure 68 – Drive Enable sequence with Proving feature	185
Figure 69 – Drive Disable sequence with Proving feature.....	186
Figure 70 – Get_Axis_Attributes_List Request format	227
Figure 71 – Get_Axis_Attributes_List Response format.....	228
Figure 72 – Get_Axis_Attributes_List Response – Single 4-byte attribute	228
Figure 73 – Get_Axis_Attributes_List Response – Single 2-byte attribute	228
Figure 74 – Get_Axis_Attributes_List Response – Byte attribute array	229
Figure 75 – Get_Axis_Attributes_List Response – Two Dimensional attribute array	229
Figure 76 – Get_Axis_Attributes_List Response – Error example	229
Figure 77 – Set_Axis_Attributes_List Request format.....	230
Figure 78 – Set_Axis_Attributes_List Request – Single 4-byte attribute	230
Figure 79 – Set_Axis_Attributes_List Request – Single 2-byte attribute	231
Figure 80 – Set_Axis_Attributes_List Request – 2-byte attribute array	231

Figure 81 – Set_Axis_Attributes_List Request – Two dimensional attribute array	231
Figure 82 – Set_Axis_Attributes_List Response format	231
Figure 83 – Set_Cyclic_Write_List Request format.....	232
Figure 84 – Set_Cyclic_Write_List Response format	232
Figure 85 – Set_Cyclic_Read_List Request format.....	233
Figure 86 – Set_Cyclic_Read_List Response format	233
Figure 87 – Motion Device Axis Object State Model	241
Figure 88 – Motion Device Axis Object State Model for Feedback Only.....	243
Figure 89 – Motion Device Axis Object State Model for Converter.....	244
Figure 90 – Command Generator	267
Figure 91 – Feedback Channels 1 and 2	272
Figure 92 – Event Capture Functionality	273
Figure 93 – No Control (Feedback Only)	275
Figure 94 – Closed Loop Position Control	276
Figure 95 – Closed Loop Velocity Control	278
Figure 96 – Open Loop Frequency Control.....	280
Figure 97 – Acceleration Control.....	282
Figure 98 – Torque Control	282
Figure 99 – Closed Loop Current Vector Control.....	286
Table 1 – Data types	32
Table 2 – Objects present in a CIP Motion device	33
Table 3 – Motion Device Axis Object content by Device Type	34
Table 4 – Object effect on behavior	34
Table 5 – Object interfaces	35
Table 6 – Time Data Set	41
Table 7 – Axis Control	45
Table 8 – Control Status	45
Table 9 – Command Data Set	46
Table 10 – Command Data Element to Motion Device Axis Object attribute mapping	46
Table 11 – Actual Data Set	47
Table 12 – Actual Data Element to Motion Device Axis Object attribute Mapping	47
Table 13 – Status Data Set	48
Table 14 – Command Control	48
Table 15 – Command Target Update vs. Update Period Ratio.....	49
Table 16 – Basic Event Cycle	51
Table 17 – Extended Event Cycle	53
Table 18 – Basic Event Cycle with Auto-rearm	55
Table 19 – Registration Data Set	57
Table 20 – Home Data Set.....	58
Table 21 – Watch Data Set	58
Table 22 – Axis Response	64
Table 23 – Event Type	67

Table 24 – Propagation of a step change in time (example 1)	81
Table 25 – Propagation of a step change in time (example 2)	83
Table 26 – CIP Motion visualisation components	92
Table 27 – Motion Device Axis Object revision history	93
Table 28 – Example for instance attribute implementation vs. Device Function Code	96
Table 29 – Instance attribute implementation vs. Device Function Code	98
Table 30 – Class attributes for the Motion Device Axis Object.....	108
Table 31 – Node Control bit definitions	111
Table 32 – Node Status bit definitions.....	112
Table 33 – Node Fault Code definitions	113
Table 34 – Node Alarm Code definitions	114
Table 35 – Dynamic Units vs. Feedback Mode	116
Table 36 – Motion Control configuration attributes	116
Table 37 –Control Mode enumeration definitions.....	117
Table 38 – Control Method enumeration definitions.....	117
Table 39 – Motion Scaling attributes	118
Table 40 – Motion Unit selection rules	120
Table 41 – Signal attributes affected by Motion Polarity	121
Table 42 – Directional Limit attributes affected by Motion Polarity.....	123
Table 43 – Connection Data attributes	124
Table 44 – Actual Data Set value determination	126
Table 45 – Command Data Set value determination	127
Table 46 – Command Target Update enumeration definition	127
Table 47 – Command Position Data Type enumeration definition	128
Table 48 – Status Data Set bit definitions	128
Table 49 – Registration Event Data format.....	129
Table 50 – Home Event Data format	129
Table 51 – Watch Event Data format.....	129
Table 52 – General Motor Info attributes.....	130
Table 53 – General Motor Configuration attributes	131
Table 54 – General PM Motor Configuration attributes	134
Table 55 – General Rotary Motor Configuration attributes	135
Table 56 – General Linear Motor Configuration attributes	136
Table 57 – Rotary PM Motor Configuration attributes	137
Table 58 – Linear PM Motor Configuration attributes.....	137
Table 59 – Induction Motor Configuration attributes	138
Table 60 – Load Transmission and Actuator Configuration attributes	139
Table 61 – Feedback Types abbreviations	140
Table 62 – Logical Feedback Channel Control functions	140
Table 63 – Logical Feedback Channel rules.....	141
Table 64 – General Feedback Info attributes.....	142
Table 65 – General Feedback Signal attributes	142
Table 66 – Feedback Configuration attributes	143

Table 67 – Feedback Mode enumeration definitions.....	149
Table 68 – Event attributes	150
Table 69 – Event Checking Control bit definitions	152
Table 70 – Event Checking Status bit definitions.....	154
Table 71 – Command Generator Signal attributes	155
Table 72 – Command Generator Configuration attributes	157
Table 73 – Position Loop Signal attributes	159
Table 74 – Position Loop Configuration attributes	160
Table 75 – Velocity Loop Signal attributes	162
Table 76 – Velocity Loop Configuration attributes	163
Table 77 – Acceleration Signal attributes	165
Table 78 – Acceleration Configuration attributes	165
Table 79 – Torque/Force Control Signal attributes	166
Table 80 – Torque/Force Control Configuration attributes	167
Table 81 – Current Control Signal attributes	169
Table 82 – Current Control Configuration attributes	171
Table 83 – Frequency Control Signal attributes.....	175
Table 84 – Frequency Control Configuration attributes.....	175
Table 85 – Drive Output attributes	176
Table 86 – Stopping/Braking attributes	177
Table 87 – Stopping Action enumeration definitions.....	180
Table 88 – Proving sub-feature attribute dependencies.....	184
Table 89 – DC Bus Control attributes	187
Table 90 – Power and Thermal Management Status attributes	190
Table 91 – Power and Thermal Management Configuration attributes	192
Table 92 – Axis Status attributes	194
Table 93 – Axis Status bit definitions	195
Table 94 – Axis Status bit vs. Axis State	198
Table 95 – Stopping Action vs. Stop Category	199
Table 96 – Axis I/O Status bit definitions.....	199
Table 97 – Exception, Fault and Alarm attributes	200
Table 98 – Standard Exception Table	202
Table 99 – Fault and Alarm Log attributes.....	207
Table 100 – Exception Factory Limit Info attributes	210
Table 101 – Exception User Limit Configuration attributes	211
Table 102 – Axis Exception Action Configuration attribute	213
Table 103 – Axis Exception Action definitions	214
Table 104 – Initialization Fault attributes.....	216
Table 105 – Standard Initialization Fault Table	216
Table 106 – Start Inhibit attributes	217
Table 107 – Standard Start Inhibit Table	217
Table 108 – APR Fault attributes	218
Table 109 – Standard APR Fault Table	219

Table 110 – Axis Statistical attributes	219
Table 111 – Axis Info attributes.....	220
Table 112 – Drive General Purpose I/O attributes	221
Table 113 – Local Mode Configuration attributes	222
Table 114 – Axis Safety Status attributes.....	223
Table 115 – Motion Device Axis Object – Common Services	224
Table 116 – Group_Sync Request Data Structure	224
Table 117 – Group_Sync Response Data Structure	225
Table 118 – Motion Device Axis Object – Object Specific Services	226
Table 119 – Run_Motor_Test Request structure	234
Table 120 – Get_Motor_Test_Data measured by Test Type	235
Table 121 – Get_Motor_Test_Data Request structure (optional)	235
Table 122 – Get_Motor_Test_Data Response standard structure (Motor Type = Induction)	236
Table 123 – Get_Motor_Test_Data Response standard structure (Motor Type = SPM)	236
Table 124 – Get_Motor_Test_Data Response standard structure (Motor Type = IPM).....	237
Table 125 – Run_Inertia_Test Request structure	237
Table 126 – Get_Inertia_Test_Data Response structure	238
Table 127 – Run_Hookup_Test Request structure	239
Table 128 – Get_Hookup_Test_Data measured by Test Type	240
Table 129 – Get_Hookup_Test_Data Response structure	240
Table 130 – Axis State Machine transitions.....	242
Table 131 – Axis State Machine conditions	243
Table 132 – Axis State Machine transitions (Feedback Only)	244
Table 133 – Axis State Machine transitions (Converter)	245
Table 134 – Axis Control Request code	246
Table 135 – Axis Response Acknowledge codes.....	246
Table 136 – Completion criteria for requested operation	247
Table 137 – Possible error conditions for requested operation	247
Table 138 – Successful Axis Control Request Cycle	248
Table 139 – Unsuccessful Axis Control Request Cycle	248
Table 140 – Pending Axis Control Request Cycle	249
Table 141 – Cancel Request Cycle	250
Table 142 – Redefine Position Reference Cycle.....	252
Table 143 – Running State – Configurable attributes	255
Table 144 – Axis state mapping to Identity Object with LED behavior	262
Table 145 – CIP Motion Device seven-segment display behavior	263
Table 146 – CIP Motion multi-character alphanumeric display behavior	264
Table 147 – Multi-axis multi-character alphanumeric display behavior	266

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power drive systems – Profile type 2 specification****FOREWORD**

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International Standard IEC 61800-7-202 has been prepared by subcommittee SC 22G: Adjustable speed electric drive systems incorporating semiconductor power converters, of IEC technical committee TC 22: Power electronic systems and equipment.

This second edition cancels and replaces the first edition published in 2007. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) update of patent information;
- b) new revision of the Drive Profile and Drive Axis specifications, with multiple clarifications and enhancements.

The text of this standard is based on the following documents:

FDIS	Report on voting
22G/308/FDIS	22G/323/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

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INTRODUCTION

0.1 General

The IEC 61800 series is intended to provide a common set of specifications for adjustable speed electrical power drive systems.

IEC 61800-7 specifies profiles for Power Drive Systems (PDS) and their mapping to existing communication systems by use of a generic interface model.

IEC 61800-7 describes a generic interface between control systems and power drive systems. This interface can be embedded in the control system. The control system itself can also be located in the drive (sometimes known as "smart drive" or "intelligent drive").

A variety of physical interfaces is available (analogue and digital inputs and outputs, serial and parallel interfaces, fieldbuses and networks). Profiles based on specific physical interfaces are already defined for some application areas (e.g. motion control) and some device classes (e.g. standard drives, positioner). The implementations of the associated drivers and application programmers interfaces are proprietary and vary widely.

IEC 61800-7 defines a set of common drive control functions, parameters, and state machines or description of sequences of operation to be mapped to the drive profiles.

IEC 61800-7 provides a way to access functions and data of a drive that is independent of the used drive profile and communication interface. The objective is a common drive model with generic functions and objects suitable to be mapped on different communication interfaces. This makes it possible to provide common implementations of motion control (or velocity control or drive control applications) in controllers without any specific knowledge of the drive implementation.

There are several reasons to define a generic interface:

For a drive device manufacturer

- less effort to support system integrators;
- less effort to describe drive functions because of common terminology;
- the selection of drives does not depend on availability of specific support.

For a control device manufacturer

- no influence of bus technology;
- easy device integration;
- independent of a drive supplier.

For a system integrator

- less integration effort for devices;
- only one understandable way of modeling;
- independent of bus technology.

Much effort is needed to design a motion control application with several different drives and a specific control system. The tasks to implement the system software and to understand the functional description of the individual components may exhaust the project resources. In some cases, the drives do not share the same physical interface. Some control devices just support a single interface which will not be supported by a specific drive. On the other hand, the functions and data structures are often specified with incompatibilities. This requires the

system integrator to write special interfaces for the application software and this should not be his responsibility.

Some applications need device exchangeability or integration of new devices in an existing configuration. They are faced with different incompatible solutions. The efforts to adapt a solution to a drive profile and to manufacturer specific extensions may be unacceptable. This will reduce the degree of freedom to select a device best suited for this application to the selection of the unit which will be available for a specific physical interface and supported by the controller.

IEC 61800-7-1 is divided into a generic part and several annexes as shown in Figure 1. The drive profiles types for CiA® 402¹, CIP Motion™², PROFIdrive³ and SERCOS®⁴ are mapped to the generic interface in the corresponding annex. The annexes have been submitted by open international network or fieldbus organizations which are responsible for the content of the related annex and use of the related trademarks.

This part of IEC 61800-7 specifies the profile type 2 (CIP Motion™).

The profile types 1, 3 and 4 are specified in IEC 61800-7-201, IEC 61800-7-203 and IEC 61800-7-204.

¹ CiA® 402 is a registered trade mark of CAN in Automation e.V (CiA). This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade mark holder or any of its products. Compliance to this profile does not require use of the registered trade mark CiA® 402. Use of the registered trade mark CiA® 402 requires permission of CAN in Automation e.V (CiA).

² CIP Motion™ is a trade mark of ODVA, Inc. This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade mark holder or any of its products. Compliance to this profile does not require use of the trade mark CIP Motion™. Use of the trade mark CIP Motion™ requires permission of ODVA, Inc.

³ PROFIdrive is a trade name of PROFIBUS & PROFINET International. This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade name holder or any of its products. Compliance to this profile does not require use of the trade name PROFIdrive. Use of the trade name PROFIdrive requires permission of PROFIBUS & PROFINET International.

⁴ SERCOS® is a registered trade mark of SERCOS International e.V. This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade mark holder or any of its products. Compliance to this profile does not require use of the registered trade mark SERCOS®. Use of the registered trade mark SERCOS® requires permission of the trade mark holder.

IEC 61800-7-301, IEC 61800-7-302, IEC 61800-7-303 and IEC 61800-7-304 specify how the profile types 1, 2, 3 and 4 are mapped to different network technologies (such as CANopen⁵, CC-Link IE⁶ Field Network⁶, EPA⁷, EtherCAT⁸, Ethernet Powerlink⁹, DeviceNet¹⁰, ControlNet¹¹, EtherNet/IP¹², PROFIBUS¹³, PROFINET¹⁴ and SERCOS[®]).

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- ⁵ CANopen[®] is a registered trade mark of CAN in Automation e.V. (CiA). This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade mark holder or any of its products. Compliance to this profile does not require use of the registered trade mark CANopen[®]. Use of the registered trade mark CANopen[®] requires permission of CAN in Automation e.V. (CiA). CANopen[®] is an acronym for Controller Area Network *open* and is used to refer to EN 50325-4.
 - ⁶ CC-Link IE[®] Field Network is a registered trade mark of Mitsubishi Electric Corporation. This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade mark holder or any of its products. Compliance to this profile does not require use of the registered trade mark CC-Link IE[®] Field Network. Use of the registered trade mark CC-Link IE[®] Field Network requires permission of Mitsubishi Electric Corporation.
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 - ¹³ PROFIBUS is a trade name of PROFIBUS & PROFINET International. This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade name holder or any of its products. Compliance to this profile does not require use of the trade name PROFIBUS. Use of the trade name PROFIBUS requires permission of PROFIBUS & PROFINET International.
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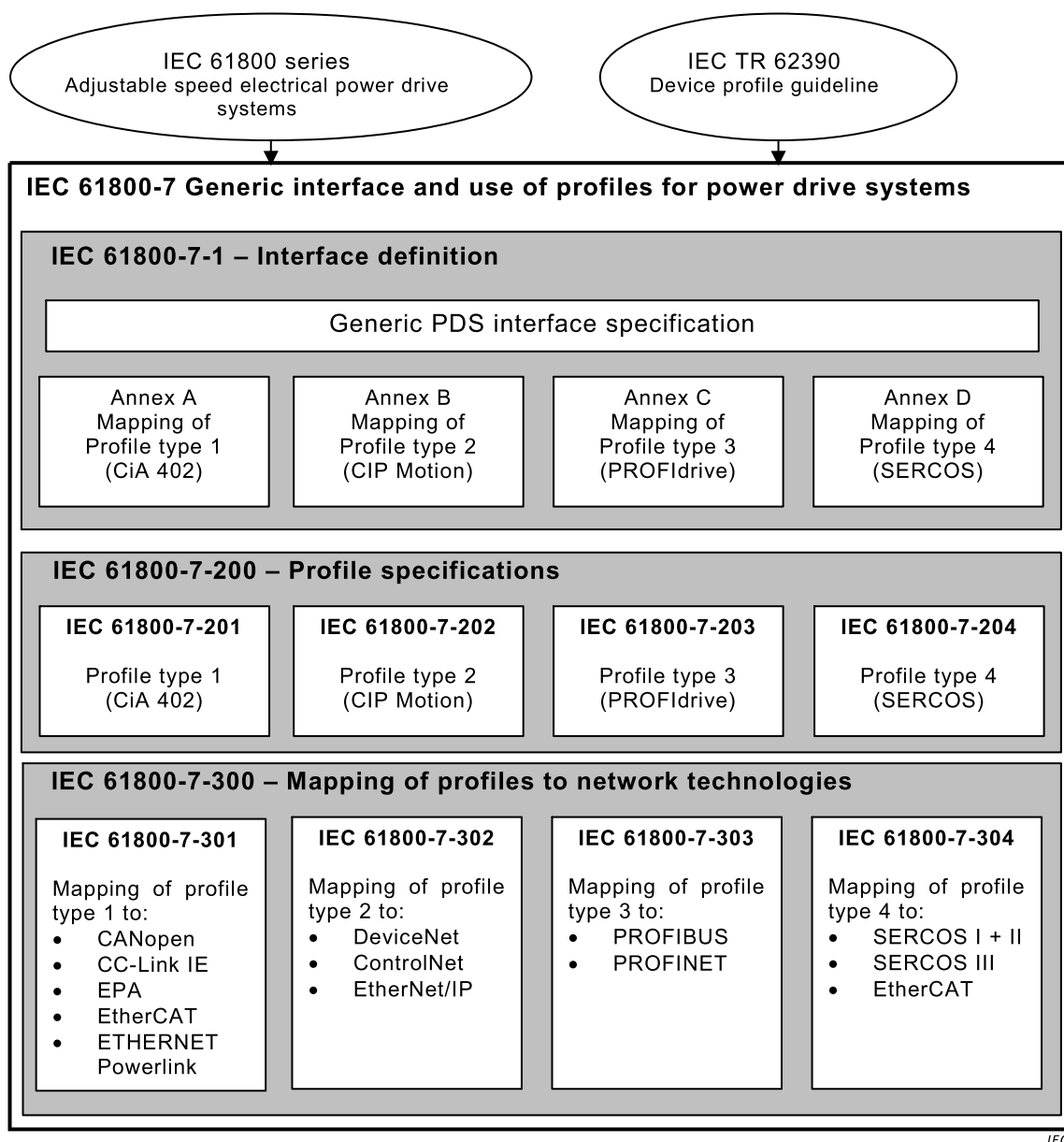


Figure 1 – Structure of IEC 61800-7

0.2 Patent declaration

The International Electrotechnical Commission (IEC) draws attention to the fact that it is claimed that compliance with this document may involve the use of a patent concerning the following. This patent is held by its inventors under license to ODVA, Inc:

Publication / Application serial number	Holder	Title
US 7,983,769 EP 1659465	[ODVA]	Time stamped motion control network protocol that enables balanced single cycle timing and utilization of dynamic data structures

IEC takes no position concerning the evidence, validity and scope of this patent right.

ODVA and the holder of this patent right have assured the IEC that ODVA is willing to negotiate licences either free of charge or under reasonable and non-discriminatory terms and

conditions with applicants throughout the world. In this respect, the statement of ODVA and the holder of this patent right is registered with IEC. Information may be obtained from:

[ODVA]	ODVA, Inc. 2370 East Stadium Boulevard #1000 Ann Arbor, Michigan 48104 USA Attention: Office of the Executive Director email: odva@odva.org
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Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights other than those identified above. IEC shall not be held responsible for identifying any or all such patent rights.

ISO (www.iso.org/patents) and IEC (<http://patents.iec.ch>) maintain on-line data bases of patents relevant to their standards. Users are encouraged to consult the data bases for the most up to date information concerning patents.

ADJUSTABLE SPEED ELECTRICAL POWER DRIVE SYSTEMS –

Part 7-202: Generic interface and use of profiles for power drive systems – Profile type 2 specification

1 Scope

This part of IEC 61800 specifies profile type 2 (CIP Motion™) for Power Drive Systems (PDS). Profile type 2 can be mapped onto different communication network technologies.

The functions specified in this part of IEC 61800-7 are not intended to ensure functional safety. This requires additional measures according to the relevant standards, agreements and laws.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60204-1, *Safety of machinery – Electrical equipment of machines – Part 1: General requirements*

IEC 61158-4-2:2014, *Industrial communication networks – Fieldbus specifications – Part 4-2: Data-link layer protocol specification – Type 2 elements*

IEC 61158-5-2:2014, *Industrial communication networks – Fieldbus specifications – Part 5-2: Application layer service definition – Type 2 elements*

IEC 61158-6-2:2014, *Industrial communication networks – Fieldbus specifications – Part 6-2: Application layer protocol specification – Type 2 elements*

IEC 61588:2009, *Precision clock synchronization protocol for networked measurement and control systems*

IEC 61800-7-1:2015, *Adjustable speed electrical power drive systems – Part 7-1: Generic interface and use of profiles for power drive systems – Interface definition*

IEEE Std 112-2004, *IEEE Standard Test Procedure for Polyphase Induction Motors and Generators*

3 Terms, definitions and abbreviated terms

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1.1

algorithm

completely determined finite sequence of operations by which the values of the output data can be calculated from the values of the input data

[SOURCE: IEC 61800-7-1:2015, 3.2.1]

3.1.2

application

software functional element specific to the solution of a problem in industrial-process measurement and control

Note 1 to entry: An application may be distributed among resources, and may communicate with other applications.

[SOURCE: IEC 61800-7-1:2015, 3.2.2]

3.1.3

application mode

type of application that can be requested from a PDS

Note 1 to entry: The different application modes reflect the control loop for torque control, velocity control, position control or other applications such as homing.

[SOURCE: IEC 61800-7-1:2015, 3.3.1.2]

3.1.4

attribute

property or characteristic of an entity

[SOURCE: IEC 61800-7-1:2015, 3.2.3]

3.1.5

axis

logical element inside an automation system (e.g. a motion control system) that represents some form of movement

Note 1 to entry: Axes can be rotary or linear, physical or virtual, controlled or simply observed.

[SOURCE: IEC 61800-7-1:2015, 3.2.4]

3.1.6

bus regulator

any method used to limit the rise in DC bus voltage level that occurs when decelerating a motor

3.1.7

CIP Motion™¹⁵

extensions to the CIP services and protocol to support motion control over CIP networks

[SOURCE: IEC 61800-7-1:2015, 3.3.3.1]

¹⁵ CIP Motion™ and CIP Sync™ are trade marks of ODVA, Inc. This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade mark holder or any of its products. Compliance to this profile does not require use of the trade marks CIP Motion™ or CIP Sync™. Use of the trade marks CIP Motion™ or CIP Sync™ requires permission of ODVA, Inc.

3.1.8**CIP Motion™ controller**

CIP compliant controller containing a Motion Control Axis Object that can interface to a CIP Motion device via a CIP Motion I/O Connection

Note 1 to entry: A description of the Motion Control Axis Object is beyond the scope of IEC 61800-7.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.2]

3.1.9**CIP Motion™ device**

CIP compliant device containing one or more Motion Device Axis Object instances that can communicate to a CIP Motion controller via a CIP Motion I/O Connection

EXAMPLE: A CIP Motion drive is a particular case of a CIP Motion device.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.3]

3.1.10**CIP Motion™ drive profile**

collection of objects used to implement a CIP Motion drive device that includes the Motion Device Axis Object, as well as standard support objects like the Identity Object and the Time Sync Object

Note 1 to entry: The Device Type assigned to the CIP Motion drive profile is 25_{hex}.

3.1.11**CIP Motion™ I/O Connection****CIP Motion™ Connection**

periodic bi-directional, class 1, CIP connection between a controller and a drive that is defined as part of the CIP Motion specification

[SOURCE: IEC 61800-7-1:2015, 3.3.3.4]

3.1.12**CIP Sync™¹⁵**

extensions to the CIP services and protocol to encapsulate IEC 61588:2009 time synchronization functionality over a CIP Network

Note 1 to entry: See Time Sync Object in IEC 61158-5-2 and IEC 61158-6-2.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.5]

3.1.13**class**

description of a set of objects that share the same attributes, operations, methods, relationships, and semantics

[SOURCE: IEC 61800-7-1:2015, 3.2.5]

3.1.14**closed loop**

methods of control where there is a feedback signal of some kind that is used to drive the actual dynamics of the motor to match the commanded dynamics by servo action

Note 1 to entry: In most cases, there is a literal feedback device to provide this signal, but in some cases, the signal is derived from the motor excitation (i.e. sensorless operation).

3.1.15

commands

set of commands from the application control program to the PDS to control the behavior of the PDS or functional elements of the PDS

Note 1 to entry: The behavior is reflected by states or operating modes.

Note 2 to entry: The different commands may be represented by one bit each.

[SOURCE: IEC 61800-7-1:2015, 3.3.1.3]

3.1.16

control

purposeful action on or in a process to meet specified objectives

[SOURCE: IEC 61800-7-1:2015, 3.2.6]

3.1.17

control device

physical unit that contains – in a module/subassembly or device – an application program to control the PDS

[SOURCE: IEC 61800-7-1:2015, 3.2.7]

3.1.18

converter

device that generally converts AC input to DC output

Note 1 to entry: A converter is also commonly called the Drive Power Supply. In the context of a drive system, the converter is responsible for converting AC main input into DC bus power.

3.1.19

Cyclic Data Block

high priority real-time data block that is transferred by a CIP Motion Connection on a periodic basis

3.1.20

data type

set of values together with a set of permitted operations

[SOURCE: IEC 61800-7-1:2015, 3.2.8]

3.1.21

device

field device

<function blocks> networked independent physical entity of an industrial automation system capable of performing specified functions in a particular context and delimited by its interfaces

[SOURCE: IEC 61800-7-1:2015, 3.2.9]

3.1.22

device

field device

<system integration> entity that performs control, actuating and/or sensing functions and interfaces to other such entities within an automation system

[SOURCE: IEC 61800-7-1:2015, 3.2.10]

3.1.23**device profile**

representation of a device in terms of its parameters, parameter assemblies and behaviour according to a device model that describes the data and behaviour of the device as viewed through a network, independent from any network technology

[SOURCE: IEC 61800-7-1:2015, 3.2.11]

3.1.24**drive**

device designed to control the dynamics of a motor

3.1.25**Event Data Block**

medium priority real-time data block that is transferred by a CIP Motion Connection only after a specified event occurs

Note 1 to entry: Registration and marker input transitions are typical drive events.

3.1.26**functional element**

entity of software or software combined with hardware, capable of accomplishing a specified function of a device

Note 1 to entry: A functional element has an interface, associations to other functional elements and functions.

Note 2 to entry: A functional element can be made out of function block(s), object(s) or parameter list(s).

[SOURCE: IEC 61800-7-1:2015, 3.2.13]

3.1.27**input data**

data transferred from an external source into a device, resource or functional element

[SOURCE: IEC 61800-7-1:2015, 3.2.14]

3.1.28**interface**

shared boundary between two entities defined by functional characteristics, signal characteristics, or other characteristics as appropriate

[SOURCE: IEC 61800-7-1:2015, 3.2.15]

3.1.29**inverter**

device that generally converts DC input to AC output

Note 1 to entry: An inverter is also commonly called the Drive Amplifier. In the context of a drive system, the inverter is responsible for controlling the application of DC bus power to an AC motor.

3.1.30**model**

mathematical or physical representation of a system or a process, based with sufficient precision upon known laws, identification or specified suppositions

[SOURCE: IEC 61800-7-1:2015, 3.2.17]

3.1.31

motion

any aspect of the dynamics of an axis

Note 1 to entry: In the context of this part of IEC 61800-7, it is not limited to servo drives but encompasses all forms of drive based motor control.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.8]

3.1.32

Motion Control Axis Object

object that defines the attributes, services, and behavior of a controller based axis according to the CIP Motion specification

3.1.33

Motion Device Axis Object

object that defines the attributes, services, and behavior of a motion device based axis according to the CIP Motion specification

Note 1 to entry: This object includes Communications, Device control, and Basic drive FE elements as defined in IEC 61800-7.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.9]

3.1.34

open loop

methods of control where there is no application of feedback to force the actual motor dynamics to match the commanded dynamics

EXAMPLE Examples of open loop control are stepper drives and variable frequency drives.

3.1.35

operating mode

characterization of the way and the extent to which the human operator intervenes in the control equipment

[SOURCE: IEC 61800-7-1:2015, 3.2.18]

3.1.36

output data

data originating in a device, resource or functional element and transferred from them to external systems

[SOURCE: IEC 61800-7-1:2015, 3.2.19]

3.1.37

parameter

data element that represents device information that can be read from or written to a device, for example through the network or a local HMI

Note 1 to entry: A parameter is typically characterized by a parameter name, data type and access direction.

[SOURCE: IEC 61800-7-1:2015, 3.2.20]

3.1.38

profile

representation of a PDS interface in terms of its parameters, parameter assemblies and behavior according to a communication profile and a device profile

[SOURCE: IEC 61800-7-1:2015, 3.2.21, modified – Note 1 to entry is deleted]

3.1.39

read

get

operation that involves the retrieving of an attribute value from the perspective of the controller side of the interface

3.1.40

registration

high speed record of motion axis position triggered by an event

3.1.41

Service Data Block

lower priority real-time data block associated with a service message from the controller that is transferred by a CIP Motion Connection on a periodic basis

Note 1 to entry: Service data includes service request messages to access Motion Device Axis Object attributes or perform various drive diagnostics.

3.1.42

set-point

value or variable used as output data of the application control program to control the PDS

Note 1 to entry: The value sent to the drive is used to directly control some aspect of the motor dynamics, which includes (but is not limited to) position, velocity, acceleration, and torque.

[SOURCE: IEC 61800-7-1:2015, 3.3.1.5, modified – Note 1 to entry is added]

3.1.43

shunt regulator

specific bus regulator method that switches the DC bus across a power dissipating resistor to dissipate the regenerative power of a decelerating motor

3.1.44

status

set of information from the PDS to the application control program reflecting the state or mode of the PDS or a functional element of the PDS

Note 1 to entry: The different status information may be coded with one bit each.

[SOURCE: IEC 61800-7-1:2015, 3.3.1.6]

3.1.45

synchronized

condition where the local clock value on the drive is locked onto the master clock of the distributed System Time

Note 1 to entry: When synchronized, the drive and controller devices may utilise time stamps associated with CIP Motion Connection data.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.11]

3.1.46

System Time

absolute time value as defined in the CIP Sync specification in the context of a distributed time system where all devices have a local clock that is synchronized with a common master clock

Note 1 to entry: In the context of CIP Motion, System Time is a 64-bit integer value in units of nanoseconds with a value of 0 corresponding to the date 1970-01-01.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.12]

3.1.47

thermal model

any algorithm that attempts to model the thermal behavior of the associated device during operation

3.1.48

time offset

System Time offset value associated with the CIP Motion Connection data that is associated with source device

Note 1 to entry: The System Time offset is a 64-bit offset value that is added to a device's local clock to generate System Time for that device

3.1.49

time stamp

System Time stamp value associated with the CIP Motion Connection data that conveys the absolute time when the associated data was captured, or that can also be used to determine when the associated data shall be applied

Note 1 to entry: CIP time stamps are always in the context of a distributed time system where all nodes on the CIP control network have clocks that are synchronized with a master clock source using CIP Sync.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.13]

3.1.50

type

hardware or software element which specifies the common attributes shared by all instances of the type

[SOURCE: IEC 61800-7-1:2015, 3.2.23]

3.1.51

variable

software entity that may take different values, one at a time

Note 1 to entry: The values of a variable as well as of a parameter are usually restricted to a certain data type.

[SOURCE: IEC 61800-7-1:2015, 3.2.25]

3.1.52

variable frequency drive

VFD

class of drive products that seek to control the speed of a motor, typically an induction motor, through a proportional relationship between drive output voltage and commanded output frequency

Note 1 to entry: Variable frequency drives are therefore sometimes referred to as a Volts/Hertz drives.

Note 2 to entry: The English abbreviation VFD is also used in French.

3.1.53

vector drive

class of drive products that seek to control the dynamics of a motor via closed loop control which includes, but is not limited to, closed loop control of both torque and flux vector components of motor stator current relative to the rotor flux vector

3.1.54**write**

set

operation that involves the setting of an attribute to a specified value from the perspective of the controller side of the interface

3.2 Abbreviated terms

Ack	acknowledge
Act	actual
Blk	block
CIP™ ¹⁶	Common Industrial Protocol (see IEC 61158 Type 2, IEC 61784-1 and IEC 61784-2 Communication Profile Family 2)
Cmd	command
Cyc	cyclic
HMI	human machine interface
ID	identifier
impl.	implementation
I/O	Input/Output
Opt.	optional
PDS	power drive system
Req.	required
RMS	root mean square
r/min	revolutions per minute
Sync'd	synchronized
TCP	Transmission Control Protocol (see IETF RFC 793)
UDP	User Datagram Protocol (see IETF RFC 768)
VFD	variable frequency drive

4 Overview**4.1 General**

CIP Motion devices control, monitor or support the motion of one or more moving component of a machine. Machine motion is typically generated by rotary or linear motion actuators, i.e.: motors, and monitored by feedback devices. Each motor is typically driven by a power structure and a motion control algorithm that comprise a CIP Motion Drive Device Type.

Drive functionality supported by the CIP Motion drive device profile can be applied to a variety of motor technologies, and can range from very simple “open loop” variable frequency drives to sophisticated “closed loop” vector controlled servo drives. In either case, motion is controlled via a command reference that can be configured for position control, velocity control, acceleration control, or current/torque control. The CIP Motion drive device profile also supports position, velocity and acceleration monitoring through multiple feedback, as does the CIP Motion encoder device profile.

¹⁶ CIP™ is a trade mark of ODVA, Inc. This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade mark holder or any of its products. Compliance to this profile does not require use of the trade mark CIP™. Use of the trade mark CIP™ requires permission of ODVA, Inc.

All the attributes, services, and state behavior of a CIP Motion device are encapsulated in one or more Motion Device Axis Object instances. In addition to motion control and feedback monitoring functionality, the Motion Device Axis Object includes support for event monitoring, such position capture on a Registration event, and DC Bus management associated with a Power Converter. A CIP Motion device can manifest any combination of these functions to create various classes of CIP Motion compliant devices ranging from full featured servo drives, to CIP Motion Encoders, to standalone Power Converters, differentiated by the CIP Motion Device Type.

This CIP Motion drive profile specification and the associated Motion Device Axis Object specification define the interface, the specific attributes, and command behaviors required to support motion control when connected to a CIP Motion compliant controller through a CIP network.

Of the applicable CIP Networks, EtherNet/IP™ is the network of choice for high performance, synchronized multi-axis control. Other CIP networks such as ControlNet™ and DeviceNet™ could be applied to lower performance non-synchronized motion device applications such as simple variable frequency drives, velocity loop drives, and indexing drives.

4.2 Control modes

4.2.1 General

The Motion Device Axis Object covers the behavior of various motion control system devices that includes feedback devices, drive devices, standalone converters and motion I/O devices. For drive devices, the Motion Device Axis Object covers a wide range of drive types from simple variable frequency (V/Hz) drives, to sophisticated position control servo drives, with or without integral converters. Indeed, many commercial drive products can be configured to operate in any one of these different motion control modes depending on the specific application requirements. The attributes of the Motion Device Axis Object are therefore organized to address this broad range of functionality and the framework for this organization is described in 4.2.

IEC 61800-7-1 and this part of the IEC 61800-7 series are organised around the general philosophy that position control is the highest form of dynamic control. That is, position control implies velocity control, and velocity control implies acceleration control. Acceleration is related to torque or force by the inertia or mass of the load, respectively, acceleration control implies torque control. And finally, since motor torque or force is generally related to motor current by a torque or force constant, respectively, torque control implies current control. The torque or force constant can be a function of the motor magnets as in a Permanent Magnet motor, or the induced flux of an Induction motor.

Since acceleration, torque/force, and current are generally related by a constant, these terms are sometimes used interchangeably in the industry. For example IEC 61800-7-1 refers to a torque control loop rather than a current control loop. This specification attempts to differentiate between these control properties where applicable. This is particularly useful when the relationship between them is not static, such as when inertia/mass changes with position or time, or when the torque/force constant changes due to temperature change or motor flux variation.

4.2.2 Control methods

Within this basic control paradigm, there is latitude for different control methods, both closed loop and open loop. By closed loop, it is generally implied that there is a feedback signal that is used to drive the actual dynamics of the motor to match the commanded dynamics by servo action. In most cases, there is a literal feedback device to provide this signal, and in some cases, the signal is derived from the motor excitation (i.e. sensorless/encoderless operation). By open loop, it is implied that there is no application of feedback to force the actual dynamics to match the commanded dynamics. While precision and performance are the hallmarks of closed loop control, simplicity and economy are the hallmarks of open loop control.

4.2.3 Control nomenclature

Finally, as evident in the description above, linear and rotary control applications can affect the control nomenclature. While rotary applications speak of torque and inertia, linear applications speak of force and mass. For the purposes of this part of the IEC 61800-7 series, when referring to rotary nomenclature, the defined behavior can generally be applied to linear applications by substituting the terms, force for torque and mass for inertia. With that understanding, Figure 2 to Figure 8 follow the IEC 61800-7-1 nomenclature using torque rather than force without loss of generality.

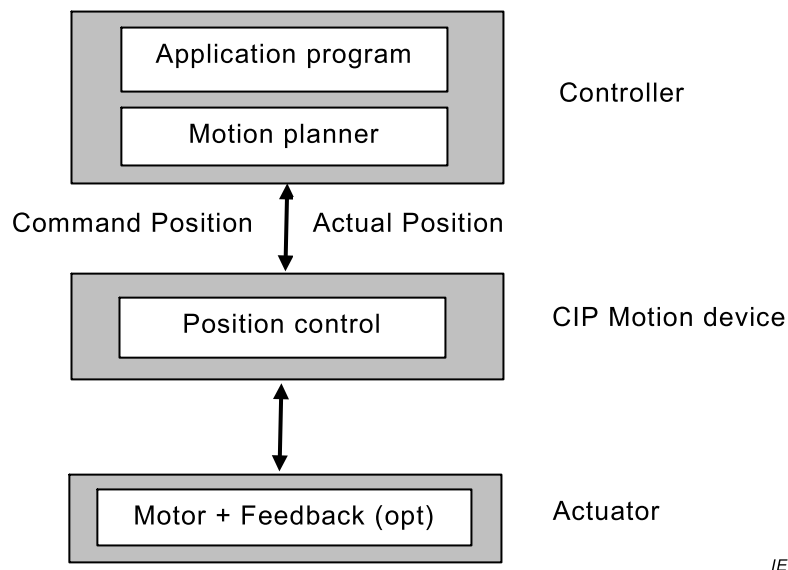
4.2.4 Position control

4.2.4.1 General concepts

In position control application mode, either the application control program (command execution function) or the motion planner (move trajectory control function) provide a set-point value to the CIP Motion device via the cyclic data connection. The position control method can be either open loop or closed loop.

4.2.4.2 Open loop position control

A device configured for open loop position control applies to a class of drive devices called stepper drives. This type of drive is illustrated in Figure 2.



IEC

Figure 2 – Open loop position control

A feedback device for this configuration is optional. In the absence of a feedback device, actual position can be estimated by the drive and returned to the controller.

4.2.4.3 Closed loop position control

A motor control device configured for closed loop position control is traditionally referred to as position loop drive or position servo drive. A position servo drive implies an inner velocity and torque control loop as shown in Figure 3. The presence of the torque/current control loop sometimes results in this kind of drive being referred to as a vector drive.

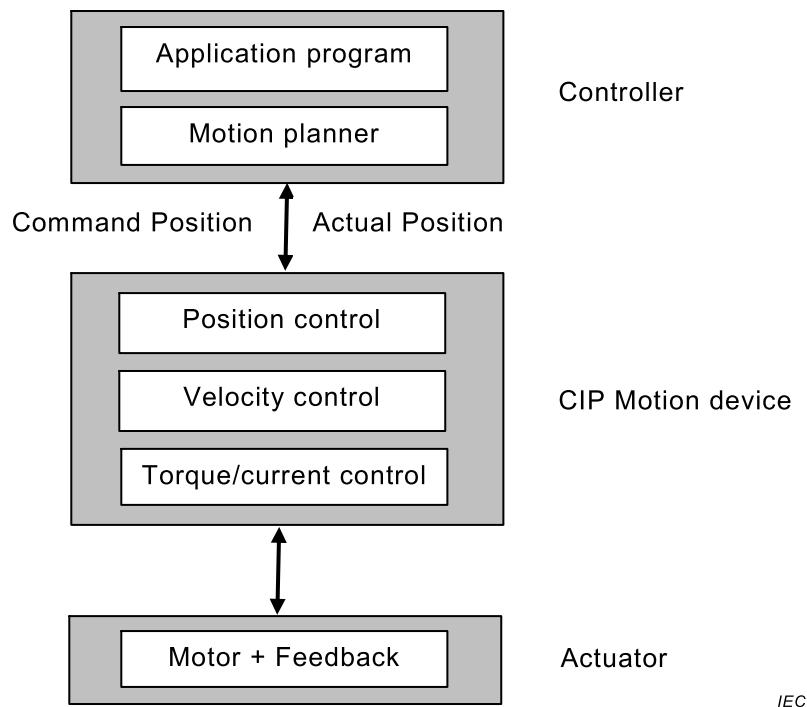


Figure 3 – Closed loop position control

A feedback device for this configuration is generally required to achieve good positioning accuracy. The feedback device may also be used to return Actual Velocity, and Actual Acceleration data to the controller via the cyclic data connection.

In addition to Command Position, the controller can pass Command Velocity and Command Acceleration for the purposes of forward control.

4.2.5 Velocity control

4.2.5.1 General concepts

In velocity control application mode, the application control program and motion planner provide a set-point value to the CIP Motion device via the cyclic data connection. The velocity control method can be either open loop or closed loop.

4.2.5.2 Open loop velocity control

A motor control device configured for open loop velocity control is traditionally referred to as variable frequency, or V/Hz, or VFD, drive. This type of drive is illustrated in Figure 4.

A feedback device for this configuration is optional. In the absence of a feedback device, actual velocity can be estimated by the drive and returned to the controller.

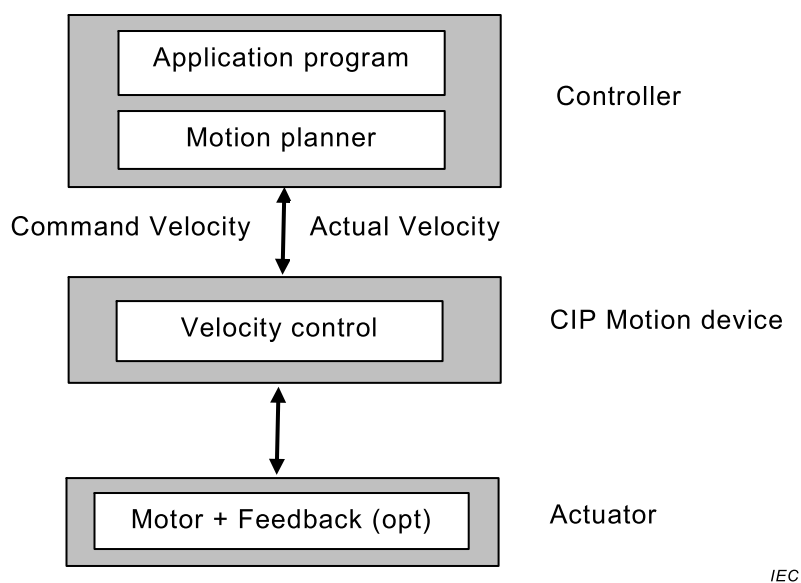


Figure 4 – Open loop velocity control

4.2.5.3 Closed loop velocity control

A motor control device configured for closed loop velocity control is traditionally referred to as velocity loop drive or velocity servo drive. A closed loop velocity control drive implies an inner torque/current control loop (see Figure 5) and therefore is sometimes referred to as a vector drive.

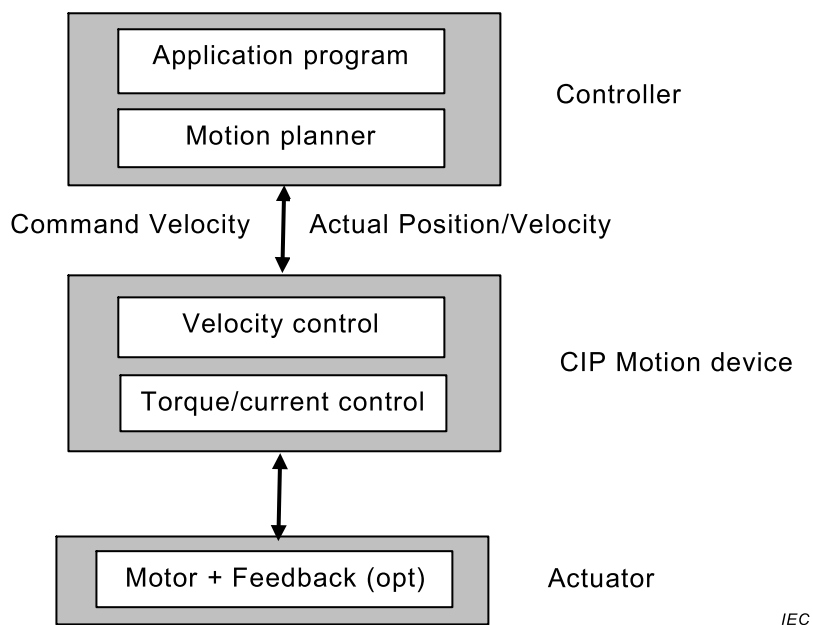


Figure 5 – Closed loop velocity control

A feedback device for the velocity loop drive configuration is optional. Tighter speed regulation is achieved when using a feedback device, particularly at low speed. When the feedback device is included, it may be used to return actual position, velocity, and acceleration data to the controller via the cyclic data connection. When the feedback device is not included, only estimated velocity can typically be returned to the controller.

In addition to Command Velocity, the controller can also pass Command Acceleration for the purposes of forward control.

4.2.6 Acceleration control

While neither a mainstream control mode in the industry, nor mentioned in IEC 61800-7-1, the acceleration control mode is included here to complete the dynamic progression from velocity control to torque control and because the Motion Device Axis Object can support an Acceleration Command, potentially derived from the controller's motion planner. In the acceleration control mode, the application control program and motion planner provide acceleration set-point values to the CIP Motion device via the cyclic data connection. The drive converts the acceleration set-point into a torque command using the estimated system inertia. Acceleration control works in concert with the inner torque/current control loop as shown in Figure 6.

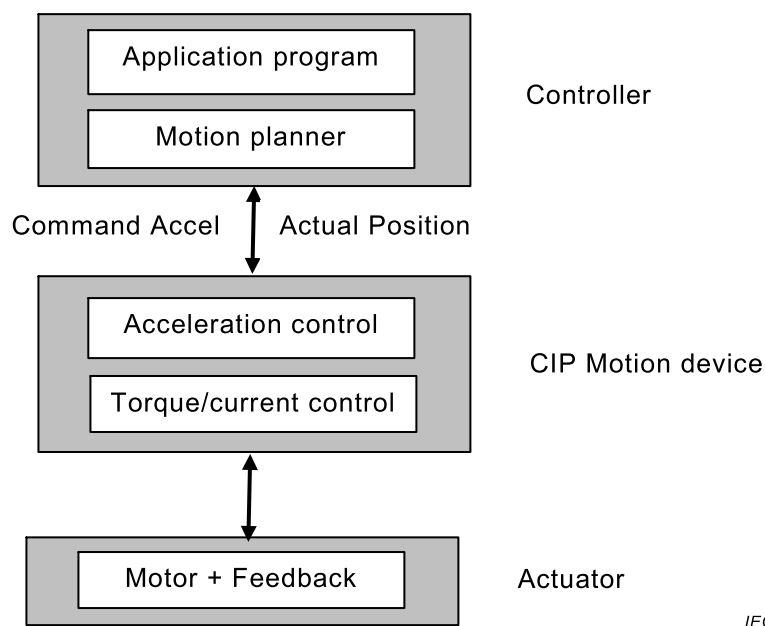


Figure 6 – Acceleration control

A feedback device for the acceleration control configuration is mandatory and may be used to return actual position, velocity, and acceleration data to the controller via the cyclic data connection.

4.2.7 Torque control

In torque control application mode, the application control program or the motion planner provide torque set-point values to the device via the cyclic data connection (see Figure 7). Because motor current and motor torque are generally related by a torque constant, K_t , torque control is often synonymous with current control.

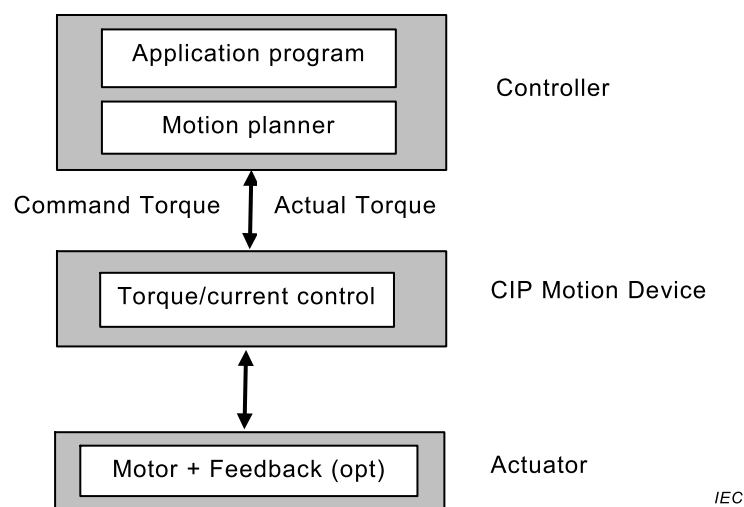


Figure 7 – Torque control

A position feedback device for this control mode is optional. If a feedback device is present, it may be used to return actual position, velocity, and acceleration data to the controller via the cyclic data connection.

4.2.8 No Control

The Motion Device Axis Object supports a “No Control” application mode where there is no dynamic motor control function. This mode is often used to support “feedback only” or “master feedback” functionality where a particular feedback channel in a CIP Motion drive device is serving as a master feedback source to the rest of the control system. This could also apply to integrated CIP Motion Encoder device types where the CIP Motion interface is applied directly to an Encoder.

In this “No Control” mode of operation, no set-point value is supplied to the CIP Motion device via the cyclic data connection, but actual position, velocity, and acceleration can be supplied by the device to the controller via the Cyclic Data Channel, if applicable. The No Control mode for Feedback Only functionality is illustrated in Figure 8.

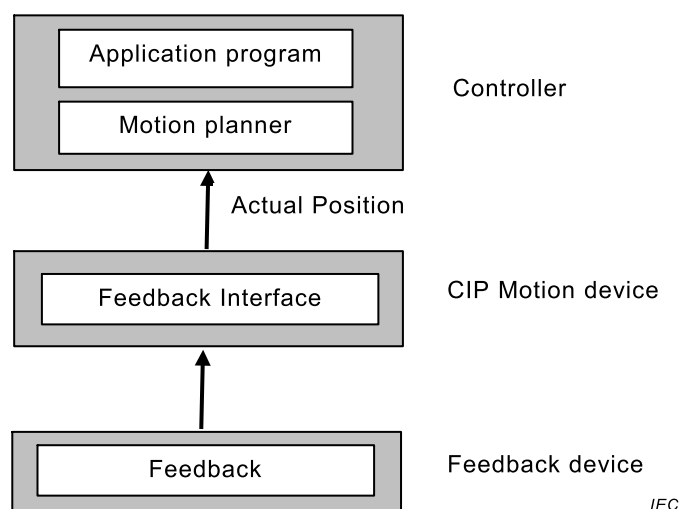


Figure 8 – No Control (Feedback Only)

No Control mode also applies to other CIP Motion device types, such as standalone Bus Power Converters and dedicated motion I/O device types. Since there is no feedback channel associated with these device types, no actual position is returned to the controller.

5 Data types

5.1 Data type overview

Table 1 shows references of data types used in this profile and the related definitions.

Table 1 – Data types

Data types used in CIP Motion	Reference to definition
BOOL	Boolean (see 5.3.1.1.2 of IEC 61158-5-2:2014)
SINT	Integer8 (see 5.3.1.4.2.2 of IEC 61158-5-2:2014)
INT	Integer16 (see 5.3.1.4.2.4 of IEC 61158-5-2:2014)
DINT	Integer32 (see 5.3.1.4.2.6 of IEC 61158-5-2:2014)
LINT	Integer64 (see 5.3.1.4.2.8 of IEC 61158-5-2:2014)
USINT	Unsigned8 (see 5.3.1.4.3.2 of IEC 61158-5-2:2014)
UINT	Unsigned16 (see 5.3.1.4.3.4 of IEC 61158-5-2:2014)
UDINT	Unsigned32 (see 5.3.1.4.3.6 of IEC 61158-5-2:2014)
ULINT	Unsigned64 (see 5.3.1.4.3.8 of IEC 61158-5-2:2014)
REAL	Float32 (see 5.3.1.4.1.2 of IEC 61158-5-2:2014)
LREAL	Float64 (see 5.3.1.4.1.4 of IEC 61158-5-2:2014)
SWORD/BYTE	Bitstring8 (see 5.3.1.2.2 of IEC 61158-5-2:2014)
WORD	Bitstring16 (see 5.3.1.2.4 of IEC 61158-5-2:2014)
DWORD	Bitstring32 (see 5.3.1.2.6 of IEC 61158-5-2:2014)
LWORD	Bitstring64 (see 5.3.1.2.8 of IEC 61158-5-2:2014)

5.2 Conventions

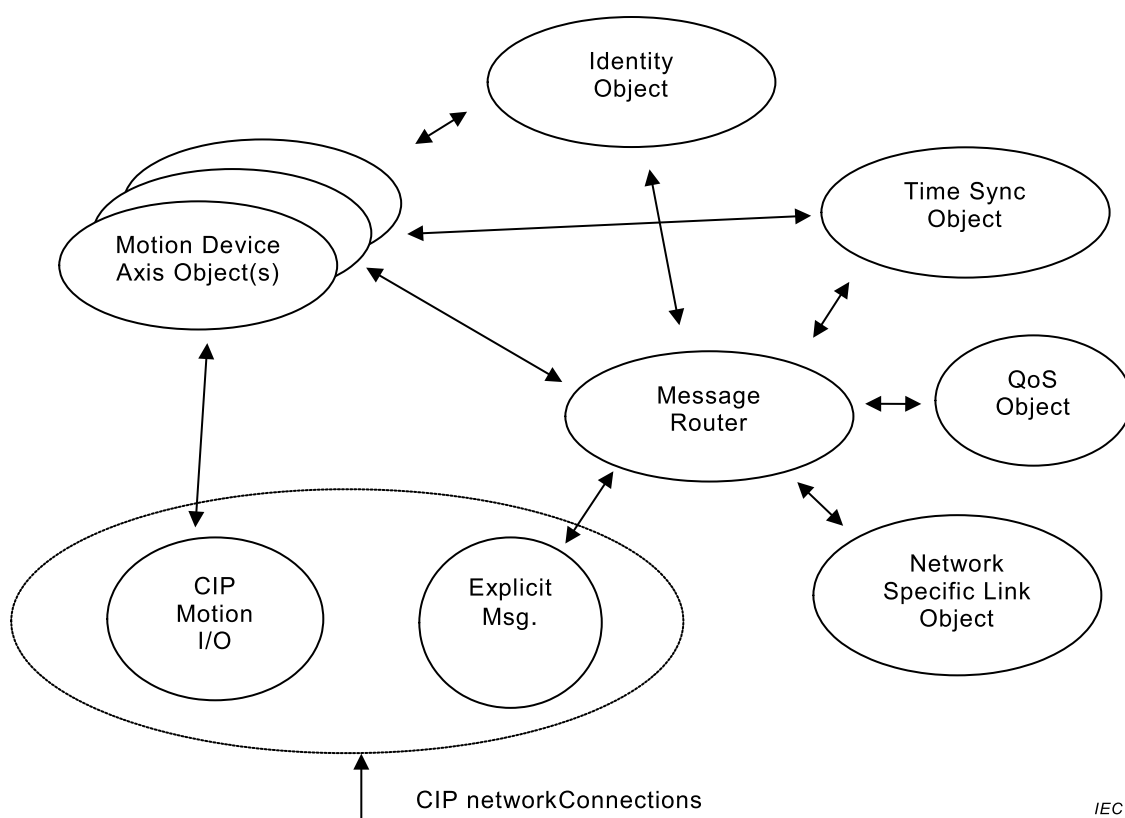
Hexadecimal numbers in this part of the IEC 61800-7 series may be represented by nn_{hex} (if their exact data type is not defined) or $0xnn$, $0xn\text{nnnn}$ (if their data type is specified).

6 CIP Motion drive profile

6.1 Object model

6.1.1 Object overview

The object model in Figure 9 represents a CIP Motion device.



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Figure 9 – Object Model for a CIP Motion device

Table 2 indicates:

- the object classes present in this device,
- whether or not the class is required,
- the number of instances present in each class.

Table 2 – Objects present in a CIP Motion device

Object class	Optional or required	Number of instances
CIP common required objects	Required	See IEC 61158-5-2 and IEC 61158-6-2
Motion Device Axis Object (0x42)	Required	1 per axis ^b
Time Sync Object (0x43)	Optional	1
QoS Object (0x48)	Conditional ^a	1
^a Required for EtherNet/IP		
^b An axis is an abstraction associated with a moving machine component, a converter power structure or a motion I/O device.		

Refer to IEC 61158-5-2, IEC 61158-6-2 and 6.4.6.1 for more details about these objects.

6.1.2 Object description

The object model in Figure 9 shows the main functional components of the CIP Motion device profile.

This object model also illustrates the use of multiple instances of a Motion Device Axis Object to implement a multi-axis motion device, such as a multi-axis drive. Each Motion Device Axis

Object instance governs the behavior of the associated axis. In this device profile, the term “axis” is synonymous with a “Motion Device Axis Object instance”. The implemented content of the Motion Device Axis Object instances is dictated by the specific CIP Motion Device Type according to Table 3.

Table 3 – Motion Device Axis Object content by Device Type

Device type	Motion Device Axis Object content
CIP Motion Drive	Support for one or more of F, P, V, T Device Function Codes
CIP Motion Encoder	Support for E, but no support for F, P, V, T Device Function Codes
CIP Motion Converter	Support for B Device Function Code only
CIP Motion I/O	Support for I/O Device Function Code only

A single bi-directional I/O connection to the Motion Device Axis Object class instance provides a cyclic data path between the controller and each individual Motion Device Axis Object instance. This connection passes on a special data structure whose self-defining format can be used to transfer cyclic, event, and service related data.

An optional Time Sync Object is included in the object model to facilitate accurate time synchronization between CIP Motion controllers and drive devices for high performance motion control. For lower performance drives such as V/Hz (VFD) drives, or simple velocity servo drives, the Time Sync Object is not necessary for drive operation.

A CIP Motion I/O Connection is a standard, cyclic, bidirectional CIP I/O connection whose packets can vary in size, based on the formats outlined in 6.4.2.2.

6.2 How objects affect behavior

The objects for this device affect the device’s behavior as shown in Table 4.

Table 4 – Object effect on behavior

Object class	Effect on behavior
CIP common required objects	See IEC 61158-5-2 and IEC 61158-6-2
Motion Device Axis Object	Provides dynamic control interface to drive, motor, and feedback components that comprise an axis.
Time Sync Object	Provide absolute time synchronization services between devices on the control network.
QoS Object	Provides configuration information associated with Ethernet QoS (Quality of Service) function.

6.3 Defining object interfaces

Objects supported for the CIP Motion device have the interfaces listed in Table 5.

Table 5 – Object interfaces

Object	Interface
CIP common required objects	See IEC 61158-5-2 and IEC 61158-6-2
Motion Device Axis Object	Message Router, CIP Motion I/O Connection, Time Sync Object, Identity Object
Time Sync Object	Message Router
QoS Object	Message Router

6.4 I/O connection messages

6.4.1 General

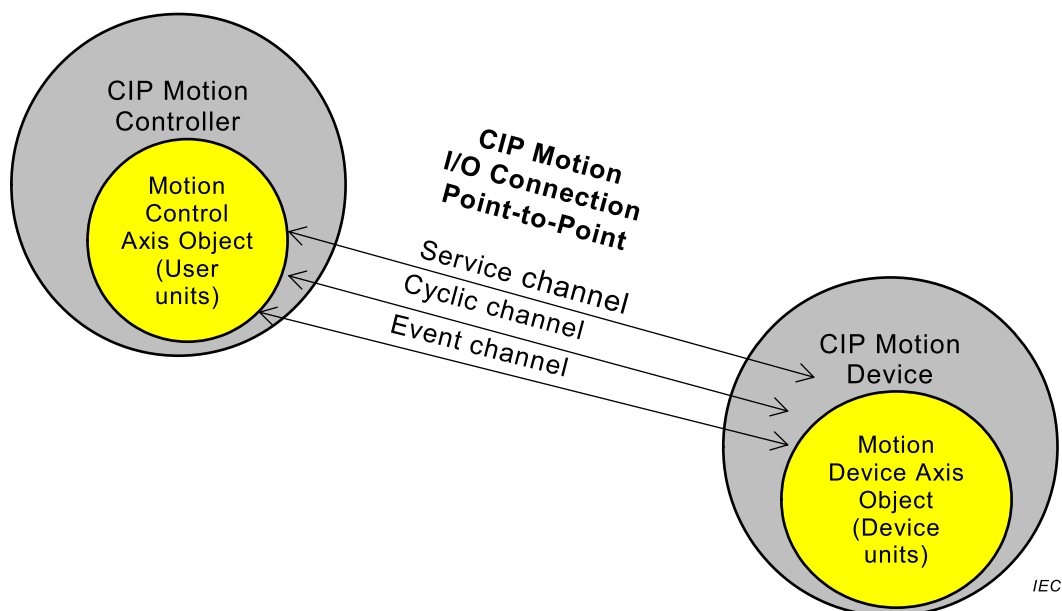
The CIP Motion device profile supports a Transport Class 1 point-to-point bi-directional I/O connection between the controller and the Motion Device Axis Object class: this I/O connection is specifically referred to as the CIP Motion I/O Connection.

The Motion Device Axis Object distributes the data in this connection to each instantiated Motion Device Axis Object instance.

6.4.2 CIP Motion I/O Connection

6.4.2.1 Overview

The following subclause specifies the CIP Motion I/O Connection format that includes the Controller-to-Device (C-to-D) Connection and the Device-to-Controller (D-to-C) Connection for bi-directional data transfer between a CIP Motion controller and a CIP Motion device, as shown in Figure 10.

**Figure 10 – CIP Motion I/O Connection model**

6.4.2.2 CIP Motion I/O Connection structure

Both CIP Motion I/O Connection data structures (Controller-to-Device and Device-to-Controller) begin with a Connection Header followed by a Time Data Block that typically includes a 64-bit time stamp. The Time Data Block is then followed by one or more Instance Data Blocks for each axis supported by the device node.

The size and contents of the Time Data Block and Instance Data Blocks vary dynamically as determined by the associated headers. This ability to vary the contents of the data blocks from update to update allows the device and controller to only send data that has changed from the last update, dramatically reducing the overall size of the typical CIP Motion Connection packet.

Each Instance Data Block within the CIP Motion I/O Connection packet consists of three sets of data blocks associated with the cyclic, event, and service data channels. From the device's perspective, these three distinct data channels have different data processing priorities as illustrated in Figure 11.

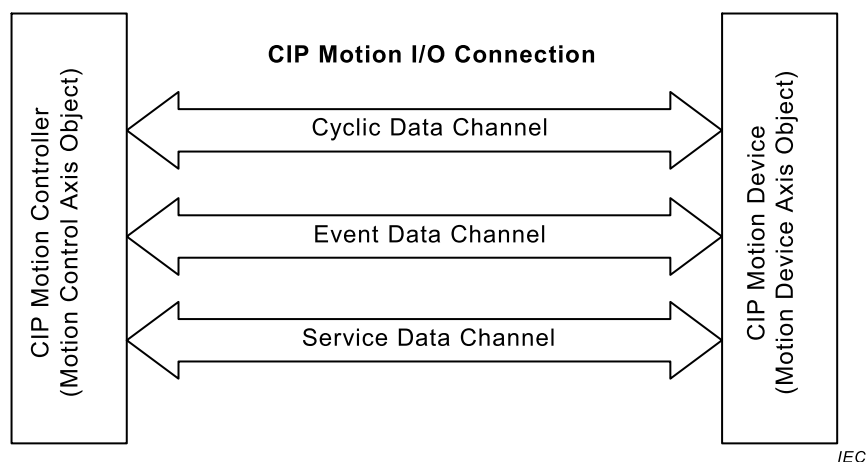


Figure 11 – CIP Motion I/O Connection channels

The specific functionality of these three data channels is as follows.

- **Cyclic Data Channel:** carries Cyclic Data Blocks that are sampled or calculated every Controller Update Period and synchronized with other nodes in the motion control system through use of distributed System Time. Cyclic data is high priority data that shall be immediately processed and applied to the device axis within one Device Update Period.
- **Event Data Channel:** carries event data associated with device event(s) (e.g. registration, homing, etc.) that have occurred within the last Controller Update Period. Event data is medium priority and shall be processed and applied within one Controller Update Period.
- **Service Data Channel:** carries data associated with service requests to read or write attribute values of the Motion Device Axis Object as part of on-line configuration and diagnostic functionality, as well as service requests to affect Motion Device Axis Object behavior as part of controller instruction execution. Service data has lowest priority and is typically buffered and processed as a background task. There is no guarantee that a service request will be processed within Controller Update Period.

Taken together, these three data channels provide a comprehensive controller to device data connection solution for industrial motion control.

6.4.2.3 I/O connection formats

An overview of the CIP Motion I/O Connection format is shown in Figure 12 and Figure 13.

Not shown in these format figures is the encapsulation associated with the Class 1 Transport that includes a Sequence Count. For a detailed description of the Class 1 Transport header refer to IEC 61158-5-2 and IEC 61158-6-2. Multi-octet data in the CIP I/O Connection data structure follows standard little-endian octet-addressing rule. In the figures, octet ordering reads left to right. The gray banners in the figures are section labels and are not part of the actual connection data structure. Unless otherwise stated, data structure elements defined as “reserved” or marked with a hyphen, “-”, shall be set to zero (0).

← 32-bit Word (octet 1 octet 2 octet 3 octet 4) →			
Controller-to-Device Connection format			
Connection Header			
Time Data Block			
Instance Data Blocks			
Connection Header and Time Data Block			
Connection Format	Format Revision	Update ID	Node Control
Instance Count	–	Last Received ID	Time Data Set
Controller Time Stamp			
Controller Time Offset			
Instance Data Block			
Instance Data Header			
Cyclic Data Block			
Cyclic Write Data Block			
Event Data Block			
Service Data Block			
Instance Data Header			
Instance Num	–	Instance Blk Size	Cyclic Blk Size
Cyc. Cmd Blk Size	Cyc. Write Blk Size	Event Blk Size	Service Blk Size
Cyclic Data Block			
Control Mode	Feedback Mode	Axis Control	Control Status
Command Data Set	Actual Data Set	Status Data Set	Command Control
Cyclic Data			
Cyclic Write Data Block			
Cyclic Write Blk ID	–	Cyclic Read Blk ID	–
Cyclic Write Data			
Event Data Block			
Event Checking Control			
Reg Data Set	Home Data Set	Watch Data Set	–
Registration Event Data			
Home Event Data			
Watch Event Data			
Event Ack. ID1	Event Ack. Status 1	Event Ack. ID2	Event Ack. Status 2
...			
Service Data Block			
Transaction ID	Service Code	–	–
Service Specific Request Data			

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Figure 12 – Controller-to-Device Connection format (Connection Point 2)

← 32-bit Word (octet 1 octet 2 octet 3 octet 4) →			
Device-to-Controller Connection format			
Connection Header			
Time Data Block			
Instance Data Blocks			
Connection Header and Time Data Block			
Connection Format	Format Revision	Update ID	Node Status
Instance Count	Node Fault/Alarm	Last Received ID	Time Data Set
Device Time Stamp			
Device Time Offset			
Lost Updates	Late Updates	–	
Timing Diagnostic Data			
Instance Data Block			
Instance Data Header			
Cyclic Data Block			
Cyclic Read Data Block			
Event Data Block			
Service Data Block			
Instance Data Header			
Instance Num	–	Instance Blk Size	Cyclic Blk Size
Cyc. Act. Blk Size	Cyc. Read Blk Size	Event Blk Size	Service Blk Size
Cyclic Data Block			
Control Mode	Feedback Mode	Axis Response	Response Status
–	Actual Data Set	Status Data Set	Axis State
Cyclic Data			
Cyclic Read Data Block			
Cyclic Write Blk ID	Cyclic Write Status	Cyclic Read Blk ID	Cyclic Read Status
Cyclic Read Data			
Event Data Block			
Event Checking Status			
Reg Data Ack	Home Data Ack	Watch Data Ack	–
Event ID	Event Status	Event Type	–
Event Position			
Event Time Stamp			
...			
Service Data Block			
Transaction ID	Service Code	General Status	Extended Status
Service Specific Response Data			

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Figure 13 – Device-to-Controller Connection format (Connection Point 2)

6.4.3 Controller-to-Device Connection

6.4.3.1 General

To facilitate a detailed description of each of its constituent data elements, the CIP Motion Controller-to-Device Connection is organised as shown in Figure 14.

CIP Motion Controller-to-Device Connection format
Connection Header
Time Data Block
Instance Data Blocks

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Figure 14 – CIP Motion Controller-to-Device Connection format

6.4.3.2 Controller-to-Device Connection Header

6.4.3.2.1 General structure

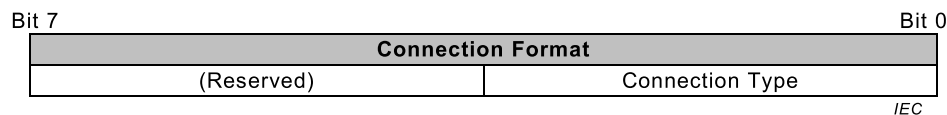
The Controller-to-Device Connection Header contains critical axis configuration information needed to parse the Instance Data Blocks. The fixed portion of the Connection Header is defined as shown in Figure 15.

Connection Header			
Connection Format	Format Revision	Update ID	Node Control

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Figure 15 – Connection Header

- **Connection Format:** This byte determines the format of the CIP Motion Connection according to the definition in Figure 16.



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Figure 16 – Connection Format

- **Connection Type:** This 4-bit value enumeration defines the CIP Motion Connection type as shown below. Valid values for a Controller-to-Device Connection are 2 for Fixed format and 6 for Variable format. Fixed connections have a fixed connection size during operation to support lower-performance CIP networks and are typically associated with a simple single axis device that does not support time synchronization services. Variable connections allow the connection data structure to vary in size during operation and are targeted for high-performance devices and CIP networks like EtherNet/IP.

- 0 = Fixed Controller Peer-to-Peer Connection
- 1 = Fixed Device Peer-to-Peer Connection
- 2 = Fixed Controller-to-Device Connection
- 3 = Fixed Device-to-Controller Connection
- 4 = Variable Controller Peer-to-Peer Connection
- 5 = Variable Device Peer-to-Peer Connection
- 6 = Variable Controller-to-Device Connection
- 7 = Variable Device-to-Controller Connection
- 8 to 15 = Reserved.

The Connection Format is not only used by the device to correctly parse the Controller-to-Device Connection data, but also by network diagnostic tools to display the individual elements of the data structure. If the device receives a Connection Format value that it cannot support, it generates a fault. In this case, the fault shall be Node Fault Error Code 4: Data Format Error. The Connection Format value shall not change for the life of the

connection and, therefore, need only be checked by the device for the first received packet.

- **Format Revision:** This edition of IEC 61800-7-202 defines Format Revision 2. Devices utilizing this edition of IEC 61800-7-202 shall only recognize Format Revision 2; Format Revision 1 is therefore rendered obsolete. This value is incremented by 1 for every revision of the Controller-to-Device Connection format that impacts the interface. The Format Revision allows newer devices to support the connection formats generated by an older controller. It also allows older devices to recognise a newer connection format from a controller that it cannot support and to generate a fault. In this case, the fault shall be Node Fault Error Code 4: Data Format Error. Network diagnostic tools can also key off the Format Revision to know how to parse the connection data packet. At the time the I/O Connection is established, the Format Revision number is also indicated by the I/O Connection Point in the Application Path of the Forward_Open service (see 6.4). The Format Revision value shall not change for the life of the connection and, therefore, need only be checked by the device for the first received packet.
- **Update ID:** This cyclic transaction number is incremented every update period. The Update ID is like the CIP sequence count and is used by the device to determine whether the connection buffer contains fresh data. In the case where the device is not synchronized, or does not support time synchronization services, the time stamp data is either not included or invalid, so the Update ID is the only way for the device to detect new connection data. In this case, the Controller-to-Device Connection Update ID is applied to the next Device-to-Controller Connection Update ID.

The Update ID also allows the device to determine if an update has been lost. Any skip in the Update ID is an indication that an update packet has been lost during transmission. If the number of consecutive missed updates exceeds the value given by the class attribute, Controller Update Delay High Limit, the Control Connection Update Fault is indicated by the Node Faults class attribute of the Motion Device Axis Object. This fault can only be cleared by closing and opening the CIP Motion I/O Connection. An optional Controller Update Delay Low Limit attribute is also available. If the number of consecutive missed updates exceeds the maximum value given by the class attribute, Controller Update Delay Low Limit, the Controller Connection Update Alarm is indicated by the Node Alarm class attribute of the Motion Device Axis Object. The alarm is cleared as soon as a fresh update is received.

- **Node Control:** This element is applied to the Motion Device Axis Object class attribute “Node Control”, which is used to control the state of the associated device communications node. See 7.2.2.1 (Motion Device Axis Object) for details.

6.4.3.2.2 Fixed Connection Header

If the Connection Format is a Fixed Controller-to-Device Connection, the above header is immediately followed by the Instance Data Block.

6.4.3.2.3 Variable Connection Header

If the Connection Format is a Variable Controller-to-Device Connection, then the Connection Header contains additional fields related to multi-axis device addressing and time stamping, as shown in Figure 17.

Connection Header			
Connection Format	Format Revision	Update ID	Node Control
Instance Count	–	Last Received ID	Time Data Set
Controller Time Stamp			
Controller Time Offset			

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Figure 17 – Connection Header

- **Instance Count:** This value reflects the number of Instance Data Blocks present in the CIP Motion Controller-to-Device Connection data structure.

- **Last Received ID:** This is the Update ID of the last Device-to-Controller Connection data block actually processed by the controller. The Last Received ID is used by the device to determine if the data that was sent was processed successfully. For example, Fault Codes passed as part of the Device-to-Controller Connections Status Data Block are sequenced into the controller's fault log. It is recommended that the Last Received ID element be read by the device to determine if the last Fault Code was entered into the Fault Log, thus allowing the device to send the next Fault Code in the sequence.
- **Time Data Set:** This bit-mapped byte contains flags that determine the usage and format of the controller and device timing information, as specified in Table 6. In general, the Time Data Set value sent by the controller in the Controller-to-Device Connection not only determines the contents of the Controller-to-Device Connection data block, but it also represents the requested Time Data Set to be sent by the device in the Device-to-Controller Connection. The value of this element is transferred to the Motion Device Axis Object attribute "Time Data Set".

Table 6 – Time Data Set

Bit	Definition	Syntax
0	Time Stamp	0 = no time stamp 1 = time stamp included
1	Time Offset	0 = no time offset – use last offset 1 = time offset included
2	Update Diagnostics	0 = no update diagnostic data in D-to-C 1 = update diagnostic data in D-to-C
3	Time Diagnostics	0 = no time diagnostic data in D-to-C 1 = time diagnostic data in D-to-C
4 to 7	(Reserved)	

- **Time Stamp** – if this bit is set, the Controller Time Stamp is included in the Time Data Set block. Setting this bit also instructs the device to send Device Time Stamps back to the controller in the Device-to-Controller Connection in the next update. It also instructs the device to send the Device Time Offset if its value has changed since the last update. If the device does not support time synchronization, the device simply returns a value of zero for the Time Stamp and does not ever return a Time Offset.
- **Time Offset** – if this bit is set, the Controller Time Offset is included in the Time Data Set block. This bit is only set when there is a change to the Controller Time Offset value, a relatively infrequent occurrence. Note that this bit does NOT determine if the device sends its Device Time Offset in the next update. That is determined by the Time Stamp bit and by whether or not the Device Time Offset has changed since the last update.
- **Update Diagnostics** – although there is no data associated with this bit in the Controller-to-Device Connection header, the Update Diagnostic bit in the Controller-to-Device Connection instructs the device to send Update Diagnostic data in the next Device-to-Controller Connection update, but only if one or more of the Update Diagnostic values have changed since the last update. This Update Diagnostic data includes the current number of Lost or Late Updates from the controller over the Controller-to-Device Connection. Update Diagnostics apply to devices regardless of the whether they support time synchronization.
- **Time Diagnostics** – although there is no data associated with this bit in the Controller-to-Device Connection header, the Time Diagnostic bit in the Controller-to-Device Connection instructs the device to send Time Diagnostic data in the next Device-to-Controller Connection update. This Time Diagnostic data includes the Controller-to-Device Data Received time stamp and the Device-to-Controller Data Transmit time stamp. If the device does not support time synchronization, the device simply returns a value of zero for the Time Diagnostic elements.
- **Controller Time Offset:** This element is included in the Connection Header if the Time Offset bit is set in the Time Data Set element. The Controller Time Offset represents the

64-bit System Time Offset value associated with the Controller Time Stamp that follows. The value of this element is transferred to the Motion Device Axis Object class attribute of the same name. The Controller Time Offset value is used by the device to determine if System Time as defined in the controller is skewed relative to System Time in the device. Normally, System Time between the device and the controller are closely matched at any given time. But every few seconds, according to the IEC 61588:2009 based CIP Sync protocol, the time master of the system can correct the System Time reference by a significant amount of time, perhaps an hour. Indeed, even the master itself can change as a higher quality time master is discovered by the system. These step changes to the System Time reference propagate through to the various devices on the network in such a way that the controller and the device are running with skewed System Time values. The Controller Time Offset value can be used together with the System Time Offset for the device to both determine if System Time between the device and the controller is skewed and if so, to compensate for the skew using the System Time Offset Compensation algorithm.

The Controller Time Offset value changes relatively infrequently, i.e. as a result of a IEC 61588 driven Sync update, which typically occurs every second. Thus, it is recommended that the controller only load and send the Time Offset data if the value has changed. The device need only parse the Controller Time Offset data if the associated Time Offset bit is set in the Time Data Set element, otherwise it continues to use the last Controller Time Offset received.

- **Controller Time Stamp:** This element is included in the Connection Header if the Time Stamp bit is set in the Time Data Set element. The Controller Time Stamp represents the 64-bit System Time value at the beginning of the Controller Update Period when the controller's update timer event occurred. In other words the Controller Time Stamp marks, in absolute time, the beginning of the current CIP Motion Connection cycle. The value of this time stamp is calculated by the controller as the sum of the controller's local clock value when update timer event occurred and the controller's System Time Offset value. The Controller Time Stamp is therefore directly associated with the command data contained in the connection. The value of this element, when read, is transferred to the Motion Device Axis Object class attribute Controller Time Stamp. Taken together with the Controller Update Period, established by the RPI of the Forward_Open service, and the Command Target Update (discussed later), the device has all the information it needs compute command interpolation polynomials to correct command data values for differences between the device and controller update timing. These differences can occur when the Controller Update Period is not an integer multiple of the Device Update Period or when the device updates are phase shifted relative to the controller.

When the connection is synchronized, the Controller Time Stamp can be used by the device along with the Controller Update Period to check for missed or late updates. If the difference between the last Controller Time Stamp and the current local Device Time Stamp exceeds the maximum value given by,

Max Fault Delay = Controller Update Delay High Limit × Controller Update Period,

the Control Connection Update Fault is indicated by the Node Faults class attribute of the Motion Device Axis Object. An optional Controller Update Delay Low Limit attribute is also available. If the difference between the last Controller Time Stamp and the current local Device Time Stamp exceeds the maximum value given by,

Max Alarm Delay = Controller Update Delay Low Limit × Controller Update Period,

the Controller Connection Update Alarm is indicated by the Node Alarm class attribute of the Motion Device Axis Object.

If either the Controller Time Offset or the Device Time Offset values have changed since the last update, System Time Offset Compensation shall be applied to the Controller Time Stamp. See 6.4.6.7 for details. If the time offsets have not changed, the typical case, the Controller Time Stamp can be directly applied to the Motion Device Axis Object.

6.4.3.3 Instance Data Blocks

6.4.3.3.1 General structure

After the Connection Header are one or more Instance Data Blocks as determined by the above Instance Count. The Instance Data Block has the basic structure shown in Figure 18.

Instance Data Block	
Instance Data Header	
Cyclic Data Block	
Event Data Block	
Service Data Block	

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Figure 18 – Instance Data Block

6.4.3.3.2 Instance Data Header

The Instance Data Header contains critical axis configuration information needed to parse and apply the data contained in the three data channels (see Figure 19). This header is only included in the Variable Connection format to accommodate multi-axis device applications. Information within the header can be used by the device communications interface to copy the individual data blocks into separate fixed memory locations for processing.

If configured for a Fixed Connection format, only the Cyclic Data Block for a single axis instance is supported so there is no need for any information to specify instance number or block sizing. The Instance Data Header is therefore not included.

Instance Data Header			
Instance Num	–	Instance Blk Size	Cyclic Blk Size
Cyc. Cmd. Blk Size	Cyc. Write Blk Size	Event Blk Size	Service Blk Size

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Figure 19 – Instance Data Header

- **Instance Number:** This is the number that identifies the specific Motion Device Axis Object instance the following Instance Data Block applies to. Motion Device Axis Object instances are created as a contiguous series of instance numbers starting with instance 1. Within the connection data structure, the Instance Numbers for each consecutive Instance Data Block shall be an ordinal sequence, i.e. 0, 1, 2, 3, etc.

NOTE 1 Instance 0 is defined as the class instance and is typically only used during initialization to configure class attributes of the Motion Device Axis Object via the Service Data Block. Otherwise, the Instance Data Block sequence starts with instance 1.

Thus, in theory, up to 255 axis instances can be serviced by the CIP Motion I/O Connection. The instance numbers sequence of the Controller-to-Device Connection determines the instance number sequence of the subsequent Device-to-Controller Connection.

- **Instance Block Size:** This value represents the size of the Instance Data Block in 32-bit words including the header. The Instance Block Size is useful when the device wants to directly access the next Instance Data Block without having to add the sizes of the cyclic, event, and service blocks. If the Instance Block Size for the Controller-to-Device Connection is zero, i.e. no Instance Data Block, then the Instance Block Size for the following Device-to-Controller Connection shall also be zero.
- **Cyclic Block Size:** This value represents the size of the Cyclic Data Block in 32-bit word units including the header.

NOTE 2 For instance 0, the class instance, the Cyclic Block Size is 0 indicating that there is no Cyclic Data Block.

- **Cyclic Command Block Size:** This value represents the size of the Cyclic Command Data Block in 32-bit word units including the header. A Cyclic Command Block Size of 0 indicates that there is no Cyclic Command Data for the device to process.

- **Cyclic Write Block Size:** This value represents the size of the Cyclic Write Data Block in 32-bit word units including the header. A Cyclic Write Block Size of 0 indicates that there is no Cyclic Write data currently configured for transfer to the device and therefore there is no Cyclic Write Block included for the device to process.
- **Event Block Size:** This value represents the size of the Event Data Block in 32-bit word units including the header. If the Event Block Size for the Controller-to-Device Connection is zero, there is no active event checking, so the Event Block Size for the Device-to-Controller Connection shall also be zero. This effectively closes the Event Channel.
- **Service Block Size:** This value represents the size of the Service Data Block in 32-bit word units including the header. A Service Block Size value of 0 indicates there is no service request to process. If the Service Block Size for the Controller-to-Device Connection is zero, then the Service Block Size for the Device-to-Controller Connection shall also be zero. This effectively closes the Service Channel.

6.4.3.3.3 Cyclic Data Block

The Cyclic Data Block consists of high priority control data that is updated every connection cycle based on the Controller Update Period. The Cyclic Data Header at the top of the Cyclic Data Block is always included regardless of the connection format. This header contains key elements related to the content of the Cyclic Data Block of both the Controller-to-Device Connection and Device-to-Controller Connection, and the context of the data as determined by the Control Mode and Feedback Mode (see Figure 20). The header also provides a mechanism to control the state of the targeted device axis.

Cyclic Data Block			
Control Mode	Feedback Mode	Axis Control	Control Status
Command Data Set	Actual Data Set	Status Data Set	Command Control
Cyclic Data			

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Figure 20 – Cyclic Data Block

- **Control Mode:** The lower 4-bits of this 8-bit element determine the control mode context of the command data as presently configured in the motion controller, as shown in Figure 21.

Bit 7	Control Mode		Bit 0
reserved – 4 bits		Control Mode – 4 bits	

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Figure 21 – Control Mode

This value can be changed while on-line and even while the device axis is in the Running state. If a particular Control Mode transition is not supported by the device, an exception is generated that can be configured to perform any one of a number of actions in response to the illegal transition.

The value of this element is transferred to the Motion Device Axis Object attribute of the same name. The complete Control Mode attribute definition can be found in the attribute tables of the Motion Device Axis Object (see 7.3.2.2.1).

- **Feedback Mode:** The lower 4-bits of this 8-bit element determine the feedback context of the command data as presently configured in the motion controller, as shown in Figure 22.

Bit 7	Feedback Mode		Bit 0
reserved – 4 bits		Feedback Mode – 4 bits	

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Figure 22 – Feedback Mode

Command position data can be referenced to feedback counts of either Feedback 1 or Feedback 2, or motor units for sensorless operation. This value can be changed while on-line and, if supported by the device, even while the device axis is in the Running state. If a particular Feedback Mode transition is not supported by the device, an exception is

generated that can be configured to perform any one of a number of actions in response to the illegal transition.

The value of this element is transferred to the Motion Device Axis Object attribute of the same name. The complete Feedback Mode attribute definition can be found in the attribute tables of the Motion Device Axis Object (see 7.3.6.2).

- **Axis Control:** An 8-bit enumerated code that can be used to directly execute Axis State change operations that do not require either passing or returning device parameters, and therefore, do not require a CIP service to initiate. This mechanism is fully described in the State Control subclause of the Motion Device Axis Object (see 7.6.1.2). Valid enumerations for this data element are shown in Table 7.

Table 7 – Axis Control

Request code	Requested operation
0	No Request
1	Enable Request
2	Disable Request
3	Shutdown Request
4	Shutdown Reset Request
5	Abort Request
6	Fault Reset Request
7	Stop Process Request
8	Change Actual Position Reference Request
9	Change Command Position Reference Request
10 to 126	(Reserved)
127	Cancel Request
128 to 255	(Vendor Specific)

- **Control Status:** A bit mapped value that can be interrogated by the device to determine the status of the control during the Initialization process. The value of this element is transferred to the optional Motion Device Axis Object attribute number 89, Control Status, if supported by the device. Valid bits for this data element are shown in Table 8.

Table 8 – Control Status

Bit	Control Status bit	Description
0	Configuration Complete	This bit is set when the controller has completed configuration of all axis instance attributes during Initialization phase.
1	(Reserved)	See Attribute 89 of the Motion Device Axis Object
2	Converter Bus Up	
3	Converter Bus Unload	
4	Converter AC Power Loss	
5 to 7	(Reserved)	

- **Command Data Set:** This bit mapped value has a bit defined for each possible real-time command reference (see Table 9). Command data appears in the same order in the Command Data Set as the bit numbers, so Command Acceleration would appear before Command Torque in the real-time data structure of the Controller-to-Device Connection.

The value of this element is transferred to the Motion Device Axis Object attribute "Command Data Set".

Table 9 – Command Data Set

Bit	Controller Command Data Element	Data type
0	Command Position	LREAL or DINT ^a
1	Command Velocity	REAL
2	Command Acceleration	REAL
3	Command Torque	REAL
4	(Reserved)	(Reserved)
5	(Reserved)	(Reserved)
6	Unwind Cycle Count	DINT
7	Position Displacement	DINT

^a Data Type for Command Position determined by Command Control element defined below. The Command Position data type can be either an LREAL or a DINT as determined by the Command Control element.

The above Controller Command Data Elements apply to the Controller-to-Device Connection's cyclic data structure and map to corresponding attributes in the Motion Device Axis Object as shown in Table 10.

The units of the Command Data Elements match the units defined for the associated Motion Device Axis Object attribute.

Table 10 – Command Data Element to Motion Device Axis Object attribute mapping

Bit	Command Data Element	Motion Device Axis Object attribute
0	Command Position (LREAL) Command Position (DINT)	Controller Position Command – Float Controller Position Command – Integer
1	Command Velocity	Controller Velocity Command
2	Command Acceleration	Controller Acceleration Command
3	Command Torque	Controller Torque Command
4	(Reserved)	(Reserved)
5	(Reserved)	(Reserved)
6	Unwind Cycle Count	Local variable
7	Position Displacement	Local variable

It is the job of the controller to ensure that the necessary Command Data Elements are included in the connection data to support the specified Control Mode.

- **Actual Data Set:** This bit mapped value has a bit defined for each possible real-time actual data attribute that is to be included in the Actual Data Set of the Device-to-Controller Connection's Instance Data Block in the next update (see Table 11). Actual data appears in the same order as the bit numbers, so Actual Position would appear before Actual Velocity in the Actual Data Set structure. Using this mechanism, the contents of the Actual Data Set may be changed at any time during device operation. The value of this element is transferred to the Motion Device Axis Object attribute "Actual Data Set".

Table 11 – Actual Data Set

Bit	Actual Data Element produced	Data type
0	Actual Position	DINT
1	Actual Velocity	REAL
2	Actual Acceleration	REAL
3	(Reserved)	(Reserved)
4	(Reserved)	(Reserved)
5	(Reserved)	(Reserved)
6	Unwind Cycle Count	DINT
7	Position Displacement	DINT

The above Actual Data Elements map to corresponding attributes in the Motion Device Axis Object as shown in Table 12. The units of the Actual Data Elements match the units defined for the associated Motion Device Axis Object attribute.

Table 12 – Actual Data Element to Motion Device Axis Object attribute Mapping

Bit	Actual Data Element	Motion Device Axis Object attribute
0	Actual Position	Position Feedback
1	Actual Velocity	Velocity Feedback
2	Actual Acceleration	Acceleration Feedback
3	(Reserved)	(Reserved)
4	(Reserved)	(Reserved)
5	(Reserved)	(Reserved)
6	Unwind Cycle Count	Local variable
7	Position Displacement	Local variable

- **Status Data Set:** This bit-mapped byte contains flags that determine what the controller requests for contents of the Status Data Set of the Device-to-Controller Connection's Instance Data Block in the next update (see Table 13). These bits may NOT solely determine if the device sends the associated Status data in the next Device-to-Controller Connection update. That may be determined by the device based on whether or not the associated Status data has changed since the last Device-to-Controller Connection update.

Status data appears in the same order as the bit numbers, so Axis Fault Type/Code data would appear before, for example, Axis Fault Time Stamp data in the Status Data Set structure. Multiple attributes comprising a selected Status Data Element are transmitted in the order listed from top to bottom, so the Axis Fault Type is transmitted before Axis Fault Code. The definitions of each of these Status Data Elements can be found by looking up the corresponding Motion Device Axis Object attribute (see 7.3) in the Motion Device Axis Object specification. The value of this element is transferred to the Motion Device Axis Object attribute "Status Data Set".

Table 13 – Status Data Set

Bit	Status Data Element produced	Data type
0	Axis Fault Type Axis Fault Code Axis Fault Sub Code Axis Fault Action Axis Fault Time Stamp	USINT USINT USINT USINT LINT
1	Axis Alarm Type Axis Alarm Code Axis Alarm Sub Code Axis Alarm State Axis Alarm Time Stamp	USINT USINT USINT USINT LINT
2	Axis Status Axis Status – Mfg	DWORD DWORD
3	Axis I/O Status Axis I/O Status – Mfg	DWORD DWORD
4	Axis Safety Status Axis Safety Status – Mfg Axis Safety State Pad[3]	DWORD DWORD USINT USINT[3]
5	(Reserved)	(Reserved)
6	(Vendor specific)	(Vendor specific)
7	(Vendor specific)	(Vendor specific)

If any of these bits are set and the associated status value has changed in the device, the device shall return the associated status value in the next available Device-to-Controller update. Status data values change relatively infrequently, so there can be many connection updates where various elements of the Status Data Set remain static. It is recommended that the device take advantage of this fact and only set the Status Data Set bit in the Device-to-Controller update when one of the associated status values has changed. Thus, the device need only load, say, the Axis Fault data associated with Status Data Set bit 0 when there is a new fault to report. Otherwise, it leaves bit 0 clear. On the controller side of the connection, the controller shall parse and process the Axis Fault data only if Status Data Set bit 0 is set. To insure that the data has been received, parsed, and processed by the controller before clearing the Status Data Set bit, the device can examine the Last Update ID of the next Controller-to-Device Connection update to confirm that it matches or exceeds the Update ID of this Device-to-Controller update.

- Command Control: The byte contains information needed to control the fine interpolation algorithm and determine the target time of the command data to the Axis Control structure (see Table 14). The value of this element is transferred to the Motion Device Axis Object attribute “Command Control”.

Table 14 – Command Control

Bit	Definition	Syntax
0 to 1	Command Target Update	0 = Immediate 1 = Extrapolate (+1 Update Period) 2 = Interpolate (+2 Update Periods) 3 = (Reserved)
2 to 3	Command Position Data Type	0 = LREAL (64-bit Floating Point) 1 = DINT (32-bit signed Integer) 2 to 3 = (Reserved)
4 to 7	(Reserved)	

Command Target Update – This 2-bit integer defines a specific time relative to the Connection Time Stamp that the Command Data is targeted for, as defined by the controller’s motion planner. The absolute Command Target Update is the sum of the

Controller Time Stamp from the controller and the product, Command Target Update $\times$ Controller Update Period.

- A Command Target Update of 0 implies that the Command Data is targeted for the beginning of the current update cycle and, thus, needs to be applied to the control structure immediately. In this case, there is no need for any fine interpolation. This situation can occur when the Controller Update Period is significantly shorter than the Device Update Period, or when the controlled motion of the axis is nearly constant during the span of the Controller Update Period.
- A Command Target Update of 1 implies that the target for the Command Data is the next Connection Update timer event. In this case, the command interpolator functions primarily as an extrapolator that estimates the next Command Data value based on the present trajectory of the axis. This is a typical setting when the Controller Update Period is comparable to the Device Update Period or when the controlled motion of the axis is relatively constant during the span of the Controller Update Period.
- A Command Target Update of 2 implies that the target for the Command Data is two Connection Update timer events from the Connection Time Stamp. In this case, the command interpolator can compute a smooth trajectory based on the current dynamics of the motor to reach the Command Data value at the targeted time. This is true fine interpolation and is applicable when the Controller Update Period is significantly larger than the Device Update Period.
- Table 15 summarizes the relationship between the Update Period Ratio and the recommended Command Target Update.

Table 15 – Command Target Update vs. Update Period Ratio

Update Period Ratio (Controller Update Period / Device Update Period)	Recommended Command Target Update
2 or more	2 = Interpolate (+2 Update Periods)
$\frac{1}{2}$ to 2	1 = Extrapolate (+1 Update Period)
$\frac{1}{2}$ or less	0 = Immediate

Command Position Data Type – This 2-bit integer defines the data type of the Command Position data element. This allows flexibility in handling the many different representations for command position in the industry.

- A Command Position Data Type of 0 corresponds to the LREAL, or double precision floating point data type. Support for this data type is preferred in the implementation since it provides fraction command count information to the device resulting in smoother motion.
 - A Command Position Data Type of 1 corresponds to the DINT, or 32-bit signed integer data type. This data type is applicable to simple drive devices that either do not require the precision of a floating point command position value or do not have sufficient hardware support for double precision floating point math.
- **Cyclic Command Data:** The Cyclic Command Data contains high priority data that needs to be applied to the associated drive axis instance during the next device update. This block consists of command data elements that are applied as references to the device's control algorithms and explicitly determined by the Command Data Set element in the Cyclic Command Data Header.

6.4.3.3.4 Cyclic Write Data Block

The Cyclic Write Data Block can be used to synchronously update one or more targeted Motion Device Axis Object configuration parameters within the device. This mechanism can be used in conjunction with a Function Block program to implement sophisticated outer loop control, gain scheduling, and dynamic limiting algorithms. Unlike service channel Set Axis Attribute service requests, which may take several device update cycles to process, the Cyclic Write Data mechanism guarantees that the targeted parameter is applied at the next available device update.

The Cyclic Write Data Block is only supported in the Variable Connection format (see Figure 23).

Cyclic Write Data Block			
Cyclic Write Blk ID	–	Cyclic Read Blk ID	–
Cyclic Write Data			

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Figure 23 – Cyclic Write Data Block

The associated header for this block contains key elements related to the content of the Cyclic Write Data and the Cyclic Read Data Block of the next Device-to-Controller Connection update.

- **Cyclic Write Block ID:** This 8-bit ID determines the pre-defined Cyclic Write Block structure to apply to the Cyclic Write Data for this update. Cyclic Write Block structures are defined using the Set Cyclic Write Data List service. The initial value for the Cyclic Write Block ID is 0, which corresponds to a null set of Cyclic Write Data. A successful response to the Set Cyclic Write Data List service includes a new, incremented, Cyclic Write Block ID that can be used in the next connection update to pass cyclic data in this format. The previous Cyclic Write Block ID and its associated structure shall be maintained until the controller requests transmission of the new Cyclic Write Block ID at which point the old Cyclic Write Block ID and structure definition are no longer required.
- **Cyclic Read Block ID:** This 8-bit ID determines the pre-defined Cyclic Read Block structure to apply to the Cyclic Read Data for next Device-to-Controller Connection update. Cyclic Read Block structures are defined using the Set Cyclic Read Data Block service. The initial value for the Cyclic Read Block ID is 0, which corresponds to a null set of Cyclic Read Data. A successful response to the Set Cyclic Read Data List service includes a new, incremented, Cyclic Read Block ID that can be used in the next connection update to allow the device to use the new Cyclic Read Data format for next available Device-to-Controller Connection update.
- **Cyclic Write Data:** The Cyclic Write Data contains high priority data that needs to be applied to the associated device axis instance during the next device update. This block consists of parametric data elements that are applied to Motion Device Axis Object attributes that are used by the control algorithms. The contents of the Cyclic Write Data are explicitly determined by the structure identified by the Cyclic Write Block ID found in the Cyclic Write Data Header. The ordering of the attribute data in the Cyclic Write Data Block is determined by the ordering of the Attr IDs in the Set_Cyclic_Write_List request. Attribute data elements shall be word aligned; 32-bit words are 32-bit aligned and 16-bit words are 16-bit aligned. Padding shall be added to maintain word alignment.

EXAMPLE Figure 24 shows a Cyclic Write Data Block with 3 attributes where Attributes 1 and 3 are 32-bit values while Attribute 2 is a 16-bit value.

Attribute value 1	
Attribute value 2	16-bits – reserved
Attribute value 3	

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Figure 24 – Cyclic Write Data Block example

6.4.3.4 Event Data Block

The Event Data Block is used to convey information regarding the event channel. In particular, the Event Data Block for the Controller-to-Device Connection is used to control the arming of event checking functions in the device as well as acknowledge receipt of event notifications from the device that are sent via the Device-to-Controller Connection's Event Data Block.

The Event Data Block for the Controller-to-Device Connection has the format shown in Figure 25.

Event Data Block			
Event Checking Control			
Reg Data Set	Home Data Set	Watch Data Set	–
Registration Event Data			
Home Event Data			
Watch Event Data			
Event Ack. ID1	Event Ack. Status 1	Event Ack. ID2	Event Ack. Status 2
...			

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Figure 25 – Event Data Block

When the Event Data Block is present, the Event Checking Control element determines which additional elements are included. The elements starting with Reg Data Set and extending through Watch Event Data constitute Extended Data and will only exist if the Extended Format bit is set in the Event Checking Control word. The Event Ack ID # and Event Ack Status # elements will be repeated from 0 to 7 times as determined by the Event Block Count field in the Event Checking Control word. If an odd number of Event Blocks are included, the Event Data Block will be padded to a 32-bit boundary with two unused bytes.

- **Event Checking Control:** This 32-bit word is copied into the Motion Device Axis Object attribute of the same name that is used to enable various device inputs, for example marker and registration inputs, to generate events. When these events occur, the device captures both the time and exact axis position when the event occurred. The last 4 bits of the Event Checking Control element is a special bit field, the least significant 3 bits of which specify the number of active events, which is literally the number of Event Acknowledge IDs listed in this Event Data Block. The most significant bit enables the extended Event Data Block format. A complete definition for the Event Checking Control attribute can be found in 7.3.7.2.1.

The basic Event Control mechanism works as specified in Table 16.

Table 16 – Basic Event Cycle

Controller action	Connection data	Device action
Controller sets the appropriate Event Checking Control bit to look for a specific event condition to occur. Multiple bits in the Event Checking Control word may be set at any given time.	→ Event Checking bit = 1 → Event Block Count = 0 → Extended Format = 0	Device detects the Event Checking Control bit is set and then initiates the requested event checking action.
Controller sees the Event Checking Status bit from the device is set indicating that specified event trigger is “armed” at the device. Starting with this update the controller can process any event notifications from the device that match the specified event condition.	← Event Checking bit = 1 ← Event Block Count = 0 ← Extended Format = 0	Device sets the corresponding Event Checking Status bit to acknowledge that the device is now actively checking for the specified event condition. Starting with this update the device can include an event notification for the specified event condition.
Controller maintains the Event Checking bit to continue checking for the specified event.	→ Event Checking bit = 1 → Event Block Count = 0 → Extended Format = 0	Device sees Event Checking Control bit set and therefore continues checking for the specified event.
Controller continues to check for new Event Notifications from device.	← Event Checking bit = 1 ← Event Block Count = 0 ← Extended Format = 0	Device continues to set the corresponding Event Checking Status bit to indicate that event checking is active.

Controller action	Connection data	Device action
Controller processes the new Event Notification, storing the Event ID, checking the Event Status, and then processing the Event Data Block based on the Event Type.	← Event Checking bit = 1 ← Event Block Count = 1 ← Extended Format = 0 ← Event ID = id ← Event Status = 0 (success) ← Event Type = type ← Event Data Block	Device detects the specified event condition, increments the Event ID, and sends an Event Notification Data Block with this ID to the controller. The specific event condition is identified by its Event Type. Since the associated Auto-rearm bit is clear in the Event Checking Control word, the device discontinues checking for the specified event. However, the Event Checking Status bit remains set until the Event Checking Control bit is cleared by the controller. Note that notifications for other events can be sent in the same update. The total number of Event Notification Data Blocks in the update is represented by the Event Block Count.
Controller sends the Event Acknowledge with the associated Event ID and Event Status to the device. Since Auto-rearm is not enabled, the controller clears the associated Event Checking Control bit.	→ Event Checking bit = 0 → Event Block Count = 1 → Extended Format = 0 → Event ID = id → Event Status = 0 (success)	Device detects the Event Checking Control bit is clear and processes the Event Acknowledge Data Block.
Controller sees the Event Checking Status bit from the device is clear indicating that the event cycle is complete. Controller is now able to set the Event Checking Control bit again. So, the minimum time between when the Event Notification was received by the controller to the time when the device is rearmed for the next event is one Controller Update Period.	← Event Checking bit = 0 ← Event Block Count = 0 ← Extended Format = 0	Device clears the corresponding Event Checking Status bit. Since the device received a successful Event Acknowledge, there is no further need to send the Event Notification Data Block for that Event ID.
Controller may now set the Event Block Size to zero for all subsequent updates until such time as there is request to check for new events.	→ Event Block Size = 0 → No Event Block	Device sees Event Block Size is zero indicating that there is no Event Block data to process.
Device sees Event Block Size is zero indicating that there is no Event Block data to process.	← Event Block Size = 0 ← No Event Block	Device may now set the Event Block Size to zero for all subsequent updates until such time as there is request to check for new events.

If the device supports Device Scaling functionality, the Extended Event Data Block format is used to exchange addition event information between the controller and the device. The Extended Event cycle works as specified in Table 17.

Table 17 – Extended Event Cycle

Controller action	Connection data	Device action
Controller sets the appropriate Event Checking Control bit to look for a specific event condition to occur. Multiple bits in the Event Checking Control word may be set at any given time. Using the Extended Event Checking Data block format, the controller passes any parameters needed by the event checking function in the Registration, Home, and Watch Event Data Blocks. The Event Format bit in the Event Checking Control element shall be set to include Extended Event Checking Data. Extended Event Checking Data is required when the device is configured for Device Scaling.	→ Event Checking bit = 1 → Event Block Count = 0 → Extended Format = 1 → Extended Event Data	Device detects the Event Checking Control bit is set, loads any associated event checking parameters from the Extended Event Checking Data block, and then initiates the requested event checking action.
Controller sees the Event Checking Status bit from the device is set indicating that specified event trigger is “armed” at the device. Starting with this update the controller can process any event notifications from the device that match the specified event condition. If the controller previously sent event checking parameters via the Extended Event Checking Data block, it also checks the returned Registration, Home, and Watch Data Set values to determine if the device has processed the Registration, Home, and Watch Event Data sent in the previous update. If the Data Ack values match the Data Set value that were sent, the Extended Event Checking Data has been processed by the device.	← Event Checking bit = 1 ← Event Block Count = 0 ← Extended Format = 1 ← Extended Event Data Ack	Device sets the corresponding Event Checking Status bit to acknowledge that the device is now actively checking for the specified event condition. If the device received event checking parameters from the Extended Event Checking Data block it acknowledges receipt of parameters passed in the Registration, Home, and Watch Event Data Blocks by returning the Registration, Home, and Watch Data Set values as the Data Ack values, respectively. Starting with this update the device can include an event notification for the specified event condition.
Controller maintains the Event Checking bit to continue checking for the specified event. With the device acknowledging receipt of the Extended Event Checking Data, the Controller clears the Extended Format bit for the Controller-to-Device Connection update.	→ Event Checking bit = 1 → Event Block Count = 0 → Extended Format = 0	Device sees the Event Checking Control bit set and therefore continues checking for the specified event.
Controller continues to check for new Event Notifications from device.	← Event Checking bit = 1 ← Event Block Count = 0 ← Extended Format = 0	Device continues to set the corresponding Event Checking Status bit to indicate that event checking is active. With no Extended Event Checking Data to acknowledge, the Device clears the Extended Format bit.

Controller action	Connection data	Device action
Controller processes the new Event Notification, storing the Event ID, checking the Event Status, and then processing the Event Data Block based on the Event Type.	← Event Checking bit = 1 ← Event Block Count = 1 ← Extended Format = 0 ← Event ID = x ← Event Status = 0 ← Event Type = y ← Event Data Block	Device detects the specified event condition, increments the Event ID, and sends an Event Notification Data Block with this ID to the controller. The specific event condition is identified by its Event Type. Since the associated Auto-rearm bit is clear in the Event Checking Control word, the device discontinues checking for the specified event. However, the Event Checking Status bit remains set until the Event Checking Control bit is cleared by the controller. Note that notifications for other events can be sent in the same update. The total number of Event Notification Data Blocks in the update is represented by the Event Block Count.
Controller sends the Event Acknowledge with the associated Event ID and Event Status to the device. Since Auto-rearm is not enabled, the controller clears the associated Event Checking Control bit.	→ Event Checking bit = 0 → Event Block Count = 1 → Extended Format = 0 → Event ID = id → Event Status = 0 (success)	Device detects the Event Checking Control bit is clear and processes the Event Acknowledge Data Block.
Controller sees the Event Checking Status bit from the device is clear indicating that the event cycle is complete. Controller is now able to set the Event Checking Control bit again. So, the minimum time between when the Event Notification was received by the controller to the time when the device is rearmed for the next event is one Controller Update Period.	← Event Checking bit = 0 ← Event Block Count = 0 ← Extended Format = 0	Device clears the corresponding Event Checking Status bit. Since the device received a successful Event Acknowledge, there is no further need to send the Event Notification Data Block for that Event ID.
Controller may now set the Event Block Size to zero for all subsequent updates until such time as there is request to check for new events.	→ Event Block Size = 0 → No Event Block	Device sees Event Block Size is zero indicating that there is no Event Block data to process.
Device sees Event Block Size is zero indicating that there is no Event Block data to process.	← Event Block Size = 0 ← No Event Block	Device may now set the Event Block Size to zero for all subsequent updates until such time as there is request to check for new events.

In the case of a Registration event where Auto-rearm Event Checking is requested, the event handling sequence is as specified in Table 18.

Table 18 – Basic Event Cycle with Auto-rearm

Controller action	Connection data	Device action
Controller sets the appropriate Event Checking Control bit to look for a specific event condition to occur and also sets the Auto-rearm bit for that event condition. Multiple bits in the Event Checking Control word may be set at any given time.	→ Event Checking bit = 1 → Event Auto-rearm bit = 1 → Event Block Count = 0 → Extended Format = 0	Device detects the Event Checking Control bit is set and then initiates the requested event checking action.
Controller sees the Event Checking Status bit from the device is set indicating that specified event trigger is “armed” at the device. Starting with this update the controller can process any event notifications from the device that match the specified event condition.	← Event Checking bit = 1 ← Event Auto-rearm bit = 1 ← Event Block Count = 0 ← Extended Format = 0	Device sets the corresponding Event Checking Status bit to acknowledge that the device is now actively checking for the specified event condition and also sets the corresponding Auto-rearm bit. Starting with this update the device can include an event notification for the specified event condition.
Controller maintains the Event Checking bit to continue checking for the specified event.	→ Event Checking bit = 1 → Event Auto-rearm bit = 1 → Event Block Count = 0 → Extended Format = 0	Device sees the Event Checking Control bit set and therefore continues checking for the specified event.
Controller continues to check for new Event Notifications from device.	← Event Checking bit = 1 ← Event Auto-rearm bit = 1 ← Event Block Count = 0 ← Extended Format = 0	Device continues to set the corresponding Event Checking Status bit to indicate that event checking is active.
Controller processes the new Event Notification, storing the Event ID, checking the Event Status, and then processing the Event Data Block based on the Event Type.	← Event Checking bit = 1 ← Event Auto-rearm bit = 1 ← Event Block Count = 1 ← Extended Format = 0 ← Event ID = id1 ← Event Status = 0 (success) ← Event Type = type ← Event Data Block	Device detects the specified event condition, increments the Event ID, and sends an Event Notification Data Block with this ID to the controller. The specific event condition is identified by its Event Type. Since the corresponding Auto-rearm bit is set, the device continues checking for the specified event. The Event Checking Status bit remains set until the Event Checking Control bit is cleared by the controller. If the Device detects another specified event condition prior to transmission of the first event to the controller, it increments the Event ID again, and appends another Event Notification with this ID to the Event Data Block to the controller. Note that notifications for other events can also be sent in the same update. The total number of Event Notification Data Blocks in the update is represented by the Event Block Count.
Controller sends the Event Acknowledge with the associated Event ID and Event Status to the device. Since the Auto-rearm bit is enabled, it leaves the associated Event Checking bit set to allow the device to continue checking for the next event.	→ Event Checking bit = 1 → Event Auto-rearm bit = 1 → Event Block Count = 1 → Extended Format = 0 → Event ID = id1 → Event Status = 0 (success)	Device detects the Event Checking Control bit is set and processes the Event Acknowledge Data Block. Since the corresponding Auto-rearm bit is set, the device continues checking for the specified event. The device can now send another Event Notification if the specified event has occurred again.

Controller action	Connection data	Device action
Controller continues to check for new Event Notifications from device.	← Event Checking bit = 1 ← Event Auto-rearm bit = 1 ← Event Block Count = 0 ← Extended Format = 0	Device sets the corresponding Event Checking and Auto-rearm Status bits indicating that the device is continuing to check for events. Since the device received a successful Event Acknowledge, there is no further need to send the Event Notification Data Block for that Event ID.
Controller processes the new Event Notification, storing the Event ID, checking the Event Status, and then processing the Event Data Block based on the Event Type.	← Event Checking bit = 1 ← Event Auto-rearm bit = 1 ← Event Block Count = 1 ← Extended Format = 0 ← Event ID = id2 ← Event Status = 0 ← Event Type = type ← Event Data Block	Device continues checking for the specified event condition and sends an Event Notification Data Block to the controller whenever the event condition occurs. The specific event condition is identified by its Event Type. The Event Checking Status bit remains set until the Event Checking Control bit is cleared by the controller. With the Auto-rearm feature, event checking is continuously enabled, insuring that no registration events are missed during the normal one cycle delay in re-arming the event checking mechanism. The down side of Auto-rearm feature is that it can generate a multitude of uninteresting events to process, in fact, multiple specified events per Controller Update Period. Therefore, these events shall either be buffered in the controller in an event array attribute or filtered based on the event data. The latter case is how Windowed Registration functionality is implemented; the Windowed Registration feature checks each registration event that arrives from the device to see if the Event Position yields an absolute position value that is within the configured Registration Window. If the computed registration position is outside the window, the event is thrown away. If the computed registration position is within the window the registration position and registration time stamp is stored in the controller
Controller sends the Event Acknowledge with the associated Event ID and Event Status to the device. Since the Auto-rearm bit is enabled, it leaves the associated Event Checking bit set to allow the device to continue checking for the next event.	→ Event Checking bit = 1 → Event Auto-rearm bit = 1 → Event Block Count = 1 → Extended Format = 0 → Event ID = id2 → Event Status = 0 (success)	Device detects the Event Checking Control bit is clear and processes the Event Acknowledge Data Block. Since the corresponding Auto-rearm bit is set, the device continues checking for the specified event. The device can now send another Event Notification if the specified event has occurred again. This Event Notification – Event Acknowledge cycle repeats until the controller clears the associated Event Checking Control bit.
...	...	...
Controller clears the Event Checking Control bit and the Auto-rearm bit to disable event checking.	→ Event Checking bit = 0 → Event Auto-rearm bit = 0 → Event Block Count = 0 → Extended Format = 0	Device detects the Event Checking Control bit is clear and disables event checking action.

Controller action	Connection data	Device action
Controller sees the Event Checking Status bit from the device is clear indicating that the event checking is disabled.	← Event Checking bit = 0 ← Event Auto-rearm bit = 0 ← Event Block Count = 0 ← Extended Format = 0	Device clears the corresponding Event Checking Status bit.
Controller may now set the Event Block Size to zero for all subsequent updates until such time as there is request to check for new events.	→ Event Block Size = 0 → No Event Block	Device sees Event Block Size is zero indicating that there is no Event Block data to process.
Device sees Event Block Size is zero indicating that there is no Event Block data to process.	← Event Block Size = 0 ← No Event Block	Device may now set the Event Block Size to zero for all subsequent updates until such time as there is request to check for new events.

In the case of an Extended Registration event where Auto-rearm Event Checking is requested, the Auto-rearm sequence above simply includes the Extended Event data as shown in Table 17.

- **Registration Data Set:** This bit mapped byte determines the contents of Registration Event Data Block that contains parameters needed to setup the registration event checking function, as specified in Table 19. The data in this block appears in the same order as the Registration Data Set bit numbers, so for example, Reg 1 Pos Window would appear before Reg 1 Neg Window in the Event Checking Data Block structure. The definitions of each of these Event Checking Data elements can be found by looking up the corresponding attribute in the Motion Device Axis Object specification. The value of this element is transferred to the Motion Device Axis Object attribute, Registration Data Set. This mechanism is not restricted to initial setup, but can also be used to change parameters associated with registration event checking function while the checking function is active.

Table 19 – Registration Data Set

Bit	Registration Data Element	Data type
0	Reg 1 Pos Window	DINT (max window) DINT (min window)
1	Reg 1 Neg Window	DINT (max window) DINT (min window)
2	Reg 2 Pos Window	DINT (max window) DINT (min window)
3	Reg 2 Neg Window	DINT (max window) DINT (min window)
4 to 7	(Reserved)	(Reserved)

- **Home Data Set:** This bit mapped byte determines the contents of Home Event Data Block that contains parameters needed to setup the home event checking function, as specified in Table 20. The data in this block appears in the same order as the Home Data Set bit numbers, so for example, Home Torque Threshold would appear before Home Torque Time. The definitions of each of these Event Checking Data elements can be found by looking up the corresponding attribute in the Motion Device Axis Object specification. The value of this element is transferred to the Motion Device Axis Object attribute, Home Data Set. This mechanism is not restricted to initial setup, but can also be used to change parameters associated with home event checking function while the checking function is active.

Table 20 – Home Data Set

Bit	Home Data Element	Data type
0	Home Torque Threshold	REAL
1	Home Torque Time	REAL
2 to 7	(Reserved)	(Reserved)

- **Watch Data Set:** This bit mapped byte determines the contents of Watch Event Data Block that contains parameters needed to setup the watch event checking function, as specified in Table 21. The data in this block appears in the same order as the Watch Data Set bit numbers, so for example, Watch 1 Position would appear before Watch 2 Position. The definitions of each of these Event Checking Data elements can be found by looking up the corresponding attribute in the Motion Device Axis Object specification. The value of this element is transferred to the Motion Device Axis Object attribute, Watch Data Set. This mechanism is not restricted to initial setup, but can also be used to change parameters associated with watch event checking function while the checking function is active.

Table 21 – Watch Data Set

Bit	Watch Data Element	Data type
0	Watch 1 Position	DINT
1	Watch 2 Position	DINT
2 to 7	(Reserved)	(Reserved)

- **Event Acknowledge ID:** Transaction number assigned to this event by the original event notification. Each event is assigned a new Event ID by incrementing the current Event ID stored in the device. Using the Event ID, the device is able to match the event acknowledgement to the appropriate event notification to complete the event data transaction.
- **Event Acknowledge Status:** Enumerated value indicating controller response to the event. A value of 0 indicates that the event was successfully processed. A non-zero value indicates that an error occurred in the event processing and the event shall be resent.

6.4.3.5 Service Data Block

The Service Data Block allows one service request per instance to be sent to the device in a given update. The service request requires a specific service response from the device indicating success or an error. In some cases, the response service contains requested data. In any case, the service request data persists in the Controller-to-Device Connection data structure until the controller receives the associated service response from the device.

Each service request is represented by a block of data organized as shown in Figure 26.

NOTE By design, the first 4 bytes of the Service Data Block do not follow the traditional CIP standard messaging format. That is primarily because this connection structure is, fundamentally, a CIP Implicit I/O connection, not an Explicit Messaging connection. However, in the case of a Fixed Connection format, the Service Specific Request Data defined below is sent via an Explicit Messaging connection and follows the CIP rules for explicit service request format.

Service Data Block			
Transaction ID	Service Code	–	–
Service Specific Request Data			

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Figure 26 – Service Data Block

- **Transaction ID:** Transaction number assigned to this service request by the controller. Each service request is assigned a new Transaction ID by incrementing the current Transaction ID stored in the controller. Using the Transaction ID, the controller is able to

match the service response to the appropriate service request and complete the service transaction.

- **Service Code:** Identifier that determines the object specific service request that follows. The list of supported Service Codes can be found in the Object Specific Services subclause of this part of the IEC 61800-7 series (see 7.5.1). CIP Common services are not applicable to the Service Data Block.
- **Service Specific Request Data:** The format and syntax of the Service Specific Request Data depends on the specified Service Code. This is true regardless of whether the service specific request data is passed in the Controller-to-Device Connection or as part of an Explicit messaging connection.

6.4.4 Device-to-Controller Connection

6.4.4.1 General

Like the Controller-to-Device Connection data structure described above, the CIP Motion Device-to-Controller Connection is organised as shown in Figure 27.

CIP Motion Device-to-Controller Connection format
Connection Header
Time Data Block
Instance Data Blocks

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Figure 27 – CIP Motion Device-to-Controller Connection format

6.4.4.2 Device-to-Controller Connection Header

6.4.4.2.1 General structure

The Device-to-Controller Connection Header contains critical axis configuration information needed to parse the Device-to-Controller Connection data block. The fixed portion of the Connection Header is defined as shown in Figure 28.

Connection Header			
Connection Format	Format Revision	Update ID	Node Status

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Figure 28 – Connection Header

- **Connection Format:** Same as Controller-to-Device definition except the required value for the Connection Type is either 3, indicating a Fixed Device-to-Controller Connection type or 7, indicating a Fixed Device-to-Controller Connection type.

0 = Fixed Controller Peer-to-Peer Connection

1 = Fixed Device Peer-to-Peer Connection

2 = Fixed Controller-to-Device Connection

3 = Fixed Device-to-Controller Connection

4 = Variable Controller Peer-to-Peer Connection

5 = Variable Device Peer-to-Peer Connection

6 = Variable Controller-to-Device Connection

7 = Variable Device-to-Controller Connection

8 to 15 = Reserved.

The Connection Format can be used to correctly parse the Device-to-Controller Connection data, not only by the device, but also by network diagnostic tools. The Connection Format value shall not change for the life of the connection and, therefore, need only be checked by the controller for the first received packet.

- **Format Revision:** The Format Revision number is 2 for any device that utilizes this edition of IEC 61800-7-202. Since controllers utilizing this edition only recognize Format Revision

2, Format Revision 1 is rendered obsolete and need not be supported by any CIP Motion device. This value is incremented by 1 for every revision of the Device-to-Controller Connection format that impacts the interface. The Format Revision allows newer controllers to support the connection formats generated by older devices. It also allows older controllers to recognize a newer connection format from a device that it cannot support and generate an appropriate error to its application. Network diagnostic tools can also key off the Format Revision to know how to parse the connection data packet. At the time the I/O Connection is established, the Format Revision number is also indicated by the I/O Connection Point in the Application Path of the Forward_Open service (see 6.4). The Format Revision value shall not change for the life of the connection and, therefore, need only be checked by the controller for the first received packet.

- **Update ID:** The Device-to-Controller Connection Update ID shall match the Update ID of the Controller-to-Device Update ID for a given cycle, and therefore shall be incremented every update period. Note that, when the device is in Synchronous Mode, this does not imply that the Controller-to-Device packet has to be processed prior to the Device-to-Controller packet being assembled to provide a matching Update ID. Maintaining matching Update IDs only requires the device increment the Device-to-Controller Update ID by one every cycle. Once the Update IDs are matched at Start-up they remain matched by incrementing every cycle thereafter. In the case where the associated Controller-to-Device packet is lost or late, the Device-to-Controller Update ID shall be incremented as if the Controller-to-Device packet had arrived on time. This allows the CIP Motion controller to ride through a lost or missed Controller-to-Device packets and maintain synchronization with matching Update IDs.

The Update ID is like the CIP message sequence count and is used by the controller to determine whether the connection buffer contains fresh data. If the Update ID, as seen by the controller, has not changed (late packet) or has skipped an Update ID (lost packet), the controller rides through the late or lost update using an extrapolation method based on the previous axis trajectory until a fresh update arrives. Like the CIP Motion device, if the number of consecutive missed updates reaches a configured limit, the controller declares a connection synchronization fault.

- **Node Status:** Contains bits used to indicate the status of the associated device communications node. The value of this element is derived from the Motion Device Axis Object class attribute of the same name. See 7.2.2.2 (Motion Device Axis Object) for details.

6.4.4.2.2 Fixed Connection Header

If the Connection Format is a Fixed Device-to-Controller Connection, the above header is immediately followed by the Instance Data Block.

6.4.4.2.3 Variable Connection Header

If the Connection Format is a Variable Device-to-Controller Connection, then the connection header contains additional fields related to multi-axis device addressing and time stamping (see Figure 29).

Connection Header			
Connection Format	Format Revision	Update ID	Node Status
Instance Count	Node Fault/Alarm	Last Received ID	Time Data Set
Device Time Stamp			
Device Time Offset			
Lost Updates	Late Updates	–	
Data Received Time Stamp			
Data Transmit Time Stamp			

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Figure 29 – Connection Header

- **Instance Count:** Same as Controller-to-Device definition.
- **Node Fault/Alarm:** This 8-bit element contains two 4-bit codes for fault and alarm conditions associated with the device communications node, as specified in Figure 30.

The values for these elements are derived from the Motion Device Axis Object class attributes of the same name. For details refer to the Motion Device Axis Object.

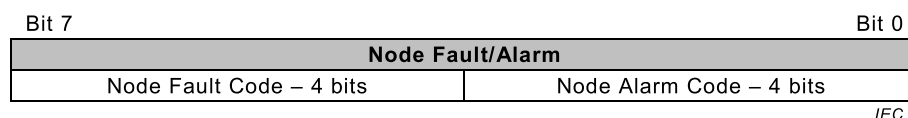


Figure 30 – Node Fault/Alarm

- **Last Received ID:** This is the Update ID of the last Controller-to-Device connection data block actually processed by the device. The Last Received ID is used by the controller to determine if the data that was sent was processed successfully by the device. For example, the Controller Time Offset is sent by the controller in the Controller-to-Device Connection only if the value has changed since the last update. It is recommended that the Last Received ID element be read by the controller to determine if the Controller Time Offset has been processed by the device and can therefore be removed from subsequent Controller-to-Device Connection updates.
- **Time Data Set:** Same as Controller-to-Device definition.
- **Device Time Offset:** This element is included in the Connection Header if the Time Stamp bit is set in the Time Data Set element. The Device Time Offset element represents the 64-bit System Time Offset value associated with the Device Time Stamp that follows. The Device Time Offset value is used by the controller to determine if System Time as defined in the device is skewed relative to System Time in the controller. Normally, System Time between the device and the controller are closely matched at any given time. But every few seconds, according to the IEC 61588:2009-based CIP Sync protocol, the time master of the system can correct the System Time reference, sometimes by a significant amount of time, perhaps as much as an hour. Indeed, even the master itself can change as a higher quality time master is discovered by the system. These step changes to the System Time reference propagate through to the various devices on the network in such a way that the controller and the device are running with skewed System Time values. The Device Time Offset value can be used together with the System Time Offset for the controller to both determine if System Time between the device and the controller is skewed and if so, to compensate for the skew.
- **Device Time Stamp:** This element is included in the Connection Header if the Time Stamp bit is set in the Time Data Set element. The Device Time Stamp represents the 64-bit System Time value, in nanoseconds, when the device's update timer event occurred that is associated with the actual data in the connection structure, for example when the actual data was captured. It is calculated by the device as the sum of the device's local clock value when update timer event occurred and the device's System Time Offset value. With the Device Time Stamp, the controller has all the information it needs to correct actual data values for differences between the device and controller update timing that result when the Controller Update Period is not an integer multiple of the Device Update Period or when the device updates are phase shifted relative to the controller. It is assumed in this timing model that the Device Time Stamp is registered to the beginning of the Device Update Period and is also the time when feedback was last captured. In the case where the Device Time Stamp does not match the local update time stamp of the controller, the controller extrapolates the actual response data value based on trajectory to correspond to the controller's time stamp. The timing diagram in Figure 31 illustrates how axis position data from the device is adjusted by the controller based on the relative time stamps between the device and the controller.

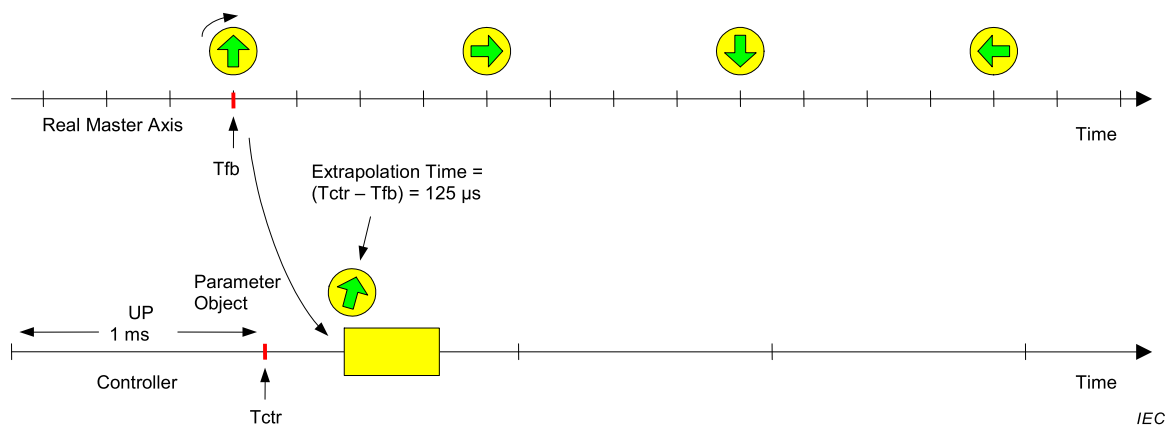


Figure 31 – Adjustment of actual position data based on Device Time Stamp

- **Lost Updates:** This element is included in the Connection Header if the Update Diagnostics bit is set in the Time Data Set element. The Lost Updates value represents the number of lost Controller-to-Device Connection packets detected since the connection was opened and synchronized. Lost packets are detected by examining the Update ID of received packets. A packet is determined to be lost when its expected Update ID is skipped. The Lost Packet class attribute is incremented for every Update ID that is skipped. The counter rolls over every 256 missed updates.
- **Late Updates:** This element is included in the Connection Header if the Update Diagnostics bit is set in the Time Data Set element. The Late Updates value represents the number of late Controller-to-Device Connection packets detected since the connection was opened and synchronized. Late packets are detected by examining the Time Stamp of received packet. A packet is determined to be late when the difference between its Controller Time Stamp and the device's Data Received Time Stamp exceeds the Controller Update Period. The counter rolls over every 256 late updates.
- **Data Received Time Stamp:** This element is included in the Connection Header if the Time Diagnostics bit is set in the Time Data Set element. The Data Received Time Stamp represents the 64-bit System Time value at the moment that the last Controller-to-Device Connection data, indicated by the Last Update ID, was written into device memory and ready for processing by the application layer. The time stamp units are nanoseconds. This value, when combined with controller's corresponding data transmit time stamp, can be used by the controller to generate Controller-to-Device Connection data delivery statistics.
- **Data Transmit Time Stamp:** This element is included in the Connection Header if the Time Diagnostics bit is set in the Time Data Set element. The Data Transmit Time Stamp represents the 64-bit System Time value at the moment that the application layer has initiated transmission of the Device-to-Controller Connection data to the controller. The time stamp units are nanoseconds. This value, when combined with controller's corresponding data received time stamp, can be used by the controller to generate Device-to-Controller Connection data delivery statistics.

6.4.4.3 Instance Data Blocks

6.4.4.3.1 General structure

After the Connection Header are one or more Instance Data Blocks as determined by the above Instance Count. The Instance Data Block is very similar to that of the Controller-to-Device Connection and has the basic structure shown in Figure 32.

Instance Data Block
Instance Data Header
Cyclic Data Block
Event Data Block
Service Data Block

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Figure 32 – Instance Data Block**6.4.4.3.2 Instance Data Header**

The Instance Data Header contains critical axis configuration information needed to parse and apply the data contained in the three data channels (see Figure 33). This header is only included in the Variable Connection format to accommodate multi-axis device applications. Information within the header can be used by the device communications interface to copy the individual data blocks into separate fixed memory locations for processing.

If configured for a Fixed Connection format, only the Cyclic Data Block for a single axis instance is supported so there is no need for any information on instance number or block sizing. Hence, the Instance Data Header is not included in the connection structure.

Instance Data Header			
Instance Num	–	Instance Blk Size	Cyclic Blk Size
Cyc. Act. Blk Size	Cyc. Read Blk Size	Event Blk Size	Service Blk Size

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Figure 33 – Instance Data Header

- Instance Number: Same as Controller-to-Device definition.
- Instance Block Size: Same as Controller-to-Device definition.
- Cyclic Block Size: Same as Controller-to-Device definition.
- Cyclic Actual Block Size: This value represents the size of the Cyclic Actual Data Block in 32-bit word units including the header. A Cyclic Actual Block Size of 0 indicates that there is no Cyclic Actual Data for the controller to process.
- Cyclic Read Block Size: This value represents the size of the Cyclic Read Data Block in 32-bit word units including the header. A Cyclic Read Block Size of 0 indicates that there is no Cyclic Read data currently configured for transfer to the controller and therefore there is no Cyclic Read Block included for the controller to process.
- Event Block Size: Same as Controller-to-Device definition. An Event Block Size of 0 indicates that there is currently no active event checking in progress.
- Service Block Size: Same as Controller-to-Device definition. A Service Block Size of 0 indicates that there is currently no service request to respond to.

6.4.4.3.3 Cyclic Data Block

The Cyclic Data Header at the top of the Cyclic Data Block of the Device-to-Controller Connection is always included regardless of the connection format. This header contains key elements related to the content of the Cyclic Data Block and the context of the data within the block with respect to the device (see Figure 34). Most of these elements are established by, and are therefore direct copies of, corresponding elements of the previous Controller-to-Device Connection Cyclic Data Block. Thus, the content of the Cyclic Data Block for the Device-to-Controller Connection is ultimately determined by the controller.

Cyclic Data Block			
Control Mode	Feedback Mode	Axis Response	Response Status
–	Actual Data Set	Status Data Set	Axis State
Cyclic Actual Data			
Cyclic Status Data			

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Figure 34 – Cyclic Data Block

- Control Mode: Same as Controller-to-Device definition.
- Feedback Mode: Same as Controller-to-Device definition.
- Axis Response: The 8-bit Axis Response is an enumerated value that is used for handshaking with the corresponding Axis Control element of the Controller-to-Device Connection to directly initiate device operations that do not require a CIP service request. Valid Acknowledge Codes match the corresponding Request Codes of the Axis Control element, and are shown in Table 22.

Table 22 – Axis Response

Acknowledge Code	Axis Response
0	No Acknowledge
1	Enable Acknowledge
2	Disable Acknowledge
3	Shutdown Acknowledge
4	Shutdown Reset Acknowledge
5	Abort Acknowledge
6	Fault Reset Acknowledge
7	Stop Process
8	Change Actual Position Reference Acknowledge
9	Change Command Position Reference Acknowledge
10 to 126	(Reserved)
127	Cancel Acknowledge
128 to 255	(Vendor Specific)

This Device Command/Axis Response mechanism for initiating state changes is fully described in the State Control subclause of the Motion Device Axis Object (see 7.6.1.2).

- Response Status: When there is a non-zero Acknowledge Code in the Axis Response, a Response Status value is also provided to indicate success or failure of the requested Axis Control operation. A Response Status of 0 indicates success, while a non-zero value indicates an error. The Response Status values comply with the CIP specification for General Status codes (see IEC 61158-6-2).
- Actual Data Set: Same as Controller-to-Device definition.
- Status Data Set: Same as Controller-to-Device definition. Status Data Set bits are only set when, 1) the corresponding bit in the Controller-to-Device connection's Status Data Set is set, and 2) there is change in the associated status data values since the last connection update.
- Axis State: This data element contains the enumerated Axis State value indicating the current state of this device axis instance according to the Motion Device Axis Object State Model.
- Cyclic Actual/Status Data: The Cyclic Actual/Status Data contains high priority data that needs to be applied to the associated device axis instance during the next device update. This block consists of actual data elements and status data elements that are consumed by the controller as explicitly determined by the Actual Data Set and the Status Data Set elements in the Cyclic Data Header. See the Controller-to-Device definitions in 6.4.3 for details of Cyclic Actual/ Status Data structure. Cyclic Actual Data shall be referenced to the Device Time Stamp.

6.4.4.3.4 Cyclic Read Data Block

The Cyclic Read Data Block can be used to synchronously update one or more targeted Motion Control Axis Object attributes within the controller based on the current value of associated attributes in the device. This mechanism can be used in conjunction with, for

example, an IEC 61131-3 based Function Block program to implement sophisticated outer loop control based on a wide variety of available Axis Control signals. Unlike service channel Get Axis Attribute service requests, which may take several device update cycles to process, the Cyclic Read Data mechanism guarantees the targeted parameter is updated every connection cycle.

The Cyclic Read Data Block is only supported in the Variable Connection format (see Figure 35).

Cyclic Read Data Block			
Cyclic Write Blk ID	Cyclic Write Status	Cyclic Read Blk ID	Cyclic Read Status
Cyclic Read Data			

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Figure 35 – Cyclic Read Data Block

The associated header for this block contains key elements related to the content of the Cyclic Read Data as well as the Cyclic Write Data from the previous Controller-to-Device connection update.

- **Cyclic Write Block ID:** This 8-bit ID determines the pre-defined Cyclic Write Block structure that was sent in the last Controller-to-Device connection update. A successful Cyclic Write Data update is indicated by echoing the associated Cyclic Write Block ID in the following Cyclic Read Data Block header. If the device determines that there is an error associated with the Cyclic Write Data, the device identifies the error with a non-zero Cyclic Write Status value.
- **Cyclic Write Status:** The Cyclic Write Status value is provided to indicate success or failure of the associated Cyclic Write Data update. A Cyclic Write Status of 0 indicates success, while a non-zero value indicates an error. The Cyclic Write Status values comply with the CIP specification for General Status codes (see IEC 61158-6-2).
- **Cyclic Read Block ID:** This 8-bit ID determines the pre-defined Cyclic Read Block structure to apply to the Cyclic Read Data for this update. Cyclic Read Block structures are defined using the Set Cyclic Read Data List service. A successful response to this service includes a new Cyclic Read Block ID that can be used in the next connection update to pass cyclic data in this format. If the device determines that there is an error associated with the Cyclic Read Data, the device identifies the error with a non-zero Cyclic Read Status value and no Cyclic Read Data.
- **Cyclic Read Status:** The Cyclic Read Status value is provided to indicate success or failure of the associated Cyclic Read Data update. A Cyclic Read Status of 0 indicates success, while a non-zero value indicates an error. The Cyclic Read Status values comply with the CIP specification for General Status codes (see IEC 61158-6-2).
- **Cyclic Read Data:** The Cyclic Read Data contains high priority data that needs to be applied to the associated controller axis instance. This block consists of signal and status data elements that are to be scaled and applied to corresponding Motion Control Axis Object attributes. The contents of the Cyclic Read Data are explicitly determined by the structure identified by the Cyclic Read Block ID found in the Cyclic Read Data Header. The ordering of the attribute data in the Cyclic Read Data Block is determined by the ordering of the Attr IDs in the Set_Cyclic_Read_List request. Attribute data elements shall be word aligned; 32-bit words are 32-bit aligned and 16-bit words are 16-bit aligned. Padding may be added to maintain word alignment.

EXAMPLE Figure 36 shows a Cyclic Read Data Block with 3 attributes where Attributes 1 and 3 are 32-bit values while Attribute 2 is a 16-bit value.

Attribute value 1	
Attribute value 2	16-bits – reserved
Attribute value 3	

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Figure 36 – Cyclic Read Data Block example

6.4.4.4 Event Data Block

The Event Data Block allows multiple event notifications to be sent to the controller in a given update. Each event notification requires a specific event acknowledge indicating success or an error. The event notification data persists in the Device-to-Controller Connection data structure until the device receives the corresponding event acknowledgement from the controller.

The Event Data Block for the Device-to-Controller Connection has the format shown in Figure 37.

Event Data Block			
Event Checking Status			
Reg Data Ack	Home Data Ack	Watch Data Ack	–
Event ID 1	Event Status 1	Event Type 1	–
Event Position 1			
Event Time Stamp 1			
Event ID 2	Event Status 2	Event Type 2	–
Event Position 2			
Event Time Stamp 2			
...			

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Figure 37 – Event Data Block

- **Event Checking Status:** This 32-bit word indicates if the device is currently checking for events based on various device inputs, for example marker, home, and registration inputs. Event checking is initiated when the corresponding Event Checking Control bit is set in the Controller-to-Device Connection. When an event occurs, the device captures both the time and exact axis position and passes this information to the controller in Event Data Blocks. But for the controller to process the event data, the corresponding Event Checking Status bit shall be set. For more detail on how this word is used in event operations, refer to the event mechanism sequence in 6.4.3.4. The last 4 bits of the Event Checking Control element is a special bit field, the least significant 3 bits of which specify the number of active events, which is literally the number of Event IDs listed in this Event Data Block. The most significant bit enables the extended Event Data Block format. A complete definition for the Event Checking Status attribute can be found in 7.3.7.2.2.

When the Event Data Block is present, the Event Checking Status element determines which additional elements are included. The word including Reg Data Ack, Home Data Ack, and Watch Data Ack are only included if the Extended Format bit is set in the Event Checking Status word. The Event ID #, Event Status #, Event Type #, Event Position #, and Event Time Stamp # elements will be repeated from 0 to 7 times as determined by the Event Block Count field in the Event Checking Status word.

- **Registration Data Ack:** This bit mapped byte is used to acknowledge receipt of any Registration Data Block parameters from a previous Controller-to-Device Connection update. There is no data in the Device-to-Controller Connection data structure associated with these bits.
- **Home Data Ack:** This bit mapped byte is used to acknowledge receipt of any Home Data Block parameters from a previous Controller-to-Device Connection update. There is no data in the Device-to-Controller Connection data structure associated with these bits.
- **Watch Data Ack:** This bit mapped byte is used to acknowledge receipt of any Watch Data Block parameters from a previous Controller-to-Device Connection update. There is no data in the Device-to-Controller Connection data structure associated with these bits.
- **Event ID:** Transaction number assigned to this specific event by the original event notification. Each event is assigned a new Event ID by incrementing the current Event ID stored in the device. Using the Event ID, the device is able to match the event acknowledgement to the appropriate event notification to complete the event data transaction.

- **Event Status:** Enumerated value indicating the status of the original event notification. A value of 0 indicates that the event was successfully detected by the device. A non-zero value indicates that an error occurred in checking for the event that prevents the event from being successfully detected. This might occur if there is a problem with event related hardware or that the required hardware is not supported by the device. The definition for Event Status error codes is currently left to the vendor discretion.
- **Event Type:** This enumerated value describes the type of event that occurred. Valid event types are as specified in Table 23.

Table 23 – Event Type

Event type	Event description
0	Registration 1 Positive Edge
1	Registration 1 Negative Edge
2	Registration 2 Positive Edge
3	Registration 2 Negative Edge
4	Marker Positive Edge
5	Marker Negative Edge
6	Home Switch Positive Edge
7	Home Switch Negative Edge
8	Home Switch-Marker ++
9	Home Switch-Marker +-
10	Home Switch-Marker -+
11	Home Switch-Marker --
12	Home Torque Threshold
13	Watch 1 Position Forward
14	Watch 1 Position Reverse
15	Watch 2 Position Forward
16	Watch 2 Position Reverse
17 to 127	(reserved)
128 to 256	(vendor specific)

- **Event Position:** 32-bit integer representation of the axis position when the designated event occurred. If the Event Status element has a non-zero error code, the Event Position is set to 0.
- **Event Time Stamp:** This element represents the 64-bit System Time value when the specified event occurred. The time stamp units are nanoseconds. Taken together with device's System Time Offset value that is part of the Device-to-Controller Connection Header, the controller has all the information it needs compute the absolute System Time when the event occurred. If the Event Status element has a non-zero error code, the Event Time is set to 0.

6.4.4.5 Service Data Block

The Service Data Block allows one service response per instance to be sent to the controller in a given update. Each service request requires a specific service response from the device indicating success or an error. In some cases, the response service contains requested data. In any case, the service response data persists in the Device-to-Controller Connection data structure until the device sees the associated service request removed from the Controller-to-Device Connection Instance Data Block (Service Block Size = 0) or a new service request is issued by the controller (incremented Transaction ID).

Each service response is represented by a block of data organised as shown in Figure 38.

NOTE Like the request structure, the structure of the service response does not follow the traditional CIP standard messaging format. That is primarily because this connection structure is, fundamentally, a CIP Implicit I/O connection, not an Explicit Messaging connection. However, the case of a Fixed Connection format, the Service Specific Request Data defined below is sent via an Explicit Messaging connection and follows the CIP rules for explicit service request format.

Service Data Block			
Transaction ID	Service Code	General Status	Extended Status
Service Specific Response Data			

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Figure 38 – Service Data Block

- **Transaction ID:** Transaction number assigned to this service response derived from the Transaction ID of the original request. Each service request is assigned a new Transaction ID by incrementing the current Transaction ID stored in the controller. Using the Transaction ID in the response, the controller is able to match the service response to the appropriate service request.
- **Service Code:** Identifier that determines the specific service response that follows, which shall match the Service code of the originating service request. A list of valid Service Codes for the Motion Device Axis Object is given in the Controller-to-Device Connection subclause.
- **General Status:** The General Status value is provided to indicate success or failure of the requested service request. A General Status of 0 indicates success, while a non-zero value indicates an error. The General Status values comply with the CIP specification for General Status codes (see IEC 61158-6-2).
- **Extended General Status:** The Extended General Status provides a method for defining vendor specific or service specific error codes. There is currently no standard definition for these codes.
- **Service Specific Response Data:** The format and syntax of the Service Specific Response Data depends on the specified Service Code.

6.4.5 Fixed Motion I/O connection format

By specifying a Fixed Connection Format, the CIP Motion I/O Connection can be reduced to a size that is readily applicable to lower-performance CIP networks or devices. In keeping with this application context, the following features have been removed from the connection structure to support the requirements of a fixed connection size and limited network bandwidth.

- Time Stamping
- Node Faults/Alarms
- Multiple instance support
- Dynamic Block Sizing
- Cyclic Read/Write Data Block
- Event Data Block
- Service Data Block

With Fixed Connection Format, service requests to the Motion Device Axis Object are supported only as an Explicit Messaging service.

Figure 39 and Figure 40 show an example of the Fixed Connection Format being used in a simple variable speed drive application requiring only a velocity command and returning actual velocity. In this case, the connection size has been reduced to 16-bytes, a size that is well suited for lower performance networks like DeviceNet.

Controller-to-Device Connection format			
Connection Format	Format Revision	Update ID	Node Control
Control Mode	Feedback Mode	Axis Control	–
Command Data Set	Actual Data Set	Status Data Set	–
Command Velocity			

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Figure 39 – Fixed Controller-to-Device Connection format (fixed size = 16 bytes)

Device-to-Controller Connection format			
Connection Format	Format Revision	Update ID	Node Status
Control Mode	Feedback Mode	Axis Response	Response Status
–	Actual Data Set	Status Data Set	Axis State
Actual Velocity			

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Figure 40 – Fixed Device-to-Controller Connection format (fixed size = 16 bytes)

6.4.6 CIP Motion I/O Connection timing model

6.4.6.1 General

The general timing model for the CIP Motion I/O Connection data exchange is described in this subclause in the context of a CIP Motion Drive device, but also applies to other CIP Motion device types. Data exchange between the drive and the controller is paced by the Controller Update Period with one Device-to-Controller data packet sent for every Controller-to-Device data packet received. The Controller-to-Device Connection packets are sent periodically according to the configured Controller Update Period. The Device Update Period, i.e. the update period at which the drive performs its control calculations, is typically much faster than the Controller Update Period. The basic CIP Motion 1-Cycle timing model is illustrated in Figure 41.

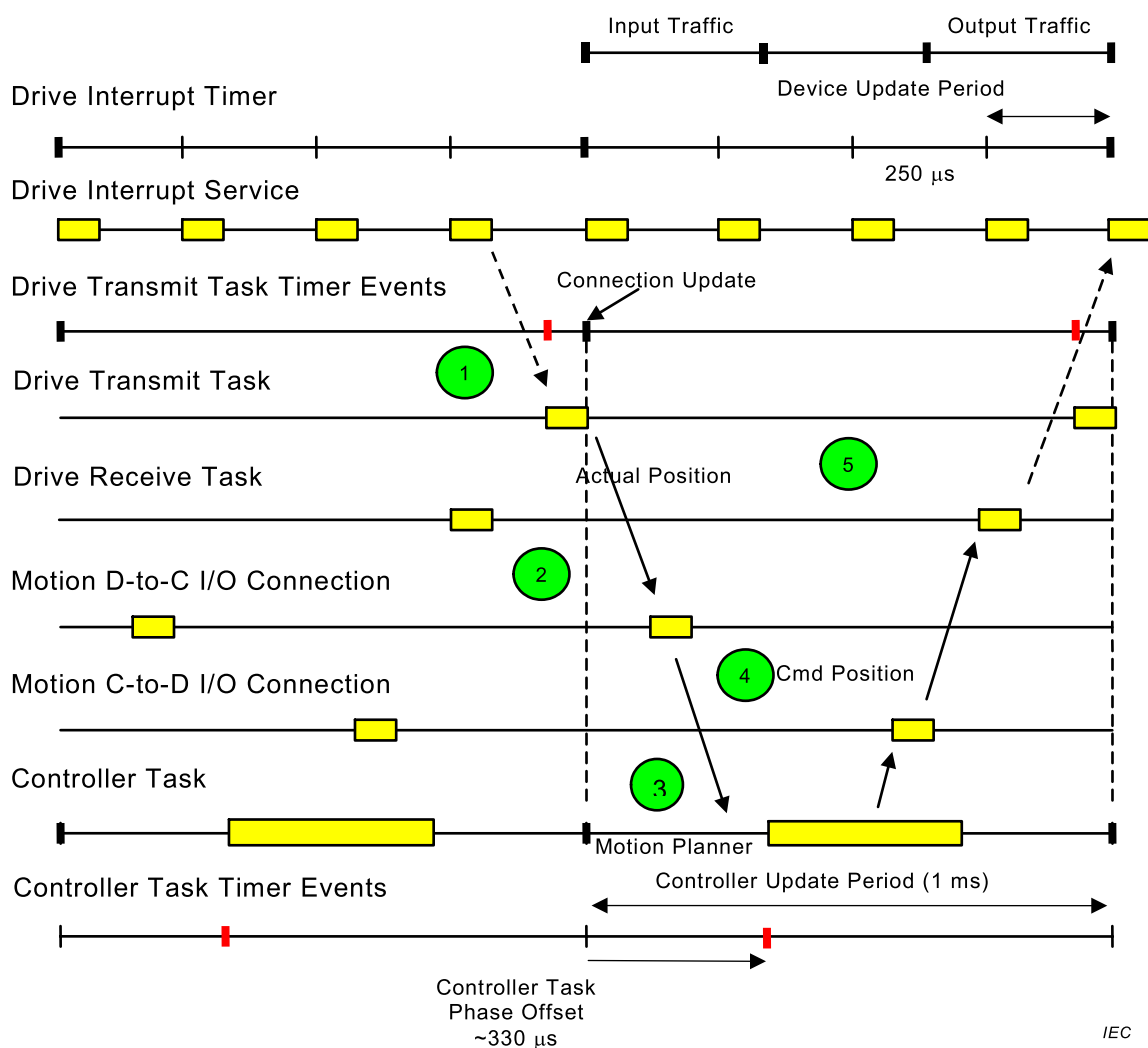


Figure 41 – CIP Motion 1-Cycle timing model

This is classified as a 1-Cycle Timing Model since a complete I/O data transaction occurs within one connection update cycle. The timing model consists of five basic steps which are shown in Figure 41 and described as follows.

- 1) A periodic Drive Transmit Timer Event (top red tick-mark shown above) scheduled just before the next Connection Update cycle starts the Drive Transmit Task that is responsible for managing the Device-to-Controller connection. The Drive Transmit Task gets the latest Actual Position value calculated by the last Drive Interrupt Service (assuming Position Control Mode in this example) and the associated Time Stamp.
- 2) Once the Actual Position and Time Stamp data is assembled, along with other axis data, into the Device-to-Controller Connection data structure, the drive transmits the packet to the controller. For best performance, it is recommended that the actual transmission be as close to the start of the Connection Update cycle as possible. The only timing constraint is that during normal operation, the drive device shall begin transmission no later than the start of the cycle. This allows time for the Device-to-Controller Connection packet to traverse the network and arrive at the controller prior to the start of the phase delayed Controller Task.
- 3) After a predetermined Phase Offset time from the start of last Controller Update cycle, which, in this model, is set to be 1/3 of the Controller Update Period, a Controller Task Timer Event triggers the Controller Task to start. The Controller Task begins by parsing the data from the drive, including the Actual Position value. The controller then runs a Motion Planner to compute a new Command Position value to send back to the drive. When gearing or camming operations are active, it may be necessary to use the Actual

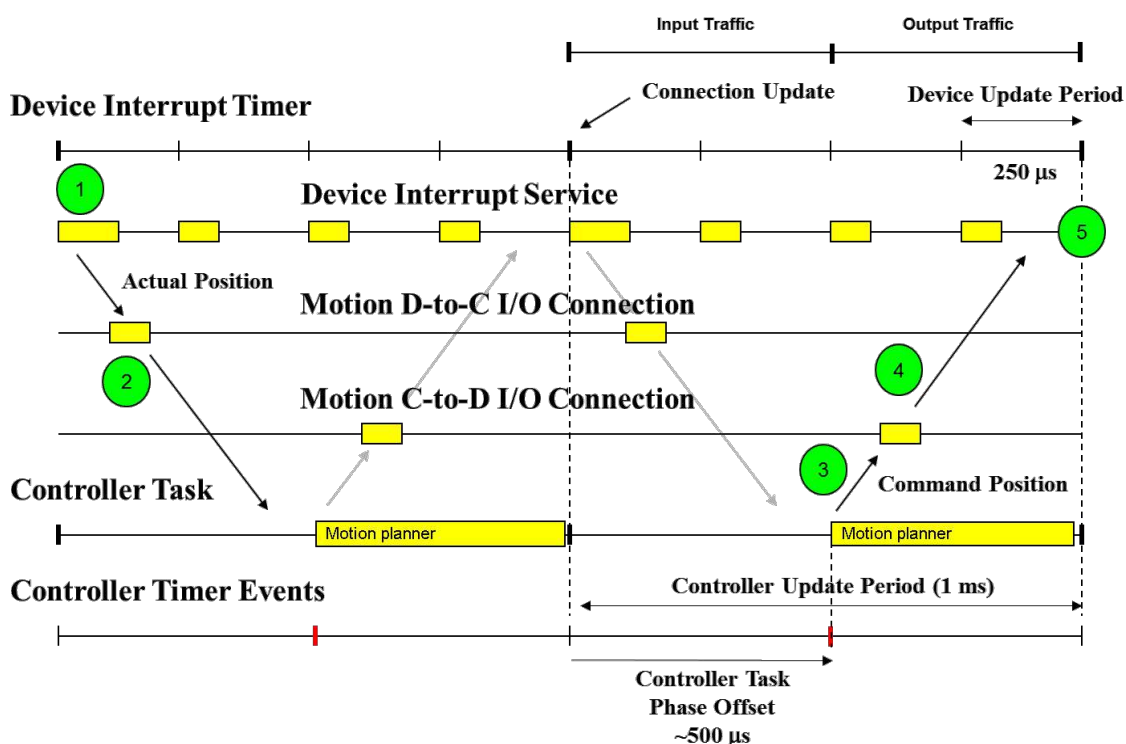
Position of a master drive axis as input to compute the Command Position of one or more slave drive axes.

- 4) Once the Command Position is calculated, it is assembled, along with other axis data, into the Controller-to-Device Connection structure, the controller transmits the data to the drive. In general, it is recommended that the transmission occur no later than 2/3rds of the way through the Controller Update Period to allow time for the packets to reach the targeted drive before the end of the current cycle. The Controller-to-Device Connection packet traverses the network and arrives at the drive prior to the start of the next Connection Update cycle. There is no hard timing constraint here; the packet may arrive much earlier in the update cycle or possibly later than the start of the next update cycle. The drive device simply processes the packet upon arrival.
- 5) A Drive Receive Task executes in response to notification from the network interface that a new Controller-to-Device packet has arrived. The Drive Receive Task begins by parsing the data from the controller, including the Controller Time Stamp and Command Position value. The drive then applies the Command Position data using the time stamp to calculate new coefficients for the fine interpolation polynomial. These coefficients are used by subsequent Drive Interrupt Services to compute the fine command position value that is applied to the input of the drive's position control loop. For best performance, it is recommended that the polynomial coefficients be applied to the fine interpolator as closely as possible to the start of the next update cycle. Applying the new coefficients early can create unnecessary error in the command trajectory when splicing the current fine interpolator segment to the new fine interpolator segment that begins with the next update cycle. Applying the new coefficients late forces the fine interpolator to run as a fine extrapolator into the next cycle, introducing unnecessary extrapolation error in the command trajectory when the extrapolated segment is spliced to the next fine interpolator segment. The Drive Receive Task concludes with the device scheduling the next Drive Transmit Timer Event relative to the start of the next Connection Update cycle, which is the sum of the Controller Time Stamp and the Controller Update Period.

The above example sequence represents an implementation that uses separate, interrupt-driven, Drive Transmit and Drive Receive Tasks. Other implementations may choose to combine one or both of these tasks with the Drive Interrupt Service that performs the control computations. In this case, processing the CIP Motion Connection data is done on a polled basis; the Drive Interrupt Service would need to check if a Controller-to-Device packet has been received and also check if it is time to assemble and send a Device-to-Controller packet.

While other timing models are possible based on the controller configuration, all CIP Motion timing models begin with the drive sending actual data to the controller at the beginning of the Controller Update cycle and end with command data arriving before a subsequent Controller Update cycle. However, it is not required that the command data sent to the drive in a given Controller Update cycle be the result of calculations performed during that cycle. These rules form the basis for multi-cycle timing models.

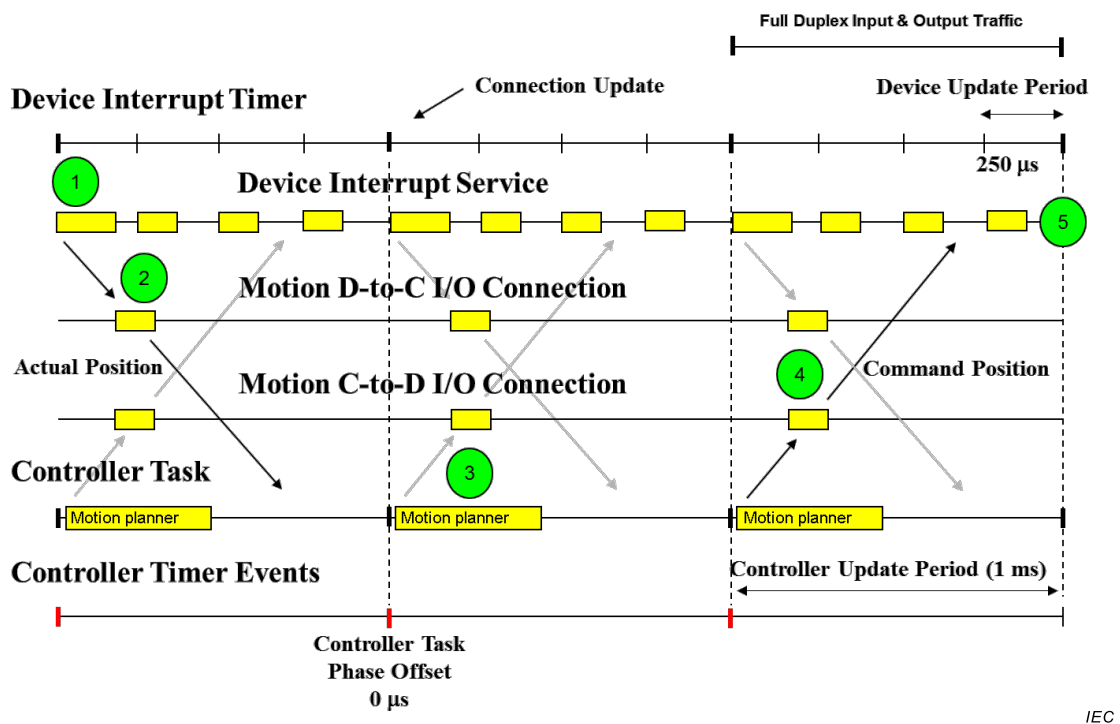
Figure 42 and Figure 43 illustrate these common CIP Motion timing model characteristics, by considering the 2-Cycle and 3-Cycle timing models.



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Figure 42 – CIP Motion 2-Cycle timing model

As with all CIP Motion timing models, the 2-Cycle Timing Model shown in Figure 42 begins with the drive transmitting the D-to-C connection packet to the controller at the beginning of the update cycle. But in this case the Controller Task does not start until half way through the update cycle, thus allowing more time for the D-to-C connection packet to reach the controller before the motion planner task runs. Unlike the 1-Cycle Timing Model, the C-to-D connection packet is not transmitted back to the drive until the next time the motion planner task runs. This again allows more time for the C-to-D connection packet to reach the drive. Thus, it takes 2 connection cycles to complete the I/O data transaction with the drive device.



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Figure 43 – CIP Motion 3-Cycle timing model

The above 3-Cycle Timing Model shown in Figure 43, once again, begins with the drive transmitting the D-to-C connection packet to the controller at the beginning of the update cycle. But in this case, the D-to-C connection packet has the entire update cycle to reach the controller before the motion planner task runs at the beginning of the next cycle. After the motion planner task runs, the resultant C-to-D connection packet is not immediately transmitted back to the drive, but rather is transmitted by the Motion Planner task at the beginning of the next cycle, again, giving the packet a full update cycle to reach the targeted drive. So, in this case, it takes 3 connection cycles to complete the I/O data transaction with the drive device. With the 3-Cycle Timing Model, CIP Motion input and output packet traffic is traversing the Full Duplex Ethernet media in both directions.

Based on these common timing model characteristics, the drive does not need to know the specific timing model the controller is applying. The drive simply transmits D-to-C packets to the controller at the beginning of the update cycle and processes C-to-D packets as they arrive from the controller.

6.4.6.2 Controller-to-Device connection timing

Most motion control protocols require the Controller Update Period to be an integer multiple of the Device Update Period, that is the drive's internal control loop update period. But because the CIP Motion I/O Connection packet includes a Time Stamp, the update period of the controller does not need to have any fixed relationship with the update period of the drive.

The following subclauses present a more detailed description of the CIP Motion timing model., presented in the context of a CIP Motion Drive device type with the understanding that the timing model applies generally to all CIP Motion device types.

One leg of the CIP Motion I/O Connection data exchange cycle is initiated by the controller via the Controller-to-Device Connection packet. The inclusion of Time Stamp and Time Offset information along with the command data in this packet relieves the stringent timing requirements imposed by other motion control network protocols. Specifically, time stamping allows the drive to compute command data values based on its own Device Updates that,

unlike other motion control network protocols, does not need be in fixed relationship to the Controller Updates.

Figure 44 illustrates how command data and time stamps delivered by the Controller-to-Device connection are applied to the drive axis when fine interpolation is required. In this example, the Drive Transmit and Receive Tasks described in above Timing Model section have been combined with the Drive Interrupt Service into a singular, periodic Drive Task that runs at the Device Update Period.

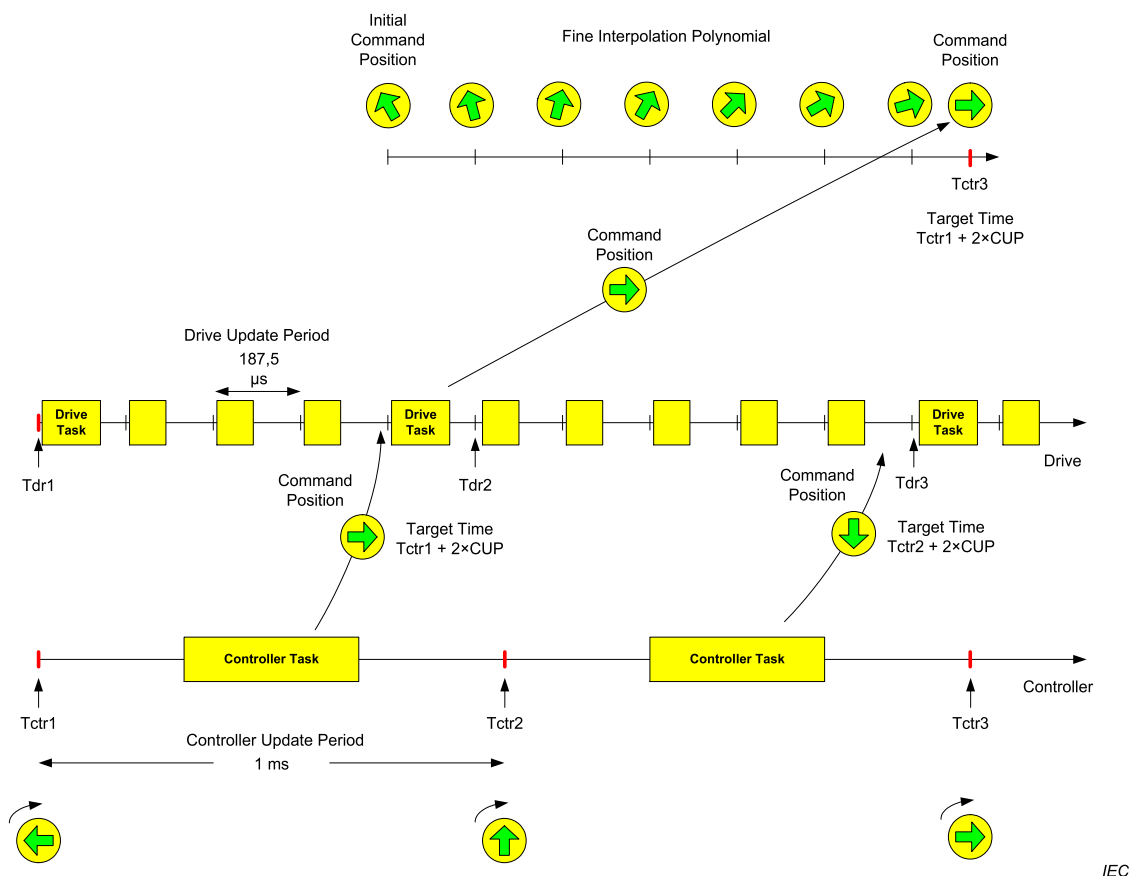


Figure 44 – Controller-to-Device Connection timing with fine interpolation

The following steps describe in detail how connection data is transferred from the controller to the drive for fine interpolation during a typical connection cycle in the general case where the Controller Update Period (CUP) is not an integer multiple of the Device Update Period.

- 1) **Controller Transmit:** As part of the Controller Task, the controller initiates transmission of a Controller-to-Device Connection packet with new command data to the targeted drive with an incremented Update ID and a new Controller System Time Stamp (and Controller System Time Offset if changed since last update) referencing the system time at the start of the current Controller Update cycle. The Instance Data Block for the targeted axis also contains the Command Target Update, which in this example is set to +2 to account for the one Controller Update Period (CUP) transport delay and the one Controller Update Period (CUP) delay for fine interpolation. It is recommended that the controller transmit the Controller-to-Device Connection packet no later than 2/3rds of the way though the Controller Update Period to allow time for the packets to reach the targeted drive. The Controller-to-Device Connection packet is received by the drive and processed by the network stack upon arrival. The network stack presents the new Controller-to-Device Connection data to the application layer's Drive Task for parsing.
- 2) **Update ID Check:** In response to new data notification by the stack, the Drive Task begins parsing the Controller-to-Device Connection packet data by first checking the Update ID. If

the Drive Task discovers that the Update ID has incremented by more than 1 since the last update, the drive increments the Lost Controller Updates value.

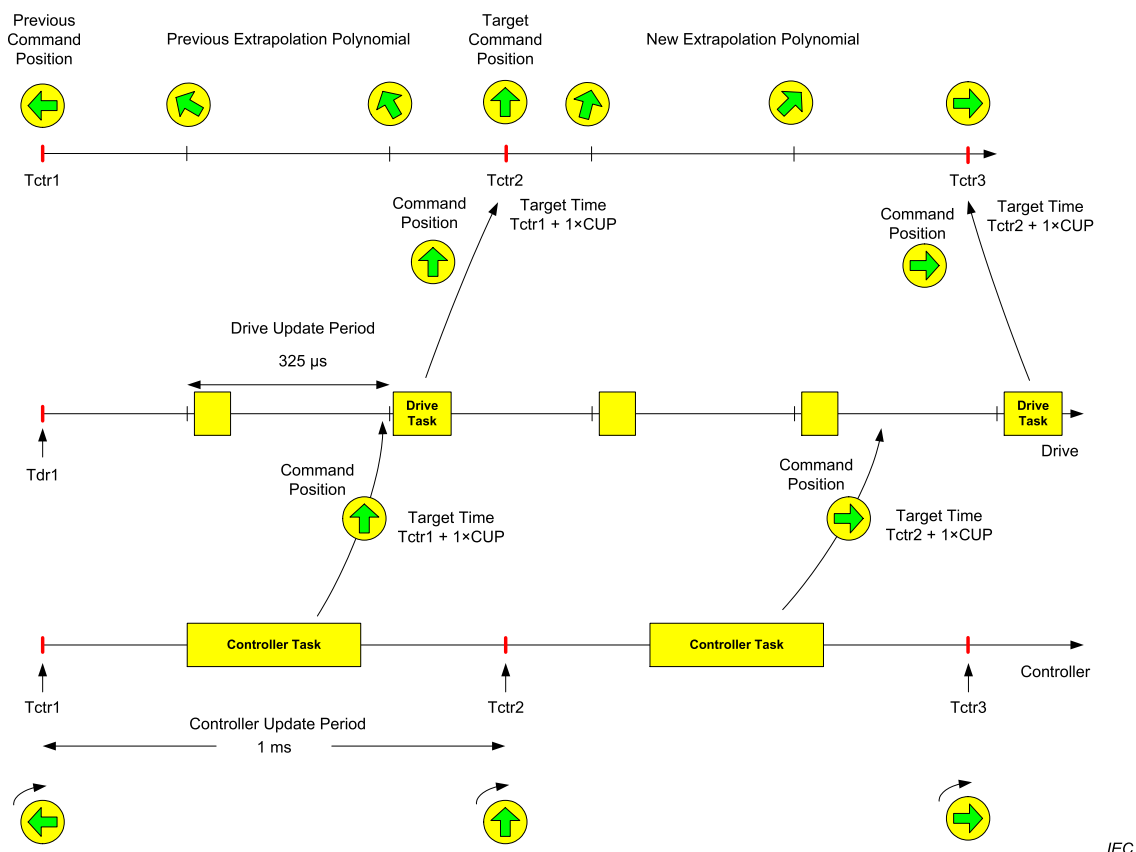
- 3) Synchronous Operation Check: Next, the Drive Task begins parsing the Controller-to-Device Connection packet data by first checking if the drive is synchronized. If not synchronized, skip to the Apply Command Data step since there is no need to perform the System Time Offset Check or the Late Update Check. Bypassing these checks allows for control of the drive during start-up or even in the case where the drive does not have any time synchronization services. This is referred to as Asynchronous Operation.
- 4) System Time Offset Check: Synchronous operation requires that System Time between the drive and the controller match to make the time stamps exchanged between the devices meaningful. Since System Time is adjusted periodically by the time master and the adjustments can propagate to the different devices in the network at different times, it is possible for System Time between the controller and the drive to be temporarily skewed. This skew shall be detected and, if present, the drive shall adjust the Controller Time Stamp to compensate for the skew. The drive does this offset compensation only if the Controller or Device System Time Offsets have changed since the last update. See 6.4.6.7 (System Time Offset Compensation) for details of this algorithm.
- 5) Late Update Check: Assuming synchronous operation, the drive then computes the difference between the current Drive Task time stamp and the Controller Time Stamp passed in the Controller-to-Device Connection packet (adjusted, if necessary). A difference of more than one Controller Update Period increments the Late Controller Updates value. If the difference is greater than (Controller Update Delay Low Limit $\times$ Controller Update Period), the drive throws a Controller Connection Update Alarm. If the difference is greater than (Controller Update Delay High Limit $\times$ Controller Update Period), the drive throws a Control Connection Update Fault.

NOTE 1 If the time difference has exceeded the Controller Update Period, the current fine interpolator polynomial has become, effectively, an extrapolator polynomial allowing the drive to ride through the late data condition until the new data arrives.

- 6) Apply Command Data: Assuming synchronous operation and a Command Target Update of +2 for fine interpolation, the drive computes coefficients for the fine interpolation polynomial based on the command reference being applied at the computed Target Time, which is the sum of the (adjusted) Controller Time Stamp (CUP), T_{ctr1} , and the product of the Command Target Update and Controller Update Period, or in this case, $2 \times CUP$. These coefficients are used by the drive's control service routine to compute the fine command position based on the time of the service. Ordinarily these new coefficients are applied for the duration of the next update cycle, providing fine interpolation until System Time has reached Target Time. If the Target Time is less than the current System Time when the Controller-to-Device Connection packet was received, i.e. this is a late packet, then new coefficients to the polynomial are still computed based on this command data to improve the accuracy of the extrapolation calculations. In general, whenever command data is late, the data still represents the freshest command data available and the coefficients in this case shall be applied as soon as possible. If asynchronous operation, the command data is applied to the drive's control service immediately.
- 7) Schedule Next Device-to-Controller Connection Update: The drive calculates System Time for the next Connection Update Cycle as the sum of the Controller Time Stamp and the Controller Update Period. This value is used to establish the Actual Update Window criteria for the next Device-to-Controller Connection Update or to directly schedule the Device-to-Controller Connection Update via a timer event.

If the Command Target Update is set to +1, the computed polynomial in step 6 is not applied for the purpose of fine interpolation but rather for extrapolation; the extrapolation polynomial allows the drive to compute an accurate command data value at the time the drive performs its control calculations based on previous axis trajectory.

Figure 45 illustrates this timing model in the general case where the Controller Update Period (CUP) is not an integer multiple of the Device Update Period and the Command Target Update is set to +1 for Extrapolation.



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Figure 45 – Controller-to-Device Connection timing with extrapolation

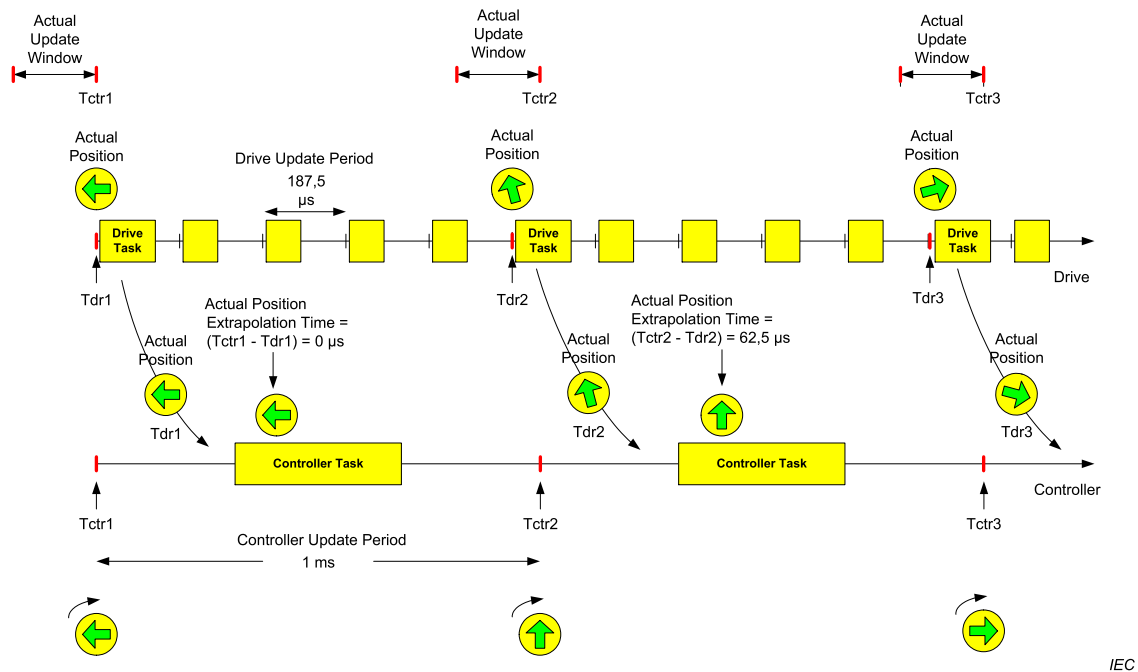
NOTE 2 In the above example, there are not many Device Update Periods in a given Controller Update Period. When this is the case, fine interpolation is not critical to drive performance and command data can be applied more directly to the drive's control structure using extrapolation without the extra delay required to support fine interpolation. Extrapolation has the disadvantage however that extrapolation error is manifested more directly to the command data resulting in rougher motion than when using fine interpolation.

All cyclic data associated with the Controller-to-Device Connection packet shall be applied in the Drive Task command update to make the earliest possible use of fresh command data, computing new interpolation/extrapolation polynomial coefficients.

6.4.6.3 Device-to-Controller connection timing

The other leg of the CIP Motion I/O Connection data exchange cycle is initiated by the drive via the Device-to-Controller Connection packet. When in synchronous mode, the CIP Motion Device-to-Controller Connection includes a Device Time Stamp (and Device Time Offset) with the actual data to allow the controller to calculate the position of the drive axis at the time the Controller Task update occurs. Time stamping allows the drive to sample feedback and compute actual data values based on its own Device Updates that, unlike other motion control network protocols, does not need be strictly related to the Controller Updates. Figure 46 illustrates how actual data and time stamps delivered by the Device-to-Controller Connection are used to adjust drive axis actual position, for example, to the controller's timebase.

Again, for simplicity, the Drive Transmit and Receive Tasks described in the above Timing Model section have been combined with the Drive Interrupt Service into a singular, periodic Drive Task that runs at the Device Update Period.



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Figure 46 – Use of Time Stamp to adjust actual position to the controller's timebase

The following steps describe in detail how connection data is transferred from the drive to the controller during a typical connection cycle in the general case where the Controller Update Period (CUP) is not an integer multiple of the Device Update Period. Of course, this sequence also applies to the special case where the Controller Update Period (CUP) is an integer multiple of the Device Update Period.

- 1) **Actual Update Check:** If the axis is synchronized, the drive compares current Drive Task time stamp with the Actual Update Window established in the last update cycle. The Actual Update Window has duration of 1 Device Update Period and ends at the latest possible time for the drive to completely assemble the Device-to-Controller Connection packet and have it ready for transmission by the start of the next update cycle. If the Drive Task time stamp is within the time window, this is an actual data update cycle. If the time stamp is before the window, then it is not time to send data to the controller. (The purpose of the Actual Update Window is to schedule the actual data transmission as close to the start of the Connection Update cycle as possible, while also minimizing time between the feedback capture and the start of the Connection Update cycle.) For timer event interrupt-driven Transmit Tasks, there is no need for the Actual Update check since the Transmit Task timer event service is already scheduled to execute at the proper time. If the axis is not synchronized and we have just received a command update via the Controller-to-Device Connection, or the time since the last command update has exceeded 1 Controller Update Period, then this is also an actual update cycle, so move on to the Drive Transmit step.
- 2) **Drive Transmit:** If this is an actual update cycle, then the Drive Task assembles and transmits the Device-to-Controller Connection packet to the controller with the latest Actual Data, incremented Update ID and, for synchronous devices, the associated Device Time Stamp (and Device Time Offset if changed since the last update). For best performance, it is recommended that the packet transmission be as close to the start of the Connection Update cycle as possible. The only timing constraint is that during normal operation, the drive device shall begin transmission no later than the start of the actual update cycle. This is the most important timing constraint associated with the CIP Motion timing model. As such, Device-to-Controller Connection packet processing shall have priority over Controller-to-Device Connection packet processing. The extra delay this may introduce to Controller-to-Device Connection packet processing does not have a significant effect on motion control quality as long as the Device Update Period is much shorter than the Controller Update Period.

- 3) Update ID Check: In the next Controller Task, the controller checks for new data from the drive by checking for a changed Update ID. If the Update ID has not changed since the last update, the controller increments the Lost Updates value associated with the drive's Device-to-Controller connection. The following steps are performed regardless of whether or not the Update ID has changed.
- 4) Sync Mode Check: Drive checks the Sync Mode bit of the drive's Node Status byte to determine if the drive axis is synchronized. If not synchronized, skip to the Apply Actual Data step to avoid Late Update checking and Time-Stamp Correction. Bypassing these subsequent steps allows the drive to operate during start-up or even in the case where the drive does not have any time synchronization services.
- 5) System Time Offset Check: Synchronous operation requires that System Time between the drive and the controller to match to make the time stamps exchanged between the devices meaningful. Since System Time can be adjusted periodically by the time master and the adjustments can propagate to the different devices in the network at different times, it is possible for System Time between the controller and the drive to be skewed. This skew shall be detected, and if present, the drive shall adjust the Device Time Stamp to compensate for the skew. The controller does this offset compensation only if the Controller or Device System Time Offsets have changed since the last update. See 6.4.6.7 (System Time Offset Compensation) for details of this algorithm.
- 6) Late Update Check: The controller computes the difference between the current Controller Task time stamp and the drive's Device Time Stamp in the Device-to-Controller Connection packet. If the difference is greater than $(\text{Controller Update Delay Low Limit} \times \text{Controller Update Period})$, the controller throws a Control Sync Alarm. If the difference is greater than $(\text{Controller Update Delay High Limit} \times \text{Controller Update Period})$, the controller throws a Control Sync Alarm.
- 7) Time-Stamp Correction: If the previously computed time stamp difference is non-zero, then extrapolate the actual data value based on previous axis actual trajectory to line up with the controller's time stamp. This correction is necessary because the motion planner assumes that actual input data is implicitly time stamped to the beginning of the Controller Update Period. Under normal operation, the computed time stamp difference is small relative to the Controller Update Period and the resultant position correction is small, but in cases where the Device-to-Controller Connection packet is late or lost, this position correction is significant and critical for continued operation as it allows the control system to "ride through" this condition based on previous axis trajectory.
- 8) Apply Actual Data: Controller applies actual data as inputs to the motion planner, which computes new command reference data for the next Controller-to-Device update.

6.4.6.4 Scheduling the next update

To insure that a Device-to-Controller Connection update occurs regardless of having received a Controller-to-Device Connection packet, during every cycle the drive shall schedule the time for next cycle's Device-to-Controller Connection update.

For synchronous operation, the drive computes the time for the next cycle's Device-to-Controller Connection update to be enough before the end of the next update cycle to allow time for Device-to-Controller Connection data to be assembled and ready to transmit before the end of the cycle. For timer event interrupt-driven Transmit Tasks, this rescheduling occurs automatically when the local periodic timer reloads. For implementations where the Device-to-Controller Connection is handled by the Drive Task, the rescheduling shall be done explicitly by the drive at the conclusion of every Device-to-Controller Connection update.

For asynchronous operation, the drive initiates a task to assemble and transmit the Device-to-Controller Connection packet immediately after receiving (and parsing) a Controller-to-Device packet. The drive then schedules the transmission of the next Device-to-Controller packet to be 1 Controller Update Period from the time this Controller-to-Device packet was received. Thus, the transmission of that Device-to-Controller packet will take place even if the next Controller-to-Device packet is lost or late. The Update ID for this Device-to-Controller packet shall be the Controller-to-Device packet's Update ID + 1 so that the Update IDs for the

Device-to-Controller and Controller-to-Device packets match for a given update cycle thereafter.

6.4.6.5 Device Update Period independence

The timing diagram in Figure 47 illustrates how two drive axes can be tightly coordinated despite having different Device Update Periods and despite having an associated Controller Update Period that is not an integer multiple of either Device Update Period.

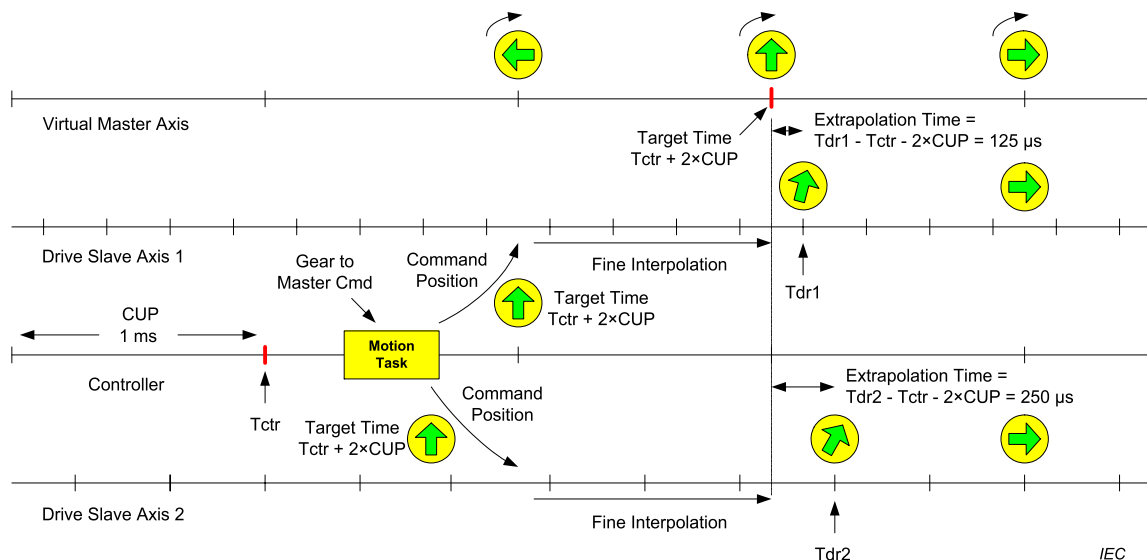


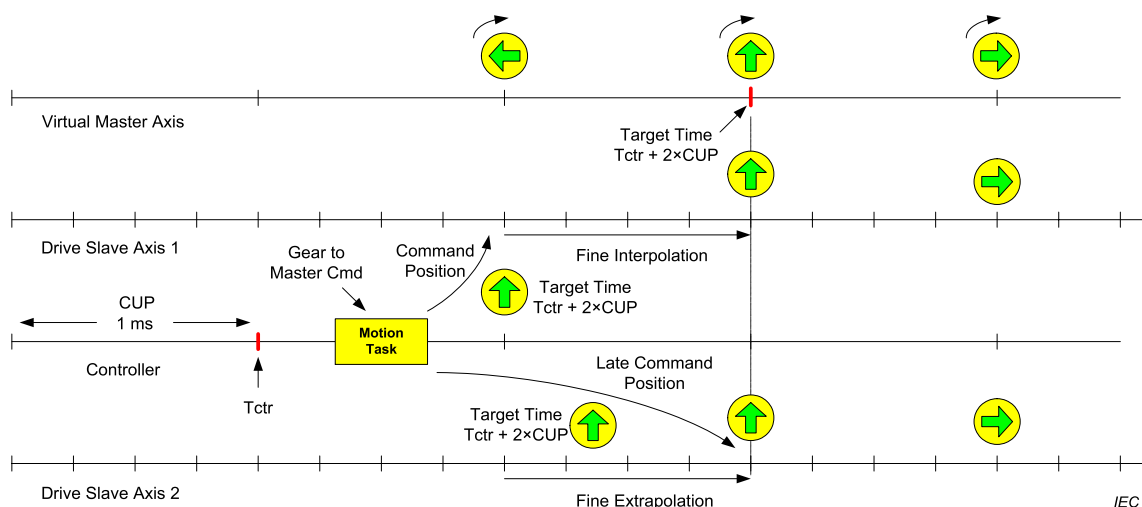
Figure 47 – Coordination of two drives with different Update Periods

In the above timing diagram, the controller's motion planner task sends identical command positions and time stamps to two slave drive axes that, while synchronized with System Time, are running at different drive update rates. When the command position data arrives at the two drives, they use the Controller Time Stamp, the Command Target Update, and the Controller Update Period to compute new coefficients to the interpolation polynomial based on the constraint that the polynomial value at time equal to (Controller Time Stamp + Command Target Update $\times$ Controller Update Period) is the specified Command Position value. Since there is no dependency on the drive update rate, the polynomial coefficients computed by each drive are identical. Since neither drive has an update that coincides with this target time, the drives use the fine interpolation polynomial to calculate the command position reference for each drive update until a fresh command position is received from the controller. If a new command position does not arrive until well after the target time, the drive continues to use the same polynomial equation to "extrapolate" command position for subsequent drive updates as shown in Figure 47. This extrapolation continues until fresh data arrives and new coefficients can be calculated. In this way, whether by interpolation or extrapolation, each slave axis runs smoothly and the two axes stay phase locked with the master axis.

6.4.6.6 Transmission latency independence

Precise coordination of multiple CIP Motion drive axes can be maintained even when the Controller-to-Device Connection packets incur significant delays while travelling across the CIP network. In Figure 48, the packet for Slave Drive Axis 2 has incurred a significant delay during transmission. As a result, the command position for this axis shall be extrapolated from the last fine interpolation polynomial. This allows the axis to move smoothly through a transmission latency disturbance. When the new command data does arrive, the new command value may not agree with extrapolated value due to extrapolation error. This error can result in a disturbance to the motion profile. The magnitude of the extrapolation error depends on the dynamics of the motion profile and the controller update rate. In most real-

world applications, transmission latencies lasting several update periods can occur without any noticeable disturbance to the associated motion profile.



**Figure 48 – Coordination of multiple drive axes
in case of delayed Controller-to-Device Connection packets**

6.4.6.7 System Time Offset compensation

6.4.6.7.1 General

CIP Motion is built on the foundation of CIP Sync and its underlying IEC 61588:2009 time synchronization protocol, which defines a mechanism to distribute and synchronize time for devices on a network.

Using the IEC 61588:2009-based CIP Sync protocol, each device on the network has a representation of System Time that is tightly synchronized with the Grand Master Clock of the network. System Time for each device is defined as the sum of the device's local clock time and the device's System Time Offset. With CIP Sync, it is important to note that while the local clock runs at the same frequency as the Grand Master Clock, it is not strictly phase locked to the Grand Master Clock. Only the device's System Time value is phase locked to the Grand Master Clock and is suitable for synchronizing data exchange between the controller and the drive device.

According to the CIP Sync protocol, System Time can incur large step changes, from a few milliseconds to many years, and these step changes are propagated through the network to each device. These step changes in time may be caused, for example, by the user manually adjusting the master's absolute time to correct for clock drift, or by a change in time mastership. IEC 61588:2009 does not address step changes in time as part of its protocol and, therefore, has no provision for handling this condition during real time control operation. The CIP Sync standard defines a concept called Time Step Compensation that, when implemented in device's Motion Device Axis Object, effectively handles this condition. The following subclause describes this algorithm in detail.

6.4.6.7.2 Time step compensation

According to CIP Sync, step changes to the master clock time shall propagate through the distributed time system; the time to propagate this step change in time through the system from the time master to the furthest time slave is a function of the sync packet interval (default is 2 s) and the number of hops (1 hop per IEC 61588:2009 boundary clock, for example a switch implementing the IEC 61588:2009 protocol). This is illustrated in Figure 49.

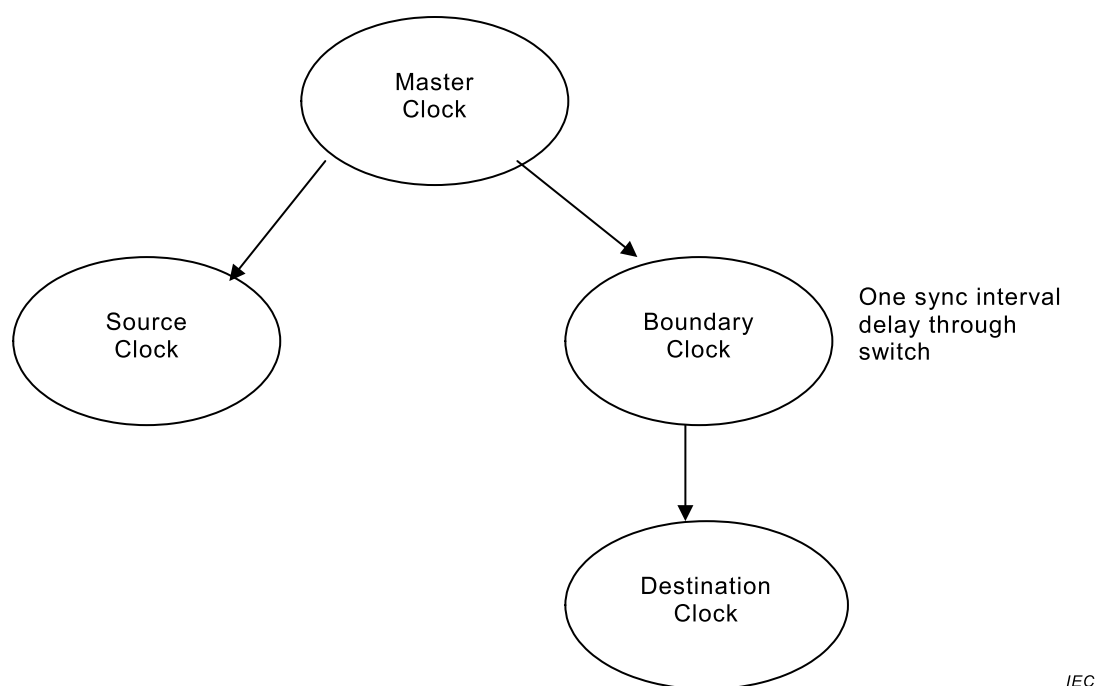


Figure 49 – Propagation of a step change in time

In Figure 49, the Source Clock is going to receive a System Time adjustment one sync interval before the Destination Clock. Thus, System Time for the Source Clock is going to be skewed relative to System Time for the Destination Clock for an entire sync packet interval.

The variability in propagating this time adjustment through the system creates a problem for time stamped based motion control in that the time stamp in the source device, say the controller, may not be applicable when received by the destination device, say a drive, because System Time for the source and destination devices is skewed. This effect is further illustrated in Table 24.

Table 24 – Propagation of a step change in time (example 1)

Time Sync Interval	Master time	Source time (controller)	Destination time (drive)
1	100	100	100
2	200 + 1 000	200	200
3	1 300	300 + 1 000	300
4	1 400	1 400	400 + 1 000
5	1 500	1 500	1 500

At Time Sync Intervals 1 and 2, both the timestamp source clock in the controller, and timestamp destination clock in the drive both have the same notion of time, i.e. System Time. At Sync Interval 2, a step change in time occurs at the master clock that propagates to the controller in Sync Interval 3, thus affecting the source clock time stamp sent to the drive device. It is not until Sync Interval 4 that the drive sees the step change. Any time stamp data sent from the controller to the drive between Sync Intervals 3 and 4 will not correlate with the drive's notion of time.

CIP Motion and CIP Sync extend the IEC 61588:2009 protocol to handle this condition. This is done by defining a run-time algorithm in the destination drive device that can detect this time step condition and adjust the value of the received timestamp so that it is accurate. Two conditions are possible:

- 1) the source device has seen a step change in time but the destination device has not;
- 2) the destination device has seen a step change in time but the source device has not.

The key component of the algorithm is the System Time Offset value defined as part of the CIP Sync's Time Sync Object. This offset value is added to the local clock time to generate System Time, that is,

$$\text{System Time} = \text{Local Clock} + \text{System Time Offset}(\text{drive})$$

When the system is synchronized, at a given moment of time, the Local Clock values and the System Time Offset values may differ from device to device, but the System Time value shall be the same within the accuracy of the IEC 61588:2009 protocol implementation.

NOTE The System Time Offset value associated with the drive device is provided by the Time Sync Object and comes pre-filtered to attenuate the influence of network traffic induced noise.

A step change in time is indicated by a change in the System Time Offset value of either the source or destination devices. The source's System Time Offset is sent to the destination device as an offset along with the source's System Time as the data time stamp. The destination device compares the offset received to the previously received offset to determine if a step change has occurred and adjusts the received timestamp value accordingly.

The Offset Compensation algorithm, as it applies to the received Timestamp from the controller via the C-to-D connection, is stated as follows:

$$\text{Timestamp}_{\text{comp}} = \text{Timestamp}_{\text{rec}} + \text{Offset Compensation}$$

$$\text{Offset Compensation} = \text{Offset}_{\text{dest}} - \text{Offset}_{\text{dest}(\text{last})} - (\text{Offset}_{\text{src}} - \text{Offset}_{\text{src}(\text{last})})$$

where,

$\text{Timestamp}_{\text{rec}}$ = received time stamp from controller.

$\text{Timestamp}_{\text{comp}}$ = compensated time stamp from controller.

$\text{Offset}_{\text{dest}}$ = the current value of the System Time Offset at the destination.

$\text{Offset}_{\text{dest}(\text{last})}$ = the previous value of the System Time Offset at the destination.

$\text{Offset}_{\text{src}}$ = the received value of the System Time Offset from the source.

$\text{Offset}_{\text{src}(\text{last})}$ = the previous value of the System Time Offset from the source.

Whenever the drive determines that either the source or destination System Time Offsets have changed, the following calculation is performed to check if there has been a system wide step change in time.

If $|\text{Offset Compensation}| \leq \text{Step Threshold}$

$$\text{Offset}_{\text{dest}(\text{last})} = \text{Offset}_{\text{dest}}$$

$$\text{Offset}_{\text{src}(\text{last})} = \text{Offset}_{\text{src}}$$

where Step Threshold is a number that defines the smallest time step that can occur in the system. For CIP Motion systems this number is typically set to the RPI of the CIP Motion Connection via the optional Motion Device Axis Object class attribute, Step Threshold. If the Step Threshold attribute is not supported in the device, the default Step Threshold value of 1 ms is used.

If Offset Compensation is greater than the Step Threshold, indicating that a system wide step change in time has occurred, the Offset Compensation algorithm will hold (rather than reinitialize) the last "pre-step change" Time Offset values for the drive and the controller until the matching Time Offset change is detected and drops the Offset Compensation below the Step Threshold. Once the step time change is complete, the Offset Compensation value is once again small and the controller System Time closely matches the drive System Time.

Depending on the network location of the Grand Master clock, the drive device might see the Time Offset change before the controller or visa versa. The algorithm handles both cases.

Table 25 provides an example of how System Time Offset Compensation works. For illustration purposes, a packet is assumed to be sent at the sync interval with no propagation delay.

Table 25 – Propagation of a step change in time (example 2)

Time Sync Interval	Master time	Timestamp source (controller)					Timestamp destination (drive)					
		Local clock	Offset	Last offset	Local time	Time stamp	Local clock	Offset	Last offset	Local time	Offset comp	Adj.time stamp
1	100	0	100	100	100	100	0	100	100	100	0	100
2	200 + 1 000	100	100	100	200	200	100	100	100	200	0	200
3	1 300	200	100 + 1 000	100	1 300	1 300	200	100	100	300	-1 000	300
4	1 400	300	1 100	100	1 400	1 400	300	100 + 1 000	100	1 400	0	1 400
5	1 500	400	1 100	1 100	1 500	1 500	400	1 100	1 100	1 500	0	1 500

At Sync Interval 2 a step change of 1 000 occurs in the master clock and is propagated to the timestamp source clock, say the controller, in Sync Interval 3. The time stamp received at the destination clock, the drive in this example, is compensated for by the Offset Compensation value of 1 000 according to the above algorithm, resulting in an Adjusted Controller Time Stamp value that is consistent with the drive's Local Time even though it has not yet seen the step change. In Sync Interval 4, the drive sees the step change and the computed Offset Compensation value is once again 0 because in the Offset Compensation algorithm the source and destination delta Offset terms cancel. Since the Offset Compensation value is now less than the Step Threshold, the Last Offset values are initialized to the current Offset values as shown in sync interval 5 and no further Offset Compensation is required.

6.4.6.7.3 Frequency step compensation

While the above algorithm works well for cases where a significant step change in System Time is propagated through the system, it does not handle the case where there is a step change to the master clock frequency as can occur when introducing a new Grand Master clock. When this occurs, a significant change to the System Time Offset occurs whenever the device receives an IEC 61588:2009 Sync message for as many Time Sync cycles as it takes to fully synchronize the device's clock to the new Grand Master clock. These Time Offsets may or may not occur for the controller, especially if the controller is assuming the role of the new Grand Master, in which case the controller Time Offset value never changes! If the controller never reports a matching Time Offset change then the Time Offset changes accumulate in the device and introduce a phase shift between the device and the controller that can be quite large, i.e. greater than 1 ms. This is generally unacceptable when running a high performance motion control application.

To mitigate this Step Frequency Change case the above Offset Compensation algorithm shall be modified to compensate for the associated phase shift. The maximum frequency change that can occur during a Grand Mastership change is limited to be 0,02 % by the IEC 61588:2009 specified crystal tolerance. This limits the magnitude of the Time Offset changes to the product of the maximum frequency differential, the Time Sync Interval, and the number of boundary clock hops between the Grand Master and the device. A practical limit for the Time Offset changes due to change in Grand Mastership is placed at around 1 ms.

The amount of phase shift that can be introduced between Sync message updates due to the frequency step can be substantial, e.g. around 100 µs. Attempting to adjust the Offset

Compensation value by this amount immediately would cause a significant motion disturbance if the device was operational at the time. So, to avoid disturbing motion, the Time Offset Compensation algorithm shall be modified to gradually reduce the Offset Compensation value to zero to eliminate the phase shift. This gradual correction is done at a rate of 1 μ s of phase correction per millisecond of running time, insuring that a 1 ms phase correction can be completed in a 1 s Time Sync Interval.

If the device is in the middle of performing such a phase correction and a new Time Offset is sent to the device from the controller, it is possible that the phase correction process is going to stop prematurely, resulting in a persistent phase shift between the controller and the device System Time clocks. Since this is unacceptable, the modified algorithm insures that any uncorrected phase shift is carried into the next Time Sync Interval.

The modified Time Offset Compensation algorithm is as follows:

$$\text{Timestamp}_{\text{comp}} = \text{Timestamp}_{\text{rec}} + \text{Offset Compensation.}$$

where,

$$\text{Offset Compensation} = \text{Variance} - \text{Phase Correction.}$$

The values for the new terms Variance, and Phase Correction are calculated based on whether or not the Time Offset values have changed. Here are the two cases.

If $\text{Offset}_{\text{dest}}$ or $\text{Offset}_{\text{src}}$ have changed from their last value:

$$\text{Variance} = ((\text{Offset}_{\text{dest}} - \text{Offset}_{\text{dest}(\text{last})}) - (\text{Offset}_{\text{src}} - \text{Offset}_{\text{src}(\text{last})}))$$

$$\text{Variance} = \text{Variance} + \text{Phase Remaining}$$

$$\text{Phase Correction} = 0$$

$$\text{Phase Remaining} = 0$$

If $|\text{Variance}| \leq \text{Step Threshold}$

Case = Normal Sync

$$\text{Offset}_{\text{dest}(\text{last})} = \text{Offset}_{\text{dest}}$$

$$\text{Offset}_{\text{src}(\text{last})} = \text{Offset}_{\text{src}}$$

Else $|\text{Variance}| > \text{Step Threshold}$

Case = Time Step Change

Else if $\text{Offset}_{\text{dest}}$ and $\text{Offset}_{\text{src}}$ have not changed:

Case = Normal Sync

If Variance = Phase Correction

No Operation

Else If Variance > Phase Correction

$$\text{Phase Correction} = \text{Phase Correction} + \text{RPI}/1000$$

If Phase Correction > Variance

$$\text{Phase Correction} = \text{Variance}$$

$$\text{Phase Remaining} = \text{Variance} - \text{Phase Correction}$$

Else If Variance < Phase Correction

$$\text{Phase Correction} = \text{Phase Correction} - \text{RPI}/1000$$

$$\text{Phase Correction} < \text{Variance}$$

$$\text{Phase Correction} = \text{Variance}$$

$$\text{Phase Remaining} = \text{Variance} - \text{Phase Correction}$$

Case = Time Step Change

No Operation

This modified version Time Offset Compensation algorithm is very robust, covering step changes in System Time, and step changes in Grand Master clock frequency. The algorithm will eliminate any persistent phase shifts between the controller and the device under these circumstances without disturbing motion during real-time operation.

6.5 Device startup procedure

6.5.1 General

In this subclause, the start-up process of a CIP Motion device is discussed. For the sake of brevity, the exact messages and their formats will not be presented. The objective is to outline the steps and the exchanges that take place between the controller and the device in order to correctly configure the CIP Motion device.

The device startup involves three distinct processes which are initiated by the CIP Motion controller and responded to by each CIP Motion device on the network. They are:

- 1) Motion I/O Connection creation;
- 2) Motion Object configuration;
- 3) time synchronization.

These processes may occur in parallel or consecutively depending on the particular implementation. Device Configuration may occur via the CIP Motion I/O Connection or separate connected explicit messaging. The startup process is outlined in the following subclauses.

6.5.2 Motion I/O Connection creation

Motion I/O Connections are created on EtherNet/IP using either the common CIP Forward_Open or Large_Forward_Open services (see IEC 61158-5-2 and IEC 61158-6-2). These services specify the maximum size of the Control-to-Device and Device-to-Controller Connection data structures and explicitly target an Input and Output Connection Point for the Motion Device Axis Object class.

In order to open this connection to the Motion Device Axis Object class, the originator constructs a Forward_Open service with the Electronic Key segment and two Application Paths that represent the Configuration data block and the set of Consuming/Producing data blocks to be used with this connection. See the Connection Manager Object in IEC 61158-5-2 and IEC 61158-6-2 for full details of the Forward_Open service.

The Configuration path is encoded as Connection Point 81 (0x51) of the Motion Device Axis Object (0x42). The I/O Connection data path is encoded as Connection Point 2 of the Motion Device Axis Object, and shall always match the Format Revision of the C-to-D and D-to-C Connection Data Structures. Next, the Data Segment (0x80) contains 4 words (8 bytes) of configuration data as defined by the Data Segment Configuration Block below. Thus, for this release of the specification the Application Path and Data Segment of the Forward_Open service should appear as follows:

20 42 2C 51 2C 02 80 04 ...followed by the 4 word Configuration Block (Rev 1)

In addition the Application Path, the following information is also included in the Forward_Open service:

T-to-O Connection Size: 220 bytes (example only)

T-to-O Connection Size Type: Variable

T-to-O Connection Type: Point-Point

O-to-T Connection Size: 220 bytes (example only)

O-to-T Connection Size Type: Variable

O-to-T Connection Type: Point-Point

The O-to-T Connection info corresponds to the Controller-to-Device Connection while the T-to-O Connection info corresponds to the Device-to-Controller Connection. In the above example, the T-to-O and O-to-T Connection Size may vary depending on the controller implementation.

As stated above, in addition to Connection Points and device Keying information, the Forward_Open service also contains a Configuration Block in the Data Segment of the service request. The Configuration Block is used to verify that the power structure associated with the device matches that of the device selected by the controller.

The Configuration Block for Format Revision 1 is an 8-byte data structure defined as shown in Figure 50.

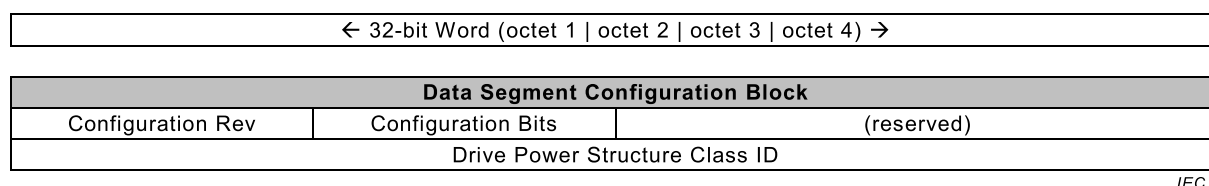


Figure 50 – Configuration Block Format Revision 1 (Connection Point 81)

The Format Revision element represents the revision of the Configuration Block data structure. If the format of the Connection Block changes in future revisions of this specification, the Format Revision number shall be incremented as shall the associated Connection Point. The Format Revision for the Configuration Block as defined above is 1, which corresponds to Connection Point 81 (0x51).

The Configuration Bits field for a CIP Motion device has only one bit currently defined, the Verify Power Rating bit as bit 0. If this bit is set, the device shall verify that the Drive Power Structure Class ID of the Forward_Open matches the Drive Power Structure Class ID attribute value of the drive device. If the values do not match, the Forward_Open service response shall indicate a Data Segment error. Specifically, the drive shall return a General Status of 0x09 (Data Segment Error) with an Extended Status of 0x04.

The Configuration Block structure for Format Revision 2, Connection Point 82 (0x52), is a variable length data structure that extends the Format Revision 1 structure to carry Drive Power Structure Axis ID information needed to verify power structures associated with each axis instance of the multi-axis drive. The Data Segment size, in words, determines the number of Drive Power Structure Axis ID elements in the array.

The Configuration Block for Format Revision 2 is an 8-byte data structure defined as shown in Figure 51.

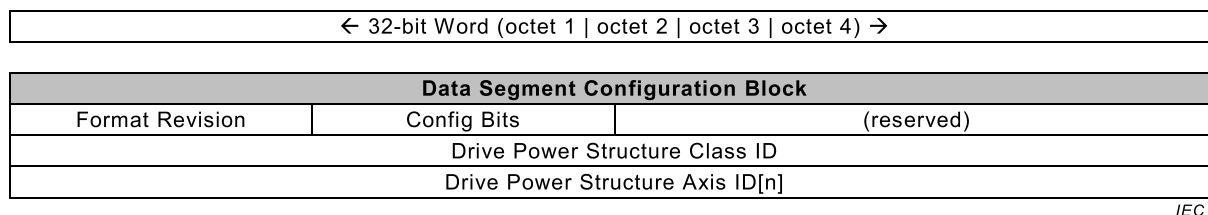


Figure 51 – Configuration Block Format Revision 2 (Connection Point 82)

In this case, if the Verify Power Rating bit is set, the device shall verify either that the Drive Power Structure Class ID of the Forward_Open matches the Drive Power Structure Class ID attribute value of the drive device or that the Drive Axis Power Structure Axis IDs match the corresponding Drive Power Structure Axis ID attribute values associated with each drive axis instance. If in either case, the values do not match, the Forward_Open service response shall indicate a Data Segment error. Specifically, the drive shall return a General Status of 0x09 (Data Segment Error) with an Extended Status of 0x04 if the Drive Power Structure Class ID values do not match. If the Drive Power Structure Axis ID values do not match the Forward_Open service returns and Extended Status of 0x05 + n, where n is the index of the first Drive Power Structure Axis ID element in the Configuration Block that failed. Single axis drives or multi-axis drives with identical power structures need only use the Drive Power Structure Class ID to verify the power rating, in which case the Revision 2 Configuration Block degenerates to the Revision 1 Configuration Block structure. But in the case of a multi-axis drive with independent power structures, the Drive Power Structure Class ID is set to 0 and the Drive Power Structure Axis ID array contains all the IDs that need to be verified against the Power Structure Axis ID attribute values associated with each drive axis instance. See descriptions for the Drive Power Structure Class ID class attribute and Drive Power Structure Axis ID instance attribute for further details.

Support for Format Revision 2 of the Configuration Block structure is required when implementing a modular multi-axis CIP Motion drive device with mixed power structures. All other types of CIP Motion devices (e.g. single axis drives, and multi-axis drives with identical power structure ratings, encoders, converters, etc) are required to either support Format Revision 1 or Format Revision 2. CIP Motion devices need only support one Format Revision. CIP Motion devices using Format Revision 1 are not required to support Format Revision 2. Similarly, CIP Motion devices using Format Revision 2 are not required to support Format Revision 1. Controllers are generally capable of sending either Configuration Block Format to the device, and shall use whatever Format Revision the specific device profile supports.

The RPI (Requested Packet Interval) of the Forward_Open for this Class 1 I/O connection is set to the Controller Update Period. This value is applied to the Controller Update Period class attribute of the Motion Device Axis Object. CIP Motion timing requires that the connection API (Actual Packet Interval) match the RPI of the controller's Forward_Open service request. Thus, in the context of a CIP Motion I/O Connection, RPI and API are synonymous.

Once the C-to-D and D-to-C connections have been successfully established using Forward_Open service, the controller and the device are each responsible for transmission of data packets at the RPI.

The first C-to-D packet transmitted by the controller shall contain a valid Connection Format and Format Revision. The Update ID, Last Update ID, Instance Count, and Time Data Set elements are all set to zero. The Node Control element is set to 0x01 since the intention is Remote control of the device. Note that the Controller Data Valid bit in the Node Control field is not set indicating that the controller is not sending any valid instance data in this update. No additional data is required for the C-to-D connection. Figure 52 shows a typical initial C-to-D connection data block.

Initial C-to-D connection data block			
Connection Format = 6	Format Revision = 2	Update ID = 0	Node Control = 1
Instance Count = 0	0	Last Received ID = 0	Time Data Set = 0

IEC

Figure 52 – Typical initial C-to-D connection data block

Similarly, the first D-to-C packet transmitted by the device, in response to the initial C-to-D packet, shall contain a valid Connection Format and Format Revision. The Update ID, Instance Count, and Time Data Set elements are all set to zero. The Node Status element is set to 0x01 since the intention is Remote control of the device. Since there is little if any CIP Motion Connection handling functionality running in this early phase of the start-up sequence, the Node Alarms and Node Fault elements are both set to 0 as well. Note that the Device Data Valid bit in the Node Status field is not set indicating that the device is not sending any valid instance data at this time. Until the controller sets the Controller Data Valid bit in the Node Control field, the device simply continues to send just the D-C Connection Header with the Update ID, Instance Count, and Time Data Set elements all set to zero. No additional data is required for the D-to-C connection in this phase. Figure 53 shows a typical initial D-to-C connection data block.

Initial D-to-C connection data block			
Connection Format = 7	Format Revision = 2	Update ID = 0	Node Status = 0
Instance Count = 0	Node Fault/Alarm = 0	Last Update ID = 0	Time Data Set = 0

IEC

Figure 53 – Typical initial D-to-C connection data block

Once the first C-to-D and D-to-C packets have been transmitted, each device keeps its part of the CIP Motion I/O Connection alive by sending Refresh packets at the RPI. Refresh packets are defined as CIP packets that have the same CIP Sequence Count as the previously transmitted CIP packet. Transmission and processing of these Refresh packets is typically handled by low-level CIP drivers and transparent to the higher level CIP Motion Connection handler.

Once the controller has the Controller Task running and is ready to manage the CIP Motion Connection, it is ready to send configuration data to the device thus marking the end of the Connecting phase of the start-up procedure and the beginning of the Configuration phase.

6.5.3 Motion Device Axis Object configuration

Motion Device Axis Object configuration is accomplished using either the common Set_Attributes_List service or the object specific Set_Axis_Attributes_List service defined by the Motion Device Axis Object and supported by the CIP Motion I/O Connection. This subclause assumes that the controller uses the CIP Motion I/O Connection to configure the device, since this would be more typical, and the method of using the Set Attribute List service is already well understood.

The first step in the configuration process is to establish the class attributes of the Motion Device Axis Object. To do this the controller sends a Set_Axis_Attributes_List service to the class (i.e. instance 0) of the object. Figure 54 illustrates the typical contents of the first C-to-D class attribute configuration packet.

1st Class instance configuration C-to-D connection data block			
Connection Format = 6	Format Revision = 2	Update ID = 1	Node Control = 5
Instance Count = 1	0	Last Received ID = 0	Time Data Set = 0
Instance Num = 0	-	Instance Blk Size = 8	Cyclic Blk Size = 0
Cmd Blk Size = 0	Write Blk Size = 0	Event Blk Size = 0	Service Blk Size = 6
Transaction ID = 1	Service Code = 76	0	0
Set_Axis_Attributes_List Specific Request Data (2 attributes)			

IEC

Figure 54 – Typical contents of first C-to-D class attribute configuration packet

Note that the Update ID is now incrementing with every packet sent by the controller and that the Node Control element is now set to 5 indicating that the controller is now sending valid instance data. The Instance Count is 1 since in this case we are only sending data to the class, i.e. Instance Number 0. Also note that the Time Data Set is 1 and the controller is sending the Controller Update Period to the device. Only the Instance Block Size and Service Block Size have non-zero values. A value of 0 for the Cyclic Block Size and Event Block Size indicates that these blocks are not included in this C-to-D Connection Data Block.

The device processes this C-to-D packet as soon as possible to provide timely D-to-C transmission. The Update ID for the D-to-C packet is also now incrementing with every packet sent by the device and the device is indicating that the instance data is now valid. This packet may or may not include the Set_Axis_Attributes_List service request depending on the priority given to the service request processing. Figure 55 illustrates a typical response to the above first C-to-D class configuration packet assuming that the service response data is included.

1st Class instance configuration D-to-C connection data block			
Connection Format = 7	Format Revision = 2	Update ID = 1	Node Status = 5
Instance Count = 1	Node Fault/Alarm = 0	Last Received ID = 1	Time Data Set = 0
Instance Num = 0	-	Instance Blk Size = 7	Cyclic Blk Size = 0
Act Blk Size = 0	Read Blk Size = 0	Event Blk Size = 0	Service Blk Size = 5
Transaction ID = 1	Service Code = 76	General Status = 0	Extended Status = 0
Set_Axis_Attributes_List Specific Response Data (2 attributes)			

IEC

Figure 55 – Typical response to first C-to-D class configuration packet

In addition to sending this response, the device also schedules the next D-to-C transmission to be 1 Controller Update Period from the time the C-to-D packet was received. If the next C-to-D packet to the device is late or lost, the device sends the next D-to-C packet as previously scheduled, and schedules the next D-to-C packet transmission. Normally, however, the next C-to-D packet arrives prior to previously scheduled D-to-C transmission, so the device processes the packet and immediately sends the D-to-C response and schedules the next transmission one Controller Update Period later. This procedure synchronizes the device D-to-C packet transmissions with the Controller Task schedule without requiring that the controller and device clocks be synchronized to each other.

NOTE In general, the Update IDs associated with the C-to-D packet and D-to-C packet for a given cycle match.

Once the class attributes are configured, the Motion Device Axis Object configuration continues with configuration of the individual object instances until all instances are configured. This is typically done one instance at a time, but could be using multiple instances to configure all axes in parallel. In fact, the initial axis instance configuration could begin in the same packet as the class instance configuration. The device is required to handle all these cases. Figure 56 illustrates the typical contents of the first C-to-D axis instance configuration packet.

1st Axis instance configuration C-to-D connection data block			
Connection Format = 6	Format Revision = 2	Update ID = 4	Node Control = 5
Instance Count = 1	0	Last Received ID = 3	Time Data Set = 0
Instance Num = 1	-	Instance Blk Size = 84	Cyclic Blk Size = 0
Cmd Blk Size = 0	Write Blk Size = 0	Event Blk Size = 0	Service Blk Size = 82
Transaction ID = 1	Service Code = 76	0	0
Set_Axis_Attributes_List Specific Request Data (40 attributes)			

IEC

Figure 56 – Typical contents of first C-to-D axis instance configuration packet

In response to this packet the device sends the data packet as shown in Figure 57.

1st Axis instance configuration D-to-C connection data block			
Connection Format = 7	Format Revision = 2	Update ID = 4	Node Status = 5
Instance Count = 1	Node Fault/Alarm = 0	Last Received ID = 4	Time Data Set = 0
Instance Num = 0	-	Instance Blk Size = 45	Cyclic Blk Size = 0
Act Blk Size = 0	Read Blk Size = 0	Event Blk Size = 0	Service Blk Size = 43
Transaction ID = 1	Service Code = 76	General Status = 0	Extended Status = 0
Set_Axis_Attributes_List Specific Response Data (40 attributes)			

IEC

Figure 57 – Typical response to first C-to-D axis configuration packet

Once all object instances are configured, the controller marks the end of the Configuration phase of the Start-up process by setting the Configuration Complete bit in the Control Status field of the next C-to-D packet sent to the device.

6.5.4 Time Synchronization

All CIP Motion device nodes that support time synchronization services are required to be synchronized with a CIP Motion controller as part of a group, commonly embodied as a Motion Group. All Time Sync capable devices in this group are sent a Group_Sync service that contains the Grand Master Clock ID associated with the controller. This is a class service, so the packet would look something like Figure 58. Even though this example only shows the class Instance Data Block, the class instance and axis Instance Data Blocks can coexist in the same packet. This allows the Time Synchronization process and the Motion Device Object Configuration process to proceed concurrently. It is a controller requirement, however, that all class attributes associated with the Motion Device Axis Object of this device shall be configured prior to the controller sending the first Group_Sync service request.

Time Sync service request C-to-D connection data block			
Connection Format = 6	Format Revision = 2	Update ID = 12	Node Control = 5
Instance Count = 1	0	Last Received ID = 11	Time Data Set = 0
Instance Num = 0	-	Instance Blk Size = 5	Cyclic Blk Size = 0
Cmd Blk Size = 0	Write Blk Size = 0	Event Blk Size = 0	Service Blk Size = 3
Transaction ID = 3	Service Code = 28	0	0
Group_Sync Specific Request Data (Grand Master Clock ID)			

IEC

Figure 58 – Typical contents of C-to-D Time Sync service request packet

The objective of this procedure is to guarantee that all devices in the Motion Group are synchronized with each other, or in other words, that they have the same notion of time based on a common Grand Master Clock. In addition to insuring that the devices all share the same Grand Master as the controller, the procedure also guarantees that no step changes to System Time have occurred at the time of synchronization that could permanently skew a device clock relative to the controller clock.

Figure 59 illustrates how the controller first waits until its local clock is synchronized to the Master System Time clock. It is the responsibility of the Time Sync object in the controller to determine if the controller is synchronized. The controller then waits a minimum delay period, MinSyncDelay, to allow time for other components in the system to synchronize to a common Grand Master. A typical value for this delay period is 10 s.

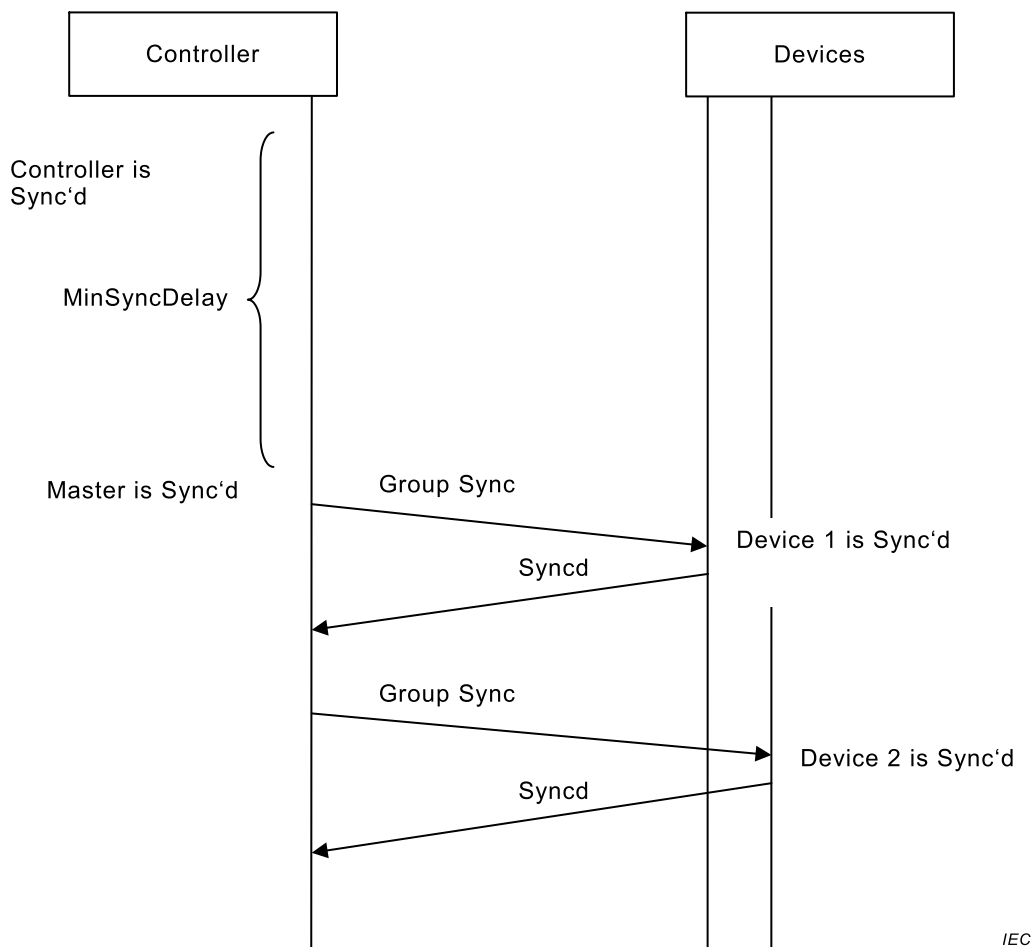


Figure 59 – Group Sync of CIP Motion devices

At the conclusion of this delay period, the controller initiates a series of Group Sync services to each of the CIP Motion devices. Each device returns a response indicating whether it is also synchronized to the controller according to the criterion described for the Motion Device Axis Object's Group_Sync service. If the Group_Sync response indicates that the device is synchronized to the controller, the controller does one last skew check and, if there's no evidence of time skew, the device and controller are considered fully synchronized. When all the devices are fully synchronized to the controller, the Motion Group is considered to be synchronized and the controller is free to begin sending cyclic data to all the CIP Motion devices associated with the Motion Group.

If the controller encounters a device that does not return a "device is synchronized" response to the Group_Sync service after repeatedly trying for at least 1 minute, the controller shall indicate that a connection fault has occurred for that device.

Once the controller and the CIP Motion devices are Group synchronized, the controller can then set the Synchronous Control bit in the Controller-to-Device Connection Header. Setting the Synchronous Control bit in the next cyclic packet sent to the devices allows the device to schedule its next Device-to-Controller Connection update based on the Controller Time Stamp contained in the connection data. Specifically, the device calculates the initial synchronous Device-to-Controller Connection update time to be:

$$\text{Controller Time Stamp} + 2 \times \text{Controller Update Period (the CIP Motion Connection RPI)} - \text{Transmission Processing Time.}$$

The Transmission Processing Time is the time necessary to assemble and prepare the Device-to-Controller Connection for transmission. When the scheduled transmission occurs, the device indicates that it is now sending synchronized Device-to-Controller Connection data

by setting the Synchronous Mode bit in the header of the Device-to-Controller packet sent to the controller.

The Device-to-Controller Connection Update ID shall be initialized for this first synchronous update cycle to match the Update ID of the Controller-to-Device Update ID for that cycle and shall be incremented by one every connection update period thereafter. Maintaining matched Update IDs per update cycle, therefore, does not require that the Controller-to-Device packet be processed, or even received, prior to the Device-to-Controller packet being assembled.

Data exchange between the controller and the device proceeds thereafter according the CIP Motion timing model. This marks the end of the Synchronization phase of the Start-up process.

If at the point of receiving the Synchronous Control request in the Controller-to-Device Connection Header the (Controller Time Stamp + Controller Update Period) time differs from the current device System Time by more than one Controller Update Period, the controller and device system time clocks are skewed.

That is,

If $|(Controller\ Time\ Stamp + CUP) - Current\ System\ Time| > 1\ CUP$

Controller and Device Clocks are Skewed

Else

Controller and Device Clocks are OK

In this unlikely case, the device simply continues asynchronous operation and the Synchronous Mode bit in the header of the Device-to-Controller packet remains clear until either the calculated update time is within acceptable tolerance or a 1-minute time-out limit is exceeded. During the 1 minute time-out interval while the clocks are skewed, the Node Alarm Code from the device indicates a Clock Skew Alarm condition. Should the 1-minute clock skew time-out period be exceeded, the device faults and sets the Node Fault Code to indicate a Clock Skew Fault condition.

6.6 Device visualisation

The CIP Motion device requires visualisation components to quickly diagnose the condition of the device. These components range from bicolor LEDs to multi-character alphanumeric displays. Table 26 defines the requirements for these visualisation components and appropriate labels for these components when employed by the device.

Table 26 – CIP Motion visualisation components

Visualisation component	Need in implementation	Associated label
Network Status LED	Required	"Net Status" "NET"
Module Status LED	Required	"Mod Status" "MOD"
Axis Status LED	Conditional ^a	"Axis Status" "XS" "Axis # Status" "XS#" ^b
Seven-Segment Display	Optional	N/A
Multi-char alphanumeric display	Optional	N/A
^a Requirement may be waived by using a multi-character alphanumeric display.		
^b Multi-axis devices need multiple Axis Status LEDs, one for each axis.		

When applied to different CIP networks, the Network Status and Module Status LED behavior shall conform to the LED behavior of the associated CIP network to which the device is connected. The mapping of the device's Module Status LED states to the Motion Device Axis Object states is fully defined in the Motion Device Axis Object section on Visualisation.

6.7 Ethernet/IP Quality of Service (QoS)

When a CIP Motion device supporting time synchronization services is targeted for operation on EtherNet/IP, standard Ethernet QoS or Quality of Service protocol shall be utilised to guarantee that CIP Sync related IEC 61588:2009 packets and CIP Motion I/O Connection packets have timely delivery through the network components. QoS for EtherNet/IP is defined in IEC 61158-4-2 (QoS behavior and optional QoS Object). While QoS mechanisms defined in IEC 61158-4-2 are specified as optional for EtherNet/IP devices, CIP Motion devices shall at least implement DSCP marking for CIP Motion packets. It is required that CIP Sync packets be passed with DSCP marking. CIP Urgent priority shall be set by default by the Forward_Open service establishing the CIP Motion Connection. It is expected that the CIP Motion device EtherNet/IP stack would give high priority to the processing of CIP Motion and IEC 61588:2009 packets over lower priority packets.

7 Motion Device Axis Object

7.1 General considerations

7.1.1 General

This subclause defines the behavior of a CIP object called the Motion Device Axis Object. This object is the main functional component of the CIP Motion device profile defined as part of the CIP Motion extensions to CIP Common. Instances of this object are required to support motion control when connected to a CIP Motion compliant controller through a CIP network. EtherNet/IP is the network of choice for high performance, synchronized multi-axis control. Other CIP networks such as ControlNet and DeviceNet could be applied to lower performance, non-synchronized drive applications such as simple variable frequency drives, and velocity loop drives and indexing drives.

7.1.2 Revision history

Clause 7 provides a complete description of the individual object attributes, their behavior, and associated object services. Since the initial release of this object class definition changes have been made that require a revision update of this object class. Table 27 represents the revision history.

Table 27 – Motion Device Axis Object revision history

Revision	Reason for object definition update
1	Initial revision (IEC 61800-7 Edition 1, obsolete)
2	Major interface changes (deprecated)
3	Modified behavior of Group_Sync service for clock skew detection. Recommended for new CIP Motion devices supporting time synchronization. Modified Get_Motor_Test_Data service to add an optional parameter request structure.

7.1.3 Object overview

The Motion Device Axis Object is one of two major object components required to support motion control under the CIP Motion control architecture. These object components are illustrated in Figure 60.

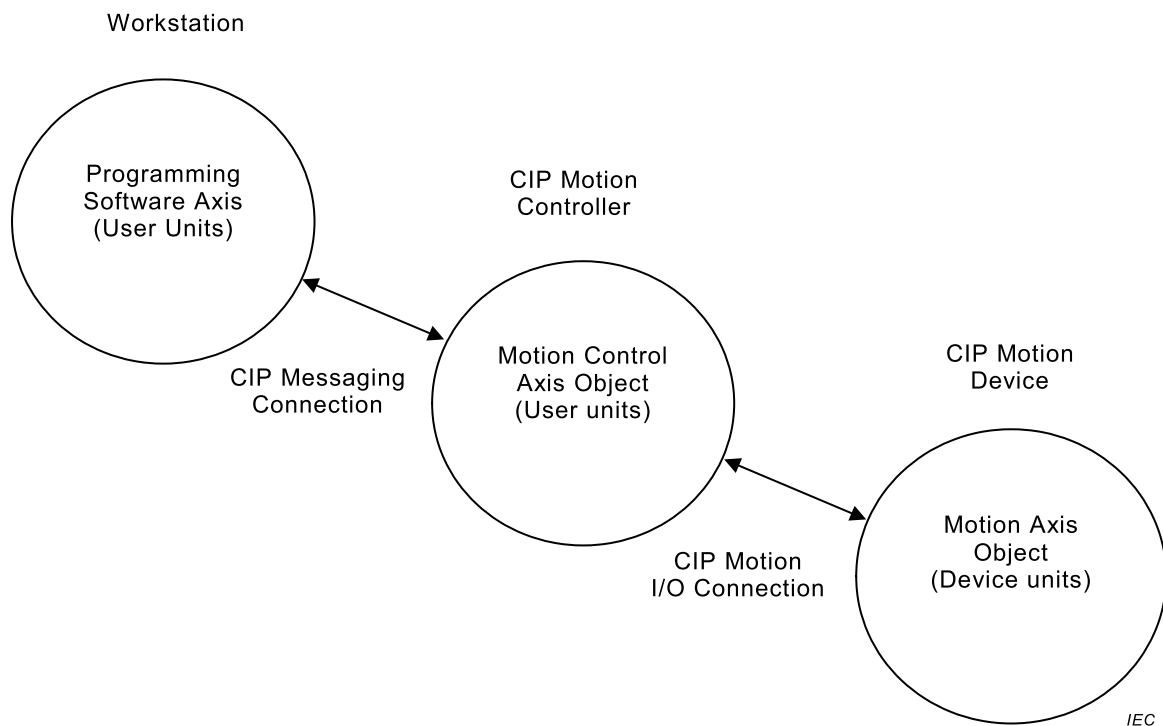


Figure 60 – Object components for CIP Motion control architecture

7.1.4 Motion Device Axis Object abstraction

The Motion Device Axis Object is an abstraction that can be applied to any motion control device function associated with a moving component of a motion control system. A given motion control device can therefore be represented by a Motion Device Axis Object instance, and since most of these device functions are associated with the motion of a machine component, a Motion Device Axis Object instance is sometimes simply referred to as an “Axis”. A Motion Device Axis Object instance generally consists of one or more of the following integral elements:

- motion actuator, for example rotary and linear motors, hydraulic and pneumatic actuators;
- motion sensor, for example encoders, resolvers, linear displacement transducers;
- actuator control, for example vector, frequency, and stepper motor control, hydraulic valve control;
- amplifier that supplies power to move actuator, for example drive power structure for motors;
- converter: converts AC line power to DC bus power for one or more associated amplifiers;
- motion I/O: registration input sensors, output cam driven actuators.

Instance attributes are included in the Motion Device Axis Object for each of these integral elements to define axis behavior. When a Motion Device Axis Object has an actuator control element, command attributes are included in the object to control the dynamics of the actuator. The object also includes attributes that describe the dynamics of the actuator, whether estimated or actual.

While the most common embodiment of a Motion Device Axis Object is a “drive” device, the Motion Device Axis Object can also be applied to other motion related devices, for example a CIP Motion encoder, a CIP Motion Power Converter, a CIP Motion registration input module, etc. For these devices, only a subset of the Motion Device Axis Object attribute set is actually implemented, with the subset consisting of all attributes that are applicable to the device’s function.

In the case of a multi-port feedback device, each feedback port is considered an axis and therefore assigned an instance of the Motion Device Axis Object. In the case of a multi-amplifier drive device, each amplifier is associated with an axis and assigned an instance of the Motion Device Axis Object. For drive axis instances, the Motion Device Axis Object can represent a complex closed loop vector controlled servo drive or a simple open loop Volts/Hertz drive (or VFD). Drive devices are often equipped with multiple feedback ports that can be dedicated to drive amplifier instances or used as independent master feedback sources. In the latter case, the master feedback channel is considered an axis and assigned an instance of the Motion Device Axis Object.

7.1.5 Motion Control Axis Object

For each device resident Motion Device Axis Object instance in the motion control system, there is a corresponding controller resident Motion Control Axis Object. Attribute values contained within the Motion Device Axis Object have matching attributes in the Motion Control Axis Object, the only difference being that the units by which these attributes are expressed. In the Motion Device Axis Object, attributes are expressed in fixed device units, for example feedback counts or r/min, as defined by the Motion Device Axis Object. Within the Motion Control Axis Object, attributes are expressed in user defined units, for example degrees, inches or centimeters. Since programming software interacts with the drive through the Motion Control Axis Object, all configuration, diagnostic, and motion programming is done using the user's preferred units. Therefore, rather than programming the motion control application in, for example, feedback counts or r/min of the motor shaft, the user can program in, for example, centimeters, feet, products per minute, or whatever makes sense for the specific application.

When configuration or command attribute values are transferred to the Motion Device Axis Object via the CIP Motion I/O Connection, these attribute values are scaled by the Motion Control Axis Object into fixed device unit values required by the Motion Device Axis Object. Similarly, when signal attributes are transferred to the Motion Control Axis Object via the CIP Motion I/O Connection, these feedback-based unit values are scaled by the Motion Control Axis Object and stored in the object in user units.

7.1.6 Device control classification

Based on the variations in Control Mode and Control Method listed in 4.2, one can define a set of basic Device Function Codes around which the many attributes of the Motion Device Axis Object can be organized. Device Function Codes are designated in the tables of Clause 7 using a single letter identifier that can be used to determine what object attributes are required (or optional) for implementation of a given CIP Motion device. The list of Device Function Codes is as follows.

Function Codes:

- B** – Bus Power Converters (No Control Mode, No Control Method)
- E** – Encoder Feedback Only (No Control Mode, No Control Method)
- P** – Position Loop (Position Control Mode, Closed Loop Vector Control Method)
- V** – Velocity Loop (Velocity Control Mode, Closed Loop Vector Control Method)
- T** – Torque Loop (Torque Control Mode, Closed Loop Vector Control Method)
- F** – Frequency Control (Velocity Control Mode, Frequency Control Method)

Device Function Code Combinations:

Combinations of these letters can be used to designate a specific class of CIP Motion devices for the purposes of identifying applicable attributes.

EXAMPLE “FV” would refer to the class of all velocity controlled drives, either closed loop vector controlled or open loop frequency controlled.

Below are some combinations that appear so frequently that special Device Function Codes have been defined for convenience:

N = BE = All Device Functions using No Control Method

O = F = All Device Functions using Open Loop Control Methods, (e.g. Frequency Control)

C = PVT = All Device Functions using Closed Loop Control Methods (e.g. PI Vector Control)

D = OC = All Device Functions using Control Methods (Control Method != No Control)

In addition to these combinations there are many attributes that are applicable or not applicable to sensorless or encoderless device operation, i.e. device axis instances operating without a feedback device. Device Function Codes can be used to specify the following conditional implementation rules for attributes. In this case the following conditional implementation rules are specified:

E = Encoder-based device operation.

IE = Encoderless or Sensorless device operation.

7.1.7 Required vs. Optional in implementation

In the subclauses that follow, attributes and services are defined as Required or Optional in the implementation of the Motion Device Axis Object. Required attributes and services shall be supported in the implementation of the object. Optional attributes and services may or may not be supported in the implementation and are left to the discretion of the vendor.

For Instance Attributes, the determination of whether a given attribute or service is Required or Optional often depends on the associated Device Function Code as defined above. If an attribute or service is marked as Required for a given Device Function Code, then a device implementation shall support that attribute or service if it is intended to operate in that mode. For example, an attribute marked as required for Device Function Code “V” shall be implemented by any CIP motion device that intends to support Velocity Loop mode.

In some cases, an instance attribute or service may not be applicable to a given Device Function Code. This situation is implied when the attribute is defined as neither Required nor Optional.

Table 29 provides a convenient list of all the Instance Attributes of this object and identifies whether the attribute is Required or Optional, Conditional or Not Applicable in the implementation based on the above Device Function Code. Attribute Enumeration and Bit definitions are also designated as Required, Optional, or Conditional, with an appropriate Device Function Code if applicable. If no designation is associated with the Enumeration or Bit definition, then it is assumed that the enumeration is Required in the implementation.

For some attributes, there are conditional implementation rules that extend beyond the Device Function Code. These rules are specified in the Conditional Implementation column of Table 29.

EXAMPLE Table 28 shows an example where the Attribute PM Motor Resistance is Required (R) in the implementation if the device supports Frequency Control, Position Control, Velocity Control, or Torque Control AND the device supports Permanent Magnet motors, i.e.: "PM Motor only". The attribute is Not Applicable (-) for a Bus Power Converter or a Feedback Only device or a drive that does not support a PM motor.

Table 28 – Example for instance attribute implementation vs. Device Function Code

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C-Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
1327	Set	PM Motor Resistance	-	-	R	R	R	R	PM Motor only

Key to Table 28 and Table 29: (see 7.1.6 and paragraphs above for details)

- (R) Required attribute – shall be supported in the implementation.
(O) Optional attribute – support is left to vendor's discretion.
(C) Conditional attribute.
(-) Not applicable attribute.
Set* Indicates the attribute is normally set by the CIP Motion Connection data block and not by a Set service.

Conditional Implementation Key:

Rotary Motor	Rotary PM motor or Rotary Induction motor (motor type)
Linear Motor	Linear PM motor or Linear Induction motor (motor type)
PM	Rotary or Linear Permanent Magnet (SPM and IPM) motor (motor type)
IM	Rotary or Linear Induction Motor (motor type)
SPM	Rotary or Linear Surface PM motor (motor type)
IPM	Rotary or Linear Interior PM motor (motor type)
TT	Digital AqB (feedback type)
TP	Digital Parallel (feedback type)
SC	Sine/Cosine (feedback type)
HI	Hiperface® (feedback type)
DSL	Hiperface DSL® (feedback type)
E21	EnDat 2.1® (feedback type)
E22	EnDat 2.2® (feedback type)
SS	SSI (feedback type)
LT	LDT or Linear Displacement Transducer (feedback type)
RS	Resolver (feedback type)
Rotary Absolute	Feedback Unit – rev; Feedback n Startup Method- absolute
Linear Absolute	Feedback Unit – meter; Feedback n Startup Method- absolute
Drive Scaling	Drive device supports drive scaling functionality.

Table 29 – Instance attribute implementation vs. Device Function Code

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
40	Set*	Control Mode	R	R	R	R	R	R	
41	Set	Control Method	R	R	R	R	R	R	
45	Set	Motion Scaling Configuration	-	R	R	R	R	R	Drive Scaling only
77	Set	Motion Unit	-	R	R	R	R	R	Drive Scaling only
78	Set	Motion Resolution	-	R	R	R	R	R	Drive Scaling only
79	Set	Motion Polarity	-	R	R	R	R	R	Drive Scaling only
83	Set	Cyclic Unwind Control	-	R	R	R	R	R	Drive Scaling only
84	Set	Cyclic Unwind Position	-	R	-	R	R	R	Drive Scaling only; E
89	Set*	Control Status	O	O	O	O	O	O	
90	Set*	Actual Data Set	-	R	R	R	R	R	
91	Set*	Command Data Set	-	-	R	R	R	R	
92	Set*	Command Control	-	-	O	R	R	R	
93	Get	Command Target Time	-	-	O	R	O	O	
94	Set*	Status Data Set	R	R	R	R	R	R	
95	Set*	Registration Data Set	-	R	R	R	R	R	Drive Scaling only; E
96	Set*	Home Data Set	-	R	R	R	R	R	Drive Scaling only; E
97	Set*	Watch Data Set	-	R	R	R	R	R	Drive Scaling only; E
1310	Get	Motor Catalog Number	-	-	O	O	O	O	
1311	Get	Motor Serial Number	-	-	O	O	O	O	
1312	Get	Motor Date Code	-	-	O	O	O	O	
1313	Set	Motor Data Source	-	-	R	R	R	R	
1314	Set	Motor Device Code	-	-	R	R	R	R	
1315	Set	Motor Type	-	-	R	R	R	R	
1316	Set	Motor Unit	-	-	R	R	R	R	
1317	Set	Motor Polarity	-	-	O	O	O	O	
1318	Set	Motor Rated Voltage	-	-	R	R	R	R	
1319	Set	Motor Rated Continuous Current	-	-	R	R	R	R	
1320	Set	Motor Rated Peak Current	-	-	C	C	C	C	R – PM; O – IM
1321	Set	Motor Rated Output Power	-	-	C	C	C	C	R – PM; O – IM
1322	Set	Motor Overload Limit	-	-	O	O	O	O	
1323	Set	Motor Integral Thermal Switch	-	-	O	O	O	O	
1324	Set	Motor Max Winding Temperature	-	-	O	O	O	O	
1325	Set	Motor Winding To Ambient Capacitance	-	-	O	O	O	O	
1326	Set	Motor Winding To Ambient Resistance	-	-	O	O	O	O	
1327	Set	PM Motor Resistance	-	-	R	R	R	R	PM Motor only
1328	Set	PM Motor Inductance	-	-	R	R	R	R	SPM Motor only
1353	Set	PM Motor Lq Inductance	-	-	R	R	R	R	IPM Motor only
1354	Set	PM Motor Ld Inductance	-	-	R	R	R	R	IPM Motor only
1329	Set	Rotary Motor Poles	-	-	R	R	R	R	Rotary Motor only

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
1330	Set	Rotary Motor Inertia	-	-	O	O	O	O	Rotary Motor only
1331	Set	Rotary Motor Rated Speed	-	-	R	R	R	R	Rotary Motor only
1332	Set	Rotary Motor Max Speed	-	-	O	O	O	O	Rotary Motor only
1333	Set	Rotary Motor Damping Coefficient	-	-	O	O	O	O	Rotary Motor only
1334	Set	Linear Motor Pole Pitch	-	-	R	R	R	R	Linear Motor only
1335	Set	Linear Motor Rated Speed	-	-	R	R	R	R	Linear Motor only
1336	Set	Linear Motor Mass	-	-	O	O	O	O	Linear Motor only
1337	Set	Linear Motor Max Speed	-	-	O	O	O	O	Linear Motor only
1338	Set	Linear Motor Damping Coefficient	-	-	O	O	O	O	Linear Motor only
1339	Set	PM Motor Rated Torque	-	-	O	O	O	O	Rotary PM Motor only
1340	Set	PM Motor Torque Constant	-	-	O	O	O	O	Rotary PM Motor only
1341	Set	PM Motor Rotary Voltage Constant	-	-	R	R	R	R	Rotary PM Motor only
1342	Set	PM Motor Rated Force	-	-	O	O	O	O	Linear PM Motor only
1343	Set	PM Motor Force Constant	-	-	O	O	O	O	Linear PM Motor only
1344	Set	PM Motor Linear Voltage Constant	-	-	R	R	R	R	Linear PM Motor only
1345	Set	Induction Motor Rated Frequency	-	-	R	R	R	R	Induction Motor only
1346	Set	Induction Motor Flux Current	-	-	R	R	R	R	Induction Motor only
1347	Set	Induction Motor Stator Resistance	-	-	R	R	R	R	Induction Motor only
1348	Set	Induction Motor Stator Leakage Reactance	-	-	O	O	O	O	Induction Motor only
1349	Set	Induction Motor Magnetization Reactance	-	-	O	O	O	O	Induction Motor only
1350	Set	Induction Motor Rotor Resistance	-	-	O	O	O	O	Induction Motor only
1351	Set	Induction Motor Rotor Leakage Reactance	-	-	O	O	O	O	Induction Motor only
1352	Set	Induction Motor Rated Slip Speed	-	-	O	O	O	O	Induction Motor only
1370	Set	Load Type	-	O	O	O	O	O	Drive Scaling only
1371	Set	Transmission Ratio Input	-	O	O	O	O	O	Drive Scaling only
1372	Set	Transmission Ratio Output	-	O	O	O	O	O	Drive Scaling only
1373	Set	Actuator Type	-	O	O	O	O	O	Drive Scaling only
1374	Set	Actuator Lead	-	O	O	O	O	O	Drive Scaling only
1375	Set	Actuator Lead Unit	-	O	O	O	O	O	Drive Scaling only
1376	Set	Actuator Diameter	-	O	O	O	O	O	Drive Scaling only
1377	Set	Actuator Diameter Unit	-	O	O	O	O	O	Drive Scaling only
1400 + o	Get	Feedback n Catalog Number	-	O	-	O	O	O	E
1401 + o	Get	Feedback n Serial Number	-	O	-	O	O	O	E
1402 + o	Get	Feedback n Position	-	R	-	R	R	R	E
1403 + o	Get	Feedback n Velocity	-	R	-	R	R	R	E
1404 + o	Get	Feedback n Acceleration	-	R	-	R	R	R	E
42	Set*	Feedback Mode	-	R	R	R	R	R	E
43	Set	Feedback Master Select	-	O	-	-	-	-	
44	Set	Feedback Unit Ratio	-	-	-	O	O	-	E

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
1410 + o	Set	Feedback n Resolution Unit	-	O	-	O	O	O	E
1411 + o	Set	Feedback n Unit	-	R	-	R	R	R	E
1412 + o	Set	Feedback n Port Select	-	O	-	O	O	O	E
1413 + o	Set	Feedback n Type	-	R	-	R	R	R	E
1414 + o	Set	Feedback n Polarity	-	O	-	O	O	O	E
1415 + o	Set	Feedback n Startup Method	-	R	-	R	R	R	E
1416 + o	Set	Feedback n Cycle Resolution	-	R	-	R	R	R	E; Not LT
1417 + o	Set	Feedback n Cycle Interpolation	-	R	-	R	R	R	E; Not LT
1418 + o	Set	Feedback n Turns	-	R	-	R	R	R	E; Rotary Absolute Only
1419 + o	Set	Feedback n Length	-	R	-	R	R	R	E; Linear Absolute Only
1420 + o	Set	Feedback n Data Length	-	O	-	O	O	O	E; TP, SS
1421 + o	Set	Feedback n Data Code	-	O	-	O	O	O	E; TP, SS
1422 + o	Set	Feedback n Resolver Trans. Ratio	-	O	-	O	O	O	E; RS
1423 + o	Set	Feedback n Resolver Excitation Voltage	-	O	-	O	O	O	E; RS
1424 + o	Set	Feedback n Resolver Excit Frequency	-	O	-	O	O	O	E; RS
1425 + o	Set	Feedback n Resolver Cable Balance	-	O	-	O	O	O	E; RS
1426 + o	Set	Feedback n LDT Type	-	R	-	R	R	R	E; LT
1427 + o	Set	Feedback n LDT Recirculations	-	R	-	R	R	R	E; LT
1434 + o	Set	Feedback n Velocity Filter Bandwidth	-	O	-	O	O	O	E
1435 + o	Set	Feedback n Accel Filter Bandwidth	-	O	-	O	O	O	E
60	Set*	Event Checking Control	-	O	-	O	O	O	E
61	Get	Event Checking Status	-	O	-	O	O	O	E
62	Get	Registration 1 Positive Edge Position	-	O	-	O	O	O	E
63	Get	Registration 1 Negative Edge Position	-	O	-	O	O	O	E
64	Get	Registration 2 Positive Edge Position	-	O	-	O	O	O	E
65	Get	Registration 2 Negative Edge Position	-	O	-	O	O	O	E
66	Get	Registration 1 Positive Edge Time	-	O	-	O	O	O	E
67	Get	Registration 1 Negative Edge Time	-	O	-	O	O	O	E
68	Get	Registration 2 Positive Edge Time	-	O	-	O	O	O	E
69	Get	Registration 2 Negative Edge Time	-	O	-	O	O	O	E
70	Get	Home Event Position	-	R	-	R	O	O	E
71	Get	Home Event Time	-	O	-	O	O	O	E
72	Get	Watch 1 Event Time	-	R	-	R	O	O	E; Drive Scaling only
73	Get	Watch 2 Event Time	-	O	-	O	O	O	E; Drive Scaling only
360	Set*	Controller Position Command – Integer	-	-	-	O	-	-	

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
361	Set*	Controller Position Command – Float	-	-	-	O	-	-	
362	Set*	Controller Velocity Command	-	-	-	O	R	-	
363	Set*	Controller Acceleration Command	-	-	-	O	O	O	Accel Control Mode – R
364	Set*	Controller Torque Command	-	-	-	-	-	R	
365	Get	Fine Command Position	-	-	-	O	-	-	
366	Get	Fine Command Velocity	-	-	-	O	O	-	
367	Get	Fine Command Acceleration	-	-	-	O	O	O	
370	Set	Skip Speed 1	-	-	O	-	-	-	
371	Set	Skip Speed 2	-	-	O	-	-	-	
372	Set	Skip Speed 3	-	-	O	-	-	-	
373	Set	Skip Speed Band	-	-	O	-	-	-	
374	Set	Ramp Velocity – Positive	-	-	O	-	O	-	
375	Set	Ramp Velocity – Negative	-	-	O	-	O	-	
376	Set	Ramp Acceleration	-	-	O	-	O	-	
377	Set	Ramp Deceleration	-	-	O	-	O	-	
378	Set	Ramp Jerk Control	-	-	O	-	O	-	
380	Set	Flying Start Enable	-	-	O	-	O	-	
430	Get	Position Command	-	-	-	R	-	-	
431	Set*	Position Trim	-	-	-	R	-	-	
432	Get	Position Reference	-	-	-	R	-	-	
433	Get	Velocity Feedforward Command	-	-	-	R	-	-	
434	Get	Position Feedback	-	R	-	R	R	R	E
436	Get	Position Error	-	-	-	R	-	-	
437	Get	Position Integrator Output	-	-	-	R	-	-	
438	Get	Position Loop Output	-	-	-	R	-	-	
440	Set	Kvff	-	-	-	R	-	-	
441	Set	Kpp	-	-	-	R	-	-	
442	Set	Kpi	-	-	-	R	-	-	
443	Set	Position Lock Tolerance	-	-	-	R	-	-	
444	Set	Position Error Tolerance	-	-	-	R	-	-	
445	Set	Position Error Tolerance Time	-	-	-	O	-	-	
446	Set	Position Integrator Control	-	-	-	R	-	-	
447	Set	Position Integrator Preload	-	-	-	O	-	-	
448	Set	Position Limit – Positive	-	O	-	O	O	O	Drive Scaling only; E
449	Set	Position Limit – Negative	-	O	-	O	O	O	Drive Scaling only; E
450	Get	Velocity Command	-	-	R	R	R	-	
451	Set*	Velocity Trim	-	-	R	R	R	-	
452	Get	Acceleration Feedforward Command	-	-	-	R	R	-	
453	Get	Velocity Reference	-	-	R	R	R	-	
454	Get	Velocity Feedback	-	R	R	R	R	R	

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
455	Get	Velocity Error	-	-	-	R	R	-	
456	Get	Velocity Integrator Output	-	-	-	R	R	-	
457	Get	Velocity Loop Output	-	-	-	R	R	-	
460	Set	Kaff	-	-	-	R	R	-	
461	Set	Kvp	-	-	-	R	R	-	
462	Set	Kvi	-	-	-	R	R	-	
464	Set	Kdr	-	-	O	O	O	-	
465	Set	Velocity Error Tolerance	-	-	-	O	O	-	
466	Set	Velocity Error Tolerance Time	-	-	-	O	O	-	
467	Set	Velocity Integrator Control	-	-	-	R	R	-	
468	Set	Velocity Integrator Preload	-	-	-	O	O	-	
469	Set	Velocity Low Pass Filter Bandwidth	-	-	-	O	O	-	
470	Set	Velocity Threshold	-	O	O	O	O	O	
471	Set	Velocity Lock Tolerance	-	-	O	O	O	-	
472	Set	Velocity Standstill Window	-	R	R	R	R	R	
473	Set	Velocity Limit – Positive	-	-	O	O	O	-	
474	Set	Velocity Limit – Negative	-	-	O	O	O	-	
480	Get	Acceleration Command	-	-	-	O	O	O	
481	Set*	Acceleration Trim	-	-	-	O	O	O	
482	Get	Acceleration Reference	-	-	-	O	O	O	
483	Get	Acceleration Feedback	-	R	-	R	R	R	
485	Set	Acceleration Limit	-	-	-	O	O	O	
486	Set	Deceleration Limit	-	-	-	O	O	O	
490	Get	Torque Command	-	-	-	R	R	R	
491	Set*	Torque Trim	-	-	-	R	R	R	
492	Get	Torque Reference	-	-	-	R	R	R	
493	Get	Torque Reference – Filtered	-	-	-	R	R	R	
494	Get	Torque Reference – Limited	-	-	-	R	R	R	
496	Set	Kj	-	-	-	R	R	O	
498	Set	Friction Compensation – Sliding	-	-	-	O	O	O	
499	Set	Friction Compensation – Static	-	-	-	O	O	O	
500	Set	Friction Compensation – Viscous	-	-	-	O	O	O	
502	Set	Torque Low Pass Filter Bandwidth	-	-	-	O	O	O	
503	Set	Torque Notch Filter Frequency	-	-	-	O	O	O	
504	Set	Torque Limit – Positive	-	-	-	R	R	R	
505	Set	Torque Limit – Negative	-	-	-	R	R	R	
506	Set	Torque Rate Limit	-	-	-	O	O	O	
507	Set	Torque Threshold	-	-	-	O	O	O	
508	Set	Overtorque Limit	-	-	O	O	O	O	
509	Set	Overtorque Limit Time	-	-	O	O	O	O	
510	Set	Undertorque Limit	-	-	O	O	O	O	

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
511	Set	Undertorque Limit Time	-	-	O	O	O	O	
520	Get	Iq Current Command	-	-	-	R	R	R	
521	Get	Operative Current Limit	-	-	-	O	O	O	
522	Get	Current Limit Source	-	-	-	O	O	O	
523	Get	Motor Electrical Angle	-	-	-	R	R	R	PM Motors only
524	Get	Iq Current Reference	-	-	-	O	O	O	
525	Get	Id Current Reference	-	-	-	O	O	O	
527	Get	Iq Current Error	-	-	-	O	O	O	
528	Get	Id Current Error	-	-	-	O	O	O	
529	Get	Iq Current Feedback	-	-	-	O	O	O	
530	Get	Id Current Feedback	-	-	-	O	O	O	
531	Get	Vq Decoupling	-	-	-	O	O	O	
532	Get	Vd Decoupling	-	-	-	O	O	O	
533	Get	Vq Voltage Output	-	-	-	O	O	O	
534	Get	Vd Voltage Output	-	-	-	O	O	O	
535	Get	U Voltage Output	-	-	-	O	O	O	
536	Get	V Voltage Output	-	-	-	O	O	O	
537	Get	W Voltage Output	-	-	-	O	O	O	
538	Get	U Current Feedback	-	-	-	O	O	O	
539	Get	V Current Feedback	-	-	-	O	O	O	
540	Get	W Current Feedback	-	-	-	O	O	O	
541	Get	U Current Offset	-	-	-	O	O	O	
542	Get	V Current Offset	-	-	-	O	O	O	
543	Get	W Current Offset	-	-	-	O	O	O	
553	Set	Current Vector Limit	-	-	O	O	O	O	
554	Set	Kqp	-	-	-	O	O	O	
555	Set	Kqi	-	-	-	O	O	O	
556	Set	Kdp	-	-	-	O	O	O	
557	Set	Kdi	-	-	-	O	O	O	
558	Set	Flux Up Control	-	-	O	O	O	O	Induction Motor only
559	Set	Flux Up Time	-	-	O	O	O	O	Induction Motor only
560	Set	Commutation Startup Method	-	-	-	O	O	O	PM Motors only
561	Set	Commutation Offset	-	-	-	R	R	R	PM Motors only
562	Set	Commutation Self-Sensing Current	-	-	-	O	O	O	PM Motors only
563	Set	Commutation Polarity	-	-	-	O	O	O	PM Motors only
564	Set	Commutation Alignment	-	-	-	O	O	O	PM Motors only
565	Get	Slip Compensation	-	-	R	-	-	-	
570	Set	Frequency Control Method	-	-	R	-	-	-	
572	Set	Maximum Voltage	-	-	R	-	-	-	
573	Set	Maximum Frequency	-	-	R	-	-	-	Basic V/Hz only
575	Set	Break Voltage	-	-	R	-	-	-	Basic V/Hz only

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
576	Set	Break Frequency	-	-	R	-	-	-	Basic V/Hz only
577	Set	Start Boost	-	-	R	-	-	-	Basic V/Hz only
578	Set	Run Boost	-	-	R	-	-	-	Basic V/Hz only
600	Get	Output Frequency	-	-	R	O	O	O	
601	Get	Output Current	-	-	R	R	R	R	
602	Get	Output Voltage	-	-	R	R	R	R	
603	Get	Output Power	-	-	R	R	R	R	
604	Set	PWM Frequency	-	-	O	O	O	O	
608	Set	Zero Speed	-	-	O	O	O	O	
609	Set	Zero Speed Time	-	-	O	O	O	O	
610	Set	Stopping Action	-	-	R	R	R	R	
611	Set	Stopping Torque	-	-	-	R	R	R	
612	Set	Stopping Time Limit	-	-	-	O	O	O	
613	Set	Resistive Brake Contact Delay	-	-	O	O	O	O	PM Motors Only
614	Set	Mechanical Brake Control	-	-	O	O	O	O	
615	Set	Mechanical Brake Release Delay	-	-	O	O	O	O	
616	Set	Mechanical Brake Engage Delay	-	-	O	O	O	O	
590	Set	Proving Configuration	-	-	O	O	O	O	
591	Set	Torque Prove Current	-	-	O	O	O	O	
592	Set	Brake Test Torque	-	-	O	O	O	O	E
593	Set	Brake Prove Ramp Time	-	-	O	O	O	O	E
594	Set	Brake Slip Tolerance	-	-	O	O	O	O	E
620	Get	DC Bus Voltage	R	-	R	R	R	R	
621	Get	DC Bus Voltage – Nominal	R	-	R	R	R	R	
622	Set	Bus Configuration	O	-	O	O	O	O	
623	Set	Bus Voltage Select	-	-	R	R	R	R	
624	Set	Bus Regulator Action	R	-	R	R	R	R	Integral Converter only
625	Set	Regenerative Power Limit	-	-	O	O	O	O	Integral Converter only
627	Set	Power Loss Action	-	-	O	O	O	O	
628	Set	Power Loss Threshold	O	-	O	O	O	O	
629	Set	Shutdown Action	O	-	O	O	O	O	
630	Set	Power Loss Time	O	-	O	O	O	O	
634	Get	Integral Converter	O	-	O	O	O	O	
635	Get	Motor Capacity	-	-	R	R	R	R	
636	Get	Inverter Capacity	-	-	R	R	R	R	
637	Get	Converter Capacity	O	-	O	O	O	O	Integral Converter only
638	Get	Bus Regulator Capacity	O	-	O	O	O	O	
639	Get	Ambient Temperature	O	-	O	O	O	O	
640	Get	Inverter Heatsink Temperature	-	-	O	O	O	O	
641	Get	Inverter Temperature	-	-	O	O	O	O	
642	Get	Motor Temperature	-	-	O	O	O	O	

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
643	Get	Feedback 1 Temperature	-	O	O	O	O	O	E
644	Get	Feedback 2 Temperature	-	O	O	O	O	O	E
645	Get	Inverter Overload Limit	-	-	O	O	O	O	
646	Set	Motor Overload Action	-	-	O	O	O	O	
647	Set	Inverter Overload Action	-	-	O	O	O	O	
648	Set	Duty Select	-	-	O	O	O	O	
650	Get	Axis State	R	R	R	R	R	R	
651	Get	Axis Status	R	R	R	R	R	R	
652	Get	Axis Status – Mfg	R	R	R	R	R	R	
653	Get	Axis I/O Status	R	R	R	R	R	R	
654	Get	Axis I/O Status – Mfg	R	R	R	R	R	R	
655	Get	Axis Exceptions	R	R	R	R	R	R	
656	Get	Axis Exceptions – Mfg	R	R	R	R	R	R	
657	Get	Axis Faults	R	R	R	R	R	R	
658	Get	Axis Faults – Mfg	R	R	R	R	R	R	
659	Get	Axis Alarms	O	O	O	O	O	O	
660	Get	Axis Alarms – Mfg	O	O	O	O	O	O	
661	Get	Axis Fault Code	R	R	R	R	R	R	
662	Get	Axis Fault Type	R	R	R	R	R	R	
663	Get	Axis Fault Time Stamp	R	R	R	R	R	R	
664	Get	Axis Alarm Code	O	O	O	O	O	O	
665	Get	Axis Alarm Type	O	O	O	O	O	O	
666	Get	Axis Fault Sub Code	O	O	O	O	O	O	
667	Get	Axis Alarm Sub Code	O	O	O	O	O	O	
668	Get	Axis Fault Action	R	R	R	R	R	R	
669	Get	Axis Alarm State	O	O	O	O	O	O	
670	Get	Axis Alarm Time Stamp	O	O	O	O	O	O	
678	Get	Rotary Motor Overspeed Factory Limit	-	-	O	O	O	O	Rotary Motor only
679	Get	Linear Motor Overspeed Factory Limit	-	-	O	O	O	O	Linear Motor only
680	Get	Motor Overtemperature Factory Limit	-	-	O	O	O	O	
681	Get	Motor Thermal Overload. Factory Limit	-	-	O	O	O	O	
682	Get	Inverter Overtemperature Factory Limit	-	-	O	O	O	O	
683	Get	Inverter Thermal Overload Factory Limit	-	-	O	O	O	O	
684	Get	Converter Overtemperature Factory Limit	O	-	O	O	O	O	
685	Get	Converter Thermal Overload Factory Limit	O	-	O	O	O	O	
693	Get	Converter Ground Current Factory Limit	O	-	O	O	O	O	

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
686	Get	Bus Reg Overtemperature Factory Limit	O	-	O	O	O	O	
687	Get	Bus Reg Thermal Overload Factory Limit	O	-	O	O	O	O	
688	Get	Bus Overvoltage Factory Limit	O	-	O	O	O	O	
689	Get	Bus Undervoltage Factory Limit	O	-	O	O	O	O	
690	Get	Feedback Noise Factory Limit	-	O	O	O	O	O	E
691	Get	Feedback Signal Loss Factory Limit	-	O	O	O	O	O	E
692	Get	Feedback Data Loss Factory Limit	-	O	O	O	O	O	E
694	Set	Motor Phase Loss Limit	-	-	O	O	O	O	
695	Set	Motor Overspeed User Limit	-	-	O	O	O	O	
696	Set	Motor Overtemperature User Limit	-	-	O	O	O	O	
697	Set	Motor Thermal Overload User Limit	-	-	O	O	O	O	
698	Set	Inverter Overtemperature User Limit	-	-	O	O	O	O	
699	Set	Inverter Thermal Overload User Limit	-	-	O	O	O	O	
700	Set	Converter Overtemperature User Limit	O	-	O	O	O	O	Integral Converter only
701	Set	Converter Thermal Overload User Limit	O	-	O	O	O	O	Integral Converter only
709	Set	Converter Ground Current User Limit	O	-	O	O	O	O	Integral Converter only
702	Set	Bus Reg Overtemperature User Limit	O	-	O	O	O	O	Integral Converter only
703	Set	Bus Reg Thermal Overload User Limit	O	-	O	O	O	O	Integral Converter only
704	Set	Bus Overvoltage User Limit	O	-	O	O	O	O	
705	Set	Bus Undervoltage User Limit	O	-	O	O	O	O	
706	Get	Feedback Noise User Limit	-	O	O	O	O	O	E
707	Get	Feedback Signal Loss User Limit	-	O	O	O	O	O	E
708	Get	Feedback Data Loss User Limit	-	O	O	O	O	O	E
672	Set	Axis Exception Action	R	R	R	R	R	R	
673	Set	Axis Exception Action – Mfg	R	R	R	R	R	R	
674	Get	Initialization Faults	R	R	R	R	R	R	
675	Get	Initialization Faults – Mfg	R	R	R	R	R	R	
676	Get	Start Inhibits	R	-	R	R	R	R	
677	Get	Start Inhibits – Mfg	R	-	R	R	R	R	
756	Get	APR Faults	-	O	-	O	O	O	E; Drive Scaling only
757	Get	APR Faults – Mfg	-	O	-	O	O	O	E; Drive Scaling only
710	Get	Control Power-up Time	O	-	O	O	O	O	
711	Get	Cumulative Run Time	O	-	O	O	O	O	
712	Get	Cumulative Energy Usage	O	-	O	O	O	O	
713	Get	Cumulative Motor Revs	-	-	O	O	O	O	
714	Get	Cumulative Main Power Cycles	O	-	O	O	O	O	Integral Converter only

Instance attribute			Implementation by Device Function Code						
Attr. ID	Access rule	Attribute name	N		O	C- Closed Loop Control			Conditional implementation
			B	E	F	P	V	T	
715	Get	Cumulative Control Power Cycles	O	-	O	O	O	O	Integral Converter only
720	Get	Inverter Rated Output Voltage	-	-	R	R	R	R	
721	Get	Inverter Rated Output Current	-	-	R	R	R	R	
722	Get	Inverter Rated Output Power	-	-	R	R	R	R	
723	Get	Converter Rated Output Current	O	-	O	O	O	O	
724	Get	Converter Rated Output Power	O	-	O	O	O	O	
725	Get	Drive Power Structure Axis ID	-	-	O	O	O	O	
730	Get	Digital Inputs	O	-	O	O	O	O	
731	Set	Digital Outputs	O	-	O	O	O	O	
732	Get	Analog Input 1	O	-	O	O	O	O	
733	Get	Analog Input 2	O	-	O	O	O	O	
734	Set	Analog Output 1	O	-	O	O	O	O	
735	Set	Analog Output 2	O	-	O	O	O	O	
736	Set	Drive Enable Input Checking	O	-	O	O	O	O	
737	Set	Hardware Overtravel Input Checking	-	-	O	O	O	O	
750	Set	Local Control	O	O	O	O	O	O	
760	Get	Axis Safety State	-	O	O	O	O	O	Safety only
761	Get	Axis Safety Status	-	O	O	O	O	O	Safety only
762	Get	Axis Safety Status – Mfg	-	O	O	O	O	O	Safety only
763	Get	Axis Safety Faults	-	O	O	O	O	O	Safety only
764	Get	Axis Safety Faults – Mfg	-	O	O	O	O	O	Safety only

7.2 Class attributes

7.2.1 General

The table of attributes in Table 30 applies to the Motion Device Axis Object class, which are referenced as instance 0. These attributes exist even before any Motion Device Axis Object instances have been created. Since they are not tied to any particular axis instance of the CIP Motion device, the class attributes are generally used to address parametric behavior that applies to all axis instances, for example the communications node behavior. As instances of this class are created, they are given consecutive instance numbers starting at 1.

NOTE 1 No attributes associated with this object require Non-Volatile storage, so a “No” is implied for all attributes under the NV column.

NOTE 2 Vendor specific bits, and enumerations provide space for device vendors to provide additional product features.

Key to Table 30: (see 7.1.6 and 7.1.7 for details)

(R) – Required bit or enumeration – shall be supported in the implementation.

(O) – Optional bit or enumeration – support is left to vendor’s discretion.

Set* – Indicates the attribute is normally set by the CIP Motion Connection data block and not by a Set service.

Table 30 – Class attributes for the Motion Device Axis Object

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1	Required	Get		Revision	UINT	Revision of this object.	The current value assigned to this attribute for this revision of the object is three (3). Revision 1 of this object is designated as obsolete. Revision 2 is no longer recommended for new devices supporting time synchronization.
2 to 7	These class attributes are optional and are described in IEC 61158-5-2 and IEC 61158-6-2.						
14	Required	Set*		Node Control	BYTE	Contains bits used to control the behavior of the associated device communications node.	See semantics in 7.2.2.1.
15	Required	Get		Node Status	BYTE	Contains bits used to indicate the status of the associated device communications node.	See semantics in 7.2.2.2.
16	Required	Get		Node Fault Code	USINT	Contains numeric code of active node fault condition. The Node Fault Code is used to provide diagnostic detail to pin-point the source of the fault condition.	See semantics in 7.2.2.3.
17	Optional	Get		Node Alarm Code	USINT	Contains numeric code of the current active alarm condition. The Node Alarm Code is used to provide diagnostic detail to pin-point the source of the alarm condition.	See semantics in 7.2.2.4.
18	Required	Set*		Controller Update Period	UDINT	Represents the period between updates of the controller that is also the period between CIP Motion C-to-D Connection updates.	Nanoseconds
19	Required	Set*		Controller Time Offset	LINT	This element represents the 64-bit value at the beginning of the Controller Update Period that is associated with the Controller Time Stamp value. The Controller Time Offset is the value that is added to the controller's local clock to produce System Time.	Nanoseconds
20	Required	Set*		Controller Time Stamp	LINT	This element represents the 64-bit System Time value at the beginning of the Controller Update Period when the controller's update timer event occurred. It is calculated by the controller as the sum of the controller's local clock value when update timer event occurred and the controller's System Time Offset value given by Controller Time Offset. The Controller Time Stamp is therefore directly associated with the command data contained in the connection. The time stamp format is absolute and follows the CIP Sync standard with 0 corresponding to 1970-01-01.	Nanoseconds (CIP Sync absolute)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
21	Optional	Set		Controller Update Delay High Limit	USINT	Represents high limit delay threshold for a Controller-to-Device (C-to-D) Connection update. This delay is specified in units of Controller Update Periods. Exceeding this limit results in a Control Connection Update Fault.	Number of Controller Update Periods
22	Optional	Set		Controller Update Delay Low Limit	USINT	Represents low limit delay threshold allowed for a Controller-to-Device (C-to-D) Connection update. This delay is specified in units of Controller Update Periods. Exceeding this limit results in a Control Connection Update Alarm.	Number of Controller Update Periods
28	Optional	Get		Sync Variance	UDINT	Represents the current statistical variation of the System Time Offset value from the mean System Time Offset value.	Nanoseconds
29	Optional	Set		Sync Threshold	UDINT	Determines the threshold for variation of System Time used to determine if the CIP Motion device is considered synchronized to a PTP master clock. This value is passed to the device's PTP subsystem to establish the synchronization criteria.	Nanoseconds Default: device dependent minimum value.
30	Optional	Get		Sync Update Delay	REAL	This value is the product of the number of networks hops the device is away from the grandmaster, as determined by the Time Sync Object attribute, Steps Removed, and the time synchronization interval, as determined by the Time Sync Object attribute, Sync Interval.	Seconds
31	Optional	Set*		Time Data Set	BYTE	This bit-mapped byte contains flags that determine the usage and format of the controller and device timing information. This value is derived from Time Configuration element sent by the controller in the Controller-to-Device (C-to-D) Connection and becomes the Time Configuration value sent by the device in the Device-to-Controller (D-to-C) Connection. Time Stamps bit: Time Stamp included in D-to-C connection. Time Offset bit: Time Offset included in D-to-C connection. Update Diagnostics bit: Device update diagnostic data included in D-to-C connection. Includes number of lost C-to-D updates, and late C-to-D updates. Time Diagnostics bit: Device timing diagnostic data included in D-to-C connection. Includes C-to-D data received time stamp, and D-to-C data transmit time stamp. Time Offset and Update Diagnostic bits need only be set when the associated data has changed value since the last D-to-C update.	Bit Field: 0 = Update Period 1 = Time Stamps 2 = Update Diagnostics 3 = Time Diagnostics 4 to 7 = (Reserved)
32	Optional	Set		Lost Controller Updates	BYTE	This byte value represents the accumulated number of lost Controller-to-Device Connection packets detected since the connection was opened and synchronized. Lost packets are detected by examining the Update ID of received packets from the controller. A packet is determined to be lost when its expected Update ID is skipped. The Lost Controller Updates attribute is incremented for every Update ID that is skipped.	Number of Lost Packets

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
33	Optional	Set		Late Controller Updates	BYTE	This byte value represents the number of late Controller-to-Device Connection packets detected since the connection was opened and synchronized. Late packets are detected by examining the Time Stamp of received packet from the controller. A packet is determined to be late when the difference between its Controller Time Stamp and the device's Data Received Time Stamp exceeds the Controller Update Period.	Number of Late Packets
34	Required	Get		Drive Power Structure Class ID	UDINT	This value represents an ID assigned by the drive vendor that uniquely identifies the power structure associated with the class instance. As a class instance attribute, the ID reflects power structure hardware which is common to all axis instances or identical for all axis instances (excluding master feedback axes). Power structure hardware that is unique to the axis instance shall be identified using the Drive Power Structure Axis ID attribute for that instance. In that case, the Drive Power Structure Class ID is simply set to 0. A single axis drive, however, shall use the Drive Power Structure Class ID to identify the power structure of the single axis instance. The Drive Power Structure Class ID is included as part of the Forward_Open service to the Motion Device Axis Object class to confirm that the drive's power structure matches the configuration in the controller.	ID number
35	Conditional ^a	Set		Step Threshold	UDINT	Determines the threshold for step changes in System Time as it applies to the Step Compensation algorithm. This value is generally set by the controller to the RPI of the CIP Motion I/O Connection. For more details, see the Step Compensation section of the Drive Device profile specification. The Step Threshold attribute value is also used by the Group_Sync service to identify clock skew. For more details on clock skew detection during device initialization refer to the Group_Sync service. Step Threshold is required for devices that implement the Group_Sync service with clock skew detection introduced in Revision 3 of this object. The controller may use successful access to this attribute to determine if the device supports the Group_Sync algorithm with clock skew detection.	Nanoseconds Default: 1 000 000
^a The Step Threshold attribute is required when the device implements the Group_Sync service, otherwise it is not allowed.							

7.2.2 Semantics

7.2.2.1 Attribute No. 14 – Node Control

Contains bits used to control the behavior of the associated motion device communications node, as specified in Table 31.

Table 31 – Node Control bit definitions

Bit	Req./Opt.	Name	Description
0	R	Remote Control	This bit is used to request that the device node turn over complete control of the device to the controller. If this bit is clear, the device is under the exclusive control of the local interface; the CIP Motion controller cannot affect the behavior of the device in any way other than to switch the device back to Remote Control mode. If the CIP Motion Connection is lost while the device is under remote control, the device may then be controlled via the local interface.
1	O	Sync Control	This bit is used by the controller to request synchronous operation of the device. Synchronous operation is defined as having the device node's local timer synchronized with System Time and that the device node start using the Controller Time Stamp and Time Offset to process the connection data and schedule future Device-to-Controller Connection updates. The Synchronous Control bit implies that the Controller Time Stamp and Time Offset values are valid. The bit shall remain set for the duration of synchronous operation. In asynchronous mode, there is no requirement for the local timer to be synchronized nor is time-stamping necessary or the associated time data even valid. Generally, this bit is set by the controller when all CIP Motion devices associated with the controller have given a successful Group_Sync response.
2	R	Controller Data Valid	This bit shall be set for the device node to process the Instance Data Blocks. During the connection initialization sequence there may be a period of time where the connection data in the Instance Data Blocks is not yet initialized. This condition can be recognized by the device by first checking this bit and, if clear, it is not necessary for the device to process any of the connection data beyond the connection header.
3	R	Node Fault Reset	This bit is used to request that the device perform a reset of the communications node and attempt to clear the Node Fault Code. When the controller detects that the Node Fault Code is clear, the Node Fault Reset bit is cleared. If the communication fault persists, the Node Fault Code may persist, in which case a power cycle or a complete reconfiguration process shall be performed.
4 to 5			(Reserved)
6 to 7			(Vendor Specific)

7.2.2.2 Attribute No. 15 – Node Status

Contains bits used to indicate the status of the associated CIP Motion device's communications node, as specified in Table 32.

Table 32 – Node Status bit definitions

Bit	Req./Opt.	Name	Description
0	R	Remote Mode	The Remote Mode bit is used to indicate that the device node has turned over total control of the device to the controller. If this bit is clear, the device does not act on any data contained in the Controller-to-Device cyclic or service channels.
1	O	Sync Mode	The Sync Mode bit indicates whether or not the device node is synchronized. Synchronous operation is defined as having the device node's local timer synchronized with System Time and that the device node is using the connection time stamps to process the connection data. The Sync Mode bit being set also implies that the Device Time Stamp is valid. The bit shall remain set for the duration of synchronous operation. If the Sync Mode bit is clear, the device is said to be in Asynchronous mode. In Asynchronous mode, there is no requirement for the local timer to be synchronized nor is time-stamping necessary or even valid. Generally, this bit is set in response to the Sync Control bit being set by the controller.
2	R	Device Data Valid	The Device Data Valid bit shall be set for the controller to process the Instance Data Blocks from the device node. During the connection initialization sequence, there may be a period of time where the connection data in the Instance Data Blocks is not yet initialized. This condition can be recognized by the controller by first checking this bit and if set, it is not necessary for the controller to process any of the connection data beyond the connection header.
3	R	Device Node Fault	The Device Node Fault bit is used to indicate that the device has detected one or more fault conditions related to the communications node. Specific fault conditions can be determined by the Node Fault Code attribute. If this bit is clear, there are no Node Fault conditions present. The Device Node Fault bit may be cleared by setting the Node Fault Reset bit in the Node Control word. All associated axes are disabled when a Node Fault condition is present.
4 to 5			(Reserved)
6 to 7			(Vendor Specific)

7.2.2.3 Attribute No. 16 – Node Fault Code

This attribute is a 4-bit fault code used to indicate the presence of specific fault condition associated with the CIP Motion device's communications node, as specified in Table 33

Table 33 – Node Fault Code definitions

Code	Name	Description
0	No Fault	Indicates that there is no fault condition currently present at the device communications node.
1	Control Connection Update Fault	The Control Connection Update Fault code is used to indicate that updates from the controller over the C-to-D connection have been excessively late or consecutively lost as determined by the Controller Update Delay High Limit attribute value.
2	Processor Watchdog Fault	The Processor Watchdog Fault code indicates that the processor associated with the device node has experienced an excessive overload condition that has tripped the associated processor watchdog mechanism.
3	Hardware Fault	The Hardware Fault code indicates that the critical support hardware (FPGA, ASIC, etc.) associated with the device node has experienced a fault condition.
4	Data Format Error	This fault code indicates that an error has occurred in the data format between the controller and the device, e.g. a Format Revision mismatch.
5	Clock Skew Fault	Clock Skew Fault code indicates that the drive has detected significant difference between the drive's System Time and the controller's System Time that prevented the drive from switching to synchronous operation after a time out period.
6	Control Connection Loss Fault	The Control Connection Loss fault code indicates that the CIP Motion C-to-D connection from the controller has timed out.
7	Clock Sync Fault	The fault condition is an indication that the local IEC 61588 clock has lost synchronization with the master and was not able to resynchronize within the allotted timeout (e.g. 40 s to 60 s).
8	Logic Watchdog	The Logic Watchdog Fault code indicates that an auxiliary logic component (e.g. FPGA, or ASIC) associated with the device node has experienced an excessive overload condition that has tripped the associated logic watchdog mechanism.
9	Duplicate Address	The Duplicate Address Fault code indicates that a device node has been detected on the network that using the same Node Address as this device node. For Ethernet, this address would be the IP Address of the device.
10 to 15	(Reserved)	

7.2.2.4 Attribute No. 17 – Node Alarm Code

This attribute is a 4-bit alarm code used to indicate specific alarm conditions of the associated CIP Motion device's communications node, as specified in Table 34. Alarms do not result in any direct action. Individual Node Alarms are automatically cleared by the drive device after 10 s. If the alarm condition persists, the associated Node Alarm is re-posted.

Table 34 – Node Alarm Code definitions

Code	Name	Description
0	No Alarm	Indicates that there is no alarm condition currently present at the device communications node.
1	Control Connection Update Alarm	The Control Connection Update Alarm code is used to indicate that updates from the controller over the C-to-D connection have been late or consecutively lost as determined by the Controller Update Delay Low Limit attribute value.
2	Processor Overload Alarm	The Processor Overload Alarm code indicates that the processor associated with device is experiencing overload conditions that could eventually lead to a fault.
3	Clock Jitter Alarm	Clock Jitter Alarm code indicates that the Sync Variance has exceeded the Sync Threshold while the device is running in Sync Mode.
4	Clock Skew Alarm	Clock Skew Alarm code indicates that the drive has detected significant difference between the drive's System Time and the controller's System Time that is preventing the drive from switching to synchronous operation.
5	Clock Sync Alarm	The Clock Sync Alarm code indicates that the device's local clock has lost synchronization with the master clock for a short period of time (e.g. 10 s to 20 s) during synchronous operation. This alarm condition can also occur when a change in the master clock source has been detected. The Clock Sync Alarm is an indication that the local IEC 61588 clock has shifted back to its start-up mode to quickly synchronize to the master clock.
6	Node Address Alarm	The Node Address Alarm code indicates that the Node Address setting of the device has been changed during device operation and may no longer be valid.
7 to 15	(Reserved)	

7.3 Instance attributes

7.3.1 General

Subclause 7.3 lists all the supported attributes of a Motion Device Axis Object instance. Because of the large number of attributes listed in 7.3, attributes have been organised by functional category. Each functional grouping may be further organised by first listing the object Status and Signal attributes, followed by the object Configuration attributes.

NOTE 1 Due to the large number of instance attributes supported by this object, 16-bit Attribute IDs are used.

NOTE 2 No attributes associated with this object require Non-Volatile storage, so a “No” is implied for all attributes under the NV column. The Non-Volatile storage function is provided by the motion controller and specifically, the Motion Control Axis Object.

Unless otherwise specified, all object attributes default to 0 at device power-up. Since it is the motion controller working in conjunction with user driven configuration software that sets device attribute values, it is the responsibility of the controller to determine appropriate default values. For this reason, specification of default attribute values is addressed in the Motion Control Axis Object and are not within the scope of this specification.

Range limits associated with configuration attributes are specified by the device manufacturer. Attempting to set an attribute to a value that is out of range shall result in a General Status Error Code 9, for ‘Invalid Attribute Value’. Since axis configuration tools interface directly to the motion controller, not to the CIP Motion device, it is the responsibility of the motion controller to enforce the attribute range limits of the device. For this reason, specification of attribute range limits is addressed in the Motion Control Axis Object and are not within the scope of this specification.

The one exception to erring an out of range configuration attribute value is when the out of range value does not impact the behavior of the device. An example of this condition would be

the Torque LP Filter Bandwidth attribute. If the drive's maximum Torque LP Filter Bandwidth is 2 000 Hz based on sample rate limitations and the controller tries to set the attribute to 5 000 Hz, the drive can simply apply 2 000 Hz with the same resultant behavior. In this case, the vendor may choose to clamp the value to the device limit or simply disable the filter.

All reserved and otherwise unused bits and enumerations are set to 0.

Key to Tables: (see 7.1.6 and 7.1.7 for details)

(R) – Required bit or enumeration – shall be supported in the implementation

(O) – Optional bit or enumeration – support is left to vendor's discretion

Device Function Codes:

B – Bus Power Converter

E – Encoder, Feedback Only

F – Frequency Control (V/Hz or VFD)

P – Position Control Loop

V – Velocity Control Loop

T – Torque Control Loop

N – All Device Functions using No Control Method

O – All Device Functions using Open Loop Control Methods (Frequency Control)

C – All Device Functions using Closed Loop Vector Control (PI Vector Control Method)

D – All Device Functions using (Control Method != No Control)

All – All Device Functions

Conditional implementation rules:

E = Encoder-based Feedback channel present

!E = Encoderless or Sensorless device operation, no Feedback channel present

All – All Control Modes

Set* – Indicates the attribute is normally set by the CIP Motion Connection Header and not by a Set service.

% Device Rated Units – defined as the percentage of the continuous rating of the device with 100 % implying operation at the continuous rated specification for the device. This unit can be applied to attributes related to speed, torque, force, current, voltage, and power. Applicable "Devices" can be Motor, Inverter, Converter, or Bus Regulator. This unit can be used independently of whether the attribute value represents an instantaneous level or a time-averaged level; the appropriate unit for the device rating is implied. As with all attributes that are in units of %, an attribute value of 100 means 100 %.

Dynamic Units: Attributes that relate to motion dynamics typically express displacement in terms of the selected feedback or motor device units. Since the CIP Motion device can use different feedback channels for different control loops depending on the Feedback Mode, the determination of which feedback device applies, if any, depends on the Feedback Mode.

Table 35 provides a cross-reference table to determine the appropriate feedback counts or units to apply to an attribute based on its Dynamic Unit type (position, velocity, or acceleration), and the configured Feedback Mode. If the CIP Motion device supports Scaling functionality, then the Position Control Unit may be expressed in Motion Counts and Motion Units, independent of the Feedback Mode. The Scaling function, if enabled, converts Motion Counts to/from Feedback Counts, and Motion Units to/from Feedback Units. When an axis

instance is configured for No Feedback, i.e. sensorless/encoderless operation (V/Hz, sensorless velocity control, etc.), feedback counts do not apply, so motion dynamics shall be expressed in terms of motor displacement.

Table 35 – Dynamic Units vs. Feedback Mode

Dynamic Units	Scaling Enabled	Feedback Mode				
		No Feedback	Master Feedback	Motor Feedback	Load Feedback	Dual Feedback
Position Control Units	Motion Counts	Motor Counts	Feedback n Counts ^a	Feedback 1 Counts	Feedback 2 Counts	Feedback 2 Counts
Velocity Control Units	Motion Units	Motor Units	Feedback n Units ^a	Feedback 1 Units	Feedback 2 Units	Feedback 1 Units
Accel Control Units	Motion Units	Motor Units	Feedback n Units ^a	Feedback 1 Units	Feedback 2 Units	Feedback 1 Units
^a Feedback Channel determined by Master Feedback Select attribute						

7.3.2 Motion Control configuration attributes

7.3.2.1 General

The following attribute table in Table 36 contains basic motion control configuration attributes associated with a Motion Device Axis Object instance. These attributes govern aspects of the overall behavior of the Motion Device Axis Object.

Set* ➔ These attributes are generally updated via the cyclic Command Data Set of the CIP Motion C-to-D Connection. When included as cyclic command data, these attributes cannot be updated via a Set service.

Table 36 – Motion Control configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
40	Required – All	Set*		Control Mode	USINT	Attribute whose lower 4-bit enumeration determines the general dynamic control behavior of the motion axis instance.	Bits 0 to 3: Enumeration: Bits 4 to 7: (Reserved) See semantics in 7.3.2.2.1
41	Required – All	Set		Control Method	USINT	Enumerated code that determines the basic motor control algorithm applied by the device to control the dynamic behavior of the motor.	See semantics in 7.3.2.2.2

7.3.2.2 Semantics

7.3.2.2.1 Attribute No. 40 – Control Mode

The Control Mode attribute is a 4-bit enumeration that determines the specific dynamic behavior of the motor that the device is to control for this axis instance. The system view of these control modes are described in detail in 4.2. Table 37 shows a brief summary of the Control Mode enumerations.

Table 37 –Control Mode enumeration definitions

Enum.	Req./Opt.	Name	Description
0	R/N	No Control	No motor control is provided in this mode but interface to a specific feedback device as a master feedback source is possible via the Feedback Mode attribute.
1	R/P	Position Control	Device seeks to control the position, or orientation, of the motor.
2	R/PV	Velocity Control	Device seeks to control the velocity of the motor.
3	O	Acceleration Control	Device seeks to control the acceleration of the motor.
4	R/C	Torque Control	Device seeks to control the torque output of the motor.
5 to 15		(Reserved)	-

7.3.2.2.2 Attribute No. 41 – Control Method

The Control Method attribute is an 8-bit enumerated code that determines the basic control algorithm applied by the device to control the dynamic behavior of the motor associated with an axis instance, as specified in Table 38.

Table 38 – Control Method enumeration definitions

Enum.	Req./Opt.	Name	Description
0	R/N	No Control	No Control is associated with a Control Mode of No Control where there is no explicit motor control provided by the device for this axis instance.
1	R/F	Frequency Control	Frequency Control is an “open loop” control method that applies voltage to the motor, generally in proportion to the commanded frequency or speed. This control method is associated with variable frequency drives (VFDs) or so-called Volts/Hertz drives.
2	R/C	PI Vector Control	PI Vector Control is a “closed loop” control method that uses actual or estimated feedback for closed loop cascaded PI control of motor dynamics, i.e. position, velocity, acceleration, and torque, and always includes independent closed loop PI control of I _q and I _d components of the motor current vector.
3 to 127		(Reserved)	-
128 to 255		(Vendor specific)	-

7.3.3 Motion Scaling attributes

7.3.3.1 General

The attribute table in Table 39 contains basic motion scaling configuration attributes associated with a Motion Device Axis Object instance. The Scaling function, if enabled, converts Motion Counts to/from Feedback Counts, and Motion Units to/from Feedback Units. Scaling functionality also encompasses cyclic unwind operation and motion polarity.

Table 39 – Motion Scaling attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
45	Required – Scaling	Set		Motion Scaling Configuration	USINT	Enumerated attribute used to determine whether the scaling function is performed by the controller or the drive. The Control Scaling Configuration selection configures the control system to perform the scaling calculations in the controller. In this mode, the controller interacts with the drive in terms of Feedback Counts or Motor Units, hence no scaling operations are required by the drive. Also, in Control Scaling mode the controller is responsible for Cyclic Unwind operations associated with Cyclic Travel Mode. The Drive Scaling selection configures the system to perform the scaling calculations in the drive device. In this mode, the controller interacts with the drive in terms of Motion Counts or Motion Units and the drive is responsible for conversion to equivalent Feedback Counts and Motor Units. Also, in Drive Scaling mode the drive is responsible for Cyclic Unwind operations associated with Cyclic Travel Mode.	Enumeration: 0 = Control Scaling (R) 1 = Drive Scaling (O) 2 to 255 = reserved
77	Required – Scaling	Set		Motion Unit	USINT	This enumerated value determines the unit of measure used to express the Motion Resolution used by motion planner functions. See semantics in 7.3.3.2.1	Enumeration: 0 = Motor Rev 1 = Load Rev 2 = Feedback Rev 3 = Motor mm 4 = Load mm 5 = Feedback mm 6 = Motor inch 7 = Load inch 8 = Feedback inch 9 = Motor r/s 10 = Load r/s 11 = Motor m/s 12 = Load m/s 13 = Motor inch/s 14 = Load inch/s 15 to 255 = (reserved)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
78	Required – Scaling	Set		Motion Resolution	DINT	An integer value that determines the number of Motion Counts per Motion Unit used by the scaling function to convert between Motion Counts and Feedback Counts. See semantics in 7.3.3.2.1	Motion Counts / Motion Unit
79	Required – Scaling	Set		Motion Polarity	USINT	When Motion Scaling Configuration is set for Drive Scaling, Motion Polarity can be used to switch the directional sense of the motion control system. (see semantics in 7.3.3.2.2 for details)	Enumeration: 0 = Normal Polarity 1 = Inverted Polarity 2 to 255 = (Reserved)
83	Required – Scaling	Set		Cyclic Unwind Control	BOOL	When this Boolean attribute is set true (1), it enables the cyclic unwind capability of the scaling function. This feature provides infinite positioning range by “unwinding” the axis position whenever the axis moves through a complete machine cycle. The number of Motion Counts per machine cycle of the axis is specified by the Cyclic Unwind Position attribute. Thus, if the axis is configured Cyclic Operation, implementation of the Cyclic Unwind Position attribute is required. If the Cyclic Unwind attribute is false (0), indicating non-cyclic operation, no “unwind” operation is performed, and unidirectional motion continues to accumulate position and can eventually exceed the range of the position representation resulting in a roll-over. For this reason, non-cyclic travel is typically limited for position control applications.	0 = Non-Cyclic Operation (R) 1 = Cyclic Operation (E)
84	Required – Scaling E	Set		Cyclic Unwind Position	DINT	This integer value is used to “unwind” command and actual position values every machine cycle. Cyclic unwind functionality provides infinite position range for cyclic axes by subtracting the Cyclic Unwind Position value from both the actual and command position whenever the axis reaches or crosses the Cyclic Unwind Position. Similarly when moving in the reverse direction, the Cyclic Unwind Position value is added to both the actual and command position whenever the axis reaches or crosses zero. Thus command and actual position values shall never be outside the range of 0 and the Cyclic Unwind Position. To avoid accumulated error due to round-off with irrational conversion constants, the unwind value is represented as an integer number of Motion Counts per Cycle.	Motion Counts / Unwind Cycle

7.3.3.2 Semantics

7.3.3.2.1 Attribute No. 77 and 78 – Motion Unit/Resolution

The Motion Resolution attribute determines how many Motion Counts there are in a Motion Unit. A Motion Count is the fundamental unit of displacement used by the Motion Planner and a Motion Unit is the standard engineering unit of measure for motion displacement. Motion Units may be configured as Revs, inches, or millimeters depending on the specific application. In general, the Motion Resolution value may be configured in Motion Counts per Motion Unit independent of the resolution of the feedback device(s) used. The drive's scaling function takes care of scaling between Feedback Counts and Motion Counts. Providing a configurable Motion Resolution value is particularly useful for addressing Fractional Unwind applications where it is necessary to have an integer number of Motion Counts per Unwind Cycle.

Valid Motion Unit attribute selections are determined by the Feedback Mode, Load Type, and Linear Actuator Unit (Lead Unit or Diameter Unit) values according to Table 40.

Table 40 – Motion Unit selection rules

Feedback mode	Load type	Linear actuator unit	Motion unit
No Feedback	Direct Rotary	-	Motor r/s
No Feedback	Rotary Transmission	-	Load r/s
No Feedback	Linear Actuator	mm/r or mm	Load m/s
No Feedback	Linear Actuator	inch/r or inch	Load inch/s
Master Feedback	Direct Rotary	-	Feedback Rev
Master Feedback	Direct Linear	-	Feedback mm or Feedback inch
Master Feedback	Rotary Transmission	-	Load Rev
Master Feedback	Linear Actuator	mm/r or mm	Load mm
Master Feedback	Linear Actuator	inch/r or inch	Load inch
Motor Feedback	Direct Rotary	-	Motor Rev
Motor Feedback	Direct Linear	-	Motor mm or Motor inch
Motor Feedback	Rotary Transmission	-	Load Rev
Motor Feedback	Linear Actuator	mm/r or mm	Load mm
Motor Feedback	Linear Actuator	inch/r or inch	Load inch
Load or Dual Feedback	Direct Rotary	-	Load Rev
Load or Dual Feedback	Direct Linear	-	Load mm or Motor inch
Load or Dual Feedback	Rotary Transmission	-	Load Rev
Load or Dual Feedback	Linear Actuator	mm/r or mm	Load mm
Load or Dual Feedback	Linear Actuator	inch/r or inch	Load inch

7.3.3.2.2 Attribute No. 79 – Motion Polarity

When Motion Scaling Configuration is set for Drive Scaling, Motion Polarity can be used to switch the directional sense of the motion control system. A Normal setting leaves the sign of the motion control command and actual signal values unchanged from their values in the drive control structure. An Inverted setting flips the sign of the command signal values to the drive control structure and flips the sign of the actual signal values coming from the drive control structure. Motion Polarity can therefore be used to adjust the sense of positive direction of the motion control system to agree with the positive direction on the machine. When the Motion Scaling Configuration is set to Drive Scaling, the Motion Polarity inversion is performed between the CIP Motion Connection interface and the drive control structure. When the

Motion Scaling Configuration is set to Controller Scaling, the Motion Polarity inversion is performed exclusively by the controller. To maintain directional consistency, the signs of all Signal Attribute values read from the drive control structure or being written to the drive control structure are determined by Motion Polarity. A comprehensive list of these Signal Attributes is defined in Table 41.

Table 41 – Signal attributes affected by Motion Polarity

Attr. ID	Access rule	Signal attribute name
1402 + o	Get	Feedback n Position
1403 + o	Get	Feedback n Velocity
1404 + o	Get	Feedback n Acceleration
62	Get	Registration 1 Positive Edge Position
63	Get	Registration 1 Negative Edge Position
64	Get	Registration 2 Positive Edge Position
65	Get	Registration 2 Negative Edge Position
70	Get	Home Event Position
360	Set*	Controller Position Command – Integer
361	Set*	Controller Position Command – Float
362	Set*	Controller Velocity Command
363	Set*	Controller Acceleration Command
364	Set*	Controller Torque Command
365	Get	Fine Command Position
366	Get	Fine Command Velocity
367	Get	Fine Command Acceleration
370	Set	Skip Speed 1
371	Set	Skip Speed 2
372	Set	Skip Speed 3
430	Get	Position Command
431	Set*	Position Trim
432	Get	Position Reference
433	Get	Velocity Feedforward Command
434	Get	Position Feedback
436	Get	Position Error
437	Get	Position Integrator Output
438	Get	Position Loop Output
450	Get	Velocity Command
451	Set*	Velocity Trim
452	Get	Acceleration Feedforward Command
453	Get	Velocity Reference
454	Get	Velocity Feedback
455	Get	Velocity Error
456	Get	Velocity Integrator Output
457	Get	Velocity Loop Output
480	Get	Acceleration Command
481	Set*	Acceleration Trim
482	Get	Acceleration Reference

Attr. ID	Access rule	Signal attribute name
483	Get	Acceleration Feedback
490	Get	Torque Command
491	Set*	Torque Trim
492	Get	Torque Reference
493	Get	Torque Reference – Filtered
494	Get	Torque Reference – Limited
520	Get	Iq Current Command
521	Get	Operative Current Limit
523	Get	Motor Electrical Angle
524	Get	Iq Current Reference
525	Get	Id Current Reference
527	Get	Iq Current Error
528	Get	Id Current Error
529	Get	Iq Current Feedback
530	Get	Id Current Feedback
531	Get	Vq Decoupling
532	Get	Vd Decoupling
533	Get	Vq Voltage Output
534	Get	Vd Voltage Output
535	Get	U Voltage Output
536	Get	V Voltage Output
537	Get	W Voltage Output
538	Get	U Current Feedback
539	Get	V Current Feedback
540	Get	W Current Feedback
541	Get	U Current Offset
542	Get	V Current Offset
543	Get	W Current Offset
565	Get	Slip Compensation
600	Get	Output Frequency
601	Get	Output Current
602	Get	Output Voltage
603	Get	Output Power

Motion Polarity can also have an impact on directional position, velocity, acceleration, and torque limit attributes. When the Motion Scaling Configuration is set to Drive Scaling, inverting Motion Polarity requires that positive and negative position, velocity, acceleration, and torque limit values be swapped between the CIP Motion Connection interface and the drive's internal control structure. When the Motion Scaling Configuration is set to Controller Scaling, inverting Motion Polarity requires that positive and negative position, velocity, acceleration, and torque limit attribute values in Motion Control Axis Object be swapped with the corresponding attributes in the Motion Device Axis Object. For example the Velocity Limit – Positive value in the controller would be mapped to the Velocity Limit – Negative value in the drive device. A comprehensive list of these Directional Limit Attributes is defined in Table 42.

Table 42 – Directional Limit attributes affected by Motion Polarity

Attr. ID	Access rule	Signal attribute name
374	Set	Ramp Velocity – Positive
375	Set	Ramp Velocity – Negative
376	Set	Ramp Acceleration
377	Set	Ramp Deceleration
448	Set	Position Limit – Positive
449	Set	Position Limit – Negative
473	Set	Velocity Limit – Positive
474	Set	Velocity Limit – Negative
485	Set	Acceleration Limit
486	Set	Deceleration Limit
504	Set	Torque Limit – Positive
505	Set	Torque Limit – Negative

7.3.4 Connection Data attributes

7.3.4.1 General

The attribute table in Table 43 contains connection data related attributes associated with a Motion Device Axis Object instance. These attributes are elements contained in the header of the CIP Motion Connection data structure that govern the format and interpretation of the connection data.

Table 43 – Connection Data attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
89	Optional – All	Set*		Control Status	BYTE	Bit mapped field that contains status bits from the controller. See semantics in 7.3.4.2.1	Bit Field: 0 = Configuration Complete 1 = (reserved) 2 = Converter Bus Up 3 = Converter Bus Unload 4 = Converter AC Power Loss 5 = (reserved) 6 = (reserved) 7 = (reserved)
90	Required – All	Set*		Actual Data Set	BYTE	Bit mapped field that determines what actual feedback data values are being transmitted to the controller in the Device-to-Controller Connection. See semantics in 7.3.4.2.1.	Bit Field: 0 = Position Feedback (R/C) 1 = Velocity Feedback (O/C) 2 = Accel Feedback (O) 3 = (reserved) 4 = (reserved) 5 = (reserved) 6 = Unwind Cycle Count ^a 7 = Position Displacement ^a
91	Required – D	Set*		Command Data Set	BYTE	Bit mapped field that determines what command data values are being updated by the controller. See semantics in 7.3.4.2.3.	Bit Field: 0 = Position Command (R/P) 1 = Velocity Command (R/V) 2 = Accel Command (O) 3 = Torque Command (R/T) 4 = (reserved) 5 = (reserved) 6 = Unwind Cycle Count ^a 7 = Position Displacement ^a
92	Required – PVT Optional – F	Set*		Command Control	BYTE	Controls the behavior of cyclic command data, including the format of the command position data and the interpolation or extrapolation method applied to position, velocity, and acceleration command data from the motion planner based on the associated time stamp.	See semantics in 7.3.4.2.4
93	Required – P Optional – FVT	Get		Command Target Time	LINT	Time Stamp associated with command data from the motion planner and used by the command fine interpolators. The Command Target Time is the sum of the Controller Time Stamp, the Device Time Stamp value, and the Command Target Update associated with this axis instance. The time stamp format is absolute and follows the CIP Sync standard with 0 corresponding to 1970-01-01.	Nanoseconds (CIP Sync absolute)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
94	Required – All	Set*		Status Data Set	BYTE	Bit mapped field that determines what status data values are to be transmitted to the controller via the Device-to-Controller Connection.	See semantics in 7.3.4.2.5
95	Required – Scaling E	Set*		Registration Data Set	BYTE	Bit mapped field that determines what Registration Event Data values are being updated by the controller over the Controller-to-Device Connection. Windowed Registration functionality is only applicable when Scaling Configuration is set to Drive Scaling. The format of the Registration Event Data is defined in 7.3.4.2.6	Bit Field: 0 = Reg 1 Pos Edge Window Position (O) 1 = Reg 1 Neg Edge Window Position (O) 2 = Reg 1 Pos Edge Window Position (O) 3 = Reg 1 Neg Edge Window Position (O) 4 = (reserved) 5 = (reserved) 6 = (reserved) 7 = (reserved)
96	Required – Scaling E	Set*		Home Data Set	BYTE	Bit mapped field that determines what Home Event Data values are being updated by the controller over the Controller-to-Device Connection. The format of the Home Event Data is defined in 7.3.4.2.6.	Bit Field: 0 = Home Torque Threshold (O) 1 = Home Torque Time (O) 2 = (reserved) 3 = (reserved) 4 = (reserved) 5 = (reserved) 6 = (reserved) 7 = (reserved)
97	Required – Scaling E	Set*		Watch Data Set	BYTE	Bit mapped field that determines what Watch Event Data values are being updated by the controller over the Controller-to-Device Connection. Watch Position functionality is only applicable when Scaling Configuration is set to Drive Scaling. The format of the Watch Event Data is defined in 7.3.4.2.6.	Bit Field: 0 = Watch 1 Position (R) 1 = Watch 2 Position (O) 2 = (reserved) 3 = (reserved) 4 = (reserved) 5 = (reserved) 6 = (reserved) 7 = (reserved)
^a Required in E with Drive Scaling.							

7.3.4.2 Semantics

7.3.4.2.1 Attribute No. 89 – Control Status

The individual bits in the Control Status Bit Field are used as indicated below.

Configuration Complete: This bit is set when the controller has completed configuration of all axis instance attributes during Initialization phase.

Converter Bus Up: This bit, when applicable, is set by the controller when all associated Converters, or CIP Motion drives with integral converters, or CIP Motion drives with external non-CIP converters, supplying DC Bus power to this device have indicated to the controller that DC Bus voltage has reached an operational level as indicated by the Converter(s) setting of the Axis Status bit, Bus Up. This bit is only applicable to drives that support DC Bus Sharing functionality and the ability to qualify the DC Bus Up status of the drive based on this bit.

Converter Bus Unload: This bit, when applicable, is set by the controller when an associated Converter, or CIP Motion drive containing an integral converter, or CIP Motion drive connected to an external non-CIP converter, supplying DC Bus power to this device has requested that this device stops drawing DC Bus power. When the Converter Bus Unload bit is set, the device shall generate a Bus Power Sharing exception if the device's power structure is enabled and drawing DC Bus power. This bit is only applicable to drives that support DC Bus Sharing functionality and have the ability to generate a Bus Power Sharing exception based on this bit.

Converter AC Power Loss: This bit, when applicable, is set by the controller when an associated Converter, or CIP Motion drive containing an integral converter, or CIP Motion drive connected to an external non-CIP converter, has detected a loss of AC input power. When the Converter AC Power Loss bit is set, the device shall generate a Converter AC Power Loss exception if the device's power structure is enabled. This bit is cleared when the controller has determined that all associated Converter(s) supplying DC Bus power to this device have sufficient AC input power for converter operation. This bit is only applicable to drives that support DC Bus Sharing functionality and have the ability to generate a Converter AC Power Loss exception based on this bit.

7.3.4.2.2 Attribute No. 90 – Actual Data Set

Generally, the Actual Data Set value is determined by the operative Control Mode as defined in Table 44.

Table 44 – Actual Data Set value determination

Bit	Actual Data	N	F	P	V	T
0	Position	X		X	() ^a	() ^a
1	Velocity		X		() ^a	() ^a
2	Acceleration					
3	Torque					X
^a Position Feedback is selected when feedback device is present, Velocity Feedback is selected when configured for sensorless or encoderless operation.						

7.3.4.2.3 Attribute No. 91 – Command Data Set

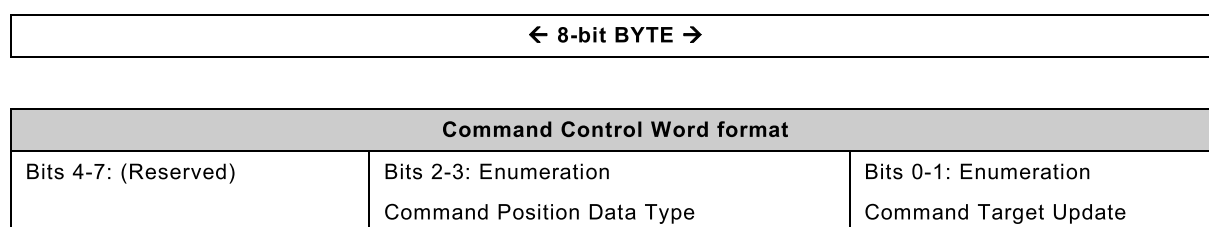
Generally, the Command Data Set value is determined by the operative Control Mode as defined in Table 45.

Table 45 – Command Data Set value determination

Bit	Cmd Data	N	F	P	V	T
0	Position			() ^a		
1	Velocity		X		X	
2	Acceleration					
3	Torque					X
^a Controller sends either Float or Integer Command Position data, but never both. Integer is only sent when drive does not support double precision floating point position command data.						

7.3.4.2.4 Attribute No. 92 – Command Control

The Command Control attribute governs the interpolation/extrapolation method applied to position, velocity, and acceleration command data from the motion planner based on the associated time stamp, as specified in Figure 61.



IEC

Figure 61 – Command Control Word field

The first two bits of this 8-bit attribute represent the Command Target Update that determines the number of Update Periods (UP) added to the command data Time Stamp to determine the absolute System Time that the command data value is targeted for, i.e. the Command Target Time (see Table 46). For more details on interpolation and extrapolation control, see 7.6.6.2.

Table 46 – Command Target Update enumeration definition

Enum.	Req./Opt.	Name	Description
0	R/P	Immediate	A Command Target Update of 0 implies that the command data is to be applied to the device control structure immediately.
1	R/P	Extrapolate (+1 UP)	A Command Target Update of 1 implies that extrapolation is to be used to apply the command data to the device control structure based on adding 1 Update Period to the command Time Stamp.
2	R/P	Interpolate (+2 UP)	A Command Target Update of 2 implies that fine interpolation is to be used to apply the command data based on adding 2 Update Periods to the command Time Stamp.
3		(Reserved)	-

Table 47 specifies the values for the Command Position Data Type field.

Table 47 – Command Position Data Type enumeration definition

Enum.	Req./Opt.	Name	Description
0	O/P ^a	LREAL (64-bit Float)	A Command Position Data Type of 0 corresponds to the LREAL, or double precision floating point data type. Support for this data type is preferred in the implementation since it provides fraction command count information to the drive resulting in smoother motion.
1	O/P ^a	DINT (32-bit Integer)	A Command Position Data Type of 1 corresponds to the DINT, or 32-bit signed integer data type. This data type is applicable to simple drive devices that either do not require the precision of a floating point command position value or do not have sufficient hardware support for double precision floating point math.
3-4		(Reserved)	-
^a If axis is configured for Position Control Mode operation, at least one of the defined Command Position Data Types shall be supported by the device.			

7.3.4.2.5 Attribute No. 94 – Status Data Set

The Status Data Set attribute is an 8-bit collection of bits indicating which Status attributes are to be transmitted to the controller over the Device-to-Controller Connection (see Table 48). Status data appears in the same order as the bit numbers, so Axis Fault Type/Code data would appear before Axis Fault Time Stamp data in the Status Data Set structure. Multiple attributes comprising a selected Status Data Element are transmitted in the order listed from top to bottom, so the Axis Fault Type is transmitted before Axis Fault Code.

Table 48 – Status Data Set bit definitions

Bit	Status Data Element produced	Data type
0	Axis Fault Type	USINT
	Axis Fault Code	USINT
	Axis Fault Sub Code	USINT
	Axis Fault Action	USINT
	Axis Fault Time Stamp	LINT
1	Axis Alarm Type	USINT
	Axis Alarm Code	USINT
	Axis Alarm Sub Code	USINT
	Axis Alarm State	USINT
	Axis Alarm Time Stamp	LINT
2	Axis Status	DWORD
	Axis Status – Mfg	DWORD
3	Axis I/O Status	DWORD
	Axis I/O Status – Mfg	DWORD
4	Axis Safety Status	DWORD
	Axis Safety Status – Mfg	DWORD
	Axis Safety State	USINT
	Pad[3]	USINT[3]
5	(Reserved)	(Reserved)
6	(Vendor Specific)	(Vendor Specific)
7	(Vendor Specific)	(Vendor Specific)

7.3.4.2.6 Attribute No. 95 to 97 – Registration, Home and Watch Event Data format

The format of the Registration Event Data is defined Table 49.

Table 49 – Registration Event Data format

Bit	Reg Data	Format
0	Reg 1 Pos Window	DINT (max) DINT (min)
1	Reg 1 Neg Window	DINT (max) DINT (min)
2	Reg 2 Pos Window	DINT (max) DINT (min)
3	Reg 2 Neg Window	DINT (max) DINT (min)

The format of the Home Event Data is defined in Table 50.

Table 50 – Home Event Data format

Bit	Home Data	Format
0	Home Torque Threshold	REAL
1	Home Torque Time	REAL

The format of the Watch Event Data is defined in Table 51.

Table 51 – Watch Event Data format

Bit	Home Data	Format
0	Watch 1 Position	DINT
1	Watch 2 Position	DINT

7.3.5 Motor attributes

7.3.5.1 General

The following attribute tables contain motor configuration attributes associated with a Motion Device Axis Object instance that apply to various motor technologies. These motor technologies include three-phase motor rotary, linear, permanent magnet and induction motors. Motor attributes are, therefore, organised according to the various motor types. The Need in Implementation category for an attribute is based on the context of the Motor Type and, thus, to the context of the table in which the attribute appears. Where needed, Standard vs. Optional can be further differentiated by abbreviations for PM (Permanent Magnet) and IM (Induction Motors). Within the PM Motor family, there is further differentiation for SPM (Surface PM) and IPM (Interior PM) motors. It is implied that these motor attributes are applicable to all drive modes, F, P, V and T, but not applicable for N, or No Control axis configurations where there is no active motor control function.

The goal of 7.3.5 is to define the minimal set of required attributes to support CIP Motion device interchangeability. This guarantees that there is sufficient parametric data provided by the controller for any CIP Motion compliant device, i.e. drive, to effectively control a given motor.

For induction motors, the Motion Device Axis Object leverages the IEEE Std 112 recommended phase-neutral equivalent circuit motor model based on “Wye” configuration (see Figure 62). Reactance values, X , are related to their corresponding Inductance values, L , by $X = \omega \times L$, where ω is the rated frequency of the motor. The prime notation, for example X_2' , R_2' , indicates that the actual rotor component values X_2 , and R_2 are referenced to the stator side of the stator-to-rotor winding ratio.

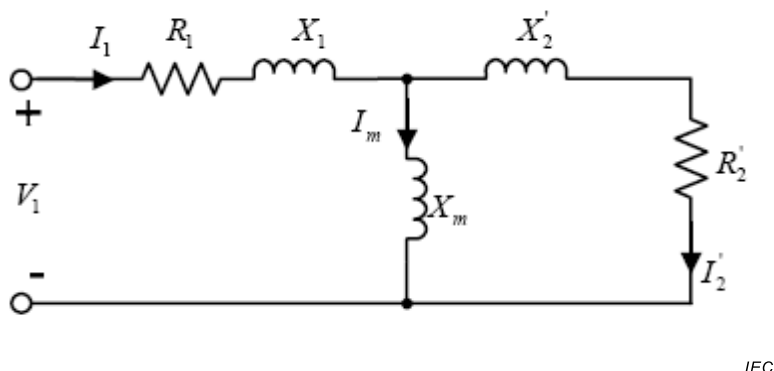


Figure 62 – IEEE Std 112 per phase motor model

For Permanent Magnet motors, the Motion Device Axis Object assumes all motor parameters are defined in the context of a phase-to-phase motor model.

7.3.5.2 General motor attributes

The attribute tables in Table 52 and Table 53 contain general motor attributes that apply to all motor technologies.

Table 52 – General Motor Info attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1310	Optional	Get		Motor Catalog Number	SHORT STRING	A 32-character string that specifies the motor catalog number. If the Motor Catalog Number is not available, the drive sets this attribute to a Null string.	For example MPL-B310F
1311	Optional	Get		Motor Serial Number	SHORT STRING	A 16-character string that specifies the serial number of the motor. If the Motor Serial Number is not available, the drive sets this attribute to a Null string.	For example 0012003400560078
1312	Optional (Motor NV)	Get		Motor Date Code	SHORT STRING	A 16-character string that specifies the manufacturing date of the motor. If the Motor Date Code is not available, the drive sets this attribute to a Null string.	For example Jan-01-2005

Table 53 – General Motor Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1313	Required	Set		Motor Data Source	USINT	An enumeration that specifies the source of motor data for the drive. Controller NV implies that the motor attributes are derived from the controller's non-volatile memory during the drive configuration process. Drive NV implies that the motor attributes are derived directly from the drive's non-volatile memory. In this mode, only a minimal set of motor and motor feedback (Feedback 1) are required to configure the drive. Motor NV implies that the motor attributes are derived from non-volatile memory of a motor-mounted smart feedback device equipped with a serial interface. Again, in this mode, only a minimal set of motor and motor feedback (Feedback 1) are required to configure the drive. Motor and motor feedback attributes sent to the drive device in Drive NV or Motor NV are merely to confirm that the controller and the drive have the agreement on the values of attributes critical to scaling operation. If the NV attribute values in the drive differ from the values set by the controller, the drive shall reject the set service with General Status indicating an Invalid Attribute Value. The current list of motor and motor feedback attributes sent to the drive in the NV modes are as follows: 1. Motor Unit 2. Feedback 1 Unit 3. Feedback 1 Type 4. Feedback 1 Startup Method 5. Feedback 1 Cycle Resolution 6. Feedback 1 Cycle Interpolation 7. Feedback 1 Turns 8. Feedback 1 Length	Bits 0 to 3: Enum 0 = Controller NV (R) 1 = Drive NV (O) 2 = Motor NV (O) 3 to 127 = (reserved) 128 to 255 = (vendor specific)
1314	Required	Set		Motor Device Code	UDINT	A unique number assigned to a motor catalogue. This value is used to ensure that the motor and integral motor mounted feedback device configuration data delivered from the controller matches the actual motor and feedback data connected to the drive. This comparison is only valid in the case where the Motor Data Source is Controller NV and the motor is equipped with a smart feedback device that positively identifies the motor. If the codes do not match, a negative acknowledge is given by the drive. Motor Device Codes are assigned by the motor manufacturer. A value of 0 for the Motor Device Code shall be accepted by the drive without comparison.	

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1315	Required	Set		Motor Type	USINT	An enumeration that specifies the motor technology. If Motor Data Source is Motor NV or Drive NV, the Motor Type may not be known to the controller but is known by the drive, so the drive can operate in this case without the controller specifying the Motor Type. In this mode, Motor Type shall not be sent by the controller. If received, it shall be rejected with General Status indicating an Invalid Attribute Value. If Motor Data Source is Datasheet or Database, an unspecified Motor Type, when received by the drive device during configuration, indicates that the motor configuration has not been defined and therefore shall be rejected with General Status indicating an Invalid Attribute Value.	Enumeration: 0 = not specified (R) 1 = rotary permanent magnet (O) 2 = rotary induction (O) 3 = linear permanent magnet (O) 4 = linear induction (O) 5 to 127 = (reserved) 128 to 255 = (vendor specific)
1316	Required	Set		Motor Unit	USINT	Unit of measure for motor displacement. This attribute is also used for encoderless or sensorless operation since the Feedback Unit in that case is not applicable.	Enumeration: 0 = Rev (R for rotary motors) 1 = Meter (R for linear motors) 2 to 127 = (reserved) 128 to 255 = (vendor specific)
1317	Optional	Set		Motor Polarity	USINT	An enumerated value used to establish the direction of motor motion when the windings are phased according to factory specification. Normal polarity is defined as the direction of motor travel when the ABC motor winding leads are hooked up according to the drives published specifications. Inverted polarity effectively switches the ABC phasing to ACB so that the motor moves in the opposite direction in response to a positive drive output. This attribute can be used to make the direction of travel agree with the user's definition of positive travel and can be used in conjunction with the Feedback Polarity bit to provide negative feedback, when closed loop control is required. When commutating a PM motor it is imperative that the commutation phase sequencing match the motor phase sequencing to properly control the motor.	Enumeration: 0 = Normal Polarity 1 = Inverted Polarity 2 to 255 = (reserved)
1318	Required	Set		Motor Rated Voltage	REAL	A float that specifies the nameplate AC voltage rating of the motor. This represents the phase-to-phase voltage applied to the motor to reach rated speed at full load.	Volts (RMS)
1319	Required	Set		Motor Rated Continuous Current	REAL	A float that specifies the nameplate AC continuous current rating of the motor. This represents the current applied to the motor under full load conditions at rated speed and voltage.	Amperes (RMS)
1320	Required – PM Optional – IM	Set		Motor Rated Peak Current	REAL	A float that specifies the peak or intermittent current rating of the motor. The peak current rating of the motor is often determined by either the thermal constraints of the stator winding or the saturation limits of PM motor magnetic material.	Amperes (RMS)
1321	Required – IM Optional – PM	Set		Motor Rated Output Power	REAL	A float that specifies the nameplate rated output power rating of the motor. This represents the power output of motor under full load conditions at rated current, speed and voltage.	Kilowatts

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1322	Optional	Set		Motor Overload Limit	REAL	A float that specifies the maximum thermal overload limit for the motor. This value is typically 100 %, corresponding to the power dissipated when operating at the continuous current rating of the motor, but can be significantly higher if cooling options are applied. For induction motors, this attribute is related to the Service Factor of the motor. When the Motor Capacity attribute value associated with the motor thermal model exceeds the Motor Overload Limit, the drive can optionally trigger a predetermined Motor Overload action. The Motor Overload Limit can also be used by the drive to determine the absolute thermal capacity limit of the motor, i.e. the Motor Thermal Overload Factory Limit, that if exceeded, generates a Motor Thermal Overload FL exception.	% Motor Rated
1323	Optional	Set		Motor Integral Thermal Switch	BOOL	A boolean that specifies if the motor has an integral thermal switch.	0 = No 1 = Yes
1324	Optional	Set		Motor Max Winding Temperature	REAL	A float that specifies the maximum winding temperature of the motor.	°C
1325	Optional	Set		Motor Winding-to-Ambient Capacitance	REAL	A float that specifies the winding-to-ambient thermal capacitance.	J/°C
1326	Optional	Set		Motor Winding-to-Ambient Resistance	REAL	A float that specifies the winding-to-ambient thermal resistance.	°C/W

7.3.5.3 General PM motor attributes

The attribute table in Table 54 contains motor configuration attributes that apply to Permanent Magnet motor types in general.

Table 54 – General PM Motor Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1327	Required	Set		PM Motor Resistance	REAL	A float that specifies the phase-to-phase, resistance of a permanent magnet motor.	Ohms
1328	Required (SPM only)	Set		PM Motor Inductance	REAL	A float that specifies the phase-to-phase, inductance of a permanent magnet motor.	Henries
1353	Required (IPM only)	Set		PM Motor Lq Inductance	REAL	A float that specifies the phase-to-phase, q-axis, inductance of an interior permanent magnet motor.	Henries
1354	Required (IPM only)	Set		PM Motor Ld Inductance	REAL	A float that specifies the phase-to-phase, d-axis, inductance of an interior permanent magnet motor.	Henries

7.3.5.4 General rotary motor attributes

The attribute table in Table 55 contains motor configuration attributes that apply specifically to rotary motor types.

Table 55 – General Rotary Motor Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1329	Required	Set		Rotary Motor Poles	UINT	An integer that specifies the number of poles per revolution for rotary motors. This value is always an even number, as poles always exist in pairs.	
1330	Optional	Set		Rotary Motor Inertia	REAL	A float that specifies the unloaded inertia of a rotary motor.	kg•m ²
1331	Required	Set		Rotary Motor Rated Speed	REAL	A float that specifies the nameplate rated speed of a rotary motor. For PM motors, this is generally specified at rated voltage based on either rated current, rated torque, or rated power. For induction motors this value is the speed of the motor driven at rated frequency under rated torque load. This value is synonymous with the term base speed.	r/min
1332	Optional	Set		Rotary Motor Max Speed	REAL	A float that specifies the absolute maximum operating speed of a rotary motor in units of r/min. This speed may be determined by the limitations of the motor or by limitations of the mechanical system. Specifically, this value can represent the maximum safe operating speed, maximum continuous “no-load” speed, maximum continuous encoder speed, or maximum continuous bearing speed of the motor. This value can be used by the drive to determine the Rotary Motor Overspeed Factory Limit.	r/min
1333	Optional	Set		Rotary Motor Damping Coefficient	REAL	A float that specifies the damping, or viscous friction, associated with a rotary motor.	N•m/(rad/s)

7.3.5.5 General linear motor attributes

The attribute table in Table 56 contains motor configuration attributes that apply specifically to linear motor types.

Table 56 – General Linear Motor Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1334	Required	Set		Linear Motor Pole Pitch	REAL	A float that specifies the pole pitch of a linear motor in units of meters, and is equivalent to the electrical cycle length.	m
1335	Required	Set		Linear Motor Rated Speed	REAL	A float that specifies the nameplate rated speed of a linear motor. For PM motors, this is generally specified at rated voltage based on either rated current, rated force, or rated power. For induction motors this value is the speed of the motor driven at rated frequency under rated force load. This value is synonymous with the term base speed.	m/s
1336	Optional	Set		Linear Motor Mass	REAL	A float that specifies the unloaded moving mass of a linear motor.	kg
1337	Optional	Set		Linear Motor Max Speed	REAL	A float that specifies the absolute maximum operating speed of a linear motor in units of m/s. This speed may be determined by the limitations of the motor or by limitations of the mechanical system. Specifically, this value can represent the maximum safe operating speed, maximum continuous “no-load” speed, maximum continuous encoder speed, or maximum continuous bearing speed of the motor. This value can be used by the drive to determine the Linear Motor Overspeed Factory Limit.	m/s
1338	Optional	Set		Linear Motor Damping Coefficient	REAL	A float that specifies the damping, or viscous friction, associated with a linear motor.	N/(m/s)

7.3.5.6 Rotary PM motor attributes

The attribute table in Table 57 contains motor configuration attributes that apply specifically to rotary motor types.

Table 57 – Rotary PM Motor Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1339	Optional	Set		PM Motor Rated Torque	REAL	A float that specifies the nameplate continuous torque rating of a rotary permanent magnet motor.	N·m
1340	Optional	Set		PM Motor Torque Constant	REAL	A float that specifies the torque constant of a rotary permanent magnet motor in newton meters per RMS ampere.	N·m/ A (RMS)
1341	Required	Set		PM Motor Rotary Voltage Constant	REAL	A float that specifies the voltage, or back-EMF, constant of a rotary permanent magnet motor in phase-to-phase RMS Volts per kr/min. If the optional PM Motor Torque Constant, K_t, is not explicitly supported in the implementation, the value may be computed from the PM Motor Rotary Voltage Constant, K_e, according to the following equation: $K_t \text{ (N·m/A)} = 0,016\ 54 \times K_e \text{ (V/kr/min)}$	Volts (RMS)/ kr/min

7.3.5.7 Linear PM motor attributes

The attribute table in Table 58 contains motor configuration attributes that apply specifically to linear PM motor types.

Table 58 – Linear PM Motor Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1342	Optional	Set		PM Motor Rated Force	REAL	A float that specifies the nameplate continuous force rating of a linear permanent magnet motor.	N
1343	Optional	Set		PM Motor Force Constant	REAL	A float that specifies the force constant of a linear permanent magnet motor in Newtons per RMS ampere.	N/A (RMS)
1344	Required	Set		PM Motor Linear Voltage Constant	REAL	A float that specifies the voltage, or back-EMF, constant of a linear permanent magnet motor in phase-to-phase RMS Volts per m/s. If the optional PM Motor Force Constant, K_f, is not explicitly supported in the implementation, the value may be computed from the PM Motor Linear Voltage Constant, K_e, according to the following equation: $K_f \text{ (N/A)} = 1,732 \times K_e \text{ (V/(m/s))}$	Volts (RMS) / (m/s)

7.3.5.8 Induction motor attributes

The attribute table in Table 59 contains motor configuration attributes that apply specifically to induction motor types.

Table 59 – Induction Motor Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1345	Required	Set		Induction Motor Rated Frequency	REAL	A float that specifies the nameplate frequency rating of an induction motor.	Hertz
1346	Required	Set		Induction Motor Flux Current	REAL	Id Current Reference that required to generate full motor flux. This value is closely approximated by the No Load Motor Rated Current commonly found in Induction Motor data sheets.	Amperes (RMS)
1347	Required	Set		Induction Motor Stator Resistance	REAL	A float that specifies the Y circuit, phase-neutral, winding resistance of the stator as shown as R_1 in the IEEE Std 112 motor model.	Ohms
1348	Required	Set		Induction Motor Stator Leakage Reactance	REAL	A float that specifies the Y circuit, phase-neutral, leakage reactance of the stator winding, at rated frequency, as shown as X_1 in the IEEE Std 112 motor model.	Ohms
1349	Optional ^a	Set		Induction Motor Magnetization Reactance	REAL	A float that specifies the Y circuit, phase-neutral, magnetizing reactance of the motor, at rated frequency, as shown as X_m in the IEEE Std 112 motor model.	Ohms
1350	Optional ^a	Set		Induction Motor Rotor Resistance	REAL	A float that specifies the phase-neutral equivalent stator-referenced winding resistance of the rotor as shown as R_2' in the IEEE Std 112 motor model.	Ohms
1351	Required	Set		Induction Motor Rotor Leakage Reactance	REAL	A float that specifies the Y circuit, phase-neutral, equivalent stator-referenced leakage inductance of the rotor winding, at rated frequency, as shown as X_2' in the IEEE Std 112 motor model.	Ohms
1352	Optional – D	Set		Induction Motor Rated Slip Speed	REAL	Represents the amount of slip at motor rated current (full load) and motor rated frequency.	r/min (rotary motor) m/s (linear motor)
^a These parameters have a strong motor temperature component that some drives circumvent through various adaption or compensation techniques.							

7.3.5.9 Load transmission and actuator attributes

The attribute table in Table 60 contains motor configuration attributes that apply specifically to rotary transmission and linear actuator mechanisms associated with the axis.

Table 60 – Load Transmission and Actuator Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1370	Optional – Scaling	Set		Load Type	USINT	Enumerated attribute used to determine how the load is mechanically linked to the motor. “Direct” enumerations indicate that the motor is directly coupled to the load. “Rotary” enumerations indicate that the load is rotating and load dynamics are measured using a rotary system of units. “Linear” enumerations indicate that the load is moving linearly and load dynamics are measured using a linear system of units.	Enumeration: 0 = Direct Rotary 1 = Direct Linear 2 = Rotary Transmission 3 = Linear Actuator 4 to 255 = reserved
1371	Optional – Scaling	Set		Transmission Ratio Input	DINT	Integer number of input shaft revolutions per transmission cycle associated with the rotary transmission.	Input Shaft Revs
1372	Optional – Scaling	Set		Transmission Ratio Output	DINT	Integer number of output shaft revolutions per transmission associated with the rotary transmission.	Output Shaft Revs
1373	Optional – Scaling	Set		Actuator Type	USINT	This enumerated value indicates the type of mechanism used for linear actuation.	Enumeration: 0 = None (R) 1 = Screw (O) 2 = Belt and Pulley (O) 3 = Chain and Sprocket (O) 4 = Rack and Pinion (O) 5 to 255 = reserved
1374	Optional – Scaling NP only	Set		Actuator Lead	REAL	This floating point value represents the lead or pitch of a screw actuator that is a measure of the linear movement of the screw mechanism per revolution of the screw shaft.	Actuator Lead Units
1375	Optional – Scaling NP only	Set		Actuator Lead Unit	USINT	This enumerated value indicates the units of the Actuator Lead attribute.	Enumeration: 0 = mm/r 1 = inch/r 2 to 255 = reserved
1376	Optional – Scaling	Set		Actuator Diameter	REAL	This floating point value represents the diameter of the pulley, sprocket, or pinion used to convert rotary motion into tangential linear displacement of the load. The Actuator Diameter is internally converted to circumference of the pulley, sprocket, or pinion to determine the amount of tangential displacement per revolution.	Actuator Diameter Units
1377	Optional-Scaling	Set		Actuator Diameter Unit	USINT	This enumerated value indicates the units of the Actuator Diameter attribute.	Enumeration: 0 = mm 1 = inch 2 to 255 = reserved

7.3.6 Feedback attributes

7.3.6.1 General

The attribute tables in Table 64, Table 65 and Table 66 contain all position feedback related attributes associated with a Motion Device Axis Object instance that apply to various feedback device and feedback interface technologies.

NOTE These feedback interface technologies include Digital AqB (digital A quad B signals), Sine/Cosine (analog A quad B signals), Digital Parallel (parallel digital bit interface), SSI (Synchronous Serial Interface), LDT (Linear Displacement Transducer) and Resolver. Other modern feedback interfaces supported are Hiperface® (by Stegmann) and EnDat 2.1® & EnDat 2.2® (by Heidenhain).

The “Need in Implementation” specification for a feedback attribute is often based on the context of the Feedback Type. To facilitate this, abbreviations for the various Feedback Types are defined in Table 61.

Table 61 – Feedback Types abbreviations

Abbreviation	Feedback type
TT	Digital AqB
TP	Digital Parallel
SC	Sine/Cosine
HI	Hiperface®
E21	EnDat 2.1®
E22	EnDat 2.2®
RS	Resolver
SS	SSI
LT	LDT

The goal of 7.3.6 is to define the minimal set of required attributes to support CIP Motion device interchangeability. This guarantees that there is sufficient parametric data provided by the controller for any CIP Motion compliant drive to effectively interface to a wide range of feedback device types.

Multiple feedback device interfaces are currently defined by the Motion Device Axis Object per axis instance to serve specific control or master feedback functions. These feedback devices are accessed via their assigned logical channels, for example, Feedback 1 and Feedback 2. Each logical feedback channel is mapped to a physical feedback interface port of the device, for example Port 1, and Port 2.

Table 62 lists Logical Feedback Channel Control functions.

Table 62 – Logical Feedback Channel Control functions

Logical Feedback Channel	Motion Control function	Master Feedback function
Feedback 1	Motor Feedback & Commutation	Master Feedback 1
Feedback 2	Load-side Feedback	Master Feedback 2

When the Control Mode is set to something other than No Control, Feedback 1 is generally associated with the motor mounted feedback device while Feedback 2 is associated with the

load-side or machine mounted feedback device. Feedback 1 is always required for Permanent Magnet motor commutation.

When Control Mode is set to No Control for a Motion Device Axis Object instance, different logical feedback channels can be used as a master feedback source, for example Feedback 1, Feedback 2, etc. Generally, Feedback 1 is used.

To minimize the length of the feedback attribute tables, the letter “n” in the generic “Feedback n” attribute name is used to specify the associated feedback channel number. Valid channel numbers for open standard feedback attributes of the Motion Device Axis Object are 1, 2, 3, and 4. Attribute IDs are assigned based on the channel number. Support for feedback interface channels 1, 2, 3, and 4 are optional in the device implementation. If no feedback interface channel is present in the device, the associated set of feedback channel attributes are not applicable. However, if hardware support for any of these feedback channels is available in a given device, these attributes are clearly applicable in the implementation and shall follow the Need in Implementation rules. An implementation rule of “Req – E” or “Opt – E” indicates that the attribute is generally applicable to all Device Function Codes where the feedback channel itself is applicable, hence the “E” for "Encoder". If a specific logical feedback channel is not applicable based on the current Feedback Mode, then attributes for feedback n are not applicable; no feedback configuration attributes for that channel are set by configuration software, nor are any such attributes sent to the drive device.

Table 63 outlines these rules.

Table 63 – Logical Feedback Channel rules

Feedback Mode	Feedback 1	Feedback 2
No Feedback	No	No
Master Feedback	Yes	No ^a
Motor Feedback	Yes	No
Load Feedback	Yes ^b	Yes
Dual Feedback	Yes	Yes
^a Feedback 2 channel only needed if Feedback Master Select supports Feedback 2 channel option.		
^b Feedback 1 channel needed for commutation of PM Motors.		

Table 64, Table 65 and Table 66 specify feedback related attributes.

Table 64 – General Feedback Info attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1400 + (n-1)x50	Optional – E	Get		Feedback n Catalog Number	SHORT STRING	A 32-character string that specifies the catalog number of the device associated with Feedback n. If the Feedback Catalog Number is not available, the drive sets this attribute to a Null string.	for example SRM-50
1401 + (n-1)x50	Optional – E	Get		Feedback n Serial Number	SHORT STRING	A 16-character string that specifies the serial number of the device associated with Feedback n. If the Feedback Serial Number is not available, the drive sets this attribute to a Null string.	for example 001200340 0560078

Table 65 – General Feedback Signal attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1402 + (n-1)x50	Required – E	Get		Feedback n Position	DINT	Actual position of the axis based on Feedback n.	Feedback n Counts
1403 + (n-1)x50	Required – E	Get		Feedback n Velocity	REAL	Actual filtered velocity of the axis based on Feedback n.	Feedback n Units/s
1404 + (n-1)x50	Required – E	Get		Feedback n Acceleration	REAL	Actual filtered acceleration of the axis based on Feedback n.	Feedback n Units/s ²

Table 66 – Feedback Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
42	Required – All	Set*		Feedback Mode	BYTE	This attribute determines how the various available feedback channels are used to implement the selected Control Mode. Currently only bits 0 to 3 are used to enumerate the Feedback Mode configuration. Bit 4 to 7 are reserved for future use.	Bits 0 to 3: Feedback Mode Enumeration: 0 = No Feedback 1 = Master Feedback 2 = Motor Feedback 3 = Load Feedback 4 = Dual Feedback 5 to 7 = (Reserved) 8 to 15 = (Vendor Specific) Bits 4 to 7: Reserved See semantics in 7.3.6.2
43	Optional – N	Set		Feedback Master Select	USINT	This attribute determines what Logical channel is assigned to this axis instance when the Feedback Mode is set to Master Feedback. Default is Feedback 1.	Enumeration: 0 = (Reserved) 1 = Feedback 1 2 = Feedback 2 2 to 255 = (reserved)
44	Optional – PV	Set		Feedback Unit Ratio	REAL	Number of Feedback 1 Units per Feedback 2 Units. This value is used to convert from feedback 2 units or counts to feedback 1 units or counts when configured for dual loop or load side feedback operation. The Feedback Ratio block that applies the Feedback Unit Ratio scaling factor appears in the Position Loop and Torque Reference block diagrams.	Feedback 1 Units per Feedback 2 Units

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1410 + (n-1)×50	Optional – E	Set		Feedback n Resolution Unit	USINT	Unit of measure for feedback resolution used by Feedback n Cycle Resolution attribute. Default selection is Cycles/Unit where resolution is expressed in feedback cycles per revolution for rotary feedback devices or per meter for linear feedback devices. If Units/Cycle is selected then Feedback n Cycle Resolution shall be expressed in Nanometers/Cycle for linear feedback devices. This selection is not applicable to rotary devices. If Bits/Unit is selected then Feedback n Cycle Resolution is expressed as 2n Cycles per revolution of a rotary feedback device, where n is the number of bits in the binary position representation of the device. This selection is not applicable for linear devices.	Enumeration: 0 = Cycles/Unit (R) 1 = Units/Cycle (O) (linear only) 2 = Bits/Unit (O) (rotary only) 3 to 127 = (reserved) 128 to 255 = (vendor specific)
1411 + (n-1)×50	Required -E	Set		Feedback n Unit	USINT	Unit of measure for the designated feedback device. The Feedback Unit for Feedback 1 shall be the same as the configured Motor Unit; if the Motor Unit is set to Rev, Feedback 1 Unit shall be set to Rev; if Motor Unit is set to Meter, Feedback 1 Unit shall be set to Meter. Feedback devices with a Feedback Unit of Rev are considered “rotary” devices, while Feedback devices with a Feedback Unit of Meter are considered “linear” devices	Enumeration: 0 = Rev 1 = Meter 2 to 127 = (reserved) 128 to 255 = (vendor specific)
1412 + (n-1)×50	Optional – E	Set		Feedback n Port Select	USINT	Maps the logical Feedback Channel “n” to a physical Feedback Port ID. A Feedback Port ID is assigned to each feedback interface port of the device by the drive vendor. If this attribute is not supported by the drive, the drive vendor shall hardcode the feedback port mapping to the logical Feedback Channels for each axis instance. Supporting the Feedback n Port Select attribute allows flexibility to map the logical Feedback Channels to different Feedback Ports. Default Feedback n Port Select = 0. A value of 0 indicates that Feedback n Channel is unmapped, hence unused.	Enumeration: 0 = Unused (R) 1 to 255 = Feedback Port ID (O)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1413 + (n-1)×50	Required – E	Set		Feedback n Type	USINT	Identifies the type of feedback interface of the associated Feedback device channel. Drive support for any individual feedback type is left to the discretion of the device manufacturer. However if a specific feedback type is supported, attributes associated with that type are generally required in the implementation. The “Integrated” Feedback Type is specified for CIP Motion compliant devices with an integral feedback transducer function, e.g. a CIP Motion encoder. In the case of a motor mounted feedback device, if Motor Data Source is Motor NV or Drive NV, the Feedback 1 Type may not be known to the controller but is known by the drive, so the drive can operate in this case without specifying the Feedback 1 Type. If the optional Commutation Startup Method attribute is not supported by the device, or the Commutation Startup Method is set to From Feedback Type, the Feedback 1 Type can be used to implicitly specify the commutation startup method. For example, by selecting the Feedback 1 Type with or without UVW commutation signals the device applies the UVW commutation startup method or the Self-Sense startup method, respectively. In this case, UVW commutation signals can be derived from UVW tracks integral to the feedback device or via separate Hall sensors in the motor. All other Feedback 1 Type selections would apply the Digital commutation startup method. In the case of a motor mounted feedback device, if the Motor Data Source is Datasheet or Database, an unspecified Feedback 1 Type, when received by the drive device during configuration, indicates that the motor feedback configuration has not been defined and therefore results in a General Status error of Invalid Attribute Value.	Enumeration: 0 = Not Specified (R) 1 = Digital AqB (O) 2 = Digital AqB with UVW (O) 3 = Digital Parallel (O) 4 = Sine/Cosine (O) 5 = Sine/Cosine with UVW (O) 6 = Hiperface (O) 7 = EnDat 2.1 (O) 8 = EnDat 2.2 (O) 9 = Resolver (O) 10 = SSI (O) 11 = LDT (O) 12 = Hiperface DSL (O) 13 = BiSS (O) 14 = Integrated (O) 15 to 127 = (Reserved) 128 to 255 = (Vendor Specific)
1414 + (n-1)×50	Optional – E	Set		Feedback n Polarity	USINT	An enumerated value used to establish the direction of change in the feedback counter in response to positive motion of the associated feedback device. Normal polarity is defined as that which results in increasing feedback counts when the feedback device is hooked up and moved in the positive direction according to the devices published specifications. Inverted polarity internally switches the polarity of the feedback accumulator so that the feedback counts decrease when the feedback device moves in the positive direction. This attribute can be used to make the direction of travel agree with the user’s definition of positive travel and can be used in conjunction with the Motor Polarity bit to provide negative feedback, when this feedback channel is used for closed loop control.	Enumeration: 0 = Normal Polarity 1 = Inverted Polarity 2 to 255 = (reserved)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1415 + (n-1)×50	Required -E	Set		Feedback n Startup Method	USINT	Determine how the device applies the feedback count value during drive startup. When configured for Incremental mode, the device zeroes the feedback count accumulator at power-up. The first Actual Position value sent to the controller in the Cyclic Data Block of the Device-to-Controller connection at power-up shall be zero. This is an indication to the controller that the drive has been power-cycled and the drive axis needs to be homed to establish a machine reference position. When configured for Absolute mode, the device initializes the feedback count accumulator at power-up to the absolute feedback position value read from the feedback device. When the feedback device's absolute position range is less than the 32-bit signed integer representation of the feedback count accumulator, the absolute position is sign extended to a 32-bit signed value. While there are many Feedback Types that support Absolute startup, there are a few strictly incremental types that do not, such as Digital AqB, and Sine/Cosine. Some device vendors tie the Feedback Start-up Method to the Feedback Type selection. In these cases, an attempt by the controller to incorrectly configured Feedback Startup Method shall generate an General Status error of Invalid Attribute Value.	Enumeration: 0 = Incremental (R) 1 = Absolute (O) 2 to 255 = (reserved)
1416 + (n-1)×50	Required – E Not LT	Set		Feedback n Cycle Resolution	UDINT	Determines the resolution capability of the associated feedback device. Units for this attribute are determined by the Feedback n Resolution Unit and the rotary or linear Feedback n Unit as shown in the Semantics column. For rotary feedback devices, this value is expressed as the number of Feedback Cycles per Revolution of the device, or alternatively by the number of bits in the binary position representation of the device per Revolution. For linear feedback devices, this value represents either the number of Feedback Cycles per Meter (m), or the number of nanometers (nm) per Feedback Cycle. Feedback Cycles for a Digital AqB device represents the “line” resolution of the encoder. Cycles for a Sin/Cos device represents the sinusoidal “cycle” resolution of the encoder. Cycles for a Resolver is the “pole” count of the device. For digital serial (e.g. SSI) or parallel absolute feedback devices Cycle represent the “step” or “count” resolution of the device.	Cycles/Unit (Rotary): Feedback Cycles / Rev (Rotary) Cycles/Unit (Linear): Feedback Cycles / m Unit/Cycle (Linear): nm / Feedback Cycle (Linear) Bits/Unit (Rotary): 2ⁿ Cycles / Rev
1417 + (n-1)×50	Required – E Not LT	Set		Feedback n Cycle Interpolation	UDINT	Number of interpolated Feedback Counts per Feedback Cycle. For a Digital AqB device the device's feedback interface hardware can generally support interpolation values of 1, 2, or 4. For a Sin/Cos, Hiperface, Endat 2.1, or Resolver feedback device the number is generally much larger and determined by the interpolation capability of the device feedback interface hardware. A value of 1 024 is typical in this case. For digital serial (e.g. SSI) or parallel absolute feedback device interfaces, this value is always 1 since there is no opportunity of device-based interpolation. The effective resolution of the feedback device in Feedback Counts per Feedback Unit is determined by combination of Feedback Cycle Resolution and Feedback Cycle Interpolation attribute values.	Feedback Counts / Feedback Cycle

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1418 + (n-1)×50	Required – E (Rotary Absolute)	Set		Feedback n Turns	UDINT	Maximum number of shaft turns specified for a rotary absolute feedback device to maintain its absolute position reference. Typical rotary absolute feedback devices specify an absolute number of turns that typically range from 1 to 4 096 in powers of 2. This attribute can be used by the control system to determine the maximum Feedback Count range of the absolute feedback device, this being the product of the feedback cycle resolution, interpolation, and turns.	Feedback Units (Rev)
1419 + (n-1)×50	Required – E (Linear Absolute)	Set		Feedback n Length	REAL	Specified length of a linear absolute feedback device. Typical linear absolute feedback devices specify length in Meters. This attribute can be used by the control system to determine the maximum travel range of absolute feedback device in Feedback Counts, this being the combination of the feedback cycle resolution, interpolation, and length.	Feedback Units (m)
1420 + (n-1)×50	Optional – E TP, SS	Set		Feedback n Data Length	USINT	Number of feedback data bits transferred over the digital serial or parallel data interface channel of a feedback device.	Number of bits
1421 + (n-1)×50	Optional – E TP, SS	Set		Feedback n Data Code	USINT	Type of feedback data bit encoding used by designated serial or parallel data interface channel of a feedback device.	Enumeration: 0 = Binary 1 = Gray 2 to 255 = (reserved)
1422 + (n-1)×50	Optional – E RS	Set		Feedback n Resolver Transformer Ratio	REAL	Transformer Ratio specification of the designated resolver feedback device.	
1423 + (n-1)×50	Optional – E RS	Set		Feedback n Resolver Excitation Voltage	REAL	Sets the sinusoidal excitation voltage applied to the rotor of the designated resolver feedback device.	Volts (RMS)
1424 + (n-1)×50	Optional – E RS	Set		Feedback n Resolver Excitation Frequency	REAL	Frequency of sinusoidal excitation signal applied to the designated resolver feedback device. Valid frequency range or values for this attribute depends on the specific device hardware interface.	Hertz
1425 + (n-1)×50	Optional – E RS	Set		Feedback n Resolver Cable Balance	REAL	Adjusts the relative amplitude of the Sine and Cosine signals from the resolver to compensate for impact of resolver cable.	%

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
1426 + (n-1)×50	Required – E LT	Set		Feedback n LDT Type	USINT	Determines the LDT type. Options are Start/Stop and PWM. Start/Stop transducers accept an input (interrogate) signal to start the measurement cycle and respond with two pulses on the Return line. Timing can be based on either the Rising or Falling edge. The time between the pulses is proportional to the position. PWM transducers respond to the interrogate signal with a single long pulse on the Return line. The pulse width is proportional to the position.	Enumeration: 0 = PWM 1 = Start/Stop Rising 2 = Start/Stop Falling 3 to 255 = (reserved)
1427 + (n-1)×50	Required – E LT	Set		Feedback n LDT Recirculations	USINT	Determines the number of recirculations associated with a PWM type LDT transducer. Multiple recirculations can be used to increase the resolution of the LDT at the expense of increasing the sample period.	Number of Recirculations
1434 + (n-1)×50	Optional – E	Set		Feedback n Velocity Filter Bandwidth.	REAL	Controls the bandwidth of the Low Pass Filter applied to the raw velocity signal from Feedback n. A value of 0 for this attribute disabled this feature.	rad/s
1435 + (n-1)×50	Optional – E	Set		Feedback n Accel Filter Bandwidth	REAL	Controls the bandwidth of the Low Pass Filter applied to the raw acceleration signal from Feedback n. A value of 0 for this attribute disabled this feature.	rad/s

7.3.6.2 Semantics: Attribute No. 42 – Feedback Mode

This attribute contains a 4-bit enumerated value that determines how the various logical feedback channels are used to implement the selected Control Mode for this axis instance.

The Feedback Mode enumeration (see Table 67) provides support for multi-feedback device control functionality for the various active device Control Modes, i.e. where the device is actively controlling the motor based on feedback. In these active device Control Modes it is assumed that logical channel, Feedback 1, is attached directly to the motor while Feedback 2 is attached to the load side of the mechanical transmission. Commutation signals for a PM motor are always derived from the Feedback 1.

Table 67 – Feedback Mode enumeration definitions

Enum.	Req./Opt.	Name	Description
0	R/S	No Feedback	No Feedback is selected when sensorless/encoderless open loop or closed loop control is desired. When performing open loop control, no feedback signal is required. In closed loop control, the required feedback signal is estimated by a sensorless control algorithm based on motor phase voltage and current signals.
1	R/N	Master Feedback	Master Feedback assigns an uncommitted feedback channel, as specified by the Feedback Master Select attribute, to this device axis instance to serve as a “master feedback” source when the device is configured for No Control mode
2	R/C	Motor Feedback	When Motor Feedback is selected, then commutation, acceleration, velocity, and position feedback signals are all derived from motor mounted Feedback 1
3	O/C	Load Feedback	When Load Feedback is selected, then motor-mounted Feedback 1 is only used for PM motor commutation while load-side Feedback 2 is used for position, velocity, and acceleration.
4	O/P	Dual Feedback	When Dual Feedback is selected, then motor mounted Feedback 1 is used for commutation, acceleration, and velocity, and load-side Feedback 2 is used strictly for position.
5 to 7	-	Reserved	
8 to 15	-	Vendor Specific	

7.3.7 Event Capture attributes

7.3.7.1 General

The attribute table in Table 68 contains all event related attributes associated with a Motion Device Axis Object instance. These include registration, marker, and homing events. The Event Capture attributes are designed to support the possibility of up to 16 active events per controller update period. The format of all Time Stamp attributes is absolute System Time and follows the CIP Sync specification with 0 corresponding to 1970-01-01.

The Motion Device Axis Object currently supports two independent registration input channels per axis instance that can be triggered on either the rising or falling edges of the signal. In other words, Registration 1 for axis instance 1 is not generally the same signal as Registration 1 for axis instance 2. If the device hardware implementation allows, event time and position data can be captured for all four event conditions simultaneously. The Event Capture attributes also support Auto-rearm for registration events. This allows for controller implementation of important features like Windowed Registration and Registration Pattern Recognition.

The Motion Device Axis Object also supports Home Switch, Marker and Switch-Marker events for homing functionality on a per axis basis. The Marker events are typically generated by the configured position feedback device for the associated axis instance.

When the Motion Scaling Configuration is set to Drive Scaling, the Event Capture functionality extends to support Windowed Registration and Watch Events.

Set* → These attributes are generally updated via the cyclic Command Data Set of the CIP Motion C-to-D connection. When included as cyclic command data, these attributes cannot be updated via a Set service.

Table 68 – Event attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
60	Optional – E	Set*		Event Checking Control	DWORD	This attribute is passed to the device by the controller as part of the Drive to Controller connection for the purpose of arming various device inputs, for example marker, home switch, and registration inputs, to generate events to the controller. When these enabled events occur, the device captures both the time and exact axis position when the event occurred. This attribute also manages the format and content of the C-to-D Event Data Block.	See Semantics in 7.3.7.2.1
61	Optional – E	Get		Event Checking Status	DWORD	This attribute is passed by the device to the controller as part of the Drive to Controller connection to indicate if the device is currently checking for events based on various device inputs, for example marker, home, and registration inputs. Event checking is initiated when the corresponding Event Checking Control bit is set in the Controller-to-Device connection. This attribute also manages the format and content of the D-to-C Event Data Block.	See Semantics in 7.3.7.2.2
62	Optional – E	Get		Registration 1 Positive Edge Position	DINT	Feedback position latched on the rising edge of the Registration Input 1.	Position Control Units
63	Optional – E	Get		Registration 1 Negative Edge Position	DINT	Feedback Position latched on the falling edge of the Registration Input 1.	Position Control Units
64	Optional – E	Get		Registration 2 Positive Edge Position	DINT	Feedback position latched on the rising edge of the Registration Input 2.	Position Control Units
65	Optional – E	Get		Registration 2 Negative Edge Position	DINT	Feedback Position latched on the falling edge of the Registration Input 2.	Position Control Units
66	Optional – E	Get		Registration 1 Positive Edge Time	LINT	System Time stamp on the rising edge of the Registration Input 1.	Nanoseconds (CIP Sync absolute)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
67	Optional – E	Get		Registration 1 Negative Edge Time	LINT	System Time stamp on the falling edge of the Registration Input 1.	Nanoseconds (CIP Sync absolute)
68	Optional – E	Get		Registration 2 Positive Edge Time	LINT	System Time stamp on the rising edge of the Registration Input 2.	Nanoseconds (CIP Sync absolute)
69	Optional – E	Get		Registration 2 Negative Edge Time	LINT	System Time stamp on the falling edge of the Registration Input 2.	Nanoseconds (CIP Sync absolute)
70	Required – NP Optional – VT E	Get		Home Event Position	DINT	Feedback Position latched on the specified home event.	Position Control Units
71	Optional – NP Optional – VT E	Get		Home Event Time	LINT	System Time stamp latched on the specified home event.	Nanoseconds (CIP Sync absolute)
72	Required – NP Optional – VT E (Drive Scaling)	Get		Watch 1 Event Time	LINT	System Time stamp latched on the specified watch event.	Nanoseconds (CIP Sync absolute)
73	Required – NP Optional – VT E (Drive Scaling)	Get		Watch 2 Event Time	LINT	System Time stamp latched on the specified watch event.	Nanoseconds (CIP Sync absolute)

7.3.7.2 Semantics

7.3.7.2.1 Attribute No. 60 – Event Checking Control

The first 28 bits of this 32-bit attribute (see Figure 63) are used to enable various device inputs, for example marker, home switch, and registration inputs, to generate events to the controller. When these enabled events occur, the device captures both the time and exact position of the associated motion axis instance. Three of the most significant 4 bits represent the number of Event Acknowledgement messages in the Event Data Block in this Controller-to-Device Connection update, with the final most, significant bit, indicating whether the Event Data Block format is Extended to support additional events associated with Drive Scaling functionality (see Motion Scaling Configuration). Support for Extended Format is optional in the device implementation. For a detailed description of the event handling protocol between the device and the controller, refer to Connection section in the appropriate CIP Motion device profile.

← 32-bit DWORD →		
Event Checking Control format		
Bit 31: Boolean Extended format	Bits 28-30: Integer Event Block Count	Bits 0-27: Bit Field Event Checking Control Bits

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Figure 63 – Event Checking Control Word field

For the Home Switch-Marker events specified in Table 69, the device first looks for level of home switch input to transition according to the first + or – symbol and then immediately look for the transition of the marker input according to the second + or – symbol.

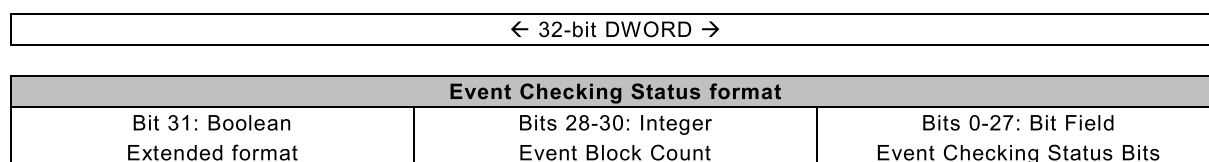
Table 69 – Event Checking Control bit definitions

Bit	Extended format only	Name	Description
0		Reg 1 Pos Edge	Enable checking for positive edge transition of Registration 1 Input
1		Reg 1 Neg Edge	Enable checking for negative edge transition of Registration 1 Input
2		Reg 2 Pos Edge	Enable checking for positive edge transition of Registration 2 Input
3		Reg 2 Neg Edge	Enable checking for negative edge transition of Registration 2 Input
4	*	Reg 1 Pos Edge Window	Enable Window check after positive edge transition of Registration 1 Input
5	*	Reg 1 Neg Edge Window	Enable Window check after negative edge transition of Registration 1 Input
6	*	Reg 2 Pos Edge Window	Enable Window check after positive edge transition of Registration 2 Input
7	*	Reg 2 Neg Edge Window	Enable Window check after negative edge transition of Registration 2 Input
8		Reg 1 Pos Auto-rearm	Enable auto-rearm after positive edge transition of Registration 1 Input
9		Reg 1 Neg Auto-rearm	Enable auto-rearm after negative edge transition of Registration 1 Input
10		Reg 2 Pos Auto-rearm	Enable auto-rearm after positive edge transition of Registration 2 Input
11		Reg 2 Neg Auto-rearm	Enable auto-rearm after negative edge transition of Registration 2 Input
12	*	Watch Position 1 Fwd	Enable checking for axis position crossing Watch Position 1 in the forward (positive) direction.
13	*	Watch Position 1 Rev	Enable checking for axis position crossing Watch Position 1 in the reverse (negative) direction.
14	*	Watch Position 2 Fwd	Enable checking for axis position crossing Watch Position 2 in the forward (positive) direction.
15	*	Watch Position 2 Rev	Enable checking for axis position crossing Watch Position 2 in the reverse (negative) direction.
16		Marker Pos Edge	Enable checking for positive edge of Marker input of associated position feedback channel.
17		Marker Neg Edge	Enable checking for negative edge of Marker input of associated position feedback channel.
18		Home Switch Pos Edge	Enable checking for positive edge of Home input associated with this axis instance.
19		Home Switch Neg Edge	Enable checking for negative edge of Home input associated with this axis instance.
20		Home Switch-Marker ++	Enable checking for event sequence specified as a positive edge transition of the Home Switch input followed by a positive edge of the Marker input.

Bit	Extended format only	Name	Description
21		Home Switch-Marker +-	Enable checking for event sequence specified as a positive edge transition of the Home Switch input followed by a negative edge of the Marker input.
22		Home Switch-Marker ++	Enable checking for event sequence specified as a negative edge transition of the Home Switch input followed by a positive edge of the Marker input.
23		Home Switch-Marker --	Enable checking for event sequence specified as a negative edge transition of the Home Switch input followed by a negative edge of the Marker input.
24	*	Home Torque Threshold	Enable checking for home torque event specified as a torque level that exceeds the Home Torque Threshold for a period of time given by the Home Torque Time.
25 to 27		(Reserved)	

7.3.7.2.2 Attribute No. 61 – Event Checking Status

The first 28 bits of this 32-bit attribute (see Figure 64) are used to indicate if the device is currently checking for events based on various device inputs, for example marker and registration inputs (see Table 70). Event checking is initiated when the corresponding Event Checking Control bit is set in the Controller-to-Device Connection. When an event occurs, the device captures both the time and exact axis position and passes this information to the controller in event notification data blocks. But for the controller to process the event data, the corresponding Event Checking Status bit shall be set. Three of the most significant 4 bits of this attribute represent the number of Event Notification messages in the Event Data Block in this Device-to-Controller Connection update, with the final, most significant, bit indicating whether the Event Data Block format is Extended to support additional events associated with Drive Scaling functionality (see Motion Scaling Configuration). Support for Extended Format is optional in the device implementation. For a detailed description of the event handling protocol between the device and the controller, refer to Connection section.



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Figure 64 – Event Checking Status word field

Table 70 – Event Checking Status bit definitions

Bit	Extended format only	Name	Description
0		Reg 1 Pos Edge	Checking for positive edge transition of Registration 1 Input
1		Reg 1 Neg Edge	Checking for negative edge transition of Registration 1 Input
2		Reg 2 Pos Edge	Checking for positive edge transition of Registration 2 Input
3		Reg 2 Neg Edge	Checking for negative edge transition of Registration 2 Input
4	*	Reg 1 Pos Edge Window	Apply Window check after positive edge transition of Registration 1 Input
5	*	Reg 1 Neg Edge Window	Apply Window check after negative edge transition of Registration 1 Input
6	*	Reg 2 Pos Edge Window	Apply Window check after positive edge transition of Registration 2 Input
7	*	Reg 2 Neg Edge Window	Apply Window check after negative edge transition of Registration 2 Input
8		Reg 1 Pos Auto-rearm	Auto-rearm after positive edge transition of Registration 1 Input active
9		Reg 1 Neg Auto-rearm	Auto-rearm after negative edge transition of Registration 1 Input active
10		Reg 2 Pos Auto-rearm	Auto-rearm after positive edge transition of Registration 2 Input active
11		Reg 2 Neg Auto-rearm	Auto-rearm after negative edge transition of Registration 2 Input active
12	*	Watch Position 1 Fwd	Checking for watch event with actual position crossing Watch Position 1 in the forward (positive) direction.
13	*	Watch Position 1 Rev	Checking for watch event with actual position crossing Watch Position 1 in the reverse (negative) direction.
14	*	Watch Position 2 Fwd	Checking for watch event with actual position crossing Watch Position 2 in the forward (positive) direction.
15	*	Watch Position 2 Rev	Checking for watch event with actual position crossing Watch Position 2 in the reverse (negative) direction.
16		Marker Pos Edge	Checking for positive edge of Marker input of associated position feedback channel
17		Marker Neg Edge	Checking for negative edge of Marker input of associated position feedback channel
18		Home Switch Pos Edge	Checking for positive edge of Home input associated with this axis instance
19		Home Switch Neg Edge	Checking for negative edge of Home input associated with this axis instance
20		Home Switch-Marker ++	Checking for event sequence specified as a positive edge transition of the Home Switch input followed by a positive edge of the Marker input
21		Home Switch-Marker +-	Checking for event sequence specified as a positive edge transition of the Home Switch input followed by a negative edge of the Marker input
22		Home Switch-Marker -+	Checking for event sequence specified as a negative edge transition of the Home Switch input followed by a positive edge of the Marker input
23		Home Switch-Marker --	Checking for event sequence specified as a negative edge transition of the Home Switch input followed by a negative edge of the Marker input
24	*	Home Torque Threshold	Checking for home torque event specified as a torque level that exceeds the Home Torque Threshold for a period of time given by the Home Torque Time.
25 to 27		(Reserved)	

7.3.8 Command reference generation attributes

The lists of attributes in Table 71 and Table 72 apply to the command reference generation functionality of the device that converts command position, velocity, acceleration, and torque data output from a controller-based planner into corresponding command references signals to the device's motor control structures. The command reference generator functionality includes fine interpolators, ramp generators, signal selector switches.

Set* → These attributes are generally updated via the cyclic Command Data Set of the CIP Motion C-to-D connection. When included as cyclic command data, these attributes shall not be updated via a Set service.

Table 71 – Command Generator Signal attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
360	Optional – P (If not supported, then Controller Position Command – Float is Required.)	Set*		Controller Position Command – Integer	DINT	Integer command position data value from controller, updated at the Controller Update Period, and feeding into the Position Fine Command Generator. This integer representation of command position is appropriate for simple positioning devices where real time processing of double precision floating point values is too time consuming and the smoothness of motion is not a driving factor.	Position Control Units
361	Optional – P (If not supported, then Controller Position Command – Integer is Required.)	Set*		Controller Position Command – Float	LREAL	Floating point command position data value from controller, updated at the Controller Update Period, and feeding into the Position Fine Command Generator. The range of this value is limited to that of a DINT by the motion planner but, but unlike a DINT, it carries a fractional component that translates to smoother motion especially for feedforward operation.	Position Control Units
362	Required – VF Optional – P	Set*		Controller Velocity Command	REAL	Command velocity data value from controller, updated at the Controller Update Period, and feeding into the Velocity Fine Command Generator.	Velocity Control Units / s
363	Optional – PVT	Set*		Controller Acceleration Command	REAL	Command acceleration data value from controller, updated at the Controller Update Period, and feeding into the Acceleration Fine Command Generator.	Accel Control Units / s ²
364	Required – T	Set*		Controller Torque Command	REAL	Commanded torque data value from controller used in torque control mode, updated at the Controller Update Period, and feeding into the Torque Fine Command Generator.	% Motor Rated

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
365	Optional – P	Get		Fine Command Position	DINT	Output value from the Position Fine Command Generator.	Position Control Units
366	Optional – PV	Get		Fine Command Velocity	REAL	Output value from the Command Velocity Fine Command Generator. When no Command Velocity signal is present when performing position control, this signal can be derived by scaling the Differential Position output value of the Position Fine Command Generator.	Velocity Control Units / s
367	Optional – C	Get		Fine Command Acceleration	REAL	Output value from the Acceleration Fine Command Generator. When no Command Acceleration signal is present when performing position or velocity control, this signal can be derived by scaling the Differential Velocity output value of the Velocity Fine Command Generator. If no Command Velocity signal is present, the Fine Command Acceleration signal can be derived by scaling the 2nd Differential Position output value of the Position Fine Command Generator.	Accel Control Units / s ²

Table 72 – Command Generator Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
370	Optional – F	Set		Skip Speed 1	REAL	Sets the central speed of a skip speed band within which the device does not operate. The skip speed value is signed.	Velocity Control Units / s
371	Optional – F	Set		Skip Speed 2	REAL	Sets the central speed of a skip speed band within which the device does not operate. The skip speed value is signed.	Velocity Control Units / s
372	Optional – F	Set		Skip Speed 3	REAL	Sets the central speed of a skip speed band within which the device does not operate. The skip speed value is signed.	Velocity Control Units / s
373	Optional – F	Set		Skip Speed Band	REAL	Determines the speed window around a skip speed that cannot be commanded. Any command set-point within this window is adjusted by the Skip Speed block to fall at either the upper or lower Skip Speed Band boundary value. The device can smoothly accelerate or decelerate through the skip speed band due to the Ramp Generator but may not operate at a set speed within the band. The Skip Speed Band is distributed ½ above and ½ below the skip speed. This Skip Speed Band attribute applies to all skip speeds supported in the device. A value of 0 for this attribute disabled this feature.	Velocity Control Units / s
374	Optional – FV	Set		Ramp Velocity – Positive	REAL	This positive value defines the maximum positive velocity command output of the Ramp Generator.	Velocity Control Units / s
375	Optional – FV	Set		Ramp Velocity – Negative	REAL	This negative value defines the maximum negative velocity command output of the Ramp Generator.	Velocity Control Units / s
376	Optional – FV	Set		Ramp Acceleration	REAL	This positive value defines the maximum acceleration (increasing speed) of the velocity command output by the Ramp Generator.	Velocity Control Units / s ²
377	Optional – FV	Set		Ramp Deceleration	REAL	This positive value defines the maximum deceleration (decreasing speed) of the velocity command output by the Ramp Generator.	Velocity Control Units / s ²

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
378	Optional – FV	Set		Ramp Jerk Control	REAL	This value sets the percentage of accel or decel time that is applied to the speed ramp as jerk limited S Curve based on a step change in velocity. The S Curve time is added ½ at the beginning and ½ at the end of the ramp. A value of 0 results in no S-Curve, i.e. a linear acceleration or deceleration ramp. A value of 100 % results in a triangular acceleration profile with the peak being the configured ramp acceleration or deceleration. As the Jerk Control value increases the derived accelerating jerk value decreases based on $(0,5 \times 0,01 \times \text{Jerk Control} \times \text{Ramp Vel Positive} / \text{Ramp Accel})$, and the decelerating Jerk limit value also decreases according to $(0,5 \times 0,01 \times \text{Jerk Control} \times \text{Ramp Vel Negative} / \text{Ramp Decel})$.	%
380	Optional – FV	Set		Flying Start Enable	BOOL	This Boolean value is used to enable or disable the Flying Start feature of the device's Ramp Generator. When Flying Start Enabled is true and the motion axis is enabled, the device determines the current velocity of the motor as part of the Starting State initialization activities. Just prior to transitioning to the Running state, the device presets the output of the Ramp Generator to the current velocity. In this way, the motor seamlessly ramps from its current velocity to the commanded velocity. When Flying Start Enabled is false, the motor velocity is irrelevant and a preset of 0 is applied to the Ramp Generator output.	0 = Flying Start Disabled 1 = Flying Start Enabled

7.3.9 Control mode attributes

7.3.9.1 Position Loop attributes

The attributes tables in Table 73 and Table 74 contain all position control related attributes associated with a Motion Device Axis Object instance.

Set* ➔ this attribute is generally updated via the cyclic Command Data Set of the CIP Motion C-to-D Connection. When included as cyclic command data, this attribute shall not be updated via a Set service.

Table 73 – Position Loop Signal attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
430	Required – P	Get		Position Command	DINT	Command position output from the Fine Command Generator (if active) into position loop when configured for position loop control.	Position Control Units
431	Required – P	Set*		Position Trim	DINT	Additional position command added to the Position Command to generate the Position Reference signal into the position loop summing junction.	Position Control Units
432	Required – P	Get		Position Reference	DINT	Command position reference signal into the position loop summing junction to be compared with a position feedback signal.	Position Control Units
433	Required – P	Get		Velocity Feedforward Command	REAL	Velocity feedforward command signal that represents a scaled version of the command velocity profile. This signal is the Velocity Fine Command signal scaled by Kvff and applied to the output of the position loop.	Velocity Control Units/s
434	Required – E	Get		Position Feedback	DINT	Position feedback value channeled into the position control summing junction. In most cases the Position Feedback signal is derived directly from the feedback device specified by the Feedback Mode selection. If the axis configured for “No Control” mode, Position Feedback represents the actual position of the feedback device specified by the Feedback Master Select.	Position Control Units
436	Required – P	Get		Position Error	REAL	Error between commanded and actual position that is the output of the position loop summing junction.	Position Control Units
437	Required – P	Get		Position Integrator Output	REAL	Output of position integrator representing the contribution of the position integrator to Position Loop Output.	Velocity Control Units/s
438	Required – P	Get		Position Loop Output	REAL	Output of the position loop forward path representing the total control effort of the position loop.	Velocity Control Units/s

Table 74 – Position Loop Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
440	Required – P	Set		Kvff	REAL	Velocity Feedforward Gain value that multiplies the Velocity Feedforward Command signal to form the Velocity Feedforward Command that is applied to the output of the position control loop. 100 % Velocity Feedforward applies the full Velocity Feed Command signal to the velocity loop.	%
441	Required – P	Set		Kpp	REAL	Position Proportional Gain value that multiplies the Position Error signal to form the proportional control signal that summed together with the integral control signal to generate the output of the position control loop. This value directly determines the bandwidth of the position loop.	rad/s
442	Required – P	Set		Kpi	REAL	Position Integral Gain value that, together with the Kpp, multiplies the Position Integrator Error signal to form the integral control signal that is summed together with the proportional control signal to generate the output of the position control loop. The reciprocal of this value, 1/Kpi, represents the integrator time constant for the position loop. A value of 0 for this attribute disables the integrator.	rad/s
443	Required – P	Set		Position Lock Tolerance	REAL	This value establishes a window around the current command position. When the actual position is within this window the Position Lock status bit is set. When actual position falls outside this window, the Position Lock status bit is cleared.	Position Control Units
444	Required – P	Set		Position Error Tolerance	REAL	Determines the absolute maximum Position Error value that can be tolerated without causing an Excessive Position Error exception.	Position Control Units
445	Optional – P	Set		Position Error Tolerance Time	REAL	Determines the maximum amount of time that the Position Error Tolerance can be exceeded without generating an exception.	Seconds

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
446	Required – P	Set		Position Integrator Control	BYTE	This bit field controls the behavior of the position loop integrator while commanding motion via the controller. When the integrator hold enable bit is set, the integrator is held while motion is being commanded with a non-zero velocity. When clear, the integrator runs without qualification. When the auto-preset bit is set, the integrator preload value is automatically loaded with the current velocity command when there is a control mode change between velocity control and position control. If clear, the integrator is loaded with the configured position integrator preload value.	Bit Field: 0: Integrator Hold Enable (R) 1: Auto-Preset (O) 2-7: (Reserved)
447	Optional – P	Set		Position Integrator Preload	REAL	Value assigned to the position integrator when the position control loop is enabled.	Velocity Control Units/s
448	Optional – E (Drive Scaling)	Set		Position Limit – Positive	REAL	This value defines the most positive position feedback value that when exceeded in the positive direction, generates a Position Overtravel Positive exception.	Position Control Units
449	Optional – E (Drive Scaling)	Set		Position Limit – Negative	REAL	This value defines the most negative position feedback value that when exceeded in the negative direction, generates a Position Overtravel Negative exception.	Position Control Units

7.3.9.2 Velocity Loop attributes

The attributes tables in Table 75 and Table 76 contain all velocity control related attributes associated with a Motion Device Axis Object instance.

Set* ➔ this attribute is generally updated via the cyclic Command Data Set of the CIP Motion C-to-D Connection. When included as cyclic command data, this attribute shall not be updated via a Set service.

Table 75 – Velocity Loop Signal attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
450	Required – FPV	Get		Velocity Command	REAL	Command velocity output from Fine Command Generator (if active) into the Velocity Loop control or the frequency controller when configured for Frequency Control operation.	Velocity Control Units/s
451	Required – FPV	Set*		Velocity Trim	REAL	Additional velocity command added to the velocity loop or frequency control summing junction.	Velocity Control Units/s
452	Required – PV	Get		Acceleration Feedforward Command	REAL	Acceleration feedforward command signal that represents a scaled version of the command acceleration profile. This signal is the Acceleration Fine Command signal scaled by Kaff and applied to the output of the velocity loop.	Velocity Control Units/s ²
453	Required – FPV	Get		Velocity Reference	REAL	Command velocity reference into velocity loop summing junction, or in the case of Frequency Control, the signal that is scaled to become the Frequency Reference.	Velocity Control Units/s
454	Required – All	Get		Velocity Feedback	REAL	Actual velocity of the axis that is applied to the velocity summing junction, if applicable, based on Control Mode selection. In most cases the Velocity Feedback signal is derived directly from the feedback device specified by the Feedback Mode selection. If the axis is configured for “No Control” mode, Velocity Feedback represents the actual velocity of the feedback device specified by the Feedback Master Select. If axis is configured for Frequency Control, the Velocity Feedback signal is derived from the Velocity Command signal. If configured for Sensorless Velocity Loop operation, Velocity Feedback is estimated by the sensorless control algorithm.	Velocity Control Units/s
455	Required – PV	Get		Velocity Error	REAL	Error between the Velocity Reference and Velocity Feedback value that is the output of the velocity loop summing junction.	Velocity Control Units/s
456	Required – PV	Get		Velocity Integrator Output	REAL	Output of velocity integrator representing the contribution of the velocity integrator to Velocity Loop Output.	Accel Control Units/s ²
457	Required – PV	Get		Velocity Loop Output	REAL	Output of velocity forward path representing the total control effort of the velocity loop.	Accel Control Units/ s ²

Table 76 – Velocity Loop Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
460	Required – PV	Set		Kaff	REAL	Acceleration Feedforward Gain value that multiplies the Acceleration Fine Command signal to form the Acceleration Feedforward Command before applying it to the output of the velocity control loop. 100 % Acceleration Fine applies the full Acceleration Feedforward Command signal to the output of the velocity loop.	%
461	Required – PV	Set		Kvp	REAL	Velocity Proportional Gain value that multiplies the Velocity Error signal to form the proportional control signal that summed together with the integral control signal to generate the output of the velocity control loop. This value directly determines the bandwidth of the velocity loop.	rad/s
462	Required – PV	Set		Kvi	REAL	Velocity Integral Gain value that, together with the Kvp, multiplies the Velocity Integrator Error signal to form the integral control signal that is summed together with the proportional control signal to generate the output of the velocity control loop. The reciprocal of this value, 1/Kvi, represents the integrator time constant for the velocity loop. A value of 0 for this attribute disables the integrator.	rad/s
464	Optional – FPV	Set		Kdr	REAL	Velocity Droop value that provides compliance to the velocity integrator by subtracting a portion of the velocity loop effort from the velocity error input to the velocity integrator. The presence of the Torque/Force scaling gain, Kj, in the droop signal path allows Velocity Droop to be specified in velocity units per % rated torque output. This parameter is also valid for V/Hz devices and its behavior is nearly identical, but instead of % rated being related to torque, % rated is related to current.	Velocity Control Units/ s/ % Rated
465	Optional – PV	Set		Velocity Error Tolerance	REAL	Determines the absolute maximum Velocity Error value that can be tolerated without causing an Excessive Velocity Error exception.	Velocity Control Units/ s
466	Optional – PV	Set		Velocity Error Tolerance Time	REAL	Determines the maximum amount of time that the Velocity Error Tolerance can be exceeded without generating an exception.	seconds
467	Required – PV	Set		Velocity Integrator Control	BYTE	This bit field controls the behavior of the velocity loop integrator while commanding motion via the controller. When the integrator hold enable bit is set, the integrator is held while motion is being commanded with a non-zero velocity. When clear, the integrator runs without qualification. When the auto-preset bit is set, the integrator preload value is automatically loaded with the current torque command when there is a control mode change between torque control and velocity control. If clear, the integrator is loaded with the configured velocity integrator preload value.	Bit Field: 0: Integrator Hold Enable (R) 1: Auto-Preset (O) 2-7: (Reserved)
468	Optional – PV	Set		Velocity Integrator Preload	REAL	Value assigned to the velocity integrator when the velocity control loop is enabled.	Accel Control Units/s ²
469	Optional – PV	Set		Velocity Low Pass Filter Bandwidth	REAL	Controls the bandwidth of the Low Pass Filter applied to the Velocity Error signal. Recommended implementation is a two pole IIR filter. A value of 0 for this attribute disables this feature.	rad/s
470	Optional – ED	Set		Velocity Threshold	REAL	Defines a minimum absolute velocity. If the magnitude of the Velocity Feedback signal is less than this value, the Velocity Threshold status bit is set.	Velocity Control Units/s

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
471	Optional – FPV	Set		Velocity Lock Tolerance	REAL	This value establishes a window around the unlimited velocity reference signal. When the Velocity Feedback signal is within this window the Velocity Lock status bit is set. When Velocity Feedback signal falls outside this window, the Velocity Lock status bit is cleared.	Velocity Control Units/s
472	Required – ED	Set		Velocity Standstill Window	REAL	This value establishes a window around zero speed. When the Velocity Feedback signal is within this window the Velocity Standstill status bit is set. When Velocity Feedback signal falls outside this window, the Velocity Standstill status bit is cleared.	Velocity Control Units/s
473	Optional – FPV	Set		Velocity Limit – Positive	REAL	This positive value defines the most positive velocity reference value into the velocity summing junction. If this velocity limit value is exceeded, the device responds by clamping the velocity reference to this limit and setting the Velocity Limit status bit.	Velocity Control Units/s
474	Optional – FPV	Set		Velocity Limit – Negative	REAL	This negative value defines the most negative velocity reference value allowed into the velocity summing junction. If this velocity limit value is exceeded, the device responds by clamping the velocity reference to this limit and setting the Velocity Limit status bit.	Velocity Control Units/s

7.3.9.3 Acceleration Control attributes

The attributes tables in Table 77 and Table 78 contain all accelerations related attributes associated with a Motion Device Axis Object instance.

Set* → this attribute is generally updated via the cyclic Command Data Set of the CIP Motion C-to-D Connection. When included as cyclic command data, this attribute shall not be updated via a Set service.

Table 77 – Acceleration Signal attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
480	Optional – C	Get		Acceleration Command	REAL	Command acceleration output from Fine Command Generator (if active) into acceleration loop when configured for acceleration control.	Accel Control Units/s ²
481	Optional – C	Set*		Acceleration Trim	REAL	Additional acceleration command added to the acceleration loop summing junction.	Accel Control Units/s ²
482	Optional – C	Get		Acceleration Reference	REAL	Command acceleration reference into acceleration loop summing junction.	Accel Control Units/s ²
483	Required – E	Get		Acceleration Feedback	REAL	Actual acceleration of the axis based on the selected feedback device. In most cases the Acceleration Feedback signal is derived directly from the feedback device specified by the Feedback Mode selection. If the axis configured for “No Control” mode Acceleration Feedback represents the actual acceleration of the feedback device specified by the Feedback Master Select.	Accel Control Units/s ²

Table 78 – Acceleration Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
485	Optional – D	Set		Acceleration Limit	REAL	This value defines the maximum acceleration value (increasing speed) into the acceleration summing junction. If this acceleration limit value is exceeded, the device responds by clamping the acceleration reference to this limit and setting the Acceleration Limit status bit.	Accel Control Units / s ²
486	Optional – D	Set		Deceleration Limit	REAL	This value defines the maximum deceleration value (decreasing speed) into the acceleration summing junction. If this deceleration limit value is exceeded, the device responds by clamping the acceleration reference to this limit and setting the Deceleration Limit status bit.	Accel Control Units / s ²

7.3.9.4 Torque/Force Control attributes

The attributes tables in Table 79 and Table 80 contain all torque/force related attributes associated with a Motion Device Axis Object instance.

Set* → this attribute is generally updated via the cyclic Command Data Set of the CIP Motion C-to-D Connection. When included as cyclic command data, this attribute shall not be updated via a Set service.

Table 79 – Torque/Force Control Signal attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
490	Required – C	Get		Torque Command	REAL	Command torque output from Fine Command Generator (if active) into torque input summing junction when configured for torque control.	% Motor Rated
491	Required – C	Set*		Torque Trim	REAL	Additional torque command added to the torque input summing junction.	% Motor Rated
492	Required – C	Get		Torque Reference	REAL	Commanded torque reference input signal before torque filter section representing the sum of the Torque Command and Torque Trim signal inputs.	% Motor Rated
493	Required – C	Get		Torque Reference – Filtered	REAL	Commanded torque reference input signal after torque filter section.	% Motor Rated
494	Required – C	Get		Torque Reference – Limited	REAL	Commanded torque reference input signal after torque limiter section.	% Motor Rated

Table 80 – Torque/Force Control Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
496	Required – PV Optional – T	Set		Kj	REAL	Acceleration to Torque/Force scaling gain value that converts commanded acceleration into equivalent rated torque/force. Properly set, this value represents the total system inertia or mass.	% Motor Rated/ (Motor Units/ s ²)
498	Optional – C	Set		Friction Compensation – Sliding	REAL	Value added to the current/torque command to offset the effects of coulomb, or sliding friction.	% Motor Rated
499	Optional – C	Set		Friction Compensation – Static	REAL	Value added to the current/torque command to offset the effects of static friction, sometimes referred to as “sticktion”.	% Motor Rated
500	Optional – C	Set		Friction Compensation – Viscous	REAL	Value added to the current/torque command to offset the effects of viscous friction, i.e. friction that is proportional to speed.	% Motor Rated / (Motor Units/ s)
502	Optional – C	Set		Torque Low Pass Filter Bandwidth	REAL	Break frequency for the low pass filter applied to torque reference signal.	rad/s
503	Optional – C	Set		Torque Notch Filter Frequency	REAL	Center frequency of the notch filter applied to the torque reference signal. A value of 0 for this attribute disables this feature.	rad/s
504	Required – C	Set		Torque Limit – Positive	REAL	This positive value determines the maximum positive torque that can be applied to the motor. If the device attempts to exceed this value, the torque command is clamped to this value.	% Motor Rated
505	Required – C	Set		Torque Limit – Negative	REAL	This negative value determines the maximum negative torque that can be applied to the motor. If the device attempts to exceed this value, the torque command is clamped to this value.	% Motor Rated
506	Optional – C	Set		Torque Rate Limit	REAL	Limits the rate of change of the torque reference signal.	% Motor Rated/s
507	Optional – C	Set		Torque Threshold	REAL	Specifies the threshold for the Filtered Torque Reference signal magnitude that, when exceeded, results in the Torque Threshold status bit being set.	% Motor Rated

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
508	Optional – D	Set		Overtorque Limit	REAL	Maximum threshold for the torque producing Iq Current Feedback signal magnitude. When the Iq Current Feedback signal is greater than this value for the duration specified by Overtorque Limit Time attribute, the results is an Overtorque Limit exception. This feature allows the device to generate an exception if there is a sudden increase in load torque magnitude during operation. This condition could occur if a bearing fails, a hard stop is reached, or there is some other mechanical failure.	% Motor Rated
509	Optional – D	Set		Overtorque Limit Time	REAL	Specifies the amount of time allowed in an Overtorque Limit condition before generating an Overtorque Limit exception. A value of 0 for this attribute disables the Overtorque feature.	seconds
510	Optional – D	Set		Undertorque Limit	REAL	Minimum threshold for the torque producing Iq Current Feedback signal magnitude. When the Iq Current Feedback signal is less than this value for the duration specified by Undertorque Limit Time attribute, the result is an Undertorque Limit exception. This feature allows the device to generate an exception if there is a sudden decrease in load torque magnitude during operation. This condition could occur if a load coupling breaks, tensioned web material breaks, etc.	% Motor Rated
511	Optional – D	Set		Undertorque Limit Time	REAL	Specifies the amount of time allowed in an Undertorque Limit condition before generating an Undertorque Limit exception. A value of 0 for this attribute disables the Undertorque feature.	seconds

7.3.9.5 Current Control attributes

The attributes tables in Table 81 and Table 82 contain all current control related attributes associated with a Motion Device Axis Object instance.

Table 81 – Current Control Signal attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
520	Required – C	Get		Iq Current Command	REAL	Represents the instantaneous commanded torque producing current signal, Iq, prior to passing through the Current Vector Limiter. It is tied directly to the output of torque reference path after the 1/Kt scaling that represents the torque effort to be applied to the drive's torque producing Iq current loop. The nominal value for 1/Kt is 1 based on 100 % rated torque being produced by 100 % rated current.	% Motor Rated
521	Optional – C	Get		Operative Current Limit	REAL	Represents the operative current limit based on multiple limit sources.	% Motor Rated
522	Optional – C	Get		Current Limit Source	USINT	Represents the operative source of a current limit shall a current limit condition occur.	Enumeration: 0 = Not Limited 1 = Inverter Peak Current Limit 2 = Motor Peak Current Limit 3 = Inverter Thermal Current Limit 4 = Motor Thermal Current Limit 5 = Shunt Regulator Limit 6 = Current Vector Limit 7 = Brake Test Limit 8 to 127 = reserved 128 to 255 = vendor specific
523	Required – C (PM Motor)	Get		Motor Electrical Angle	REAL	Calculated electrical angle of the motor based on motor pole count, commutation offset, and selected feedback device.	Degrees
524	Optional – C	Get		Iq Current Reference	REAL	Current reference signal, Iq, into the torque producing current loop summing junction.	% Motor Rated
525	Optional – C	Get		Id Current Reference	REAL	Command current reference, Id, into the flux producing current loop summing junction.	% Motor Rated
527	Optional – C	Get		Iq Current Error	REAL	Error between commanded and actual current that is the output of the torque producing, q-axis, current loop summing junction.	% Motor Rated

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
528	Optional – C	Get		Id Current Error	REAL	Error between commanded and actual current that is the output of the flux producing, d-axis, current loop summing junction.	% Motor Rated
529	Optional – C	Get		Iq Current Feedback	REAL	Actual torque current applied to the axis based on current sensor feedback.	% Motor Rated
530	Optional – C	Get		Id Current Feedback	REAL	Actual flux current applied to the axis based on current sensor feedback.	% Motor Rated
531	Optional – C	Get		Vq Decoupling	REAL	Instantaneous voltage signal added to the Iq control loop output to compensate for the effects of Id.	Volts
532	Optional – C	Get		Vd Decoupling	REAL	Instantaneous voltage signal added to the Id control loop output to compensate for the effects of Iq.	Volts
533	Optional – C	Get		Vq Voltage Output	REAL	Instantaneous Torque/Force producing output voltage.	Volts
534	Optional – C	Get		Vd Voltage Output	REAL	Instantaneous Flux producing output voltage.	Volts
535	Optional – C	Get		U Voltage Output	REAL	Instantaneous voltage applied to U phase.	Volts
536	Optional – C	Get		V Voltage Output	REAL	Instantaneous voltage applied to V phase.	Volts
537	Optional – C	Get		W Voltage Output	REAL	Instantaneous voltage applied to W phase.	Volts
538	Optional – C	Get		U Current Feedback	REAL	Instantaneous current measured on U phase.	Amperes
539	Optional – C	Get		V Current Feedback	REAL	Instantaneous current measured on V phase.	Amperes
540	Optional – C	Get		W Current Feedback	REAL	Instantaneous current measured on W phase.	Amperes
541	Optional – C	Get		U Current Offset	REAL	Offset for U Phase current transducer.	Amperes
542	Optional – C	Get		V Current Offset	REAL	Offset for V Phase current transducer.	Amperes
543	Optional – C	Get		W Current Offset	REAL	Offset for W Phase current transducer.	Amperes

Table 82 – Current Control Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
553	Optional – D	Set		Current Vector Limit	REAL	Current Vector Limit value applied to current vector limiter to provide a configurable limit the magnitude of the current vector.	% Motor Rated
554	Optional – C	Set		Kqp	REAL	Iq Proportional Gain value that multiplies the Iq Current Error signal. This value directly determines the bandwidth of the torque producing current loop.	rad/s
555	Optional – C	Set		Kqi	REAL	Iq Integral Gain value that, together with Kdp, multiplies the Iq Current Error signal before applying it to the Iq Integrator Error accumulator. The reciprocal of this value, 1/Kdi, represents the integrator time constant for the flux current loop. A value of 0 for this attribute disables the integrator.	rad/s
556	Optional – C	Set		Kdp	REAL	Id Proportional Gain value that multiplies the Id Current Error signal. This value directly determines the bandwidth of the flux producing current loop.	rad/s
557	Optional – C	Set		Kdi	REAL	Id Integral Gain value that, together with Kdp, multiplies the Id Current Error signal before applying it to the Id Integrator Error accumulator. The reciprocal of this value, 1/Kdi, represents the integrator time constant for the flux current loop. A value of 0 for this attribute disables the integrator.	rad/s
558	Optional – D (Induction Motor)	Set		Flux Up Control	USINT	When the motion axis is enabled, DC current is applied to an induction motor to build stator flux before transitioning to the Running state. This attribute controls how an induction motor is to be fluxed in the Starting state prior to transitioning to the Running state. If No Delay is selected the axis transitions immediately to the Running state while the motor flux is building. With Manual Delay, the axis remains in the Starting state for the Flux Up Time to allow time for the motor to be fully fluxed. With Automatic Delay, the drive device determines the amount of time to delay to fully flux the motor based on motor configuration attribute data or measurements.	Enumeration: 0 = No Delay (R) 1 = Manual Delay (O) 2 = Automatic Delay (O) 3 to 255 = (reserved)
559	Optional – D (Induction Motor)	Set		Flux Up Time	REAL	Sets the amount of time the drive device allows to build full motor flux before transitioning to the Running state.	seconds

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
560	Optional – CE (PM Motor)	Set		Commutation Startup Method	USINT	Specifies the method used by the drive to establish absolute rotor (or moving coil) alignment relative to stator windings for the purposes of PM motor commutation when starting up the drive. If this attribute is not supported, the commutation startup method is determined implicitly by the Feedback 1 Type selection. Likewise, if this attribute is supported and set to From Feedback Type, the commutation startup method is also determined implicitly by the Feedback 1 Type selection. The UVW startup method uses UVW signals from motor mounted encoder tracks or Hall sensors together with the Commutation Offset to properly align the rotor with stator windings or, in the case of a linear motor, the moving coil with magnet track. Once aligned, commutation is maintained via position signals from the motor mounted feedback device, i.e. Feedback 1. The Digital startup method uses Digital signals from a motor mounted absolute feedback device together with the Commutation Offset to align the rotor with stator windings or, in the case of a linear motor, the moving coil with the magnet track. The Self-Sensing start-up method applies current to the motor stator during the initial Starting state to force the rotor to the Null position and thereby achieve proper commutation alignment. Once aligned, commutation is maintained via position signals from the motor mounted feedback device, Feedback 1. This method is used when there is no absolute feedback available to align the motor, e.g. a motor equipped with an incremental encoder.	Enumeration: 0 = From Feedback Type (R) 1 = UVW (O) 2 = Digital (O) 3 = Self-Sense (O) 4 to 255 = (Reserved)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
561	Required – CE (PM Motor)	Set		Commutation Offset	REAL	A value that specifies the commutation offset of the PM motor mounted feedback device in units of electrical degrees. This attribute specifies the offset from a commutation reference position defined by applying DC current into the A terminal and out of the shorted B and C terminals of the motor and allowing the rotor to move to its magnetic null position relative to the stator. On an absolute encoder or resolver, the offset is the difference from the device's zero absolute position and the commutation reference position. On an incremental encoder or Hall sensor with UVW signals, the offset is the difference between the position corresponding to a transition of the commutation device's W (S3) channel with the U (S1) channel high and the V (S2) channel low, and the commutation reference position. The commutation offset is only applicable to the motor mounted Feedback 1 device. When the optional Commutation Alignment attribute is supported and set to Controller Offset, the drive shall apply the Commutation Offset value from the controller to determine the electrical angle of the motor. In this case, a valid Commutation Offset value shall be established by the controller. In the unusual case where the commutation offset is also stored in the motor, and differs significantly from Commutation Offset value from the controller, the drive shall transition to the Start Inhibited state. If the Commutation Alignment attribute is not set to Controller Offset, the Commutation Offset value from the controller is ignored by the drive and the drive shall determine its internal commutation offset value by other means. Without a valid commutation offset, the drive shall be Start Inhibited.	Electrical Degrees
562	Optional – CE (PM Motor)	Set		Commutation Self-Sensing Current	REAL	When a PM motor feedback drive device is an incremental encoder without UVW tracks for commutation, a Self-Sensing algorithm shall be run during the Starting state that determines the Commutation Offset to apply to the position feedback. This algorithm applies a current to the motor stator to orient the rotor to establish the motor commutation phasing.	% Motor Rated
563	Optional – CE (PM Motor)	Set		Commutation Polarity	USINT	When a PM motor is using UVW signals for commutation start up, it is critical that the UVW phases of the commutation device follow the phasing of the motor. Normal polarity implies UVW phasing according to factory specification when the commutation device is moving in the factory defined positive direction. Inverted polarity effectively switches the UVW phasing to UWV thus reversing the directional sense of the commutation device. If it is determined via a Commutation Test that the phasing of the motor and the phasing of the commutation device have opposite polarity, this attribute can be used to compensate for the mismatch.	Enumeration: 0 = Normal Polarity 1 = Inverted Polarity 2 to 255 = (reserved)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
564	Optional – CE (PM Motor)	Set		Commutation Alignment	USINT	This enumerated parameter is set to Controller Offset (1) when the motor mounted absolute feedback device is to be aligned with the stator windings of the PM motor according to the Commutation Offset value. In some cases the Commutation Offset can be preset to a value established by factory alignment of the motor feedback device relative to the motor stator windings. A setting of Not Aligned (0) indicates that the motor is not aligned, and that the Commutation Offset value is not valid. If the Commutation Offset is not valid, it cannot be used by the drive to determine the commutation angle. Any attempt to enable the drive with an invalid commutation angle shall result in a Start Inhibit condition. Alignment can be achieved via a Commutation Test that measures and sets the Commutation Offset for the motor or by direct user entry. If this attribute is set to Motor Offset (2) the drive derives the commutation offset directly from the motor. If set to Self-Sense (4) the drive automatically measures the commutation offset when it transitions to the Starting state for the first time after a power cycle. This generally applies to a PM motor equipped with a simple incremental feedback device.	Enumeration: 0 = Not Aligned (R) 1 = Controller Offset (R) 2 = Motor Offset (O) 3 = Self-Sense (O) 4 to 255 = Reserved

7.3.9.6 Frequency Control attributes

The attributes tables in Table 83 and Table 84 contain all related attributes associated with the Frequency Control method of operation of a Motion Device Axis Object instance.

Table 83 – Frequency Control Signal attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
565	Required – F	Get		Slip Compensation	REAL	Indicates the actual amount of slip compensation currently being applied.	r/min

Table 84 – Frequency Control Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
570	Required – F	Set		Frequency Control Method	SINT	The Basic Volts/Hertz control method applies voltage to the motor generally in direct proportion to the commanded frequency or speed.	Enumeration 0 = Basic Volts/Hertz (R) 1 to 127 = reserved 128 to 255 = vendor specific
572	Required – F	Set		Maximum Voltage	REAL	Sets the highest phase-to-phase voltage the drive device can output.	Volts (RMS)
573	Required – F	Set		Maximum Frequency	REAL	Sets the highest frequency the drive device can output.	Hertz
575	Required – F	Set		Break Voltage	REAL	Sets the phase-to-phase output voltage of the drive device at the Break Frequency where boost ends.	Volts (RMS)
576	Required – F	Set		Break Frequency	REAL	Sets the output frequency of the drive device at the Break Voltage where boost ends.	Hertz
577	Required – F	Set		Start Boost	REAL	Sets phase-to-phase voltage boost level for starting and accelerating.	Volts (RMS)
578	Required – F	Set		Run Boost	REAL	Sets the phase-to-phase voltage boost level for steady-state speed or deceleration.	Volts (RMS)

7.3.9.7 Drive Output attributes

The attributes table in Table 85 contains all inverter output related attributes associated with a Motion Device Axis Object instance.

Table 85 – Drive Output attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
600	Required – F Optional – C	Get		Output Frequency	REAL	Output frequency applied to motor. The frequency value is in terms of electrical cycles.	Hertz
601	Required – D	Get		Output Current	REAL	Output current applied to motor.	Amperes (RMS)
602	Required – D	Get		Output Voltage	REAL	Phase-to-phase output voltage applied to motor.	Volts (RMS)
603	Required – D	Get		Output Power	REAL	Output power of the motor. This value is based on the product of the Torque Reference signal and the Velocity Feedback.	kW
604	Optional – D	Set		PWM Frequency	UINT	Sets the carrier frequency for the Pulse Width Modulation output to the motor. Drive derating is required at higher PWM frequencies due to switching losses. Current loop update time is tied directly to the PWM frequency so loop performance generally increases with increasing PWM rate.	Hertz

7.3.10 Stopping & Braking attributes

7.3.10.1 General

The attributes table in Table 86 contains all active stopping and braking related attributes associated with a Motion Device Axis Object instance.

Table 86 – Stopping/Braking attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
610	Required – D	Set		Stopping Action	USINT	When disabling or aborting an axis, via a Disable Request or an Abort Request, this value determines the stopping method to apply to the motor. The final state after the stopping method is applied is the Stopped state. In the Stopped state, the device's inverter power structure shall either be Disabled (Disable selection) and free of torque or actively held (Hold selection) in a static condition. This attribute does not, in any way, determine the stopping actions applied in response to fault conditions.	See Semantics in 7.3.10.2.1
611	Required – C	Set		Stopping Torque	REAL	When disabling or aborting an axis, this value determines the maximum amount of torque producing current available to stop the motor when the Stopping Action is set to Current Decel.	% Motor Rated
612	Optional – C	Set		Stopping Time Limit	REAL	When disabling or aborting an axis, this parameter determines the maximum amount of time the drive allows to reach zero speed.	seconds
613	Optional – D (PM Motor)	Set		Resistive Brake Contact Delay	REAL	When an external resistive brake is used, the Resistive Brake Contact Delay can be set to delay the enabling of the device power structure until after the resistive brake has had time to connect the motor to the drive device.	seconds See Semantics in 7.3.10.2.2 for details.
614	Optional – D	Set		Mechanical Brake Control	USINT	This attribute governs the operation of the drive's Mechanical Brake Output that controls the mechanical brake mechanism. When set to Automatic, the Mechanical Brake is under the control of the axis state machine. The sequencing for the brake is described in detail by the Mechanical Brake Engage Delay and Mechanical Brake Release Delay attributes. When set to Brake Release, the brake is unconditionally released, and no longer under the control of the axis state machine.	Enumeration: 0 = Automatic 1 = Brake Release 2 to 255 = (reserved)
615	Optional – D	Set		Mechanical Brake Release Delay	REAL	When enabling the axis, the Mechanical Brake Release Delay value determines the amount of time the device shall delay transition from the Starting state to the Running or Testing states. This delay prevents any commanded motion of the motion axis until the external mechanical brake has had enough time to disengage.	seconds See Semantics in 7.3.10.2.3 for details.
616	Optional – D	Set		Mechanical Brake Engage Delay	REAL	When disabling the motion axis, the Mechanical Brake Engage Delay value determines the amount of time the device power structure shall remain enabled after the axis has decelerated to stop. This allows time for an external mechanical brake to engage. The configured Stopping Action determines the type of stopping sequence applied.	seconds See Semantics in 7.3.10.2.4 for details.
608	Optional – D	Set		Zero Speed	REAL	This attribute sets the speed threshold associated with the zero speed criteria of the stop sequence. Zero Speed is specified as a percent of motor rated speed. When Zero Speed Time attribute is supported, this attribute sets the speed threshold where the zero speed timer starts. When the axis speed has been below the Zero Speed threshold for Zero Speed Time the axis has satisfied the zero speed criteria. In all but Category 2 stops, this results in action to engage the mechanical brake. If this attribute is not supported, the zero speed threshold is left to the vendor's discretion and typically set to 1% of motor rated speed. Axis speed in the above description is based on the Velocity Feedback signal, or in the case of a Frequency Control drive, axis speed is based on Velocity Reference signal.	% Motor Rated

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
609	Optional – D	Set		Zero Speed Time	REAL	This attribute sets the amount of time that the axis speed shall be below the zero speed threshold, set by the Zero Speed attribute or established by the drive vendor, before satisfying the zero speed criteria. In all but Category 2 stops, this results in action to engage the mechanical brake. If this attribute is not supported, the amount of time needed to satisfy the zero speed criteria is left to the vendor's discretion and typically immediate (0). Axis speed in the above description is based on the Velocity Feedback signal, or in the case of a Frequency Control drive, axis speed is based on Velocity Reference signal.	seconds
590	Optional – D	Set		Proving Configuration	USINT	This attribute enables the operation of the drive's Torque Proving and Brake Proving functions that work in conjunction with mechanical brake control. When Proving is enabled, the mechanical brake shall be set as soon as the drive is disabled. When the brake is under the control of the axis state machine this is automatic. But when controlled externally, failure to set the brake when the drive is disabled can cause a free fall condition on a vertical application. When enabled, the drive performs a Torque Prove test of the motor current while in the Starting state to “prove” that current is properly flowing through each of the motor phases before releasing the brake. Should the Torque Prove test fail, a Motor Phase Loss exception is generated. While Torque Proving functionality is applicable to drive Control Modes that are not capable of generating reliable holding torque based on a feedback device, e.g. Frequency Control and Sensorless Velocity Control, it should not be used in applications where holding torque is critical to safe operation, such as in a typical lift or crane application. If the optional Brake Test Torque attribute is supported, the Torque Prove test also includes a proactive Brake Test to insure the mechanical brake is functioning properly. Should the Brake Test detect brake slip, a Brake Slip exception is generated. When Proving is enabled, the drive also performs a Brake Prove test while in the Stopping or Aborting states to “prove” proper mechanical brake function before the drive power structure is disabled. Should the Brake Prove test detect brake slip a Brake Slip exception is generated. Unless another vendor specific method is used to address a Brake Slip condition in the Stopping or Aborting state, the appropriate Fault Action for the Brake Slip exception is Torque Limited Stop and Hold. This Fault Action applies holding torque to arrest the brake slip and transitions the axis to the Major Faulted state. In general, Brake Proving functionality is only applicable to drive Control Modes that are capable of generating holding torque based on a feedback device. Brake Proving is therefore is not applicable to Frequency Control or Sensorless Velocity Control modes. The sequencing of the torque and brake “prove” tests are described in detail by the Mechanical Brake Engage Delay and Mechanical Brake Release Delay attribute semantic sections below.	Enumeration: 0 = Disabled 1 = Enabled 2 to 255 = (reserved)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
591	Optional – D	Set		Torque Prove Current	REAL	This attribute sets the percent of motor rated current applied to the motor by the Torque Prove test as part of the Torque Proving function executed in the Starting state. The Torque Prove test applies current to the motor to “prove” that current is properly flowing through each of the motor phases before releasing the brake.	% Motor Rated
592	Optional – D E	Set		Brake Test Torque	REAL	This attribute sets the percent of motor rated torque applied to the motor by the Brake Test as part of the Torque Proving function executed in the Starting state. This Brake Test proactively tests the ability of the mechanical brake to hold the maximum anticipated load before releasing the brake and allowing operation. Should the Brake Test detect brake slip, a Brake Slip exception is generated. If the Brake Test Torque attribute value is 0 the Brake Test shall not be performed in the Starting state.	% Motor Rated
593	Optional – D E	Set		Brake Prove Ramp Time	REAL	This attribute determines the amount of time the drive shall take to ramp the applied torque of the motor down to zero during the Brake Proving test in the Stopping or Aborting state. The Brake Prove Ramp Time determines the ramp down rate of the applied torque output by dividing the Torque Limit by the Brake Prove Ramp Time. The Torque Limit in this case is the maximum of the configured Torque Limit Positive and Torque Limit Negative values. The Brake Prove test is performed to check for brake slip before the power structure is disabled.	seconds
594	Optional – D E	Set		Brake Slip Tolerance	REAL	This attribute determines the amount of brake slip allowed after the brake is engaged. If this tolerance is exceeded while the brake is engaged, a Brake Slip exception is generated. Brake slip can therefore be monitored in any axis state where the brake is engaged.	Position Control Units

7.3.10.2 Semantics

7.3.10.2.1 Attribute No. 610 – Stopping Action

When disabling or aborting an axis, this attribute determines the stopping method to apply to the motor (see Table 87). The final state after the stopping method is applied is the Stopped state. In the Stopped state the device's inverter power structure shall either be Disabled (Disable selection) and free of torque or actively held (Hold selection) in a static condition. This attribute has no impact or relationship to the planner generated acceleration and deceleration profiles.

Table 87 – Stopping Action enumeration definitions

Enum.	Req./Opt.	Name	Description
0	R/D	Disable & Coast	Disable & Coast immediately disables the device power structure and active control loops, which causes the motor to coast unless some form of external braking is applied. This is equivalent to an IEC 60204-1 Category 0 Stop.
1	R/C O/F	Current Decel & Disable	Current Decel & Disable leaves the power structure and any active control loops enabled while stopping. If configured for position control mode, the drive forces the position reference to hold its current value until the axis reaches zero speed. Once at zero speed the position reference is immediately set equal to the actual position to hold the axis at standstill. If in velocity control mode, the drive forces the velocity reference to zero. In either case, forcing the position or velocity reference signals to a fixed value results in a rapid increase in torque reference of the moving axis that saturates the torque producing current of the drive to the configured Stopping Torque that brings the motor to a stop. Once stopped, or the configured Stopping Time limit expires, the drive disables the power structure and control loops. This stop mode complies with the IEC 60204-1 Category 1 Stop. In frequency control mode the Current Vector Limit attribute, rather than the Stopping Torque attribute, is used to regulate the stopping current.
2	O/FV	Ramped Decel & Disable	Ramped Decel & Disable also leaves the power structure and any active control loops enabled while stopping but uses the Ramp Generator associated with the Velocity Fine Command Generator block to decelerate the motor to a stop. When initiating a Ramped Decel & Disable Stop, the Ramp Generator is immediately activated and the drive no longer follows command from the controller. The Ramp Generator input is initialized to zero and the output is initialized to the current speed of the motor, thus causing the Ramp Generator output to ramp the motor from its current speed down to zero according to the ramp control parameters. Once stopped, or the configured Stopping Time or factory timeout limit expires, the device disables the power structure and control loops. This stop mode also complies with the IEC 60204-1 Category 1 Stop.
3	O/PV	Current Decel & Hold	Current Decel & Hold behaves like Current Decel & Disable, but leaves the power structure active with holding torque to maintain the stopped condition. The method for generating holding torque is left to the drive vendor's discretion. This stop mode complies with the IEC 60204-1 Category 2 Stop.
4	O/PV	Ramped Decel & Hold	Ramped Decel & Hold behaves like Ramped Decel & Disable, but leaves the power structure active with holding torque to maintain the stopped condition. The method for generating holding torque is left to the drive vendor's discretion. This stop mode also complies with the IEC 60204-1 Category 2 Stop.
5-127		(Reserved)	-
128-255		(Vendor Specific)	-

7.3.10.2.2 Attribute No. 613 – Resistive Brake Contact Delay

When an external resistive brake is used, an external contactor switches the ABC motor leads from the inverter power structure to an energy dissipating resistor to stop the motor. This switching does not occur instantaneously and so enabling the power structure too early can cause electrical arcing across the contactor. To prevent this condition, the Resistive Brake Contact Delay can be set to the maximum time that it takes to fully close the contactor across the ABC motor lines so when the axis is enabled, the inverter power structure is not enabled until after the Resistive Brake Contact Delay Time has expired. Resistive Brake operation is only applicable to PM Motor types. The following sequence further defines how the Resistive Brake Contact Delay factors into the overall Enable Sequence that may also include the operation of a Mechanical Brake.

Enable Sequence:

- 1) Switch to Starting state.
- 2) Activate Resistive Brake contactor to connect motor to inverter power structure.
- 3) Wait for “Resistive Brake Contact Delay” while Resistive Brake contacts close.
- 4) Enable inverter power structure.
- 5) Activate Mechanical Brake output to release brake.
- 6) Wait for “Mechanical Brake Release Delay” while brake releases.
- 7) Transition to Running state.

7.3.10.2.3 Attribute No. 615 – Mechanical Brake Release Delay

When enabling the axis with an engaged mechanical brake, the Mechanical Brake Release Delay value determines the amount of time the drive shall delay transition from the Starting state to the Running or Testing states. This delay prevents any commanded motion of the motion axis until the external mechanical brake has had enough time to disengage. If supported, a Torque Proving operation is included in this sequence prior to releasing the brake.

Enable Sequence:

- 1) Switch to Starting state.
- 2) Activate Resistive Brake contactor to connect motor to inverter power structure.
- 3) Wait for “Resistive Brake Contact Delay” while Resistive Brake contacts close.
- 4) Enable inverter power structure.
- 5) Perform (optional) Torque Proving operation to verify motor control of load.
- 6) Activate Mechanical Brake output to release brake.
- 7) Wait for “Mechanical Brake Release Delay” while brake releases.
- 8) Transition to Running (or Testing) state.

7.3.10.2.4 Attribute No. 616 – Mechanical Brake Engage Delay

7.3.10.2.4.1 General

When disabling the motion axis using a Category 1 Stopping Action, the Mechanical Brake Engage Delay value determines the amount of time the inverter power structure shall remain enabled after the axis has decelerated to a stop. This allows time for an external mechanical brake to engage. The configured Stopping Action determines the type of stopping sequence applied. If supported, a Brake Proving operation is included in the Category 1 stopping sequence prior to disabling the power structure.

7.3.10.2.4.2 Disable Sequence (Category 0 Stop)

Inverter is immediately disabled. Brake Proving is not applicable. The corresponding sequence is specified below and illustrated in Figure 65.

- 1) Switch to Stopping state.
- 2) Disable inverter power structure.
- 3) Wait for zero speed (see 7.3.10.2.4.5) or "Stopping Time Limit" or a factory set timeout, whichever occurs first.
- 4) Transition to Stopped state.
- 5) Deactivate Mechanical Brake output to engage brake.
- 6) Deactivate Resistive Brake contactor to disconnect motor from inverter power structure.

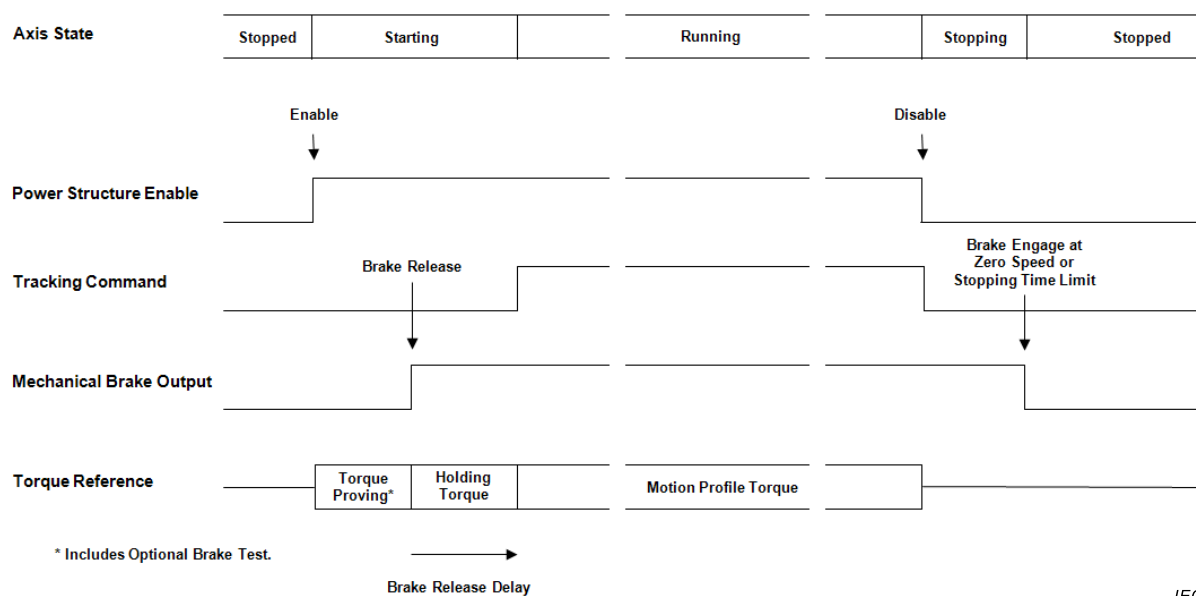
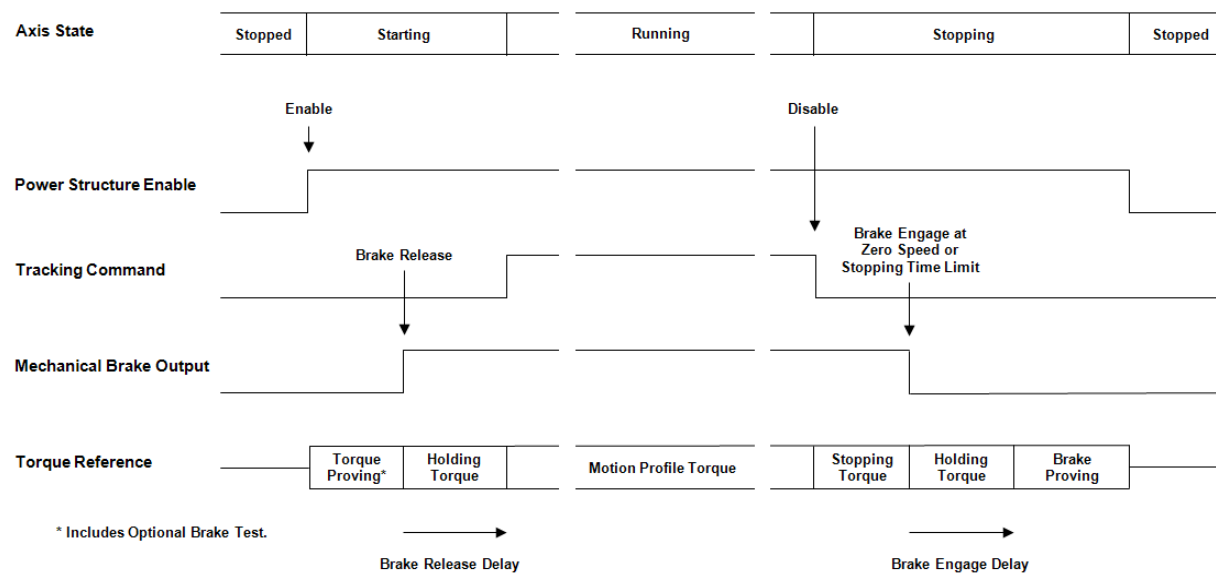


Figure 65 – Brake Control Sequence (Category 0 Stop)

7.3.10.2.4.3 Disable Sequence (Category 1 Stop)

Torque is applied to stop motor. Brake Proving is applicable. The corresponding sequence is specified below and illustrated in Figure 66.

- 1) Switch to Stopping state.
- 2) Apply "Current Decel" or "Ramp Decel" method to stop motor.
- 3) Wait for zero speed (see 7.3.10.2.4.5) or "Stopping Time Limit" or factory set timeout, whichever occurs first.
- 4) Deactivate Mechanical Brake output to engage brake.
- 5) Wait for "Mechanical Brake Engage Delay" while brake engages.
- 6) Perform (optional) Brake Proving operation to verify brake control of load.
- 7) Disable inverter power structure.
- 8) Transition to Stopped state.
- 9) Deactivate Resistive Brake contactor to disconnect motor from inverter power structure.



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Figure 66 – Brake Control Sequence (Category 1 Stop)**7.3.10.2.4.4 Disable Sequence (Category 2 Stop)**

The mechanical brake is not used. Brake Proving is not applicable. The corresponding sequence is specified below and illustrated in Figure 67.

- 1) Switch to Stopping state.
- 2) Apply “Current Decel” or “Ramp Decel” method to stop motor.
- 3) Wait for zero speed (see 7.3.10.2.4.5) or “Stopping Time Limit” or factory set timeout, whichever occurs first.
- 4) Transition to Stopped state.

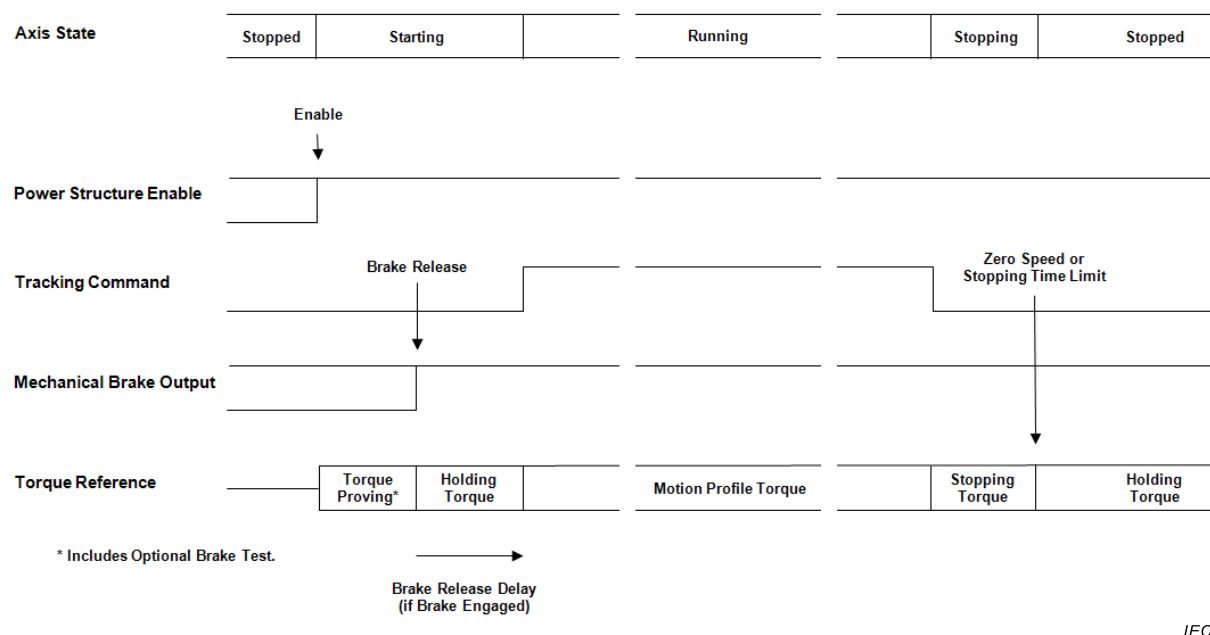


Figure 67 – Brake Control Sequence (Category 2 Stop)

7.3.10.2.4.5 Zero speed criteria

Recommended criteria for zero speed is based on Velocity Feedback, or in the case of a Frequency Control drive, based on Velocity Reference. Zero speed criteria can be established explicitly via optional Zero Speed and Zero Speed Time attributes or implicitly as 1 % of motor rated speed or left to the drive vendor's discretion.

7.3.10.2.5 Attribute No. 590 – Proving Configuration

The Proving feature includes a number of optional sub-features, many of which depend on support of other Proving feature attributes. Table 88 defines these attribute dependencies.

Table 88 – Proving sub-feature attribute dependencies

Proving sub-feature	Controlling attributes	Attribute prerequisites
Torque Prove		Proving Configuration
Brake Test	Brake Test Torque Brake Slip Tolerance	Proving Configuration
Brake Prove	Brake Prove Ramp Time Brake Slip Tolerance	Proving Configuration

Proving tests are performed when enabling or disabling the drive axis. During these state transitions a series of operations are performed by the drive to insure the proper function of the motor (Torque Proving) and the brake (Brake Proving). The flow charts in Figure 68 Figure 67and Figure 69 define these operational sequences in the context of a drive enable transition and a drive disable or abort transition.

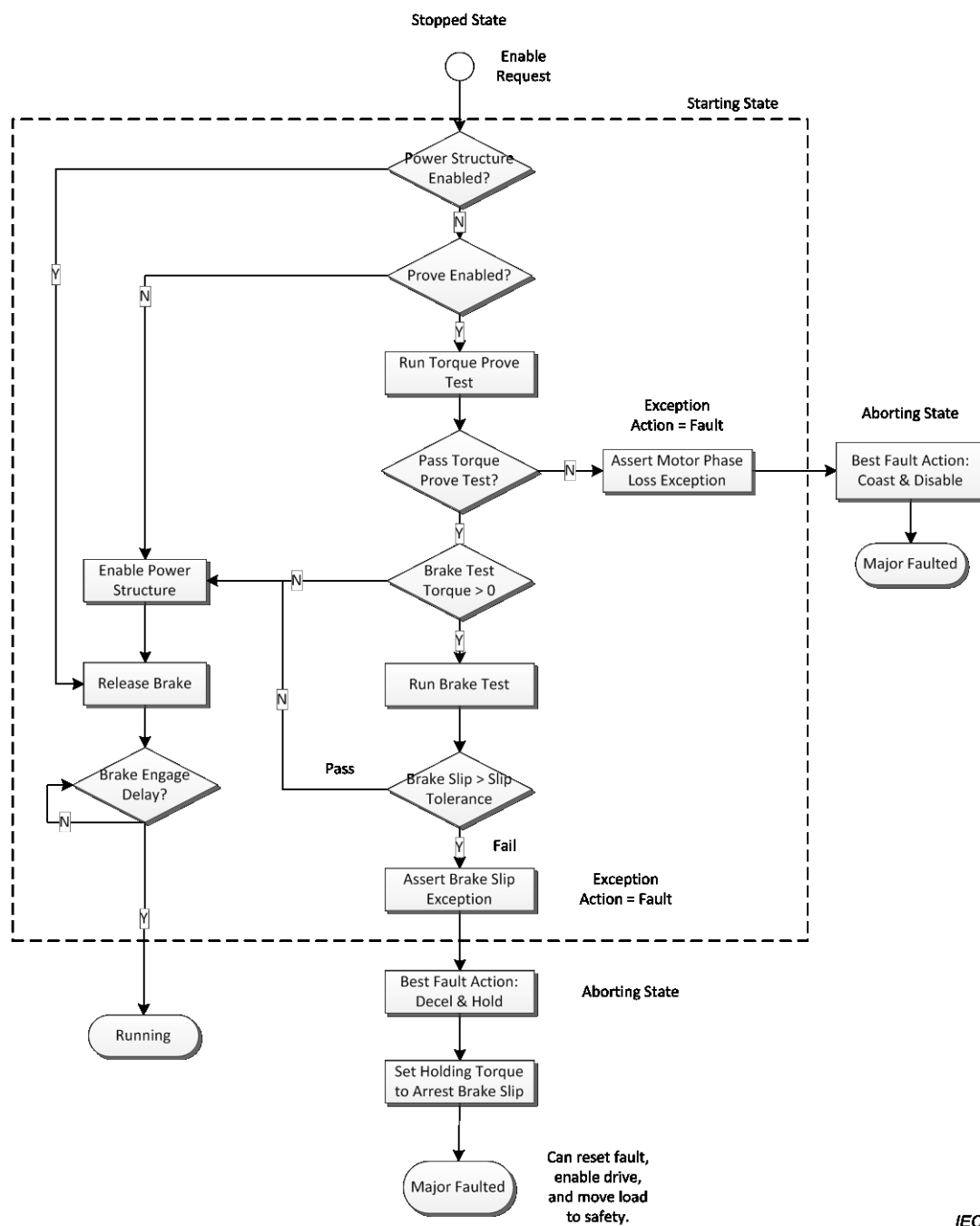
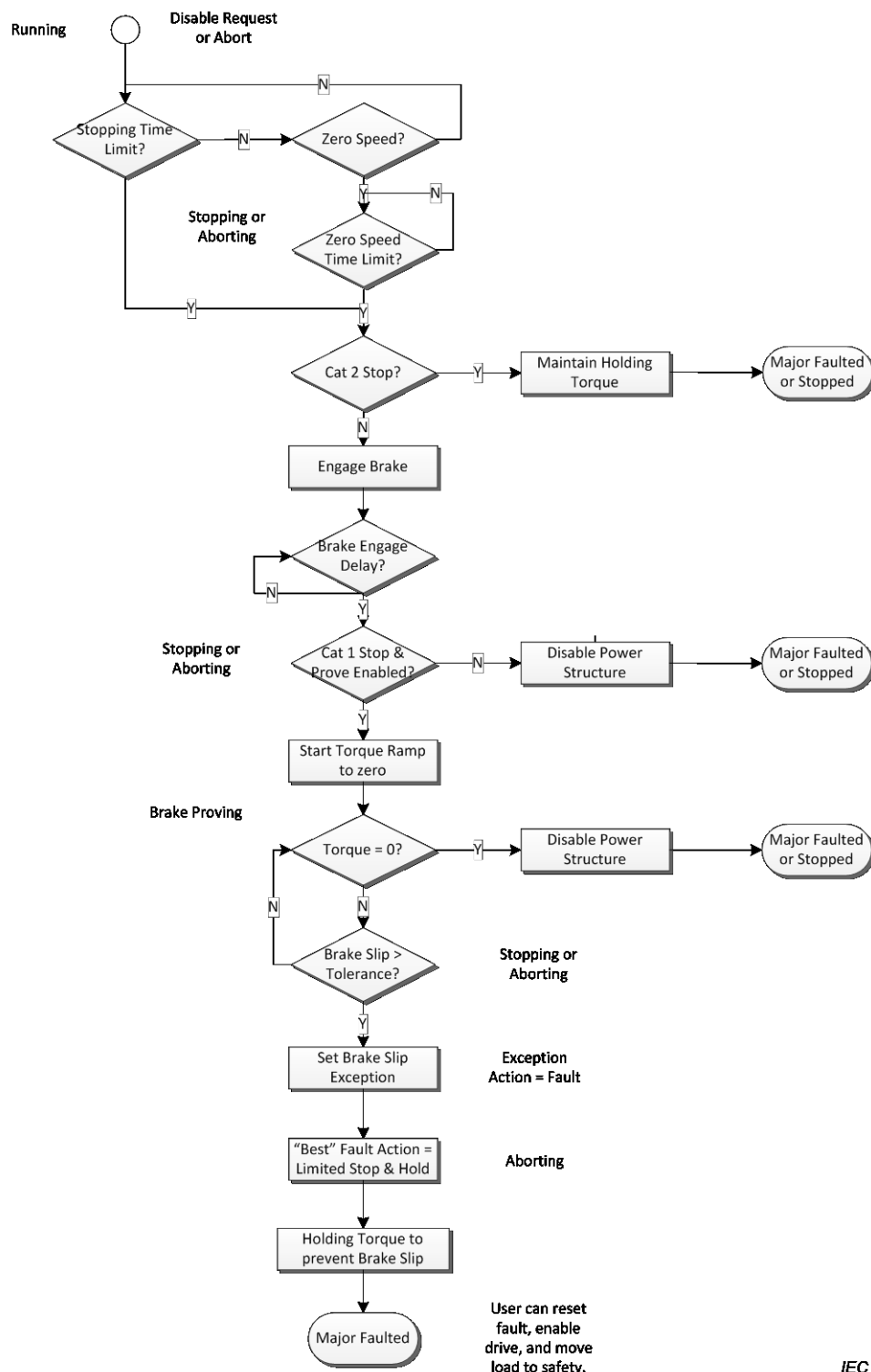


Figure 68 – Drive Enable sequence with Proving feature



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Figure 69 – Drive Disable sequence with Proving feature

7.3.11 DC Bus Control attributes

The attributes table in Table 89 contains Motion Device Axis Object instance attributes associated with DC Bus control including functionality to address both under-voltage and over-voltage conditions.

Table 89 – DC Bus Control attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
620	Required – BD	Get		DC Bus Voltage	REAL	Measured DC Bus Voltage.	Volts
621	Required – BD	Get		DC Bus Voltage – Nominal	REAL	Normal DC Bus Voltage during operation as determined by averaging the DC Bus Voltage over a device specific time interval. This value is used as the basis for Bus Overvoltage and Undervoltage limits. NOTE If the device does not support this bus voltage averaging concept, the alternative is to hard code this value.	Volts
622	Optional – BD	Set		Bus Configuration	USINT	This enumerated selection indicates how the DC Bus is going to be used. Standalone indicates that DC Bus power supplied by the drive's converter section is applied only to this drive's power structure. Shared AC/DC indicates that the converter associated with this CIP Motion device is to supply and share DC Bus power with other drives. This would typically result in de-rating of the continuous current rating for the Shared AC/DC drive. Shared DC indicates that this drive is sharing DC bus power generated by another Shared AC/DC drive. Shared DC – Non CIP Converter indicates that this drive is receiving DC bus power generated by an external AC/DC converter that is not CIP Motion compliant and distributing its DC bus power to other CIP Motion drives. A drive configured for Shared DC – Non CIP Converter is responsible for communicating the status of the external converter to the control system as if the external converter were integrated with the drive. Specifically, this communication includes the Bus Up and DC Bus Unload status bits reflecting the current state of associated external converter.	Enumeration: 0 = Standalone 1 = Shared AC/DC 2 = Shared DC 3 = Shared DC – Non-CIP Converter 4 to 255 = (reserved)
623	Optional – D	Set		Bus Voltage Select	USINT	This value indicates the expected bus voltage level of the drive application. High bus voltage selection is usually associated with drive running on the North American power grid, while operating in Europe a Low Bus Voltage selection would be appropriate. This parameter can be used to compensate for these different bus voltage levels in the current loop.	Enumeration: 0 = High (115 V, 230 V, 460 V) 1 = Low (100 V, 200 V, 400 V) 2 to 255 = (reserved)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
624	Required – BD (Integral Converter)	Set		Bus Regulator Action	USINT	Controls the method of operation of the DC Bus Regulator that addresses the regenerative over-voltage conditions that can occur when decelerating a motor. If Disabled, no regulation is applied to the DC Bus level to control regenerative energy sourced by the motor. This selection is only valid when applied to an Induction Motor where the motor may be allowed to coast to a stop. When Shunt Regulator is selected the associated shunt regulation hardware is applied to the DC Bus to dissipate regenerative energy via an internal or external resistor.	Enumeration: 0 = Disabled (R) 1 = Shunt Regulator (R) 3 to 127 = (reserved) 128 to 255 = (vendor specific)
625	Optional – D (Integral Converter)	Set		Regenerative Power Limit	REAL	Limits the amount of power allowed to transfer between the motor and the DC Bus during regenerative braking of the motor load. When using an external shunt resistor, set this value to its maximum value. Since this is regenerative power, the value of the limit is negative.	% Motor Rated
627	Optional – D	Set		Power Loss Action	USINT	Set the reaction to a DC Bus under-voltage condition when the DC Bus drops below a hard-coded threshold in the device or the configured Power Loss Threshold. This provides a specific (configured) response to an incoming power loss while the drive/motor is running. A continue action selection configures the drive to ignore the power loss condition and continue to run for as long as possible. A Bus Undervoltage exception may occur if the DC Bus Voltage falls below the Factory or User Limits. Otherwise, operation will continue until the low voltage power supplies drop out. There may be concerns operating the power structure below the point where the gate drives start to loose power where this could possibly result in drive damage. The Bus Undervoltage Exception Actions shall be set accordingly. A coast thru action selection configures the drive to zero the PWM output of the drive while leaving the axis in the Running state. If the incoming power returns before the timeout period, given by the Power Loss Time, the drive automatically starts to control the motor again. If, however, the power doesn't return before Power Loss Timeout period expires, a Bus Power Loss Exception is generated. The decel regen action selection configures the drive to regeneratively charge the DC bus by decelerating the motor using the bus regulator to regulate the bus voltage at a predetermined level. When incoming power is restored the drive returns to normal operation. If, however, the drive reaches zero speed or the Power Loss Time period expires before the incoming power has restored, the drive power structure is disabled and a Bus Power Loss exception is generated.	Enumeration: 0 = Continue (Ignore) (R) 1 = Coast Thru (R) 2 = Decel Regen (O) 3 to 127 = (reserved) 128 to 255 = (vendor specific)
628	Optional – BD	Set		Power Loss Threshold	REAL	Sets the Level for Power Loss as percent of nominal DC Bus Voltage. If this value is 0, the hard-coded threshold is used.	%

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
629	Optional – BD	Set		Shutdown Action	USINT	Selects the device's action when transitioning to the Shutdown state as a result of a Shutdown Request or in response to an Exception Action where the best available stopping action is Shutdown. Default action is to immediately disable the device's power structure. If Drop DC Bus is selected, action can be taken to drop the DC Bus voltage as well. This is generally done by opening an AC Contactor Enable output provided by the drive that controls power to the converter.	Enumeration: 0 = Disable (R) 1 = Drop DC Bus (O) 2 to 127 = (reserved) 128 to 255 = (vendor specific)
630	Optional – BD	Set		Power Loss Time	REAL	Sets the timeout value before a Bus Power Loss Exception is generated by the drive in response to a Power Loss condition. See Power Loss Action for details.	seconds

7.3.12 Power and thermal management attributes

The attribute tables in Table 90 and Table 91 contain all power and thermal related attributes associated with a Motion Device Axis Object instance.

Table 90 – Power and Thermal Management Status attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
634	Optional – BD	Get		Integral Converter	BOOL	This boolean value indicates whether the axis is associated with an integral converter (power supply) that provides DC Bus Power and Bus Regulation.	0 = No 1 = Yes
635	Required – D	Get		Motor Capacity	REAL	Real-time estimate of the continuous rated motor thermal capacity utilized during operation based on the motor thermal model. A value of 100 % would indicate that the motor is being used at 100 % of rated thermal capacity as determined by the continuous current rating of the motor.	% Motor Rated
636	Required – D	Get		Inverter Capacity	REAL	Real-time estimate of the continuous rated inverter thermal capacity utilized during operation based on the inverter thermal model. A value of 100 % would indicate that the inverter is being used at 100 % of rated thermal capacity as determined by the continuous current rating of the inverter.	% Inverter Rated
637	Optional – BD (Integral Converter)	Get		Converter Capacity	REAL	Real-time estimate of the continuous rated converter thermal capacity utilized during operation based on the converter thermal model. A value of 100 % would indicate that the converter is being used at 100 % of rated thermal capacity as determined by the continuous current rating of the converter.	% Converter Rated
638	Optional – BD (Integral Converter)	Get		Bus Regulator Capacity	REAL	Real-time estimate of the continuous rated bus regulator thermal capacity utilized during operation based on the bus regulator thermal model. A value of 100 % would indicate that the bus regulator is being used at 100 % of rated thermal capacity as determined by the continuous current rating of the bus regulator.	% Regulator Rated
639	Optional – BD	Get		Ambient Temperature	REAL	Current internal ambient temperature of the device enclosure.	°C
640	Optional – D	Get		Inverter Heatsink Temperature	REAL	Current temperature of the device's inverter heatsink, typically based on an embedded temp sensor.	°C
641	Optional – D	Get		Inverter Temperature	REAL	Current temperature of the power block used in the inverter's power structure, sometime referred to as the semiconductor junction temperature.	°C

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
642	Optional – D	Get		Motor Temperature	REAL	Current temperature of the motor stator, typically based on an embedded temp sensor.	°C
643	Optional – E	Get		Feedback 1 Temperature	REAL	Current temperature of the Feedback 1 device.	°C
644	Optional – E	Get		Feedback 2 Temperature	REAL	Current temperature of the Feedback 2 device.	°C
645	Optional – D	Get		Inverter Overload Limit	REAL	Factory set maximum limit for Inverter Capacity that when exceeded triggers the selected Inverter Overload action.	% Inverter Rated

Table 91 – Power and Thermal Management Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
646	Optional – D	Set		Motor Overload Action	USINT	Selects the device's response to a motor overload condition based on an I2t based motor thermal model. The motor overload condition occurs when the motor thermal model indicates that the Motor Capacity has exceeded the Motor Overload Limit. This motor overload action functionality is independent of the motor overload exception action functionality. No explicit action is taken by the device in the overload condition if None is the selected overload action. Selecting the Current Foldback action, however, results in a reduction of the current reference via the Motor Thermal Current Limit attribute value that is reduced in proportion to the percentage difference between Motor Capacity and the Motor Overload Limit.	Enumeration : 0 = None (R) 1 = Current Foldback (O) 2 to 127 = (reserved) 128 to 255 = (vendor specific)
647	Optional – D	Set		Inverter Overload Action	USINT	Selects the device's response to an inverter overload alarm condition based on an I2t based inverter thermal model. The inverter overload alarm condition occurs when the inverter thermal model indicates that the Inverter Capacity has exceeded the Inverter Overload Limit. The Inverter Overload Action provides opportunities to mitigate the overload condition without stopping operation. This inverter overload action functionality is independent of the motor overload exception action functionality. No explicit action is taken by the device in the overload condition if None is the selected overload action. Selecting the Current Foldback action, however, results in a reduction of the current command via the Inverter Thermal Current Limit attribute value that is reduced in proportion to the percentage difference between Inverter Capacity and the Inverter Overload Limit.	Enumeration : 0 = None (R) 1 = Current Foldback (O) 2 to 127 = (reserved) 128 to 255 = (vendor specific)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
648	Optional – D	Set		Duty Select	USINT	This value indicates the duty level of the drive application and balances the continuous and intermittent overload capacity of the drive and motor accordingly. Normal Duty provides nominal continuous rating and nominal overload capacity. Heavy Duty provides highest overload capacity at the expense of a lower continuous rating. Light Duty provides highest continuous rating at the expense of lower overload capacity Specification for the continuous and overload ratings understr Normal, Heavy and Light Duty are left to the discretion of the drive vendor. Duty Select is used to determine the level of thermal protection for the motor and the inverter during drive operation.	Enumeration : 0 = Normal (R) 1 = Heavy (O) 2 = Light (O) 3 to 255 = (reserved)

7.3.13 Axis Status attributes

7.3.13.1 General

The attributes table in Table 92 contains all status attributes associated with a Motion Device Axis Object instance.

Table 92 – Axis Status attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
650	Required – All	Get		Axis State	USINT	Enumerated value indicating the state of the motion axis.	Enumeration: 0 = Initializing 1 = Pre-Charge 2 = Stopped 3 = Starting 4 = Running 5 = Testing 6 = Stopping 7 = Aborting 8 = Major Faulted 9 = Start Inhibited 10 = Shutdown 11 to 255: Reserved
651	Required – All	Get		Axis Status	DWORD	Collection of standard bits indicating the internal status of the motion axis.	See Semantics in 7.3.13.2.1
652	Required – All	Get		Axis Status – Mfg	DWORD	Collection of vendor specific bits indicating the internal status of the motion axis.	Bitmap: 0 to 31: Vendor Specific (Published in Product Manual)
653	Required – All	Get		Axis I/O Status	DWORD	The Axis I/O Status attribute is a 32-bit collection of bits indicating the state of standard digital inputs and outputs associated with the operation of this motion axis. A value of zero for a given input bit indicates a logical 0 value, while a value of 1 indicates a logical 1 value.	See Semantics in 7.3.13.2.2
654	Required – All	Get		Axis I/O Status – Mfg	DWORD	Collection of bits indicating the state of vendor specific digital inputs associated with the operation of this motion axis. A value of zero for a given input bit indicates a logical 0 value, while a value of 1 indicates a logical 1 value.	Bitmap: 0 to 31: Vendor Specific (Published in Product Manual)

7.3.13.2 Semantics

7.3.13.2.1 Attribute No. 651 – Axis Status

The Axis Status attribute is a 32-bit collection of bits indicating various internal status conditions of the axis instance, as specified in Table 93. Any status bits that are not applicable shall be set to 0.

Table 93 – Axis Status bit definitions

Bit	Req./Opt.	Name	Description
0	R	Local Control	This bit is set if axis is taking command reference and services from local interface instead of the remote (CIP Motion) interface. This bit is based on the current state of the Remote Mode bit of the Node Status attribute (see 7.2.2.2).
1	R	Alarm	This bit is set if the axis has detected one or more exception conditions configured to generate an alarm. This bit is clear if there are no current axis alarm conditions (see 7.3.13).
2	R/BD	DC Bus Up	This bit is set if DC Bus has charged up to an operational voltage level based on direct measurement and, if applicable, the Converter Bus Up Status bit associated with external converter(s) supplying DC Bus power to this device. When an axis is in the Pre-charge state, transition of the DC Bus Up status bit from 0 to 1 initiates state transition to the Stopped State (drive axis) or to the Running Up state (converter axis). Once set, the DC Bus Up bit is cleared when the DC Bus voltage has dropped below an operational voltage level, independent of the state of the Converter Bus Up Status bit (see 7.6.2.5).
3	R/D	Power Structure Enabled	This bit is set if the axis amplifier is energized and capable of generating motor flux and torque. The value of the Power Structure Enabled bit is determined by the Axis State (see 7.6.1) and the configured Stopping Action attribute value (see 7.3.10.2.1).
4	R/D	Motor Flux Up	This bit is set if motor flux for an induction motor has reached an operational level. Transition of the Motor Flux Up bit is initiated in the Starting State (see 7.6.2.7) according to the configured Flux Up Control attribute value (see 7.3.9.5). This bit is only applicable to Induction Motor types.
5	R/D	Tracking Command	This bit is set if the axis control structure is actively tracking the command reference from motion planner. The Tracking Command bit is directly associated with the Running state of the Axis State Model (see 7.6.2.8).
6	R/P	Position Lock	This bit is set if the actual position is within Position Lock Tolerance of command position (see 7.6.9.3).
7	R/PV	Velocity Lock	This bit is set if the velocity feedback signal is within Velocity Lock Tolerance of the unlimited velocity reference (see 7.6.9.5).
8	R/ED	Velocity Standstill	This bit is set if the velocity feedback signal is within Velocity Standstill Window of 0. For a Frequency Control drive this bit is set if the velocity reference signal is within Velocity Standstill Window of 0 (see 7.6.9.5).
9	R/ED	Velocity Threshold	This bit is set if the absolute velocity feedback signal is below Velocity Threshold. For a Frequency Control drive this bit is set if the absolute velocity reference signal is below Velocity Threshold (see 7.6.9.5).
10	R/FPV	Velocity Limit	This bit is set if the velocity reference signal is currently being limited by the Velocity Limiter (see 7.6.9.6.2).
11	R/C	Acceleration Limit	This bit is set if the acceleration reference signal is currently being limited by the Acceleration Limiter (see 7.6.9.8.2).
12	R/C	Deceleration Limit	This bit is set if the acceleration reference signal is currently being limited by the Deceleration Limiter (see 7.6.9.8.2).
13	R/C	Torque Threshold	This bit is set if the absolute filtered torque reference is above the Torque Threshold (see 7.6.9.9).
14	R/C	Torque Limit	This bit is set if the filtered torque reference is currently being limited by the Torque Limiter (see 7.6.9.9.7).
15	R/C	Current Limit	This bit is set if the command current, I_q , is currently being limited by the Current Vector Limiter (see 7.6.9.10.2).
16	R/C	Thermal Limit	This bit is set if Current Vector Limit condition of the axis is being limited by any of the axis's Thermal Models (see 7.6.9.10.2).

Bit	Req./Opt.	Name	Description
17	R/E	Feedback Integrity	This bit, when set, indicates that the feedback device is accurately reflecting changes to axis position, and there have been no conditions detected that would compromise the quality of the feedback position value. The bit is set at power-up assuming that the feedback device passes any power-up self test required. If during operation a feedback exception occurs that could impact the fidelity of axis position, the bit is immediately cleared. The bit remains clear until either a fault reset is executed by the drive or the drive is power cycled. Note that the Feedback Integrity bit behavior applies to both absolute and incremental feedback device operation.
18	R/BD	Shutdown	This bit is set when the axis is in the shutdown state or in the faulted state but would transition to the shutdown state if the faults were cleared. Therefore, the Shutdown bit is closely associated with the Shutdown State of the Axis State Model (see 7.6.2.14).
19	R	In Process	This bit is set for the duration of an active process. An example of active process would be an operation initiated by a service request such as, Run_Motor_Test, Run_Hookup_Test, or Run_Inertia_Test. The In Process bit should be set as soon as the process is initiated and should remain set until the process is complete. An active process that requires the enabling of the axis power structure results in a transition to the Testing State of the Axis State Model (see 7.6.2.9).
20	O/BD	DC Bus Unload	This bit is set by a CIP Motion converter or a CIP Motion drive, containing an integral converter, or a CIP Motion drive connected to an external non-CIP converter, to indicate that the converter cannot continue supplying DC Bus power to other drives on a shared DC Bus. This is usually the result of a shutdown fault action initiated by the drive or converter, or a shutdown request from the controller. Thus, when the DC Bus Unload bit is set, the Shutdown bit (bit 18) should also be set. When there is no AC Contactor Enable output to drop the DC Bus, a method is needed to unload the converter from all other drives sharing the DC Bus. By monitoring the DC Bus Unload status bit, the control system can propagate a Bus Power Sharing exception to all drives on the shared DC Bus that are configured for Shared AC/DC or Shared DC operation. This Bus Power Sharing exception invokes the configured Exception Action that, by default, disables the drive power structure, thereby unloading the bus. The Bus Power Sharing Fault on these drives is a persistent fault that cannot be cleared as long as the DC Bus Unload bit is set on this originating drive or converter. The control system shall simply regenerate the Bus Power Sharing Faults based on the DC Bus Unload status bit still being set. Note that only the originating drive or converter with the DC Bus Unload condition can cause Bus Power Sharing Faults on other shared drives. In other words, no device with a Bus Power Sharing Fault can cause a Bus Power Sharing exception on other shared drives by setting its DC Bus Unload bit. This qualification prevents DC Bus recovery deadlock. To recover full DC Bus operation, the originating drive with the DC Bus Unload condition shall first be reset via a Shutdown Reset Request. Once clear, the Bus Power Sharing Faults on the shared drives can then be successfully cleared by either a Fault Reset Request, or a Shutdown Reset Request, allowing these drives to become operational.

Bit	Req./Opt.	Name	Description
21	O/BD	AC Power Loss	This bit is set when a CIP Motion converter, or a CIP Motion drive containing an integral converter, or a CIP Motion drive connected to an external non-CIP converter, has detected a loss of AC input power. This bit is cleared when AC input power is determined to be sufficient for converter operation. When an AC Power Loss condition is detected by a converter supplying power to other drives over a shared DC Bus, a method is needed to generate a Converter AC Power Loss exception on any drive whose power structure is enabled. To accomplish this, the control system monitors the AC Power Loss status bits of converters supplying DC Bus power and propagates AC Power Loss status to all drives on the shared DC Bus, i.e. drives that are configured for Shared AC/DC or Shared DC operation. Upon notification of AC Power Loss, drives that have enabled power structures shall assert a Converter AC Power Loss exception and invoke the programmed exception action. Disabled drives shall not generate an exception action on AC Power Loss. Thus, no drive faults shall occur on removal of AC Power from a converter as long as no drive power structures drawing power from that converter are enabled.
22 to 31		Reserved	-

Many of the Axis Status bits defined in Table 93 are related to the current Axis State as shown in Table 94.

Table 94 – Axis Status bit vs. Axis State

Bit	Axis Status bit name	Initializing	Pre-Charge	Shutdown	Start Inhibit	Stopped	Starting	Running	Testing	Stopping	Aborting	Major Faulted
0	Local Control	x	x	x	x	x	x	x	x	x	x	x
1	Alarm	x	x	x	x	x	x	x	x	x	x	x
2	DC Bus Up	x	0	x	1	1	1	1	1	1	1	1
3	Power Structure Enabled	0	0	0	0	0/1	1	1	1	1	1	0/1
4	Motor Flux Up (IM only)	0	0	0	0	0/1	1	1	1	1	1	0/1
5	Tracking Command	0	0	0	0	0	0	1	0	0	0	0
6	Position Lock	0	0	0	0	0/x	x	x	0	x	x	0/x
7	Velocity Lock	0	0	0	0	0/x	x	x	0	x	x	0/x
8	Velocity Standstill	0	0	0	0	0/x	x	x	0	x	x	0/x
9	Velocity Threshold	0	0	0	0	0/x	x	x	0	x	x	0/x
10	Velocity Limit	0	0	0	0	0/x	x	x	0	x	x	0/x
11	Acceleration Limit	0	0	0	0	0/x	x	x	0	x	x	0/x
12	Deceleration Limit	0	0	0	0	0/x	x	x	0	x	x	0/x
13	Torque Threshold	0	0	0	0	0/x	x	x	0	x	x	0/x
14	Torque Limit	0	0	0	0	0/x	x	x	0	x	x	0/x
15	Current Limit	0	0	0	0	0/x	x	x	x	x	x	0/x
16	Thermal Limit	x	x	x	x	x	x	x	x	x	x	x
17	Feedback Integrity	x	x	x	x	x	x	x	x	x	x	x
18	Shutdown	0	0	1	0	0	0	0	0	0	x	x
19	In Process	0	x	x	x	x	0	0	1	0	0	0
20	DC Bus Unload	0	0	x	0	0	0	0	0	0	x	x
Key: x = 0 or 1 0/1 = 0 for Category 0 or 1 Stop, 1 for Category 2 Stop 0/x = 0 for Category 0 or 1 Stop, x for Category 2 Stop												

The correspondence between Stopping Action and Stop Category is given in Table 95.

Table 95 – Stopping Action vs. Stop Category

Stopping Action enum.	Stopping Action name	IEC-60204-1
0	Disable & Coast	Category 0 Stop.
1	Current Decel & Disable	Category 1 Stop.
2	Ramped Decel & Disable	Category 1 Stop.
3	Current Decel & Hold	Category 2 Stop.
4	Ramped Decel & Hold	Category 2 Stop.

7.3.13.2.2 Attribute No. 653 – Axis I/O Status

The Axis I/O Status attribute is a 32-bit collection of bits indicating the state of standard digital inputs and outputs associated with the operation of this motion axis, as specified in Table 96.

A value of zero for a given input bit indicates a logical 0 value, while a value of 1 indicates a logical 1 value. Any status bits that are not applicable shall be set to 0.

Table 96 – Axis I/O Status bit definitions

Bit	Req./Opt.	Name
0	R	Enable Input
1	R/E	Home Input
2	R/E	Registration 1 Input
3	O/E	Registration 2 Input
4	R/P	Positive Overtravel Input
5	R/P	Negative Overtravel Input
6	O/E	Feedback 1 Thermostat
7	O/D	Resistive Brake Output
8	O/D	Mechanical Brake Output
9 to 31		Reserved

7.3.14 Exception, fault, and alarm attributes

7.3.14.1 General

The attribute table in Table 97 contains all exception, fault, and alarm related attributes associated with a Motion Device Axis Object instance. Exceptions are conditions that can occur during motion axis operation that have the potential of generating faults or alarms.

Table 97 – Exception, Fault and Alarm attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
655	Required – All	Get		Axis Exceptions	LWORD	A bit map that represents the current state of all standard exception conditions. See the Standard Exception Table bit map definition in 7.3.14.2 for detail on the bit locations. Each exception has a corresponding Axis Exception Action. Exceptions that are configured to be Ignored are only visible in this attribute.	See Semantics in 7.3.14.2.
656	Required – All	Get		Axis Exceptions – Mfg	LWORD	A bit map that represents the current state of all manufacturer specific exception conditions. See the Mfg. Exception Table published in drive product manual. Each exception has a corresponding Axis Exception Action. Exceptions that are configured to be Ignored are only visible in this attribute.	See Mfg. Exception Table (Published in Product Manual)
657	Required – All	Get		Axis Faults	LWORD	A bit map that represents the state of all standard runtime faults. The bit map is identical to that of the Axis Exceptions attribute. Fault bits when set are latched until a fault reset occurs. A fault reset clears the runtime fault bits, but the bits set again immediately if the underlying exception condition is still present. Any exceptions whose Axis Exception Action is configured to ignore or report as alarms do not appear in this attribute.	See Semantics in 7.3.14.2.
658	Required – All	Get		Axis Faults – Mfg	LWORD	A bit map that represents the state of all manufacturer specific runtime faults. The bit map is identical to that of the Mfg. Axis Exceptions attribute. Fault bits when set are latched until a fault reset occurs. A fault reset clears the runtime fault bits, but the bits set again immediately if the underlying exception condition is still present. Any exceptions whose Axis Exception Action is configured to ignore or report as alarms do not appear in this attribute.	See Mfg. Exception Table (Published in Product Manual)
659	Required – All	Get		Axis Alarms	LWORD	A bit map that represents the current state of all standard alarm conditions. The bit map is identical to that of the Std. Axis Exceptions attribute. Only exception conditions whose Axis Exception Action is configured to report as an alarm appear in this attribute, and will not be reported in the Axis Faults attribute. Alarm bits when set are not latched and will clear as soon as the underlying exception condition is corrected.	See Semantics in 7.3.14.2.

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
660	Required – All	Get		Axis Alarms – Mfg	LWORD	A bit map that represents the current state of all manufacturer specific alarm conditions. The bit map is identical to that of the Mfg. Axis Exceptions attribute. Only exception conditions whose Axis Exception Action is configured to report as an alarm appear in this attribute, and will not be reported in the Axis Faults attribute. Alarm bits when set are not latched and will clear as soon as the underlying exception condition is corrected.	See Mfg. Exception Table. (Published in Product Manual)

7.3.14.2 Semantics: Standard Exception Table

Table 98 defines a list of standard exception conditions associated with the Axis Exceptions, Axis Faults, and Axis Alarms attributes. While the Axis Exceptions, Axis Faults, and Axis Alarms attributes are all Required in the CIP Motion device implementation, support for each of the individual exception conditions therein is left Optional. The Rule column in Table 98 indicates the Device Function Codes where the associated exception is applicable.

Table 98 – Standard Exception Table

Bit	Rule	Exception	Description
0	-	-- Reserved --	This bit cannot be used since the Alarm Codes and Fault Code are defined by the associated exception bit number and an Alarm Code or Fault Code of 0 means no alarm or fault condition is present.
1	D	Motor Overcurrent	Motor current has exceeded its rated peak or instantaneous current limit.
2	D	Motor Commutation	Permanent magnet motor commutation problem detected. Example would be an illegal state “111” or “000” for a UVW commutation device.
3	D	Motor Overspeed FL	Motor speed has exceeded its maximum limit given by the Motor Overspeed Factory Limit attribute associated with the motor type.
4	D	Motor Overspeed UL	Motor speed has exceeded the user defined speed limit given by Motor Overspeed User Limit.
5	D	Motor Overtemperature FL	Motor temperature has exceeded its factory set temperature limit given by Motor Overtemperature Factory Limit, or the integral motor thermal switch has tripped.
6	D	Motor Overtemperature UL	Motor temperature has exceeded the user defined temperature limit given by Motor Overtemperature User Limit.
7	D	Motor Thermal Overload FL	Motor thermal model has exceeded its factory set thermal capacity limit given by Motor Thermal Overload Factory Limit.
8	D	Motor Thermal Overload UL	Motor thermal model has exceeded its user defined thermal capacity given by Motor Thermal Overload User Limit.
9	D	Motor Phase Loss	The current in one or more motor phases has been lost or has been below a factory set threshold or, if supported, the configured Motor Phase Loss Limit. This exception is also associated with the optional Torque Prove function that tests motor current against an engaged mechanical brake. During normal operation in the Running state, the motor phase loss test cycles through the three motor currents checking that current in each motor phase exceeds the threshold level. When the phase being checked exceeds the level, the check is advanced to the next phase. If any phase fails to exceed the level within a vendor specific time period, e.g. one second, this exception is issued. The motor phase loss test only runs when the motor is running above a vendor specified speed. When Torque Proving is enabled, the motor phase current is checked during the Starting state. The current is applied to the motor at a fixed angle that is known to produce current in all three motor phases; hence this test takes very little time to execute. The Motor Phase Loss Limit is used to determine if the drive can produce torque. The measured current in all three phases need to exceed this level for a pass to occur.
10	D	Inverter Overcurrent	Inverter current has exceeded the factory set peak or instantaneous current limit.
11	D	Inverter Overtemperature FL	Inverter temperature has exceeded its factory set temperature limit given by Inverter Overtemperature Factory Limit.
12	D	Inverter Overtemperature UL	Inverter temperature has exceeded the user defined temperature limit given by Inverter Overtemperature User Limit.
13	D	Inverter Thermal Overload FL	Inverter thermal model has exceeded its factory set thermal capacity limit given by Inverter Thermal Overload Factory Limit.
14	D	Inverter Thermal Overload UL	Inverter thermal model has exceeded its user defined thermal capacity given by Inverter Thermal Overload User Limit.
15	BD	Converter Overcurrent	Converter current has exceeded the factory set peak or instantaneous current limit
16	BD	Converter Ground Current FL	Ground Current has exceeded its factory set current limit given by Converter Ground Current Factory Limit.
17	BD	Converter Ground Current UL	Ground Current has exceeded user defined limit given by Converter Ground Current User Limit.

Bit	Rule	Exception	Description
18	BD	Converter Overtemperature FL	Converter temperature has exceeded its factory set temperature limit given by Converter Overtemperature Factory Limit.
19	BD	Converter Overtemperature UL	Converter temperature has exceeded the user defined temperature limit given by Converter Overtemperature User Limit.
20	BD	Converter Thermal Overload FL	Converter thermal model has exceeded its factory set thermal capacity limit given by Converter Thermal Overload Factory Limit.
21	BD	Converter Thermal Overload UL	Converter thermal model has exceeded its user defined thermal capacity given by Converter Thermal Overload User Limit.
22	BD	Converter AC Power Loss	Multiple AC phases have been lost on the AC line to the converter, usually as a result of opening an AC line contactor. When associated with an external converter in a Shared AC/DC or Shared DC bus configuration, the AC Power Loss condition detected by the converter can be conveyed via the CIP Motion Connection's Control Status element. Generally, this exception is not asserted unless the device's power structure is enabled.
23	BD	Converter AC Single Phase Loss	One AC phase has been lost on the AC line to the converter.
24	BD	Converter AC Phase Short	A short has been detected between an AC phase and another AC phase or ground.
25	BD	Converter Pre-Charge Failure	A problem has been detected in the Converter's Pre-Charge circuitry preventing the DC Bus from charging to an acceptable voltage level.
26	-	-- Reserved --	
27	BD	Bus Regulator Overtemperature FL	Bus Regulator temperature has exceeded its factory set temperature limit given by Bus Regulator Overtemperature Factory Limit.
28	BD	Bus Regulator Overtemperature UL	Bus Regulator temperature has exceeded the user defined temperature limit given by Bus Regulator Overtemperature User Limit.
29	BD	Bus Regulator Thermal Overload FL	Bus Regulator thermal model has exceeded its factory set thermal capacity limit given by Bus Regulator Thermal Overload Factory Limit.
30	BD	Bus Regulator Thermal Overload UL	Bus Regulator thermal model has exceeded its user defined thermal capacity given by Bus Regulator Thermal Overload User Limit.
31	BD	Bus Regulator Failure	The bus regulator (shunt) module in a multi-axis system has a failed.
32	-	-- Reserved --	
33	BD	Bus Undervoltage FL	DC Bus voltage level is below the factory set limit given by Bus Undervoltage Factory Limit.
34	BD	Bus Undervoltage UL	DC Bus voltage level is below user defined limit given by Bus Undervoltage User Limit.
35	BD	Bus Overvoltage FL	DC Bus voltage level is above the factory set limit given by Bus Overvoltage Factory Limit.
36	BD	Bus Overvoltage UL	DC Bus voltage level is above user defined limit given by Bus Overvoltage User Limit.
37	BD	Bus Power Loss	DC Bus voltage level is below the Bus Power Loss Threshold for more than the timeout period specified Bus Power Loss Time value.
38	BD	Bus Power Blown Fuse	DC Bus power loss due to blown fuse.
39	D	Bus Power Leakage	DC Bus power leak has been detected when configured for Standalone operation. This can occur when the drive, configured for Standalone operation, is incorrectly wired to share DC bus power.

Bit	Rule	Exception	Description
40	D	Bus Power Sharing	An external converter sharing DC Bus power with this drive in a Shared AC/DC or Shared DC configuration has requested that this drive stop consuming power from the shared DC Bus. This may require that the drive be disabled to remove its DC Bus Power load from the failed converter. When there is no communication link between this drive and the external converter, the controller can monitor the DC Bus Unload bit of the converter axis and, if set, it can initiate Bus Power Sharing exceptions on all drives associated with the converter. See the DC Bus Unload status bit definition associated with the Axis Status attribute for a detailed description of this behavior.
41	E	Feedback Signal Noise FL	Noise induced A/B channel state changes (illegal states) from a feedback device were detected by the drive. Specifically, the number of these noise events that have occurred on this channel has exceeded the Feedback Noise Factory Limit. The offending feedback channel number is encoded in the associated Fault/Alarm Sub Code.
42	E	Feedback Signal Noise UL	Noise induced A/B channel state changes (illegal states) from a feedback device were detected on a feedback channel. Specifically, the number of these noise events that have occurred on this channel has exceeded the Feedback Noise User Limit. The offending feedback channel is encoded in the associated Fault/Alarm Sub Code.
43	E	Feedback Signal Loss FL	One or more A/B channel signals from a feedback device are open, shorted, missing, or severely attenuated. Specifically, the detected voltage levels of the signals are below the Feedback Signal Loss Factory Limit. The offending feedback channel is encoded in the associated Fault/Alarm Sub Code.
44	E	Feedback Signal Loss UL	One or more A/B channel signals from a feedback device are open, shorted, missing, or severely attenuated. Specifically, the detected voltage levels of the signals are below the Feedback Signal Loss User Limit. The offending feedback channel is encoded in the associated Fault/Alarm Sub Code.
45	E	Feedback Data Loss FL	The number of consecutive missed or corrupted serial data packets over the serial data channel from a feedback device has exceeded the Feedback Data Loss Factory Limit. The offending feedback channel is encoded in the associated Fault/Alarm Sub Code.
46	E	Feedback Data Loss UL	The number of consecutive missed or corrupted serial data packets over the serial data channel from a feedback device has exceeded the Feedback Data Loss User Limit. The offending feedback channel is encoded in the associated Fault/Alarm Sub Code.
47	E	Feedback Device Failure	The feedback device has detected an internal error.
48	-	-- Reserved --	
49	D	Brake Slip	Motor displacement has exceeded the Brake Slip Tolerance while the mechanical brake is engaged.
50	D	Hardware Overtravel Positive	Axis moved beyond the physical travel limits in the positive direction and activated the Positive Overtravel limit switch.
51	D	Hardware Overtravel Negative	Axis moved beyond the physical travel limits in the negative direction and activated the Negative Overtravel limit switch.
52	E	Position Overtravel Positive	Axis actual position exceeded the configured Position Limit – Positive attribute value in the positive direction. (Drive Scaling only)
53	E	Position Overtravel Negative	Axis actual position exceeded the configured Position Limit – Negative attribute value in the negative direction. (Drive Scaling only)
54	D	Excessive Position Error	The Position Error value of the position control loop has exceeded the configured value for Position Error Tolerance.
55	D	Excessive Velocity Error	The Velocity Error value of the velocity control loop has exceeded the configured value for Velocity Error Tolerance.
56	D	Overtorque Limit	Motor torque has risen above user defined maximum torque level given by Overtorque Limit.

Bit	Rule	Exception	Description
57	D	Undertorque Limit	Motor torque has dropped below user defined minimum torque level given by Undertorque Limit.
58	-	-- Reserved --	
59	-	-- Reserved --	
60	All	Illegal Control Mode	Controller has specified an unsupported Control Mode or Feedback Mode
61	BD	Enable Input Deactivated	Enable Input has been deactivated while the axis power structure is enabled and supplying current to the DC Bus or motor.
62	All	Controller Initiated Exception	Exception generated specifically by controller.
63	All	External Exception Input	Exception generated by external input to device.

7.3.15 Fault and alarm Log attributes

The attributes in Table 99 are used in conjunction with device based fault and alarm logs. These logs are basically a list of data records with each record corresponding to a fault or alarm related condition detected by the device. The Motion Device Axis Object specification does not specifically define the Fault Log or Alarm Log structures; the log structure is left to the device vendor's discretion. The specification does, however, require that certain information exist in the Fault Log and Alarm Log records.

For the purposes of the discussion that follows, a representative structure for a Fault Log could look like the following:

```
Struct
{
Index;           // Index to the latest fault record element
Fault Log Record Struct // Array of fault records
{
Fault Type; // Type of fault
Fault Code; // Specific fault identifier
Fault Sub Code; // Extended fault diagnostic info
Fault Action; // Drives response to the fault condition
Time Stamp; // Time when this fault occurred
} [25] // Store last 25 fault records in this circular buffer.
}
```

Similarly an alarm log could look like the following:

```
Struct
{
Index;           // Index to the latest alarm record element
Alarm Log Record Struct // Array of alarm records
{
Alarm Type; // Type of alarm
Alarm Code; // Specific alarm identifier
Alarm Sub Code; // Extended alarm diagnostic info
Alarm State; // Alarm turned on (1) or turned off (0)
Time Stamp; // Time stamp when this alarm transition occurred.
} [25] // Store last 25 alarm records in circular buffer
}
```

Fault and alarm logs serve several purposes in the CIP Motion device. First, they provide a mechanism to queue up multiple fault and alarm events with their associated time stamps for subsequent transfer to the controller for the purpose of building a user accessible replica of the fault and alarm log in the controller. Second, the log record elements themselves provide an efficient way to communicate fault and alarm info to the controller. Instead of sending 24

bytes of fault attributes to the controller every update, the device only needs to send a 1 byte Fault Code corresponding to a fault that has occurred.

To facilitate transfer of fault and alarm records to the controller, attributes are defined for each of the associated record elements that map directly to elements in the CIP Motion Device-to-Controller connection's Status Data Set and to corresponding attributes of the Motion Control Axis Object.

One fault log record and one alarm log record can be transferred to the controller per connection update. The Last Received ID is used by the drive to determine if the data that was sent was processed successfully, i.e. that the Fault and Alarm Codes passed as part of the Device-to-Controller connection's Status Data Block have been entered into the controller's fault and alarm logs. This condition allows the device to send the next Fault Code and Alarm Code in the sequence. When the device has successfully transmitted the most recent fault or alarm event, the device then sends the No Faults for no new faults or No Alarms for no new alarm conditions to report. Alternatively, the device may clear the associated Status Data Set bits indicating to the controller that there are no faults or alarms to report.

Table 99 – Fault and Alarm Log attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
661	Required – All	Get		Axis Fault Code	USINT	An 8-bit enumeration that, when the Axis Fault Type is non-zero, specifies the bit number of a fault condition in an axis fault log record. Possible sources of faults include standard and manufacturer specific Initialization Faults, Axis Faults and Axis Safety Faults. In the case of a Fault Type = 0, this value indicates the source of a fault clearing event. Possible fault clearing events are a device reset, a fault reset request, a shutdown reset request, or a reset resulting from opening a CIP Motion Connection. A Reset shall be recorded regardless of whether or not the axis is currently faulted. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' fault attributes and fault log. A value of 0 indicates that all faults have been cleared and there are no new fault log records to report.	If Axis Fault Type != 0 Enumeration: 0 to 255: Bit Number of Fault If Axis Fault Type = 0 Enumeration: 0 = No Faults 1 = Device Reset (Power Cycle) 2 = Fault Reset 3 = Shutdown Reset 4 = Connection Reset 5 to 255 = (reserved)
662	Required – All	Get		Axis Fault Type	USINT	An 8-bit enumeration that specifies the source of a fault condition or a fault clearing event in an axis fault log record. Possible sources of faults include standard and manufacturer specific Initialization Faults, Axis Faults and Axis Safety Faults. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' fault attributes and fault log. A value of 0 indicates that faults have been cleared. In this case the Axis Fault Code indicates what specific fault clearing event took place to clear the faults.	Enumeration: 0 = Faults Cleared 1 = Initialization Fault 2 = Initialization Fault – Mfg 3 = Axis Fault 4 = Axis Fault – Mfg 5 = APR Fault 6 = APR Fault – Mfg 7 = Axis Safety Fault 8 = Axis Safety Fault – Mfg 9 to 127: Reserved 128 to 255: Vendor Specific
663	Required – All	Get		Axis Fault Time Stamp	LINT	A 64-bit Time Stamp of the axis fault condition or fault clearing event in an axis fault log record. The time stamp format follows the CIP Sync standard with 0 corresponding to 1970-01-01. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' fault log. If the fault log record's Axis Fault Type and Axis Fault Code attributes are 0, the Axis Fault Time Stamp indicates time that the axis faults were last cleared.	Nanoseconds (CIP Sync)

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
664	Optional – All	Get		Axis Alarm Code	USINT	An 8-bit enumeration that specifies the bit number of an alarm condition in an axis alarm log record. Possible sources of alarms include standard and manufacturer specific Axis Alarms. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' alarm status and alarm log. A value of 0 indicates that no alarm conditions are currently present for this axis instance.	Enumeration: 0 to 255: Bit Number of Alarm
665	Optional – All	Get		Axis Alarm Type	USINT	An 8-bit enumeration that specifies the source of an alarm condition in an axis alarm log record. Possible sources of alarms include standard and manufacturer specific Axis Alarms. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' alarm attributes and alarm log. A value of 0 indicates that no alarm conditions are currently present for this axis instance.	Enumeration: 0 = No Alarms 1 = Start Inhibit 2 = Start Inhibit – Mfg 3 = Axis Alarm 4 = Axis Alarm – Mfg 5 to 127: Reserved 128 to 255: Vendor Specific
666	Optional – All	Get		Axis Fault Sub Code	USINT	An 8-bit enumeration that specifies the fault sub code of a fault condition in an axis fault log record. This attribute allows additional detail to be provided in the case of some faults, and has a different interpretation for each fault. Non-zero values indicate that additional detail is present. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' fault log.	Enumeration: specific to fault condition.
667	Optional – All	Get		Axis Alarm Sub Code	USINT	An 8-bit enumeration that specifies the alarm sub code of an alarm condition in an axis alarm log record. This attribute allows additional detail to be provided in the case of some alarms, and has a different interpretation for each alarm. Non-zero values indicate that additional detail is present. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' alarm log.	Enumeration: specific to alarm condition.

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
668	Required – All	Get		Axis Fault Action	USINT	An 8-bit value consisting of two nibbles whose enumerations indicate the action taken by the device in response to the fault condition listed in an axis fault log record. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' fault attributes and fault log. The action taken by the device in response to the fault may also be used by the controller to modify motion planner and motion instruction operation. The Fault Action applied by the device is defined by a Stop Action followed by a State Change Action.	Bit 0 to 3: Stopping Action Enumeration: 0 = Minor Fault (No Action) 1 = (reserved) 2 = Ramped Stop 3 = Torque Limited Stop 4 = Coast 5 to 15 = reserved. Bit 4 to 7: State Change Enumeration: 0 = Minor Fault (No Action) 1 = Hold 2 = Disable 3 = Shutdown 4 to 15 = reserved.
669	Optional – All	Get		Axis Alarm State	USINT	An 8-bit enumeration that indicates the current state of an alarm condition associated with an axis alarm log record. Alarm records are logged when an alarm condition occurs, i.e. when an alarm bit turns on, and when an alarm condition clears, i.e. when an alarm bit turns off. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' alarm attributes and alarm log. If the alarm log record's Axis Alarm Code is 0, i.e. no Alarm is active, the value of this attribute is also 0.	Enumeration: 0 = Alarm Bit Off – 0 1 = Alarm Bit On – 1 2 to 255 = reserved.
670	Optional – All	Get		Axis Alarm Time Stamp	LINT	A 64-bit Time Stamp of the axis alarm condition associated with an axis alarm log record. The time stamp format follows the CIP Sync standard with 0 corresponding to 1970-01-01. This attribute is designed to be passed to the controller via the CIP Motion Connection's Cyclic Data Block for the purpose of replicating the devices' alarm log. If the alarm log record's Axis Alarm Type and Axis Alarm Code attributes are 0, the Axis Alarm Time Stamp value is irrelevant and may also be set to 0.	Nanoseconds (CIP Sync)

7.3.16 Exception limit attributes

The attribute tables in Table 100 and Table 101 contain all exception limit related attributes associated with a Motion Device Axis Object instance. Exception Limit attributes define the conditions under which a corresponding exception is generated during motion axis operation that has the potential of generating either a fault or alarm. They are typically associated with temperature, current, and voltage conditions of the device that are continuous in nature. Factory Limits (FL) for exceptions are usually hard coded in the device and typically result in a major fault condition. User Limits (UL) for exceptions are configurable and typically used to generate a minor fault, or alarm condition. For this reason, the User Limits are generally set inside the corresponding Factory Limits. The triggering of a User Limit exception does not preclude triggering of the corresponding Factory Limit exception; the two exception trigger conditions are totally independent of one another.

Table 100 – Exception Factory Limit Info attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
678	Optional – D	Get		Rotary Motor Overspeed Factory Limit	REAL	Returns the Factory Limit for the Motor Overspeed FL exception based on a factory set value determined by the Rotary Motor Rated Speed or Rotary Motor Max Speed attribute values.	r/min
679	Optional – D	Get		Linear Motor Overspeed Factory Limit	REAL	Returns the Factory Limit for the Motor Overspeed FL exception based on a factory set value determined by the Linear Motor Rated Speed or Linear Motor Max Speed attribute values.	m/s
680	Optional – D	Get		Motor Overtemperature Factory Limit	REAL	Returns the Factory Limit for the Motor Overtemperature FL exception based on a factory set value related to the Motor Max Winding Temperature of the motor.	°C
681	Optional – D	Get		Motor Thermal Overload Factory Limit	REAL	Returns the Factory Limit for the Motor Thermal Overload FL exception based on a factory set value related to the Motor Thermal Overload rating of the motor.	% Motor Rated
682	Optional – D	Get		Inverter Overtemperature Factory Limit	REAL	Returns the Factory Limit for the Inverter Overtemperature FL exception.	°C
683	Optional – D	Get		Inverter Thermal Overload Factory Limit	REAL	Returns the Factory Limit for the Inverter Thermal Overload FL exception.	% Inverter Rated
684	Optional – BD	Get		Converter Overtemperature Factory Limit	REAL	Returns the Factory Limit for the Converter Overtemperature FL exception.	°C
685	Optional – BD	Get		Converter Thermal Overload Factory Limit	REAL	Returns the Factory Limit for the Converter Thermal Overload FL exception.	% Converter Rated
693	Optional – BD	Get		Converter Ground Current Factory Limit	REAL	Returns the Factory Limit for the Converter Ground Current FL exception	Amperes
686	Optional – BD	Get		Bus Regulator Overtemperature Factory Limit	REAL	Returns the Factory Limit for the Bus Regulator Overtemperature FL exception.	°C

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
687	Optional – BD	Get		Bus Regulator Thermal Overload Factory Limit	REAL	Returns the Factory Limit for the Bus Regulator Thermal Overload FL exception.	% Regulator Rated
688	Optional – BD	Get		Bus Overvoltage Factory Limit	REAL	Returns the Factory Limit for the Bus Overvoltage FL exception.	Volts
689	Optional – BD	Get		Bus Undervoltage Factory Limit	REAL	Returns the Factory Limit for the Bus Undervoltage FL exception.	Volts
690	Optional – E	Get		Feedback Noise Factory Limit	UDINT	Returns the Factory Limit for the Feedback Noise FL exception.	Noise Counts
691	Optional – E	Get		Feedback Signal Loss Factory Limit	REAL	Returns the Factory Limit for the Feedback Signal Loss FL exception.	% Nominal Voltage
692	Optional – E	Get		Feedback Data Loss Factory Limit	UDINT	Returns the Factory Limit for the Feedback Data Loss FL exception.	Consecutive Lost Data Packets

Table 101 – Exception User Limit Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
694	Optional – D	Set		Motor Phase Loss Limit	REAL	Sets the minimum motor phase current for the Motor Phase Loss exception. The current in each motor phase shall exceed this value during the motor phase loss test or a Motor Phase Loss exception occurs. Decreasing this attribute's value lowers sensitivity to phase loss conditions. A value of 0 shall effectively disable the motor phase loss test.	% Motor Rated
695	Optional – D	Set		Motor Overspeed User Limit	REAL	Sets the Overspeed User Limit relative to Rotary Motor Rated Speed or Linear Motor Max Speed that is allowable before throwing a Motor Overspeed UL exception.	% Motor Rated
696	Optional – D	Set		Motor Overtemperature User Limit	REAL	Sets User Limit for the Motor Overtemperature UL exception.	°C
697	Optional – D	Set		Motor Thermal Overload User Limit	REAL	Sets User Limit for the Motor Thermal Overload UL exception.	% Motor Rated
698	Optional – D	Set		Inverter Overtemperature User Limit	REAL	Sets User Limit for the Inverter Overtemperature UL exception.	°C
699	Optional – D	Set		Inverter Thermal Overload User Limit	REAL	Sets User Limit for the Inverter Thermal Overload UL exception.	% Inverter Rated
700	Optional – BD (Integral Converter only)	Set		Converter Overtemperature User Limit	REAL	Sets User Limit for the Converter Overtemperature UL exception.	°C

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
701	Optional – BD (Integral Converter only)	Set		Converter Thermal Overload User Limit	REAL	Sets User Limit for the Converter Thermal Overload UL exception.	% Converter Rated
709	Optional – BD (Integral Converter only)	Set		Converter Ground Current User Limit	REAL	Sets User Limit for the Converter Ground Current UL exception	% Factory Limit
702	Optional – BD (Integral Converter only)	Set		Bus Regulator Overtemperature User Limit	REAL	Sets User Limit for the Bus Regulator Overtemperature UL exception.	°C
703	Optional – BD (Integral Converter only)	Set		Bus Regulator Thermal Overload User Limit	REAL	Sets User Limit for the Bus Regulator Thermal UL exception.	% Regulator Rated
704	Optional – BD	Set		Bus Overvoltage User Limit	REAL	Sets User Limit for the Bus Overvoltage UL exception. Unlike the corresponding Factory Limit, which is specified in Volts, the User Limit is based on percent of Nominal Bus Voltage during operation.	% Nominal Bus Voltage
705	Optional – BD	Set		Bus Undervoltage User Limit	REAL	Sets User Limit for the Bus Undervoltage UL exception. Unlike the corresponding Factory Limit, which is specified in Volts, the User Limit is based on percent of Nominal Bus Voltage during operation.	% Nominal Bus Voltage
706	Optional – E	Set		Feedback Noise User Limit	UDINT	Sets User Limit for the Feedback Noise Overload UL exception. Example of Noise Counts would be simultaneous transitions of the A and B channel of a quadrature encoder feedback device.	Noise Counts
707	Optional – E	Set		Feedback Signal Loss User Limit	REAL	Sets User Limit for the Feedback Signal Loss UL exception. Feedback interface hardware typically monitor average voltage levels on incoming signals. Feedback Signal Loss conditions occur when the average voltage levels drop below a percentage of the voltage drop allowed by the Feedback Signal Loss Factory Limit.	% FL Voltage Drop

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
708	Optional – E	Set		Feedback Data Loss User Limit	UDINT	Sets User Limit for the Feedback Data Loss UL exception. For digital feedback devices, feedback interface hardware monitors the integrity of data transferred over the serial connection to the feedback device. Feedback Data Loss conditions occur when the two or more consecutive data packets are lost or corrupted.	Consecutive Lost Data Packets

7.3.17 Axis exception action configuration attribute

7.3.17.1 General

The configuration attribute in Table 102 controls the action performed by the device as a result of an exception condition. A unique exception action is defined for each supported exception condition.

Table 102 – Axis Exception Action Configuration attribute

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
672	Required – All	Set		Axis Exception Action	USINT [64]	The Axis Exception Action attribute is a 64-element array of enumerated bytes that specifies the action for the associated standard exception. See Semantics in 7.3.17.2 for details.	Enumeration: 0 = Ignore (O) 1 = Alarm (O) 2 = Minor Fault (O) 3 = Major Fault (R) 4 to 254 = reserved 255 = Unsupported
673	Required – All	Set		Axis Exception Action – Mfg	USINT [64]	The Axis Exception Action attribute is a 64-element array of enumerated bytes that specifies the action for the associated manufacturer specific exception. See Semantics in 7.3.17.2 for details	Enumeration: 0 = Ignore (O) 1 = Alarm (O) 2 = Minor Fault (O) 3 = Major Fault (R) 4 to 254 = reserved 255 = Unsupported

7.3.17.2 Semantics: Attribute No. 672 and 673 – Axis Exception Action

The Axis Exception Action and Axis Exception Action–Mfg. attributes are 64-element arrays of enumerated bytes that specifies the action for the associated standard or manufacturer specific exception, respectively (see Table 103). For a given exception, certain exception actions may not be supported. Attempting to do so shall result in an error. Each device product shall specify the available actions for each exception that is supported. If a specific exception is not supported by the device, the only valid exception action enumeration is “Unsupported”. Attempting to write any other value to the element associated with an unsupported exception results in an error.

Table 103 – Axis Exception Action definitions

Enum.	Need in implementation	Name	Description
0	Optional	Ignore	Ignore instructs the device to completely ignore the exception condition. For some exceptions that are fundamental to the operation of the axis, it may not be possible to ignore the condition.
1	Optional	Alarm	Alarm action simply sets the associated bit in the Axis Alarms word but does not otherwise affect axis behavior. For some exceptions that are fundamental to the operation of the device it may not be possible to select this action or any other action that leaves device operation unaffected.
2	Optional	Minor Fault	Minor Fault action reports the exception by setting the associated bit in the Axis Faults word but does not affect axis behavior. Minor Faults allow the controller to take application specific action in response to the exception condition. For some exceptions that are fundamental to the operation of the device, it may not be possible to simply indicate the condition as a Minor Fault and leave device operation unaffected. Converters (B) executing a Minor Fault exception action continue to supply DC Bus Power and do not set the DC Bus Unload bit in Axis Status attribute.
3	Required	Major Fault	Major Fault action results in both setting the associated bit in the Faults word and bringing the axis to a stop based on the factory set “best” available stopping method. This “best” stopping method includes both the method of decelerating the motor to a stop and the final state of the axis given the expected level of control still available. The level of axis control available depends on the specific exception condition and on the configured control mode. The available deceleration methods are defined by the Stopping Action attribute. Standard stopping actions, listed in decreasing levels of deceleration control, are as follows: 1. Ramp Decel 2. Current Limit Decel 3. Coast In general, the “best” stopping action is the most controlled deceleration method still available given the exception condition. While the final Axis State after a Major Fault is always the Major Faulted state the final state of the power structure in response to the Major Fault exception action can be any one of the following states that are listed in decreasing levels of control functionality: 1. Hold (stopped with Holding Torque) 2. Disable (stopped with Power Structure Disabled) 3. Shutdown (stopped with Shutdown Action) The “best” final state of the power structure is the state with the most control functionality still available given the exception condition. The specific stopping action and final state associated with a given Major Fault exception action is captured in the Axis Fault Action attribute that is included in the Fault Log record. Axis Fault Action enumerations are as follows: Stop Action Enumerations: 0 = No Action 1 = (reserved) 2 = Ramped Stop 3 = Current Limited Stop 4 = Coast State Change Enumerations: 0 = No Action 1 = Hold 2 = Disable 3 = Shutdown

Enum.	Need in implementation	Name	Description
			If the application requires exception action that is a more severe stopping action than the factory set “best” method, the controller shall initiate that action. If the application requires exception action that is less severe than the factory set “best” method, the controller shall configure the device axis instance for a Minor Fault exception action and handle the fault directly. This may put device and motor components at risk and shall only be allowed by the device when there is an opportunity, albeit temporal, for the device to remain operational. This is important in applications where the value of the product is higher than the value of the motor or device. When the Major Fault exception action is applied to a converter device (B), stopping action is not applicable (0 = No Action). The final states of Disable or Shutdown for the converter are applicable, however, with Shutdown executing the configured Shutdown Action. In either case, the DC Bus Unload bit of the converter's Axis Status attribute is set allowing the control system to generate a Bus Sharing exception on all drives that consume DC Bus power from this converter. When multiple major faults occur with different stopping actions, the most severe of the associated stopping actions is applied, i.e. the stopping action that requires the lowest level of control functionality. This rule also applies to the stopping action associated with the Stopping Action associated with a Disable Request.
4 to 254		Reserved	-
255	Required	Unsupported	The Unsupported Exception Action is the value assigned to Exceptions that are not supported by the device. Trying to assign an Exception Action other than Unsupported to an exception that is not supported by the device results in an error.

7.3.18 Initialization fault attributes

7.3.18.1 General

The attribute table in Table 104 contains all initialization fault related attributes associated with a Motion Device Axis Object instance. Initialization Faults are conditions that can occur during the device initialization process that prevent normal operation of the device.

Table 104 – Initialization Fault attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
674	Required – All	Get		Initialization Faults	DWORD	A bit map that represents the state of all standard initialization faults. These faults prevent any motion, and do not have configurable fault actions. Examples of initialization faults are corrupted memory data, calibration errors, firmware startup problems, or an invalid configuration attribute value. Initialization faults cannot be cleared with a Fault Reset request, although a power-cycle provides a new attempt at initialization.	See Semantics in 7.3.18.2
675	Required – All	Get		Initialization Faults – Mfg	DWORD	A bit map that represents the state of all manufacturer specific initialization faults. These faults prevent any motion, and do not have configurable fault actions. Examples of initialization faults are corrupted memory data, calibration errors, firmware startup problems, or an invalid configuration attribute value. Initialization faults cannot be cleared with a Fault Reset request, although a power-cycle provides a new attempt at initialization.	See Mfg. Initialization Fault Table (Published in Product Manual)

7.3.18.2 Semantics: Standard Initialization Fault Table

Table 105 defines a list of standard faults associated with the Initialization Faults attribute and also applicable to the Initialization Fault Code attribute.

Table 105 – Standard Initialization Fault Table

Bit	Exception	Description
0	-- Reserved --	This bit cannot be used since the Fault Code is defined by the associated exception bit number and Fault Code of 0 means no fault condition is present.
1	Boot Block Checksum Fault	Checksum or CRC error for Boot Block of CIP Motion device detected as part of Self-Test.
2	Main Block Checksum Fault	Checksum or CRC error for Main Block of CIP Motion device detected as part of Self-Test.
3	Nonvolatile Memory Checksum Fault	Checksum or CRC error for NV Memory of CIP Motion device detected as part of Self-Test.
4 to 31	-- Reserved --	

7.3.19 Start inhibit attributes

7.3.19.1 General

The attribute table in Table 106 contains all Start Inhibit related attributes associated with a Motion Device Axis Object instance. Start Inhibits are conditions that prevent transition of the motion axis from the Stopped State into any of the operational states.

Table 106 – Start Inhibit attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
676	Required – D	Get		Start Inhibits	WORD	A bit map that specifies the current state of all standard conditions that inhibit starting of the axis.	See Semantics in 7.3.19.2
677	Required – D	Get		Start Inhibits – Mfg	WORD	A bit map that specifies the current state of all manufacturer specific conditions that inhibit starting of the axis.	See Mfg. Start Inhibit Table (Published in Product Manual)

7.3.19.2 Semantics: Standard Start Inhibit Table

Table 107 defines a list of standard start inhibits associated with the Start Inhibits attribute.

Table 107 – Standard Start Inhibit Table

Bit	Exception	Description
0	-- Reserved --	This bit cannot be used since the Start Inhibit Code is defined by the associated bit number and Start Inhibit Code of 0 means no fault condition is present.
1	Axis Enable Input	Axis Enable Input is not active.
2	Motor Not Configured	The associated motor has not been configured for use.
3	Feedback Not Configured	The associated feedback device has not been configured for use
4	Commutation Not Configured	The associated PM motor commutation function has not been configured for use.
5	Safe Torque Off Active	The integrated Safe Torque Off safety function is active based on the Safe Torque Off Active bit (bit 3) of the Axis Safety Status (Attribute No. 761) being set.
6 to 15	-- Reserved --	

7.3.20 APR fault attributes

7.3.20.1 General

The attribute table in Table 108 contains all APR (Absolute Position Recovery) fault related attributes associated with a Motion Device Axis Object instance. APR Faults are conditions that can occur during the device initialization process when trying to restore the absolute position of an axis. Unlike Initialization Faults, these faults are recoverable and may be cleared with a Fault Reset request.

Table 108 – APR Fault attributes

Attr id	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
756	Optional – E Drive Scaling	Get		APR Faults	WORD	A bit map that represents the state of all standard APR (Absolute Position Recovery) faults. An APR fault is generated when the system fails to recover the absolute position of the axis after power cycle, reset, or reconnection. APR faults are detected during the initial configuration or initialization of the drive axis. When an APR fault occurs, the actual position of the axis is no longer correlated to the position of the axis prior to the power cycle, reset, or reconnect. Examples of standard APR faults are feedback serial number mismatch, Axis ID mismatch, scaling configuration change, etc. APR faults are recoverable and can be cleared with a Fault Reset request.	See semantics in 7.3.20.2
757	Optional – E Drive Scaling	Get		APR Faults – Mfg	WORD	A bit map that represents the state of all manufacturer APR (Absolute Position Recovery) faults. An APR fault is generated when the system fails to recover the absolute position of the axis after power cycle, reset, or reconnection. APR faults are detected during the initial configuration or initialization of the drive axis. When an APR fault occurs, the actual position of the axis is no longer correlated to the position of the axis prior to the power cycle, reset, or reconnect. APR faults are recoverable and can be cleared with a Fault Reset request.	See Mfg. APR Fault Table (Published in Product Manual)

7.3.20.2 Semantics: Standard APR Fault Table

Table 109 defines a list of standard faults associated with the APR Faults attribute and also applicable to the APR Fault Code attribute.

Table 109 – Standard APR Fault Table

Bit	Exception	Description
0	-- Reserved --	This bit cannot be used since the Fault Code is defined by the associated exception bit number and Fault Code of 0 means no fault condition is present.
1	Memory Write Error	Error in saving absolute position data to NV memory.
2	Memory Read Error	Error in reading absolute position data from NV memory
3	Feedback Serial Number Mismatch	Position Feedback Serial Number does not match saved Feedback Serial Number
4	Buffer Allocation Fault	Caused when there is not enough RAM memory left to save APR data.
5	Scaling Configuration Changed	Scaling attribute configuration for this axis does not match the saved scaling configuration.
6	Feedback Mode Changed	Feedback Mode has changed and does not match the saved Feedback Mode configuration.
7 to 15	-- Reserved --	

7.3.21 Axis statistical attributes

Table 110 includes attributes that provide useful statistics on motion axis operation.

Table 110 – Axis Statistical attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
710	Optional – BD	Get		Control Power-up Time	REAL	Elapsed time since axis control power was last applied.	Seconds
711	Optional – BD	Get		Cumulative Run Time	REAL	Accumulated time that the axis has been powering the Running state.	Hours
712	Optional – BD	Get		Cumulative Energy Usage	REAL	Accumulated output energy of the axis.	kW•h
713	Optional – D	Get		Cumulative Motor Revs	LINT	Cumulative number of times motor shaft has turned. (Rotary Motors Only)	
714	Optional – BD	Get		Cumulative Main Power Cycles	DINT	Cumulative number of times AC Mains has been cycled.	
715	Optional – BD	Get		Cumulative Control Power Cycles	DINT	Cumulative number of times Control Power has been cycled.	

7.3.22 Axis info attributes

Table 111 includes attributes that provide information about the associated hardware capabilities of Motion Device Axis Object instance.

Table 111 – Axis Info attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
720	Required – D	Get		Inverter Rated Output Voltage	REAL	Drive inverter output voltage rating. This value is hard coded in the device.	Volts (RMS)
721	Required – D	Get		Inverter Rated Output Current	REAL	Drive inverter output current rating. This value is hard coded in the device.	Amperes (RMS)
722	Required – D	Get		Inverter Rated Output Power	REAL	Drive inverter output power rating. This value is hard coded in the device.	kW
723	Optional – BD (Integral Converter Only)	Get		Converter Rated Output Current	REAL	Drive converter output current rating. This value is determined by the motion axis from the associated converter.	Amperes
724	Optional – BD (Integral Converter Only)	Get		Converter Rated Output Power	REAL	Drive converter output power rating. This value is determined by the motion axis from the associated converter.	kW
725	Optional – D	Get		Drive Power Structure Axis ID	UDINT	This value represents an ID assigned by the drive vendor that uniquely identifies the power structure associated with this axis instance. By contrast, power structure hardware that is common or identical to all axis instances of the drive (excluding master feedback axes) can be identified using the Drive Power Structure Class ID attribute. The Drive Power Structure Axis ID can be included as part of the data segment in the CIP Motion Connection's Forward_Open service to confirm that the power structure for this axis instance matches the configuration in the controller. Axis instances with power structures that are not configured are set to 0 indicating to the device that the Drive Power Structure Axis ID for such an axis instance does not need to be checked.	ID#

7.3.23 General purpose I/O attributes

Table 112 includes attributes that provide to general purpose analog and digital I/O associated with the Motion Device Axis Object instance.

Table 112 – Drive General Purpose I/O attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
730	Optional – BD	Get		Digital Inputs	DWORD	A 32-bit word with whose bits can be assigned by the vendor to general purpose digital inputs.	Vendor Specific Bit Map
731	Optional – BD	Set		Digital Outputs	DWORD	A 32-bit word with whose bits can be assigned by the vendor to general purpose digital outputs.	Vendor Specific Bit Map
732	Optional – BD	Get		Analog Input 1	REAL	General purpose analogue input 1 level.	% Full Scale
733	Optional – BD	Get		Analog Input 2	REAL	General purpose analogue input 2 level.	% Full Scale
734	Optional – BD	Set		Analog Output 1	REAL	General purpose analogue output 1 level.	% Full Scale
735	Optional – BD	Set		Analog Output 2	REAL	General purpose analogue output 2 level.	% Full Scale
736	Optional – BD	Set		Drive Enable Input Checking	BOOL	Boolean value that controls whether or not the drive shall regularly check the state of the Drive Enable input. When Drive Enable Input Checking is enabled, an inactive Drive Enable input results in a Start Inhibit condition. If the Drive Enable input is deactivated while the drive is enabled, a Drive Enable Input Deactivated exception is generated. If Drive Enable Input Checking is disabled, the drive shall not check the state of the Drive Enable input.	0 = Disabled 1 = Enabled
737	Optional – D	Set		Hardware Overtravel Input Checking	BOOL	Boolean value that controls whether or not the drive shall regularly check the state of the positive and negative Hardware Overtravel inputs. When Hardware Overtravel Input Checking is enabled, an inactive Hardware Overtravel input results in an associated Hardware Overtravel Positive or Negative exception. If Drive Hardware Overtravel Checking is disabled, the drive shall not check the state of the Hardware Overtravel inputs.	0 = Disabled 1 = Enabled

7.3.24 Local Mode attributes

Table 113 contains all local mode attributes associated with a Motion Device Axis Object instance. These attributes govern the behavior of the Motion Device Axis Object instance when a local drive (i.e. non-CIP Motion) device interface requests control of axis behavior.

Table 113 – Local Mode Configuration attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
750	Optional	Set		Local Control	USINT	Mechanism for controller to allow a local interface to request local control. Control implies, in this case, the ability to change the state or behavior of the axis. Local Control Not Allowed configures the device to prevent a local interface from taking control of the drive. Local Control Conditionally Allowed configures the device to prevent a local interface from taking control of the axis while the axis is in an Operational State, such as Running or Testing. Local Control Allowed configures the device to allow a local interface to take control of the axis, even in an Operational State. Some devices may not support this state.	Enumeration: 0 = Not Allowed (R) 1 = Conditionally Allowed (O) 2 = Allowed (O) 3 to 255 = reserved

7.3.25 Axis Safety attributes

The following attribute table contains axis attributes associated with the “Networked” Safety functionality associated with a Motion Device Axis Object instance included in a CIP Motion Safety Drive. These attributes reflect the current state of an embedded Safety Core within the device that is designed to interoperate with an external Safety Controller via a CIP Safety™¹⁷ connection.

The Axis Safety State, Axis Safety Status, and Axis Safety Fault attributes defined in Table 114 are based on the values read from attributes resident in objects associated with the Safety Core and are used by the motion control system to monitor the behavior of the Safety Core via the CIP Motion Connection.

NOTE The CIP Motion Safety Drive, Safety Core, CIP Safety™ connection, as well as associated safety related objects and their attributes are outside the scope of this standard. Their specification can be found in [30].

¹⁷ CIP Safety™ is a trade mark of ODVA, Inc. This information is given for the convenience of users of this International Standard and does not constitute an endorsement by IEC of the trade mark holder or any of its products. Compliance to this profile does not require use of the trade mark CIP Safety™. Use of the trade mark CIP Safety™ requires permission of ODVA, Inc.

Table 114 – Axis Safety Status attributes

Attr ID	Need in impl.	Access rule	NV	Attribute name	Data type	Description of attribute	Semantics of values
760	Optional – DE Safety only	Get		Axis Safety State	USINT	8-bit enumeration that indicates the state of the associated Safety Supervisor object of the device. There is only one Safety Supervisor object servicing the CIP Motion device so its state applies to all applicable Axis instances of the device. This means that all instances of this object will have the same state for this attribute.	Enum
761	Optional – DE Safety only	Get		Axis Safety Status	DWORD	Collection of bits indicating the status of the standard safety functions for the axis as reported by the embedded Safety Core.	Bitmap
762	Optional – DE Safety only	Get		Axis Safety Status – Mfg	DWORD	Collection of bits indicating the status of the manufacturer specific safety functions for the axis as reported by the embedded Safety Core.	Bitmap: 0 to 31: Vendor Specific
763	Optional – DE Safety only	Get		Axis Safety Faults	DWORD	Collection of bits indicating the safety fault status of the axis associated with standard safety functionality as reported by the embedded Safety Core.	Bitmap
764	Optional – DE Safety only	Get		Axis Safety Faults – Mfg	DWORD	Collection of bits indicating the safety fault status of the axis associated with manufacturer specific safety functionality as reported by the embedded Safety Core.	Bitmap: 0 31: Vendor specific
NOTE See [30] for complete definition of enumerations, bitmaps, and corresponding state behavior.							

7.4 Common services

7.4.1 Supported services

The Motion Device Axis Object provides the CIP Common Services specified in Table 115.

Table 115 – Motion Device Axis Object – Common Services

Service code (Hex)	Need in implementation		Service name	Description of service
	Class	Instance		
03 _{hex}	Required	Required	Get_Attributes_List	Returns the contents of the selected gettable attributes for given instance.
04 _{hex}	Required	Required	Set_Attributes_List	Sets the content of the selected settable attributes for given instance.
0E _{hex}	Required	Required	Get_Attribute_Single	Returns the contents of the selected gettable attribute for given instance.
10 _{hex}	Required	Required	Set_Attribute_Single	Sets the content of the selected settable attribute for given instance.
1C _{hex}	Conditional ^a	N/A	Group_Sync	Passes controller's Grand Master Clock ID to device and reports if this device can be synchronized with the controller as part of a group of associated devices.
^a Group_Sync service is Required for CIP Motion devices that support Time Synchronization service and include the Time Sync Object.				

7.4.2 Service specific data

7.4.2.1 General

Common Services supported by the Motion Device Axis Object have associated Request Data and Response Data. The format and syntax of the Service Specific Data depends on the specified Service Code. Descriptions of the Service Specific Data for the Set/Get_Attributes_List and Set/Get_Attribute_Single services are already defined in CIP (see IEC 61158-5-2 and IEC 61158-6-2). A description of the Service Specific Data associated with the Group_Sync service (part of the CIP Sync extensions to CIP specifications) is detailed in 7.4.2.2.

7.4.2.2 Group_Sync

The Group_Sync service is used to synchronize the Motion Device Axis Object associated with this device node to the controller issuing the service request and, ultimately, to all other devices comprising the controller's Motion Group. The service response indicates “device synchronized” if the device is presently synchronized to the same IEC 61588:2009 Grand Master as the controller.

The format and syntax of Group_Sync Request Data is specified in Table 116.

Table 116 – Group_Sync Request Data Structure

Name	Type	Description of request parameter	Semantics of values
Grand Master Clock ID	SINT[8]	Clock ID of Grandmaster Clock associated with the Controller.	See Time Sync Object in IEC 61158-5-2 and IEC 61158-6-2

The Grand Master Clock ID is used to verify that the IEC 61588:2009 PTP Grand Master Clock for the Controller is the same Grand Master Clock associated with the device.

The format and syntax of Group_Sync Response Data is specified in Table 117.

Table 117 – Group_Sync Response Data Structure

Name	Type	Description of response parameter	Semantics of values
Sync Status	SINT	Enumerated value that indicates whether or not the device is presently group synchronized or has different time master than controller.	0 = device synchronized 1 = device not synchronized 2 = different Grand Master 3 = clock skew detected 4 to 255 = reserved

When the Group_Sync service request is executed, the device first checks if the Grand Master associated with the controller matches the associated Grand Master of the device. This is an important prerequisite to successfully synchronizing device's local clock with the controller. If the Grand Master IDs do not match, the Group_Sync replies with a Sync Status of 2, indicating "different grand master". If the Grand Master Clock IDs match, then the Group_Sync service moves on to check the state of the device's clock synchronization.

Next, the device shall query the Time Sync Object to determine if the device is synchronized. This requires that the System Time Offset changes have been less than the Time Sync Threshold (typically, 10 μ s) for a predetermined period of time. If the device's clock is synchronized with a common Grand Master clock, then the Group_Sync service moves on to check for possible clock skew. Otherwise the service returns a Sync Status of "device not synchronized" (1).

If the above conditions are met, the device then checks for possible time skew between the controller's System Time and device's System Time. This check is made possible by the controller's device configuration process insuring that regular Time Stamp and Time Offset values are sent to the device prior to sending a Group_Sync service request, which in turn requires that regular Time Stamp and Time Offset values are sent back to the controller prior sending a Group_Sync service response. The device performs a clock skew check by loading the Controller Time Stamp from the most recent Control-to-Device packet. If the sum of the Controller Time Stamp and the Controller Update Period (CUP) differs from the current device System Time by more than one Controller Update Period (CUP), the controller and device System Time clocks are skewed.

That is:

If $|(Controller\ Time\ Stamp + CUP) - Current\ Device\ System\ Time| > 1\ CUP$

Controller and Device Clocks are Skewed

Else

Controller and Device Clocks are OK

If the device clock is not skewed, then the controller and device System Time clocks are considered fully synchronized at this instant in time. Based on this assumption, the Group_Sync service initializes the Controller's Last System Time Offset value with the Controller Time Stamp. Similarly, the service initializes the Device's Last System Time Offset value with the value associated with the Current Device Time Stamp. These two variables shall be used later when executing the Time Step Compensation algorithm. Finally, the Group_Sync returns with Sync Status of "device synchronized" (0). Otherwise, if clock skew is detected, the service returns a Sync Status of "clock skew detected" (3).

When the controller receives a "device synchronized" response from the device, it performs its own skew check and Time Offset initialization. If no clock skew is detected, the device is then considered Group Synchronized. If the controller receives a non-zero Sync Status response from the device or fails its skew test, the controller shall send another Group_Sync service request. This process shall continue for at least 1 min until the device is Group Synchronized or the process timeout occurs resulting in a controller initiated connection fault.

Once the controller and all the CIP Motion devices are group synchronized, the controller then sets the Synchronous Control bit in the Controller-to-Device Connection Header of the next cyclic packet sent to the device indicating that the device can schedule its next Device-to-Controller Connection update based on the Controller Time Stamp and Controller Update Period contained in the connection data. The device then indicates that it is sending scheduled connection data based on the Controller Update Period by setting the Sync Mode bit in the Device-to-Controller Connection Header of the next cyclic packet sent to the controller. Data exchange between the controller and the device proceed thereafter according the CIP Motion timing model.

7.5 Object specific services

7.5.1 Supported services

The Motion Device Axis Object provides the Object Specific Services specified in Table 118.

Table 118 – Motion Device Axis Object – Object Specific Services

Service code (Hex)	Need in implementation		Service name	Description of service
	Class	Instance		
4B _{hex}	n/a	Required	Get_Axis_Attributes_List	Returns the contents of the selected gettable Motion Device Axis Object attributes.
4C _{hex}	n/a	Required	Set_Axis_Attributes_List	Sets the content of the selected settable Motion Device Axis Object attributes.
4D _{hex}	n/a	Required	Set_Cyclic_Write_List	Sets the attribute contents of the Cyclic Write Data block.
4E _{hex}	n/a	Required	Set_Cyclic_Read_List	Sets the attribute contents of the Cyclic Read Data block.
4F _{hex}	n/a	Optional – D	Run_Motor_Test	Initiates the selected test on the motor to measure various motor parameters.
50 _{hex}	n/a	Optional – D	Get_Motor_Test_Data	Returns the results of the preceding Run_Motor_Test service.
51 _{hex}	n/a	Required – C	Run_Inertia_Test	Initiates the selected test on the motor to measure the inertia.
52 _{hex}	n/a	Required – C	Get_Inertia_Test_Data	Returns the results of the preceding Run_Inertia_Test service.
53 _{hex}	n/a	Required	Run_Hookup_Test	Initiates the selected test to determine the condition of the motor and feedback device connections.
54 _{hex}	n/a	Required	Get_Hookup_Test_Data	Returns the results of the preceding Run_Hookup_Test service.
55 to 63 _{hex}			Reserved	

7.5.2 Service specific data

7.5.2.1 General

Object Specific Services supported by the Motion Device Axis Object have associated Request Data and Response Data. The format and syntax of the Service Specific Data depends on the specified Service Code. All parameters contained in the Service Specific Data are word aligned. This is true regardless of whether the service specific request data is passed in the Controller-to-Device Connection or as part of an Explicit messaging connection. A description of the Service Specific Data associated with each service is shown in 7.5.2.2 to 7.5.2.11.

7.5.2.2 Get_Axis_Attributes_List

The Get_Axis_Attributes List service provides a mechanism to read the current value of one or more Motion Device Axis Object attributes, including attributes having a multi-dimensional array data type. The buffer/array transfer mechanism can be used to transfer large data log arrays from the device to facilitate features like high speed trending.

The format of the Request Data Block for this service is shown in Figure 70.

← 32-bit Word →		
Get_Axis_Attributes_List Request format		
Number of Attributes	-	
Attr ID 1	Attr 1 Dimension	Attr 1 Element Size
Attr Start Index 1 (array only)	Attr Data Elements 1 (array only)	
...	...	
Attr ID 2	Attr 2 Dimension	Attr 2 Element Size
Attr Start Index 2 (array only)	Attr Data Elements 2 (array only)	
...	...	
Attr ID n	Attr n Dimension	Attr n Element Size
Attr Start Index n (array only)	Attr Data Elements n (array only)	
...	...	

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Figure 70 – Get_Axis_Attributes_List Request format

Definitions of the individual parameters in this data structure are as follows.

- **Number of Attributes:** represents the number of attributes contained in the Get Axis Attribute service request.
- **Attr ID:** identifies the targeted Motion Device Axis Object attribute to get.
- **Attr Dimension:** determines the dimension of the attribute array. A dimension of zero means the attribute is a singular data element and, therefore, not really an array at all. Multidimensional arrays (dimension > 1) are supported by adding additional Attr Start Index and Attr Data Elements values. If there is an error associated with a specific requested get attribute operation, the device shall indicate this by setting the Attr Dimension to 0xFF, in which case the Element Size field contains the specific error code. When an error code is present, neither the array index parameters nor the attribute data fields are returned.
- **Attr Element Size:** determines the size, in bytes, of the attribute data element. Data elements shall be word aligned; 32-bit words are 32-bit aligned and 16-bit words are 16-bit aligned. Padding may be added to maintain word alignment. If there is an error associated with a specific requested get attribute operation, i.e. Attr Dimension = 0xFF, the Element Size field then contains the CIP Common General Status error code, as defined in IEC 61158-6. When an error code is present neither the array index parameters nor the attribute data fields are returned.
- **Attr Start Index:** identifies the starting index for the array of attribute values in the Attr Data section. This field is only present when the attribute data type is an array (i.e., dimension > 0).
- **Attr Data Elements:** determines the number of data element values in the Attr Data section for the associated index. This field is only present when the attribute data type is an array (i.e., dimension > 0).

The structure of Response Data Block for this service is as shown in Figure 71.

← 32-bit Word →		
Get_Axis_Attributes_List Response format		
Number of Attributes	-	
Attr ID 1	Attr 1 Dimension	Attr 1 Element Size
Attr Start Index 1 (array only)	Attr Data Elements 1 (array only)	
...	...	
Attr Data 1		
...		
Attr ID 2	Attr 2 Dimension	Attr 2 Element Size
Attr Start Index 2 (array only)	Attr Data Elements 2 (array only)	
...	...	
Attr Data 2		
...		
Attr ID n	Attr n Dimension	Attr n Element Size
Attr Start Index n (array only)	Attr Data Elements n (array only)	
...	...	
Attr Data n		
...		

IEC

Figure 71 – Get_Axis_Attributes_List Response format

Definitions of the individual parameters in this data structure are identical to the request data structure with the addition of the Attribute Data element, which is defined as follows.

- **Attr Data:** contains the current value(s) of the targeted Motion Device Axis Object attribute indicated by the Attr ID. If the attribute is an array, the value(s) are listed according to the Attr Start Index and the number of Attr Data Elements. For multidimensional arrays (dimension > 1), the sequence of data move sequentially through the indices from right to left. Examples are provided below.

EXAMPLE 1 Figure 72 defines a Get_Axis_Attributes_List Response for a simple 4-byte DINT scalar attribute. Since the Attr Dim is 0, the Attr Start Index and Attr Data Element fields are omitted.

Get_Axis_Attributes_List Response – 0 Dimensional Data Example		
Number of Attributes = 1	-	
Attr ID = 25	Attr Dim = 0	Attr Elem Size = 4
Attr Data = ?		

IEC

Figure 72 – Get_Axis_Attributes_List Response – Single 4-byte attribute

EXAMPLE 2 Figure 73 defines a Get_Axis_Attributes_List Response for a simple 2-byte scalar attribute showing the pad byte that is added to maintain 32-bit word alignment.

Get_Axis_Attributes_List Response – 0 Dimensional Data Example		
Number of Attributes = 1	-	
Attr ID = 20	Attr Dim = 0	Attr Elem Size = 2
Attr Data = ?	(Pad)	

IEC

Figure 73 – Get_Axis_Attributes_List Response – Single 2-byte attribute

EXAMPLE 3 Figure 74 defines a Get_Axis_Attributes_List Response for the first three elements of a one dimensional array attribute that is a UINT. Since the Attr Elem Size for this attribute is a 2-byte value, a Pad byte is added to maintain word alignment for any connection data to follow.

Get_Axis_Attributes_List Response – 1 Dimensional Array Example		
Number of Attributes = 1	-	
Attr ID = 43	Attr Dim = 1	Attr Elem Size = 2
Attr Start Index 1 = 0	Attr Data Elements 1 = 3	
Attr Data (0) = ?	Attr Data (1) = ?	
Attr Data (2) = ?	(Pad)	

IEC

Figure 74 – Get_Axis_Attributes_List Response – Byte attribute array

EXAMPLE 4 Figure 75 defines a Get_Axis_Attributes_List Response for six elements of a two dimensional array of REALs. The Attr Data sequences through the leftmost array index first beginning with the Attr Start Index 1 of 0.

Get_Axis_Attributes_List Response – 2 Dimensional Array Example		
Number of Attributes = 1	-	
Attr ID = 27	Attr Dim = 2	Attr Elem Size = 4
Attr Start Index 1 = 0	Attr Data Elements 1 = 2	
Attr Start Index 2 = 2	Attr Data Elements 2 = 3	
Attr Data (0, 2) = ?		
Attr Data (0, 3) = ?		
Attr Data (0, 4) = ?		
Attr Data (1, 2) = ?		
Attr Data (1, 3) = ?		
Attr Data (1, 4) = ?		

IEC

Figure 75 – Get_Axis_Attributes_List Response – Two Dimensional attribute array

If there is an error associated with a specific requested get attribute operation, the device shall indicate this by setting the Attr Dimension to 0xFF, in which case the Element Size field may contain the specific error code. When an error code is present, neither the array index parameters nor the attribute data fields are returned.

EXAMPLE 5 Figure 76 defines such a case, where Attribute 29 is not supported by the targeted motion axis, as indicated by the General Status Code of 0x14.

Get_Axis_Attributes_List Response – Attribute Error Example		
Number of Attributes = 3	-	
Attr ID = 25	Attr Dim = 0	Attr Elem Size = 4
Attr Data = 43,5		
Attr ID = 29	Attr Dim = 0xFF	Error Code = 0x14
Attr ID = 43	Attr Dim = 1	Attr Elem Size = 2
Attr Start Index 1 = 0	Attr Data Elements 1 = 3	
Attr Data [0] = 55	Attr Data [1] = 45	
Attr Data [2] = 35	(Pad)	

IEC

Figure 76 – Get_Axis_Attributes_List Response – Error example

7.5.2.3 Set_Axis_Attributes_List

The Set_Axis_Attributes_List service provides a mechanism to write a value to one or more settable Motion Device Axis Object attributes, including attributes having a multi-dimensional array data type. The buffer/array transfer mechanism can be used to build parameter tables in the device for indexing, or camming applications.

The format of the Request Data Block for this service is shown in Figure 77.

← 32-bit Word →		
Set_Axis_Attributes_List Request format		
Number of Attributes	-	
Attr ID 1	Attr 1 Dimension	Attr 1 Element Size
Attr Start Index 1 (array only)	Attr Data Elements 1 (array only)	
...	...	
Attr Data 1		
...		
Attr ID 2	Attr 2 Dimension	Attr 2 Element Size
Attr Start Index 2 (array only)	Attr Data Elements 2 (array only)	
...	...	
Attr Data 2		
...		
...		
Attr ID 3	Attr n Dimension	Attr n Element Size
Attr Start Index n (array only)	Attr Data Elements n (array only)	
...	...	
Attr Data 3		
...		

IEC

Figure 77 – Set_Axis_Attributes_List Request format

Definitions of the individual parameters in this data structure are as follows.

- **Number of Attributes:** represents the number of attributes contained in the Set Axis Attribute service request.
- **Attr ID:** identifies the targeted Motion Device Axis Object configuration attribute to set.
- **Attr Dimension:** determines the dimension of the attribute array. A dimension of zero means the attribute is a singular data element and, therefore, not really an array at all. Multidimensional arrays (dimension > 1) are supported by adding additional Attr Start Index and Addr Data Elements values prior to the Attr Data sequence.
- **Attr Element Size:** determines the size, in bytes, of the attribute data element. Data elements shall be word aligned; 32-bit words are 32-bit aligned and 16-bit words are 16-bit aligned. Padding may be added to maintain word alignment.
- **Attr Start Index:** identifies the starting index for the array of attribute values in the Attr Data section. This field is only present when the attribute data type is an array (i.e., dimension > 0).
- **Attr Data Elements:** determines the number of data element values based on the start index that included in the Attr Data section. This field is only present when the attribute data type is an array (i.e., dimension > 0).
- **Attr Data:** contains the new value(s) that are to be applied to the targeted device configuration attribute indicated by the Attr ID. If the attribute is an array, the new value(s) are applied according to the Attr Start Index and the number of Attr Data Elements. For multidimensional arrays (dimension > 1), the list of Addr Data elements moves sequentially through the indices from left to right. The data type of the Attr Data shall match the data type of the attribute specified by the Attr ID.

EXAMPLE 1 Figure 78 defines a Set_Axis_Attributes_List Request for a simple 4-byte scalar attribute. Since the Attr Dim is 0, the Attr Start Index and Attr Data Element fields are omitted.

Set_Axis_Attributes_List Request – 0 Dimensional Data Example		
Number of Attributes = 1	-	
Attr ID = 25	Attr Dim = 0	Attr Elem Size = 4
Attr Data = ?		

IEC

Figure 78 – Set_Axis_Attributes_List Request – Single 4-byte attribute

EXAMPLE 2 Figure 79 defines a Set_Axis_Attributes_List Request for a simple 2-byte scalar attribute showing the pad byte that is added to maintain 32-bit word alignment.

Set_Axis_Attributes_List Request – 0 Dimensional Data Example		
Number of Attributes = 1	-	
Attr ID = 20	Attr Dim = 0	Attr Elem Size = 2
Attr Data = ?	(Pad)	

IEC

Figure 79 – Set_Axis_Attributes_List Request – Single 2-byte attribute

EXAMPLE 3 Figure 80 defines a Set_Axis_Attributes_List Request for the first three elements of a one dimensional array of UINTs. Since the Attr Elem Size is 2-bytes, a Pad byte is added to maintain word alignment for any connection data to follow.

Set_Axis_Attributes_List Request – 1 Dimensional Array Example		
Number of Attributes = 1	-	
Attr ID = 43	Attr Dim = 1	Attr Elem Size = 2
Attr Start Index 1 = 0	Attr Data Elements 1 = 3	
Attr Data (0) = ?	Attr Data (1) = ?	
Attr Data (2) = ?	(Pad)	

IEC

Figure 80 – Set_Axis_Attributes_List Request – 2-byte attribute array

EXAMPLE 4 Figure 81 defines a Set_Axis_Attributes_List Request for six elements of a two dimensional array of REALs. The Attr Data sequences through the leftmost array index first beginning with the Attr Start Index 1 of 0.

Set_Axis_Attributes_List Request – 2 Dimensional Array Example		
Number of Attributes = 1	-	
Attr ID = 27	Attr Dim = 2	Attr Elem Size = 4
Attr Start Index 1 = 0	Attr Data Elements 1 = 2	
Attr Start Index 2 = 2	Attr Data Elements 2 = 3	
Attr Data (0, 2) = ?		
Attr Data (0, 3) = ?		
Attr Data (0, 4) = ?		
Attr Data (1, 2) = ?		
Attr Data (1, 3) = ?		
Attr Data (1, 4) = ?		

IEC

Figure 81 – Set_Axis_Attributes_List Request – Two dimensional attribute array

The structure of Response Data Block for this service is as shown in Figure 82.

← 32-bit Word →		
Set_Axis_Attributes_List Response format		
Number of Attributes	-	
Attr ID 1	Attr Status 1	Attr Index 1 (array only)
Attr ID 2	Attr Status 2	Attr Index 2 (array only)
...		
Attr ID n	Attr Status n	Attr Index n (array only)

IEC

Figure 82 – Set_Axis_Attributes_List Response format

- Attr ID: identifies the targeted Motion Device Axis Object configuration attribute of the set list request.
- Attr Status: indicates whether the set list action for the targeted attribute was successful. An Attribute Status value is zero if the targeted attribute was successfully written. A non-zero Attr Status value indicates that an error occurred that prevented the attribute from being updated. The error codes follow the CIP Common General Status Codes (see

IEC 61158-6-2). A non-zero Attr Status value for any element in the list shall also be reflected in this service's General Status element by indicating an Attribute List Error, 0x0A.

- Attr Index: indicates the array index of the array element that is responsible for the non-zero Attr Status error code value. This value is zero when the Attr Status value is 0 (success) or when the attribute data type is not an array.

7.5.2.4 Set_Cyclic_Write_List

The Set_Cyclic_Write_List service provides a mechanism to determine the list of attributes to be passed as part of the Cyclic Write Data Block of the Controller-to-Device Connection.

The format of the Request Data Block for this service is shown in Figure 83.

← 32-bit Word →	
Set_Cyclic_Write_List Request format	
Number of Attributes	-
Attr ID 1	Attr ID 2
...	

IEC

Figure 83 – Set_Cyclic_Write_List Request format

- Number of Attributes: represents the number of attributes contained in the Cyclic Write List.
- Attr ID: identifies the specific device configuration attribute that is to be updated via the Cyclic Write Data Block of the Controller-to-Device Connection. The Attr ID determines the data type and implied semantics of the data based on the specifications for the associated attribute.

The ordering of the attribute data in the Cyclic Write Data Block is determined by the ordering of the Attr IDs in the Set_Cyclic_Write_List request. Attribute data elements shall be word aligned; 32-bit words are 32-bit aligned and 16-bit words are 16-bit aligned. Padding may be added to maintain word alignment.

The structure of Response Data Block for this service is as shown in Figure 84.

← 32-bit Word →		
Set_Cyclic_Write_List Response format		
Number of Attributes	Cyclic Write Block ID	
Attr ID 1	Attr Status 1	-
Attr ID 2	Attr Status 2	-
...		
Attr ID n	Attr Status n	-

IEC

Figure 84 – Set_Cyclic_Write_List Response format

- Number of Attributes: represents the number of attributes contained in the requested Cyclic Write List.
- Cyclic Write Block ID: If all attributes in the requested list can be supported in the Set Cyclic Write Data Block, the Set_Cyclic_Write_List response provides a new Cyclic Write Block ID that is simply the increment of the current Cyclic Write Block ID. This new Cyclic Write Block ID can be used in the next Controller-to-Device Connection update. If the Set_Cyclic_Write_List is not successful, the Cyclic Write Block ID remains the current value.
- Attr ID: identifies the specific axis configuration attribute that was requested to be updated via the Cyclic Write Data Block of the Controller-to-Device Connection.

- **Attr Status:** The Attr Status value is zero if the associated attribute indicated by the Attr ID can be supported in the Set Cyclic Write Data Block. A non-zero Attr Status code indicates the specific reason why the associated attribute cannot be supported. These codes follow the CIP Common General Status Codes Codes (see IEC 61158-6-2). If the Attr Status of one or more Attribute IDs in the list indicates an error, the Set_Cyclic_Write_List request is considered unsuccessful.

7.5.2.5 Set_Cyclic_Read_List

The Set_Cyclic_Read_List service provides a mechanism to determine the list of attributes to be passed as part of the Cyclic Read Data Block of the Device-to-Controller Connection.

The format of the Request Data Block for this service is shown in Figure 85.

← 32-bit Word →	
Set_Cyclic_Read_List Request format	
Number of Attributes	-
Attr ID 1	Attr ID 2
...	

IEC

Figure 85 – Set_Cyclic_Read_List Request format

- **Number of Attributes:** represents the number of attributes contained in the Cyclic Read List.
- **Attr ID:** identifies the specific device status or signal attribute that is to be updated via the Cyclic Read Data Block of the Device-to-Controller Connection. The Attr ID determines the data type and implied semantics of the data based on the specifications for the associated attribute.

The ordering of the attribute data in the Cyclic Read Data Block is determined by the ordering of the Attr IDs in the Set_Cyclic_Read_List request. Attribute data elements shall be word aligned; 32-bit words are 32-bit aligned and 16-bit words are 16-bit aligned. Padding may be added to maintain word alignment.

The structure of Response Data Block for this service is as shown in Figure 86.

← 32-bit Word →		
Set_Cyclic_Read_List Response format		
Number of Attributes	Cyclic Read Block ID	
Attr ID 1	Attr Status 1	-
Attr ID 2	Attr Status 2	-
...		
Attr ID n	Attr Status n	-

IEC

Figure 86 – Set_Cyclic_Read_List Response format

- **Number of Attributes:** represents the number of attributes contained in the requested Cyclic Read List.
- **Cyclic Read Block ID:** If all attributes in the requested list can be supported in the Set Cyclic Read Data Block, the Set_Cyclic_Read_List response provides a new Cyclic Read Block ID that is simply the increment of the current Cyclic Read Block ID. This new Cyclic Read Block ID can be used in the next Device-to-Controller Connection update. If the Set_Cyclic_Read_List is not successful, the Cyclic Read Block ID remains the current value.
- **Attr ID:** identifies the specific axis configuration attribute that was requested to be updated via the Cyclic Read Data Block of the Device-to-Controller Connection.

- **Attr Status:** The Attr Status value is zero if the associated attribute indicated by the Attr ID can be supported in the Set Cyclic Read Data Block. A non-zero Attr Status code indicates the specific reason why the associated attribute cannot be supported. These codes follow the CIP Common General Status Codes (see IEC 61158-6-2). If the Attr Status of one or more Attribute IDs in the list indicates an error, the Set_Cyclic_Read_List request is considered unsuccessful.

7.5.2.6 Run_Motor_Test

The Run_Motor_Test service initiates various test operations on the motor to determine important motor model parameters. For active tests that enable the power structure, the axis shall be in the Stopped state for this service request to be accepted. For passive tests that do not enable the power structure, the axis may be in the Stopped state or in the following Standby states: Shutdown, Pre-charge, or Start Inhibit. The Run_Motor_Test initiates the specified test process and sets the In Process bit in the Axis Status attribute. For active tests, the axis state transitions to the Testing state. Since an active test process can take a considerable amount of time to complete, a service response is sent back to the controller as soon as the test process is initiated. This frees the service channel of the CIP Motion Connection to be used for other service requests for the duration of the process. When an active test process completes, the axis state transitions from the Testing state to either the Stopped state or the Major Faulted state. Once this state transition is complete, the In Process bit is cleared.

There are several different Test Types supported by this service(see Table 119). The Static Test applies a directionally static flux to the motor, while the Dynamic Test performs dynamic tests on the motor that produce motion. When Test Type is Calculated, the drive estimates motor model parameters based on the assumption that the motor is well matched to the power rating of the drive. The parameters measured by these tests are retrieved by the controller via the Get_Motor_Test_Data service.

Table 119 – Run_Motor_Test Request structure

Name	Type	Description of request parameter	Semantics of values
Test Type	USINT	Enumeration that specifies the type of motor test to perform:	Enumeration: 0 = static test 1 = dynamic test 2 = calculated 3 to 255 = reserved

There are no specific response parameters required for this service.

7.5.2.7 Get_Motor_Test_Data

The Get_Motor_Test_Data service provides access to the motor parameter measurements generated by the last Run_Motor_Test command. The controller uses this data to update the device with the best estimates of critical motor parameters. Parameters that are returned depend on the configured Motor Type at the time of the last Run_Motor_Test service. The Get_Motor_Test_Data service returns a 0 for parameters that were not explicitly measured or calculated by the last Run_Motor_Test procedure. Table 120 lists motor parameters that are typically measured or calculated by the Run_Motor_Test process according to the Test Type.

Table 120 – Get_Motor_Test_Data measured by Test Type

Motor parameters	Test Type		
	Static	Dynamic	Calculated
Stator Resistance (All)	yes	yes	yes
Leakage Inductance (IM, SPM)	yes	yes	yes
Flux Current (IM)	no	yes	yes
Slip Speed (IM)	no	yes	yes
Counter EMF (PM)	no	yes	no
Max Speed (IPM)	no	yes	no
Lq Inductance (IPM)	no	yes	no
Ld Inductance (IPM)	no	yes	no

To segregate standard motor test data from vendor specific motor test data, a Motor Test Data Set parameter is optionally included in the Get_Motor_Test_Data service request. If the Motor Test Data Set value is not included in the service request, the service request is for standard motor test data.

The (optional) request parameters for this service are defined in Table 121.

Table 121 – Get_Motor_Test_Data Request structure (optional)

Name	Type	Description of request parameter	Semantics of values
Motor Test Data Set	USINT	Enumeration that specifies the motor test data set to the service shall return based on the configured Motor Type.	Enumeration: 0 = standard data 1 = vendor specific data 2 to 255 = reserved

The response parameters for this service are defined in Table 122, Table 123 and Table 124.

Should the Motor Test be aborted, time-out, or fail for whatever reason, the device shall return parameter values that have been successfully measured during the test and set any parameters that were not successfully measured to zero.

**Table 122 – Get_Motor_Test_Data Response standard structure
(Motor Type = Induction)**

Name	Type	Description of response parameter	Semantics of values
Test Status	USINT	Indicates the result of the last Motor Test command.	Enumeration: 0 = test process successful 1 = test in progress 2 = test process aborted 3 = test process timed-out 4 = test process faulted
(Reserved)	USINT[3]	Pad bytes for 32-bit word alignment	
Stator Resistance	REAL	Measured phase-to-phase stator resistance of the motor.	Ohms
Leakage Inductance	REAL	Measured phase-to-phase leakage inductance of the motor.	Henries
Flux Current	REAL	Measured current to achieve full induction motor flux.	Amperes
Slip Speed	REAL	Measured slip at rated induction motor current.	r/min (rotary motor) m/s (linear motor)
(Reserved)	REAL	Pad for backward compatibility.	

Table 123 – Get_Motor_Test_Data Response standard structure (Motor Type = SPM)

Name	Type	Description of response parameter	Semantics of values
Test Status	USINT	Indicates the result of the last Motor Test command.	Enumeration: 0 = test process successful 1 = test in progress 2 = test process aborted 3 = test process timed-out 4 = test process faulted
(Reserved)	USINT[3]	Pad bytes for 32-bit word alignment	
Stator Resistance	REAL	Measured phase-to-phase stator resistance of the motor.	Ohms
Leakage Inductance	REAL	Measured phase-to-phase leakage inductance of the motor.	Henries
(Reserved)	REAL	Pad for backward compatibility.	
(Reserved)	REAL	Pad for backward compatibility.	
Counter EMF	REAL	Measured CEMF voltage drop at rated speed.	Volts

Table 124 – Get_Motor_Test_Data Response standard structure (Motor Type = IPM)

Name	Type	Description of response parameter	Semantics of values
Test Status	USINT	Indicates the result of the last Motor Test command.	Enumeration: 0 = test process successful 1 = test in progress 2 = test process aborted 3 = test process timed-out 4 = test process faulted
(Reserved)	USINT[3]	Pad bytes for 32-bit word alignment	
Stator Resistance	REAL	Measured phase-to-phase stator resistance of the motor.	Ohms
Lq Inductance	REAL	Measured phase-to-phase q-axis stator inductance of the motor.	Henries
Ld Inductance	REAL	Measured phase-to-phase d-axis stator inductance of the motor.	Henries
Counter EMF	REAL	Measured CEMF voltage drop at rated speed.	Volts

7.5.2.8 Run_Inertia_Test

The Run_Inertia_Test service performs an acceleration and deceleration ramp on the axis and makes timing measurements in the process (see Table 125).

Since this is an active test that enables the power structure, the axis shall be in the Stopped state for this service request to be accepted. The Run_Inertia_Test initiates the specified test process and sets the In Process bit in the Axis Status attribute. For this active test, the axis state transitions to the Testing state. Since an active test process can take a considerable amount of time to complete, a service response is sent back to the controller as soon as the test process is initiated. This frees the service channel of the CIP Motion Connection to be used for other service requests for the duration of the process. When the active test process completes, the axis state transitions from the Testing state to either the Stopped state or the Major Faulted state. Once this state transition is complete, the In Process bit is cleared.

The resultant timing measurements are accessed by the controller via the Get_Inertia_Test_Data command and ultimately used to calculate an accurate inertia value for the motor and load.

Table 125 – Run_Inertia_Test Request structure

Name	Type	Description of request parameter	Semantics of values
Test Direction	USINT	Enumeration that selects test direction.	Enumeration: 0 = forward 1 = reverse 2 to 255 = reserved
(Reserved)	USINT[3]	Pad bytes for 32-bit word alignment	
Test Velocity	REAL	Determines the maximum velocity that will be reached during the test profile.	Motor Units / s For linear motors, Motor Unit = Meters, for rotary motors, Motor Unit = Revs.
Test Torque	REAL	Determines the maximum torque that is applied during the test profile.	% Motor Rated
Test Travel Limit	REAL	Establishes the maximum excursion of the axis in Motor Units allowed in the test direction.	Motor Units For linear motors, Motor Unit = Meters, for rotary motors, Motor Unit = Revs.

There are no specific response parameters required for this service.

7.5.2.9 Get_Inertia_Test_Data

The Get_Inertia_Test_Data service provides access to the timing measurements generated by the last Run_Inertia_Test service. The controller uses this data to ultimately calculate an accurate inertia value for the motor and load. The Get_Inertia_Test_Data service returns a 0 for parameters that are not successfully measured by the last Run_Inertia_Test procedure.

There are no specific request parameters required for this service.

The response parameters for this service are defined in Table 126. Should the Inertia Test be aborted, time-out, or fail for whatever reason, the device shall return parameter values that have been successfully measured during the test and set any parameters that were not successfully measured to zero.

Table 126 – Get_Inertia_Test_Data Response structure

Name	Type	Description of response parameter	Semantics of values
Test Status	USINT	Enumeration that indicates the result of the last Run_Inertia_Test service:	Enumeration: 0 = test process successful 1 = test in progress 2 = test process aborted 3 = test process timed-out 4 = test process fault 5 = test reached limit 6 = test polarity incorrect 7 = test measurement error 8 = test configuration error 9 to 255 = reserved
(Reserved)	USINT[3]	Pad bytes for 32-bit word alignment	
Accel Time	REAL	Measured acceleration time to reach test velocity.	Seconds
Decel Time	REAL	Measured deceleration time to reach zero speed from test velocity.	Seconds

7.5.2.10 Run_Hookup_Test

The Run_Hookup_Test service performs a number of different tests to check for proper interface to the motor and/or feedback device. For active tests that enable the drive power structure, the axis shall be in the Stopped state for this service request to be accepted. For passive tests that do not enable the power structure, the axis may be in the Stopped state or in the following Standby states: Shutdown, Pre-charge, or Start Inhibit. The Run_Hookup_Test initiates the specified test process and sets the In Process bit in the Axis Status attribute. For active tests, the axis state transitions to the Testing state. Since an active test process can take a considerable amount of time to complete, a service response is sent back to the controller as soon as the test process is initiated. This frees the service channel of the CIP Motion Connection to be used for other service requests for the duration of the process. When an active test process completes, the axis state transitions from the Testing state to either the Stopped state or the Major Faulted state. Once this state transition is complete, the In Process bit is cleared.

There are several different Test Types supported by this service. The following is a brief description of each of these Test Types.

- The Motor/Feedback test is an active test that attempts to move the motor in the forward direction for a specified distance given by the Test Distance, or for a specified time given by the Test Time. Forward direction is defined as the direction the motor moves when following the normal UVW motor phase sequencing. During the test the device monitors the change in feedback counts over the duration of the test for applicable feedback

channels. When the Feedback Mode is set for Dual Feedback or Load Feedback, both feedback channels are tested.

- The Feedback test is a passive test that monitors the axis while an external agent moves it and indicates success if the axis feedback count exceeds the distance specified by the Test Distance. Which feedback channel to monitor is determined by the Feedback Channel parameter.
- The Marker test is a passive test that monitors the axis positioning feedback channel's marker signal while an external agent moves the axis through the marker location and indicates success if a marker pulse is detected. Which marker signal to test is determined by the Feedback Channel parameter.
- The Commutation Test, which applies only to PM motors, is an active test that applies current to the motor to align the rotor, check for proper phasing of a UVW encoder or Hall sensor if applicable, and measure the Commutation Offset. The Commutation test always targets Feedback 1, the motor feedback channel.

A time-out limit for the Run_Hookup_Test service is applied by the device vendor to handle the case where the success criteria cannot be met. This is typically around 30 s.

The resultant data generated by these test can be accessed by the Get Test Data service and used by the controller to automatically set the proper polarities of the feedback interface and the motor interface.

The request parameters for this service are defined in Table 127.

Table 127 – Run_Hookup_Test Request structure

Name	Type	Description of request parameter	Semantics of values
Test Type	USINT	Determines the specific hookup test to run:	Enumeration: 0 = motor & feedback test 1 = feedback test 2 = marker test 3 = commutation test (O) 4 to 255 = reserved
Feedback Channel	USINT	Determines the logical channel number targeted for the feedback test. When not applicable, this parameter shall be set to 0.	Channel Number
(Reserved)	USINT[2]	Pad bytes for 32-bit word alignment	
Test Distance	REAL	Establishes the distance that the axis needs to travel, in Motor or Feedback n Units to indicate a successful test. Test Type determines units. A Test Type of "motor & feedback test" uses Motor Units. A Test Type of "feedback test" uses Feedback n Units where n is determined by Feedback Channel. Test Distance is valid if the axis is not configured Encoderless/Sensorless operation. When not applicable, this parameter shall be set to 0.	Motor or Feedback n Units
Test Time	REAL	Establishes the time duration of axis motion during the test. Test Time is valid if the axis is configured Encoderless/Sensorless operation. When not applicable, this value shall be set to 0.	seconds

There are no specific response parameters required for this service request.

7.5.2.11 Get_Hookup_Test_Data

The Get_Hookup_Test_Data service provides access to the hookup test results generated by the last Run_Hookup_Test service (see Table 128). The controller uses this data to flag a wiring problem to the user and to calculate polarity configuration bit parameters for the axis. The results of this test are independent of the configured values for motor, feedback, and

commutation polarity. The Get_Hookup_Test_Data service returns a 0 for parameters that are not determined by the last Run_Hookup_Test procedure.

Table 128 – Get_Hookup_Test_Data measured by Test Type

Get_Hookup_Test parameters	Test Type			
	Motor/Feedback	Feedback	Marker	Commutation
Commutation Offset	no	no	no	yes
Commutation Polarity	no	no	no	yes
Feedback 1 Direction	Yes ^a	yes ^b	no	no
Feedback 2 Direction	yes ^a	yes ^b	no	no
^a When running a Motor/Feedback Hookup Test, Feedback 1 Direction is always measured unless the Feedback Mode is set to No Feedback (encoderless/sensorless operation). Feedback 2 Direction is measured when the Feedback Mode is set to either Dual Feedback or Load Feedback.				
^b When running a Feedback Hookup Test, either Feedback 1 Direction or Feedback 2 Direction are measured based on Feedback Channel specified by last Run_Hookup_Test.				

There are no specific request parameters required for this service.

The response parameters for this service are defined in Table 129. Should the Hookup Test be aborted, time-out, or fail for whatever reason, the device shall return parameter values that have been successfully measured during the test and set any parameters that were not successfully measured to zero.

Table 129 – Get_Hookup_Test_Data Response structure

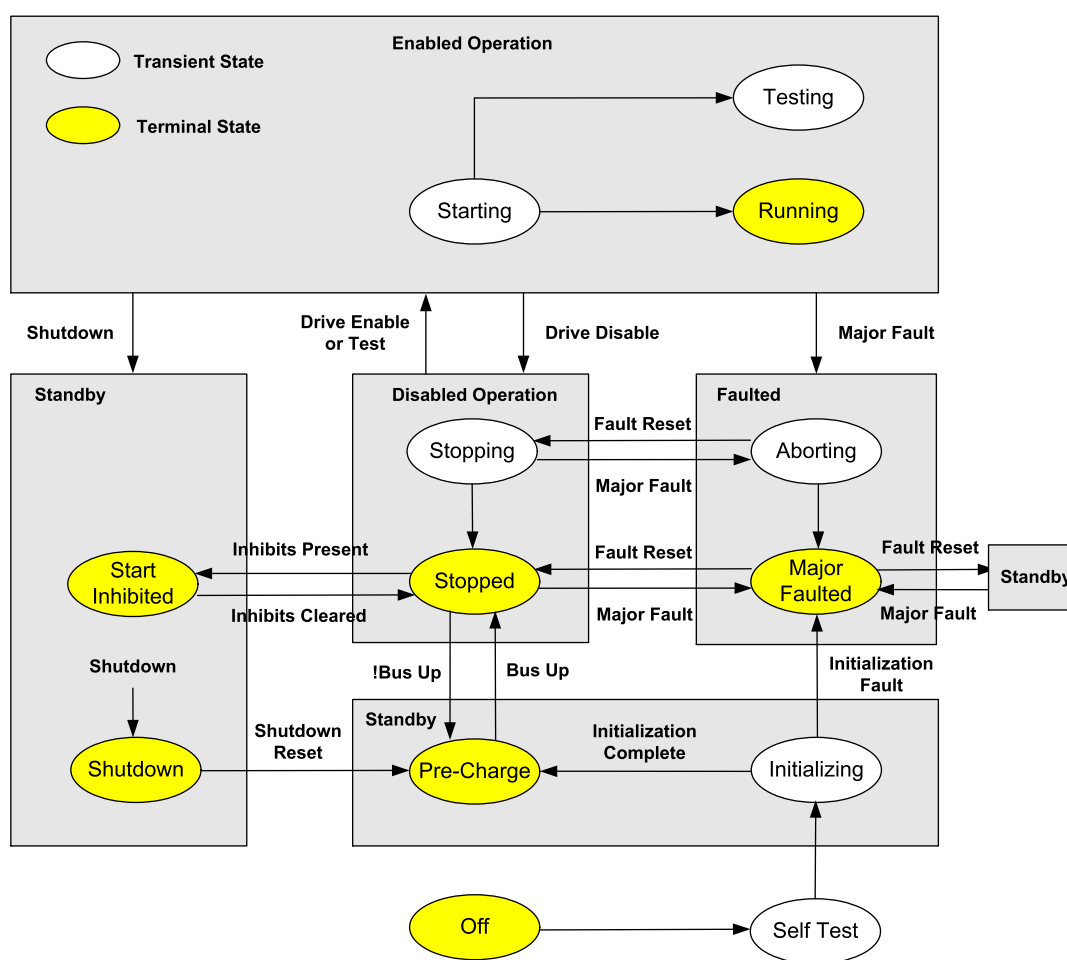
Name	Type	Description of response parameter	Semantics of values
Test Status	USINT	Enumeration that indicates the result of the last Hookup Test command:	Enumeration: 0 = test process successful 1 = test in progress 2 = test process aborted 3 = test process timed-out 4 = test process faulted 5 = test failed no feedback 1 6 = test failed no feedback 2 7 to 255 = reserved
Feedback 1 Direction	USINT	Enumeration that indicates the direction of axis motion associated with the last Hookup Test command during a Feedback 1 or Motor/Feedback Test, if applicable. Positive direction indicates the test generated positive incrementing feedback 1 counts.	Enumeration: 0 = Positive 1 = Negative 2 to 255 = reserved
Feedback 2 Direction	USINT	Enumeration that indicates the direction of axis motion associated with the last Hookup Test command during a Feedback 2 or Motor/Feedback Test, if applicable. Positive direction indicates the test generated positive incrementing feedback 2 counts.	Enumeration: 0 = Positive 1 = Negative 2 to 255 = reserved
Commutation Polarity	USINT	When performing a Commutation Test on a PM motor the test checks if the UVW phasing of the Encoder or Hall Sensor match the phasing of the Motor. If the phasing matches, Commutation Polarity is Normal. If it is determined that the phasing for the motor and commutation device do not match, this parameter reports that the UVW phasing is Inverted. See Commutation Polarity attribute for further details.	Enumeration: 0 = Normal 1 = Inverted 2 to 255 = reserved
Commutation Offset	REAL	Offset measured during the Commutation Test that shall be applied to the motor position accumulator in order to align the Electrical Angle signal with motor stator windings. See Commutation Offset attribute.	Electrical Degrees

7.6 Behavior

7.6.1 State model

7.6.1.1 General

The Motion Device Axis Object State Model is based on elements of the S88 and Pack/ML standard state models. The current state of the Motion Device Axis Object instance is indicated by the Axis State attribute (Attribute ID = 650). State transitions can be initiated either directly via the Axis Control request mechanism or by conditions that occur during device operation. Figure 87 shows the basic operating states of the Motion Device Axis Object when actively controlling axis motion (Control Mode != No Control). Shaded regions show mapping of Axis States to corresponding Identity Object states.



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Figure 87 – Motion Device Axis Object State Model

Valid transitions for the Axis State Model are explicitly defined in Table 130.

Table 130 – Axis State Machine transitions

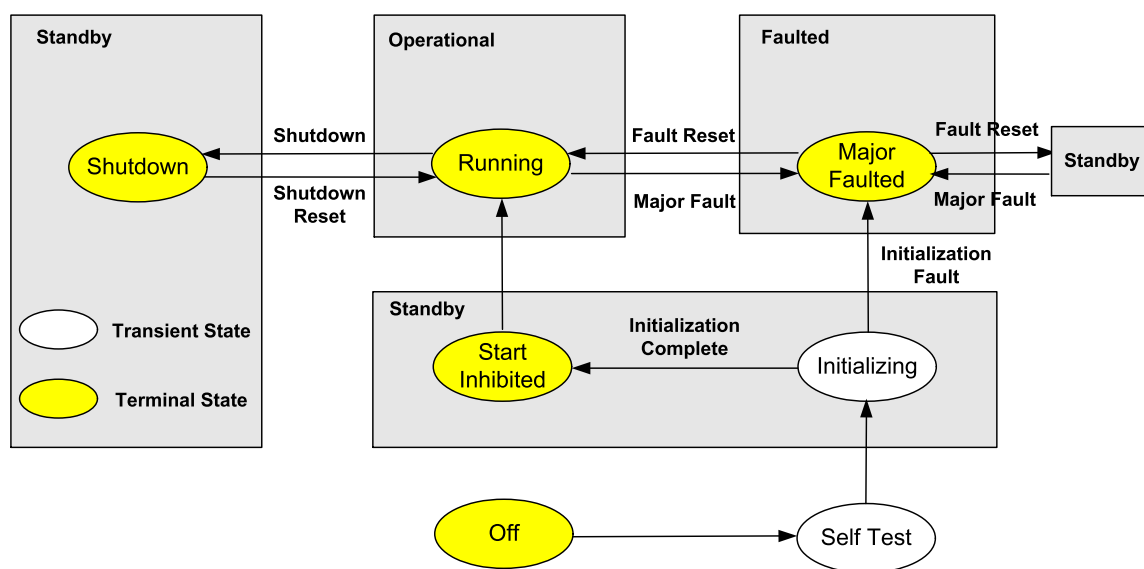
Current state	Event	Conditions	Next state
Off	Power Up		Self Test
Self Test	Self Test Complete		Initializing
Initializing	Initialization Fault		Major Faulted
Initializing	Initialization Complete		Pre-Charge
Shutdown	Major Fault or Abort		Major Faulted
Shutdown	Shutdown Reset		Pre-Charge
Pre-Charge	Shutdown		Shutdown
Pre-Charge	Major Fault		Major Faulted
Pre-Charge	Bus Up		Stopped
Start Inhibited	Shutdown		Shutdown
Start Inhibited	Major Fault		Major Faulted
Start Inhibited	Inhibits Cleared		Stopped
Major Faulted	Fault Reset	SD = 1	Shutdown
Major Faulted	Fault Reset	SD = 0, BU = 0	Pre-Charge
Major Faulted	Fault Reset	SD = 0, BU = 1, SI > 0	Start Inhibited
Major Faulted	Fault Reset	SD = 0, BU = 1, SI = 0	Stopped
Major Faulted	Reconnection		Initializing
Stopped	!Bus Up		Pre-Charge
Stopped	Shutdown		Shutdown
Stopped	Major Fault		Major Faulted
Stopped	Start Inhibit		Start Inhibit
Stopped	Enable		Starting
Stopped	Test		Starting
Starting	Shutdown		Shutdown
Starting	Major Fault or Abort		Aborting
Starting	Disable		Stopping
Starting	Start Complete	IP = 0	Running
Starting	Start Complete	IP = 1	Testing
Stopping	Shutdown		Shutdown
Stopping	Stop Complete		Stopped
Stopping	Major Fault		Aborting
Stopping	Enable	Flying Start Enabled	Starting
Aborting	Stop Complete		Major Faulted
Aborting	Fault Reset		Stopping
Testing	Shutdown		Shutdown
Testing	Major Fault or Abort		Aborting
Testing	Disable		Stopping
Running	Shutdown		Shutdown
Running	Major Fault or Abort		Aborting
Running	Disable		Stopping
Any State	Connection Close		Initializing
Any State	Connection Loss		Major Faulted

Some of the axis state machine transitions in Table 130 have dependencies on the current status conditions of the axis defined in Table 131.

Table 131 – Axis State Machine conditions

Condition	Symbol	Description
Bus Up	BU	This Axis Status bit is set if the DC bus is charged within the operating range.
Shutdown	SD	This Axis Status bit is set if there is an active shutdown. This is different than the shutdown state. The Shutdown status bit can be active in the faulted state.
In Process	IP	This Axis Status bit indicates a test process associated with service request is active. Note that the In Process bit may be active without the axis state machine being in the testing state.
Start Inhibit	SI	This represents the set of active Start Inhibits. An SI value of 0 indicates no active Start Inhibits. Note that start inhibits may be present without the axis state machine being in the start inhibit state.

When the Motion Device Axis Object is not actively controlling axis motion (Control Mode = No Control), the state diagram in Figure 87 reduces to the one shown in Figure 88 for a Feedback Only axis or CIP Motion Encoder device type.



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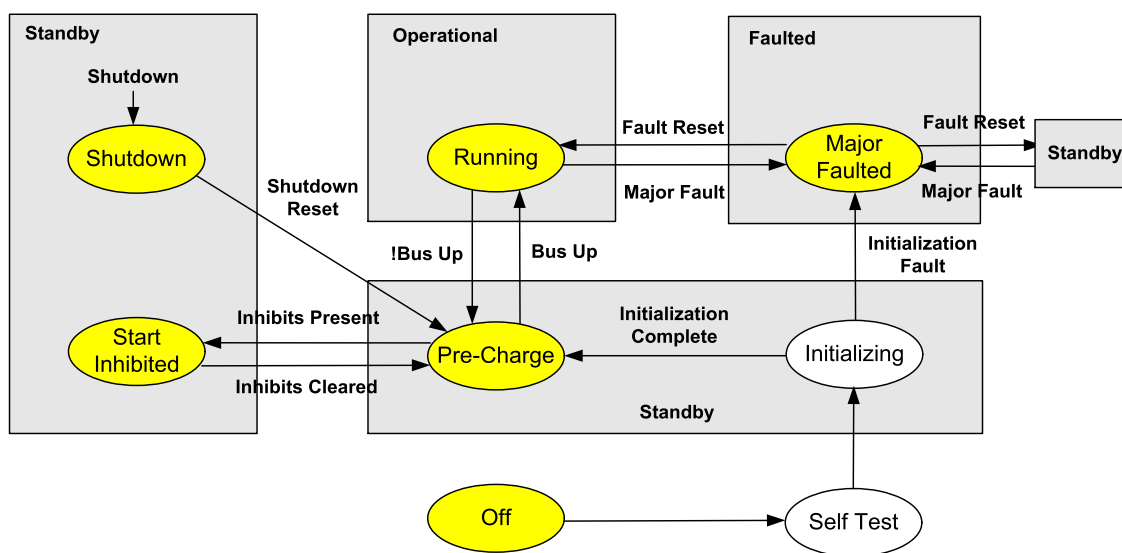
Figure 88 – Motion Device Axis Object State Model for Feedback Only

Valid transitions for the Axis State Model of a Feedback Only axis or CIP Motion are explicitly defined in Table 132.

Table 132 – Axis State Machine transitions (Feedback Only)

Current state	Event	Conditions	Next state
Off	Power Up		Self Test
Self Test	Self Test Complete		Initializing
Initializing	Initialization Fault		Major Faulted
Initializing	Initialization Complete		Start Inhibited
Shutdown	Major Fault		Major Faulted
Shutdown	Shutdown Reset		Running
Start Inhibited	Major Fault		Major Faulted
Start Inhibited	Inhibits Cleared		Running
Major Faulted	Fault Reset	SD = 1	Shutdown
Major Faulted	Fault Reset	SD = 0, SI > 0	Start Inhibited
Major Faulted	Fault Reset	SD = 0, SI = 0	Running
Major Faulted	Reconnection		Initializing
Running	Shutdown		Shutdown
Running	Major Fault		Major Faulted
Any State	Connection Close		Initializing
Any State	Connection Loss		Major Faulted

When the Motion Device Axis Object is associated with a CIP Motion Converter, the Active Control state diagram in Figure 87 reduces to the one shown in Figure 89.



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Figure 89 – Motion Device Axis Object State Model for Converter

Valid transitions for the Axis State Model of a CIP Motion Converter axis are explicitly defined in Table 133.

Table 133 – Axis State Machine transitions (Converter)

Current state	Event	Conditions	Next state
Off	Power Up		Self Test
Self Test	Self Test Complete		Initializing
Initializing	Initialization Fault		Major Faulted
Initializing	Initialization Complete		Pre-Charge
Shutdown	Major Fault		Major Faulted
Shutdown	Shutdown Reset		Pre-Charge
Start Inhibited	Inhibits Cleared		Pre-Charge
Pre-Charge	Start Inhibit		Start Inhibited
Pre-Charge	Shutdown		Shutdown
Pre-Charge	Major Fault		Major Faulted
Pre-Charge	Bus Up		Running
Major Faulted	Fault Reset	SD = 1	Shutdown
Major Faulted	Fault Reset	SD = 0	Pre-Charge
Major Faulted	Reconnection		Initializing
Running	!Bus Up		Pre-Charge
Running	Shutdown		Shutdown
Running	Major Fault		Major Faulted
Any State	Connection Close		Initializing
Any State	Connection Loss		Major Faulted

Fault Reset State transition precedence

In the Major Faulted state shown in the above Axis State Diagrams, the axis may transition to one of several different states in response to a Fault Reset event or the axis may transition right back to the Major Faulted state if there is a persistent fault condition present. Which state the axis transitions to depends on other state/status conditions of the axis. In fact, it may be possible for more than one state condition to be present at the same time, e.g. Shutdown, Start Inhibited, etc. Since the axis state model can only represent one state at any given time, the state of the axis is determined according to the following precedence:

- Major Faulted
- Shutdown
- Pre-Charge
- Start Inhibited
- Stopped

This and other state transition behavior is discussed in more detail in 7.6.1.2 and 7.6.2.

7.6.1.2 State Control

The primary method for changing the state of a Motion Device Axis Object instance is via the Axis Control/Axis Response mechanism that is built into the CIP Motion I/O Connection Header. Changing the state of the motion axis is simply performed by placing the appropriate Request Code in the Axis Control element of the Controller-to-Device Connection Header. The Control Request codes are defined in the CIP Motion device profile and reproduced in Table 134 for convenience.

Table 134 – Axis Control Request code

Request Code	Requested operation
0	No Request
1	Enable Request
2	Disable Request
3	Shutdown Request
4	Shutdown Reset Request
5	Abort Request
6	Fault Reset Request
7	Stop Process Request
8	Change Actual Position Reference Request
9	Change Command Position Reference Request
10 to 126	(Reserved)
127	Cancel Request
128 to 255	(Vendor Specific)

When a state transition is requested via the Axis Control element of the Controller-to-Device Connection, the device initiates the state transition and then acknowledges the transition request via the Axis Response attribute when the requested state transition completes or is determined unsuccessful. The Acknowledge Codes for the Axis Response element are the same as the Request Codes for the Axis Control element as defined in the CIP Motion device profile (see Table 135).

Table 135 – Axis Response Acknowledge codes

Acknowledge Code	Axis Response
0	No Acknowledge
1	Enable Acknowledge
2	Disable Acknowledge
3	Shutdown Acknowledge
4	Shutdown Reset Acknowledge
5	Abort Acknowledge
6	Fault Reset Acknowledge
7	Stop Process Acknowledge
8	Change Actual Position Reference Acknowledge
9	Change Command Position Reference Acknowledge
127	Cancel Acknowledge

The criteria for successful completion of the requested operation depend on the specific Request Code as shown in Table 136.

Table 136 – Completion criteria for requested operation

Request Code	Completion criteria
1	Axis successfully transitions to the Running state with the Tracking Command bit set in the Axis Status attribute.
2	Axis successfully transitions out of Enabled Operation with the Tracking Command bit clear in the Axis Status attribute.
3	Axis successfully transitions to Shutdown state or the Faulted state, with the Shutdown bit set in the Axis Status attribute.
4	Axis successfully transitions to Pre-charge, Stopped, or Inhibited state with the Shutdown bit clear in the Axis Status attribute and the Axis Faults attribute also momentarily clear. ^a If the axis is not in the Faulted or Shutdown states, the Shutdown Reset operation has no effect on the axis.
5	Axis successfully transitions to Aborting or Faulted state with the Controller Initiated Exception bit set in the Axis Faults attribute.
6	Axis successfully transitions to Stopped, Stopping, Inhibited, or Shutdown state with Axis Faults attribute momentarily clear. ^a If the axis is not in the Faulted state, the Fault Reset operation has no effect on the axis.
7	Axis successfully stops an active process, clearing the In Process bit in the Axis Status attribute.
8	Axis successfully applied Displacement to Actual Position to transition to new position reference system.
9	Axis successfully applied Displacement to Command Position to transition to new position reference system.
127	Always completes.
^a In the case of a Fault Reset or a Shutdown Reset, if any fault generating exception conditions still exist, the axis transitions out of the Faulted state only momentarily since the axis shall immediately fault again and return to the Faulted state.	

If the device determines that the requested state transition cannot be successfully completed, the device acknowledges the Control Request with a non-zero Error Code in the Response Status element. Possible error conditions are shown in Table 137.

Table 137 – Possible error conditions for requested operation

Request Code	Possible error conditions
1	Axis enters Faulted or Start Inhibited state after the request was received, before reaching the Running state.
2	None
3	None
4	None
5	None
6	None
7	None
8	None
127	None.

When the controller receives the Acknowledge Code, and there is no pending Control Request to initiate, it then zeroes the matching Request Code. When the device sees the Request Code zeroed, it clears the Acknowledge Code and the associated Response Status.

If the axis cannot transition to the requested state, the specific error condition is indicated by a non-zero Response Status attribute value that is passed along with the Axis Response at part of the Device-to-Controller Connection.

This behavior is further described in Table 138 and Table 139.

Table 138 – Successful Axis Control Request Cycle

Controller action	Connection data	Device action
Controller sets Request Code in the Axis Control element to request a state change.	→ Axis Control = x	Device sees non-zero Request Code, x, in Axis Control element and initiates the requested state change.
	← Axis Response = x ← Response Status = 0	If state change operation is successful, the device acknowledges the state change by setting the Acknowledge Code in the Axis Response element to the originating Request Code, x. The Response Status element is set to 0 indicating success.
Controller sees the non-zero Acknowledge Code in Axis Response element and zeroes the originating Request Code in the Axis Control element. The requested state change was successful as evidenced by the zero Response Status.	→ Axis Control = 0	
The state change request transaction is now complete. Another Axis Control request can be processed at this time.	← Axis Response = 0 ← Response Status = 0	Device sees a zero Request Code in the Axis Control element and zeroes the associated Acknowledge Code in the Axis Response element.

Table 139 – Unsuccessful Axis Control Request Cycle

Controller action	Connection	Device action
Controller sets Request Code in the Axis Control element to request a state change.	→ Axis Control = x	Device sees non-zero Request Code, x, in Axis Control element and initiates the requested state change.
	← Axis Response = x ← Response Status = Error Code	If state change operation is unsuccessful, the device indicates the problem via a non-zero Response Status Error Code along with the Acknowledge Code matching the originating Request Code, x.
Controller sees the non-zero Response Status associated with the Request Code, x, in the Axis Response, handles the error condition based on the Error Code, and clears the originating Request Code in the Axis Control element.	→ Axis Control = 0	
The state change request transaction is now complete. Another Axis Control request can be processed at this time.	← Axis Response = 0 ← Response Status = 0	Device sees a zero Request Code in the Axis Control element and zeroes the associated Acknowledge Code in the Axis Response element.

If there is a pending Control Request when the Acknowledge Code for the previous Control Request cycle is processed by the controller, the controller can immediately initiate the pending Control Request. In this way, the delay in executing the pending Control Request is minimized. This behavior is further described in Table 140.

Table 140 – Pending Axis Control Request Cycle

Controller action	Connection data	Device action
Controller sets Request Code in the Axis Control element to request a state change x.	→ Axis Control = x	Device sees non-zero Request Code, x, in Axis Control element and initiates the requested state change.
Controller has a pending state change request y as it receives the Acknowledge Code from state change request x	← Axis Response = x ← Response Status = 0	If state change operation is successful, the device acknowledges the state change by setting the Acknowledge Code, x, in the Axis Response element to the originating Request Code, x. The Response Status element is set to 0 indicating success.
Based on the non-zero Acknowledge Code in Axis Response element, the Controller immediately sets Request Code y in the Axis Control element to request the pending state change.	→ Axis Control = y	Device sees non-zero Request Code, y, in Axis Control element and initiates the requested state change.
	← Axis Response = y ← Response Status = 0	If state change operation is successful, the device acknowledges the state change by setting the Acknowledge Code, y, in the Axis Response element to the originating Request Code, x. The Response Status element is set to 0 indicating success.
Controller sees the non-zero Acknowledge Code in Axis Response element and, with no pending state change requests, zeroes the originating Request Code in the Axis Control element. The requested state change was successful as evidenced by the zero Response Status.	→ Axis Control = 0	
The last state change request transaction is now complete. Another Axis Control request can be processed at this time.	← Axis Response = 0 ← Response Status = 0	Device sees a zero Request Code in the Axis Control element and zeroes the associated Acknowledge Code in the Axis Response element.

Object state transitions can also be initiated by service requests such as the Test service requests. In general, these state change requests use services because they require one or more parameters to be passed to the device in order to initiate the state change or are not real-time performance critical.

In some cases the Control Request operation can take a considerable amount of time to perform, for example an Enable Request. To prevent such a request from monopolizing the Control Request mechanism, a special Cancel Request code is defined. The Cancel Request is immediately acknowledged by the device to complete the handshaking cycle, thereby, allowing a subsequent Control Request (see Table 141).

Table 141 – Cancel Request Cycle

Controller action	Connection data	Device action
Controller sets Request Code in the Axis Control element to request a state change, an Enable Request in this example.	→ Axis Control = Enable Req	Device sees the Enable Request Code in Axis Control element and initiates the requested state change. This state change happens to require significant time to complete.
Meanwhile, the controller determines that it is necessary to override the last Enable state change request with a Disable Request. To free up the Control Request mechanism to send the Disable Request, the Axis Control element shall first be set to Cancel Request	→ Axis Control = Cancel Req ← Axis Response = Cancel Ack ← Response Status = 0	Device sees the Cancel Request Code in Axis Control element and immediately sends the Cancel Acknowledge Code in the Axis Response element. The device shall continue the original state change operation until another Control Request is received.
Controller sees the non-zero Acknowledge Code in Axis Response element and zeroes the originating Request Code in the Axis Control element. The Cancel Request transaction is now complete. Another Axis Control request can now be processed at this time.	→ Axis Control = 0 ← Axis Response = 0 ← Response Status = 0	Device sees a zero Request Code in the Axis Control element and zeroes the associated Acknowledge Code in the Axis Response element.
Controller sets Request Code in the Axis Control element to request a Disable Request.	→ Axis Control = Disable Req	Device sees non-zero Request Code in Axis Control element and initiates the requested Disable state change. This may override any cancelled state change operation that is in process.
	← Axis Response = Disable Ack ← Response Status = 0	Once the Disable state change is successful, the device acknowledges the state change by setting the Acknowledge Code in the Axis Response element to the originating Request Code. The Response Status element is set to 0 indicating success.
Controller sees the non-zero Acknowledge Code in Axis Response element and zeroes the originating Request Code in the Axis Control element. The requested state change was successful as evidenced by the zero Response Status.	→ Axis Control = 0	
The state change request transaction is now complete. Another Axis Control request can be processed at this time.	← Axis Response = 0 ← Response Status = 0	Device sees a zero Request Code in the Axis Control element and zeroes the associated Acknowledge Code in the Axis Response element.

7.6.1.3 Changing position reference system

When the scaling function is enabled in the drive (see Control Mode definition) the Axis Control/Axis Response mechanism is used to coordinate transition of Actual Position and Command Position in both the controller and the drive to a new absolute position reference system. The ability to change the absolute position reference system is fundamental to Homing and Redefine Position operations associated with Position Control Systems.

Homing and Redefine Position operations begin by determining a Displacement value between the current position reference system and the new, or desired, position reference system. The trick is to apply the Displacement value to the Actual Position and Command Position values in the drive and the controller in such a way that it doesn't disrupt motion in progress.

The process of coordinating the transition to the new position reference system begins by placing the Change Actual Position Request Code in the Axis Control element of the Controller-to-Device Connection and a Position Displacement element in the Cyclic Data. When the drive receives the Change Actual Position Request, it applies the Position Displacement to the Actual Position in the drive. The next Device-to-Controller update passes Actual Position based on the new position reference system and the Position Displacement to the controller along with the Change Actual Position Acknowledge Code in the Axis Response element. The controller recognizes the Change Actual Position Acknowledge Code, so when computing the actual position delta (used in gearing and camming), the Position Displacement value is applied to compensate for the position reference change.

After the Change Actual Position Acknowledge Code is received by the controller, it then applies the Position Displacement value to the Command Position and sends the Command Position based on the new position reference system to the drive in the next Controller-to-Device update along with the Change Command Position Request Code in the Axis Control element. Again the Position Displacement is included as an element of the Cyclic data. Based on receiving the Change Command Position Request, the drive applies the Position Displacement to compensate for the position reference change when computing the command position delta.

The change-over to the new position reference system is officially completed with the Change Command Position Acknowledge passed back to the controller in the next Drive-to-Controller update. No Position Displacement needs to be passed to the controller in this update. Table 142 further defines the complete redefine position reference cycle.

Table 142 – Redefine Position Reference Cycle

Controller action	Connection data	Device action
Controller sets Request Code in the Axis Control element to request a Change Actual Position Request, chg actpos, and updates drive with Displacement.	→ Axis Control = chg actpos → Cyclic Data = displacement	Device sees non-zero Request Code, a Change Actual Position Request, in Axis Control element applies Displacement to Actual Position in the drive device.
Controller receives the Change Actual Position Acknowledge from the drive and applies Displacement to delta actual position calculation to correct for position reference change. Controller applies Displacement to shift Command Position to the new position reference system.	← Axis Response = chg actpos ← Response Status = 0 ← Cyclic Data = new actpos ← Cyclic Data = displacement	The drive device acknowledges the request by passing the Change Actual Position Acknowledge Code, chg actpos, back to the controller along with the new Actual Position in the new position reference system. The Response Status element is set to 0 indicating success.
Controller sets the Request Code to Change Command Position Request, chg compos, and updates drive along with the new Command Position based on the new position reference system and the Displacement.	→ Axis Control = chg compos → Cyclic Data = new compos → Cyclic Data = displacement	Drive receives the Change Command Position Acknowledge from the controller and applies Displacement to delta command position calculation to correct for position reference change
Controller indicates that the originating Home or Redefine Position operation is complete.	← Axis Response = chg compos ← Response Status = 0	The drive device acknowledges the Change Command Position Request change by passing the Acknowledge Code, chg compos, back to the controller in the Axis Response element. The Response Status element is set to 0 indicating success.
Controller sees the non-zero Acknowledge Code in Axis Response element and, with no pending state change requests, zeroes the originating Request Code in the Axis Control element.	→ Axis Control = 0	
The last state change request transaction is now complete. Another Axis Control request can be processed at this time.	← Axis Response = 0 ← Response Status = 0	Device sees a zero Request Code in the Axis Control element and zeroes the associated Acknowledge Code in the Axis Response element.

7.6.2 State behavior

7.6.2.1 General

Subclauses 7.6.2.2 to 7.6.2.14 offer a detailed description of each of the states and state transitions of the Motion Device Axis Object state model.

7.6.2.2 Off state

This is the state of the Motion Device Axis Object with power off.

7.6.2.3 Self Test state

When power is applied to the device, or the device is reset, the device typically goes through a series of self-test diagnostics and internal device parameters are set to their power-up default values. Once completed successfully, the device and all its associated axis instances transition to the Initializing state and are ready for initialization by the associated controller. If unsuccessful, the device and all its associated axis instances transition immediately to the Major Faulted state by declaring an Initialization Fault that is classified as Unrecoverable according to the terminology defined by the Identity Object. Clearing this fault can only be accomplished through a power cycle and is most likely the result of a device hardware problem.

If the device supports stand-alone operation under local control with local configuration data, the device is free to transition from the Self-test state to the Pre-charge state and on to the Stopped state. If the device receives a subsequent Forward_Open service to open a CIP Motion Connection, the drive device shall disable all axes and transition back to the Initializing state, following the state sequence outlined below.

If the device does not support stand-alone operation and depends on remote configuration data to be supplied over a CIP connection, the device shall transition to the Initializing state and wait (Standby) for the Forward_Open service from the controller to open the CIP Motion Connection.

7.6.2.4 Initializing state

During the Initializing state, the device waits for the CIP Motion Connections to the device to be established by the controller via a Forward_Open service. Once the Forward_Open service is successfully processed, the device initializes all attributes to their factory default values, resets all active faults, resets applicable axis status conditions including the shutdown bit, in preparation for device attribute configuration.

Once connections are established, the controller sends Set services to the device to set the Motion Device Axis Object configuration attributes to values stored in the controller. Any configuration error encountered during this process, for example due to value out of range or value not applicable, shall be handled by erring the Set service response; not by generating an Initialization Fault. The controller shall not complete the configuration process unless all configuration attributes have been successfully acknowledged.

If the device supports synchronous operation, the controller then synchronizes with the device using the Group_Sync service. Once this entire process has been completed successfully, the device and all its associated axis instances transition to the Pre-charge state. If a problem is found during this initialization process that is beyond the scope of a Set service error, the device generates an Initialization Fault. An Initialization Fault is viewed as an unrecoverable fault so clearing the fault can only be accomplished through a power cycle or a device reset service to the associated Identity Object.

If the CIP Motion Connection is intentionally closed for any reason during operation via a Forward_Close service, the device clears all active faults and alarms and returns to the Initializing state. If the CIP Motion Connection is lost for any other reason during operation, the device generates a Node Fault and transitions to the Major Faulted state. In either case the device shall wait for the CIP Motion Connections to the device to be re-established by the controller via a Forward_Open service.

The Initializing state is classified as an Identity Object Standby state and, therefore, the device shall insure that all associated power structures are disabled.

7.6.2.5 Pre-Charge state

In the Pre-Charge state, when applicable, the device is waiting for the DC Bus to fully charge (DC Bus Up status bit is clear). Once the DC Bus reaches an operational voltage level (DC Bus Up status bit is set) the axis either transitions to the Stopped state (drive axis) or to the Running state (converter axis). The drive device's inverter power structure is always disabled in this state (Power Structure Enabled status bit clear). Any attempt for the controller to enable a drive via the Axis Control mechanism while it is in the Pre-charge state is reported back to the controller as an error in the Response Status and the axis remains in the Pre-charge state.

The Pre-Charge state is classified as an Identity Object Standby state and, therefore, requires that the associated inverter power structure, if applicable, is disabled.

7.6.2.6 Stopped state

In the Stopped state, the device's inverter power structure shall either be disabled and free of torque (Power Structure Enabled status bit clear) or held in a static condition via an active control loop (Power Structure Enabled status bit set). No motion can be initiated by the device in the Stopped State nor can the device respond to a planner generated command references (Tracking Command status bit clear). In general, the axis shall be at rest, but if an external force or torque is applied to the load, a brake may be needed to maintain the rest condition. In the Stopped state, main power is applied to the device and the DC Bus is at an operational voltage level. If there are any Start Inhibit conditions detected while in this state, the axis transitions to the Start Inhibited state. If an Enable request or one of the Run Test service requests is applied to an axis in the Stopped state, the motion axis transitions to the Starting state.

7.6.2.7 Starting state

When an Enable request is given to an axis in the Stopped, or Stopping state when executing a Flying Start, the axis immediately transitions to the Starting state. In this state, the device checks various conditions before transitioning to the Running state. These conditions can include Brake Release delay time and Induction Motor flux level. The device control and power structures are activated during the Starting state (Power Structure Enabled status bit set) but the command reference is set to a local static value and will not track the command reference derived from the motion planner (Tracking Command status bit clear). If all the starting conditions are met, the axis state transitions to either the Running state or the Testing state.

7.6.2.8 Running state

The Running state is where the work gets done. In this state, the device's power structure is active (Power Structure Enabled status bit set) and the selected Control Mode is enabled and actively tracking command data from the controller based motion planner output to affect axis motion (Tracking Command status bit set). The motion axis remains in the Running state until either a fault occurs or it is explicitly commanded to stop via an Axis Control request.

In the case of an axis with no active control function (Control Mode, = No Control), the Running state simply indicates that the device is fully operational (Power Structure Enabled status bit and the Tracking Command status bit are both clear). The motion axis remains in the Running state until either a fault occurs or it is explicitly commanded to Shutdown via an Axis Control request.

In the Running state, only a subset of all axis instance configuration attributes are allowed by the device to be modified. Table 143 lists all attributes that the device is required to support in the Running state. Access to all other configuration attributes in the Running state is left to the device vendor's discretion.

Table 143 – Running State – Configurable attributes

Attr. ID	Attribute name	B	E	F	P	V	T
370	Skip Speed 1	-	-	O	-	-	-
371	Skip Speed 2	-	-	O	-	-	-
372	Skip Speed 3	-	-	O	-	-	-
373	Skip Speed Band	-	-	O	-	-	-
374	Ramp Velocity – Positive	-	-	O	-	O	-
375	Ramp Velocity – Negative	-	-	O	-	O	-
376	Ramp Acceleration	-	-	O	-	O	-
377	Ramp Deceleration	-	-	O	-	O	-
378	Ramp Jerk Control	-	-	O	-	O	-
440	Kvff	-	-	-	R	-	-
441	Kpp	-	-	-	R	-	-
442	Kpi	-	-	-	R	-	-
443	Position Lock Tolerance	-	-	-	R	-	-
444	Position Error Tolerance	-	-	-	R	-	-
445	Position Error Tolerance Time	-	-	-	O	-	-
446	Position Integrator Control	-	-	-	R	-	-
447	Position Integrator Preload	-	-	-	O	-	-
460	Kaff	-	-	-	R	R	-
461	Kvp	-	-	-	R	R	-
462	Kvi	-	-	-	R	R	-
464	Kdr	-	-	O	O	O	-
465	Velocity Error Tolerance	-	-	-	O	O	-
466	Velocity Error Tolerance Time	-	-	-	O	O	-
467	Velocity Integrator Control	-	-	-	R	R	-
468	Velocity Integrator Preload	-	-	-	O	O	-
469	Velocity Low Pass Filter Bandwidth	-	-	-	O	O	-
470	Velocity Threshold	-	O	O	O	O	O
471	Velocity Lock Tolerance	-	-	O	O	O	-
472	Velocity Standstill Window	-	R	R	R	R	R
473	Velocity Limit – Positive	-	-	O	O	O	-
474	Velocity Limit – Negative	-	-	O	O	O	-
485	Acceleration Limit	-	-	-	O	O	O
486	Deceleration Limit	-	-	-	O	O	O
496	Kj	-	-	-	R	R	O
498	Friction Compensation – Sliding	-	-	-	O	O	O
499	Friction Compensation – Static	-	-	-	O	O	O
500	Friction Compensation – Viscous	-	-	-	O	O	O
502	Torque Low Pass Filter Bandwidth	-	-	-	O	O	O
503	Torque Notch Filter Frequency	-	-	-	O	O	O
504	Torque Limit – Positive	-	-	-	R	R	R
505	Torque Limit – Negative	-	-	-	R	R	R
506	Torque Rate Limit	-	-	-	O	O	O
507	Torque Threshold	-	-	-	O	O	O

Attr. ID	Attribute name	B	E	F	P	V	T
508	Overtorque Limit	-	-	O	O	O	O
509	Overtorque Limit Time	-	-	O	O	O	O
510	Undertorque Limit	-	-	O	O	O	O
511	Undertorque Limit Time	-	-	O	O	O	O
553	Current Vector Limit	-	-	O	O	O	O
554	Kqp	-	-	-	O	O	O
555	Kqi	-	-	-	O	O	O
556	Kdp	-	-	-	O	O	O
557	Kdi	-	-	-	O	O	O
731	Digital Outputs	-	-	O	O	O	O
734	Analog Output 1	-	-	O	O	O	O
735	Analog Output 2	-	-	O	O	O	O

7.6.2.9 Testing state

When any one of the Run Test request services is sent to the motion axis while in the Stopped state, i.e. services that require an active power structure to execute, the axis immediately transitions to the Starting State (Power Structure Enabled status bit set), and then once the Starting conditions are met, the axis transitions to the Testing state. This Testing state is like the Running state in that the device's power structure is active (Power Structure Enabled status bit set), but in the Testing state one of the device's built-in test algorithms is controlling the motor, not command data from a motion planner (Tracking Command status bit clear). In the Testing state, the device excites the motor in various ways while performing measurements to determine characteristics of the motor and load. The motion axis remains in this state for the duration of the requested test procedure and then returns to the Stopped state. The motion axis can also exit the Testing state by either a fault or an explicit Axis Control request.

7.6.2.10 Start Inhibited state

The Start Inhibited state is the same as the Stopped state with the exception that the axis has one or more "start inhibit" conditions that prevent it from successfully transitioning to the Starting state. These conditions can be found in the Start Inhibit attributes. Once corrected, the axis state automatically transitions back to the Stopped state.

For an axis with no active control function (Control Mode = No Control) the Start Inhibited axis state prevents transitioning to the Running state until specific Start Inhibit conditions are resolved, such as when the associated device is not fully configured for operation. Again, once these conditions are corrected, the axis state automatically transitions to the Running state.

The Start Inhibited state is classified as an Identity Object Standby state and, therefore, requires that the associated power structure, if applicable, is disabled.

7.6.2.11 Stopping state

When a Disable request is issued to the Motion Device Axis Object in the Starting, Running, or Testing states, the axis immediately transitions to the Stopping state. In this state, the axis is in the process of stopping and is no longer tracking command data from the motion planner (Tracking Command status bit clear). There are a number of different Stopping Actions supported by the Motion Device Axis Object. Most of these Stopping Actions actively decelerate the axis to a stop. The power structure may remain active (Power Structure Enabled status bit set) as long as the Stopping Action procedure takes to complete. Once the selected Stopping Action procedure has completed, the axis transitions to the Stopped state.

When the Stopping Action is Disable and Coast, however, the power structure is immediately disabled (Power Structure Enabled status bit clear) and the axis coasts to a stop while in the Stopping state. In any case, the drive device shall wait until the axis has reached zero speed before transitioning to the Stopped state. In some cases, such as when the axis is stationary, this transition can be immediate. Recommended criteria for zero speed is based on Velocity Feedback, or in the case of Frequency Control drive device, this implies Velocity Reference being less than 1 % of motor rated speed. Ultimately this criteria is left to the vendors discretion.

When an Enable Request is given to an axis in the Stopping state with Flying Start Enabled, the axis shall immediately transition to the Starting state.

7.6.2.12 Aborting state

When a Major Fault occurs in the motion device while the axis is in the Starting, Running, Testing, or Stopping states, the axis immediately transitions to the Aborting state. In this state, the axis is in the process of stopping and is no longer tracking command data from the motion planner (Tracking Command status bit clear). The Aborting state executes the appropriate stopping action as specified by the device vendor. When actively stopping the axis in the Aborting state, the power structure remains active (Power Structure Enabled status bit set) as long as the stopping action takes to complete. In some cases the power structure shall be immediately disabled so the axis may coast to a stop while in the Aborting state. In any case, the drive shall wait until the axis has reached zero speed before transitioning to the Major Faulted state. Once the stopping procedure is complete and the axis has reached zero speed, the axis transitions to the Major Faulted state. In some cases, such as when the axis is stationary, this transition can be immediate.

When an Abort Request is issued to the Motion Device Axis Object a Controller Initiated Exception is generated. If the associated Axis Exception Action is set to generate a Major Fault the drive stops the axis according to the configured Stopping Action before transitioning to the Major Faulted state. Recommended criteria for zero speed is based on Velocity Feedback, or in the case of Frequency Control drive, this implies Velocity Reference being less than 1 % of motor rated speed. Ultimately this criteria is left to vendor discretion and can be application specific.

7.6.2.13 Major Faulted state

The Major Faulted state is identical to the Stopped state (or, if a Shutdown fault action was initiated, the Shutdown state) with the exception that there are one or more Major Faults active. In other words, a Major Faulted axis is a Stopped (or Shutdown) axis with a Major Fault condition present. Since faults are latched conditions, a Fault Reset request from the controller is required to clear the fault and, assuming the original fault condition has been removed, the axis transitions to the Stopped (or Shutdown) state.

There are four different sources of Major Faults: Node Faults, Initialization Faults, Axis Faults and Axis Safety Faults.

Initialization Faults

This kind of faults can only occur during the Initializing state. You cannot generate an Initialization fault in any other state of the device, i.e. faults occurring during operation of the device after transitioning out of the Initializing state. Initialization Faults can apply to a specific axis or to the entire device, in which case all axis instances would indicate the Initialization Fault. The device power structure, if applicable, is disabled when there is an Initialization Fault present.

Node Faults

These Faults always apply to the entire device and affect all axes the same. These faults can occur at any time during device operation. Node Faults are primarily communication faults, but can include general hardware faults where these fault conditions are checked during run-time. A CPU watchdog fault would be an example of a Hardware Node Fault. The device power structure, if applicable, is disabled when there is a Node Fault present.

Axis Faults

As the name implies, Axis Faults apply to a specific axis instances. Axis Faults are the direct result of Axis Exceptions that are configured to generate a Fault response. These exception conditions may apply to individual axis instances or to all axis instances. In any case, applications may require the device be configured to handle these exceptions differently for different axes. Run time conditions related to Motor, Inverter, Converter, Bus Regulator, and Feedback components, in general, shall be handled as Axis Exceptions. The device power structure, if applicable, may or may not be disabled when there is an Axis Fault present depending on the specific stopping action applied by the device in response to the fault condition.

Axis Safety Faults

Axis Safety Faults also apply to specific axis instances. Safety Faults are reported by the embedded Safety Core of the device that is responsible for monitoring the condition of various critical safety functions associated with the axis.

NOTE See [30] for more information.

Faults that occur after the device's axis state has transitioned out of the Initializing state shall be defined as either Node Faults, Axis Faults or Axis Safety Faults.

7.6.2.14 Shutdown state

When a Shutdown request is executed by the device, the targeted axis transitions to the Shutdown state. In the case of a Shutdown request, the axis immediately transitions from whatever state it is currently in to the Shutdown state. The Shutdown state has the same basic characteristics as the Stopped state except that the device's inverter power structure shall be disabled and free of torque (Power Structure Enabled status bit clear), and the Shutdown Action attribute can be configured to drop the DC Bus power to the device's power structure (DC Bus Up status bit clear).

NOTE This is generally done by opening an AC Contactor Enable, if applicable, output provided by the device that controls power to the converter.

Regardless of whether or not DC Bus power is disconnected, this state requires an explicit Shutdown Reset request from the controller to transition to the Pre-Charge state. If the device is configured to keep the DC Bus power active while in the Shutdown state then the motion axis transitions through the Pre-Charge state to the Stopped state. The Shutdown state offers an extra level of safety against unexpected motion.

In the case where a Shutdown fault action is initiated by the device in response to an exception condition that is configured to be a Major Fault, the device executes the Shutdown action, but the axis goes to the Major Faulted state, not the Shutdown state. Similarly, when the axis is in the Shutdown state and a major fault condition occurs, the axis transitions to the Major Faulted state. In other words, the major fault condition has precedence over the shutdown condition and the shutdown condition can be considered a sub-state. In either of these cases a Fault Reset request from the controller clears the fault and, assuming the original fault condition has been removed, the axis then transitions to the Shutdown state. A Shutdown Reset request from the controller, however, both clears the fault and performs a

shutdown reset so, assuming the original fault condition has been removed, the axis transitions to the Pre-Charge state as described above.

In addition to the Shutdown action functionality, the Shutdown state can also be used by the controller to disable any slave gearing or camming motion planner functions that reference this device axis as a master axis. For this reason, the Shutdown state is applicable to a Feedback Only where the axis instance is simply associated with a feedback device that has no active control function.

The Shutdown state is classified as an Identity Object Standby state and, therefore, requires that the associated drive power structure is disabled.

7.6.3 Fault and alarm behavior

7.6.3.1 General

The Motion Device Axis Object's Fault and Alarm handling functionality addresses both the need for a large and ever-expanding number of specific faults and alarms, the need for programmable actions, and the need for timely reporting of those faults and alarms to the controller. Additionally, no compromises are made to restrict the resolution of the reported faults and alarms, so that the controller always has access to the unique axis condition and a meaningful diagnosis. Numerous Fault and Alarm related attributes can be included in the fixed portion of the cyclic Device-to-Controller Connection so the controller can monitor the condition of the motion axis in real-time, without cumbersome polling.

The Axis Status attribute contains bits to indicate whether an alarm condition is present. The Axis State enumeration indicates when the axis has a major fault, which could be a regular runtime Axis Fault, or an Initialization Fault. The Axis Fault Code and related attributes are provided to report the specific fault condition, time stamp, and fault action to the controller for the purposes of building a fault log. But before going into detail on this, the terms used to describe the Fault and Alarm functionality of the Motion Device Axis Object need to be carefully defined.

7.6.3.2 Exceptions

Exceptions are runtime conditions that the device continually checks that might indicate improper behavior of the motion axis or operation outside of an allowable range. An exception can result in an alarm, a minor fault, or a major fault, depending on how the associated Axis Exception Action has been configured – an exception can even be configured to be ignored. Exceptions are automatically cleared by the device when the underlying exception condition is no longer present.

7.6.3.3 Exception Actions

For each exception, the motion axis can be programmed a variety of actions via the Axis Exception Action attribute. Exception Actions range from generating a major fault that results in the stopping of the motion axis all the way to taking no action at all. The Axis Faults attribute allows the controller to have immediate access to any exceptions that have been configured to generate a major or minor fault. The Axis Alarms attribute allows the controller to have immediate access to any exceptions that have been configured to be reported as alarms.

7.6.3.4 Alarms

Alarms are runtime exception conditions for which the device is to take no action other than to report as an alarm. Alarms and warnings, therefore, are basically synonymous. On a given device product, some exception conditions may not be able to simply be reported as an alarm without any associated action; for example an IPM fault in which the power module automatically shuts off without software intervention. Alarm conditions are automatically cleared when the underlying exception condition is no longer present.

7.6.3.5 Major Faults

Major Faults can be initialization faults, safety faults or runtime exception conditions that the device has been configured to regard as a major fault. If a runtime fault occurs during an operational state, for example Running or Testing, it results in the device stopping (or aborting) all axis motion. Major Faults ultimately transition the axis state to the Major Faulted state. A Major Fault that results from an exception condition is latched, and does not clear when the exception condition clears. A fault can only be cleared with a Fault Reset service request from the controller. If the fault condition is classified as an “unrecoverable fault”, only a power cycle or a device reset can clear the fault condition.

7.6.3.6 Minor Faults

Minor Faults are exception conditions that the device has been configured to report to the controller as a fault but not take any direct action. This provides the controller an opportunity to perform an application specific fault action that may not be supported by the device as one of the defined exception actions. Like alarms, Minor Faults do not initiate a state change nor does a Minor Faulted state even exist; a Minor Fault allows the motion axis to continue operation in the state that it is presently in. But unlike alarms, a Minor Fault is latched, i.e. the fault does not clear when the exception condition clears. Both Major and Minor Faults can only be cleared with a Fault Reset service request from the controller.

7.6.3.7 Initialization Faults

Initialization Faults are faults that are generated during the power-up or device reset procedure when the device detects a problem that prevents normal device operation. This could be a hardware or firmware problem detected as part of its self-diagnostic tests or a problem with the attribute configuration process. These faults are not sourced by exception conditions and, therefore, they do not have configurable actions. Examples of initialization faults are corrupted memory data, calibration errors, or firmware startup problems. Initialization Faults that result in the Faulted state are considered “unrecoverable faults” and cannot be cleared with a Fault Reset service request, so any kind of motion is impossible in this state; only a power-cycle or a Device Reset has a chance of clearing this kind of fault.

7.6.3.8 Safety Faults

Safety Faults are faults that are reported by the drive’s built-in Safety Core that is an integral part of a safety system that includes this device, a Safety Controller, and a CIP Safety network connection. When the safety system detects a problem that prevents normal safe operation of the drive it generates a safety fault. These faults are not sourced by exception conditions and, therefore, they do not have configurable actions. Like run-time faults, Safety Faults that result in the Faulted state are considered “recoverable faults” and can be cleared with a Fault Reset request.

EXAMPLE Safety feedback faults, safe stop faults, or safe speed faults.

7.6.3.9 Node Faults

Node Faults are faults that are generated by the drive’s communications interface. Such faults are not axis specific but apply to the Motion Device Axis Object class. If a Node Fault occurs during an operational state, for example Running or Testing, it results in the device stopping (or aborting) all controlled axes associated with the device. Node Faults ultimately transition the axis state to the Major Faulted state. A Major Fault that results from a Node Fault is latched. A fault can only be cleared with a Node Fault Reset service request from the controller. If the node fault condition is classified as an “unrecoverable fault”, only a power cycle or a device reset can clear the fault condition.

7.6.3.10 Fault codes

The Fault Code attribute is an enumeration that indicates the runtime exception condition, initialization condition or safety condition that generated the fault. The source of the condition

is specified by the Fault Type attribute. Like the fault status bits, this attribute value is latched and does not change unless, 1) the fault is “recoverable” and a Fault Reset request is initiated, or 2) another fault condition occurs. The Fault Code value corresponds to the bit position of the associated fault attribute. A value of 0 indicates that no fault condition currently exists for any of the potential fault sources. Fault Codes are primarily used for Visualization purposes and to build a Fault Log in the associated controller.

7.6.4 Start Inhibit behavior

A Start Inhibit is a condition that inhibits the axis from starting, i.e. transitioning to the Starting state for enabled axis operation. This condition does not generate an exception if a start attempt is made. If the circumstances that led to the Start Inhibit are no longer present, the start inhibit condition is automatically cleared by the device, returning the axis to the Stopped State.

If the motion axis is in the Start Inhibit state, it indicates that one or more conditions are present that prevent the axis from transitioning to enabled operation. The Start Inhibits attribute reports the specific condition that is inhibiting the axis.

7.6.5 Visualization behavior

7.6.5.1 General

Motion Device Axis Object state behavior has a direct impact on motion device visualization components. These components range from bicolor LED, or LED equivalent indicators to multi-character alphanumeric displays. This section defines how the Motion Device Axis Object states affect the behavior of these visualization components.

7.6.5.2 Module Status LED

Motion Device Axis Object states have a relationship to the state behavior of the Identity Object of the CIP Motion device and to its associated Module Status LED. Table 144 maps the states of the primary Motion Device Axis Object instance to the appropriate states of the Identity Object. CIP Motion compliant devices are required to support the Module Status LED.

For more information regarding the Identity Object state model, refer to the Identity Object specification (see IEC 61158–5-2).

7.6.5.3 Axis Status LED

To further augment the visual information provided by the standard Module Status LED, the CIP Motion device profile also defines the behavior of a second LED. This so-called Axis Status LED provides visual indication of, for example, whether or not the DC Bus is energized, whether the axis is enabled or disabled, and even provides indication of active alarms or minor fault conditions (see Table 144). This LED is also required by CIP Motion compliant devices unless the device is equipped with a multi-character alphanumeric display.

The Axis Status LED uses three colors that can be generated by a standard bicolor Red/Green LED, namely Red, Green, and Amber. Amber (or Yellow) is the color produced when both the Red and Green junctions of the bicolor LED are on. The general meaning of these three colors are as follows:

- green – indicates a normal power-up or operational state;
- amber – indicates the presence of an alarm or start inhibiting condition;
- red – indicates presence of some form of fault condition.

The normal power up or device reset, both the Module Status and Axis Status LEDs shall start in the Red state (under hardware control) while the device processor is booting and then switch to the Green state for approximately 1 s once the device begins executing its self test. Ideally, both LEDs shall be Red for approximately 1 s and then Green for approximately 1 s

prior to transitioning to a Standby state indication. This Red-Green sequence confirms proper operation of the LED indicators.

Table 144 – Axis state mapping to Identity Object with LED behavior

Identity Object state	Module Status LED	Motion Device Axis Object state	Axis Status LED
Nonexistent – Power Off	Off	Off	Off
Device Self-Testing	Flash Red/Green	Self Test	Flash Red/Green
Standby	Flashing Green	Initialization – Bus not Up	Off
		Initialization – Bus Up	Flashing Green
		Shutdown – Bus not Up	Off
		Shutdown – Bus Up	Flashing Amber ^a
		Pre-Charge – Bus not Up	Off
		Start Inhibit	Flashing Amber ^a
Operational	Solid Green	Stopped	Flashing Green ^{a b}
		Stopping	Solid Green ^{a b}
		Starting	Solid Green ^{a b}
		Running	Solid Green ^{a b}
		Testing	Solid Green ^{a b}
Major Recoverable Fault	Flashing Red	Aborting	Flashing Red
		Major Faulted	Flashing Red
Major Unrecoverable Fault	Solid Red	Aborting	Solid Red
		Major Faulted	Solid Red
^a The Motion Device Axis Object and the Identity Object define minor fault conditions. While a minor fault does not affect the Module Status LED, it does affect the Axis Status LED. When a minor fault condition is detected, a normally Solid Green LED indication changes to alternating Red-Green-Red-Green, a normally Flashing Green LED indication changes to alternating Red-Off-Green-Off, and a normally Flashing Amber indications changes to Red-Off-Amber-Off.			
^b The Motion Device Axis Object also defines alarm conditions. When an alarm condition is detected, a normally Solid Green LED indication changes to alternating Amber-Green-Amber-Green, while a normally Flashing Green LED indication changes to alternating Amber-Off-Green-Off.			

7.6.5.4 Alphanumeric display

7.6.5.4.1 General

In addition to the required LED visualization provided by the Module Status LED and Axis Status LED, the Motion Device Axis Object also defines the behavior of an optional alphanumeric display to more explicitly indicate the condition of the device. Such a display is particularly useful for monitoring progress through the initialization process and providing detailed diagnostic information concerning fault, alarm, and inhibit conditions. Alphanumeric displays can range from simple seven-segment displays to multi-character alphanumeric displays.

7.6.5.4.2 Seven-segment display

If the device is equipped with a seven-segment display, the display can be used to indicate progress through the Initializing state and various fault, alarm, and inhibit conditions. As a minimum, the display shall support the mapping to various conditions of the device specified in Table 145.

Table 145 – CIP Motion Device seven-segment display behavior

Display digit	Device condition
8	Executing device Self-Test
0	Waiting for connection to controller
1	Configuring device attributes
2	Waiting for group synchronization
3	Waiting for DC Bus to charge
4	Device is operational
I##	Initialization Fault Code
Ic##	Initialization Fault Code – Manufacturer's custom extensions
F##	Axis Fault Code
Fc##	Axis Fault Code – Manufacturer's custom extensions
A##	Alarm Code
Ac##	Axis Alarm Code – Manufacturer's custom extensions
S##	Start Inhibit Code
Sc##	Start Inhibit Code – Manufacturer's custom extensions
SF##	Safety Fault Code
SFc##	Safety Fault Code – Manufacturer's custom extensions
nF##	Node Fault Code
nA##	Node Alarm Code

7.6.5.4.3 Multi-character alphanumeric display

If the device is equipped with a multi-character alphanumeric display, even more useful device information can be conveyed to the user via scrolling or static character fields. The capabilities of the display generally dictate the length of the character strings that can be effectively displayed. Nevertheless, for the purpose of enforcing consistent behavior among CIP Motion compliant device products, the Motion Device Axis Object dictates exactly what is displayed for the conditions outlined in Table 146.

Table 146 – CIP Motion multi-character alphanumeric display behavior

Display string	Device condition
SELF-TEST	Executing device Self-Test
STANDBY	Waiting for Forward_Open Service
CONNECTING	Waiting for 1st Set Attribute Service. (Refreshing)
CONFIGURING	Configuring Class & Axis Instance Attributes
SYNCING	Waiting for Successful Group_Sync Service.
PRE-CHARGE	Waiting for DC Bus Up
SHUTDOWN	Axis has been Shutdown
STOPPED	Axis has stopped
STARTING	Axis is Starting
RUNNING	Axis is Running
TESTING	Axis is executing a Test procedure
STOPPING	Decelerating to a stop as a result of a disable
ABORTING	Decelerating to a stop as a result of a fault
INIT FLT S##	Initialization Fault – Std Fault Code
INIT FLT M##	Initialization Fault – Mfg Fault Code
FLT S##	Axis Fault – Std. Fault Code
FLT M##	Axis Fault – Mfg Fault Code
ALARM S##	Axis Alarm – Std Alarm Code
ALARM M##	Axis Alarm – Mfg Alarm Code
INHIBIT S##	Start Inhibit – Std Inhibit Code
INHIBIT M##	Start Inhibit – Mfg Inhibit Code
SAFE FLT S##	Axis Safety Fault – Std Safety Fault Code
SAFE FLT M##	Axis Safety Fault – Mfg Safety Fault Code
NODE FLT ##	Node Fault Code
NODE ALARM ##	Node Alarm Code

In addition to this basic functionality, the alphanumeric display can also provide detailed text descriptions of fault, alarm, and inhibit conditions. Definition of these specific strings is well beyond the scope of this object and is left to the discretion of the device vendor.

7.6.5.4.4 Multi-axis device visualization

In the case where there are multiple motion axis instances supported by the device, the above behavior needs further explanation.

First of all, there is only one Module Status LED per device node, so its condition is a roll-up of the states of all the Motion Device Axis Object instances. By contrast, one Axis Status LED is associated with each Motion Device Axis Object instance in the device that has a power structure. A single multi-character alphanumeric display can easily manage multiple motion axis instances.

In the case of a motion axis instance configured with no active control function (Control Mode = No Control), there is very little state information to visualize; the motion axis is in the operational state called Running and remains in that state unless there is a fault that transitions the axis to Major Faulted. A No Control axis, therefore, has no effect on the Module Status LED unless a fault occurs, in which case the Module Status shows Flashing

Red (Major Recoverable Fault). A No Control axis does not require a separate Axis Status LED.

In the case of multiple motion axis instances, each with a separate power structure, the behavior of the Module Status LED is a roll-up of the states of the device axes. When the device axes are in disparate states, the Module Status LED condition is based on the following precedence:

- 1) Major Unrecoverable Fault;
- 2) Major Recoverable Fault;
- 3) Standby;
- 4) Operational.

In other words, as far as the Module Status LED is concerned, a Major Recoverable Fault on Axis 1 trumps Axis 2 that is in the Standby state, Start Inhibit.

NOTE Minor faults and alarms are not recognized by the Module Status LED.

Multi-character displays easily handle multiple axis instances by adding a “X#” prefix to the display string to specify the associated motion axis instance number. For Master Feedback axis instances, no specific display string is shown unless the Master Axis has a fault or alarm condition.

Table 147 specifies the mapping between the display and the device condition in the case of multiple motion device axis instances.

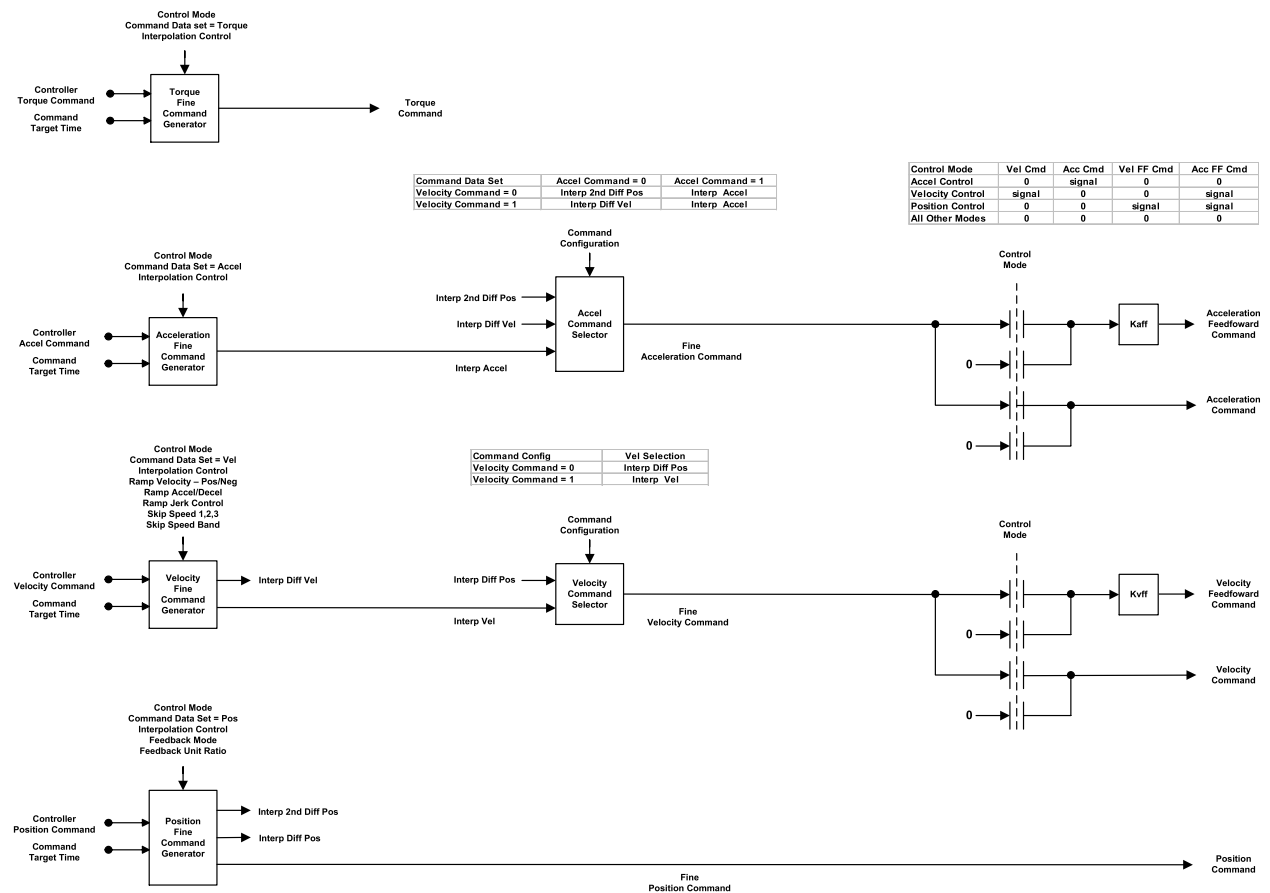
Table 147 – Multi-axis multi-character alphanumeric display behavior

Display string ^a	Device condition
SELF-TEST	Executing device Self-Test
STANDBY	Waiting for Forward_Open Service
CONNECTING	Waiting for 1 st Set Attribute Service. (Refreshing)
CONFIGURING	Configuring Class & Axis Instance Attributes
SYNCING	Waiting for Successful Group_Sync Service.
PRE-CHARGE	Waiting for DC Bus Up
X#:SHUTDOWN	Axis has been Shutdown
X#:STOPPED	Axis has stopped
X#:INHIBITED	Axis is Start Inhibited
X#:STARTING	Axis is Starting
X#:RUNNING	Axis is Running
X#:TESTING	Axis is executing a Test procedure
X#:STOPPING	Decelerating to a stop as a result of a disable
X#:ABORTING	Decelerating to a stop as a result of a fault
X#:INIT FLT S##	Initialization Fault – Std Fault Code
X#:INIT FLT M##	Initialization Fault – Mfg Fault Code
X#:FLT S##	Axis Fault – Std. Fault Code
X#:FLT M##	Axis Fault – Mfg Fault Code
X#:ALARM S##	Axis Alarm – Std Alarm Code
X#:ALARM M##	Axis Alarm – Mfg Alarm Code
X#:INHIBIT S##	Start Inhibit– Std Inhibit Code
X#:INHIBIT M##	Start Inhibit– Mfg Inhibit Code
X#:SAFE FLT S##	Axis Safety Fault – Std Inhibit Code
X#:SAFE FLT M##	Axis Safety Fault – Mfg Inhibit Code
NODE FLT ##	Node Fault Code
NODE ALARM ##	Node Alarm Code
^a X# = Axis Instance Number (0, 1, 2, 3 ...) → 0 = class instance	

7.6.6 Command generation behavior

7.6.6.1 Command data sources

Command data that affects axis motion can come from a variety of sources. The most common command data source is from a controller-based motion planner via the CIP Motion C-to-D Connection. In this context, command data can take the form of Controller Position, Velocity, Acceleration, and Torque Commands generated by the motion planner (see Figure 90). The command data elements provided are specified by the Command Data Set attribute, which is based on the selected Control Mode. The primary command data element can be augmented by higher order command elements for the purposes of generating high quality feed-forward signals. Alternatively, these higher order command elements can be derived by the device from the primary command data. In either case, a Fine Command Interpolator is generally applied to the Command Data to generate command reference signals to the devices' control structure at the devices' update rate.



IEC

Figure 90 – Command Generator

7.6.6.2 Command fine interpolation

For synchronized, high-performance applications using CIP Motion, command data is received from the CIP Motion C-to-D Connection and based on the connection's Command Target Update element being set to "Interpolate", processed by the Fine Interpolator functionality of the Command Generator blocks. The job of the Fine Interpolator is to compute coefficients to a trajectory polynomial that is designed to reach the command data at its associated Command Target Time. Depending on the specific command data element, the trajectory can follow a 1st, 2nd, or 3rd order polynomial trajectory with initial conditions based on current axis dynamics. Since the polynomial is a function of time, a new fine command value can be calculated any time the CIP Motion device needs to perform a control calculation. As a result, it is not necessary that the device's control calculation period be integrally divisible into the Controller Update Period.

To improve device interchangeability, the Motion Device Axis Object recommends a minimum order for the fine interpolators. Since contemporary motion planners typically generate their trajectories based on 3rd order polynomials in position, it is important that the fine interpolators reproduce these trajectories with high fidelity. Therefore, the position fine interpolator is defined as 3rd order, the velocity interpolator is 2nd order, and the acceleration and torque interpolators are both 1st order. Higher order fine interpolators are possible and are left to the vendor's discretion.

Position Fine Interpolation Polynomial:

$$P(t) = a_0 + a_1 \times (t - t_0) + a_2 \times (t - t_0)^2 + a_3 \times (t - t_0)^3$$

Velocity Fine Interpolation Polynomial:

$$V(t) = b_0 + b_1 \times (t - t_0) + b_2 \times (t - t_0)^2$$

Acceleration Fine Interpolation Polynomial:

$$A(t) = c_0 + c_1 \times (t - t_0)$$

Torque Fine Interpolation Polynomial:

$$T(t) = d_0 + d_1 \times (t - t_0)$$

In these equations, time t_0 represents the Command Target Time for the previous motion planner update such that when $t = t_0$ the position, velocity, acceleration, and torque command values are equal to the values sent in the previous motion planner update, i.e. P_{-1} , V_{-1} , A_{-1} , and T_{-1} . This establishes the 0th order coefficients of the polynomials.

$$P(t_0) = P_{-1} = a_0$$

$$V(t_0) = V_{-1} = b_0$$

$$A(t_0) = A_{-1} = c_0$$

$$T(t_0) = T_{-1} = d_0$$

The higher order polynomial coefficients are calculated such that by the next motion planner update, corresponding to Command Target Time, t_1 , the position, velocity, acceleration, and torque command values are the values sent in the latest motion planner update, i.e. P_0 , V_0 , A_0 , and T_0 .

$$P(t_1) = P_0$$

$$V(t_1) = V_0$$

$$A(t_1) = A_0$$

$$T(t_1) = T_0$$

Using the above polynomial interpolation equations, the CIP Motion device can compute position, velocity, acceleration, and torque command values at any time by plugging in the current System Time value of the device into the variable, t . This allows the device's control calculation to be performed according to a schedule that is independent of the controller's update schedule.

One thing that shall be done, however, is to adjust the Command Target Time, t_0 , should there be a shift in the System Time Offset for the device; t_0 and t shall always be based on the same System Time reference system. For example, assume the device's System Time Offset when the control command timestamp, t_0 , was received is Offset_0 . If the command interpolation equation is to be applied at $t = t_1$ and the current System Time Offset is defined as Offset_1 then t_0 shall be adjusted as follows before executing the polynomial:

$$\text{Adjusted } t_0 = t_0 + (\text{Offset}_1 - \text{Offset}_0)$$

Alternatively, the values for t , t_0 and t_1 can be based on local time rather than system time by using the current System Time Offset to convert between System Time to local time. This may be more convenient for the interpolator implementation and is left to the vendors discretion.

The polynomial coefficients are computed based on standard formulas that are a function of the history of command values over the last few updates. The number of historical command values used in the formula depends on the order of the polynomial. For example, the third order command position polynomial uses the three previous command position values. For convenience, the interpolator polynomial coefficient formulae are as follows:

Position Fine Interpolation Polynomial Coefficients:

$$a_0 = P_{-1}$$

$$a_1 = 1/T \times (\Delta P_0 - 1/2 \times \Delta V_0 - 1/6 \times \Delta A_0)$$

$$a_2 = 1/T^2 \times (1/2 \times \Delta V_0)$$

$$a_3 = 1/T^3 \times (1/6 \times \Delta A_0)$$

Velocity Fine Interpolation Polynomial Coefficients:

$$b_0 = V_{-1}$$

$$b_1 = 1/T \times (\Delta V_0 - 1/2 \times \Delta A_0)$$

$$b_2 = 1/T^2 \times (1/2 \times \Delta A_0)$$

Acceleration Fine Interpolation Polynomial Coefficients (Torque is same form as Accel):

$$c_0 = A_{-1}$$

$$c_1 = 1/T \times \Delta A_0$$

The above equations are based on the following nomenclature:

T = Controller Update Period

$$\Delta P_0 = (P_0 - P_{-1})$$

$$\Delta V_0 = (V_0 - V_{-1}) = (P_0 - 2P_{-1} + P_{-2})$$

$$\Delta A_0 = (A_0 - A_{-1}) = (V_0 - 2V_{-1} + V_{-2}) = (P_0 - 3P_{-1} + 3P_{-2} - P_{-3})$$

The above polynomial coefficients should be applied to the fine interpolator as soon possible after t is equal to or greater than t_0 . Applying the new coefficients too early, i.e. with t significantly less than t_0 , can create unnecessary error in the command trajectory when connecting the last fine interpolator segment to the new fine interpolator segment at t_0 .

When $t > t_1$, the fine interpolation polynomial becomes an extrapolation polynomial. In the absence of a fresh update from the motion planner, the extrapolation polynomial can be used to provide estimated command data to the device control structure until fresh motion planner command data is available. Once fresh command data is made available, new polynomial coefficients shall be computed without delay. In this way, the motion control can be maintained (“ride-thru”) despite occasional late or lost connection data packets resulting in a robust distributed motion control network solution. To be clear, late connection data is always applied and never thrown away; late data still represents the freshest data available from the controller and the extrapolation polynomial ensures that the command data is applied in such a way as to maintain a smooth motion trajectory despite variations in command data delivery.

When the update period of the motion planner is short enough relative to the dynamics of the command trajectory, or is comparable to the device control calculation period, fine interpolation may not be necessary. The motion planner can make this determination by comparing the planner update period to that of the device control calculation period. When fine interpolation is used, the planner shall add additional planner update periods to the planner time stamp, so it is advantageous to eliminate this planner update period delay if interpolation is not necessary.

Even though fine interpolation may not be necessary in some cases, it does not mean that the command data is to be applied directly to the device control structure. It still may be necessary to calculate the above polynomials so the device can extrapolate the command value when the device’s control update occurs. That is because, in general, the device’s control update time stamp does not need to match the time stamp of the command data.

Finally, there are applications and CIP Motion device types that do not require the dynamic accuracy that time-stamped interpolation and extrapolation provide. Various velocity and torque control applications, for example, may fall in this category. In general, command data can also be applied to the control structures of variable frequency drives without interpolation or extrapolation.

7.6.6.3 Command selectors

The Velocity and Acceleration Selectors are used to select the source of the Fine Velocity Command and Fine Acceleration Command signals. Selection is based on the Command Data Set value from the controller.

For example, if Velocity Command is provided by the controller, the Fine Velocity Command is sourced by the output of the Velocity Fine Command Generator. When configured for fine interpolation or extrapolation, the polynomial used to calculate the Velocity Fine Command is given by:

$$V(t) = b_0 + b_1 \times (t - t_0) + b_2 \times (t - t_0)^2$$

The polynomial coefficients are:

$$b_0 = V_{-1}$$

$$b_1 = 1/T \times (\Delta V_0 - 1/2 \times \Delta A_0)$$

$$b_2 = 1/T^2 \times (1/2 \times \Delta A_0)$$

In this case, values for ΔV_0 and ΔA_0 are calculated in terms of Velocity Command values, V_0 , V_{-1} , and V_{-2} :

$$\Delta V_0 = (V_0 - V_{-1})$$

$$\Delta A_0 = (V_0 - 2V_{-1} + V_{-2})$$

If the Velocity Command is not provided by the controller, then the Fine Velocity Command signal is the Interpolated Differential Position signal from the Position Fine Command Generator that is the time derivative of the Fine Position Command signal. When configured for fine interpolation or extrapolation, the polynomial used to calculate the Velocity Fine Command are the same as above, but the polynomial coefficients are calculated using ΔV_0 and ΔA_0 values derived from Position Command values, P_0 , P_{-1} , P_{-2} , and P_{-3} :

$$\Delta V_0 = (P_0 - 2P_{-1} + P_{-2})$$

$$\Delta A_0 = (P_0 - 3P_{-1} + 3P_{-2} - P_{-3})$$

7.6.6.4 Command ramp generator

The Ramp Generator feature of the Command Generator block is applied to the Command Data value sent by the controller when the Command Target Update element of the connection is set to “Immediate” mode. In Immediate mode, the Command Data is applied immediately to the devices’ control structure. Since there is generally no motion planner generating the Command Data in this mode, the Command Data value from the controller can change drastically from one update to the next. To address this condition, a Ramp Generator function is needed to ramp the motor to the new Command Data value within the dynamic limitations of system. An example of if the Controller Velocity Command value suddenly changed from 0 to 30 revolutions per second in Immediate Mode, the Ramp Generator would produce a Fine Velocity Command signal that accelerates the motor to the Controller Velocity Command value based on the configured Ramp Acceleration and Jerk Control attribute values. The Ramp Jerk Control attribute determines what percentage of the acceleration or deceleration ramps is S-Curve with the remaining portion of the ramp governed by the fixed Ramp Acceleration or Deceleration attribute values.

While a Ramp Generator function could be included in each of the Fine Command Generator blocks for position, velocity, and acceleration commands, this version of the Motion Device Axis Object specification only supports a Ramp Generator in the Velocity Fine Command Generator block.

The Ramp Generator enforces directional velocity limits on the Command Data, insuring that the Velocity Command never exceeds the configured Maximum Velocity Pos/Neg values.

The Ramp Generator also supports Flying Start functionality. When enabling the drive while the motor is still moving, the Ramp Generator output is initialized to the current speed of the motor. From there, the Ramp Generator smoothly accelerates or decelerates the motor to the current Controller Velocity Command.

Finally, the Ramp Generator supports Skip Bands that are most frequently used in Frequency Control applications when certain speeds excite mechanical resonance frequencies of the motor and load. The Skip Band feature allows three separate Skip Speeds to be defined that shift the Velocity Command signal to avoid, or skip, these problematic speeds. The Skip Speed Band determines the range of speeds centred on the three Skip Speeds that the device avoids. If the Velocity Command falls within the Skip Band but below the Skip Speed the Velocity Command output is set to the Skip Speed, minus $\frac{1}{2}$ the Skip Speed Band. If the Fine Velocity Command falls within the Skip Band but above the Skip Speed the Velocity Command output is set to the Skip Speed, plus $\frac{1}{2}$ the Skip Speed Band.

7.6.6.5 Feed-Forward signal selection

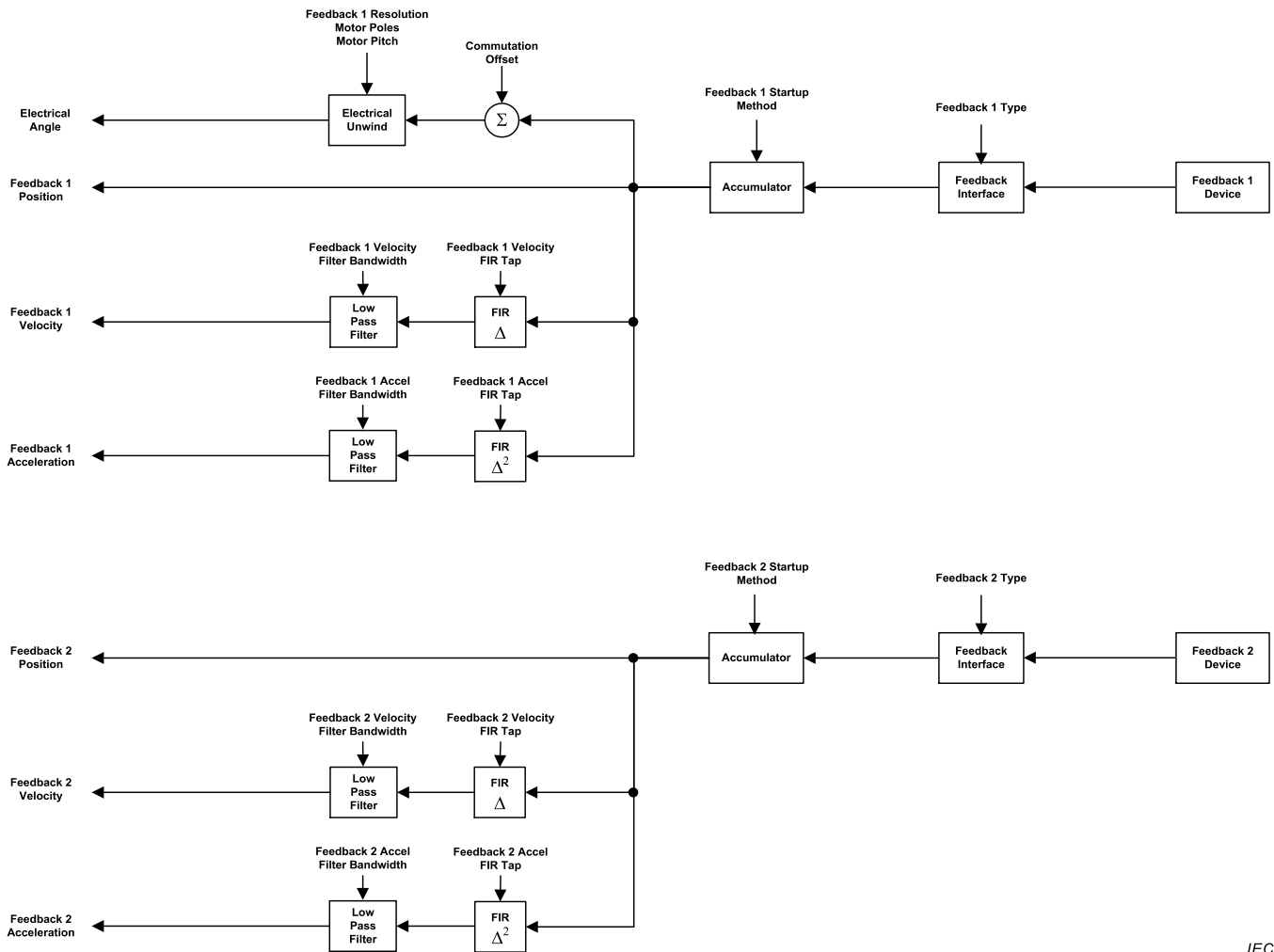
The Fine Command Generators defined as part of the Motion Device Axis Object can generate higher derivatives of the command data input to serve as feedforward signals. The units for the velocity and acceleration feedforward signals are generally different than the derivative units, hence the derivative signals shall be scaled appropriately. Superior signal quality, however, can be provided by the motion planner trajectory generators. The feedforward selection blocks pick the best feed-forward signal to apply based on the bits set in the Command Data Set attribute. The best signal is defined as the signal derived using the fewest differencing operations. The fine command position is applied directly to the position control loop without any of the typical de-referencing and offsets. It is assumed that these operations are performed by the controller based motion planner.

7.6.7 Feedback interface behavior

7.6.7.1 Feedback sources

Feedback signals defined by the Motion Device Axis Object can be derived from any of 4 different feedback interface channels (see Figure 91). The two primary feedback channels employed by the various closed loop control modes are designated Feedback 1 and Feedback 2. This allows the control loops to operate with either a motor based feedback device that is typically attached to the Feedback 1 channel or a load-side feedback device that is connected to the Feedback 2 channel. Which feedback source is used by the loop is governed by the Feedback Mode attribute.

Each feedback interface is capable of supporting a number of different feedback device types as enumerated by the Feedback Type attribute. The feedback interface output is the number of feedback counts that the feedback device has moved since the last time the device was sampled. If the feedback device is an absolute device, the feedback interface also determines the absolute position of the feedback device at power-up and communicates that value to the Feedback Accumulator to preset the accumulator.



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Figure 91 – Feedback Channels 1 and 2

7.6.7.2 Feedback accumulator

The role of the Feedback Accumulator depends on the configured Feedback n Startup Method which can be either Incremental or Absolute. If Incremental is selected, the accumulator simply accumulates changes to the feedback count value, a 32-bit signed integer, with every device update. If Absolute is selected, the Feedback Accumulator works basically the same way as in Incremental mode. The only difference is the initialization of the accumulator at device power-up. In Incremental mode, the Feedback Accumulator is set to zero, while in Absolute mode, the accumulator is initialized to the absolute position of the feedback device. This allows for the recovery of absolute position through a power-cycle as long as power-off movement of the absolute feedback device is limited to half of the absolute feedback range of the device. There is no device requirement to extend the absolute position range of the feedback device through non-volatile storage of the accumulator. This simple absolute feedback handling mechanism is due to the fact that the CIP Motion normally places the responsibility of extending the absolute position range of the axis, and establishing the absolute machine position reference, on the controller.

7.6.7.3 Commutation unwind and offset

An Electronic Unwind block is also connected to the Feedback 1 interface. This block is designed to unwind, or modulo, the position accumulator output to generate a signal that is proportional to the electrical angle of a Permanent Magnet motor based on the Pole Count or Pole Pitch of the motor. To align this signal with the physical ABC windings of

the motor rather than the zero of the feedback device, a configurable Commutation Offset is added prior to the Electrical Unwind block.

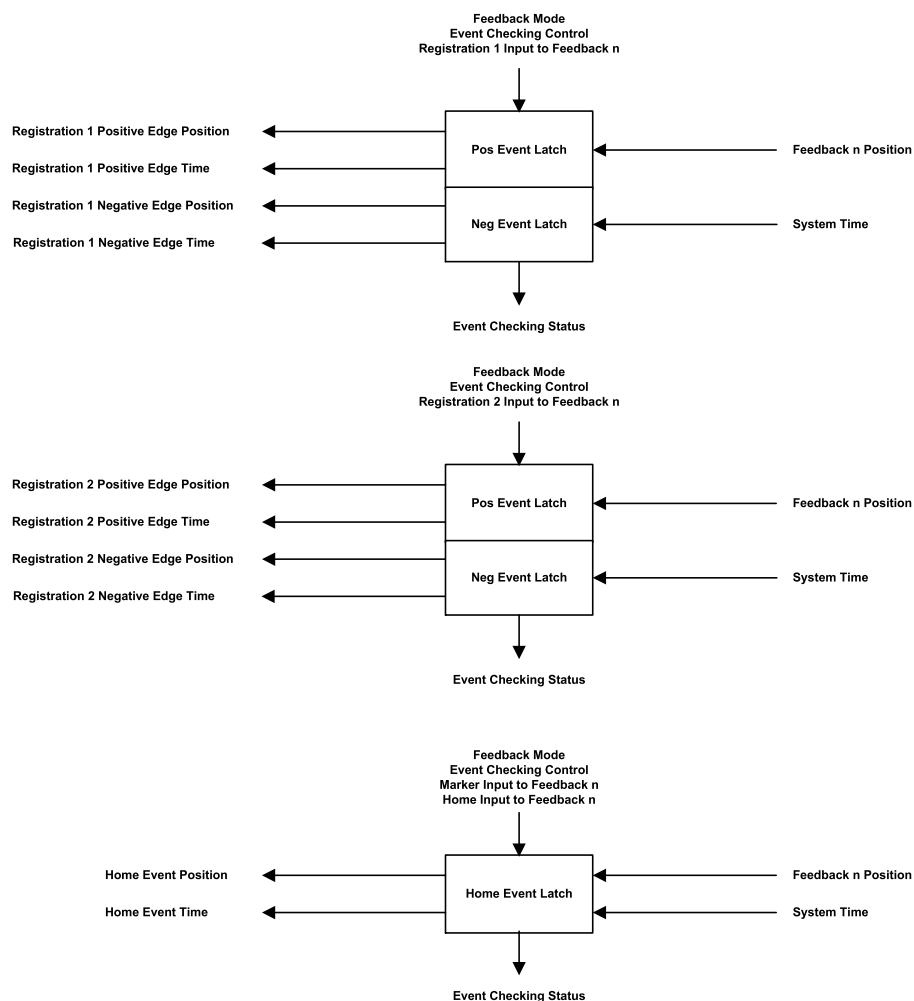
7.6.7.4 Feedback filtering

A configurable low-pass IIR filter is defined by the Motion Device Axis Object for filtering the velocity and acceleration estimates for each feedback channel. These filters can be used to reduce the level of quantization noise associated with differencing digital feedback signals. The bandwidth of the velocity and acceleration IIR filters for each feedback channel are individually programmable.

7.6.8 Event Capture Behavior

7.6.8.1 Event input sources

The Motion Device Axis Object defines a mechanism to capture both the feedback position and time stamp associated with specific state transitions of selected event input sources. Event input sources currently supported by the object are Registration 1, Registration 2, Marker, and Home Switch. These 4 event input sources apply to each supported feedback channel (see Figure 92).



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Figure 92 – Event Capture Functionality

7.6.8.2 Event latches

To facilitate accurate capture of both feedback position and time, hardware event latches are typically implemented as shown in the following block diagram. Two independent latches are defined for each registration input, one latch to capture positive edge transition events and one to capture negative edge transition events. This design enables capture of both registration events in applications with narrow registration pulses where the rising and falling edges occur nearly simultaneously. In addition to the registration latches, a separate latch is also defined for the home event capture. The home input event that triggers the Home Event Latch can be any of a number of different combinations of home switch and marker input events, i.e. marker transitions, switch transitions, or switch transitions followed by a marker transition.

With hardware based event latches, event capture accuracy is, in general, only limited by the latency of the associated event input. Registration and Marker event inputs are lightly filtered so event capture accuracy is in the order of 1 μ s. In terms of position capture accuracy, that would be calculated as the product of the event capture accuracy and the speed of the axis. Home switch inputs are heavily filtered, in general, and therefore limited to an event capture accuracy of 1 ms to 10 ms. Thus, to get an accurate position capture based on a home switch input transition, a homing sequence with a slow homing speed is required.

7.6.8.3 Event time stamps

Since the registration time stamp is passed to the controller as part of the Event Notification data, the controller can apply the event time stamp to the position history of other axes in the system to interpolate their positions. This is particularly useful in applications where it is necessary to determine the location of several axes at the time of a single registration event. The more accurate the time stamp, the more accurately the controller can determine these positions.

One thing that shall be done, however, is to adjust the event time stamp, t_0 , should there be a shift in the System Time Offset for the device prior to transmitting to the controller; the event time stamp shall always be based on the same System Time reference system at the time of transmission. For example, if Offset_0 is the device's System Time Offset when the event timestamp t_0 occurred, and t_1 is the System Time Offset at the time that the event is to be transmitted to the controller, then t_0 shall be adjusted to be t_1 prior to transmission with the rest of the associated event data to the controller, i.e.: $t_1 = t_0 + (\text{Offset}_1 - \text{Offset}_0)$.

7.6.9 Control Mode behavior

7.6.9.1 General

The instance attributes defined in 7.3 affect device behavior in the context of the Control Mode, Control Method, and Feedback Mode. As discussed in 4.2 concerning the scope of the Motion Device Axis Object, there are basically 4 Control Modes common to devices, position control, velocity control, torque control, and no control. A new control mode, acceleration control, is added to this list to complete the progression from velocity control to torque control. Subclause 7.6.9 provides a block diagram for each of the control modes in an effort to further define the collective behavior of the various Motion Device Axis Object attributes.

7.6.9.2 No Control (feedback only) mode

A Motion Device Axis Object instance can be configured for No Control mode. This Control Mode selection applies to several different Device Functions that include Bus Power Converters and Feedback Only devices. Feedback Only axis functionality allows the position, velocity, and acceleration of any one of four possible feedback channels to be accessed by the controller via the Device-to-Controller Connection. These signals can then be distributed across the motion control system as a master axis for gearing and camming operations. In this mode, the Feedback Master Select attribute, if supported, determines which feedback channel produces the Position, Velocity, and Acceleration Feedback signals (see Figure 93).

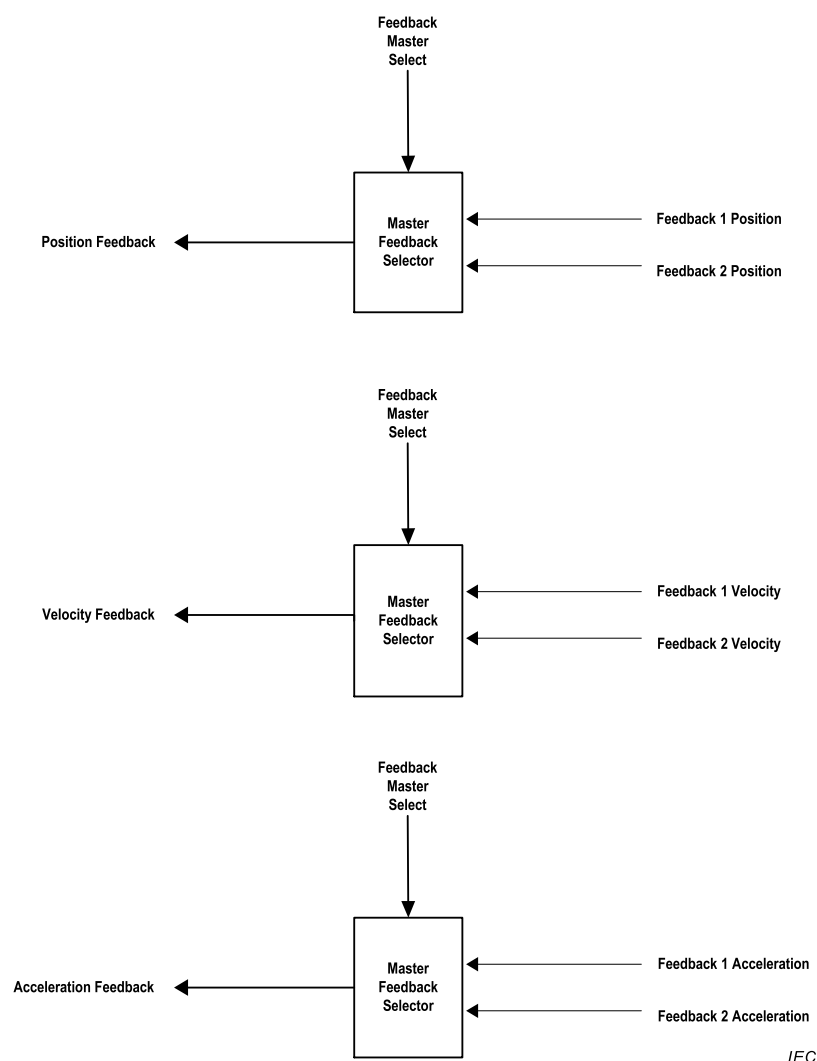


Figure 93 – No Control (Feedback Only)

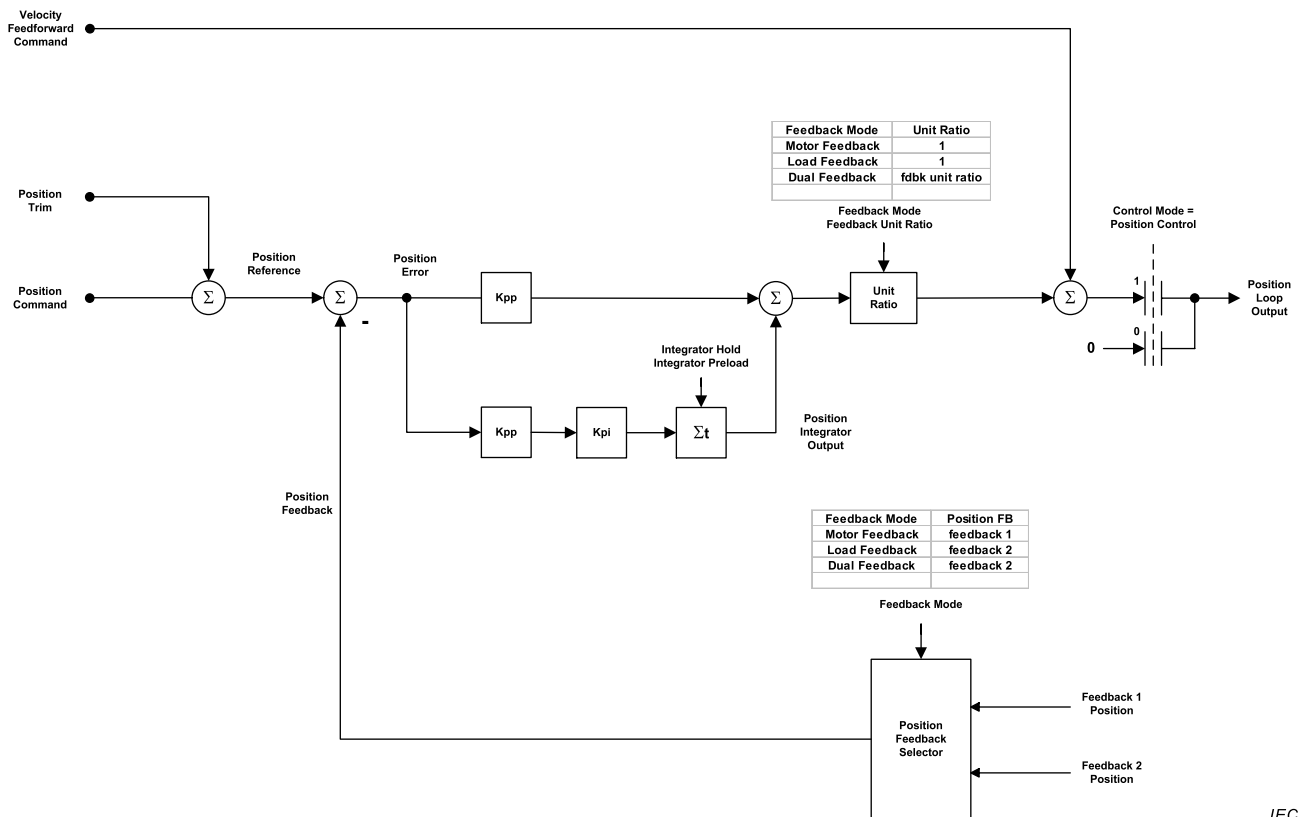
7.6.9.3 Position Control

In Position Control mode, the only operative Control Method supported by the object currently is Closed Loop servo control. At a later date, the object could be expanded to include a Stepper based position Control Method.

7.6.9.4 Closed Loop Position Control

7.6.9.4.1 General

When performing closed loop position control, the device applies the Position Command signal output of the Command Generator to the position loop summing junction. In addition to the Position Command, a Position Trim input is provided which can be used to provide an offset to the position loop. The classic PI control loop generates a Position Loop Output signal to an inner velocity loop (see Figure 94).



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Figure 94 – Closed Loop Position Control

7.6.9.4.2 Position feedback selection

Feedback to the PI regulator can be derived from two different feedback channels. This flexibility allows the position loop to operate with either a motor based feedback device that is typically attached to the Feedback 1 channel or a load-side feedback device that is connected to the Feedback 2 channel. Which feedback source is used by the loop is governed by the Feedback Mode attribute.

When the Feedback Mode calls for Dual Feedback operation, the position loop utilizes the Feedback 2 channel and the velocity loop uses the Feedback 1 channel. Since the two feedback channels may not have the same feedback resolution, it is necessary to convert position loop output from Feedback 1 units to Feedback 2 units prior to applying the output to the velocity loop summing junction. This is done by scaling the position loop output via the Unit Scaling block using the Feedback Unit Ratio.

7.6.9.4.3 Position PI gains

The Proportional Gain of the classic PI controller sets the unity gain bandwidth of the position loop in rad/s, while the Integral Gain is used to drive the Position Error signal to zero to compensate for the effect of any static and quasi-static torque or forces applied to the load.

7.6.9.4.4 Velocity Feedforward

The inner velocity loop requires a non-zero command input to generate steady-state axis motor velocity. To provide the non-zero output from the device to the motor, a non-zero position loop output is required, which translates to a non-zero position error. This dynamic error between command position and actual position while moving is often called “following error”. Most closed loop motion control applications desire zero following error – all the time. This could be achieved to some extent through use of the position integral gain control as described above, but typically, the response time of the integrator action is too slow to be

effective in high-performance motion control applications. An alternative approach that has superior dynamic response is to use Velocity Feedforward.

The Velocity Feedforward feature is used in Position Control Mode to provide the bulk of the Velocity Reference input necessary to generate the desired motor velocity. It does this by scaling the Fine Velocity Command signal output of the Command Generator by the Velocity Feedforward Gain and adding the resultant Velocity Feedforward Command signal to the Position Loop Output generated by the position loop to form the Velocity Reference signal. With this feature, the position loop does not need to generate much effort to produce the required velocity command level, hence the Position Error value is significantly reduced. The Velocity Feedforward Command signal allows the following error of the position control loop to be reduced to nearly zero when running at a constant velocity. This is important in applications such as electronic gearing and synchronization applications where it is necessary that the actual axis position does not significantly lag behind the commanded position at any time.

The optimal value for Velocity Feedforward Gain is theoretically 100 %. In reality, however, the value may need to be tweaked to accommodate velocity loops with finite loop gain. One aspect that may force a smaller Velocity Feedforward value is that increasing amounts of feedforward tends to exacerbate axis overshoot. For this reason, feedforward is not recommended for point-to-point positioning applications.

7.6.9.5 Velocity Control

In Velocity Control mode, there are two operative control methods supported by the object, Closed Loop Velocity Control and Open Loop Frequency Control.

7.6.9.6 Closed Loop Velocity Control

7.6.9.6.1 General

The Closed Loop velocity control method is targeted for applications that require tight speed regulation. The command input to the velocity loop can be derived directly from the Velocity Command of the Command Generator when configured for Velocity Control Mode or from the Position Loop Output when configured for Position Control Mode as described in 7.6.9.3 (see Figure 95).

When serving as an outer velocity loop in Velocity Control Mode, the device applies the Velocity Command input to the velocity command summing junction to generate the Velocity Reference signal into a classic PI regulator. Also contributing to the velocity command summing junction is the Velocity Trim input, which can be used in conjunction with an outer control loop to make minor adjustments to the velocity of the motor.

When serving as an inner velocity loop in Position Control Mode, the device applies the Position Loop Output signal to the input of the velocity command summing junction. Input signals that are not applicable to the configured control mode are generally set to zero.

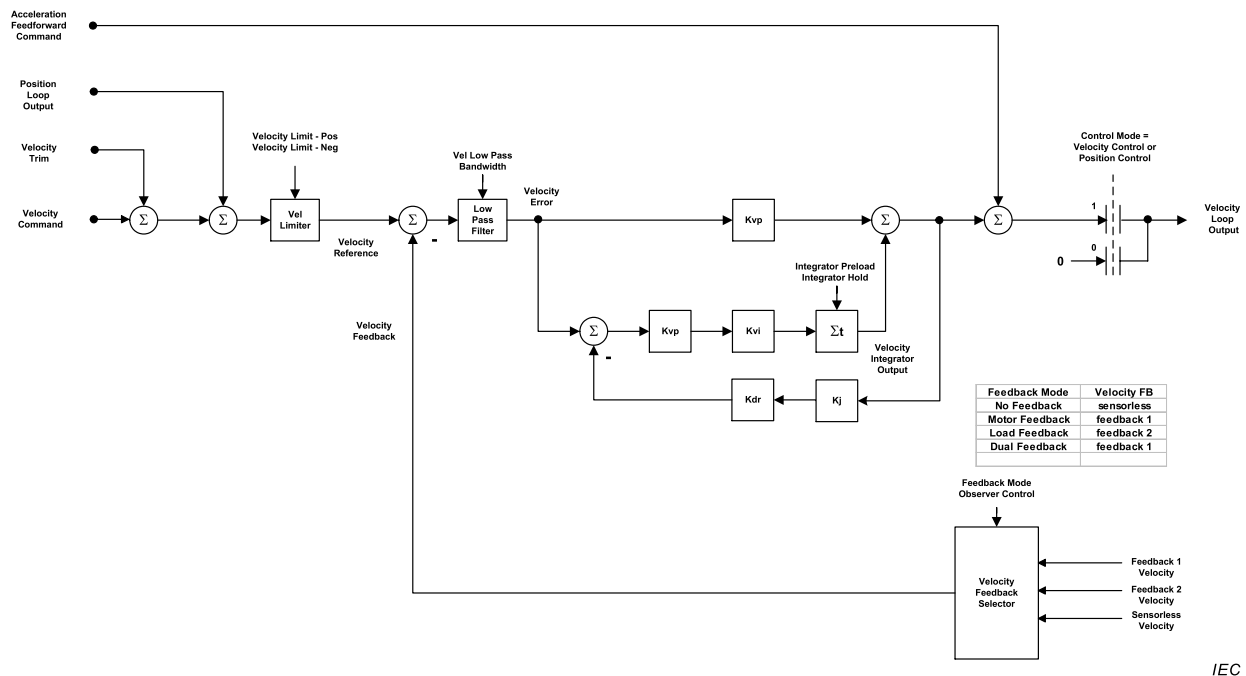


Figure 95 – Closed Loop Velocity Control

7.6.9.6.2 Velocity Limiter

The output of the velocity command summing junction signal passes through a classic limiter block to produce the Velocity Reference signal into the velocity loop. The Velocity Limiter block applies a directional velocity limit, either Velocity Limit – Pos or Velocity Limit – Neg, to the velocity command signal input that is based on the sign.

7.6.9.6.3 Velocity Feedback Selection

Feedback to the PI regulator can be derived from either of the two available feedback transducers, Feedback 1 or Feedback 2. Which feedback source is used by the loop is governed by the Feedback Mode enumeration. If Feedback Mode is No Feedback, indicating sensorless operation, the Velocity Feedback signal is the Sensorless Velocity signal generated by the sensorless control algorithm.

7.6.9.6.4 Velocity Error Filter

A low pass filter can be optionally applied to the velocity error signal generated by velocity loop summing junction. The output of this filter becomes the Velocity Error signal that is subsequently operated on by the velocity loop PI control algorithm. When used, the filter is typically set between 5 to 10 times the velocity loop bandwidth. It is recommended that this filter be a two pole IIR filter to maximum its effectiveness at quantization noise filtering.

7.6.9.6.5 Velocity PI gains

The velocity loop generates a Velocity Loop Output signal to the next inner loop via a classic PI control loop structure. The Proportional Gain of the controller sets the unity gain bandwidth of the velocity loop in rad/s, while the Integral Gain is used to drive the Velocity Error signal to zero to compensate for any static and quasi-static torque or forces applied to the load. The integrator path includes a Proportional Gain so that units of the Integral Gain represent the bandwidth of the integrator in rad/s.

The integral section of the velocity regulator includes an anti-windup feature. The anti-windup feature automatically holds the regulator's integral term when a limit condition is reached in the forward path. The anti-windup feature is conditioned by the arithmetic sign of the

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integrator's input. The integrator is held when the input's sign is such that the integrator output moves further into the active limit. In other words, the integrator is allowed to operate (not held) when the input would tend to bring the integrator output value off the active limit.

The integrator may also be configured for integrator hold operation. When the Integrator Hold attribute is set true, the regulator holds the integrator from accumulating while the axis is being commanded to move. This behavior is helpful in point-to-point positioning applications.

An automatic preset feature of the velocity regulator's integral term occurs when a transition is made from a torque control mode to speed control, using the Control Mode selection parameter. Upon transition to speed mode, the speed regulator's integral term is preset to the motor torque reference parameter. If the speed error is small, this provides a 'bumpless' transition from the last torque reference value present just prior to entering speed mode.

7.6.9.6.6 Velocity droop

Another feature of the velocity regulator is the velocity droop function. The velocity error input to the integral term is reduced by a fraction of the velocity regulator's output, as controlled by the droop gain setting, Kdr. As torque loading on the motor increases, actual motor speed is reduced in proportion to the droop gain. This is helpful when some level of compliance is required due to rigid mechanical coupling between two motors.

7.6.9.6.7 Acceleration Feedforward

The velocity loop requires a non-zero velocity loop output to generate steady-state axis motor acceleration. To provide the non-zero output from the drive to the motor, a non-zero velocity error is generally required. In position control applications, this non-zero velocity error translates to a non-zero position loop error. Since many closed loop motion control applications require near zero control loop error, this behavior is not desirable. Again, the position and velocity loop error could be reduced by applying the velocity integral gain control as described above, but the integrator action is still too slow to be very effective. The preferred approach with superior dynamic response is to use Acceleration Feedforward.

The Acceleration Feedforward feature is used to generate the bulk of the Acceleration Reference necessary to generate the commanded acceleration. It does this by scaling the Fine Acceleration Feedforward generated by the Command Generator by the Acceleration Feedforward Gain and adding the resultant Acceleration Feedforward Command Signal as an offset to the output of the velocity loop. With this feature, the velocity loop does not need to generate much control effort, thus reducing the amount of control loop error.

The optimal value for Acceleration Feedforward is 100 % theoretically. In reality, however, the value may need to be tweaked to accommodate variations in load inertia and the torque constant of the motor. Like Velocity Feedforward, Acceleration Feedforward can result in overshoot behavior and therefore shall not be used in point-to-point positioning applications.

When used in conjunction with the Velocity Feedforward, the Acceleration Feedforward allows the following error of the position or velocity control loop to be reduced to nearly zero during the acceleration and deceleration phases of motion. This is important in tracking applications that use electronic gearing and camming operations to precisely synchronize a slave axis to the movements of a master axis.

7.6.9.7 Open Loop Frequency Control

7.6.9.7.1 General

Another Velocity Control method is the open loop Frequency Control method associated with so called Volts/Hertz or variable frequency drives (VFDs) that do not have a current control loop and typically drive an induction motor. Velocity control with this method is achieved by controlling the voltage and frequency output of the drive device in some manner where voltage is generally proportional to frequency. For an induction motor, the velocity of the

motor is determined by the Output Frequency of the drive device divided by the Motor Pole count. This control method is applicable to velocity control applications that do not require tight speed regulation and therefore do not require a feedback device. Figure 96 further defines this open loop velocity control method.

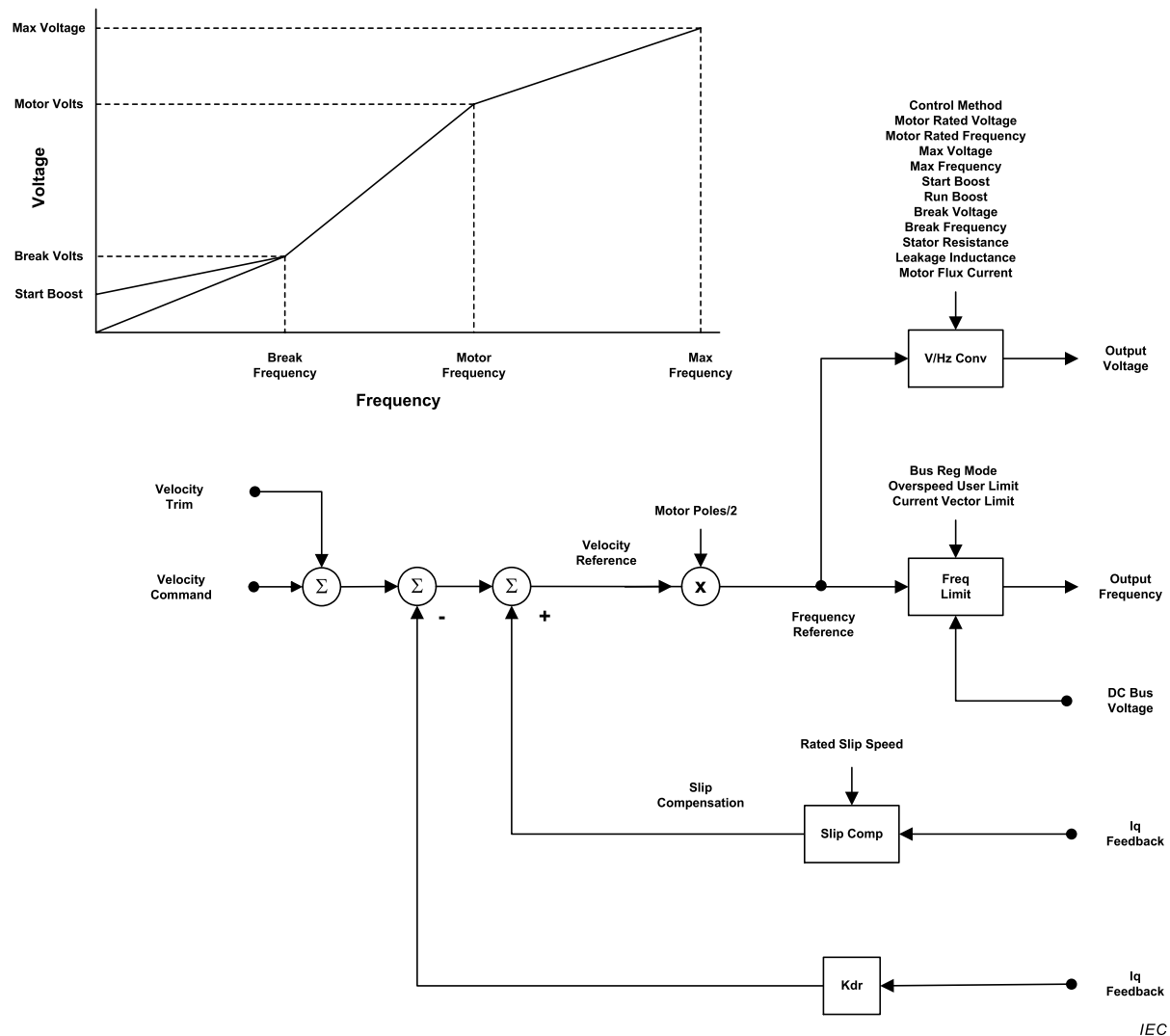


Figure 96 – Open Loop Frequency Control

7.6.9.7.2 Basic Volts/Hertz Operation

The Motion Device Axis Object provides a number attributes that are used to specify the relationship the drive device uses between output frequency (speed) and output voltage for a given (induction) motor. The Break Frequency and Break Voltage attributes define the point on the Volts/Hertz curve below which the Start Boost feature is applied. As the name indicates, Start Boost is used to provide a non-zero output voltage to the motor at stand-still to assist start-up. The contribution of the Start Boost to the output voltage of the drive device tapers off to zero when the motor reaches the Break Frequency. Above the break point, output voltage and output frequency follow a linear slope to the point defined by the Motor Rated Frequency and Motor Rated Voltage. From this point on, the Volts/Hertz curve follows another linear slope to the point defined by the Max Frequency and Max Voltage attributes. This segment of the Volts/Hertz curve allows for operation above the rated frequency and voltage of the motor in applications where that is required.

7.6.9.7.3 Sensorless Vector Operation

Sensorless Vector is an alternative Velocity Control Method that does not require configuration of a Volts/Hertz curve. Instead, by knowing the Stator Resistance and Leakage Inductance of the motor, the drive device can calculate the appropriate Output Voltage required for a given Output Frequency. This method provides better low speed velocity control behavior than using the Basic Volts/Hertz method.

7.6.9.7.4 Slip Compensation

When driving an induction motor at a specific frequency, the actual motor velocity is generally less than the command speed, given by the output frequency divided by the motor pole count, by an amount that is proportional to the load torque applied to the motor. This difference in speed is called “Slip” and is a configuration attribute associated with the motor. The Motion Device Axis Object supports a Slip Compensation feature that is common to variable frequency drives. The amount of Slip Compensation applied to the Velocity Reference is the product of the measured torque producing current, I_q , and the configured Induction Motor Rated Slip Speed.

7.6.9.7.5 Velocity Droop

Another feature defined for the Frequency Control method is the droop function. The droop function reduces the velocity reference by a scaled fraction of the torque producing current, I_q , as controlled by the droop gain setting, K_{dr} . As torque loading on the motor is increased, actual motor speed is reduced in proportion to the droop gain. This is helpful when some level of compliance is required when performing torque sharing between two motors on a common load.

7.6.9.8 Acceleration Control

7.6.9.8.1 General

While dynamic motor control via an acceleration command is not common in the industry, Acceleration Control was added to the Motion Device Axis Object to complete the dynamic progression from velocity control to torque control. The output of the velocity loop, Velocity Loop Output, also has units of acceleration. So, like the other control modes, the contributions of the Acceleration Command, Acceleration Trim, and Velocity Loop Output are summed to form the Acceleration Reference signal that serves as one of the primary inputs to the Torque Control section.

7.6.9.8.2 Acceleration Limiter

The output of the acceleration command summing junction signal passes through a classic limiter block to produce the Acceleration Reference signal. The Accel Limiter block applies a directional acceleration limit, either the Acceleration Limit and Deceleration Limit, to the input command signal based on the sign of the signal (see Figure 97).

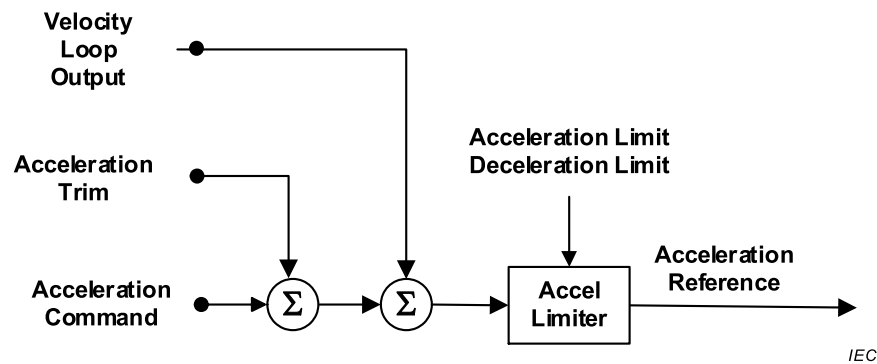


Figure 97 – Acceleration Control

7.6.9.9 Torque Control

7.6.9.9.1 General

Torque is generally proportional to acceleration and to the torque producing motor current, I_q . The purpose of the Torque Control structure is to combine input signals to create a Torque Reference from a number of different sources based on the Control Mode and apply various filters and compensation algorithms to the Torque Reference to create a Filtered Torque Reference. The Filtered Torque Reference signal is scaled by the reciprocal of the torque constant K_t of the motor to become the I_q Current Command input to the current loop. Since the motor current is also per unitized to the % Rated current of the motor, the torque constant K_t is nominally 1. In other words, in general it is assumed that 100 % rated current produces 100 % rated torque.

7.6.9.9.2 Torque input sources

The Torque Control section can take input from a variety of sources depending on the Control Mode. Input to the Torque Reference path can come via the cyclic Torque Command or Torque Trim signal in Torque Control mode. In Position or Velocity Control mode, torque input is derived from the outer velocity loop by bringing in the resulting acceleration signals and scaling these signals into equivalent torque (see Figure 98).

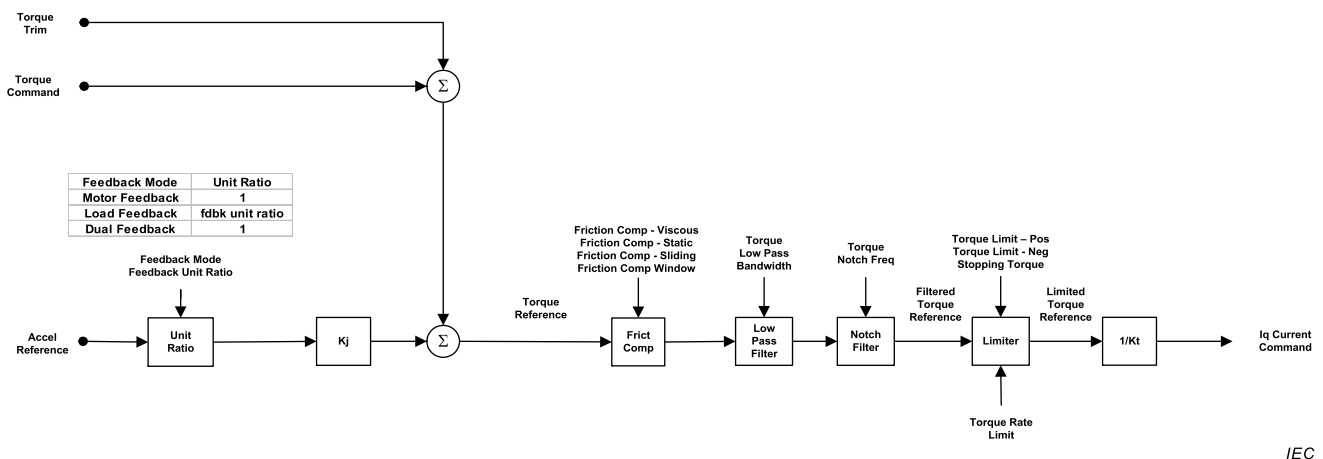


Figure 98 – Torque Control

7.6.9.9.3 Acceleration to torque scaling

Since the acceleration input signals into the Torque Control section are expressed in units of acceleration, a scaling factor K_j is needed to convert acceleration units to torque % Rated Torque units. This scaling factor, when properly configured, represents the total System

Inertia or mass of the system that includes the motor and the load and has the effect of cancelling the effects the system inertia/mass has on control loop response and loop gain settings. Since the torque units are expressed as % of Rated Torque of the motor, the units for the System Inertia attribute are % Rated per Motor Units/s². However, acceleration units can be expressed in Feedback 1 or Feedback 2 Units based on the Feedback Mode setting. Therefore, in the case where Feedback 2 applies, the acceleration signal needs to be scaled by the Feedback Unit Ratio as shown by the Unit Ratio block in Figure 98.

7.6.9.9.4 Friction compensation

7.6.9.9.4.1 General

Friction Compensation applies a compensating directional torque or force to the motor to overcome the effects of friction in the mechanical system, thus minimizing the amount of control effort required. Individual attributes have been defined to support compensation for static friction, sliding (Coulomb) friction, and viscous friction. A compensation window attribute is also provided to mitigate motor dithering associated with conventional friction compensation methods.

7.6.9.9.4.2 Static friction compensation

It is not unusual for an axis to have enough static friction, commonly called “sticktion”, in position control applications that even with a significant position error, the mechanical system refuses to budge. Of course, position integral gain can be used to generate enough output to the drive to correct the error, but this approach may not be responsive enough for the application. An alternative is to use Static Friction Compensation to break the sticktion in the presence of a non-zero position error. This is done by adding or subtracting a fixed torque level, as determined by the Static Friction Compensation attribute, to the Torque Reference signal value based on its current sign. This form of friction compensation shall only be applied when the axis is static, i.e. when there is no change in the position command.

The Static Friction Compensation value shall be just under the value that would overcome the sticktion. A larger value results in axis “dither”, a phenomena describing a rapid back and forth motion of the axis centred on the commanded position as it overcompensates for the sticktion.

To address the issue of dither when applying Static Friction Compensation, a Friction Compensation Window is applied around the current command position when the axis is at rest. If the actual position is within the Friction Compensation Window the Static Friction Compensation value is applied to the Servo Output but scaled by the ratio of the Position Error signal to the Friction Compensation Window. Within the window, the position loop and velocity loop integrators are also disabled to avoid the hunting effect that occurs when the integrators wind up. Thus, once the position error reaches or exceeds the value of the Friction Compensation Window attribute, the full Static Friction Compensation value is applied. Of course, shall the Friction Compensation Window be set to zero, this feature is effectively disabled.

A non-zero Friction Compensation Window has the effect of softening the Static Friction Compensation as its applied to the Torque Reference and reducing the dithering and hunting effects that it can create. This feature generally allows higher values of Static Friction Compensation to be applied resulting in better point-to-point positioning.

7.6.9.9.4.3 Sliding friction compensation

Sliding friction or Coulomb friction, by definition, is the component of friction that is independent of speed as long as the mechanical system is moving. Sliding friction is always less than static friction for a given mechanical system. The method of compensating for sliding friction is basically the same as that for static friction but the torque level added to the Torque Reference is less than that applied to overcome static friction and is determined by

the Sliding Friction Compensation attribute. Sliding Friction Compensation is only applied when the axis is being commanded to move.

7.6.9.9.4 Viscous friction compensation

Viscous friction, by definition, is the component of friction that increases linearly with the speed of the mechanical system. The method of compensating for viscous friction is to multiply the configured Viscous Friction Compensation value by the speed of the motor and apply the result to the Torque Reference signal. Viscous Friction Compensation is only applied when the axis is being commanded to move.

7.6.9.9.5 Low Pass filter

The Low Pass Filter is effective in resonance control when the natural resonance frequency is much higher ($>5x$) than the control loop bandwidth. This filter works by reducing the amount of high-frequency energy in the device output that excite the natural resonance. The Low Pass Filter design can be single pole or multiple poles. Care shall be taken, however, to limit the amount of phase lag introduced by this filter to the control loop to avoid potential instability.

7.6.9.9.6 Notch filter

The Notch Filter is effective in resonance control when the natural resonance frequency is higher than the control loop bandwidth. Like the Low Pass filter, the notch filter works by significantly reducing the amount of energy in the device output that can excite the natural resonance. It can be used even when the natural resonance frequency is relatively close to the control loop bandwidth. That is because the phase lag introduced by the notch filter is localized around the notch frequency. For the notch filter to be effective, the Notch Filter Frequency has to be set very close to the natural resonance frequency of the load.

A typical equation for the notch filter is as follows:

$$G(s) = \frac{s^2 + \omega_n^2}{s^2 + s \times \omega_n/Q + \omega_n^2}$$

In this equation, Q represents the sharpness of the notch. In most implementations, the sharpness, Q, is typically hard-coded in the device. The attenuation depth of the notch filter is infinite.

7.6.9.9.7 Torque limiter

The Filtered Torque Reference signal passes through a limiter block to produce the Limited Torque Reference signal. The Torque Limiter block applies a torque limit to the signal that is based on the sign of the torque reference signal input and the state of the axis. During normal operation it is the Torque Limit – Positive and Torque Limit – Negative attributes, set by the user, that are applied to the torque reference signal. When the axis is commanded to stop as part of a disable request or major fault condition, the device applies the Stopping Torque Limit.

Also included with the torque limit block is a built in Torque Rate of Change Limit. This feature limits the rate of change of the torque reference output.

7.6.9.9.8 Torque to current scaling

The final result of all this torque signal filtering, compensation, and limiting functionality is the Filtered Torque Reference signal, which when scaled by the reciprocal of the Torque Constant of the motor, $1/K_t$, becomes the torque producing I_q Current Command signal to the current loop. Ideally, the relationship between motor torque and motor current is independent of position, time, current, and environmental conditions. In other words, the $1/K_t$ scaling in

Figure 98 has a nominal value of 1, i.e. 100 % rated torque translates to 100 % rated current. In practice, this may not be the case. Compensation can be applied to the $1/K_t$ value to address these issues at the drive vendors' discretion.

7.6.9.10 Current Control

7.6.9.10.1 General

In general, motor torque is controlled by controlling the orientation and magnitude of the motor stator current vector with respect to the rotor magnetic flux vector. The Current Control loop is responsible for providing this control and is actually composed of two PI loops, one that controls the torque producing current, I_q , and one that controls the flux producing current, I_d . It is the quadrature component of current, I_q , that is used for dynamic torque control (see Figure 99).

In the case of an induction motor, the flux producing current, I_d , is solely responsible for generating rotor flux. In the case of permanent magnet motors, rotor flux is generated by the rotor magnets and I_d is only used to in some cases to extend the speed range of the motor by changing the angle of stator field relative to the rotor field. In this case, the angle of I_q relative to the rotor field remains the same, i.e. at quadrature. But since the vector combination of I_q and I_d determines the stator flux angle relative to the rotor, increasing amounts of I_d can shift the stator flux away from quadrature to extend the speed range of the motor at the expense of torque.

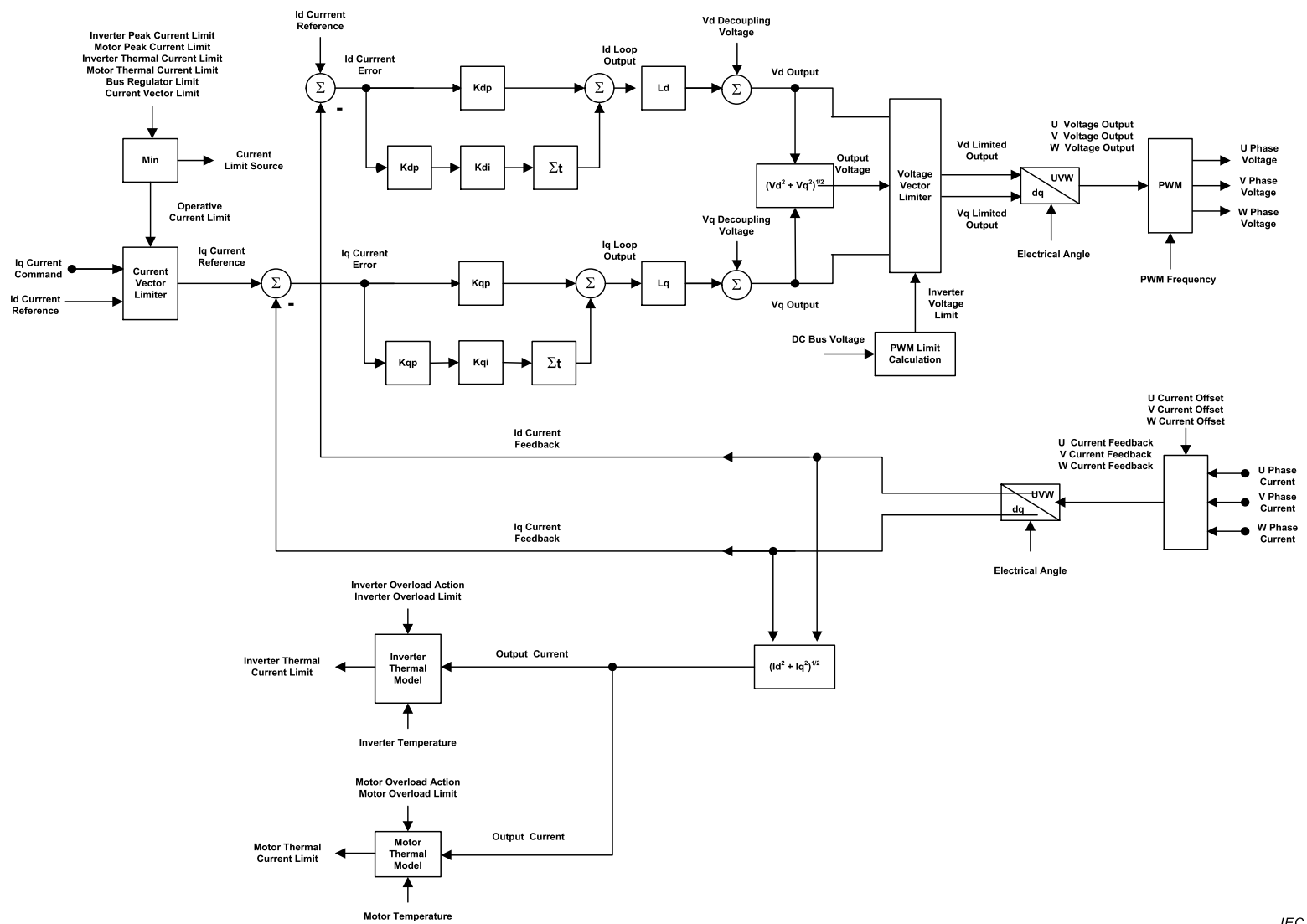


Figure 99 – Closed Loop Current Vector Control

IEC

7.6.9.10.2 Current vector limiter

The Iq Current Command passes through a Current Vector Limiter block before becoming the Iq Current Reference signal. This limiter block computes the combined vector magnitude of the Iq Current Reference and the Id Current Reference signals. The resultant current vector magnitude is compared to the Operative Current Limit that represents the minimum current limit from among a set of potential current limits of the drive device and motor. If the vector magnitude exceeds the Operative Current Limit, the Iq Current Reference is reduced so the vector magnitude equals the Operative Current Limit. Potential current limit sources can be the Peak Current Limit ratings as well as the Thermal Limits for the Motor and Drive Inverter. Another possible limit source is the user configurable Current Vector Limit attribute. Some of these limits are conditional and dynamic, such as the Motor and Inverter Thermal Current Limits derived from the thermal models for these devices. These limits are only active when the corresponding Motor and Inverter Overload Action attributes are set to provide current fold-back. The thermal current limits in this case would decrease as the simulated temperature of the modeled devices increases. The Bus Regulator Limit is only applied when the motor is regenerating power onto the DC Bus and is based on the Regenerative Power Limit.

With all these potential current limit sources that could be operative, a Current Limit Source attributes was included with the Motion Device Axis Object to identify the source of the active current limit.

7.6.9.10.3 Voltage output

The output of each current loop is scaled by the motor inductance to generate a voltage command to the vector transformation block. It is the job of the vector transformation block to transform the torque producing, Vq, and flux producing, Vd, command signals from the rotating synchronous reference frame to the stationary stator reference frame. The resultant U, V, and W Output Voltage values are then applied to the motor by Pulse Width Modulation (PWM). The PWM Frequency is also a configurable attribute of the Motion Device Axis Object.

The magnitude of the Vq, Vd vector is calculated in real time as the total Output Voltage signal. The maximum Output Voltage signal that can be applied to the motor is ultimately limited by the DC Bus Voltage and enforced by the Voltage Vector Limiter. Any attempt to exceed this value results in an Inverter Voltage Limit condition.

7.6.9.10.4 Current feedback

Current feedback signals to the current loop are provided by two or three current sensors. The signals from these sensors are conditioned and corrected for device specific offsets to become the U, V, and W Current Feedback signals associated with the stationary motor stator frame. These three signals are transformed back to the synchronous reference frame to generate the Iq and Id Current Feedback signals. The magnitude of the Iq, Id current vector is calculated in real-time and used as an input to the thermal models for the inverter and motor.

7.6.9.10.5 Motor commutation

Motor commutation is critical to closed loop motor control. The orientation of the motor rotor can be determined from a feedback source mounted to the motor. The actual commutation source is the motor feedback device assigned to Feedback 1. Once the feedback device is calibrated to the absolute orientation of the rotor using the Commutation Offset attribute, the commutation block can then generate the true Electrical Angle of the rotor. This signal is used to perform the vector transforms between the rotary and stationary motor frames and can also be used for any other algorithms that require knowledge of rotor position.

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SOMMAIRE

AVANT-PROPOS	299
INTRODUCTION.....	301
0.1 Généralités.....	301
0.2 Déclaration de propriété	305
1 Domaine d'application	307
2 Références normatives.....	307
3 Termes, définitions et abréviations	308
3.1 Termes et définitions	308
3.2 Abréviations.....	315
4 Présentation générale	316
4.1 Généralités	316
4.2 Modes de commande.....	317
4.2.1 Généralités.....	317
4.2.2 Méthodes de commande.....	317
4.2.3 Nomenclature de commande	318
4.2.4 Asservissement de position	318
4.2.5 Commande de vitesse	319
4.2.6 Commande d'accélération	321
4.2.7 Asservissement de couple	321
4.2.8 Pas d'asservissement	322
5 Types de données	323
5.1 Type de données – Présentation.....	323
5.2 Conventions.....	323
6 Profil d'entraînement CIP Motion	323
6.1 Modèle d'objet	323
6.1.1 Présentation de l'objet	323
6.1.2 Description de l'objet	324
6.2 Influence des objets sur le comportement	325
6.3 Définition des interfaces d'objet	325
6.4 Messages de connexion E/S	326
6.4.1 Généralités.....	326
6.4.2 Connexion E/S CIP Motion	326
6.4.3 Connexion Contrôleur-Dispositif	331
6.4.4 Connexion Dispositif – Contrôleur.....	354
6.4.5 Format de connexion E/S de mouvement fixe	364
6.4.6 Modèle de synchronisation de Connexion E/S CIP Motion	365
6.5 Procédure de démarrage du dispositif.....	385
6.5.1 Généralités.....	385
6.5.2 Création d'une connexion E/S de mouvement	386
6.5.3 Configuration de l'Objet Axe de dispositif de mouvement.....	389
6.5.4 Synchronisation temporelle.....	391
6.6 Visualisation du dispositif.....	393
6.7 Qualité de Service (Quality of Service ou QoS) Ethernet/IP	394
7 Objet Axe de dispositif de mouvement.....	394
7.1 Considérations d'ordre général	394

7.1.1	Généralités	394
7.1.2	Historique des révisions	394
7.1.3	Présentation de l'objet	395
7.1.4	Abstraction de l'Objet Axe de dispositif de mouvement	395
7.1.5	Objet Axe de commande de mouvement	396
7.1.6	Classification de commande de dispositif	397
7.1.7	Obligatoire par rapport à Facultatif dans une mise en œuvre	398
7.2	Attributs de classe	412
7.2.1	Généralités	412
7.2.2	Sémantique	417
7.3	Attributs d'instances	420
7.3.1	Généralités	420
7.3.2	Attributs de configuration de commande de mouvement	422
7.3.3	Attributs d'étalonnage du mouvement	424
7.3.4	Attributs de données de connexion	431
7.3.5	Attributs du moteur	438
7.3.6	Attributs de réaction	452
7.3.7	Attributs Capture d'événement	462
7.3.8	Attributs de génération de référence de consigne	469
7.3.9	Attributs du mode de commande	472
7.3.10	Attributs Arrêt et freinage	491
7.3.11	Attributs Commande de bus à courant continu	506
7.3.12	Attributs de gestion de l'alimentation et de gestion thermique	511
7.3.13	Attributs Statut d'axe	514
7.3.14	Attributs d'exception, de défaut et d'alarme	521
7.3.15	Attributs Journaux de défauts et d'alarmes	527
7.3.16	Attributs de limite d'exception	534
7.3.17	Attribut de configuration d'action d'exception d'axe	538
7.3.18	Attributs de défaut d'initialisation	541
7.3.19	Attributs Inhibition de démarrage	543
7.3.20	Attributs Défauts APR	544
7.3.21	Attributs de statistiques d'axe	546
7.3.22	Attributs d'informations sur l'axe	546
7.3.23	Attributs E/S à usage général	548
7.3.24	Attributs Mode local	550
7.3.25	Attributs Sécurité d'axe	551
7.4	Services communs	552
7.4.1	Services pris en charge	552
7.4.2	Données spécifiques au service	552
7.5	Services spécifiques à l'objet	554
7.5.1	Services pris en charge	554
7.5.2	Données spécifiques au service	555
7.6	Comportement	570
7.6.1	Modèle d'état	570
7.6.2	Comportement d'état	583
7.6.3	Comportement de défaut et d'alarme	590
7.6.4	Comportement Inhibition de démarrage	592
7.6.5	Comportement de visualisation	592
7.6.6	Comportement de génération de consigne	598

7.6.7	Comportement de l'interface de réaction.....	605
7.6.8	Comportement de capture d'événement.....	607
7.6.9	Comportement du mode de commande.....	610
Bibliographie.....		631
Figure 1	– Structure de l'IEC 61800-7	305
Figure 2	– Asservissement de position en boucle ouverte	318
Figure 3	– Asservissement de position en boucle fermée	319
Figure 4	– Commande de vitesse en boucle ouverte	320
Figure 5	– Commande de vitesse en boucle fermée	320
Figure 6	– Commande d'accélération	321
Figure 7	– Asservissement de couple.....	322
Figure 8	– Pas d'asservissement (Réaction uniquement)	322
Figure 9	– Modèle d'objet d'un dispositif CIP Motion	324
Figure 10	– Modèle de Connexion E/S CIP Motion.....	327
Figure 11	– Canaux de Connexion E/S CIP Motion	327
Figure 12	– Format de connexion Contrôleur-Dispositif (Point de connexion 2).....	329
Figure 13	– Format de connexion Dispositif-Contrôleur (Point de connexion 2).....	330
Figure 14	– Format de connexion Contrôleur-Dispositif CIP Motion.....	331
Figure 15	– En-tête de connexion	331
Figure 16	– Format de connexion.....	331
Figure 17	– En-tête de connexion	333
Figure 18	– Bloc de données d'instance.....	335
Figure 19	– En-tête de données d'instance	336
Figure 20	– Bloc de données cycliques	337
Figure 21	– Mode de commande	337
Figure 22	– Mode de réaction	337
Figure 23	– Bloc de données d'écriture cycliques.....	343
Figure 24	– Bloc de données d'écriture cycliques.....	344
Figure 25	– Bloc de données d'événement.....	344
Figure 26	– Bloc de données de service	354
Figure 27	– Format de connexion Dispositif-Contrôleur CIP Motion.....	354
Figure 28	– En-tête de connexion	355
Figure 29	– En-tête de connexion	356
Figure 30	– Défaut/Alarme de nœud	356
Figure 31	– Ajustement des données de position réelles en fonction de la datation du dispositif	357
Figure 32	– Bloc de données d'instance.....	358
Figure 33	– En-tête de données d'instance	359
Figure 34	– Bloc de données cycliques	359
Figure 35	– Bloc de données de lecture cycliques.....	361
Figure 36	– Bloc de données de lecture cycliques.....	362
Figure 37	– Bloc de données d'événement.....	362
Figure 38	– Bloc de données de service	364

Figure 39 – Format de connexion Contrôleur-Dispositif fixe (taille fixe = 16 octets)	365
Figure 40 – Format de connexion Dispositif-Contrôleur fixe (taille fixe = 16 octets)	365
Figure 41 – Modèle de synchronisation à 1 cycle CIP Motion	366
Figure 42 – Modèle de synchronisation à 2 cycles CIP Motion	368
Figure 43 – Modèle de synchronisation à 3 cycles CIP Motion	369
Figure 44 – Synchronisation de connexion Contrôleur-Dispositif avec interpolation fine	371
Figure 45 – Synchronisation de connexion Contrôleur-Dispositif avec extrapolation	374
Figure 46 – Utilisation de la datation pour ajuster la position réelle à la base de temps du contrôleur	375
Figure 47 – Coordination des deux dispositifs d'entraînement avec différentes périodes de mise à jour	378
Figure 48 – Coordination de plusieurs axes de dispositif d'entraînement en cas de retard des paquets de connexion Contrôleur-Dispositif	379
Figure 49 – Propagation d'une réponse à un échelon dans le temps	381
Figure 50 – Révision de format 1 de bloc de configuration (Point de connexion 81)	387
Figure 51 – Révision de format 2 de bloc de configuration (Point de connexion 82)	387
Figure 52 – Bloc de données de connexion C-D initial type	388
Figure 53 – Bloc de données de connexion D-C initial type	389
Figure 54 – Contenu type du premier paquet de configuration d'attributs de classe C-D	389
Figure 55 – Réponse type au premier paquet de configuration de classe C-D	390
Figure 56 – Contenu type du premier paquet de configuration d'instance d'axe C-D	390
Figure 57 – Réponse type au premier paquet de configuration d'axe C-D	391
Figure 58 – Contenu type du paquet de demande de service de synchronisation temporelle C-D	391
Figure 59 – Synchronisation de groupe de dispositifs CIP Motion	392
Figure 60 – Composants d'objet de l'architecture de commande CIP Motion	395
Figure 61 – Champ de mot Commande de consigne	436
Figure 62 – IEEE Std 112 par modèle de moteur phasé	439
Figure 63 – Champ de mot Commande de vérification d'événement	465
Figure 64 – Champ de mot État de vérification d'événement	467
Figure 65 – Séquence de commande de frein (Arrêt de catégorie 0)	499
Figure 66 – Séquence de commande de frein (Arrêt de catégorie 1)	500
Figure 67 – Séquence de commande de frein (Arrêt de catégorie 2)	501
Figure 68 – Séquence d'activation du dispositif d'entraînement avec propriété de Contrôle	504
Figure 69 – Séquence de désactivation du dispositif d'entraînement avec propriété de Contrôle	506
Figure 70 – Format de demande Get_Axis_Attributes_List	555
Figure 71 – Format de réponse Get_Axis_Attributes_List	556
Figure 72 – Réponse Get_Axis_Attributes_List – Attribut de 4 octets simple	557
Figure 73 – Réponse Get_Axis_Attributes_List – Attribut de 2 octets simple	557
Figure 74 – Réponse Get_Axis_Attributes_List – Matrice d'attributs d'octets	557
Figure 75 – Réponse Get_Axis_Attributes_List – Matrice d'attributs bidimensionnelle	557
Figure 76 – Réponse Get_Axis_Attributes_List – Exemple d'erreur	558
Figure 77 – Format de demande Set_Axis_Attributes_List	558

Figure 78 – Demande Set_Axis_Attributes_List – Attribut de 4 octets simple	559
Figure 79 – Demande Set_Axis_Attributes_List – Attribut de 2 octets simple	559
Figure 80 – Demande Set_Axis_Attributes_List – Matrice d'attributs à 2 octets	559
Figure 81 – Demande Set_Axis_Attributes_List – Matrice d'attributs bidimensionnelle	560
Figure 82 – Format de réponse Set_Axis_Attributes_List	560
Figure 83 – Format de demande Set_Cyclic_Write_List	560
Figure 84 – Format de réponse Set_Cyclic_Write_List	561
Figure 85 – Format de demande Set_Cyclic_Read_List	562
Figure 86 – Format de réponse Set_Cyclic_Read_List	562
Figure 87 – Modèle d'état de l'Objet Axe de dispositif de mouvement	571
Figure 88 – Modèle d'état de l'Objet Axe de dispositif de mouvement pour le mode Réaction uniquement	573
Figure 89 – Modèle d'état de l'Objet Axe de dispositif de mouvement pour le Convertisseur	575
Figure 90 – Générateur de consigne	600
Figure 91 – Canaux de réaction 1 et 2	607
Figure 92 – Fonctionnalité de capture d'événement	609
Figure 93 – Pas d'asservissement (Réaction uniquement)	611
Figure 94 – Asservissement de position en boucle fermée	613
Figure 95 – Commande de vitesse en boucle fermée	616
Figure 96 – Contrôle de fréquence en boucle ouverte	619
Figure 97 – Commande d'accélération	621
Figure 98 – Asservissement de couple	622
Figure 99 – Commande vectorielle de courant en boucle fermée	628
 Tableau 1 – Types de données	 323
Tableau 2 – Objets présents dans un dispositif CIP Motion	324
Tableau 3 – Contenu de l'Objet Axe de dispositif de mouvement par Type de dispositif	325
Tableau 4 – Influence de l'objet sur le comportement	325
Tableau 5 – Interfaces d'objet	326
Tableau 6 – Ensemble de données temporelles	333
Tableau 7 – Commande d'axe	338
Tableau 8 – État de commande	338
Tableau 9 – Ensemble de données de consigne	339
Tableau 10 – Mise en correspondance de l'élément de données de consigne avec l'attribut d'Objet Axe de dispositif de mouvement	339
Tableau 11 – Ensemble de données réelles	340
Tableau 12 – Mise en correspondance de l'élément de données réelles avec l'attribut d'Objet Axe de dispositif de mouvement	340
Tableau 13 – Ensemble de données d'état	341
Tableau 14 – Commande de consigne	341
Tableau 15 – Mise à jour de consigne cible et rapport de période de mise à jour	342
Tableau 16 – Cycle d'événements de base	345
Tableau 17 – Cycle d'événements étendu	346

Tableau 18 – Cycle d'événements de base avec réarmement automatique	349
Tableau 19 – Ensemble de données d'enregistrement	352
Tableau 20 – Ensemble de données d'origine	353
Tableau 21 – Ensemble de données d'observation.....	353
Tableau 22 – Réponse d'axe.....	360
Tableau 23 – Type d'événement	363
Tableau 24 – Propagation d'une réponse à un échelon dans le temps (exemple 1)	381
Tableau 25 – Propagation d'une réponse à un échelon dans le temps (exemple 2)	383
Tableau 26 – Composants de visualisation CIP Motion	394
Tableau 27 – Historique des révisions de l'Objet Axe de dispositif de mouvement	395
Tableau 28 – Exemple de mise en œuvre d'attributs d'instance par rapport au Code de fonction de dispositif	398
Tableau 29 – Mise en œuvre d'attributs d'instances par rapport au Code de fonction de dispositif	400
Tableau 30 – Attributs de classe de l'Objet Axe de dispositif de mouvement	413
Tableau 31 – Définitions de bit Contrôle de nœud.....	417
Tableau 32 – Définitions du bit État du nœud.....	418
Tableau 33 – Définitions du code de défauts de nœud	419
Tableau 34 – Définitions du code Alarme de nœud	420
Tableau 35 – Unités dynamiques par rapport au mode de réaction	422
Tableau 36 – Attributs de configuration de dispositif de commande de mouvement.....	423
Tableau 37 – Définitions de l'énumération du mode de commande	423
Tableau 38 – Définitions de l'énumération de la méthode de commande.....	424
Tableau 39 – Attributs de mise à l'échelle du mouvement	425
Tableau 40 – Règles de sélection de l'unité de mouvement	428
Tableau 41 – Attributs de signal affectés par la polarité de mouvement	429
Tableau 42 – Attributs de limite de direction affectés par la polarité de mouvement	431
Tableau 43 – Attributs de données de connexion	432
Tableau 44 – Détermination de la valeur Ensemble de données réelles	435
Tableau 45 – Détermination de la valeur Ensemble de données de consigne.....	436
Tableau 46 – Définition de l'énumération de mise à jour de consigne cible	436
Tableau 47 – Définition de l'énumération de type de données de position de consigne	437
Tableau 48 – Définitions du bit Ensemble de données d'état.....	437
Tableau 49 – Format de données d'événement d'enregistrement	438
Tableau 50 – Format de données d'événement d'origine.....	438
Tableau 51 – Format de données d'événement d'observation	438
Tableau 52 – Attributs généraux d'informations sur le moteur	440
Tableau 53 – Attributs généraux de configuration du moteur.....	441
Tableau 54 – Attributs généraux de configuration du moteur à aimant permanent.....	445
Tableau 55 – Attributs généraux de configuration du moteur rotatif.....	446
Tableau 56 – Attributs généraux de configuration du moteur linéaire	447
Tableau 57 – Attributs de configuration du moteur à aimant permanent rotatif	448
Tableau 58 – Attributs de configuration du moteur à aimant permanent linéaire	449
Tableau 59 – Attributs de configuration du moteur à induction	450

Tableau 60 – Attributs de transmission de charge et de configuration d'actionneur	451
Tableau 61 – Abréviations des types de réaction	452
Tableau 62 – Fonctions de commande de canaux de réaction logiques	453
Tableau 63 – Règles applicables aux canaux de réaction logiques	454
Tableau 64 – Attributs généraux d'informations sur la réaction	454
Tableau 65 – Attributs généraux de signaux de réaction	454
Tableau 66 – Attributs de Configuration de réaction.....	455
Tableau 67 – Définitions de l'énumération Mode de réaction.....	462
Tableau 68 – Attributs d'événement	463
Tableau 69 – Définitions de bit Commande de vérification d'événement.....	466
Tableau 70 – Définitions de bit État de vérification d'événement.....	468
Tableau 71 – Attributs Signal du générateur de consigne.....	469
Tableau 72 – Attributs Configuration du générateur de consigne	471
Tableau 73 – Attributs Signal de boucle de position	473
Tableau 74 – Attributs Configuration de boucle de position	474
Tableau 75 – Attributs Signal de boucle de vitesse	476
Tableau 76 – Attributs Configuration de boucle de vitesse	477
Tableau 77 – Attributs Signal d'accélération	479
Tableau 78 – Attributs Configuration d'accélération	480
Tableau 79 – Attributs Signal d'asservissement de couple/force	481
Tableau 80 – Attributs Configuration d'asservissement de couple/force	482
Tableau 81 – Attributs Signal d'asservissement de courant.....	484
Tableau 82 – Attributs Configuration de contrôle de courant	486
Tableau 83 – Attributs Signal de contrôle de fréquence	490
Tableau 84 – Attributs Configuration de contrôle de fréquence	490
Tableau 85 – Attributs Sortie de dispositif d'entraînement.....	491
Tableau 86 – Attributs Arrêt/Freinage	492
Tableau 87 – Définitions de l'énumération Action d'arrêt.....	496
Tableau 88 – Dépendances d'attributs de sous-propriété de contrôle.....	502
Tableau 89 – Attributs Commande de bus à courant continu	507
Tableau 90 – Attributs État de gestion de l'alimentation et de gestion thermique.....	511
Tableau 91 – Attributs Configuration de gestion de l'alimentation et de gestion thermique	512
Tableau 92 – Attributs Statut d'axe	515
Tableau 93 – Définitions du bit Statut d'axe	516
Tableau 94 – Bit Statut d'axe par rapport à l'État d'axe.....	520
Tableau 95 – Action d'arrêt par rapport à la Catégorie d'arrêt.....	521
Tableau 96 – Définitions du bit Statut d'E/S d'axe	521
Tableau 97 – Attributs Exception, défaut et alarme	522
Tableau 98 – Tableau Exceptions standard.....	524
Tableau 99 – Attributs Journaux de défauts et d'alarmes	529
Tableau 100 – Attributs Informations de tolérance de fabrication d'exception.....	534
Tableau 101 – Attributs Configuration de limite utilisateur d'exception	536

Tableau 102 – Attribut Configuration d'action d'exception d'axe	538
Tableau 103 – Définitions de l'action d'exception d'axe	539
Tableau 104 – Attributs Défaut d'initialisation	542
Tableau 105 – Tableau Défaut d'initialisation standard	543
Tableau 106 – Attributs d'inhibition de démarrage	543
Tableau 107 – Tableau Inhibition de démarrage standard	544
Tableau 108 – Attributs Défauts APR	545
Tableau 109 – Tableau Défauts APR standard	546
Tableau 110 – Attributs Statistiques d'axe	546
Tableau 111 – Attributs Informations sur l'axe	547
Tableau 112 – Attributs E/S à usage général de dispositif d'entraînement	549
Tableau 113 – Attributs Configuration du mode local	550
Tableau 114 – Attributs Statut de sécurité d'axe	551
Tableau 115 – Objet Axe de dispositif de mouvement – Services communs	552
Tableau 116 – Structure des données de demande Group_Sync	552
Tableau 117 – Structure des données de réponse Group_Sync	553
Tableau 118 – Objet Axe de dispositif de mouvement – Services spécifiques à l'objet	554
Tableau 119 – Structure de demande Run_Motor_Test	563
Tableau 120 – Get_Motor_Test_Data mesurées par le type d'essai	563
Tableau 121 – Structure de demande Get_Motor_Test_Data (facultatif)	564
Tableau 122 – Structure normalisée de réponse Get_Motor_Test_Data (Type de moteur = à induction)	564
Tableau 123 – Structure normalisée de réponse Get_Motor_Test_Data (Type de moteur = SPM)	565
Tableau 124 – Structure normalisée de réponse Get_Motor_Test_Data (Type de moteur = IPM)	565
Tableau 125 – Structure de demande Run_Inertia_Test	566
Tableau 126 – Structure de réponse Get_Inertia_Test_Data	567
Tableau 127 – Structure de demande Run_Hookup_Test	568
Tableau 128 – Get_Hookup_Test_Data mesurées par le type d'essai	569
Tableau 129 – Structure de réponse Get_Hookup_Test_Data	569
Tableau 130 – Transitions de diagramme d'états d'axe	572
Tableau 131 – Conditions de diagramme d'états d'axe	573
Tableau 132 – Transitions de diagramme d'états d'axe (Réaction uniquement)	574
Tableau 133 – Transitions de diagramme d'états d'axe (Convertisseur)	576
Tableau 134 – Code de demande de Commande d'axe	577
Tableau 135 – Codes d'acquittement de réponse d'axe	577
Tableau 136 – Critères d'exécution réussie du fonctionnement demandé	578
Tableau 137 – Conditions d'erreur possibles du fonctionnement demandé	578
Tableau 138 – Cycle de demande de commande d'axe réussie	579
Tableau 139 – Cycle de demande de commande d'axe non réussie	579
Tableau 140 – Cycle de demande de commande d'axe en attente	580
Tableau 141 – Cycle de demande d'annulation	581
Tableau 142 – Cycle de redéfinition de la référence de position	583

Tableau 143 – État Exécution – Attributs configurables.....	586
Tableau 144 – Mise en correspondance de l'état d'axe avec l'Objet Identité grâce au comportement de la LED.....	594
Tableau 145 – Comportement de l'affichage à sept segments du dispositif CIP Motion	595
Tableau 146 – Comportement de l'affichage alphanumérique à plusieurs caractères CIP Motion.....	596
Tableau 147 – Comportement de l'affichage alphanumérique à plusieurs caractères multiaxe	598

COMMISSION ÉLECTROTECHNIQUE INTERNATIONALE

**ENTRAÎNEMENTS ÉLECTRIQUES
DE PUISSANCE À VITESSE VARIABLE –****Partie 7-202: Interface générique et utilisation de profils
pour les entraînements électriques de puissance –
Spécification de profil de type 2****AVANT-PROPOS**

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La Norme internationale IEC 61800-7-202 a été établie par le sous-comité SC 22G: Systèmes d'entraînement électrique à vitesse variable, comprenant des convertisseurs à semi-conducteurs, du comité d'études 22 de l'IEC: Systèmes et équipements électroniques de puissance.

Cette deuxième édition annule et remplace la première édition parue en 2007. Cette édition constitue une révision technique.

Cette édition inclut les modifications techniques majeures suivantes par rapport à l'édition précédente:

- a) mise à jour de l'information sur les brevets;
- b) nouvelle révision des spécifications de Profil d'Entraînement et d'Axe d'Entraînement avec multiples clarifications et améliorations.

Le texte de cette norme est issu des documents suivants:

FDIS	Rapport de vote
22G/308/FDIS	22G/323/RVD

Le rapport de vote indiqué dans le tableau ci-dessus donne toute information sur le vote ayant abouti à l'approbation de cette norme.

Cette publication a été rédigée selon les Directives ISO/IEC, Partie 2.

Une liste de toutes les parties de la série IEC 61800, publiées sous le titre général *Entraînements électriques de puissance à vitesse variable*, peut être consultée sur le site web de l'IEC.

Le comité a décidé que le contenu de cette publication ne sera pas modifié avant la date de stabilité indiquée sur le site web de l'IEC sous "<http://webstore.iec.ch>" dans les données relatives à la publication recherchée. A cette date, la publication sera

- reconduite,
- supprimée,
- remplacée par une édition révisée, ou
- amendée.

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INTRODUCTION

0.1 Généralités

La série IEC 61800 est destinée à fournir un ensemble commun de spécifications dédiées aux entraînements électriques de puissance à vitesse variable.

L'IEC 61800-7 spécifie les profils dédiés aux entraînements électriques de puissance (PDS) et leur mise en correspondance avec les systèmes de communication existants grâce à un modèle d'interface générique.

L'IEC 61800-7 décrit une interface générique entre les systèmes de commande et les entraînements électriques de puissance. Cette interface peut être intégrée au système de commande. Le système de commande proprement dit peut également être situé dans le dispositif d'entraînement (parfois appelé "dispositif d'entraînement intelligent").

Il existe un grand nombre d'interfaces physiques disponibles (entrées et sorties analogiques et numériques, interfaces séries et parallèles, bus de terrain et réseaux). Les profils établis sur des interfaces physiques spécifiques sont déjà définis pour certains domaines d'application (par exemple, commande de mouvement) et certaines classes de dispositifs (par exemple, dispositifs d'entraînement classiques, positionneur). Les implémentations correspondantes des interfaces de programmes de commande et de programmeurs d'application associées sont de nature propriétaire et varient de manière importante.

L'IEC 61800-7 définit un ensemble de fonctions, paramètres et diagrammes d'états communs pour la commande d'entraînement ou une description de séquences d'opérations à mettre en correspondance avec les profils d'entraînement.

L'IEC 61800-7 fournit une procédure d'accès aux fonctions et données d'un dispositif d'entraînement, indépendante du profil d'entraînement et de l'interface de communication employés. Il s'agit de définir un modèle commun d'entraînement comportant des fonctions génériques et des objets qui peuvent être mis en correspondance avec des interfaces de communication différentes. Ceci permet de prévoir des implémentations communes de commande de mouvement (ou applications de commande de vitesse ou de commande d'entraînement) dans les contrôleurs sans aucune connaissance spécifique de la mise en œuvre du dispositif d'entraînement.

Il y a plusieurs raisons de définir une interface générique:

Pour un constructeur de dispositif d'entraînement

- assistance plus aisée des intégrateurs de systèmes;
- description plus aisée des fonctions d'entraînement du fait d'une terminologie commune;
- le choix des dispositifs d'entraînement ne dépend pas de la disponibilité d'une assistance spécifique.

Pour un constructeur de dispositif de commande

- aucune influence de la technologie de bus;
- intégration aisée des dispositifs;
- indépendance par rapport à un fournisseur de dispositifs d'entraînement.

Pour un intégrateur de systèmes

- effort moindre d'intégration des dispositifs;
- méthode intelligible unique de modélisation;
- indépendance par rapport à la technologie de bus.

Concevoir une application de commande de mouvement avec plusieurs dispositifs d'entraînement différents et un système de commande spécifique nécessite un effort certain. Les tâches de mise en œuvre des logiciels systèmes et de compréhension de la description fonctionnelle des composants individuels peuvent conduire à l'épuisement des ressources d'un projet. Dans certains cas, les dispositifs d'entraînement ne partagent pas la même interface physique. Certains dispositifs de commande ne prennent en charge qu'une interface unique qui n'est pas prise en charge par un dispositif d'entraînement spécifique. D'autre part, les fonctions et les structures de données sont souvent spécifiées avec des incompatibilités. Cela exige de l'intégrateur de systèmes d'établir des interfaces spéciales pour le logiciel d'application alors que cette opération ne relève pas vraiment de sa responsabilité.

Certaines applications nécessitent de pouvoir échanger des dispositifs, voire intégrer de nouveaux dispositifs dans une configuration existante. Elles sont alors confrontées à différentes solutions incompatibles. Les efforts nécessaires pour adapter une solution relative à un profil d'entraînement et aux extensions spécifiques au constructeur peuvent se révéler inacceptables. Ceci réduit le degré de liberté concernant le choix d'un dispositif le mieux adapté à cette application à la simple sélection du dispositif disponible pour une interface physique spécifique et pris en charge par le contrôleur.

L'IEC 61800-7-1 est divisée en une partie générique et en plusieurs annexes comme le représente la Figure 1. Les types de profils d'entraînement pour CiA® 402¹, CIP Motion™², PROFIdrive³ et SERCOS®⁴ sont mis en correspondance avec l'interface générique dans l'annexe correspondante. Les annexes ont été soumises par des organismes internationaux indépendants spécialisés dans les réseaux ou les bus de terrain, et responsables du contenu de l'annexe qui y est associée, ainsi que de l'utilisation des marques connexes.

Cette partie de l'IEC 61800-7 spécifie le type de profil 2 (CIP Motion™).

Les types de profils 1, 3 et 4 sont spécifiés dans l'IEC 61800-7-201, l'IEC 61800-7-203 et l'IEC 61800-7-204.

¹ CiA® 402 est une marque déposée de CAN in Automation, e.V (CiA). Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque déposée CiA® 402. L'utilisation de la marque déposée CiA® 402 nécessite l'autorisation de CAN in Automation e.V (CiA).

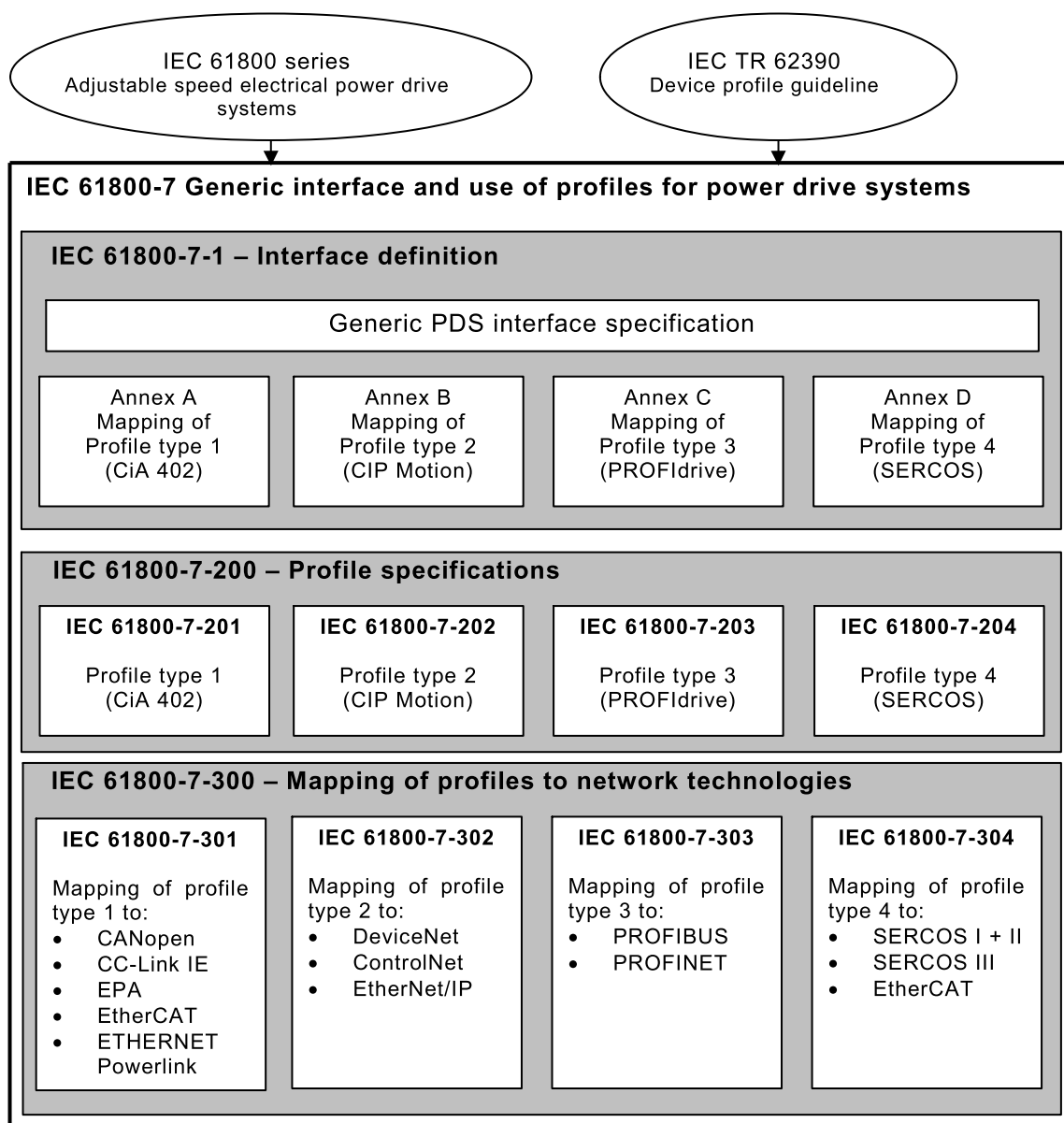
² CIP Motion™ est une marque de ODVA, Inc. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque CIP Motion™. L'utilisation de la marque CIP Motion™ nécessite l'autorisation de ODVA, Inc.

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⁴ SERCOS® est une marque déposée de SERCOS International e.V. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque déposée SERCOS®. L'utilisation de la marque déposée SERCOS® nécessite l'autorisation de son détenteur.

L'IEC 61800-7-301, l'IEC 61800-7-302, l'IEC 61800-7-303 et l'IEC 61800-7-304 spécifient la méthode de mise en correspondance des types de profil 1, 2, 3 et 4 avec différentes technologies de réseau (telles que CANopen⁵, CC-Link IE⁶ Field Network⁶, EPA⁷, EtherCAT⁸, Ethernet Powerlink⁹, DeviceNet¹⁰, ControlNet¹¹, EtherNet/IP¹², PROFIBUS¹³, PROFINET¹⁴ et SERCOS[®]).

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- ⁵ CANopen[®] est une marque déposée de CAN in Automation, e.V (CiA). Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque déposée CANopen[®]. L'utilisation de la marque déposée CANopen[®] nécessite l'autorisation de CAN in Automation e.V (CiA). CANopen[®] est l'acronyme de "Controller Area Network open (*Gestionnaire de réseau de communication ouvert*) et fait référence à l'EN 50325-4.
- ⁶ CC-Link IE[®] Field Network est une marque déposée de Mitsubishi Electric Corporation. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque déposée CC-Link IE[®] Field Network. L'utilisation de la marque déposée CC-Link IE[®] Field Network nécessite l'autorisation de Mitsubishi Electric Corporation.
- ⁷ EPA[™] est une marque de SUPCON Group Co. Ltd. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque EPA[™]. L'utilisation de la marque EPA[™] nécessite l'autorisation de son détenteur.
- ⁸ EtherCAT[®] est une marque déposée de Beckhoff, Verl. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque déposée EtherCAT[®]. L'utilisation de la marque déposée EtherCAT[®] nécessite l'autorisation de son détenteur.
- ⁹ Ethernet Powerlink[™] est une marque de Bernecker & Rainer Industrieelektronik Ges.m.b.H., le contrôle de son utilisation est confié à l'organisme à but non lucratif EPSG. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque Ethernet Powerlink[™]. L'utilisation de la marque Ethernet Powerlink[™] nécessite l'autorisation de son détenteur.
- ¹⁰ DeviceNet[™] est une marque de ODVA, Inc. Cette information est fournie pour la commodité des utilisateurs de la présente Norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque DeviceNet[™]. L'utilisation de la marque DeviceNet[™] nécessite l'autorisation de ODVA, Inc.
- ¹¹ ControlNet[™] est une marque de ODVA, Ltd. Cette information est fournie pour la commodité des utilisateurs de la présente Norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque ControlNet[™]. L'utilisation de la marque ControlNet[™] nécessite l'autorisation de ODVA, Inc.
- ¹² EtherNet/IP[™] est une marque de ODVA, Inc. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque EtherNet/IP[™]. L'utilisation de la marque EtherNet/IP[™] nécessite l'autorisation de ODVA, Inc.
- ¹³ PROFIBUS est une marque de PROFIBUS & PROFINET International. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque PROFIBUS. L'utilisation de la marque PROFIBUS nécessite l'autorisation de PROFIBUS & PROFINET International.
- ¹⁴ PROFINET est une marque de PROFIBUS & PROFINET International. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque PROFINET. L'utilisation de la marque PROFINET nécessite l'autorisation de PROFIBUS & PROFINET International.



IEC

Anglais	Français
IEC 61800 series Adjustable speed electrical power drive systems	Série IEC 61800 Entraînements électriques de puissance à vitesse variable
IEC TR 62390 Device profile guideline	IEC TR 62390 Device profile guideline (disponible en anglais seulement)
IEC 61800-7 Generic interface and use of profiles for power drive systems	IEC 61800-7 Interface générique et utilisation de profils pour les entraînements électriques de puissance
IEC 61800-7-1 Interface definition	IEC 61800-7-1 Définition de l'interface
Generic PDS interface specification	Spécification d'interface PDS générique
Annex A, Mapping of Profile type 1 (CiA 402)	Annexe A, Mise en correspondance de profil de type 1 (CiA 402)
Annex B, Mapping of Profile type 2 (CIP Motion)	Annexe B, Mise en correspondance du profil de type 2 (CIP Motion)
Annex C, Mapping of Profile type 3 (PROFIdrive)	Annexe C, Mise en correspondance du profil de type 3 (PROFIdrive)
Annex D, Mapping of Profile type 4 (SERCOS)	Annexe D, Mise en correspondance du profil de type 4 (SERCOS)

Anglais	Français
IEC 61800-7-200 – Profile specifications	IEC 61800-7-200 – Spécifications des profils
IEC 61800-7-201 Profile type 1 (CiA 102)	IEC 61800-7-201 Profil de type 1 (CiA 102)
IEC 61800-7-202 Profile type 2 (CIP Motion)	IEC 61800-7-202 Profil de type 2 (CIPMotion)
IEC 61800-7-203 Profile type 3 (PROFIdrive)	IEC 61800-7-203 Profil de type 3 (PROFIdrive)
IEC 61800-7-204 Profile type 4 (PROFIdrive)	IEC 61800-7-204 Profil de type 4 (SERCOS)
IEC 61800-7-300 – Mapping of profiles to network technologies	IEC 61800-7-300 – Mise en correspondance des profils avec les technologies de réseaux
IEC 61800-7-301 Mapping of profile type 1 to CANopen CC-Link IE EPA EtherCAT ETHERNET Powerlink	IEC 61800-7-301 Mise en correspondance du profil de type 1 avec CANopen CC-Link IE EPA EtherCAT ETHERNET Powerlink
IEC 61800-7-302 Mapping of profile type 2 to DeviceNet ControlNet EtherNet/IP	IEC 61800-7-302 Mise en correspondance du profil de type 2 avec DeviceNet ControlNet EtherNet/IP
IEC 61800-7-303 Mapping of profile type 3 to PROFIBUS PROFINET	IEC 61800-7-303 Mise en correspondance du profil de type 3 avec PROFIBUS PROFINET
IEC 61800-7-304 Mapping of profile type 4 to SERCOS I + II SERCOS III EtherCAT	IEC 61800-7-304 Mise en correspondance du profil de type 4 avec SERCOS I + II SERCOS III EtherCAT

Figure 1 – Structure de l'IEC 61800-7

0.2 Déclaration de propriété

La Commission électrotechnique internationale (IEC) attire l'attention sur le fait qu'il est déclaré que la conformité avec les dispositions du présent document peut impliquer l'utilisation d'un brevet intéressant ce qui suit. Ce brevet est détenu par ses inventeurs dans le cadre d'une licence octroyée à ODVA, Inc.

Numéro de série de la publication / application	Détenteur	Titre
US 7,983,769 EP 1659465	[ODVA]	Time stamped motion control network protocol that enables balanced single cycle timing and utilization of dynamic data structures

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[ODVA]	ODVA, Inc. 2370 East Stadium Boulevard #1000 Ann Arbor, Michigan 48104 USA Attention: Office of the Executive Director email: odva@odva.org
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L'attention est d'autre part attirée sur le fait que certains des éléments du présent document peuvent faire l'objet de droits de propriété intellectuelle autres que ceux identifiés ci-dessus. L'IEC ne saurait être tenue pour responsable de ne pas avoir identifié de tels droits de propriété et de ne pas avoir signalé leur existence.

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ENTRAÎNEMENTS ÉLECTRIQUES DE PUISSANCE À VITESSE VARIABLE –

Partie 7-202: Interface générique et utilisation de profils pour les entraînements électriques de puissance – Spécification de profil de type 2

1 Domaine d'application

La présente partie de l'IEC 61800 spécifie le type de profil 2 (CIP Motion™) pour les entraînements électriques de puissance (PDS). Le type de profil 2 peut être mis en correspondance avec différentes technologies de réseau de communication.

Les fonctions spécifiées dans la présente partie de l'IEC 61800-7 ne sont pas destinées à assurer la sécurité fonctionnelle. Ceci exige l'application de mesures supplémentaires conformes aux normes, conventions et lois pertinentes.

2 Références normatives

Les documents suivants sont cités en référence de manière informative, en intégralité ou en partie, dans le présent document et sont indispensables pour son application. Pour les références datées, seule l'édition citée s'applique. Pour les références non datées, la dernière édition du document de référence s'applique (y compris les éventuels amendements).

IEC 60204-1, *Sécurité des machines – Équipement électrique des machines – Partie 1: Règles générales*

IEC 61158-4-2:2014, *Réseaux de communication industriels – Spécifications des bus de terrain – Partie 4-2: Spécification du protocole de la couche liaison de données – Éléments de type 2*

IEC 61158-5-2:2014, *Réseaux de communication industriels – Spécifications des bus de terrain – Partie 5-2: Définition des services de la couche application – Éléments de type 2*

IEC 61158-6-2:2014, *Réseaux de communication industriels – Spécifications des bus de terrain – Partie 6-2: Spécification du protocole de la couche application – Éléments de type 2*

IEC 61588:2009, *Precision clock synchronization protocol for networked measurement and control systems* (disponible en anglais seulement)

IEC 61800-7-1:2015, *Entraînements électriques de puissance à vitesse variable – Partie 7-1: Interface générique et utilisation de profils pour les entraînements électriques de puissance – Définition de l'interface*

IEEE Std 112-2004, *IEEE Standard Test Procedure for Polyphase Induction Motors and Generators*

3 Termes, définitions et abréviations

3.1 Termes et définitions

Pour les besoins du présent document, les termes et définitions suivants s'appliquent.

3.1.1

algorithme

séquence finie d'opérations complètement déterminée par laquelle la valeur des données de sortie peut être calculée à partir de la valeur des données d'entrée

[SOURCE: IEC 61800-7-1:2015, 3.2.1]

3.1.2

application

élément fonctionnel logiciel spécifique à la résolution d'un problème en termes de mesure et de commande de procédés industriels

Note 1 à l'article: Une application peut être répartie entre les ressources, et peut communiquer avec d'autres applications.

[SOURCE: IEC 61800-7-1:2015, 3.2.2]

3.1.3

mode d'application

type d'application pouvant être demandé à par un PDS

Note 1 à l'article: Les différents modes d'application reflètent la boucle d'asservissement dédiée à l'asservissement de couple ou de position, à la commande de vitesse ou à d'autres applications comme le retour à la position de référence.

[SOURCE: IEC 61800-7-1:2015, 3.3.1.2]

3.1.4

attribut

propriété ou caractéristique d'une entité

[SOURCE: IEC 61800-7-1:2015, 3.2.3]

3.1.5

axe

élément logique interne à un système d'automatisation (par exemple, système de commande de mouvement) qui représente une certaine forme de mouvement

Note 1 à l'article: Les axes peuvent être rotatifs ou linéaires, physiques ou virtuels, commandés ou simplement observés.

[SOURCE: IEC 61800-7-1:2015, 3.2.4]

3.1.6

régulateur de bus

méthode permettant de limiter la hausse du niveau de tension de bus à courant continu suite à la décélération d'un moteur

3.1.7

CIP Motion™¹⁵

extensions des services et protocole CIP afin de prendre en charge la commande de mouvement sur les réseaux CIP

[SOURCE: IEC 61800-7-1:2015, 3.3.3.1]

3.1.8

contrôleur CIP Motion™

contrôleur à conformité CIP contenant un Objet Axe de commande de mouvement (Motion Control Axis Object) pouvant être en interface avec un dispositif CIP Motion via une Connexion E/S CIP Motion

Note 1 à l'article: Une description de l'Objet Axe de commande de mouvement (Motion Control Axis Object) ne relève pas du domaine d'application de l'IEC 61800-7.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.2]

3.1.9

dispositif CIP Motion™

dispositif à conformité CIP contenant une ou plusieurs instances d'Objet Axe de dispositif de mouvement (Motion Device Axis Object) pouvant communiquer avec un contrôleur CIP Motion via une Connexion E/S CIP Motion

EXEMPLE: Un variateur CIP Motion est un cas particulier d'un dispositif CIP Motion.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.3]

3.1.10

profil d'entraînement CIP Motion™

ensemble d'objets permettant de mettre en œuvre un dispositif d'entraînement (variateur) CIP Motion comprenant l'Objet Axe de dispositif de mouvement, ainsi que des objets de support standard tels que l'Objet Identité (Identity Object) et l'Objet Synchronisation temporelle (Time Sync Object)

Note 1 à l'article: Le type de dispositif (Device Type) assigné au profil d'entraînement CIP Motion est 25_{hex}.

3.1.11

Connexion E/S CIP Motion™

Connexion CIP Motion™

connexion CIP bidirectionnelle périodique de classe 1 entre un contrôleur et un variateur, définie comme partie intégrante de la spécification de CIP Motion

[SOURCE: IEC 61800-7-1:2015, 3.3.3.4]

3.1.12

CIP Sync™¹⁵

extensions des services et du protocole CIP visant à intégrer la fonctionnalité de synchronisation temporelle définie dans l'IEC 61588:2009 dans un réseau CIP

Note 1 à l'article: Voir "Objet Synchronisation temporelle" ('Time Sync Object') dans l'IEC 61158-5-2 et l'IEC 61158-6-2

[SOURCE: IEC 61800-7-1:2015, 3.3.3.5]

¹⁵ CIP Motion™ et CIP Sync™ sont des marques de ODVA, Inc. Cette information est fournie pour la commodité des utilisateurs de la présente Norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation des marques CIP Motion™ ou CIP Sync™. L'utilisation de la marque CIP Motion™ ou CIP Sync™ nécessite l'autorisation de ODVA, Inc.

3.1.13**classe**

description d'un ensemble d'objets qui partagent les mêmes attributs, opérations, méthodes, relations et sémantique

[SOURCE: IEC 61800-7-1:2015, 3.2.5]

3.1.14**boucle fermée**

méthodes de commande dans lesquelles un signal de réaction d'un type donné est utilisé pour entraîner la dynamique réelle du moteur afin de correspondre à la dynamique commandée par asservissement

Note 1 à l'article: Dans la plupart des cas, un dispositif de réaction littéral fournit ce signal, mais dans certains cas, le signal est dérivé de l'excitation du moteur (c'est-à-dire fonctionnement sans capteur).

3.1.15**commandes****consignes**

ensemble de commandes entre le programme de commande d'application et le PDS permettant de contrôler le comportement du PDS ou les éléments fonctionnels de celui-ci

Note 1 à l'article: Les états ou les modes de fonctionnement reflètent le comportement.

Note 2 à l'article: Les différentes commandes peuvent être représentées par un bit chacune.

[SOURCE: IEC 61800-7-1:2015, 3.3.1.3]

3.1.16**commande****régulation**

action délibérée sur (ou dans) un processus, en vue d'atteindre des objectifs définis

[SOURCE: IEC 61800-7-1:2015, 3.2.6]

3.1.17**dispositif de commande/régulation**

unité physique contenant – dans un module/sous-ensemble ou dispositif – un programme d'application de commande du PDS

[SOURCE: IEC 61800-7-1:2015, 3.2.7]

3.1.18**convertisseur**

dispositif qui convertit généralement une entrée en courant alternatif en sortie en courant continu

Note 1 à l'article: Un convertisseur est également communément appelé Alimentation d'entraînement. Dans le contexte d'un système d'entraînement, le convertisseur est chargé de convertir l'alimentation en courant alternatif en alimentation de bus à courant continu.

3.1.19**Bloc de données cycliques**

bloc de données en temps réel à priorité élevée transférées par une Connexion CIP Motion de manière périodique

3.1.20**type de données**

ensemble de valeurs associé à un ensemble d'opérations autorisées

[SOURCE: IEC 61800-7-1:2015, 3.2.8]

3.1.21**dispositif**

dispositif de terrain

<blocs de fonction> entité physique indépendante en réseau dans un système d'automatisation industriel, capable d'accomplir des fonctions spécifiées dans un contexte particulier et délimitée par ses interfaces

[SOURCE: IEC 61800-7-1:2015, 3.2.9]

3.1.22**dispositif**

dispositif de terrain

<intégration système> entité qui permet le contrôle, la servocommande et/ou la détection et s'interface à d'autres entités de ce type à l'intérieur d'un système d'automatisation

[SOURCE: IEC 61800-7-1:2015, 3.2.10]

3.1.23**profil de dispositif**

représentation d'un dispositif en termes de ses paramètres, ensembles de paramètres et comportement selon un modèle de dispositif qui décrit les données et le comportement du dispositif tel que perçus par l'intermédiaire d'un réseau, indépendamment de toute technologie de réseau

[SOURCE: IEC 61800-7-1:2015, 3.2.11]

3.1.24**dispositif d'entraînement**

entraînement

variateur

dispositif conçu pour commander la dynamique d'un moteur

3.1.25**Bloc de données d'événement**

bloc de données en temps réel à priorité moyenne transférées par une Connexion CIP Motion uniquement après occurrence d'un événement spécifié

Note 1 à l'article: Les transitions des entrées registration et marqueur constituent des événements typiques des variateurs.

3.1.26**élément fonctionnel**

entité de logiciel ou logiciel combiné au matériel, capable d'accomplir une fonction spécifiée d'un dispositif

Note 1 à l'article: Un élément fonctionnel comporte une interface et des associations à d'autres éléments fonctionnels et fonctions.

Note 2 à l'article: Un élément fonctionnel peut être constitué de bloc(s) de fonctions, d'objet(s) ou de liste(s) de paramètres.

[SOURCE: IEC 61800-7-1:2015, 3.2.13]

3.1.27**données d'entrée**

données transférées d'une source externe dans un dispositif, une ressource ou un élément fonctionnel

[SOURCE: IEC 61800-7-1:2015, 3.2.14]

3.1.28**interface**

frontière partagée entre deux entités, définie par des caractéristiques fonctionnelles, des caractéristiques de signal ou d'autres caractéristiques appropriées

[SOURCE: IEC 61800-7-1:2015, 3.2.15]

3.1.29**onduleur**

dispositif qui convertit généralement une entrée en courant continu en sortie en courant alternatif

Note 1 à l'article: Un onduleur est également communément appelé Amplificateur d'entraînement. Dans le contexte d'un système d'entraînement, l'onduleur est chargé de contrôler l'application de l'alimentation de bus à courant continu à un moteur à courant alternatif.

3.1.30**modèle**

représentation mathématique ou physique d'un système ou d'un processus, basée, avec une précision suffisante, sur des lois connues, sur une identification ou sur des hypothèses spécifiées

[SOURCE: IEC 61800-7-1:2015, 3.2.17]

3.1.31**mouvement**

tout aspect de la dynamique d'un axe

Note 1 à l'article: Dans le contexte de la présente partie de l'IEC 61800-7, cette définition ne se limite pas aux entraînements asservis, mais englobe toutes les formes de commande de moteur par entraînement.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.8]

3.1.32**Objet Axe de commande de mouvement****Motion Control Axis Object**

objet qui définit les attributs, services et comportement d'un axe de contrôleur selon la spécification de CIP Motion

3.1.33**Objet Axe de dispositif de mouvement****Motion Device Axis Object**

objet qui définit les attributs, services et comportement d'un axe de dispositif de mouvement selon la spécification de CIP Motion

Note 1 à l'article: Cet objet comprend les éléments fonctionnels Communication, Commande de dispositif et Dispositif d'entraînement de base, tels que définis dans l'IEC 61800-7.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.9]

3.1.34**boucle ouverte**

méthodes de commande dans lesquelles il n'existe aucune application de réaction pour forcer la dynamique réelle du moteur afin de correspondre à la dynamique commandée

EXEMPLE Les exemples de commande à boucle ouverte sont les entraînements pas à pas et les dispositifs d'entraînement à fréquence variable.

3.1.35**mode de fonctionnement**

caractérisation de la manière et du degré avec lequel l'opérateur humain intervient sur l'équipement de commande

[SOURCE: IEC 61800-7-1:2015, 3.2.18]

3.1.36

données de sortie

données provenant d'un dispositif, d'une ressource ou d'un élément fonctionnel et transférées de ces derniers vers des systèmes externes

[SOURCE: IEC 61800-7-1:2015, 3.2.19]

3.1.37

paramètre

élément de donnée qui représente les informations d'un dispositif qui peuvent être lues ou saisies dans un dispositif, par exemple, par le biais du réseau ou d'une IHM locale

Note 1 à l'article: Un paramètre est caractérisé généralement par son nom, le type de données et la direction d'accès.

[SOURCE: IEC 61800-7-1:2015, 3.2.20]

3.1.38

profil

représentation d'une interface PDS en termes de ses paramètres, ensembles de paramètres et comportement selon un profil de communication et un profil de dispositif

[SOURCE: IEC 61800-7-1:2015, 3.2.21, modifiée – La Note 1 à l'article est supprimée]

3.1.39

lire

obtenir

opération qui implique la recherche d'une valeur d'attribut à partir de la partie contrôleur de l'interface

3.1.40

enregistrement

registration

sauvegarde à haute vitesse de la position d'un axe de mouvement, déclenchée par un événement

3.1.41

Bloc de données de service

bloc de données en temps réel de priorité moins élevée associées à un message de service du contrôleur, transférées par une Connexion CIP Motion de manière périodique

Note 1 à l'article: Les données de service incluent les messages de demande de service qui permettent d'accéder aux attributs d'Objet Axe de dispositif de mouvement (Motion Device Axis Object) ou d'effectuer divers diagnostics du variateur.

3.1.42

point de consigne

valeur ou variable utilisée comme donnée de sortie du programme de commande d'application afin de commander le PDS

Note 1 à l'article: La valeur envoyée au dispositif d'entraînement est utilisée pour commander directement certains aspects liés à la dynamique du moteur, comprenant (sans toutefois s'y limiter) la position, la vitesse, l'accélération et le couple

[SOURCE: IEC 61800-7-1:2015, 3.3.1.5, modifiée – la Note 1 à l'article est ajoutée]

3.1.43**régulateur de shunt**

méthode de régulation de bus spécifique qui fait basculer le bus à courant continu sur une résistance de dissipation d'énergie afin de dissiper la force régénératrice d'un moteur en cours de décélération

3.1.44**état****statut**

ensemble d'informations entre le PDS et le programme de commande d'application, qui reflète l'état ou le mode du PDS ou un élément fonctionnel de ce dernier

Note 1 à l'article: Les différentes informations d'état peuvent être codées avec un bit chacune.

[SOURCE: IEC 61800-7-1:2015, 3.3.1.6]

3.1.45**synchronisé**

état de verrouillage de la valeur d'horloge locale du variateur sur l'horloge maîtresse du temps système réparti

Note 1 à l'article: Lorsqu'ils sont synchronisés, le variateur et le contrôleur peuvent utiliser les datations associées aux données de Connexion CIP Motion.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.11]

3.1.46**temps système****System Time**

valeur temporelle absolue définie dans la spécification de synchronisation CIP Sync dans le contexte d'un système temporel réparti où tous les dispositifs comportent une horloge locale synchronisée avec une horloge maîtresse commune

Note 1 à l'article: Dans le contexte de CIP Motion, le temps système est une valeur entière à 64 bits en nanosecondes, avec une valeur de 0 correspondant à la date 1970-01-01.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.12]

3.1.47**modèle thermique**

algorithme visant à modéliser le comportement thermique du dispositif concerné en cours de fonctionnement

3.1.48**décalage temporel**

valeur de décalage du temps système associée aux données de Connexion CIP Motion liées au dispositif source

Note 1 à l'article: La valeur de décalage du temps système est une valeur de 64 bits ajoutée à l'horloge locale d'un dispositif afin de générer son temps système.

3.1.49**datation****horodatage**

valeur de datation système associée aux données de Connexion CIP Motion qui achemine le temps absolu lorsque les données associées ont été saisies, ou qui peut également être utilisée pour déterminer le moment où les données associées doivent être appliquées

Note 1 à l'article: Les datations CIP sont toujours utilisées dans le contexte d'un système temporel réparti où tous les nœuds du réseau de commande CIP comportent des horloges synchronisées avec une source d'horloge maîtresse utilisant la spécification de synchronisation CIP Sync.

[SOURCE: IEC 61800-7-1:2015, 3.3.3.13]

3.1.50

type

élément matériel ou logiciel qui spécifie les attributs communs partagés par toutes les instances du type

[SOURCE: IEC 61800-7-1:2015, 3.2.23]

3.1.51

variable

entité logicielle qui peut prendre différentes valeurs, mais une seule valeur à la fois

Note 1 à l'article: Les valeurs d'une variable, ainsi que d'un paramètre, se limitent habituellement à un certain type de données.

[SOURCE: IEC 61800-7-1:2015, 3.2.25]

3.1.52

dispositif d'entraînement à fréquence variable

VFD

classe de produits d'entraînement visant à réguler le régime d'un moteur, généralement un moteur à induction, au moyen d'une relation proportionnelle entre la tension de sortie du dispositif d'entraînement et la fréquence de sortie commandée

Note 1 à l'article: Par conséquent, les dispositifs d'entraînement à fréquence variable sont parfois désignés dispositifs d'entraînement à Volts/Hertz.

Note 2 à l'article: L'abréviation "VFD" est dérivée du terme anglais développé correspondant "Variable Frequency Drive".

3.1.53

dispositif d'entraînement vectoriel

classe de produits d'entraînement visant à commander les éléments dynamiques d'un moteur par l'intermédiaire d'une boucle fermée comprenant, sans toutefois s'y limiter, le contrôle de boucle fermée du couple et des composantes vectorielles de flux du courant du stator par rapport au vecteur de flux du rotor

3.1.54

écrire

définir

opération qui implique le réglage d'un attribut à une valeur spécifiée à partir de la partie contrôleur de l'interface

3.2 Abréviations

Ack	acknowledge (Acquittement)
Act	actual (Réel)
Blk	block (Bloc)
CIP™ ¹⁶	Common Industrial Protocol (Protocole industriel commun) (voir IEC 61158 Type 2, IEC 61784-1 et IEC 61784-2 Famille de Profils de Communication 2)
Cmd	command (commande, consigne)
Cyc	cyclique

¹⁶ CIP™ est une marque de ODVA, Inc. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque CIP™. L'utilisation de la marque CIP™ nécessite l'autorisation de ODVA, Inc

IHM	interface homme-machine
ID	identificateur
Impl.	implementation (mise en œuvre)
E/S	entrée/sortie
Opt.	optional (Facultatif ou F)
PDS	power drive system (Entraînement électrique de puissance)
Req.	required (Obligatoire ou O)
RMS	root mean square (Moyenne quadratique)
r/min	tours par minute
Sync'd	synchronized (Synchronisé)
TCP	Transmission Control Protocol (Protocole de contrôle de transmission) (voir IETF RFC 793)
UDP	User Datagram Protocol (Protocole de datagramme utilisateur (voir IETF RFC 768)
VFD	variable frequency drive (dispositif d'entraînement à fréquence variable)

4 Présentation générale

4.1 Généralités

Les dispositifs d'entraînement CIP Motion permettent de commander, contrôler ou prendre en charge le mouvement d'un ou de plusieurs composants mobiles d'une machine. En règle générale, le mouvement de la machine est généré par des actionneurs de mouvement rotatifs ou linéaires, c'est-à-dire des moteurs, et contrôlé par des dispositifs de réaction. Chaque moteur est généralement entraîné par une structure de puissance et un algorithme de commande de mouvement comprenant un type de dispositif d'entraînement CIP Motion.

La fonctionnalité d'entraînement prise en charge par le profil de dispositif d'entraînement CIP Motion peut être appliquée à un grand nombre de technologies de motorisation pouvant couvrir les dispositifs d'entraînement à fréquence variable en "boucle ouverte" les plus simples jusqu'aux entraînements asservis à commande vectorielle en "boucle fermée" plus complexes. Dans tous les cas, le mouvement est commandé par une référence de consigne qui peut être configurée pour l'asservissement de position, la commande de vitesse, le contrôle d'accélération ou l'asservissement de courant/couple. Le profil de dispositif d'entraînement CIP Motion prend également en charge le contrôle de position, de vitesse et d'accélération par le biais de plusieurs réactions comme le fait le profil de dispositif d'entraînement à codeur CIP Motion.

Tous les attributs, services et comportement d'état d'un dispositif CIP Motion sont intégrés dans une ou plusieurs instances d'Objet Axe de dispositif de mouvement. Outre la fonctionnalité de commande de mouvement et de contrôle de réaction, l'Objet Axe de dispositif de mouvement comprend la prise en charge du contrôle d'événement, tel que la capture de position sur un événement d'Enregistrement, et la gestion du bus à courant continu associée à un Convertisseur de puissance. Un dispositif CIP Motion peut présenter toute forme de combinaison de ces fonctions afin de créer diverses classes de dispositifs à conformité CIP Motion allant des entraînements asservis jusqu'aux Codeurs CIP Motion en passant par les Convertisseurs de puissance autonomes, différenciés par le Type de dispositif CIP Motion.

Cette spécification du profil d'entraînement CIP Motion et la spécification de l'Objet Axe de dispositif de mouvement associée définissent l'interface, les attributs spécifiques et les comportements de consigne exigés pour la prise en charge de la commande de mouvement lors de la connexion à un contrôleur à conformité CIP Motion par l'intermédiaire d'un réseau CIP.

Parmi les réseaux CIP applicables, EtherNet/IP™ est le réseau privilégié pour les commandes multiaxe synchronisées haute performance. Les autres réseaux CIP tels que ControlNet™ et DeviceNet™ peuvent être réservés aux applications de dispositif non synchronisé à faible performance tels que les dispositifs d'entraînement à fréquence variable simples, les dispositifs d'entraînement à boucle de vitesse et les dispositifs d'entraînement indexés.

4.2 Modes de commande

4.2.1 Généralités

L'Objet Axe de dispositif de mouvement porte sur le comportement des différents dispositifs de système de commande de mouvement, y compris les dispositifs de réaction, les dispositifs d'entraînement, les convertisseurs autonomes et les dispositifs E/S de mouvement. Concernant les dispositifs d'entraînement, l'Objet Axe de dispositif de mouvement couvre un large éventail de types d'entraînement, des simples dispositifs d'entraînement à fréquence variable (V/Hz) aux entraînements d'asservissement de position les plus sophistiqués avec ou sans convertisseurs intégrés. En effet, la plupart des produits d'entraînement disponibles dans le commerce peuvent être configurés pour fonctionner dans l'un quelconque de ces différents modes de commande de mouvement, en fonction des exigences d'application spécifiques. Les attributs de l'Objet Axe de dispositif de mouvement sont par conséquent organisés pour traiter ce large éventail de fonctionnalités dont le cadre d'organisation est décrit en 4.2.

L'IEC 61800-7-1 et la présente partie de la série IEC 61800-7 s'appuient sur la philosophie générale selon laquelle l'asservissement de position est la forme la plus élevée de commande dynamique. En d'autres termes, l'asservissement de position implique la commande de vitesse, qui elle-même implique la commande d'accélération. L'accélération est liée au couple ou à la force par l'inertie ou la masse de la charge, respectivement, la commande d'accélération impliquant l'asservissement de couple. Enfin, dans la mesure où le couple moteur ou la force du moteur est généralement lié(e) au courant du moteur par une constante de couple ou de force, respectivement, l'asservissement de couple implique l'asservissement de courant. La constante de couple ou de force peut être fonction des aimants de moteur (dans les moteurs à aimant permanent, par exemple) ou du flux induit d'un moteur à induction.

L'accélération, le couple/la force et le courant étant généralement liés par une constante, ces termes sont parfois utilisés les uns pour les autres dans l'industrie. Par exemple, l'IEC 61800-7-1 fait référence à une boucle d'asservissement de couple plutôt qu'à une boucle d'asservissement de courant. Cette spécification tente de distinguer ces propriétés de commande, le cas échéant. Cela est particulièrement utile si leur relation n'est pas statique (lorsque l'inertie/la masse change en fonction de la position ou de la durée, ou lorsque la constante de couple/force change en raison des variations de la température ou du flux du moteur, par exemple).

4.2.2 Méthodes de commande

Ce paradigme de commande de base offre toute la latitude nécessaire aux différentes méthodes de commande, en boucle fermée et en boucle ouverte. Une boucle fermée implique généralement la présence d'un signal de réaction utilisé pour entraîner les éléments dynamiques réels du moteur afin de correspondre aux éléments dynamiques commandés par l'action asservie. Dans la plupart des cas, un dispositif de réaction littéral fournit ce signal, mais dans d'autres cas, le signal est déduit de l'excitation du moteur (c'est-à-dire d'un fonctionnement sans capteur/sans codeur). Une boucle ouverte implique l'absence d'application de réaction pour forcer les éléments dynamiques réels à correspondre aux éléments dynamiques commandés. La précision et les performances étant les marques du contrôle de boucle fermée, la simplicité et l'économie sont celles du contrôle de boucle ouverte.

4.2.3 Nomenclature de commande

Enfin, comme le montre la description ci-dessus, les applications de commande linéaires et rotatives peuvent avoir un impact sur la nomenclature de commande. Les applications rotatives parlent de couple et d'inertie, alors que les applications linéaires parlent de force et de masse. Pour les besoins de la présente partie de la série IEC 61800-7, s'il est fait référence à la nomenclature rotative, le comportement défini peut en général être appliqué aux applications linéaires en remplaçant les termes utilisés ("force" pour "couple" et "masse" pour "inertie"). À cet égard, les expressions de la Figure 2 à la Figure 8 suivent la nomenclature de l'IEC 61800-7-1 en utilisant le terme "couple" plutôt que le terme "force" sans réduire le caractère général.

4.2.4 Asservissement de position

4.2.4.1 Concepts généraux

En mode d'application d'asservissement de position, le programme de commande d'application (fonction d'exécution de commande) ou le planificateur de mouvement (fonction de commande de la trajectoire de déplacement) fournit une valeur de point de consigne au dispositif CIP Motion par l'intermédiaire d'une connexion de données cycliques. La méthode d'asservissement de position peut être une boucle ouverte ou une boucle fermée.

4.2.4.2 Asservissement de position en boucle ouverte

Un dispositif configuré pour l'asservissement de position en boucle ouverte s'applique à une classe de dispositifs d'entraînement appelés "entraînements pas à pas". Ce type d'entraînement est représenté à la Figure 2.

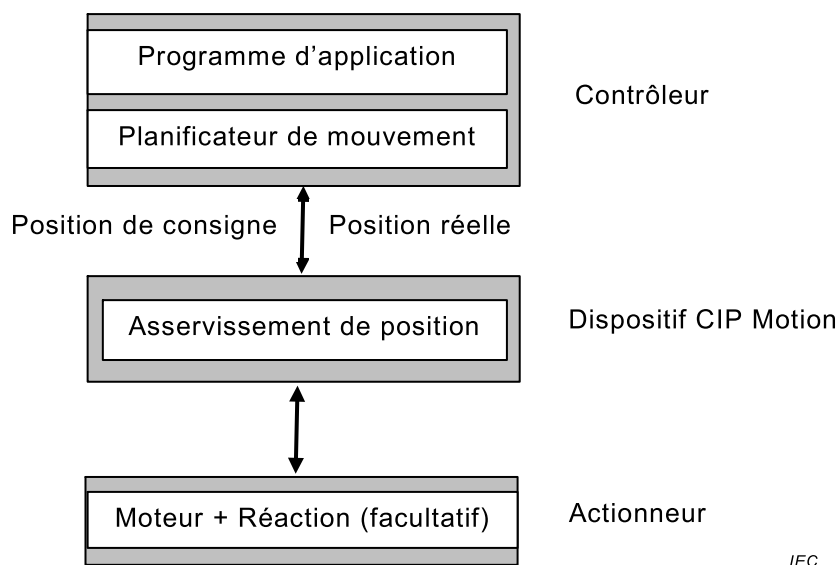


Figure 2 – Asservissement de position en boucle ouverte

Dans cette configuration, la présence d'un dispositif de réaction est facultative. En l'absence de dispositif de réaction, la position réelle peut être estimée par le dispositif d'entraînement et renvoyée au contrôleur.

4.2.4.3 Asservissement de position en boucle fermée

Un dispositif de commande de moteur configuré pour l'asservissement de position en boucle fermée est en général appelé dispositif d'entraînement de boucle de position ou entraînement asservi de position. Un entraînement asservi de position implique une vitesse interne et une boucle d'asservissement de couple comme représenté à la Figure 3. La présence d'une

boucle d'asservissement de couple/courant donne parfois lieu à ce type de dispositif d'entraînement, appelé dispositif d'entraînement vectoriel.

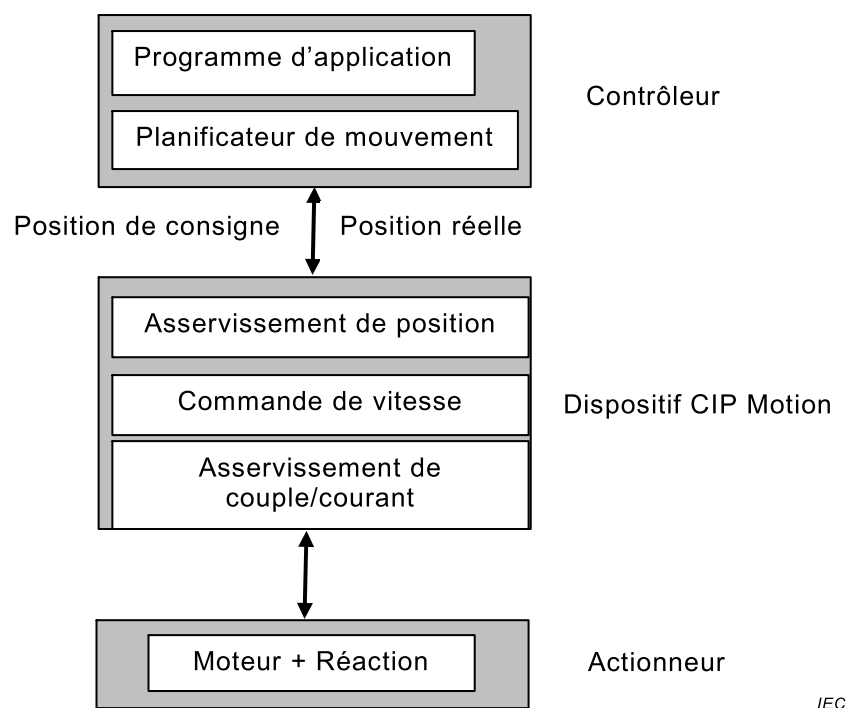


Figure 3 – Asservissement de position en boucle fermée

Dans cette configuration, un dispositif de réaction est en général exigé pour obtenir une bonne exactitude de positionnement. Le dispositif de réaction peut également être utilisé pour renvoyer les données de vitesse réelle et d'accélération réelle au contrôleur par l'intermédiaire de la connexion de données cycliques.

Outre la position de consigne, le contrôleur peut transmettre la vitesse de consigne et l'accélération de consigne pour les besoins de l'asservissement avancé.

4.2.5 Commande de vitesse

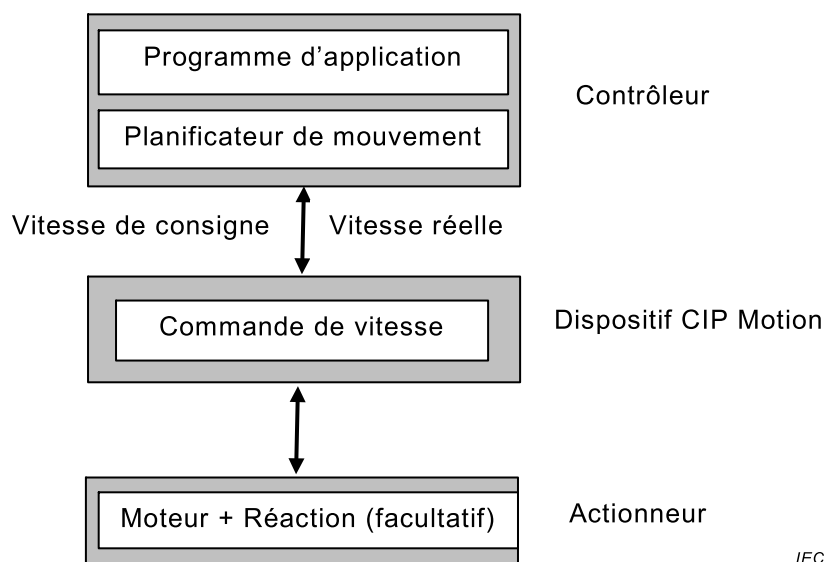
4.2.5.1 Concepts généraux

En mode d'application de commande de vitesse, le programme de commande d'application et le planificateur de mouvement fournissent une valeur de point de consigne au dispositif CIP Motion par l'intermédiaire de la connexion de données cycliques. La méthode de commande de vitesse peut être une boucle ouverte ou une boucle fermée.

4.2.5.2 Commande de vitesse en boucle ouverte

Un dispositif de commande de moteur configuré pour la commande de vitesse en boucle ouverte est en général appelé dispositif d'entraînement à fréquence variable, V/Hz ou VFD. Ce type d'entraînement est représenté à la Figure 4.

Dans cette configuration, la présence d'un dispositif de réaction est facultative. En l'absence de dispositif de réaction, la vitesse réelle peut être estimée par le dispositif d'entraînement et renvoyée au contrôleur.

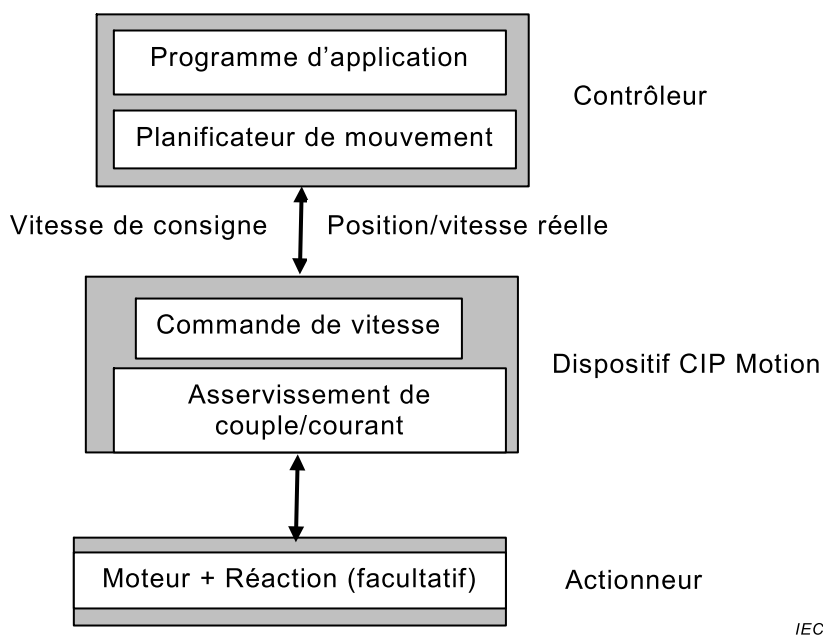


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Figure 4 – Commande de vitesse en boucle ouverte

4.2.5.3 Commande de vitesse en boucle fermée

Un dispositif de commande de moteur configuré pour la commande de vitesse en boucle fermée est en général appelé dispositif d'entraînement de boucle de vitesse ou entraînement asservi de vitesse. Un entraînement de commande de vitesse en boucle fermée implique une boucle d'asservissement de couple/courant interne (voir Figure 5). Il est donc parfois appelé dispositif d'entraînement vectoriel.



IEC

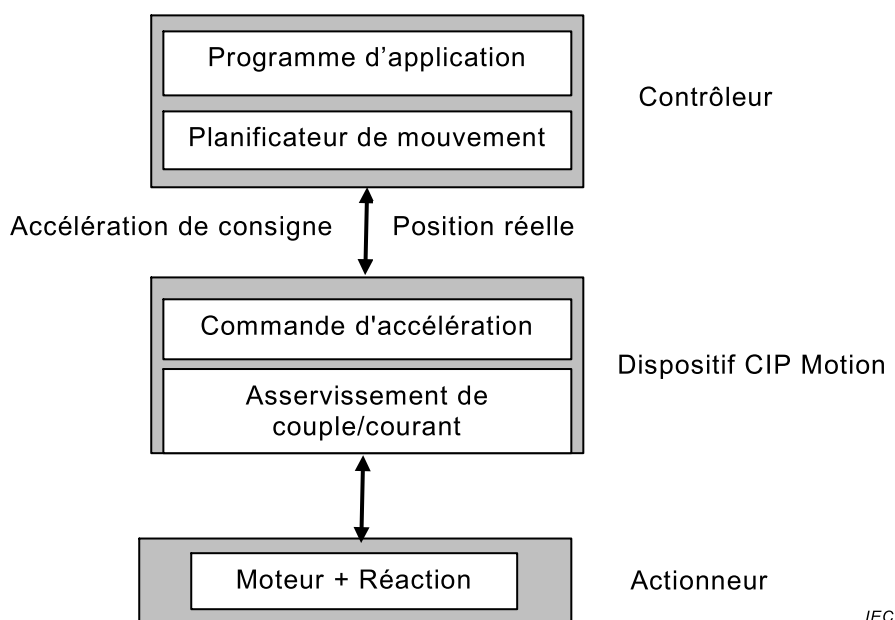
Figure 5 – Commande de vitesse en boucle fermée

Dans cette configuration, la présence d'un dispositif de réaction pour entraînement de commande de vitesse est facultative. Une régulation de vitesse plus stricte est obtenue à l'aide d'un dispositif de réaction, particulièrement à faible vitesse. Si le dispositif de réaction est inclus, il peut être utilisé pour renvoyer les données de position, de vitesse et d'accélération réelles au contrôleur par l'intermédiaire de la connexion de données cycliques. Si le dispositif de réaction n'est pas inclus, seule la vitesse estimée peut en général être renvoyée au contrôleur.

Outre la vitesse de consigne, le contrôleur peut également transmettre l'accélération de consigne pour les besoins de l'asservissement avancé.

4.2.6 Commande d'accélération

Le mode de commande principal n'étant mentionné ni dans l'industrie ni dans l'IEC 61800-7-1, le mode de commande d'accélération est présenté ici pour compléter la progression dynamique entre la commande de vitesse et l'asservissement de couple, et parce que l'Objet Axe de dispositif de mouvement peut prendre en charge une consigne d'accélération, potentiellement déduite du planificateur de mouvement du contrôleur. En mode de commande d'accélération, le programme de commande d'application et le planificateur de mouvement fournissent des valeurs de point de consigne d'accélération au dispositif CIP Motion par l'intermédiaire de la connexion de données cycliques. Le dispositif d'entraînement convertit le point de consigne d'accélération en consigne de couple à l'aide de l'inertie estimée du système. La commande d'accélération fonctionne de concert avec la boucle d'asservissement de couple/courant interne comme représenté à la Figure 6.



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Figure 6 – Commande d'accélération

La présence d'un dispositif de réaction de commande d'accélération est obligatoire. Il peut être utilisé pour renvoyer les données de position, de vitesse et d'accélération réelles au contrôleur par l'intermédiaire de la connexion de données cycliques.

4.2.7 Asservissement de couple

En mode d'application d'asservissement de couple, le programme de commande d'application et le planificateur de mouvement fournissent des valeurs de point de consigne de couple au dispositif par l'intermédiaire de la connexion de données cycliques (voir Figure 7). Le courant et le couple du moteur étant généralement liés par une constante de couple (K_t), l'asservissement de couple est souvent synonyme d'asservissement de courant.

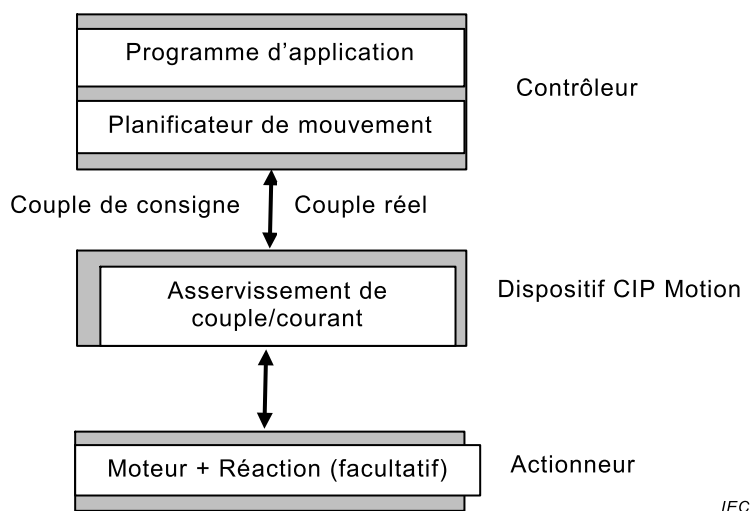


Figure 7 – Asservissement de couple

La présence d'un dispositif de retour en position est facultative pour ce mode d'asservissement. Si un dispositif de réaction est présent, il peut être utilisé pour renvoyer les données de position, de vitesse et d'accélération réelles au contrôleur par l'intermédiaire de la connexion de données cycliques.

4.2.8 Pas d'asservissement

L'Objet Axe de dispositif de mouvement prend en charge un mode d'application "Pas d'asservissement" en l'absence de fonction de contrôle dynamique du moteur. Ce mode est souvent utilisé pour prendre en charge la fonctionnalité "réaction uniquement" ou "réaction maître" lorsqu'une voie particulière de réaction dans un dispositif d'entraînement CIP Motion fait office de source de réaction maître vers le reste du système de commande. Ceci peut également être appliqué aux types de dispositifs de Codeur CIP Motion intégrés où l'interface CIP Motion est directement appliquée à un Codeur.

Dans ce mode de fonctionnement "Pas d'asservissement", aucune valeur de point de consigne n'est fournie au dispositif CIP Motion par l'intermédiaire de la connexion de données cycliques, mais les données de position, de vitesse et d'accélération réelles peuvent être fournies par le dispositif au contrôleur via le canal de données cycliques le cas échéant. Le mode "Pas d'asservissement" pour la fonctionnalité "réaction uniquement" est représenté à la Figure 8.

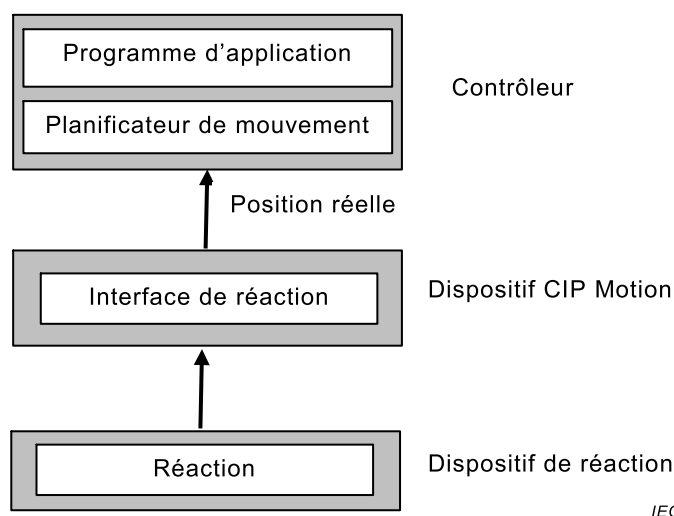


Figure 8 – Pas d'asservissement (Réaction uniquement)

Le mode "Pas d'asservissement" s'applique également à d'autres types de dispositifs CIP Motion, tels que les convertisseurs de puissance de bus autonomes et les types de dispositifs E/S de mouvement dédiés. Dans la mesure où il n'y a aucun canal de réaction associé à ces types de dispositifs, aucune position réelle n'est renvoyée au contrôleur.

5 Types de données

5.1 Type de données – Présentation

Le Tableau 1 présente les références de types de données utilisées dans ce profil et leurs définitions connexes.

Tableau 1 – Types de données

Types de données utilisés dans CIP Motion	Référence à la définition
BOOL	Booléen (voir 5.3.1.1.2 de l'IEC 61158-5-2:2014)
SINT	Integer8 (voir 5.3.1.4.2.2 de l'IEC 61158-5-2:2014)
INT	Integer16 (voir 5.3.1.4.2.4 de l'IEC 61158-5-2:2014)
DINT	Integer32 (voir 5.3.1.4.2.6 de l'IEC 61158-5-2:2014)
LINT	Integer64 (voir 5.3.1.4.2.8 de l'IEC 61158-5-2:2014)
USINT	Unsigned8 (voir 5.3.1.4.3.2 de l'IEC 61158-5-2:2014)
UINT	Unsigned16 (voir 5.3.1.4.3.4 de l'IEC 61158-5-2:2014)
UDINT	Unsigned32 (voir 5.3.1.4.3.6 de l'IEC 61158-5-2:2014)
ULINT	Unsigned64 (voir 5.3.1.4.3.8 de l'IEC 61158-5-2:2014)
REAL	Float32 (voir 5.3.1.4.1.2 de l'IEC 61158-5-2:2014)
LREAL	Float64 (voir 5.3.1.4.1.4 de l'IEC 61158-5-2:2014)
SWORD/BYTE	Bitstring8 (voir 5.3.1.2.2 de l'IEC 61158-5-2:2014)
WORD	Bitstring16 (voir 5.3.1.2.4 de l'IEC 61158-5-2:2014)
DWORD	Bitstring32 (voir 5.3.1.2.6 de l'IEC 61158-5-2:2014)
LWORD	Bitstring64 (voir 5.3.1.2.8 de l'IEC 61158-5-2:2014)

5.2 Conventions

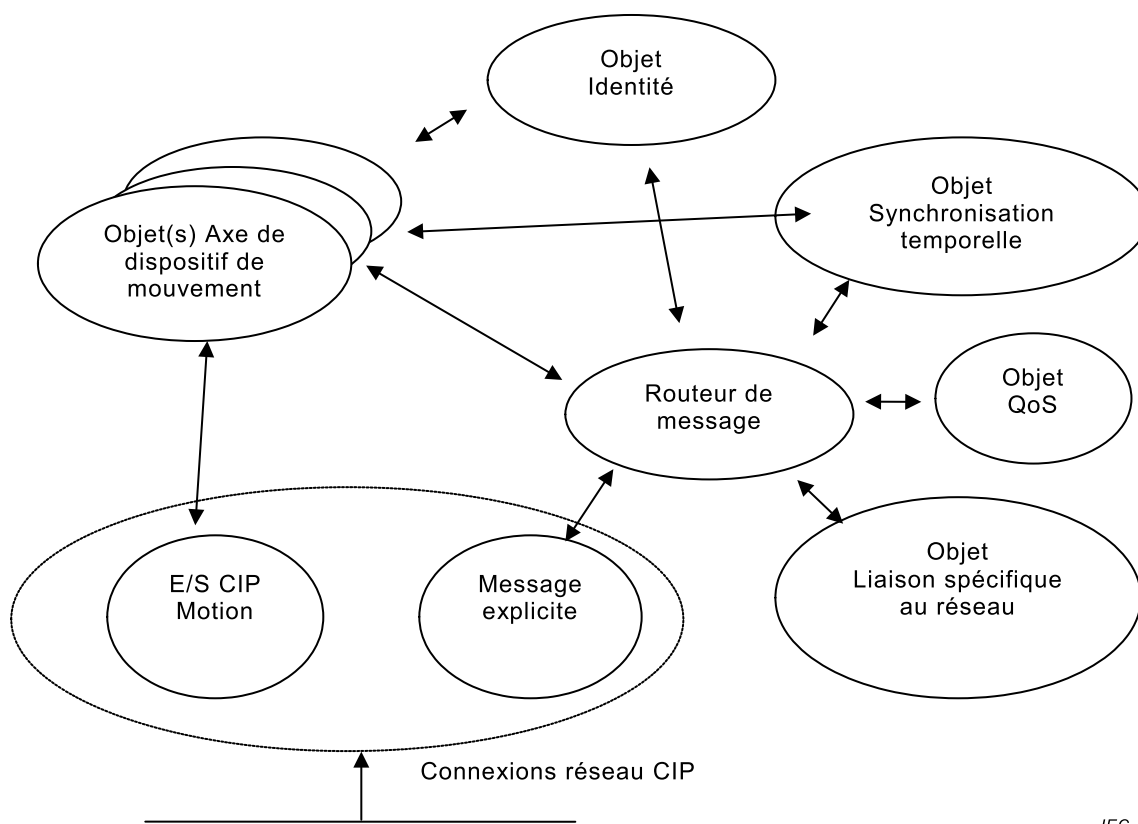
Les nombres hexadécimaux de la présente partie de la série IEC 61800-7 peuvent être représentés par nn_{hex} (si leur type de données exact n'est pas défini) ou $0xnn$, $0xn\text{nnnn}$ (si leur type de données est spécifié).

6 Profil d'entraînement CIP Motion

6.1 Modèle d'objet

6.1.1 Présentation de l'objet

Le modèle d'objet de la Figure 9 représente un dispositif CIP Motion.



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Figure 9 – Modèle d'objet d'un dispositif CIP Motion

Le Tableau 2 indique:

- les classes d'objets présentes dans ce dispositif,
- le caractère obligatoire ou non de la classe,
- le nombre d'instances présentes dans chaque classe.

Tableau 2 – Objets présents dans un dispositif CIP Motion

Classe d'objets	Facultatif ou obligatoire	Numéro d'instance
Objets exigés communs CIP	Obligatoire	Voir IEC 61158-5-2 et IEC 61158-6-2
Objet Axe de dispositif de mouvement (0x42)	Obligatoire	1 par axe ^b
Objet Synchronisation temporelle (0x43)	Facultatif	1
Objet QoS (0x48)	Conditionnel ^a	1
^a Obligatoire pour EtherNet/IP		
^b Un axe est une abstraction associée à un composant mobile de machine, une structure de puissance de convertisseur ou un dispositif E/S de mouvement.		

Se reporter à l'IEC 61158-5-2, l'IEC 61158-6-2 et 6.4.6.1 pour plus de détails sur ces objets.

6.1.2 Description de l'objet

Le modèle d'objet de la Figure 9 représente les principaux composants fonctionnels du profil de dispositif CIP Motion.

Ce modèle d'objet représente également l'utilisation de plusieurs instances d'un Objet Axe de dispositif de mouvement pour mettre en œuvre un dispositif de mouvement multiaxe, tel qu'un

entraînement multiaxe. Chaque instance de l'Objet Axe de dispositif de mouvement gère le comportement d'axe associé. Dans ce profil de dispositif, le terme "axe" est synonyme d'une "instance Objet Axe de dispositif de mouvement". Le contenu mis en œuvre des instances Objet Axe de dispositif de mouvement est régi par le type de dispositif CIP Motion spécifique selon le Tableau 3.

Tableau 3 – Contenu de l'Objet Axe de dispositif de mouvement par Type de dispositif

Type de dispositif	Contenu de l'Objet Axe de dispositif de mouvement
Entraînement CIP Motion	Prise en charge d'un ou plusieurs codes de fonctionnement de dispositif F, P, V, T
Codeur CIP Motion	Prise en charge de E, mais pas des codes de fonctionnement de dispositif F, P, V, T
Convertisseur CIP Motion	Prise en charge du code de fonctionnement de dispositif B uniquement
E/S CIP Motion	Prise en charge du code de fonctionnement de dispositif E/S uniquement

Une seule connexion E/S bidirectionnelle à l'instance de la classe Objet Axe de dispositif de mouvement offre un chemin d'accès aux données cycliques entre le contrôleur et chaque instance Objet Axe de dispositif de mouvement. Cette connexion transmet une structure de données particulière dont le format à définition automatique peut être utilisé pour transférer des données cycliques, d'événement et de service.

Un Objet Synchronisation temporelle facultatif est inclus dans le modèle d'objet. Il s'agit de faciliter la synchronisation temporelle exacte entre les contrôleurs CIP Motion et les dispositifs d'entraînement pour une commande de mouvement haute performance. Pour les dispositifs d'entraînement présentant des performances moins élevées (les dispositifs d'entraînement V/Hz (VFD), ou les entraînements asservis de vitesse simples par exemple), l'Objet Synchronisation temporelle n'est pas nécessaire pour le fonctionnement du dispositif d'entraînement.

Une Connexion E/S CIP Motion est une connexion E/S CIP bidirectionnelle cyclique standard dont la taille des paquets peut varier, en fonction des formats présentés en 6.4.2.2.

6.2 Influence des objets sur le comportement

Les objets de ce dispositif influent sur son comportement comme indiqué dans le Tableau 4.

Tableau 4 – Influence de l'objet sur le comportement

Classe d'objets	Influence sur le comportement
Objets exigés communs CIP	Voir IEC 61158-5-2 et IEC 61158-6-2
Objet Axe de dispositif de mouvement	Offre une interface de commande dynamique aux composants d'entraînement, de moteur et de réaction comprenant un axe.
Objet Synchronisation temporelle	Offre des services de synchronisation temporelle absolue entre les dispositifs sur le réseau de commande.
Objet QoS	Offre des informations de configuration associées à la fonction QoS (Qualité de Service) Ethernet.

6.3 Définition des interfaces d'objet

Les objets pris en charge pour le dispositif CIP Motion comportent les interfaces figurant au Tableau 5.

Tableau 5 – Interfaces d'objet

Objet	Interface
Objets exigés communs CIP	Voir IEC 61158-5-2 et IEC 61158-6-2
Objet Axe de dispositif de mouvement	Routeur de message, Connexion E/S CIP Motion, Objet Synchronisation temporelle, Objet Identité
Objet Synchronisation temporelle	Routeur de message
Objet QoS	Routeur de message

6.4 Messages de connexion E/S

6.4.1 Généralités

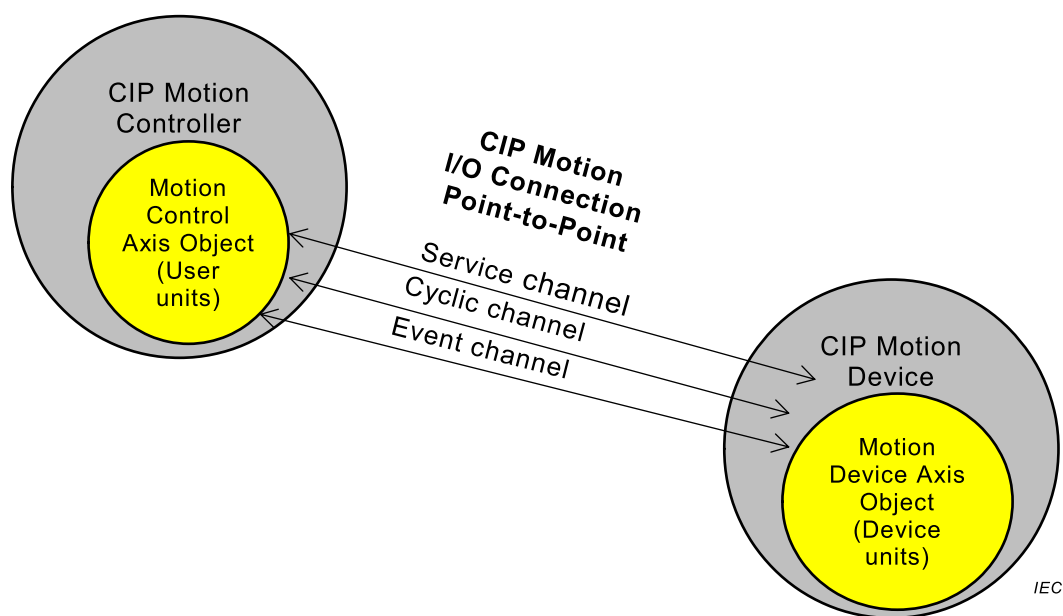
Le profil de dispositif CIP Motion prend en charge une connexion E/S bidirectionnelle point à point de classe de transport 1 entre le contrôleur et la classe Objet Axe de dispositif de mouvement: cette connexion E/S est spécifiquement appelée Connexion E/S CIP Motion.

L'Objet Axe de dispositif de mouvement distribue les données de cette connexion à chaque instance Objet Axe de dispositif de mouvement instanciée.

6.4.2 Connexion E/S CIP Motion

6.4.2.1 Présentation générale

Le paragraphe ci-dessous spécifie le format de Connexion E/S CIP Motion comprenant la connexion C-D et la connexion D-C dans le cadre d'un transfert bidirectionnel de données entre un contrôleur CIP Motion et un dispositif CIP Motion, comme représenté à la Figure 10.



Anglais	Français
CIP Motion Controller	Contrôleur CIP Motion
Motion Control Axis Object (User units)	Objet Axe de commande de mouvement (Unités utilisateur)
CIP Motion Device	Dispositif CIP Motion
Motion Device Axis Object (Device Units)	Objet Axe de dispositif e mouvement (Unités du dispositif)
CIP Motion I/O Connection	Connexion E-S CIP Motion

Anglais	Français
Point-to-Point	Point à point
Service channel	Canal de service
Cyclic channel	Canal cyclique
Event channel	Canal d'événements

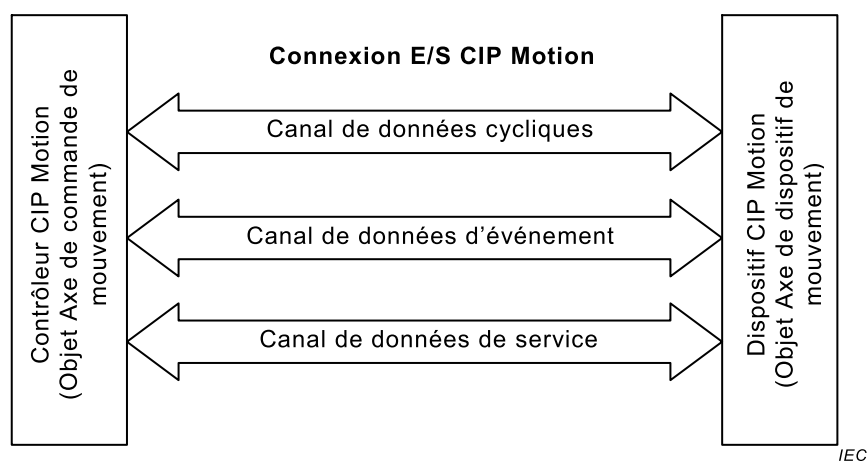
Figure 10 – Modèle de Connexion E/S CIP Motion

6.4.2.2 Structure de Connexion E/S CIP Motion

Les deux structures de données de Connexion E/S CIP Motion (C-D et D-C) commencent par un en-tête de connexion suivi d'un bloc de données temporelles qui comporte généralement une datation à 64 bits. Le bloc de données temporelles est ensuite suivi par un ou plusieurs blocs de données d'instance pour chaque axe pris en charge par le nœud de dispositif.

La taille et le contenu du bloc de données temporelles et des blocs de données d'instance varient de façon dynamique comme déterminé par les en-têtes associés. Cette capacité de variation du contenu des blocs de données d'une mise à jour à l'autre permet au dispositif et au contrôleur d'envoyer uniquement les données modifiées depuis la dernière mise à jour, en réduisant considérablement la taille globale du paquet de Connexion CIP Motion type.

Chaque Bloc de données d'instance du paquet de Connexions E/S CIP Motion est composé de trois ensembles de blocs de données associés aux canaux de données cycliques, d'événement et de service. Du point de vue du dispositif, ces trois canaux de données distincts font l'objet de différentes priorités de traitement de données comme représenté à la Figure 11.

**Figure 11 – Canaux de Connexion E/S CIP Motion**

Les fonctions spécifiques de ces trois canaux de données sont les suivantes:

- Canal de données cycliques – transporte des Blocs de données cycliques échantillonnés ou calculés à chaque Période de mise à jour du contrôleur, et synchronisés avec d'autres nœuds dans le système de commande de mouvement grâce au Temps système réparti. Les données cycliques sont des données à priorité élevée qui doivent être immédiatement traitées et appliquées à l'axe de dispositif dans une Période de mise à jour du dispositif.
- Canal de données d'événement – transporte les données d'événement associées à l' (aux) événement(s) de dispositif (enregistrement, retour à la position de référence, etc.) qui se sont produits dans la dernière Période de mise à jour du contrôleur. Les données d'événement sont des données de priorité moyenne et doivent être traitées et appliquées dans la Période de mise à jour du contrôleur.

- Canal de données de service – transporte les données associées aux demandes de service de lecture et d'écriture des valeurs d'attribut d'Objet Axe de dispositif de mouvement dans le cadre d'une fonctionnalité de configuration et de diagnostic en ligne, ainsi que les demandes de service affectant le comportement de l'Objet Axe de dispositif de mouvement dans le cadre de l'exécution d'une instruction du contrôleur. Les données de service sont des données de priorité basse et sont en général mises en mémoire tampon et traitées en arrière-plan. Rien ne peut garantir qu'une demande de service sera traitée dans la Période de mise à jour du contrôleur.

Considérés ensemble, ces trois canaux de données fournissent un contrôleur exhaustif à la solution de connexion des données de dispositif pour la commande de mouvement industrielle.

6.4.2.3 Formats de connexion E/S

Une présentation générale du format de Connexion E/S CIP Motion est représentée à la Figure 12 et à la Figure 13.

L'encapsulage associé au transport de classe 1 qui inclut un compte de séquences n'est pas représenté dans ces figures de format. Pour une description détaillée de l'en-tête de transport de classe 1, se reporter à l'IEC 61158-5-2 et à l'IEC 61158-6-2. Les données à octets multiples dans la structure de données de connexion E/S CIP suivent la règle d'adressage d'octets petit-boutiste standard. Dans les figures, l'ordonnancement des octets se lit de gauche à droite. Les bandeaux grisés qui apparaissent dans les figures sont des libellés de section et ne font pas partie de la structure de données de connexion réelle. Sauf indication contraire, les éléments de structure de données définis comme "réservés" ou marqués avec un tiret, "-", doivent être mis à zéro (0).

← Mot à 32 bits (octet 1 octet 2 octet 3 octet 4) →			
Format de connexion Contrôleur-Dispositif			
En-tête de connexion			
Bloc de données temporelles			
Blocs de données d'instance			
En-tête de connexion et Bloc de données temporelles			
Format de connexion	Révision de format	ID de mise à jour	Contrôle de nœud
Compte d'instances	–	Dernier ID reçu	Ensemble de données temporelles
Datation du contrôleur			
Décalage temporel du contrôleur			
Bloc de données d'instance			
En-tête de données d'instance			
Bloc de données cycliques			
Bloc de données d'écriture cycliques			
Bloc de données d'événement			
Bloc de données de service			
En-tête de données d'instance			
Numéro d'instance	–	Taille de bloc d'instance	Taille de bloc cyclique
Taille de bloc de consigne cyclique	Taille de bloc d'écriture cyclique	Taille de bloc d'événement	Taille de bloc de service
Bloc de données cycliques			
Mode de commande	Mode de réaction	Commande d'axe	État de commande
Ensemble de données de consigne	Ensemble de données réelles	Ensemble de données d'état	Commande d'interpolation
Données cycliques			
Bloc de données d'écriture cycliques			
ID de bloc d'écriture cyclique		ID de bloc de lecture cyclique	
Données d'écriture cycliques			
Bloc de données d'événement			
Commande de vérification d'événement			
Ensemble de données d'enregistrement	Ensemble de données d'origine	Ensemble de données d'observation	–
Données d'événement d'enregistrement			
Données d'événement d'origine			
Données d'événement d'observation			
Acquittement d'événement ID1	Acquittement d'événement État 1	Acquittement d'événement ID2	Acquittement d'événement État 2
...			
Bloc de données de service			
ID de transaction	Code de service	–	–
Données de demande spécifique au service			

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Figure 12 – Format de connexion Contrôleur-Dispositif (Point de connexion 2)

← Mot à 32 bits (octet 1 octet 2 octet 3 octet 4) →			
Format de connexion Dispositif-Contrôleur			
En-tête de connexion			
Bloc de données temporelles			
Blocs de données d'instance			
En-tête de connexion et Bloc de données temporelles			
Format de connexion	Révision de format	ID de mise à jour	État de nœud
Compte d'instances	Défaut/Alarme de nœud	Dernier ID reçu	Ensemble de données temporelles
Datation du dispositif			
Décalage temporel du dispositif			
Perte de mises à jour	Mises à jour retardées	—	
Données de diagnostic de synchronisation			
Bloc de données d'instance			
En-tête de données d'instance			
Bloc de données cycliques			
Bloc de données de lecture cycliques			
Bloc de données d'événement			
Bloc de données de service			
En-tête de données d'instance			
Numéro d'instance	-	Taille de bloc d'instance	Taille de bloc cyclique
Taille de bloc cyclique réelle	Taille de bloc de lecture cyclique	Taille de bloc d'événement	Taille de bloc de service
Bloc de données cycliques			
Mode de commande	Mode de réaction	Réponse d'axe	État de réponse
-	Ensemble de données réelles	Ensemble de données d'état	État d'axe
Données cycliques			
Bloc de données de lecture cycliques			
ID de bloc d'écriture cyclique	État d'écriture cyclique	ID de bloc de lecture cyclique	État de lecture cyclique
Données de lecture cycliques			
Bloc de données d'événement			
État de vérification d'événement			
Acquittement de données d'enregistrement	Acquittement de données d'origine	Acquittement de données d'observation	—
ID d'événement	État d'événement	Type d'événement	—
Position d'événement			
Datation de l'événement			
...			
Bloc de données de service			
ID de transaction	Code de service	État général	État étendu
Données de réponse spécifique au service			

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Figure 13 – Format de connexion Dispositif-Contrôleur (Point de connexion 2)

6.4.3 Connexion Contrôleur-Dispositif

6.4.3.1 Généralités

Pour faciliter la description détaillée de chacun de ses éléments de données constitutants, la connexion Contrôleur-Dispositif CIP Motion est organisée comme indiqué à la Figure 14.

Format de connexion Contrôleur-Dispositif CIP Motion
En-tête de connexion
Bloc de données temporelles
Blocs de données d'instance

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Figure 14 – Format de connexion Contrôleur-Dispositif CIP Motion

6.4.3.2 En-tête de connexion Contrôleur-Dispositif

6.4.3.2.1 Structure générale

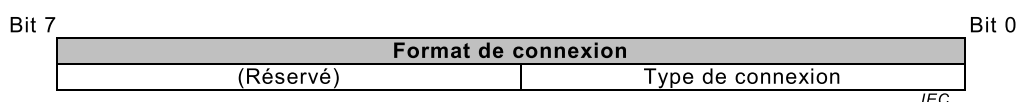
L'en-tête de connexion Contrôleur-Dispositif contient les informations essentielles de configuration d'axe nécessaires à l'analyse des blocs de données d'instance. La partie fixe de l'en-tête de connexion est définie comme indiqué à la Figure 15.

En-tête de connexion			
Format de connexion	Révision de format	ID de mise à jour	Contrôle de nœud

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Figure 15 – En-tête de connexion

- Format de connexion: cet octet détermine le format de la Connexion CIP Motion conformément à la définition de la Figure 16.



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Figure 16 – Format de connexion

- Type de connexion: cette énumération à valeur de 4 bits définit le type de Connexion CIP Motion comme indiqué ci-dessous. Les valeurs valides d'une connexion Contrôleur-Dispositif sont 2 pour le format fixe et 6 pour le format variable. Les connexions fixes présentent une taille de connexion fixe pendant le fonctionnement, afin de prendre en charge les réseaux CIP basse performance et sont généralement associées à un seul dispositif d'axe simple ne prenant pas en charge les services de synchronisation temporelle. Les connexions variables permettent à la taille de la structure de données de connexion de varier pendant le fonctionnement et sont destinées aux dispositifs haute performance et aux réseaux CIP tels qu'EtherNet/IP.

0 = Connexion pair-à-pair fixe entre Contrôleurs

1 = Connexion pair-à-pair fixe entre Dispositifs

2 = Connexion fixe Contrôleur-vers-Dispositif

3 = Connexion fixe Dispositif-vers-Contrôleur

4 = Connexion pair-à-pair variable entre Contrôleurs

5 = Connexion pair-à-pair variable entre Dispositifs

6 = Connexion variable Contrôleur-vers-Dispositif

7 = Connexion variable Dispositif-vers-Contrôleur

8 à 15 = Réservé.

Le format de connexion est utilisé non seulement par le dispositif pour analyser correctement les données de connexion contrôleur-dispositif, mais également par les outils de diagnostic de réseau afin d'afficher les éléments individuels de la structure de données. Lorsque le dispositif reçoit une valeur de format de connexion qu'il ne peut pas prendre en charge, il génère un défaut. Dans ce cas, le défaut doit être le code d'erreur de défaut de nœud 4: Erreur de format de données. La valeur de format de connexion ne doit

pas varier pendant toute la durée de vie de la connexion, et sa vérification par le dispositif n'est par conséquent nécessaire que pour le premier paquet reçu.

- Révision de format: La présente édition de l'IEC 61800-7-202 définit la révision de format 2. Les dispositifs utilisant la présente édition de l'IEC 61800-7-202 doivent reconnaître uniquement la révision de format 2; la révision de format 1 est par conséquent rendue obsolète. Cette valeur est incrémentée de 1 à chaque révision du format de connexion Contrôleur-Dispositif ayant un impact sur l'interface. La révision de format permet aux dispositifs les plus récents de prendre en charge les formats de connexion générés par un contrôleur plus ancien. Elle permet également aux dispositifs plus anciens de reconnaître un format de connexion plus récent à partir d'un contrôleur qu'il ne peut pas prendre en charge, et de générer un défaut. Dans ce cas, le défaut doit être le code d'erreur de défaut de nœud 4: Erreur de format de données. Les outils de diagnostic de réseau peuvent également définir la révision de format afin de savoir comment analyser le paquet de données de connexion. Lorsque la connexion E/S est établie, le numéro de révision de format est également indiqué par le point de connexion E/S dans le chemin d'application du service Forward_Open (voir 6.4). La valeur de révision de format ne doit pas varier pendant toute la durée de vie de la connexion, et sa vérification par le dispositif n'est par conséquent nécessaire que pour le premier paquet reçu.
- ID de mise à jour: Ce numéro de transaction cyclique est incrémenté à chaque période de mise à jour. L'ID de mise à jour s'apparente au nombre de séquences CIP et permet au dispositif de déterminer si la mémoire tampon de connexion contient de nouvelles données. Si le dispositif n'est pas synchronisé ou ne prend pas en charge les services de synchronisation temporelle, les données de datation ne sont pas incluses ou ne sont pas valides. L'ID de mise à jour est donc le seul moyen dont dispose le dispositif pour détecter les nouvelles données de connexion. Dans ce cas, l'ID de mise à jour de connexion Contrôleur-Dispositif est appliqué à l'ID de mise à jour de la connexion Dispositif-Contrôleur suivant.

L'ID de mise à jour permet également au dispositif de déterminer si une mise à jour a été perdue. Tout saut dans l'ID de mise à jour constitue une indication de perte d'un paquet de mises à jour lors d'une transmission. Lorsque le nombre de mises à jour manquées consécutives dépasse la valeur donnée par l'attribut de classe, à savoir Limite supérieure du délai de mise à jour du contrôleur, le défaut de mise à jour de la connexion de commande est indiqué par l'attribut de classe de défauts de nœud de l'Objet Axe de dispositif de mouvement. Ce défaut peut être supprimé par les seules fermeture et ouverture de la Connexion E/S CIP Motion. Un attribut facultatif Limite inférieure du délai de mise à jour du contrôleur est également disponible. Lorsque le nombre de mises à jour manquées consécutives dépasse la valeur maximale donnée par l'attribut de classe, à savoir Limite inférieure du délai de mise à jour du contrôleur, l'alarme de mise à jour de la connexion du contrôleur est indiquée par l'attribut de classe d'alarme de nœud de l'Objet Axe de dispositif de mouvement. L'alarme est interrompue dès la réception d'une nouvelle mise à jour.

- Contrôle de nœud: Cet élément est appliqué à l'attribut de classe d'Objet Axe de dispositif de mouvement "Contrôle de nœud", utilisé pour contrôler l'état du nœud de communication de dispositif associé. Voir 7.2.2.1 (Objet Axe de dispositif de mouvement) pour plus de détails.

6.4.3.2.2 En-tête de connexion fixe

Si le format de connexion est une connexion Contrôleur-Dispositif fixe, l'en-tête ci-dessus est immédiatement suivi du Bloc de données d'instance.

6.4.3.2.3 En-tête de connexion variable

Si le format de connexion est une connexion Contrôleur-Dispositif variable, l'en-tête de connexion contient alors des champs supplémentaires relatifs à l'adressage et à la datation de dispositif multiaxe comme représenté à la Figure 17.

En-tête de connexion			
Format de connexion	Révision de format	ID de mise à jour	Contrôle de nœud
Compte d'instances	–	Dernier ID reçu	Ensemble de données temporelles
Datation du contrôleur			
Décalage temporel du contrôleur			

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Figure 17 – En-tête de connexion

- **Compte d'instances:** cette valeur reflète le nombre de Blocs de données d'instance présents dans la structure de données de connexion Contrôleur-Dispositif CIP Motion.
- **Dernier ID reçu:** ID de mise à jour du dernier bloc de données de connexion Dispositif-Contrôleur effectivement traité par le contrôleur. Le dernier ID reçu est utilisé par le dispositif pour déterminer si les données envoyées ont été traitées avec succès. Par exemple, les codes de défaut transmis comme partie intégrante du bloc de données d'état des connexions Dispositif-Contrôleur sont répartis en séquences dans le journal des défauts du contrôleur. Il est recommandé que le dispositif lise le dernier élément ID reçu afin de déterminer si le dernier code de défaut a été saisi dans le journal des défauts, permettant ainsi au dispositif d'envoyer le prochain code de défaut dans la séquence.
- **Ensemble de données temporelles:** cet octet de matriciel contient des indicateurs qui déterminent l'utilisation et le format des informations de synchronisation du contrôleur et du dispositif comme spécifié dans le Tableau 6. En général, la valeur de l'ensemble de données temporelles envoyée par le contrôleur dans le cadre d'une connexion Contrôleur-Dispositif détermine non seulement le contenu du bloc de données de connexion Contrôleur-Dispositif, mais représente également l'ensemble de données temporelles demandé à envoyer par le dispositif dans le cadre de la connexion Dispositif-Contrôleur. La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement "Ensemble de données temporelles".

Tableau 6 – Ensemble de données temporelles

Bit	Définition	Syntaxe
0	Datation (horodatage)	0 = pas de datation 1 = datation incluse
1	Décalage temporel	0 = pas de décalage temporel – utiliser le dernier décalage 1 = décalage temporel inclus
2	Diagnostic de mise à jour	0 = pas de données de diagnostic de mise à jour dans le cadre de la connexion D-C 1 = données de diagnostic de mise à jour dans le cadre de la connexion D-C
3	Diagnostic temporel	0 = pas de données de diagnostic temporel dans le cadre de la connexion D-C 1 = données de diagnostic temporel dans le cadre de la connexion D-C
4 à 7	(Réservé)	

- **Datation** – Lorsque ce bit est défini, la datation du contrôleur est incluse dans le bloc de l'ensemble de données temporelles. La définition de ce bit indique également au dispositif de renvoyer les datations de dispositif au contrôleur dans le cadre de la connexion Dispositif-Contrôleur dans la mise à jour suivante. Elle indique également au dispositif d'envoyer le décalage temporel de dispositif si sa valeur a changé depuis la dernière mise à jour. Si le dispositif ne prend pas en charge la synchronisation temporelle, il renvoie simplement une valeur nulle pour la datation, mais ne renvoie jamais un décalage temporel.
- **Décalage temporel** – Lorsque ce bit est défini, le décalage temporel du contrôleur est inclus dans le bloc de l'ensemble de données temporelles. Ce bit est défini uniquement en cas de changement de la valeur de décalage temporel du contrôleur, ce qui se produit relativement peu souvent. Noter que ce bit NE détermine PAS si le dispositif envoie son décalage temporel de dispositif dans la mise à jour suivante. Ceci est déterminé par le bit de datation et par le changement effectif ou non du décalage temporel de dispositif depuis la dernière mise à jour.

- Diagnostic de mise à jour – Bien qu'aucune donnée ne soit associée à ce bit dans l'en-tête de connexion Contrôleur-Dispositif, le bit de diagnostic de mise à jour dans le cadre de ladite Connexion indique au dispositif d'envoyer des données de diagnostic de mise à jour dans la mise à jour suivante de la connexion Dispositif-Contrôleur, mais uniquement lorsqu'une ou plusieurs valeurs de diagnostic de mise à jour ont changé depuis la dernière mise à jour. Ces données de diagnostic de mise à jour comportent le nombre actuel de mises à jour perdues ou retardées à partir du contrôleur dans le cadre de la connexion Contrôleur-Dispositif. Le diagnostic de mise à jour s'applique aux dispositifs indépendamment du fait qu'ils prennent ou non en charge la synchronisation temporelle.
- Diagnostic temporel – Bien qu'aucune donnée ne soit associée à ce bit dans l'en-tête de connexion Contrôleur-Dispositif, le bit de diagnostic temporel dans le cadre de ladite Connexion indique au dispositif d'envoyer des données de diagnostic temporel dans la mise à jour suivante de la connexion Dispositif-Contrôleur. Ces données de diagnostic temporel comportent la datation reçue des données Contrôleur-Dispositif, ainsi que la datation de transmission Dispositif-Contrôleur. Si le dispositif ne prend pas en charge la synchronisation temporelle, il renvoie simplement une valeur nulle pour les éléments de diagnostic temporel.
- Décalage temporel du contrôleur: Cet élément est inclus dans l'en-tête de connexion si le bit de décalage temporel est défini dans l'élément de l'ensemble de données temporelles. Le décalage temporel du contrôleur représente la valeur de décalage du temps système de 64 bits associée à la datation du contrôleur qui suit. La valeur de cet élément est transférée vers l'attribut de classe d'Objet Axe de dispositif de mouvement du même nom. La valeur Décalage temporel du contrôleur permet au dispositif de déterminer si le temps système défini dans le contrôleur est étalé par rapport à celui du dispositif. En principe, le temps système entre le dispositif et le contrôleur sont proches, quel que soit le temps considéré. Mais, conformément au protocole de synchronisation CIP de l'IEC 61588:2009, l'horloge maîtresse du système peut corriger toutes les secondes la référence du temps système d'une durée significative (une heure, par exemple). En effet, le maître lui-même peut changer si une horloge maîtresse de qualité supérieure est détectée par le système. Ces modifications brusques apportées à la référence du temps système se propagent vers les différents dispositifs présents sur le réseau, de sorte que le contrôleur et le dispositif fonctionnent avec des valeurs de temps système étalées. La valeur Décalage temporel du contrôleur peut être utilisée conjointement avec la valeur Décalage du temps système du dispositif, afin de déterminer si le temps système entre le dispositif et le contrôleur est étalé et, si c'est le cas, pour compenser cet étalement en utilisant l'algorithme de compensation de décalage du temps système.

La valeur Décalage temporel du contrôleur change relativement peu souvent, c'est-à-dire suite à une mise à jour de synchronisation entraînée définie dans l'IEC 61588, qui se produit généralement chaque seconde. Ainsi, il est recommandé que le contrôleur charge et envoie les données de décalage temporel uniquement en cas de changement de la valeur. Il est nécessaire que le dispositif analyse uniquement les données de décalage temporel du contrôleur si le bit de décalage temporel associé est défini dans l'élément de l'ensemble de données temporelles. Dans le cas contraire, il continue à utiliser le dernier décalage temporel du contrôleur reçu.

- Datation du contrôleur: Cet élément est inclus dans l'en-tête de connexion si le bit de datation est défini dans l'élément de l'ensemble de données temporelles. La datation du contrôleur représente la valeur Temps système de 64 bits au début de la période de mise à jour du contrôleur lorsqu'un événement de synchronisation de la mise à jour du contrôleur s'est produit. Autrement dit, la datation du contrôleur marque, en temps absolu, le début du cycle de Connexion CIP Motion actuel. La valeur de cette datation est calculée par le contrôleur comme la somme de la valeur d'horloge locale du contrôleur en cas d'événement de synchronisation de mise à jour et de la valeur Décalage du temps système du contrôleur. La datation du contrôleur est donc directement liée aux données de consigne contenues dans la connexion. La valeur de cet élément, lorsqu'elle est lue, est transférée vers la datation du contrôleur de l'attribut de classe d'Objet Axe de dispositif de mouvement. Outre la période de mise à jour du contrôleur, établie par le RPI du service Forward_Open, et la mise à jour de consigne cible (présentée ci-après), le dispositif dispose de toutes les informations dont il a besoin pour calculer les polynômes d'interpolation de consigne, afin de corriger les différences de valeurs des données de

consigne entre la synchronisation de mise à jour du dispositif et du contrôleur. Ces différences peuvent apparaître lorsque la période de mise à jour du contrôleur n'est pas un entier multiple de la période de mise à jour du dispositif ou lorsque les mises à jour du dispositif présentent un déphasage par rapport au contrôleur.

Lors de la synchronisation de la connexion, la datation du contrôleur peut être utilisée par le dispositif avec la période de mise à jour du contrôleur pour vérifier les mises à jour manquées ou retardées. Si la différence entre la dernière datation du contrôleur et la datation en cours du dispositif local dépasse la valeur maximale donnée par,

Délai de défaut max = Limite supérieure du délai de mise à jour du contrôleur × Période de mise à jour du contrôleur,

le défaut de mise à jour de connexion de commande est indiqué par l'attribut de classe Défauts de nœud de l'Objet Axe de dispositif de mouvement. Un attribut facultatif Limite inférieure du délai de mise à jour du contrôleur est également disponible. Si la différence entre la dernière datation du contrôleur et la datation en cours du dispositif local dépasse la valeur maximale donnée par,

Délai d'alarme max = Limite inférieure du délai de mise à jour du contrôleur × Période de mise à jour du contrôleur,

L'Alarme de mise à jour de connexion du contrôleur est indiquée par l'attribut de classe Alarme de nœud de l'Objet Axe de dispositif de mouvement.

Lorsque les valeurs Décalage temporel du contrôleur ou les valeurs Décalage temporel du dispositif ont changé depuis la dernière mise à jour, la compensation de décalage du temps système doit être appliquée à la datation du contrôleur. Voir 6.4.6.7 pour plus de détails. Lorsque les décalages temporels n'ont pas changé, ce qui est généralement le cas, la datation du contrôleur peut être appliquée directement à l'Objet Axe de dispositif de mouvement.

6.4.3.3 Blocs de données d'instance

6.4.3.3.1 Structure générale

L'En-tête de connexion est suivi d'un ou de plusieurs Blocs de données d'instance, comme le montre le Compte d'instances ci-dessus. La structure de base du Bloc de données d'instance est représentée à la Figure 18.

Bloc de données d'instance
En-tête de données d'instance
Bloc de données cycliques
Bloc de données d'événement
Bloc de données de service

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Figure 18 – Bloc de données d'instance

6.4.3.3.2 En-tête de données d'instance

L'En-tête de données d'instance contient les informations essentielles de configuration d'axe nécessaires à l'analyse et à l'application des données contenues dans les trois canaux de données (voir Figure 19). Cet en-tête est uniquement inclus dans le format Connexion variable afin de s'adapter aux applications de dispositif multiaxe. Les informations qu'il contient peuvent être utilisées par l'interface de communication du dispositif pour copier les blocs de données individuels dans des mémoires fixes séparées pour le traitement.

S'il est configuré pour un format Connexion fixe, seul le Bloc de données cycliques d'une seule instance d'axe est pris en charge. Il n'est donc pas utile que les informations précisent le numéro d'instance ou la taille du bloc. L'en-tête de données d'instance n'est donc pas inclus.

En-tête de données d'instance			
Numéro d'instance	-	Taille de bloc d'instance	Taille de bloc cyclique
Taille de bloc de consigne cyclique	Taille de bloc cyclique	Taille de bloc d'événement	Taille de bloc de service

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Figure 19 – En-tête de données d'instance

- **Numéro d'instance:** Il s'agit du numéro d'identification de l'instance Objet Axe de dispositif de mouvement spécifique à laquelle s'applique le Bloc de données d'instance suivant. Les instances Objet Axe de dispositif de mouvement sont créées comme une série contiguë de numéros d'instance commençant par 1. Dans la structure de données de connexion, les numéros d'instance de chaque Bloc de données d'instance consécutif doivent être une séquence ordinale, c'est-à-dire 0, 1, 2, 3, etc.

NOTE 1 L'instance 0 est définie comme étant l'instance de classe et est généralement utilisée uniquement lors de l'initialisation afin de configurer les attributs de classe de l'Objet Axe de dispositif de mouvement via le Bloc de données de service. Dans le cas contraire, la séquence de Blocs de données d'instance commence avec l'instance 1.

Par conséquent, en théorie, la Connexion E/S CIP Motion peut proposer jusqu'à 255 instances d'axe. La séquence des numéros d'instance de la Connexion Contrôleur-Dispositif détermine la séquence des numéros d'instance de la connexion Dispositif-Contrôleur ultérieure.

- **Taille de bloc d'instance:** Cette valeur représente la taille du Bloc de données d'instance, en mots à 32 bits, en-tête compris. La taille de bloc d'instance est utile lorsque le dispositif souhaite accéder directement au Bloc de données d'instance suivant sans qu'il soit nécessaire d'ajouter les tailles de blocs cyclique, d'événement et de service. Lorsque la taille de bloc d'instance pour la connexion Contrôleur-Dispositif est égale à zéro, c'est-à-dire qu'il n'y a pas de bloc de données d'instance, la taille de bloc d'instance pour la connexion Dispositif-Contrôleur suivante doit également être égale à zéro.
- **Taille de bloc cyclique:** Cette valeur représente la taille du Bloc de données cycliques, en unités de mot à 32 bits, en-tête compris.

NOTE 2 Pour l'instance 0, à savoir l'instance de classe, la taille de bloc cyclique est égale à 0, indiquant l'absence de bloc de données cycliques.

- **Taille de bloc de consigne cyclique:** Cette valeur représente la taille du Bloc de données de consigne cycliques, en unités de mot à 32 bits, en-tête compris. Une taille de bloc de consigne cyclique de 0 indique que le dispositif n'a à traiter aucune donnée de consigne cyclique.
- **Taille de bloc d'écriture cyclique:** Cette valeur représente la taille du Bloc de données d'écriture cycliques, en unités de mot à 32 bits, en-tête compris. Une taille de bloc d'écriture cyclique de 0 indique qu'aucune donnée d'écriture cyclique n'est actuellement configurée pour un transfert vers le dispositif et, par conséquent, le dispositif n'a à traiter aucun bloc d'écriture cyclique.
- **Taille de bloc d'événement:** Cette valeur représente la taille du Bloc de données d'événement, en unités de mot à 32 bits, en-tête compris. Lorsque la taille de bloc d'événement pour la connexion Contrôleur-Dispositif est égale à zéro, c'est-à-dire qu'il n'y a pas de vérification d'événement active, la taille de bloc d'événement pour la connexion Dispositif-Contrôleur suivante doit également être égale à zéro. Ceci ferme effectivement le canal d'événement.
- **Taille de bloc de service:** Cette valeur représente la taille du Bloc de données de service, en unités de mot à 32 bits, en-tête compris. Si la valeur Taille de bloc de service est nulle, il n'existe aucune demande de service à traiter. Lorsque la taille de bloc de service pour la connexion Contrôleur-Dispositif est égale à zéro, la taille de bloc de service pour la connexion Dispositif-Contrôleur doit également être égale à zéro. Ceci ferme effectivement le canal de service.

6.4.3.3.3 Bloc de données cycliques

Le Bloc de données cycliques comprend des données de commande à priorité élevée qui sont mises à jour à chaque cycle de connexion sur la base de la Période de mise à jour du contrôleur. L'en-tête de données cycliques, placé en haut du bloc de données cycliques, est toujours inclus, quel que soit le format de connexion. Cet en-tête contient les éléments essentiels relatifs au contenu du bloc de données cycliques de la connexion Contrôleur-Dispositif et de la connexion Dispositif-Contrôleur, et au contexte des données déterminé par le mode de commande et la Configuration de réaction (voir Figure 20). Il fournit également un mécanisme de contrôle de l'état d'axe de dispositif ciblé.

Bloc de données cycliques			
Mode de commande	Mode de réaction	Commande d'axe	État de commande
Ensemble de données de consigne	Ensemble de données réelles	Ensemble de données d'état	Commande d'interpolation
Données cycliques			

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Figure 20 – Bloc de données cycliques

- Mode de commande: Les 4 bits inférieurs de cet élément à 8 bits déterminent le contexte de mode de commande des données de consigne actuellement configurées dans le contrôleur de mouvement, comme représenté à la Figure 21.

Bit 7	Bit 0
Mode de commande	
réservé – 4 bits	Mode de commande – 4 bits

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Figure 21 – Mode de commande

Cette valeur peut être modifiée en ligne, voire lorsque l'axe de dispositif est à l'état Exécution. Si une transition Mode de commande particulière n'est pas prise en charge par le dispositif, une exception est générée, qui peut être configurée pour exécuter l'une des actions en réponse à une transition illégale.

La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement du même nom. La définition de l'attribut Mode de commande complet peut être consultée dans les tables d'attribut d'Objet Axe de dispositif de mouvement (voir 7.3.2.2.1).

- Mode de réaction: Les 4 bits inférieurs de cet élément à 8 bits déterminent le contexte de réaction des données de consigne actuellement configurées dans le contrôleur de mouvement, comme représenté à la Figure 22.

Bit 7	Bit 0
Mode de réaction	
réservé – 4 bits	Mode de réaction – 4 bits

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Figure 22 – Mode de réaction

Les données de position de consigne peuvent faire référence aux nombres de réactions de Réaction 1 ou de Réaction 2, ou aux unités moteur pour les opérations sans capteur. Cette valeur peut être modifiée en ligne, et si elle est prise en charge par le dispositif, voire lorsque l'axe de dispositif est à l'état Exécution. Si une transition Mode de réaction particulière n'est pas prise en charge par le dispositif, une exception est générée, qui peut être configurée pour exécuter l'une des actions en réponse à une transition illégale.

La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement du même nom. La définition de l'attribut Mode de réaction complet peut être consultée dans les tables d'attribut d'Objet Axe de dispositif de mouvement (voir 7.3.6.2).

- **Commande d'axe:** Un code énuméré à 8 bits qui peut être utilisé pour exécuter directement les opérations de changement d'état d'axe n'impliquant pas de transmettre ou de renvoyer les paramètres du dispositif, et donc, de lancer un service CIP. Ce mécanisme est décrit entièrement dans le paragraphe Commande d'état de l'Objet Axe de dispositif de mouvement (voir 7.6.1.2). Les énumérations valides de cet élément de données sont présentées dans le Tableau 7.

Tableau 7 – Commande d'axe

Code de demande	Opération demandée
0	Pas de demande
1	Demande d'activation
2	Demande de désactivation
3	Demande d'interruption
4	Demande de réinitialisation suite à une interruption
5	Demande d'abandon
6	Demande de réinitialisation suite à un défaut
7	Demande d'arrêt de processus
8	Demande de changement de référence de position réelle
9	Demande de changement de référence de position de consigne
10 à 126	(Réservé)
127	Demande d'annulation
128 à 255	(Spécifique au fournisseur)

- **État de commande:** Valeur de matriciel qui peut être interrogée par le dispositif afin de déterminer l'état de la commande au cours du processus d'initialisation. La valeur de cet élément est transférée au numéro d'attribut 89 de l'Objet Axe de dispositif de mouvement facultatif, État de commande, s'il est pris en charge par le dispositif. Les bits valides de cet élément de données sont présentés dans le Tableau 8.

Tableau 8 – État de commande

Bit	Bit d'État de commande	Description
0	Configuration achevée	Ce bit est défini lorsque le contrôleur a achevé la configuration de tous les attributs d'instance d'axe au cours de la phase d'initialisation.
1	(Réservé)	Voir Attribut 89 de l'Objet Axe de dispositif de mouvement
2	Bus de convertisseur activé	
3	Bus de convertisseur sans charge	
4	Perte de puissance en courant alternatif du convertisseur	
5 à 7	(Réservé)	

- **Ensemble de données de consigne:** Cette valeur de matriciel comporte un bit défini pour chaque référence de consigne en temps réel potentielle (voir Tableau 9). Les données de consigne apparaissent dans le même ordre dans l'ensemble de données de consigne que les numéros de bit. Ainsi, l'accélération de consigne apparaît avant le Couple de consigne dans la structure de données en temps réel de la connexion Contrôleur-Dispositif.

La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement "Ensemble de données de consigne".

Tableau 9 – Ensemble de données de consigne

Bit	Élément de données de consigne du contrôleur	Type de données
0	Position de consigne	LREAL ou DINT ^a
1	Vitesse de consigne	REAL
2	Accélération de consigne	REAL
3	Couple de consigne	REAL
4	(Réservé)	(Réservé)
5	(Réservé)	(Réservé)
6	Compte des cycles de déroulement	DINT
7	Déplacement de position	DINT
^a Type de données pour la position de consigne déterminé par l'élément de commande de consigne défini ci-dessous. Le type de données de position de consigne peut être LREAL ou DINT tel que déterminé par l'élément de commande de consigne.		

Les Éléments de données de consigne du contrôleur ci-dessus s'appliquent à la structure de données cycliques de la connexion Contrôleur-Dispositif et sont mis en correspondance avec les attributs correspondants de l'Objet Axe de dispositif de mouvement comme représenté au Tableau 10.

Les unités des Éléments de données de consigne correspondent à celles définies pour l'attribut d'Objet Axe de dispositif de mouvement associé.

Tableau 10 – Mise en correspondance de l'élément de données de consigne avec l'attribut d'Objet Axe de dispositif de mouvement

Bit	Élément de données de consigne	Attribut d'Objet Axe de dispositif de mouvement
0	Position de consigne (LREAL) Position de consigne (DINT)	Commande de position du contrôleur – Virgule flottante Commande de position du contrôleur – Entier
1	Vitesse de consigne	Consigne de vitesse du contrôleur
2	Accélération de consigne	Consigne d'accélération du contrôleur
3	Couple de consigne	Consigne de couple du contrôleur
4	(Réservé)	(Réservé)
5	(Réservé)	(Réservé)
6	Compte des cycles de déroulement	Variable locale
7	Déplacement de position	Variable locale

Il appartient au contrôleur de s'assurer que les données de connexion contiennent les Éléments de données de consigne nécessaires à la prise en charge du mode de commande spécifié.

- Ensemble de données réelles: Cette valeur de matriciel comporte un bit défini pour chaque attribut potentiel de données réelles en temps réel à inclure dans l'ensemble de données réelles du bloc de données d'instance de la connexion Dispositif-Contrôleur dans la mise à jour suivante (voir Tableau 11). Les données réelles apparaissent dans le même ordre que les numéros de bit. Ainsi, la Position réelle apparaît avant la Vitesse réelle dans la structure d'ensemble de données réelles. Ce mécanisme permet de modifier le contenu de l'ensemble de données réelles à tout moment pendant le fonctionnement du dispositif. La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement "Ensemble de données réelles".

Tableau 11 – Ensemble de données réelles

Bit	Élément de données réelles produit	Type de données
0	Position réelle	DINT
1	Vitesse réelle	REAL
2	Accélération réelle	REAL
3	(Réservé)	(Réservé)
4	(Réservé)	(Réservé)
5	(Réservé)	(Réservé)
6	Compte des cycles de déroulement	DINT
7	Déplacement de position	DINT

Les éléments de données réelles ci-dessus sont mis en correspondance avec les attributs correspondants de l'Objet Axe de dispositif de mouvement comme représenté au Tableau 12. Les unités des éléments de données réelles correspondent à celles définies pour l'attribut d'Objet Axe de dispositif de mouvement associé.

Tableau 12 – Mise en correspondance de l'élément de données réelles avec l'attribut d'Objet Axe de dispositif de mouvement

Bit	Élément de données réelles	Attribut d'Objet Axe de dispositif de mouvement
0	Position réelle	Retour en position
1	Vitesse réelle	Retour en vitesse
2	Accélération réelle	Retour en accélération
3	(Réservé)	(Réservé)
4	(Réservé)	(Réservé)
5	(Réservé)	(Réservé)
6	Compte des cycles de déroulement	Variable locale
7	Déplacement de position	Variable locale

- Ensemble de données d'état: Cet octet de matriciel contient des indicateurs qui déterminent ce que le contrôleur demande concernant le contenu de l'ensemble de données d'état du bloc de données d'instance de la connexion Dispositif-Contrôleur dans la mise à jour suivante (voir Tableau 13). Ces bits peuvent NE PAS déterminer uniquement si le dispositif envoie les données d'état associées dans la mise à jour suivante de la connexion Dispositif-Contrôleur. Cette opération peut être déterminée par le dispositif sur la base du changement ou non des données d'état associées depuis la dernière mise à jour de la connexion Dispositif-Contrôleur.

Les données d'état apparaissent dans le même ordre que les numéros de bit. Ainsi, le type/code de défaut d'axe apparaît avant, par exemple, les données de datation de défaut d'axe dans la structure Ensemble de données d'état. Plusieurs attributs comprenant un élément de données d'état sélectionné sont transmis dans l'ordre de haut en bas. Ainsi, le type de défaut d'axe est transmis avant le Code de défaut d'axe. La consultation de l'attribut correspondant de l'Objet Axe de dispositif de mouvement fournit les définitions de chacun de ces éléments de données d'état (voir 7.3) dans la spécification de l'Objet Axe de dispositif de mouvement. La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement "Ensemble de données d'état".

Tableau 13 – Ensemble de données d'état

Bit	Élément de données d'état produit	Type de données
0	Type de défaut d'axe Code de défaut d'axe Sous-code de défaut d'axe Action de défaut d'axe Datation de défaut d'axe	USINT USINT USINT USINT LINT
1	Type d'alarme d'axe Code d'alarme d'axe Sous-code d'alarme d'axe État d'alarme d'axe Datation d'alarme d'axe	USINT USINT USINT USINT LINT
2	État d'axe État d'axe – Mfg	DWORD DWORD
3	État E/S d'axe État E/S d'axe – Mfg	DWORD DWORD
4	État de sécurité d'axe État de sécurité d'axe – Mfg État de sécurité d'axe Remplissage[3]	DWORD DWORD USINT USINT[3]
5	(Réservé)	(Réservé)
6	(Spécifique au fournisseur)	(Spécifique au fournisseur)
7	(Spécifique au fournisseur)	(Spécifique au fournisseur)

Lorsque ces bits sont définis et lorsque la valeur d'état associée a changé dans le dispositif, ce dernier doit renvoyer ladite valeur dans la mise à jour Dispositif-Contrôleur suivante. Les valeurs de données d'état changent relativement peu souvent, et de ce fait il peut y avoir de nombreuses mises à jour de connexion où divers éléments de l'ensemble de données d'état restent statiques. Il est recommandé que le dispositif tire profit de ce fait et définisse uniquement le bit de l'ensemble de données d'état dans la mise à jour Dispositif-Contrôleur lorsque l'une des valeurs d'état associées a changé. Ainsi, il est nécessaire que le dispositif charge uniquement, par exemple, les données de défaut d'axe associées au bit 0 Ensemble de données d'état, lorsqu'un nouveau défaut est à consigner dans un rapport. Dans le cas contraire, il annule le bit 0. Côté contrôleur de la connexion, le contrôleur doit analyser et traiter les données de défaut d'axe uniquement si le bit 0 Ensemble de données d'état est défini. Afin de s'assurer que les données ont été reçues, analysées et traitées par le contrôleur avant d'annuler le bit Ensemble de données d'état, le dispositif peut examiner le dernier ID de mise à jour de la mise à jour suivante de la Connexion Contrôleur-Dispositif de manière à confirmer qu'il correspond ou dépasse l'ID de mise à jour de cette mise à jour Dispositif-Contrôleur.

- Commande de consigne: L'octet contient les informations nécessaires au contrôle de l'algorithme d'interpolation fine et à la détermination du temps cible des données de consigne de la structure Commande d'axe (voir Tableau 14). La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement "Commande de consigne".

Tableau 14 – Commande de consigne

Bit	Définition	Syntaxe
0 à 1	Mise à jour de consigne cible	0 = Immédiate 1 = Extrapoler (+1 Période de mise à jour) 2 = Interpoler (+2 Périodes de mise à jour) 3 = (Réservé)
2 à 3	Type de données de position de consigne	0 = LREAL (Virgule flottante de 64 bits) 1 = DINT (entier signé à 32 bits) 2 to 3 = (Réservé)
4 à 7	(Réservé)	

Mise à jour de consigne cible – Cet entier à 2 bits définit un temps spécifique par rapport à la datation de connexion pour lequel les données de consigne sont ciblées, comme défini par le planificateur de mouvement du contrôleur. La mise à jour de consigne cible absolue est la somme de la datation du contrôleur depuis le contrôleur et le produit (Mise à jour de consigne cible × Période de mise à jour du contrôleur).

- Une mise à jour de consigne cible nulle implique que les données de consigne sont ciblées pour le début du cycle de mise à jour en cours et, par conséquent, nécessite d'être immédiatement appliqué à la structure de commande. Dans ce cas, il n'est pas utile de procéder à une interpolation fine. Cette situation peut se produire lorsque la période de mise à jour du contrôleur est plus courte que celle du dispositif ou que le mouvement contrôlé d'axe est presque constant pendant la durée de la période de mise à jour du contrôleur.
- Une mise à jour de consigne cible de 1 implique que la cible des données de consigne est l'événement de synchronisation Mise à jour de connexion suivant. Dans ce cas, l'interpolateur de consigne fonctionne principalement comme un extrapolateur qui estime la valeur suivante des données de consigne en fonction de la trajectoire présente d'axe. Il s'agit d'un paramètre classique lorsque la période de mise à jour du contrôleur est comparable à celle du dispositif ou que le mouvement contrôlé d'axe est relativement constant pendant la durée de la période de mise à jour du contrôleur.
- Une mise à jour de consigne cible de 2 implique que la cible des données de consigne est composée de deux événements de synchronisation Mise à jour de connexion issus de la datation de connexion. Dans ce cas, l'interpolateur de consigne peut calculer une trajectoire lissée en fonction de la dynamique courante du moteur pour atteindre la valeur Données de consigne au moment ciblé. Il s'agit de la véritable interpolation fine, qui s'applique lorsque la période de mise à jour du contrôleur est bien plus longue que celle du dispositif.
- Le Tableau 15 résume la relation entre le rapport de période de mise à jour et la mise à jour de consigne cible recommandée.

Tableau 15 – Mise à jour de consigne cible et rapport de période de mise à jour

Rapport de période de mise à jour (Période de mise à jour du contrôleur / Période de mise à jour du dispositif)	Mise à jour de consigne cible recommandée
2 ou plus	2 = Interpoler (+2 Périodes de mise à jour)
$\frac{1}{2}$ à 2	1 = Extrapoler (+1 Période de mise à jour)
$\frac{1}{2}$ ou moins	0 = Immédiate

Type de données de position de consigne – Cet entier à 2 bits définit le type de données de l'élément de données de position de consigne. Ceci permet une souplesse de traitement du très grand nombre de représentations différentes de la position de consigne dans l'industrie.

- Un type de données de position de consigne de 0 correspond au type LREAL, ou à un type de données à virgule flottante en double précision. Il est préférable que la mise en œuvre prenne en charge ce type de données dans la mesure où il fournit au dispositif une information de commande fractionnée, ce qui produit un mouvement régulier.
- Un type de données de position de consigne de 1 correspond au type DNIT, ou à un type de données à entier signé à 32 bits. Ce type de données est applicable aux dispositifs d'entraînement simples qui n'exigent pas la précision d'une valeur de position de consigne à virgule flottante ou ne comportent pas de prise en charge matérielle suffisante pour une valeur mathématique à virgule flottante en double précision.
- Données de consigne cycliques: Les données de consigne cycliques contiennent des données à priorité élevée qu'il est nécessaire d'appliquer à l'instance d'axe d'entraînement associée lors de la mise à jour suivante du dispositif. Ce bloc est composé d'éléments de données de consigne appliqués en référence aux algorithmes de commande du dispositif,

et déterminés de manière explicite par l'élément Ensemble de données de consigne dans l'En-tête de données de consigne cycliques.

6.4.3.3.4 Bloc de données d'écriture cycliques

Le bloc de données d'écriture cycliques peut être utilisé pour mettre à jour de manière synchrone un ou plusieurs paramètres de configuration d'Objet Axe de dispositif de mouvement ciblés dans le dispositif. Ce mécanisme peut être utilisé conjointement avec un programme Bloc de fonctions pour mettre en œuvre des algorithmes sophistiqués de commande de boucle externe, de séquençement de gain et de limitation dynamique. À l'inverse des demandes de service Définition d'attribut d'axe du canal de service, dont le traitement peut durer plusieurs cycles de mise à jour de dispositif, le mécanisme de données d'écriture cycliques garantit l'application du paramètre ciblé à la mise à jour suivante de dispositif disponible.

Le bloc de données d'écriture cycliques est uniquement pris en charge dans le format Connexion variable (voir Figure 23).

Bloc de données d'écriture cycliques			
ID de bloc d'écriture cyclique	–	ID de bloc de lecture cyclique	–
Données d'écriture cycliques			

IEC

Figure 23 – Bloc de données d'écriture cycliques

L'en-tête associé pour ce bloc contient les éléments essentiels relatifs au contenu des blocs de données d'écriture et de lecture cycliques de la mise à jour suivante de la Connexion Dispositif-Contrôleur.

- ID de bloc d'écriture cyclique: Cet ID de 8 bits détermine la structure prédéfinie du bloc d'écriture cyclique à appliquer aux données d'écriture cycliques pour cette mise à jour. Les structures de bloc d'écriture cyclique sont définies à l'aide du service Liste de définition des données d'écriture cycliques. La valeur initiale pour l'ID du bloc d'écriture cyclique est 0, ce qui correspond à un ensemble nul de données d'écriture cycliques. Une réponse correcte au service Liste de définition des données d'écriture cycliques inclut un nouvel ID augmenté de bloc d'écriture cyclique qui peut être utilisé dans la mise à jour de connexion suivante pour transmettre les données cycliques dans ce format. L'ID de bloc d'écriture cyclique précédent et sa structure associée doivent être conservés tant que le contrôleur ne demande pas la transmission du nouvel ID de bloc d'écriture cyclique, l'ancien ID et la définition de structure du bloc d'écriture cyclique n'étant alors plus obligatoires.
- ID de bloc de lecture cyclique: Cet ID de 8 bits détermine la structure prédéfinie du bloc de lecture cyclique à appliquer aux données de lecture cycliques pour la mise à jour suivante de la connexion Dispositif-Contrôleur. Les structures de bloc de lecture cyclique sont définies à l'aide du service Définition du bloc de données de lecture cycliques. La valeur initiale pour l'ID du bloc de lecture cyclique est 0, ce qui correspond à un ensemble nul de données de lecture cycliques. Une réponse correcte au service Liste de définition des données de lecture cycliques inclut un nouvel ID de bloc de lecture cyclique augmenté qui peut être utilisé dans la mise à jour de la connexion suivante pour permettre au dispositif d'utiliser le nouveau format de données de lecture cycliques pour la mise à jour suivante disponible de la connexion Dispositif-Contrôleur.
- Données d'écriture cycliques: Les données d'écriture cycliques contiennent des données à priorité élevée qu'il faut appliquer à l'instance d'axe de dispositif associée lors de la mise à jour suivante du dispositif d'entraînement. Ce bloc est composé d'éléments de données paramétriques appliqués aux attributs d'Objet Axe de dispositif de mouvement utilisés par les algorithmes de commande. Le contenu des données d'écriture cycliques est déterminé de manière explicite par la structure identifiée par l'ID de bloc d'écriture cyclique se trouvant dans l'en-tête de données d'écriture cycliques. L'ordre des données d'attribut dans le Bloc de données d'écriture cycliques est déterminé par celui des ID d'attribut de la demande Set_Cyclic_Write_List. Les éléments de données d'attribut doivent être du texte

aligné. Les mots à 32 bits et à 16 bits sont respectivement alignés sur 32 et 16 bits. Le remplissage doit être ajouté pour conserver l'alignement des mots.

EXEMPLE La Figure 24 représente un bloc de données d'écriture cycliques avec 3 attributs où les Attributs 1 et 3 sont des valeurs à 32 bits, tandis que l'Attribut 2 est une valeur à 16 bits.

Valeur d'attribut 1	
Valeur d'attribut 2	16 bits – réservé
Valeur d'attribut 3	

IEC

Figure 24 – Bloc de données d'écriture cycliques

6.4.3.4 Bloc de données d'événement

Le bloc de données d'événement permet d'acheminer des informations concernant le canal d'événements. En particulier, le bloc de données d'événement de la connexion Contrôleur-Dispositif permet de contrôler l'armement des fonctions de vérification des événements dans le dispositif, et d'acquitter la réception des notifications d'événement provenant du dispositif envoyées par l'intermédiaire du bloc de données d'événement de la connexion Dispositif-Contrôleur.

Le format du bloc de données d'événement de la connexion Contrôleur-Dispositif est représenté à la Figure 25.

Bloc de données d'événement			
Commande de vérification d'événement			
Ensemble de données d'enregistrement	Ensemble de données d'origine	Ensemble de données d'observation	—
Données d'événement d'enregistrement			
Données d'événement d'origine			
Données d'événement d'observation			
Acquittement d'événement ID1	Acquittement d'événement État 1	Acquittement d'événement ID2	Acquittement d'événement État 2
...			

IEC

Figure 25 – Bloc de données d'événement

En présence du bloc de données d'événement, l'élément Commande de vérification d'événement détermine quels sont les éléments supplémentaires inclus. Les éléments qui commencent avec l'ensemble de données d'enregistrement et s'étendent par l'intermédiaire des données d'événement d'observation constituent des données étendues et existent uniquement si le bit Format Étendu est défini dans le mot Commande de vérification d'événement. Les éléments # d'ID d'acquittement d'événement et # d'état d'acquittement d'événement sont répétés entre 0 et 7 fois, comme déterminé par le champ Compte des blocs d'événement dans le mot Commande de vérification d'événement. Lorsqu'un nombre impair de blocs d'événements est inclus, le bloc de données d'événement est rempli jusqu'à une limite de 32 bits avec deux octets non utilisés.

- Commande de vérification d'événement: Ce mot à 32 bits est copié dans l'attribut d'Objet Axe de dispositif de mouvement du même nom utilisé pour activer différentes entrées du dispositif (les entrées de marqueur et d'enregistrement, par exemple) afin de générer des événements. Lorsque ces événements se produisent, le dispositif capture l'heure et la position exacte d'axe lors du déroulement de l'événement. Les 4 derniers bits de l'élément Commande de vérification d'événement constituent un champ de bit spécial, dont les 3 bits de poids faible spécifient le nombre d'événements actifs, et qui est littéralement le nombre d'ID d'acquittement d'événement figurant dans ce bloc de données d'événement. Le bit de poids fort active le format du Bloc de données d'événement étendu. Une

définition exhaustive de l'attribut Commande de vérification d'événement peut être consultée en 7.3.7.2.1.

Le mécanisme de base de commande d'événement fonctionne comme spécifié dans le Tableau 16.

Tableau 16 – Cycle d'événements de base

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur définit le bit Commande de vérification d'événement approprié pour rechercher une condition d'erreur spécifique. Plusieurs bits du mot Commande de vérification d'événement peuvent être définis à un instant donné.	→ Bit Vérification d'événement = 1 → Compte de blocs d'événement = 0 → Format étendu = 0	Le dispositif détecte que le bit Commande de vérification d'événement est défini, puis exécute l'action de vérification d'événement demandée.
Le contrôleur détermine que le bit État de vérification d'événement provenant du dispositif est défini, indiquant que le déclencheur d'événements spécifié est "armé" au niveau du dispositif. En commençant par cette mise à jour, le contrôleur peut traiter toutes les notifications d'événement provenant du dispositif qui correspondent à la condition d'événement spécifiée.	← Bit Vérification d'événement = 1 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif définit le bit Statut de vérification d'événement correspondant pour confirmer qu'il procède à présent à la vérification active de la condition d'événement spécifiée. En commençant par cette mise à jour, le dispositif peut inclure une notification d'événement pour la condition d'événement spécifiée.
Le contrôleur maintient le bit Vérification d'événement afin de poursuivre la vérification de l'événement spécifié.	→ Bit Vérification d'événement = 1 → Compte de blocs d'événement = 0 → Format étendu = 0	Le dispositif détermine que le bit Commande de vérification d'événement est défini et poursuit par conséquent la vérification de l'événement spécifié.
Le contrôleur poursuit sa vérification des nouvelles notifications d'événement provenant du dispositif.	← Bit Vérification d'événement = 1 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif continue à définir le bit État de vérification d'événement correspondant afin d'indiquer que la vérification d'événement est active.
Le contrôleur traite la nouvelle Notification d'événement, en archivant l'ID de l'événement, vérifiant l'état de l'événement, puis en traitant le bloc de données d'événement sur la base du type d'événement.	← Bit Vérification d'événement = 1 ← Compte de blocs d'événement = 1 ← Format étendu = 0 ← ID d'événement = id ← État d'événement = 0 (succès) ← Type d'événement = type ← Bloc de données d'événement	Le dispositif détecte la condition d'événement spécifiée, incrémente l'ID d'événement, puis envoie au contrôleur un bloc de données de notification d'événement avec cet ID. La condition d'événement spécifique est identifiée par son Type d'événement. Étant donné que le bit Réarmement automatique associé est annulé dans le mot Commande de vérification d'événement, le dispositif interrompt la vérification de l'événement spécifié. Toutefois, le bit État de vérification d'événement est toujours défini tant que le bit Commande de vérification d'événement n'est pas annulé par le contrôleur. Remarquer que les notifications des autres événements peuvent être envoyées dans la même mise à jour. Le nombre total de blocs de données de notification d'événement dans la mise à jour est représenté par le compte des blocs d'événement.
Le contrôleur envoie au dispositif l'acquittement d'événement avec l'ID et l'état d'événement associés. Étant donné que le réarmement automatique n'est pas activé, le contrôleur annule le bit Commande de vérification d'événement associé	→ Bit Vérification d'événement = 0 → Compte de blocs d'événement = 1 → Format étendu = 0 → ID d'événement = id → État d'événement = 0 (succès)	Le dispositif détecte que le bit Commande de vérification d'événement est annulé et traite le bloc de données d'acquittement d'événement.

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur détermine que le bit État de vérification d'événement provenant du dispositif est annulé, indiquant que le cycle d'événements est achevé. Le contrôleur est à présent capable de redéfinir le bit Commande de vérification d'événement. Ainsi, la durée minimale entre la réception de la notification d'événement par le contrôleur et le réarmement du dispositif pour l'événement suivant est une période de mise à jour du contrôleur.	← Bit Vérification d'événement = 0 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif annule le bit État de vérification d'événement correspondant. Dans la mesure où le dispositif a reçu un acquittement d'événement positif, il n'est plus nécessaire d'envoyer le bloc de données de notification d'événement pour l'ID de cet événement.
Le contrôleur peut à présent régler la taille des blocs d'événement sur zéro pour toutes les mises à jour ultérieures tant qu'une demande de vérification des nouveaux événements existe.	→ Taille de bloc d'événement = 0 → Pas de bloc d'événement	Le dispositif détermine que la taille des blocs d'événement est zéro, en indiquant qu'il n'existe aucune donnée de bloc d'événement à traiter.
Le dispositif détermine que la taille des blocs d'événement est zéro, en indiquant qu'il n'existe aucune donnée de bloc d'événement à traiter.	← Taille de bloc d'événement = 0 ← Pas de bloc d'événement	Le dispositif peut à présent régler la taille des blocs d'événement sur zéro pour toutes les mises à jour ultérieures tant qu'une demande de vérification des nouveaux événements existe.

Lorsque le dispositif prend en charge la fonctionnalité de mise à l'échelle du dispositif, le format de bloc de données d'événement étendu permet d'échanger des informations d'événement supplémentaire entre le contrôleur et le dispositif d'entraînement. Le cycle d'événements étendu fonctionne comme spécifié dans le Tableau 17.

Tableau 17 – Cycle d'événements étendu

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur définit le bit Commande de vérification d'événement approprié pour rechercher une condition d'erreur spécifique. Plusieurs bits du mot Commande de vérification d'événement peuvent être définis à un instant donné. À l'aide du format de bloc de données de vérification d'événement étendu, le contrôleur transmet les paramètres dont a besoin la fonction de vérification d'événement dans les blocs de données d'événement d'enregistrement, d'origine et d'observation. Le bit Format d'événement dans l'élément Commande de vérification d'événement doit être défini afin qu'il comporte les données de vérification d'événement étendu. Les données de vérification d'événement étendu sont exigées lorsque le dispositif est configuré pour la mise à l'échelle du dispositif.	→ Bit Vérification d'événement = 1 → Compte de blocs d'événement = 0 → Format étendu = 1 → Données d'événement étendu	Le dispositif détecte que le bit Commande de vérification d'événement est défini, charge les paramètres de vérification d'événement associés provenant du bloc de données de vérification d'événement étendu, puis lancer l'action de vérification d'événement demandé.

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur détermine que le bit État de vérification d'événement provenant du dispositif est défini, indiquant que le déclencheur d'événements spécifié est "armé" au niveau du dispositif. En commençant par cette mise à jour, le contrôleur peut traiter toutes les notifications d'événement provenant du dispositif qui correspondent à la condition d'événement spécifiée. Lorsque le contrôleur a envoyé précédemment les paramètres de vérification d'événement via le bloc de données de vérification d'événement étendu, il vérifie également les valeurs Ensemble de données d'enregistrement, de référence et d'observation renvoyées, afin de déterminer si le dispositif a traité les données d'événement d'enregistrement, d'origine et d'observation envoyées dans la mise à jour précédente. Lorsque les valeurs Acquittance de données correspondent à la valeur Ensemble de données envoyée, les données de vérification d'événement étendu ont été traitées par le dispositif.	← Bit Vérification d'événement = 1 ← Compte de blocs d'événement = 0 ← Format étendu = 1 ← Acquittance des données d'événement étendu	Le dispositif définit le bit État de vérification d'événement correspondant pour confirmer qu'il procède à présent à la vérification active de la condition d'événement spécifiée. Lorsque le dispositif reçoit des paramètres de vérification d'événement du bloc de données de vérification d'événement étendu, il acquitte les paramètres transmis dans les blocs de données d'événement d'enregistrement, d'origine et d'observation en renvoyant les valeurs Ensemble de données d'enregistrement, d'origine et d'observation comme valeurs Acquittance des données, respectivement. En commençant par cette mise à jour, le dispositif peut inclure une notification d'événement pour la condition d'événement spécifiée.
Le contrôleur maintient le bit Vérification d'événement afin de poursuivre la vérification de l'événement spécifié. Dans la mesure où le dispositif acquitte les données de vérification d'événement étendu, le contrôleur annule le bit Format étendu pour la mise à jour de la connexion Contrôleur-Dispositif.	→ Bit Vérification d'événement = 1 → Compte de blocs d'événement = 0 → Format étendu = 0	Le dispositif détermine que le bit Commande de vérification d'événement est défini et poursuit par conséquent la vérification de l'événement spécifié.
Le contrôleur poursuit sa vérification des nouvelles notifications d'événement provenant du dispositif.	← Bit Vérification d'événement = 1 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif continue à définir le bit État de vérification d'événement correspondant afin d'indiquer que la vérification d'événement est active. Aucune donnée de vérification d'événement étendu n'étant à acquitter, le dispositif annule le bit Format étendu.
Le contrôleur traite la nouvelle Notification d'événement, en archivant l'ID de l'événement, vérifiant l'état de l'événement, puis en traitant le bloc de données d'événement sur la base du type d'événement.	← Bit Vérification d'événement = 1 ← Compte de blocs d'événement = 1 ← Format étendu = 0 ← ID d'événement = x ← État d'événement = 0 ← Type d'événement = y ← Bloc de données d'événement	Le dispositif détecte la condition d'événement spécifiée, incrémente l'ID d'événement, puis envoie au contrôleur un bloc de données de notification d'événement avec cet ID. La condition d'événement spécifique est identifiée par son Type d'événement. Étant donné que le bit Réarmement automatique associé est annulé dans le mot Commande de vérification d'événement, le dispositif interrompt la vérification de l'événement spécifié. Toutefois, le bit État de vérification d'événement est toujours défini tant que le bit Commande de vérification d'événement n'est pas annulé par le contrôleur. Remarquer que les notifications des autres événements peuvent être envoyées dans la même mise à jour. Le nombre total de blocs de données de notification d'événement dans la mise à jour est représenté par le compte des blocs d'événement.

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur envoie au dispositif l'acquittement d'événement avec l'ID et l'état d'événement associés. Étant donné que le réarmement automatique n'est pas activé, le contrôleur annule le bit Commande de vérification d'événement associé	→ Bit Vérification d'événement = 0 → Compte de blocs d'événement = 1 → Format étendu = 0 → ID d'événement = id → État d'événement = 0 (succès)	Le dispositif détecte que le bit Commande de vérification d'événement est annulé et traite le bloc de données d'acquittement d'événement.
Le contrôleur détermine que le bit État de vérification d'événement provenant du dispositif est annulé, indiquant que le cycle d'événements est achevé. Le contrôleur est à présent capable de redéfinir le bit Commande de vérification d'événement. Ainsi, la durée minimale entre la réception de la notification d'événement par le contrôleur et le réarmement du dispositif pour l'événement suivant est une période de mise à jour du contrôleur.	← Bit Vérification d'événement = 0 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif annule le bit État de vérification d'événement correspondant. Dans la mesure où le dispositif a reçu un acquittement d'événement positif, il n'est plus nécessaire d'envoyer le bloc de données de notification d'événement pour l'ID de cet événement.
Le contrôleur peut à présent régler la taille des blocs d'événement sur zéro pour toutes les mises à jour ultérieures tant qu'une demande de vérification des nouveaux événements existe.	→ Taille de bloc d'événement = 0 → Pas de bloc d'événement	Le dispositif détermine que la taille des blocs d'événement est zéro, en indiquant qu'il n'existe aucune donnée de bloc d'événement à traiter.
Le dispositif détermine que la taille des blocs d'événement est zéro, en indiquant qu'il n'existe aucune donnée de bloc d'événement à traiter.	← Taille de bloc d'événement = 0 ← Pas de bloc d'événement	Le dispositif peut à présent régler la taille des blocs d'événement sur zéro pour toutes les mises à jour ultérieures tant qu'une demande de vérification des nouveaux événements existe.

Dans le cas d'un événement d'enregistrement où la vérification de l'événement de réarmement automatique est demandée, la séquence de gestion d'événement se présente comme spécifié dans le Tableau 18.

Tableau 18 – Cycle d'événements de base avec réarmement automatique

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur définit le bit Commande de vérification d'événement approprié pour rechercher une condition d'événement spécifique à déclencher, et définit également le bit Réarmement automatique pour cette condition d'événement. Plusieurs bits du mot Commande de vérification d'événement peuvent être définis à un instant donné.	→ Bit Vérification d'événement = 1 → Bit Réarmement automatique d'événement = 1 → Compte de blocs d'événement = 0 → Format étendu = 0	Le dispositif détecte que le bit Commande de vérification d'événement est défini, puis exécute l'action de vérification d'événement demandée.
Le contrôleur détermine que le bit État de vérification d'événement provenant du dispositif est défini, indiquant que le déclencheur d'événements spécifié est "armé" au niveau du dispositif. En commençant par cette mise à jour, le contrôleur peut traiter toutes les notifications d'événement provenant du dispositif qui correspondent à la condition d'événement spécifiée.	← Bit Vérification d'événement = 1 ← Bit Réarmement automatique d'événement = 1 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif définit le bit État de vérification d'événement correspondant pour confirmer qu'il procède à présent à la vérification active de la condition d'événement spécifiée, et définit également le bit Réarmement automatique correspondant. En commençant par cette mise à jour, le dispositif peut inclure une notification d'événement pour la condition d'événement spécifiée.
Le contrôleur maintient le bit Vérification d'événement afin de poursuivre la vérification de l'événement spécifié.	→ Bit Vérification d'événement = 1 → Bit Réarmement automatique d'événement = 1 → Compte de blocs d'événement = 0 → Format étendu = 0	Le dispositif détermine que le bit Commande de vérification d'événement est défini et poursuit par conséquent la vérification de l'événement spécifié.
Le contrôleur poursuit sa vérification des nouvelles notifications d'événement provenant du dispositif.	← Bit Vérification d'événement = 1 ← Bit Réarmement automatique d'événement = 1 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif continue à définir le bit État de vérification d'événement correspondant afin d'indiquer que la vérification d'événement est active.
Le contrôleur traite la nouvelle Notification d'événement, en archivant l'ID de l'événement, vérifiant l'état de l'événement, puis en traitant le bloc de données d'événement sur la base du type d'événement.	← Bit Vérification d'événement = 1 ← Bit Réarmement automatique d'événement = 1 ← Compte de blocs d'événement = 1 ← Format étendu = 0 ← ID d'événement = id1 ← État d'événement = 0 (succès) ← Type d'événement = type ← Bloc de données d'événement	Le dispositif détecte la condition d'événement spécifiée, incrémente l'ID d'événement, puis envoie au contrôleur un bloc de données de notification d'événement avec cet ID. La condition d'événement spécifique est identifiée par son Type d'événement. Étant donné que le bit Réarmement automatique correspondant est défini, le dispositif poursuit la vérification de l'événement spécifié. Le bit État de vérification d'événement est toujours défini tant que le bit Commande de vérification d'événement n'est pas annulé par le contrôleur. Si le dispositif détecte une autre condition d'événement spécifiée avant la transmission du premier événement au contrôleur, il incrémente de nouveau l'ID d'événement, puis ajoute une autre notification d'événement avec cet ID au bloc de données d'événement du contrôleur. Remarquer que les notifications des autres événements peuvent être également envoyées dans la même mise à jour. Le nombre total de blocs de données de notification d'événement dans la mise à jour est représenté par le compte des blocs d'événement.

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur envoie au dispositif l'acquittement d'événement avec l'ID et l'état d'événement associés. Étant donné que le bit Réarmement automatique est activé, il maintient le bit Vérification d'événement associé défini pour permettre au dispositif de poursuivre la vérification de l'événement suivant.	→ Bit Vérification d'événement = 1 → Bit Réarmement automatique d'événement = 1 → Compte de blocs d'événement = 1 → Format étendu = 0 → ID d'événement = id1 → État d'événement = 0 (succès)	Le dispositif détecte que le bit Commande de vérification d'événement est défini et traite le bloc de données d'acquittement d'événement. Étant donné que le bit Réarmement automatique correspondant est défini, le dispositif poursuit la vérification de l'événement spécifié. Le dispositif peut désormais envoyer une autre notification d'événement si l'événement spécifié s'est de nouveau produit.
Le contrôleur poursuit sa vérification des nouvelles notifications d'événement provenant du dispositif.	← Bit Vérification d'événement = 1 ← Bit Réarmement automatique d'événement = 1 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif définit les bits Vérification d'événement et État de réarmement automatique correspondants en indiquant qu'il poursuit sa vérification des événements. Dans la mesure où le dispositif a reçu un acquittement d'événement positif, il n'est plus nécessaire d'envoyer le bloc de données de notification d'événement pour l'ID de cet événement.

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur traite la nouvelle Notification d'événement, en archivant l'ID de l'événement, vérifiant l'état de l'événement, puis en traitant le bloc de données d'événement sur la base du type d'événement.	← Bit Vérification d'événement = 1 ← Bit Réarmement automatique d'événement = 1 ← Compte de blocs d'événement = 1 ← Format étendu = 0 ← ID d'événement = id2 ← État d'événement = 0 ← Type d'événement = type ← Bloc de données d'événement	Le dispositif continue la vérification de la condition d'événement spécifiée et envoie un bloc de données de notification d'événement au contrôleur à chaque fois que la condition d'événement se produit. La condition d'événement spécifique est identifiée par son Type d'événement. Le bit État de vérification d'événement est toujours défini tant que le bit Commande de vérification d'événement n'est pas annulé par le contrôleur. Avec la fonction de réarmement automatique, la vérification d'événement est activée en permanence, ce qui assure de ne manquer aucun événement d'enregistrement pendant le délai normal d'un cycle de réarmement du mécanisme de vérification d'événement. La fonction de réarmement automatique présente l'inconvénient de pouvoir générer une multitude d'événements dont le traitement ne présente aucun intérêt, en fait, plusieurs événements spécifiés par période de mise à jour du contrôleur. Par conséquent, ces événements doivent être placés dans la mémoire tampon du contrôleur dans un attribut de matrice d'événement ou filtrés en fonction des données d'événement. Cette dernière possibilité porte sur la manière dont la fonction Enregistrement fenêtré est mise en œuvre. Cette fonction vérifie tous les événements d'enregistrement provenant du dispositif pour voir si la Position d'événement donne une valeur de position absolue dans les limites de la fenêtre d'enregistrement configurée. Si la position d'enregistrement calculée n'entre pas dans la fenêtre, l'événement est ignoré. Si la position d'enregistrement calculée entre dans la fenêtre, elle est archivée dans le contrôleur avec la datation d'enregistrement.
Le contrôleur envoie au dispositif l'acquiescement d'événement avec l'ID et l'état d'événement associés. Étant donné que le bit Réarmement automatique est activé, il maintient le bit Vérification d'événement associé défini pour permettre au dispositif de poursuivre la vérification de l'événement suivant.	→ Bit Vérification d'événement = 1 → Bit Réarmement automatique d'événement = 1 → Compte de blocs d'événement = 1 → Format étendu = 0 → ID d'événement = id2 → État d'événement = 0 (succès)	Le dispositif détecte que le bit Commande de vérification d'événement est annulé et traite le bloc de données d'acquiescement d'événement. Étant donné que le bit Réarmement automatique correspondant est défini, le dispositif poursuit la vérification de l'événement spécifié. Le dispositif peut désormais envoyer une autre notification d'événement si l'événement spécifié s'est de nouveau produit. Cette notification d'événement – le cycle Acquiescement d'événement se répète tant que le contrôleur n'annule pas le bit Commande de vérification d'événement associé.
...	...	...

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur annule le bit Commande de vérification d'événement et le bit Réarmement automatique afin de désactiver la vérification d'événement.	→ Bit Vérification d'événement = 0 → Bit Réarmement automatique d'événement = 0 → Compte de blocs d'événement = 0 → Format étendu = 0	Le dispositif détecte que le bit Commande de vérification d'événement est annulé, puis désactive l'action de vérification d'événement.
Le contrôleur détermine que le bit État de vérification d'événement provenant du dispositif est annulé, indiquant que la vérification d'événement est désactivée.	← Bit Vérification d'événement = 0 ← Bit Réarmement automatique d'événement = 0 ← Compte de blocs d'événement = 0 ← Format étendu = 0	Le dispositif annule le bit État de vérification d'événement correspondant.
Le contrôleur peut à présent régler la taille des blocs d'événement sur zéro pour toutes les mises à jour ultérieures tant qu'une demande de vérification des nouveaux événements existe.	→ Taille de bloc d'événement = 0 → Pas de bloc d'événement	Le dispositif détermine que la taille des blocs d'événement est zéro, en indiquant qu'il n'existe aucune donnée de bloc d'événement à traiter.
Le dispositif détermine que la taille des blocs d'événement est zéro, en indiquant qu'il n'existe aucune donnée de bloc d'événement à traiter.	← Taille de bloc d'événement = 0 ← Pas de bloc d'événement	Le dispositif peut à présent régler la taille des blocs d'événement sur zéro pour toutes les mises à jour ultérieures tant qu'une demande de vérification des nouveaux événements existe.

Dans le cas d'un événement Enregistrement étendu où la vérification d'événement de réarmement automatique est demandée, la séquence de réarmement automatique ci-dessus inclut simplement les données Événement étendu comme indiqué dans le Tableau 17.

- Ensemble de données d'enregistrement: Cet octet de matriciel détermine le contenu du bloc de données d'événement d'enregistrement qui contient les paramètres nécessaires pour la configuration de la fonction de vérification des événements d'enregistrement, comme spécifié dans le Tableau 19. Les données dans ce bloc apparaissent dans le même ordre que les numéros de bit Ensemble de données d'enregistrement. Ainsi, par exemple, la fenêtre positive d'enregistrement 1 apparaîtrait avant la fenêtre négative d'enregistrement 1 dans la structure du bloc de données de vérification d'événement. La consultation de l'attribut correspondant fournit les définitions de chacun de ces éléments Données de vérification d'événement dans la spécification de l'Objet Axe de dispositif de mouvement. La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement "Ensemble de données d'enregistrement". Ce mécanisme n'est pas limité à une configuration initiale, mais peut également être utilisé pour changer les paramètres associés à la fonction de vérification des événements d'enregistrement, ladite fonction étant active.

Tableau 19 – Ensemble de données d'enregistrement

Bit	Élément de données d'enregistrement	Type de données
0	Fenêtre Pos d'enreg 1	DINT (fenêtre max DINT (fenêtre min)
1	Fenêtre Neg d'enreg 1	DINT (fenêtre max DINT (fenêtre min)
2	Fenêtre Pos d'enreg 2	DINT (fenêtre max DINT (fenêtre min)
3	Fenêtre Neg d'enreg 2	DINT (fenêtre max DINT (fenêtre min)
4 à 7	(Réservé)	(Réservé)

- **Ensemble de données d'origine:** Cet octet de matriciel détermine le contenu du bloc de données d'événement d'origine qui contient les paramètres nécessaires pour la configuration de la fonction de vérification des événements de référence, comme spécifié dans le Tableau 20. Les données contenues dans ce bloc apparaissent dans le même ordre que les numéros de bit Ensemble de données d'origine; ainsi, par exemple, le seuil de couple d'origine (Home Torque Threshold) apparaît avant le temps de couple d'origine (Home Torque Time). La consultation de l'attribut correspondant fournit les définitions de chacun de ces éléments Données de vérification d'événement dans la spécification de l'Objet Axe de dispositif de mouvement. La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement "Ensemble de données d'origine". Ce mécanisme n'est pas limité à une configuration initiale, mais peut également être utilisé pour changer les paramètres associés à la fonction de vérification des événements d'origine, ladite fonction étant active.

Tableau 20 – Ensemble de données d'origine

Bit	Élément de données d'origine	Type de données
0	Seuil de couple d'origine	REAL
1	Temps de couple d'origine	REAL
2 à 7	(Réservé)	(Réservé)

- **Ensemble de données d'observation:** Cet octet de matriciel détermine le contenu du bloc de données d'événement d'observation qui contient les paramètres nécessaires pour la configuration de la fonction de vérification des événements d'observation, comme spécifié dans le Tableau 21. Les données contenues dans ce bloc apparaissent dans le même ordre que les numéros de bit Ensemble de données d'observation; ainsi, par exemple, la position d'observation 1 (Watch Position 1) apparaît avant la position d'observation 2 (Watch Position 2). La consultation de l'attribut correspondant fournit les définitions de chacun de ces éléments Données de vérification d'événement dans la spécification de l'Objet Axe de dispositif de mouvement. La valeur de cet élément est transférée vers l'attribut d'Objet Axe de dispositif de mouvement "Ensemble de données d'observation". Ce mécanisme n'est pas limité à une configuration initiale, mais peut également être utilisé pour changer les paramètres associés à la fonction de vérification des événements d'observation, ladite fonction étant active.

Tableau 21 – Ensemble de données d'observation

Bit	Élément de données d'observation	Type de données
0	Position d'observation 1	DINT
1	Position d'observation 2	DINT
2 à 7	(Réservé)	(Réservé)

- **ID d'acquiescement d'événement:** Numéro de transaction affecté à cet événement par la notification d'événement d'origine. Un ID d'événement est affecté à chaque événement par incrémentation de l'ID d'événement en cours archivé dans le dispositif. Grâce à l'ID d'événement, le dispositif est capable de mettre en correspondance l'acquiescement d'événement et la notification d'événement appropriée pour procéder à la transaction de données d'événement.
- **État d'acquiescement d'événement:** Valeur énumérée indiquant la réponse du contrôleur à l'événement. Une valeur nulle indique que le traitement de l'événement a abouti. Une valeur non nulle indique qu'une erreur s'est produite dans le traitement de l'événement et que l'événement doit être de nouveau envoyé.

6.4.3.5 Bloc de données de service

Le Bloc de données de service permet d'envoyer une demande de service par instance au dispositif dans une mise à jour donnée. La demande de service exige une réponse de service

spécifique de la part du dispositif, indiquant un succès ou une erreur. Dans certains cas, le service de réponse contient les données demandées. Dans tous les cas, les données de demande de service sont conservées dans la structure de données de connexion Contrôleur-Dispositif tant que le contrôleur ne reçoit pas la réponse de service associée de la part du dispositif.

Chaque demande de service est représentée par un bloc de données organisé comme indiqué à la Figure 26.

NOTE Par nature, les 4 premiers octets du Bloc de données de service ne suivent pas le format de messagerie CIP standard traditionnel. La raison principale est que cette structure de connexion est fondamentalement une connexion E/S CIP implicite et pas une connexion de messagerie explicite. Toutefois, dans le cas d'un format de connexion fixe, les données de demande spécifique au service définies ci-dessous sont envoyées par l'intermédiaire d'une connexion de messagerie explicite et suivent les règles CIP en matière de format de demande de service explicite.

Bloc de données de service			
ID de transaction	Code de service	–	–
Données de demande spécifique au service			

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Figure 26 – Bloc de données de service

- ID de transaction: Numéro de transaction affecté à cette demande de service par le contrôleur. Un nouvel ID de transaction est affecté à chaque demande de service par incrémentation de l'ID de transaction en cours archivé dans le contrôleur. Grâce à cet ID de transaction, le contrôleur est en mesure d'associer la réponse de service à la demande de service appropriée et de procéder à la transaction de service.
- Code de service: Identificateur qui détermine la demande de service spécifique à l'objet suivante. La liste des codes de service pris en charge peut être consultée dans le paragraphe Services spécifiques à l'objet de la présente partie de la série IEC 61800-7 (voir 7.5.1). Les services communs CIP ne s'appliquent pas au bloc de données de service.
- Données de demande spécifique au service: Le format et la syntaxe des données de demande spécifique au service dépendent du code de service spécifié. Cela est vrai, que les données de demande spécifique au service soient transmises dans le cadre de la connexion Contrôleur-Dispositif ou d'une connexion de messagerie explicite.

6.4.4 Connexion Dispositif – Contrôleur

6.4.4.1 Généralités

À l'instar de la structure de données de connexion Contrôleur-Dispositif décrite ci-dessus, la connexion Dispositif-Contrôleur CIP Motion est organisée comme indiqué à la Figure 27.

Format de connexion Dispositif-Contrôleur CIP Motion
En-tête de connexion
Bloc de données temporelles
Blocs de données d'instance

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Figure 27 – Format de connexion Dispositif-Contrôleur CIP Motion

6.4.4.2 En-tête de connexion Dispositif-Contrôleur

6.4.4.2.1 Structure générale

L'en-tête de connexion Dispositif-Contrôleur contient les informations essentielles de configuration d'axe nécessaires à l'analyse du bloc de données de connexion Dispositif-Contrôleur. La partie fixe de l'en-tête de connexion est définie comme indiqué à la Figure 28.

En-tête de connexion			
Format de connexion	Révision de format	ID de mise à jour	État de nœud

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Figure 28 – En-tête de connexion

- Format de connexion: Identique à la définition Contrôleur-Dispositif, sauf que la valeur exigée pour le type de connexion est 3 (type de connexion Dispositif-Contrôleur fixe) ou 7 (type de connexion Dispositif-Contrôleur fixe).
 - 0 = Connexion pair-à-pair fixe entre Contrôleurs
 - 1 = Connexion pair-à-pair fixe entre Dispositifs
 - 2 = Connexion fixe Contrôleur-vers-Dispositif
 - 3 = Connexion fixe Dispositif-vers-Contrôleur
 - 4 = Connexion pair-à-pair variable entre Contrôleurs
 - 5 = Connexion pair-à-pair variable entre Dispositifs
 - 6 = Connexion variable Contrôleur-vers-Dispositif
 - 7 = Connexion variable Dispositif-vers-Contrôleur
 - 8 à 15 = Réserve.

Le format de connexion peut être utilisé pour une analyse correcte des données de connexion Dispositif-Contrôleur, non seulement par le dispositif, mais également par les outils de diagnostic de réseau. La valeur Format de connexion ne doit pas varier pendant toute la durée de vie de la connexion, et sa vérification par le contrôleur n'est par conséquent nécessaire que pour le premier paquet reçu.

- Révision de format: Le numéro de révision de format est 2 pour tout dispositif qui utilise la présente édition de l'IEC 61800-7-202. Dans la mesure où les contrôleurs qui utilisent la présente édition reconnaissent uniquement la révision de format 2, la révision de format 1 est rendue obsolète et il n'est plus nécessaire que les dispositifs CIP Motion la prennent en charge. Cette valeur est incrémentée de 1 à chaque révision du format de connexion Dispositif-Contrôleur ayant un impact sur l'interface. La révision de format permet aux contrôleurs les plus récents de prendre en charge les formats de connexion générés par des dispositifs plus anciens. Elle permet également aux anciens contrôleurs de reconnaître un format de connexion récent provenant d'un dispositif qu'il ne peut pas prendre en charge et de générer une erreur appropriée à cette application. Les outils de diagnostic de réseau peuvent également définir la révision de format afin de savoir comment analyser le paquet de données de connexion. Lorsque la connexion E/S est établie, le numéro de révision de format est également indiqué par le point de connexion E/S dans le chemin d'application du service Forward_Open (voir 6.4). La valeur Révision de format ne doit pas varier pendant toute la durée de vie de la connexion, et sa vérification par le contrôleur n'est par conséquent nécessaire que pour le premier paquet reçu.
- ID de mise à jour: L'ID de mise à jour de la connexion Dispositif-Contrôleur doit être mis en correspondance avec l'ID de mise à jour de la connexion Contrôleur-Dispositif pour un cycle donné, et doit donc être incrémenté à chaque période de mise à jour. Noter que si le dispositif est en mode synchrone, ceci n'implique pas de devoir traiter le paquet Contrôleur-Dispositif avant d'assembler le paquet Dispositif-Contrôleur pour obtenir un ID de mise à jour correspondant. Le maintien des ID de mise à jour correspondants nécessite uniquement que le dispositif incrémente de un l'ID de mise à jour Dispositif-Contrôleur à chaque cycle. Une fois que les ID de mise à jour sont mis en correspondance au démarrage, ils le restent par l'incrémentation correspondante à chaque cycle. En cas de perte ou de retard du paquet Contrôleur-Dispositif associé, l'ID de mise à jour Dispositif-Contrôleur doit être incrémenté comme si ce paquet était arrivé à temps. Cela permet au contrôleur CIP Motion de passer sur les paquets Contrôleur-Dispositif perdus ou manqués et de maintenir la synchronisation avec les ID de mise à jour correspondants.

L'ID de mise à jour s'apparente au nombre de séquences de messages CIP et permet au contrôleur de déterminer si la mémoire tampon de connexion contient de nouvelles données. Si l'ID de mise à jour, tel que vu par le contrôleur, n'a pas été modifié (paquet retardé) ou a été manqué (paquet perdu), le contrôleur passe sur la mise à jour retardée

ou perdue en utilisant une méthode d'extrapolation en fonction de la précédente trajectoire d'axe, tant que la nouvelle mise à jour n'est pas arrivée. Comme le dispositif CIP Motion, lorsque le nombre de mises à jour manquées consécutives atteint une limite configurée, le contrôleur déclare un défaut de synchronisation de connexion.

- **État de nœud:** Contient les bits utilisés pour indiquer l'état du nœud de communication du dispositif associé. La valeur de cet élément est déduite de l'attribut de classe de l'Objet Axe de dispositif de mouvement du même nom. Voir 7.2.2.2 (Objet Axe de dispositif de mouvement) pour plus de détails.

6.4.4.2.2 En-tête de connexion fixe

Si le format de connexion est une connexion Dispositif-Contrôleur fixe, l'en-tête ci-dessus est immédiatement suivi du Bloc de données d'instance.

6.4.4.2.3 En-tête de connexion variable

Si le format de connexion est une connexion Dispositif-Contrôleur variable, l'en-tête de connexion contient alors des champs supplémentaires relatifs à l'adressage et à la datation de dispositif multiaxe (voir Figure 29).

En-tête de connexion			
Format de connexion	Révision de format	ID de mise à jour	État de nœud
Compte d'instances	Défaut/Alarme de nœud	Dernier ID reçu	Ensemble de données temporelles
Datation du dispositif			
Décalage temporel du dispositif			
Perte de mises à jour	Mises à jour retardées	—	
Datation reçue des données			
Datation de transmission des données			

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Figure 29 – En-tête de connexion

- **Compte d'instances:** Identique à la définition Contrôleur-Dispositif.
- **Défaut/Alarme de nœud:** Cet élément à 8 bits contient deux codes à 4 bits pour les conditions de défaut et d'alarme associées au nœud de communication du dispositif, comme spécifié à la Figure 30. Les valeurs de ces éléments sont déduites des attributs de classe de l'Objet Axe de dispositif de mouvement du même nom. Pour plus de détails, se reporter à l'Objet Axe de dispositif de mouvement.

Bit 7		Bit 0
Défaut/Alarme de nœud		
Code de défaut de nœud – 4 bits		Code d'alarme de nœud – 4 bits

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Figure 30 – Défaut/Alarme de nœud

- **Dernier ID reçu:** ID de mise à jour du dernier bloc de données de connexion contrôleur-dispositif effectivement traité par le dispositif. Le dernier ID reçu est utilisé par le contrôleur pour déterminer si les données envoyées ont été traitées avec succès par le dispositif. Par exemple, le décalage temporel du contrôleur est envoyé par ce dernier dans le cadre de la connexion Contrôleur-Dispositif uniquement si la valeur a changé depuis la dernière mise à jour. Il est recommandé que l'élément Dernier ID reçu soit lu par le contrôleur afin de déterminer si le décalage temporel de ce dernier a été traité par le dispositif et peut par conséquent être supprimé des mises à jour ultérieures de la connexion Contrôleur-Dispositif.
- **Ensemble de données temporelles:** Identique à la définition Contrôleur-Dispositif.
- **Décalage temporel du dispositif:** Cet élément est inclus dans l'en-tête de connexion si le bit Datation est défini dans l'élément Ensemble de données temporelles. L'élément Décalage temporel du dispositif représente la valeur de décalage du temps système de

64 bits associée à la datation du dispositif qui suit. La valeur Décalage temporel du dispositif permet au contrôleur de déterminer si le temps système défini dans le dispositif est étalé par rapport à celui du contrôleur. En principe, les temps système entre le dispositif et le contrôleur sont proches, quel que soit le temps considéré. Mais, conformément au protocole de synchronisation CIP défini dans l'IEC 61588:2009, l'horloge maîtresse du système peut corriger toutes les secondes la référence du temps système, parfois d'une durée significative (une heure, par exemple). En effet, le maître lui-même peut changer si une horloge maîtresse de qualité supérieure est détectée par le système. Ces modifications brusques apportées à la référence du temps système se propagent vers les différents dispositifs présents sur le réseau, de sorte que le contrôleur et le dispositif fonctionnent avec des valeurs de temps système étalées. La valeur Décalage temporel du dispositif peut être utilisée conjointement avec la valeur Décalage du temps système du contrôleur, afin de déterminer si le temps système entre le dispositif et le contrôleur est étalé et, si tel est le cas, afin de compenser cet étalement.

- **Datation du dispositif:** Cet élément est inclus dans l'en-tête de connexion si le bit Datation est défini dans l'élément Ensemble de données temporelles. La datation du dispositif représente la valeur de temps système de 64 bits, en nanosecondes, lorsque l'événement de synchronisation de mise à jour du dispositif associé aux données réelles de la structure de connexion s'est produit (lorsque les données réelles ont été capturées, par exemple). Calculée par le dispositif, il s'agit de la somme de la valeur d'horloge locale du dispositif en cas d'événement de synchronisation de mise à jour et de la valeur Décalage du temps système du dispositif. Avec la datation du dispositif, le contrôleur dispose de toutes les informations nécessaires à la correction des valeurs de données réelles pour connaître les différences entre la synchronisation de mise à jour du dispositif et celle du contrôleur lorsque la période de mise à jour du contrôleur n'est pas un entier multiple de la période de mise à jour du dispositif ou que les mises à jour de dispositif présentent un déphasage par rapport au contrôleur. Ce modèle de synchronisation suppose d'enregistrer la datation du dispositif au début de la période de mise à jour du dispositif, qui est également le moment où la réaction a été capturée pour la dernière fois. Si la datation du dispositif ne correspond pas à la datation de mise à jour locale du contrôleur, ce dernier extrapole la valeur instantanée de données de réponse en fonction de la trajectoire de manière à correspondre à la datation du contrôleur. Le diagramme de synchronisation de la Figure 31 représente comment le contrôleur ajuste les données de position d'axe provenant du dispositif en fonction des datations relatives entre le dispositif et le contrôleur.

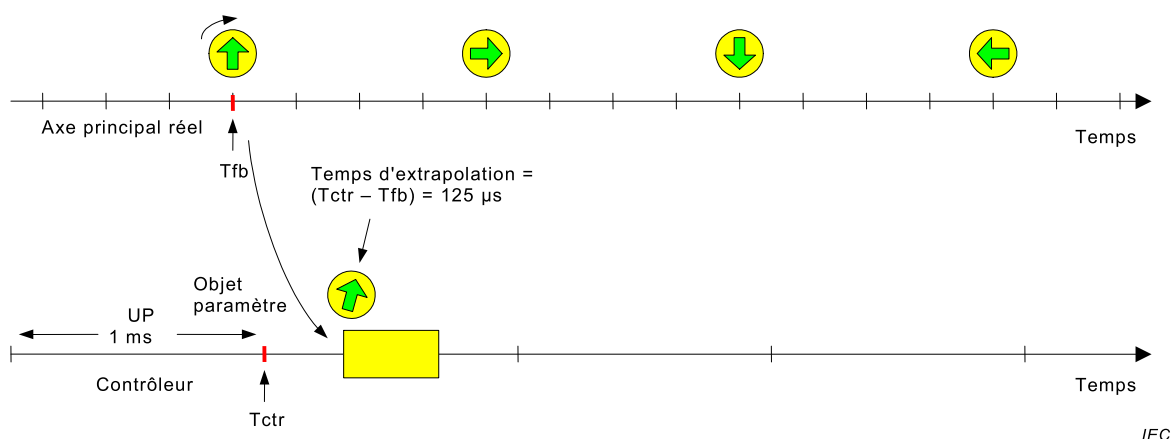


Figure 31 – Ajustement des données de position réelles en fonction de la datation du dispositif

- **Perte de mises à jour:** Cet élément est inclus dans l'en-tête de connexion si le bit Diagnostic de mise à jour est défini dans l'élément Ensemble de données temporelles. La valeur Perte de mise à jour représente le nombre de paquets perdus de connexion Contrôleur-Dispositif détectés depuis l'ouverture et la synchronisation de la connexion. Les paquets perdus sont détectés par examen de l'ID de mise à jour des paquets reçus. Un paquet est déterminé comme perdu lorsque son ID de mise à jour prévu est omis. L'attribut de classe Perte de paquet est incrémenté pour chaque ID de mise à jour omis. Le compteur se renouvelle toutes les 256 mises à jour manquées.

- **Mises à jour retardées:** Cet élément est inclus dans l'en-tête de connexion si le bit Diagnostic de mise à jour est défini dans l'élément Ensemble de données temporelles. La valeur Mises à jour retardées représente le nombre de paquets retardés de connexion Contrôleur-Dispositif détectés depuis l'ouverture et la synchronisation de la connexion. Les paquets retardés sont détectés par examen de la datation du paquet reçu. Un paquet est déterminé comme retardé lorsque la différence entre la datation de son contrôleur et la datation reçue des données du dispositif dépasse la période de mise à jour du contrôleur. Le compteur se renouvelle toutes les 256 mises à jour retardées.
- **Datation reçue des données:** Cet élément est inclus dans l'en-tête de connexion si le bit Diagnostic temporel est défini dans l'élément Ensemble de données temporelles. La datation reçue des données représente la valeur de temps système à 64 bits au moment de la saisie des données de la dernière connexion Contrôleur-Dispositif, indiquées par l'ID de dernière mise à jour, dans la mémoire du dispositif et au moment où ces mêmes données sont prêtes à être traitées par la couche application. La datation est exprimée en nanosecondes. Cette valeur, combinée à la datation de transmission de données correspondante du contrôleur, peut être utilisée par le contrôleur pour générer des statistiques de livraison de données de connexion Contrôleur-Dispositif.
- **Datation de transmission des données:** Cet élément est inclus dans l'en-tête de connexion si le bit Diagnostic temporel est défini dans l'élément Ensemble de données temporelles. La datation de transmission des données représente la valeur de temps système à 64 bits au moment du lancement par la couche application de la transmission des données de connexion Dispositif-Contrôleur au contrôleur. La datation est exprimée en nanosecondes. Cette valeur, combinée à la datation reçue de données correspondante du contrôleur, peut être utilisée par le contrôleur pour générer des statistiques de livraison de données de connexion Dispositif-Contrôleur.

6.4.4.3 Blocs de données d'instance

6.4.4.3.1 Structure générale

L'En-tête de connexion est suivi d'un ou de plusieurs Blocs de données d'instance, comme le montre le Compte d'instances ci-dessus. Le Bloc de données d'instance s'apparente beaucoup à celui de la connexion Contrôleur-Dispositif, sa structure de base étant présentée à la Figure 32.

Bloc de données d'instance
En-tête de données d'instance
Bloc de données cycliques
Bloc de données d'événement
Bloc de données de service

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Figure 32 – Bloc de données d'instance

6.4.4.3.2 En-tête de données d'instance

L'En-tête de données d'instance contient les informations essentielles de configuration d'axe nécessaires à l'analyse et à l'application des données contenues dans les trois canaux de données (voir Figure 33). Cet en-tête est uniquement inclus dans le format Connexion variable afin de s'adapter aux applications de dispositif multiaxe. Les informations qu'il contient peuvent être utilisées par l'interface de communication du dispositif pour copier les blocs de données individuels dans des mémoires fixes séparées pour le traitement.

S'il est configuré pour un format Connexion fixe, seul le Bloc de données cycliques d'une seule instance d'axe est pris en charge. Les informations relatives au numéro d'instance ou à la taille du bloc ne sont donc pas utiles. Par conséquent, l'En-tête de données d'instance n'est pas inclus dans la structure de connexion.

En-tête de données d'instance			
Numéro d'instance	–	Taille de bloc d'instance	Taille de bloc cyclique
Taille de bloc réel cyclique	Taille de bloc de lecture cyclique	Taille de bloc d'événement	Taille de bloc de service

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Figure 33 – En-tête de données d'instance

- Numéro d'instance: Identique à la définition Contrôleur-Dispositif.
- Taille de bloc d'instance: Identique à la définition Contrôleur-Dispositif.
- Taille de bloc cyclique: Identique à la définition Contrôleur-Dispositif.
- Taille de bloc réel cyclique: Cette valeur représente la taille du bloc de données réelles cycliques, en unités de mot à 32 bits, en-tête compris. Une taille de bloc réel cyclique de 0 indique que le contrôleur n'a à traiter aucune donnée réelle cyclique.
- Taille de bloc de lecture cyclique: Cette valeur représente la taille du bloc de données de lecture cycliques, en unités de mot à 32 bits, en-tête compris. Une taille de bloc de lecture cyclique de 0 indique qu'aucune donnée de lecture cyclique n'est actuellement configurée pour un transfert vers le contrôleur et, par conséquent, le contrôleur n'a à traiter aucun bloc de lecture cyclique.
- Taille de bloc d'événement: Identique à la définition Contrôleur-Dispositif. Une taille de bloc d'événement de 0 indique qu'aucune vérification d'événement active n'est actuellement en cours.
- Taille de bloc de service: Identique à la définition Contrôleur-Dispositif. Une taille de bloc de service de 0 indique qu'il n'y a actuellement lieu de répondre à aucune demande de service.

6.4.4.3.3 Bloc de données cycliques

L'en-tête de données cycliques, placé en haut du bloc de données cycliques de la connexion Dispositif-Contrôleur, est toujours inclus, quel que soit le format de connexion. Cet en-tête contient les éléments essentiels relatifs au contenu du bloc de données cycliques et au contexte des données dans le bloc concernant le dispositif (voir Figure 34). Pour la plupart, ces éléments sont établis par les éléments correspondants du précédent bloc de données cycliques de la connexion Contrôleur-Dispositif, et en sont donc des copies directes. Par conséquent, le contenu du bloc de données cycliques de la connexion Dispositif-Contrôleur est en dernier lieu déterminé par le contrôleur.

Bloc de données cycliques			
Mode de commande	Mode de réaction	Réponse d'axe	État de réponse
–	Ensemble de données réelles	Ensemble de données d'état	État d'axe
Données réelles cycliques			
Données d'état cycliques			

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Figure 34 – Bloc de données cycliques

- Mode de commande: Identique à la définition Contrôleur-Dispositif.
- Mode de réaction: Identique à la définition Contrôleur-Dispositif.
- Réponse d'axe: La réponse d'axe à 8 bits est une valeur énumérée utilisée pour établir une liaison avec l'élément Commande d'axe correspondant de la connexion Contrôleur-Dispositif afin d'initier directement le fonctionnement du dispositif qui n'exige pas de demande de service CIP. Les codes d'acquiescement valides correspondent aux codes de demande correspondants de l'élément Commande d'axe et sont présentés dans le Tableau 22.

Tableau 22 – Réponse d'axe

Code d'acquiescement	Réponse d'axe
0	Pas d'acquiescement
1	Acquiescement d'activation
2	Acquiescement de désactivation
3	Acquiescement d'interruption
4	Acquiescement de réinitialisation suite à une interruption
5	Acquiescement d'abandon
6	Acquiescement de réinitialisation suite à un défaut
7	Arrêter le processus
8	Acquiescement de changement de référence de position réelle
9	Acquiescement de changement de référence de position de consigne
10 à 126	(Réservé)
127	Acquiescement d'annulation
128 à 255	(Spécifique au fournisseur)

Ce mécanisme Consigne de dispositif/Réponse d'axe permettant d'initier les modifications d'état est décrit de manière exhaustive dans le paragraphe Commande d'état de l'Objet Axe de dispositif de mouvement (voir 7.6.1.2).

- **État de réponse:** En présence d'un Code d'acquiescement différent de zéro dans la Réponse d'axe, une valeur État de réponse est également fournie pour indiquer le succès ou l'échec de l'opération Commande d'axe demandée. Un État de réponse nul (0) indique le succès, une valeur différente de zéro indiquant une erreur. Les valeurs État de réponse sont conformes à la spécification CIP des codes État général (voir IEC 61158-6-2).
- **Ensemble de données réelles:** Identique à la définition Contrôleur-Dispositif.
- **Ensemble de données d'état:** Identique à la définition Contrôleur-Dispositif. Les bits Ensemble de données d'état sont définis uniquement lorsque, 1) le bit correspondant dans l'ensemble de données d'état de la connexion Contrôleur-Dispositif est défini, et 2) il y a variation des valeurs de données d'état associées depuis la dernière mise à jour de connexion.
- **État d'axe:** Cet élément de données contient la valeur énumérée État d'axe indiquant l'état en cours d'axe d'instance de ce dispositif en fonction du modèle d'état de l'Objet Axe de dispositif de mouvement.
- **Données réelles/d'état cycliques:** Les données réelles/d'état cycliques contiennent des données à priorité élevée qu'il faut appliquer à l'instance d'axe de dispositif associée lors de la mise à jour suivante du dispositif. Ce bloc est composé des éléments de données réelles et des éléments de données d'état utilisés par le contrôleur, comme le déterminent de manière explicite les éléments Ensemble de données réelles et Ensemble de données d'état de l'En-tête de données cycliques. Voir les définitions Contrôleur-Dispositif de 6.4.3 pour plus de détails de la structure Données réelles/d'état cycliques. Les données réelles cycliques doivent être référencées par rapport à la datation de dispositif.

6.4.4.3.4 Bloc de données de lecture cycliques

Le bloc de données de lecture cycliques peut être utilisé pour mettre à jour de manière synchrone un ou plusieurs attributs de l'Objet Axe de commande de mouvement ciblés dans le contrôleur en fonction de la valeur en cours des attributs associés dans le dispositif. Ce mécanisme peut par exemple être utilisé conjointement avec un programme de bloc de fonction conforme à l'IEC 61131-3 afin de mettre en œuvre un contrôle de boucle externe sophistiqué en fonction d'un large éventail de signaux de commande d'axe disponibles. À l'inverse des demandes de service Extraction d'attribut d'axe du canal de service, dont le

traitement peut durer plusieurs cycles de mise à jour de dispositif, le mécanisme de données de lecture cycliques garantit la mise à jour du paramètre ciblé à chaque cycle de connexion.

Le bloc de données de lecture cycliques est uniquement pris en charge dans le format Connexion variable (voir Figure 35).

Bloc de données de lecture cycliques			
ID de bloc d'écriture cyclique	État d'écriture cyclique	ID de bloc de lecture cyclique	État de lecture cyclique
Données de lecture cycliques			

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Figure 35 – Bloc de données de lecture cycliques

L'en-tête associé pour ce bloc contient les éléments essentiels relatifs au contenu des blocs de données de lecture et d'écriture cycliques de la mise à jour précédente de la connexion Contrôleur-Dispositif.

- **ID de bloc d'écriture cyclique:** Cet ID de 8 bits détermine la structure prédéfinie du bloc d'écriture cyclique envoyée dans la dernière mise à jour de la connexion Contrôleur-Dispositif. Une mise à jour réussie du bloc de données d'écriture cycliques est indiquée par la mise en écho de l'ID du bloc d'écriture cyclique associé dans l'en-tête de bloc de données de lecture cycliques suivant. Lorsque le dispositif détermine qu'une erreur est associée aux données d'écriture cycliques, il l'identifie au moyen d'une valeur État d'écriture cyclique différente de zéro.
- **État d'écriture cyclique:** La valeur État d'écriture cyclique est destinée à indiquer le succès ou l'échec de la mise à jour associée des données d'écriture cycliques. Un État d'écriture cyclique nul (0) indique le succès, une valeur différente de zéro indiquant une erreur. Les valeurs État d'écriture cyclique sont conformes à la spécification CIP des codes État général (voir IEC 61158-6-2).
- **ID de bloc de lecture cyclique:** Cet ID à 8 bits détermine la structure prédéfinie du bloc de lecture cyclique à appliquer aux données de lecture cycliques pour cette mise à jour. Les structures de bloc de lecture cyclique sont définies à l'aide du service Définition de la liste de données de lecture cycliques. Une réponse correcte à ce service inclut un nouvel ID de bloc de lecture cyclique qui peut être utilisé dans la mise à jour de connexion suivante pour transmettre les données cycliques dans ce format. Lorsque le dispositif détermine qu'une erreur est associée aux données de lecture cycliques, il l'identifie au moyen d'une valeur État de lecture cyclique différente de zéro et sans données de lecture cycliques.
- **État de lecture cyclique:** La valeur État de lecture cyclique est destinée à indiquer le succès ou l'échec de la mise à jour associée des données de lecture cycliques. Un État de lecture cyclique nul (0) indique le succès, une valeur différente de zéro indiquant une erreur. Les valeurs État de lecture cyclique sont conformes à la spécification CIP des codes État général (voir IEC 61158-6-2).
- **Données de lecture cycliques:** Les Données de lecture cycliques contiennent des données à priorité élevée qu'il faut appliquer à l'instance d'axe de contrôleur associée. Ce bloc est composé d'un signal et d'éléments de données d'état à adapter et appliquer aux attributs d'Objet Axe de commande de mouvement correspondants. Le contenu des données de lecture cycliques est déterminé de manière explicite par la structure identifiée par l'ID de bloc de lecture cyclique se trouvant dans l'en-tête de données de lecture cycliques. L'ordre des données d'attribut dans le Bloc de données de lecture cycliques est déterminé par celui des ID d'attribut de la demande Set_Cyclic_Read_List. Les éléments de données d'attribut doivent être du texte aligné. Les mots à 32 bits et à 16 bits sont respectivement alignés sur 32 bits et 16 bits. Le remplissage peut être ajouté pour conserver l'alignement des mots.

EXEMPLE La Figure 36 représente un bloc de données de lecture cycliques avec 3 attributs où les Attributs 1 et 3 sont des valeurs à 32 bits, tandis que l'Attribut 2 est une valeur à 16 bits.

Valeur d'attribut 1	
Valeur d'attribut 2	16 bits – réservé
Valeur d'attribut 3	

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Figure 36 – Bloc de données de lecture cycliques**6.4.4.4 Bloc de données d'événement**

Le bloc de données d'événement permet d'envoyer plusieurs notifications d'événement au contrôleur dans le cadre d'une mise à jour donnée. Chaque notification d'événement implique un acquittement d'événement spécifique indiquant le succès ou une erreur. Les données de notification d'événement restent dans la structure de données de connexion Dispositif-Contrôleur tant que le dispositif n'a pas reçu l'acquiescement d'événement correspondant de la part du contrôleur.

Le format du bloc de données d'événement de la connexion Dispositif-Contrôleur est représenté à la Figure 37.

Bloc de données d'événement			
État de vérification d'événement			
Acquittement de données d'enregistrement	Acquittement de données d'origine	Acquittement de données d'observation	–
ID d'événement 1	État d'événement 1	Type d'événement 1	–
Position d'événement 1			
Datation d'événement 1			
ID d'événement 2	État d'événement 2	Type d'événement 2	–
Position d'événement 2			
Datation d'événement 2			
...			

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Figure 37 – Bloc de données d'événement

- **État de vérification d'événement:** Ce mot à 32 bits indique si le dispositif recherche actuellement les événements en fonction des différentes entrées du dispositif (les entrées de marqueur, d'origine et d'enregistrement, par exemple). La vérification d'événement est initiée lorsque le bit Commande de vérification d'événement correspondant est défini dans la connexion Contrôleur-Dispositif. Lorsqu'un événement se produit, le dispositif capture l'heure et la position exacte d'axe, puis transmet ces informations au contrôleur dans des Blocs de données d'événement. Mais pour que le contrôleur puisse traiter les données d'événement, le bit d'État de vérification d'événement correspondant doit être défini. Pour plus de détails sur la manière d'utiliser ce mot dans les opérations d'événement, voir la séquence de mécanisme d'événement en 6.4.3.4. Les 4 derniers bits de l'élément Commande de vérification d'événement constituent un champ de bit spécial, dont les 3 bits de poids faible spécifient le nombre d'événements actifs, et qui est littéralement le nombre d'ID d'événement figurant dans ce bloc de données d'événement. Le bit de poids fort active le format du Bloc de données d'événement étendu. Une définition exhaustive de l'attribut État de vérification d'événement peut être consultée en 7.3.7.2.2.

En présence du bloc de données d'événement, l'élément État de vérification d'événement détermine quels sont les éléments supplémentaires inclus. Le mot incluant les acquittements des données d'enregistrement, d'origine et d'observation est inclus uniquement si le bit Format Étendu est défini dans le mot État de vérification d'événement. Les éléments # d'ID, d'état, de type, de position et de datation d'événement sont répétés entre 0 et 7 fois, comme déterminé par le champ Compte des blocs d'événement dans le mot État de vérification d'événement.

- **Acquittement de données d'enregistrement:** Cet octet de matriciel sert à l'acquiescement des paramètres de bloc de données d'enregistrement issus d'une mise à jour précédente de la connexion Contrôleur-Dispositif. La structure des données de connexion Dispositif - Contrôleur ne comporte aucune donnée associée à ces bits.

- **Acquittement de données d'origine:** Cet octet de matriciel sert à l'acquittement des paramètres de bloc de données d'origine issus d'une mise à jour précédente de la connexion Contrôleur-Dispositif. La structure des données de connexion Dispositif-Contrôleur ne comporte aucune donnée associée à ces bits.
- **Acquittement de données d'observation:** Cet octet de matriciel sert à l'acquittement des paramètres de bloc de données d'observation issus d'une mise à jour précédente de la connexion Contrôleur-Dispositif. La structure des données de connexion Dispositif-Contrôleur ne comporte aucune donnée associée à ces bits.
- **ID d'événement:** Numéro de transaction affecté à cet événement spécifique par la notification d'événement d'origine. Un ID d'événement est affecté à chaque événement par incrémentation de l'ID d'événement en cours archivé dans le dispositif. Grâce à l'ID d'événement, le dispositif est capable de mettre en correspondance l'acquittement d'événement et la notification d'événement appropriée pour procéder à la transaction de données d'événement.
- **État d'événement:** Valeur énumérée indiquant l'état de la notification d'événement d'origine. Une valeur nulle indique que l'événement a été détecté avec succès par le dispositif. Une valeur différente de zéro indique qu'une erreur s'est produite dans la vérification de l'événement qui empêche que l'événement soit détecté avec succès. Ceci pourrait se produire en cas de problème avec le matériel lié à l'événement ou lorsque le matériel exigé n'est pas pris en charge par le dispositif. La définition applicable aux codes d'erreur État d'événement est laissée actuellement au choix du fournisseur.
- **Type d'événement:** Cette valeur énumérée décrit le type d'événement qui s'est produit. Des types d'événements valides sont spécifiés dans le Tableau 23.

Tableau 23 – Type d'événement

Type d'événement	Description de l'événement
0	Front positif de l'enregistrement 1
1	Front négatif de l'enregistrement 1
2	Front positif de l'enregistrement 2
3	Front négatif de l'enregistrement 2
4	Front positif du marqueur
5	Front négatif du marqueur
6	Front positif de l'interrupteur d'origine
7	Front négatif de l'interrupteur d'origine
8	Marqueur – Interrupteur d'origine ++
9	Marqueur – Interrupteur d'origine +-
10	Marqueur – Interrupteur d'origine -+
11	Marqueur – Interrupteur d'origine --
12	Seuil de couple d'origine
13	Position d'observation 1 Avant
14	Position d'observation 1 Inverse
15	Position d'observation 2 Avant
16	Position d'observation 2 Inverse
17 à 127	(réservé)
128 à 256	(spécifique au fournisseur)

- **Position d'événement:** Représentation entière à 32 bits de la position d'axe lorsque l'événement désigné s'est produit. Lorsque l'élément État d'événement a un code d'erreur différent de zéro, la Position d'événement est mise à 0.

- **Datation de l'événement:** Cet élément représente la valeur de temps système à 64 bits lorsque l'événement spécifié s'est produit. La datation est exprimée en nanosecondes. S'il est accompagné de la valeur Décalage du temps système du dispositif, qui fait partie intégrante de l'en-tête connexion Dispositif-Contrôleur, le contrôleur dispose de toutes les informations nécessaires pour calculer le temps système absolu lorsque l'événement s'est produit. Lorsque l'élément État d'événement a un code d'erreur différent de zéro, l'heure d'événement est mise à 0.

6.4.4.5 Bloc de données de service

Le Bloc de données de service permet d'envoyer une réponse de service par instance au contrôleur dans une mise à jour donnée. Chaque demande de service requiert une réponse de service spécifique de la part du dispositif, indiquant un succès ou une erreur. Dans certains cas, le service de réponse contient les données demandées. Dans tous les cas, les données de réponse de service restent dans la structure de données de connexion Dispositif-Contrôleur tant que le dispositif n'a pas vu que la demande de service associée a été supprimée du Bloc de données d'instance de connexion Contrôleur-Dispositif (Taille de bloc de service = 0) ou qu'une nouvelle demande de service n'a pas été émise par le contrôleur (ID de transaction incrémenté).

Chaque réponse de service est représentée par un bloc de données organisé comme indiqué à la Figure 38.

NOTE À l'instar de la structure de demande, la structure de la réponse de service ne suit pas le format de messagerie standard CIP traditionnel. La raison principale est que cette structure de connexion est fondamentalement une connexion E/S CIP implicite et pas une connexion de messagerie explicite. Toutefois, dans le cas d'un format de connexion fixe, les données de demande spécifique au service définies ci-dessous sont envoyées par l'intermédiaire d'une connexion de messagerie explicite et suivent les règles CIP en matière de format de demande de service explicite.

Bloc de données de service			
ID de transaction	Code de service	État général	État étendu
Données de réponse spécifique au service			

IEC

Figure 38 – Bloc de données de service

- **ID de transaction:** Numéro de transaction affecté à cette réponse de service déduite de l'ID de transaction de la demande d'origine. Un nouvel ID de transaction est affecté à chaque demande de service par incrémentation de l'ID de transaction en cours archivé dans le contrôleur. Grâce à l'ID de transaction de la réponse, le contrôleur est en mesure d'associer la réponse de service à la demande de service appropriée.
- **Code de service:** Identificateur qui détermine la réponse de service spécifique qui suit, qui doit correspondre au code de service de la demande de service d'origine. Une liste des codes de service valides pour l'Objet Axe de dispositif de mouvement est donnée dans le paragraphe Connexion Contrôleur-Dispositif.
- **État général:** La valeur État général indique le succès ou l'échec de la demande de service. Un État général nul (0) indique le succès, une valeur différente de zéro indiquant une erreur. Les valeurs État général sont conformes à la spécification CIP des codes État général (voir IEC 61158-6-2).
- **État général étendu:** L'état général étendu offre une méthode de définition des codes d'erreur spécifiques au fournisseur ou au service. Ces codes ne font actuellement l'objet d'aucune définition standard.
- **Données de réponse spécifiques au service:** Le format et la syntaxe des données de réponse spécifiques au service dépendent du code de service spécifié.

6.4.5 Format de connexion E/S de mouvement fixe

En spécifiant un format de connexion fixe, la connexion E/S CIP Motion peut être réduite à une taille aisément applicable aux réseaux ou aux dispositifs CIP basse performance. En conformité avec ce contexte d'application, les fonctions suivantes ont été retirées de la

structure de connexion pour prendre en charge les exigences d'une taille de connexion fixe et d'une bande passante de réseau limitée.

- Datation
- Défaut/Alarme de nœud
- Prise en charge de plusieurs instances
- Taille de bloc dynamique
- Bloc de données de lecture/écriture cycliques
- Bloc de données d'événement
- Bloc de données de service

Avec le format de connexion fixe, les demandes de service adressées à l'Objet Axe de dispositif de mouvement sont prises en charge uniquement en tant que service de Messagerie explicite.

La Figure 39 et la Figure 40 présentent un exemple de format de connexion fixe utilisé dans une simple application de dispositif d'entraînement à vitesse variable exigeant une seule consigne de vitesse et donnant la vitesse réelle. Dans ce cas, la taille de connexion a été réduite à 16 octets, taille bien adaptée aux réseaux basse performance (DeviceNet, par exemple).

Format de connexion contrôleur-dispositif			
Format de connexion	Révision de format	ID de mise à jour	Contrôle de nœud
Mode de commande	Mode de réaction	Commande d'axe	–
Ensemble de données de consigne	Ensemble de données réelles	Ensemble de données d'état	–
Vitesse de consigne			

IEC

Figure 39 – Format de connexion Contrôleur-Dispositif fixe (taille fixe = 16 octets)

Format de connexion dispositif-contrôleur			
Format de connexion	Révision de format	ID de mise à jour	État de nœud
Mode de commande	Mode de réaction	Réponse d'axe	État de réponse
–	Ensemble de données réelles	Ensemble de données d'état	État d'axe
Vitesse réelle			

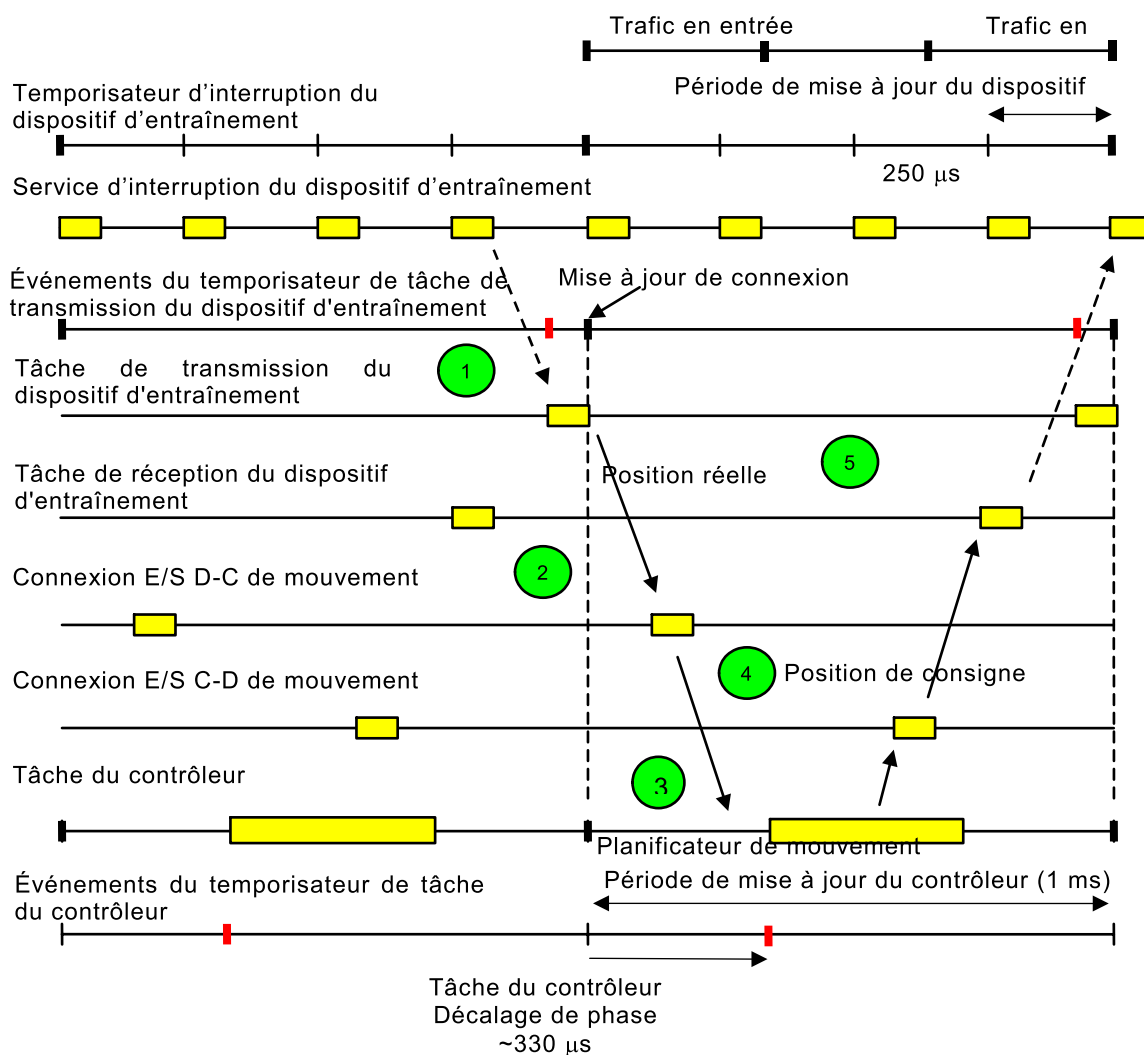
IEC

Figure 40 – Format de connexion Dispositif-Contrôleur fixe (taille fixe = 16 octets)

6.4.6 Modèle de synchronisation de Connexion E/S CIP Motion

6.4.6.1 Généralités

Le modèle de synchronisation général de l'échange de données de connexion E/S CIP Motion est décrit dans ce paragraphe dans le contexte d'un dispositif d'entraînement CIP Motion mais il s'applique également à d'autres types de dispositifs CIP Motion. L'échange de données entre le dispositif d'entraînement et le contrôleur est assuré par le contrôleur avec un paquet de données Dispositif-Contrôleur envoyé par paquet de données Contrôleur-Dispositif reçu. Les paquets de connexion Contrôleur-Dispositif sont envoyés périodiquement en fonction de la période de mise à jour du contrôleur configurée. La période de mise à jour du dispositif, c'est-à-dire la période de mise à jour à laquelle le dispositif d'entraînement exécute ses calculs de commande, est en général plus rapide que la période de mise à jour du contrôleur. Le modèle de synchronisation de connexion CIP Motion de base est représenté à la Figure 41.



IEC

Figure 41 – Modèle de synchronisation à 1 cycle CIP Motion

Ce modèle est classé comme modèle de synchronisation à 1 cycle étant donné qu'une transaction de données E/S exhaustive a lieu dans un cycle de mise à jour de la connexion. Le modèle de synchronisation est composé de cinq étapes de base représentées à la Figure 41 et décrites ci-après.

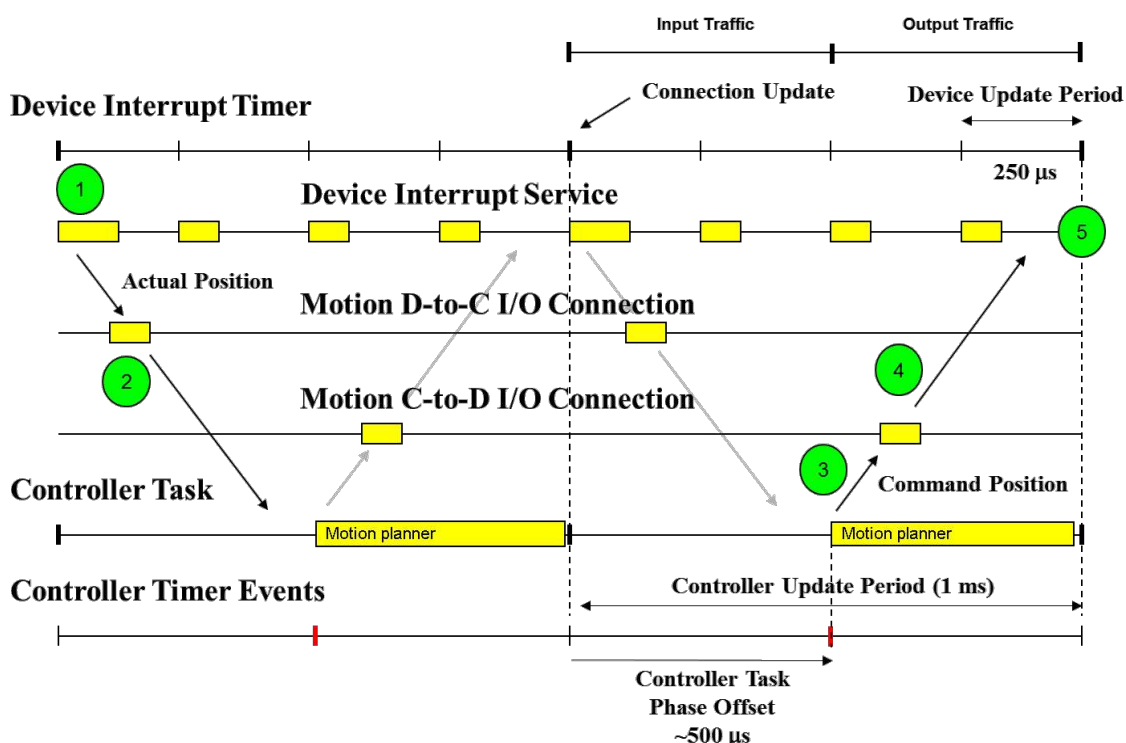
- 1) Événement périodique de synchronisation de transmission du dispositif d'entraînement (marque de graduation rouge supérieure indiquée ci-dessus) programmé juste avant que le cycle suivant de mise à jour de la connexion ne déclenche la tâche de transmission du dispositif d'entraînement chargée de gérer la connexion Dispositif-Contrôleur. La tâche de transmission du dispositif d'entraînement obtient la toute dernière valeur Position réelle calculée par le dernier Service d'interruption du dispositif d'entraînement (à savoir le mode d'asservissement de position dans cet exemple) et la datation associée.
- 2) Une fois les données Position réelle et Datation assemblées, avec d'autres données d'axe, dans la structure des données de connexion Dispositif-Contrôleur, le dispositif d'entraînement transmet le paquet au contrôleur. Pour de meilleures performances, il est recommandé que la transmission réelle soit le plus proche possible du début du cycle Mise à jour de connexion. La seule contrainte de synchronisation réside dans le fait que, en fonctionnement normal, le dispositif d'entraînement doit lancer la transmission au plus tard au début du cycle. Ceci prévoit une durée de traversée du réseau par le paquet connexion Dispositif-Contrôleur qui arrive au contrôleur avant le début de la tâche de contrôleur retardée par phase.

- 3) Après une durée prédéterminée de décalage de phase par rapport au début du dernier cycle de mise à jour du contrôleur qui, dans ce modèle, est réglée pour correspondre à $1/3$ de la période de mise à jour dudit contrôleur, un événement de synchronisation de tâche du contrôleur déclenche le lancement de ladite tâche. La tâche du contrôleur commence par l'analyse des données provenant du dispositif d'entraînement, y compris la valeur Position réelle. Ensuite, le contrôleur lance un planificateur de mouvement pour calculer une nouvelle valeur Position de consigne à renvoyer au dispositif d'entraînement. Lorsque les systèmes d'engrenage ou de came sont actifs, il peut être nécessaire d'utiliser la position réelle d'un axe d'entraînement maître en entrée pour calculer la position de consigne d'un ou de plusieurs axes d'entraînement esclaves.
- 4) Une fois la Position de consigne calculée, elle est assemblée avec d'autres données d'axe, dans la structure des données de connexion Contrôleur-Dispositif, le contrôleur transmet les données au dispositif d'entraînement. En général, il est recommandé que la transmission se produise au plus tard aux $2/3$ de la durée de la période de mise à jour du contrôleur afin de permettre aux paquets d'atteindre le dispositif d'entraînement ciblé avant la fin du cycle en cours. Le paquet Connexion Contrôleur-Dispositif traverse le réseau et arrive au dispositif d'entraînement avant le début de la période de mise à jour de connexion suivante. Aucune contrainte de synchronisation sévère ne s'applique ici; le paquet peut arriver bien plus tôt dans le cycle de mise à jour ou éventuellement après le début du cycle de mise à jour suivant. Le dispositif d'entraînement traite simplement le paquet à son arrivée.
- 5) Une tâche de réception de dispositif d'entraînement répond à la notification par l'interface réseau de l'arrivée d'un nouveau paquet Contrôleur-Dispositif. La tâche de réception de dispositif d'entraînement commence par l'analyse des données provenant du contrôleur, y compris la valeur Datation du contrôleur et Position de consigne. Le dispositif d'entraînement applique ensuite les données Position de consigne en utilisant la datation pour calculer de nouveaux coefficients applicables au polynôme d'interpolation fine. Ces coefficients sont utilisés par les services Interruption du dispositif d'entraînement suivants pour calculer la valeur de position de consigne fine qui est appliquée à l'entrée de la boucle d'asservissement de la position du dispositif d'entraînement. Pour de meilleures performances, il est recommandé d'appliquer les coefficients polynomiaux à l'interpolateur fin le plus près possible du début du cycle de mise à jour suivant. Une application précoce des nouveaux coefficients peut générer une erreur inutile dans la trajectoire de consigne lors de la jonction du segment d'interpolateur fin actuel avec le nouveau segment qui commence avec le cycle de mise à jour suivant. Une application retardée des nouveaux coefficients contraint l'interpolateur fin à adopter le cycle suivant en tant qu'extrapolateur fin, en introduisant une erreur d'extrapolation inutile dans la trajectoire de consigne lors de la jonction du segment extrapolé avec le segment d'interpolateur fin suivant. La tâche de réception de dispositif d'entraînement se termine par la programmation, par le dispositif, de l'événement suivant de synchronisation de transmission du dispositif d'entraînement par rapport au début du cycle de mise à jour de la connexion suivante, qui correspond à la somme de la datation du contrôleur et de la période de mise à jour de ce dernier.

L'exemple de séquence ci-dessus représente une mise en œuvre qui utilise des tâches distinctes de transmission et de réception de dispositif d'entraînement entraînées par interruption. D'autres mises en œuvre peuvent choisir de combiner une ou les deux tâches avec le service d'interruption de dispositif d'entraînement qui effectue les calculs de commande. Dans ce cas, le traitement des données de connexion CIP Motion s'effectue par interrogation; le service d'interruption de dispositif d'entraînement a besoin de vérifier si un paquet Contrôleur-Dispositif a été reçu et également de vérifier si le moment est venu d'assembler et d'envoyer un paquet Dispositif-Contrôleur.

Dans la mesure où les autres modèles de synchronisation peuvent éventuellement reposer sur la configuration du contrôleur, tous les modèles de synchronisation CIP Motion commencent par les données réelles envoyées par le dispositif d'entraînement au contrôleur au début du cycle de mise à jour du contrôleur, et se terminent par les données de consigne arrivant avant un cycle de mise à jour du contrôleur suivant. Toutefois, il n'est pas obligatoire que les données de consigne envoyées au dispositif d'entraînement dans un cycle de mise à jour donné du contrôleur soient le résultat des calculs exécutés pendant ce cycle. Ces règles constituent la base des modèles de synchronisation multicycle.

La Figure 42 et la Figure 43 représentent ces caractéristiques communes de modèle de synchronisation CIP Motion, en tenant compte des modèles de synchronisation à 2 cycles et à 3 cycles.



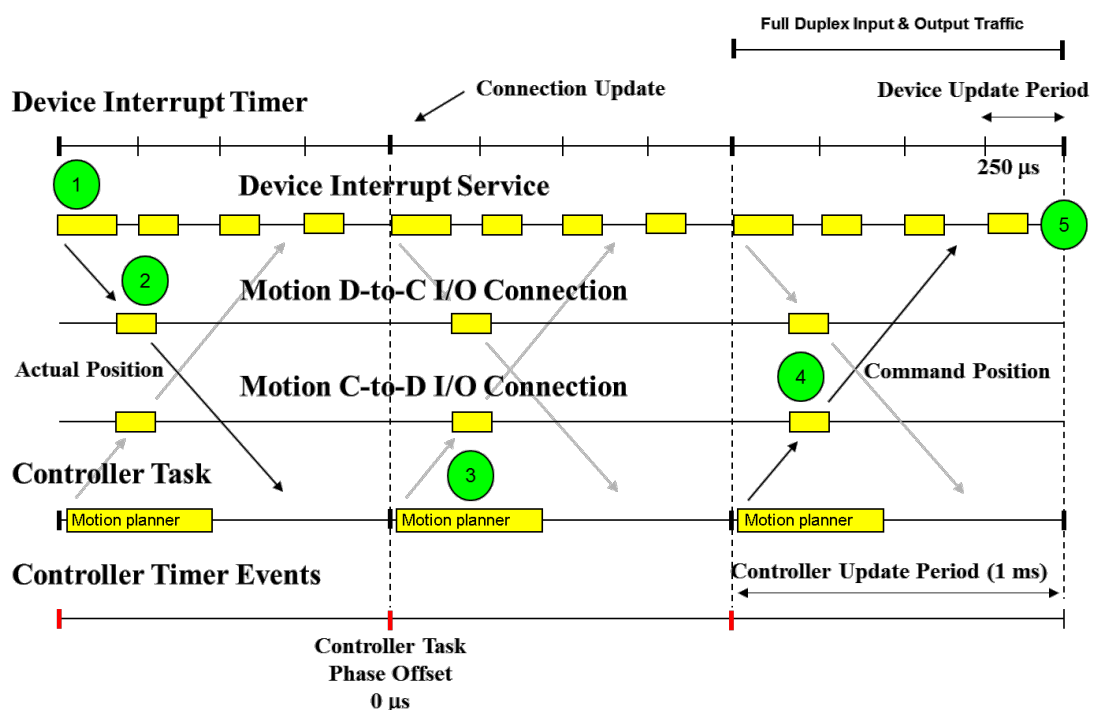
IEC

Anglais	Français
Device Interrupt Timer	Temporisateur d'interruption du dispositif
Input Traffic	Trafic en entrée
Output Traffic	Trafic en sortie
Connection Update	Mise à jour de connexion
Device Update Period	Période de mise à jour du dispositif
Device Interrupt Service	Service d'interruption du dispositif
Actual Position	Position réelle
Motion D-to-C I/O Connection	Connexion E/S D-C CIP Motion
Motion C-to-D I/O Connection	Connexion E/S C-D CIP Motion
Controller Task	Tâche du contrôleur
Command Position	Position de consigne
Motion planner	Planificateur de mouvement
Controller Timer Events	Événements de temporisateur du contrôleur
Controller Update Period	Période de mise à jour du contrôleur
Phase Offset	Décalage de phase

Figure 42 – Modèle de synchronisation à 2 cycles CIP Motion

Comme avec tous les modèles de synchronisation CIP Motion, le modèle de synchronisation à 2 cycles présenté à la Figure 42 commence par la transmission du paquet de connexion D-C au contrôleur au début du cycle de mise à jour. Toutefois, dans ce cas, la tâche du contrôleur ne commence pas avant que se soit écoulée la moitié du cycle de mise à jour, en accordant ainsi une plus longue durée au paquet de connexion D-C pour atteindre le

contrôleur avant l'exécution de la tâche Planificateur de mouvement. Contrairement au modèle de synchronisation à 1 cycle, le paquet de connexion C-D n'est pas retransmis au dispositif d'entraînement avant la prochaine exécution de la tâche Planificateur de mouvement. Cette opération accorde une nouvelle fois une plus longue durée au paquet de connexion C-D pour atteindre le dispositif d'entraînement. Ainsi, 2 cycles de connexion sont nécessaires pour accomplir la transaction de données E/S avec le dispositif d'entraînement.



IEC

Anglais	Français
Device Interrupt Timer	Temporisateur d'interruption du dispositif
Full Duplex Input & Output traffic	Trafic en entrée & en sortie Full Duplex
Connection Update	Mise à jour de connexion
Device Update Period	Période de mise à jour du dispositif
Device Interrupt service	Service d'interruption du dispositif
Actual Position	Position réelle
Motion D-to-C I/O Connection	Connexion E/S D-C CIP Motion
Motion C-to-D I/O Connection	Connexion E/S C-D CIP Motion
Controller Task	Tâche du contrôleur
Command Position	Position de consigne
Motion Planner	Planificateur de mouvement
Controller Timer Events	Événements de temporisateur du contrôleur
Controller Update Period	Période de mise à jour du contrôleur
Phase Offset	Décalage de phase

Figure 43 – Modèle de synchronisation à 3 cycles CIP Motion

Le modèle de synchronisation à 3 cycles ci-dessus présenté à la Figure 43 commence, une nouvelle fois, par la transmission par le dispositif d'entraînement du paquet de connexion D-C au contrôleur au début du cycle de mise à jour. Toutefois, dans ce cas, le paquet de connexion D-C dispose du cycle de mise à jour complet pour atteindre le contrôleur avant l'exécution de la tâche Planificateur de mouvement au début du cycle suivant. Après

exécution de la tâche Planificateur de mouvement, le paquet de connexion C-D résultant n'est pas retransmis immédiatement au dispositif d'entraînement, mais est transmis en revanche par ladite tâche au début du cycle suivant, en accordant une nouvelle fois un cycle plus long au paquet pour atteindre le dispositif d'entraînement ciblé. Ainsi, dans ce cas, 3 cycles de connexion sont nécessaires pour accomplir la transaction de données E/S avec le dispositif d'entraînement. Avec le modèle de synchronisation à 3 cycles, le trafic de paquets d'entrée et de sortie CIP Motion traverse le support Ethernet Full Duplex dans les deux directions.

Sur la base de ces caractéristiques communes de modèle de synchronisation, il n'est pas nécessaire que le dispositif d'entraînement connaisse le modèle de synchronisation spécifique appliqué par le contrôleur. Le dispositif d'entraînement transmet simplement les paquets D-C au contrôleur au début du cycle de mise à jour et traite les paquets C-D provenant du contrôleur.

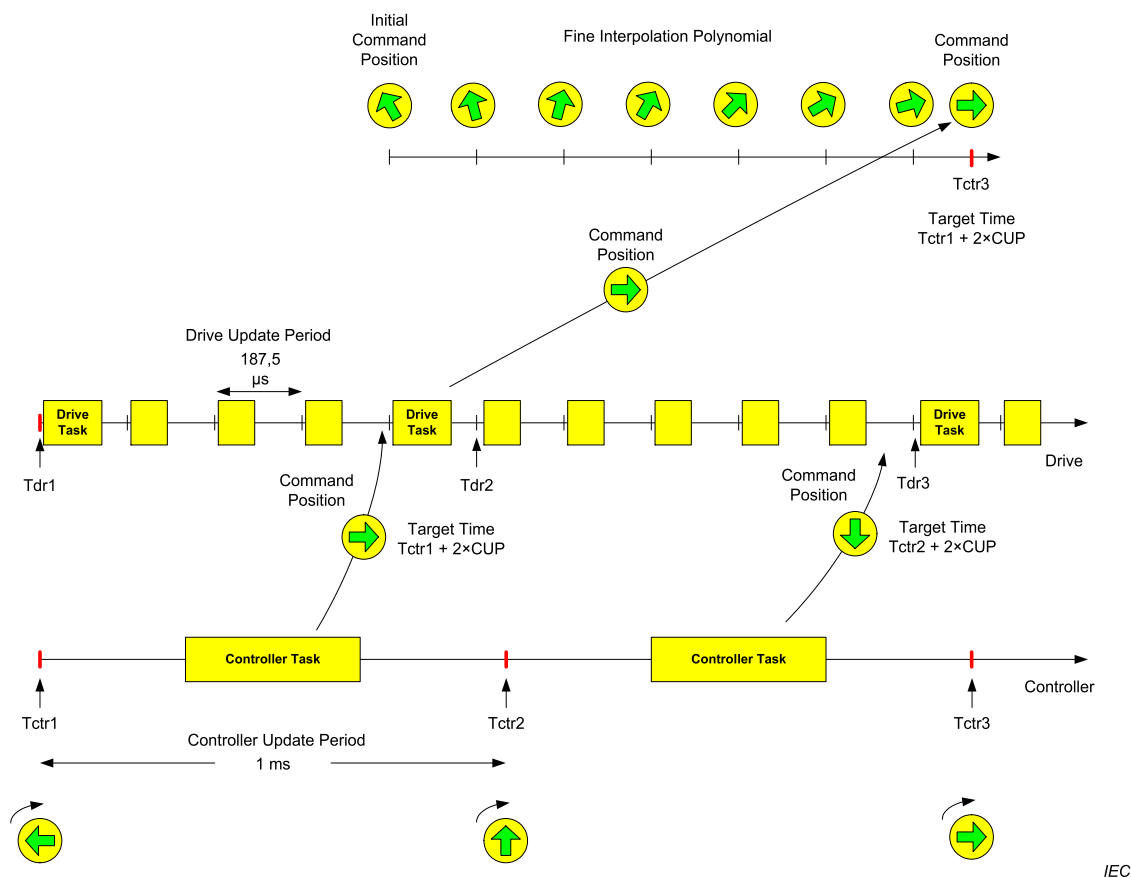
6.4.6.2 Synchronisation de connexion Contrôleur-Dispositif

La plupart des protocoles de commande de mouvement exigent que la période de mise à jour du contrôleur soit un entier multiple de la période de mise à jour du dispositif, à savoir la période de mise à jour de la boucle d'asservissement interne du dispositif d'entraînement. Mais étant donné que le paquet de Connexion E/S CIP Motion contient une datation, il n'est pas utile que la période de mise à jour du contrôleur entretienne une relation fixe avec la période de mise à jour du dispositif d'entraînement.

Les paragraphes suivants présentent une description plus détaillée du modèle de synchronisation CIP Motion, présentée dans le contexte d'un type de dispositif d'entraînement CIP Motion étant entendu que le modèle de synchronisation s'applique en général à tous les types de dispositifs CIP Motion.

Une branche du cycle d'échange de données de connexion E/S CIP Motion est initiée par le contrôleur via le paquet de connexion Contrôleur-Dispositif. L'inclusion des informations de datation et de décalage temporel, ainsi que des données de consigne dans ce paquet, supprime les exigences de synchronisation strictes imposées par d'autres protocoles de réseau de commande de mouvement. Plus particulièrement, la datation permet au dispositif d'entraînement de calculer les valeurs de données de consigne sur la base de ses propres mises à jour de dispositif. Contrairement à d'autres protocoles de réseau de commande de mouvement, il n'est pas nécessaire que la relation du dispositif avec les mises à jour du contrôleur soit fixe.

La Figure 44 représente la méthode d'application des datations et données de consigne, fournies par la connexion Contrôleur-Dispositif, à l'axe d'entraînement lorsqu'une interpolation fine est exigée. Dans cet exemple, les tâches de transmission et de réception de dispositif d'entraînement décrites dans la section de modèle de synchronisation ci-dessus ont été combinées avec le service d'interruption de dispositif d'entraînement dans une tâche du dispositif d'entraînement périodique singulière exécutée au cours de la Période de mise à jour du dispositif.



Anglais	Français
Initial Command Position	Position de consigne initiale
Fine Interpolation Polynomial	Polynôme d'interpolation fine
Command Position	Position de consigne
Target Time	Temps cible
Drive Update Period	Période de mise à jour du dispositif d'entraînement
Drive	Dispositif d'entraînement
Controller Update Period	Période de mise à jour du contrôleur
Controller	Contrôleur
Drive Task	Tâche du dispositif d'entraînement
Controller Task	Tâche du contrôleur

**Figure 44 – Synchronisation de connexion
Contrôleur-Dispositif avec interpolation fine**

Les étapes suivantes décrivent en détail la manière dont les données de connexion sont transférées du contrôleur au dispositif d'entraînement pour l'interpolation fine lors d'un cycle de connexion classique, dans le cas général d'une période de mise à jour du contrôleur (CUP) qui n'est pas un entier multiple de la période de mise à jour du dispositif.

- 1) Transmission du contrôleur: Dans le cadre de la tâche du contrôleur, ce dernier lance la transmission d'un paquet de connexion Contrôleur-Dispositif avec de nouvelles données de consigne au dispositif d'entraînement ciblé, avec un ID de mise à jour incrémenté et une nouvelle datation de système de contrôleur (ainsi qu'un décalage du temps système

du contrôleur en cas de changement depuis la dernière mise à jour) faisant référence au temps système au début du cycle de mise à jour du contrôleur en cours. Le Bloc de données d'instance d'axe ciblé contient également la mise à jour de consigne cible qui est, dans cet exemple, établi à +2 afin de tenir compte du délai de transport de la période de mise à jour du contrôleur et du délai de la période de mise à jour du contrôleur pour l'interpolation fine. Il est recommandé que le contrôleur transmette le paquet de connexion Contrôleur-Dispositif au plus tard aux 2/3 de la durée de la période de mise à jour du contrôleur afin de permettre aux paquets d'atteindre le dispositif d'entraînement ciblé. Le paquet de connexion Contrôleur-Dispositif est reçu par le dispositif d'entraînement et traité par la pile réseau à l'arrivée. La pile réseau présente les nouvelles données de la connexion Contrôleur-Dispositif à la tâche du dispositif d'entraînement de la couche application pour analyse.

- 2) Vérification de l'ID de mise à jour: En réponse à une nouvelle notification de données par la pile, la tâche du dispositif d'entraînement commence à analyser les données du paquet de connexion Contrôleur-Dispositif en vérifiant tout d'abord l'ID de mise à jour. Lorsque la tâche du dispositif d'entraînement découvre que l'incrément de l'ID de mise à jour est supérieure à 1 depuis la dernière mise à jour, le dispositif d'entraînement incrémente la valeur Perte de mises à jour du contrôleur.
- 3) Vérification de fonctionnement synchrone: Ensuite, la tâche du dispositif d'entraînement commence à analyser les données du paquet de connexion Contrôleur-Dispositif en vérifiant tout d'abord si le dispositif d'entraînement est synchronisé. Si ce n'est pas le cas, passer à l'étape Appliquer les données de consigne, puisqu'il n'est pas utile de procéder à une vérification de décalage du temps système ou à une vérification de mise à jour retardée. Éviter ces vérifications permet de contrôler le dispositif d'entraînement lors du démarrage, voire dans le cas où le dispositif d'entraînement ne dispose d'aucun service de synchronisation temporelle. Cette opération est appelée fonctionnement asynchrone.
- 4) Vérification de décalage du temps système: Un fonctionnement synchrone implique une mise en correspondance du temps système du dispositif d'entraînement et du contrôleur, de manière à rendre significatives les datations échangées entre les dispositifs. Étant donné que le temps système est ajusté de façon périodique par l'horloge maîtresse et que les ajustements peuvent se propager vers les différents dispositifs du réseau à différents moments, il est possible que le temps système entre le contrôleur et le dispositif d'entraînement soit provisoirement étalé. Cet étalement doit être détecté, auquel cas, le dispositif d'entraînement doit ajuster la datation du contrôleur afin de le compenser. Le dispositif d'entraînement effectue cette compensation de décalage uniquement si les décalages de temps système du contrôleur ou du dispositif ont varié depuis la dernière mise à jour. Voir 6.4.6.7 (compensation de décalage du temps système) pour plus de détails sur cet algorithme.
- 5) Vérification de mise à jour retardée: En supposant le caractère synchrone du fonctionnement, le dispositif d'entraînement calcule alors la différence entre la datation de la tâche d'entraînement en cours et la datation du contrôleur transmise dans le paquet de connexion Contrôleur-Dispositif (ajusté, si nécessaire). Une différence de deux périodes ou plus de mises à jour du contrôleur incrémente la valeur Mises à jour retardées du contrôleur. Si la différence est supérieure à (Limite inférieure du délai de mise à jour du contrôleur x Période de mise à jour du contrôleur), le dispositif d'entraînement émet une alarme de mise à jour de connexion du contrôleur. Si la différence est supérieure à (Limite supérieure du délai de mise à jour du contrôleur x Période de mise à jour du contrôleur), le dispositif d'entraînement émet un Défaut de mise à jour de connexion de commande.

NOTE 1 Si la différence de temps a dépassé la période de mise à jour du contrôleur, le polynôme d'interpolation fine en cours est effectivement devenu un polynôme d'extrapolation, permettant au dispositif d'entraînement de passer à la condition de données retardées tant que les nouvelles données ne sont pas arrivées.

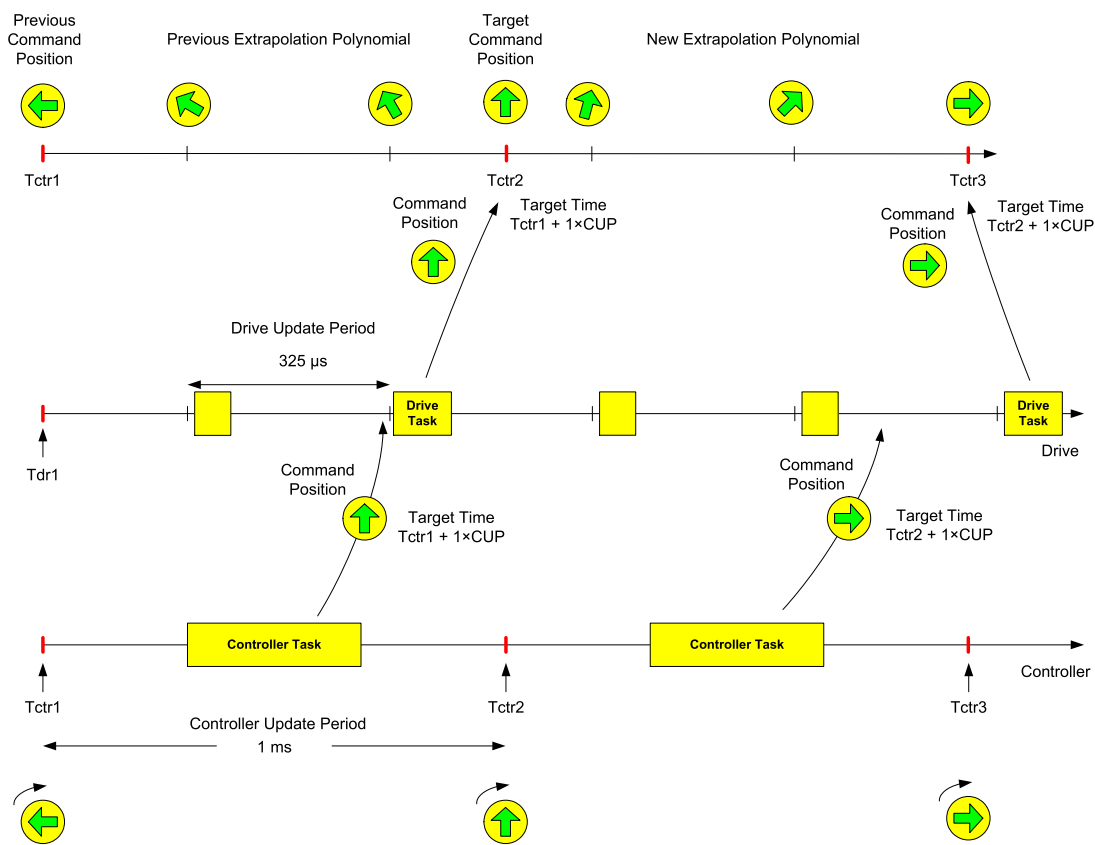
- 6) Appliquer les données de consigne: En supposant le caractère synchrone du fonctionnement et une mise à jour de consigne cible de +2 pour l'interpolation fine, le dispositif d'entraînement calcule les coefficients du polynôme d'interpolation fine en fonction de la référence de consigne appliquée au Temps cible calculé, qui est la somme de la datation du contrôleur (ajusté) (CUP), Tctr1, et du produit de la Mise à jour de consigne cible et de la période de mise à jour du contrôleur ou, dans ce cas, 2 x CUP. Ces coefficients sont utilisés par le programme de service de commande du dispositif

d'entraînement pour calculer la position de consigne fine basée sur la durée du service. D'ordinaire, ces nouveaux coefficients sont appliqués pendant toute la durée du cycle de mise à jour suivant, en prévoyant une interpolation fine jusqu'à ce que le temps système ait atteint le temps cible. Si le temps cible est inférieur au temps système en cours au moment de la réception du paquet de connexion Contrôleur-Dispositif, c'est-à-dire qu'il s'agit d'un paquet retardé, de nouveaux coefficients par rapport au polynôme sont alors toujours calculés en fonction de ces données de consigne, afin d'améliorer l'exactitude des calculs d'extrapolation. En général, à chaque retard des données de consigne, les données représentent toujours les données de consigne les plus récentes disponibles et les coefficients doivent dans ce cas être appliqués dès que possible. Dans le cas d'un fonctionnement asynchrone, les données de consigne sont immédiatement appliquées au service de commande du dispositif d'entraînement.

- 7) Planifier la mise à jour de la connexion Dispositif-Contrôleur suivante: Le dispositif d'entraînement calcule le temps système pour le cycle de mise à jour de connexion suivant comme la somme de la datation du contrôleur et la période de mise à jour du contrôleur. Cette valeur sert à établir les critères de fenêtre de mise à jour réelle pour la mise à jour de la connexion Dispositif-Contrôleur suivante, ou pour planifier directement cette mise à jour via un événement de synchronisation.

Lorsque la mise à jour de consigne cible est établie à +1, le polynôme calculé à l'étape 6 n'est pas appliqué à des fins d'interpolation fine, mais davantage à des fins d'extrapolation; le polynôme d'extrapolation permet au dispositif d'entraînement de calculer une valeur de données de consigne précise au moment où ce même dispositif effectue ses calculs de contrôle en fonction de la trajectoire d'axe précédente.

La Figure 45 représente ce modèle de synchronisation dans le cas général où la période de mise à jour du contrôleur (CUP) n'est pas un entier multiple de la période de mise à jour du dispositif et où la mise à jour de consigne cible est établie à +1 à des fins d'extrapolation.



IEC

Anglais	Français
Previous Command Position	Position de consigne précédente
Previous Extrapolation Polynomial	Polynôme d'extrapolation précédent
Target Command Position	Position de consigne cible
New Extrapolation Polynomial	Nouveau polynôme d'extrapolation
Command Position	Position de consigne
Target Time	Temps cible
Drive Update Period	Période de mise à jour du dispositif d'entraînement
Drive Task	Tâche du dispositif d'entraînement
Drive	Dispositif d'entraînement
Controller Update Period	Période de mise à jour du contrôleur
Controller	Contrôleur
Controller Task	Tâche du contrôleur

Figure 45 – Synchronisation de connexion Contrôleur-Dispositif avec extrapolation

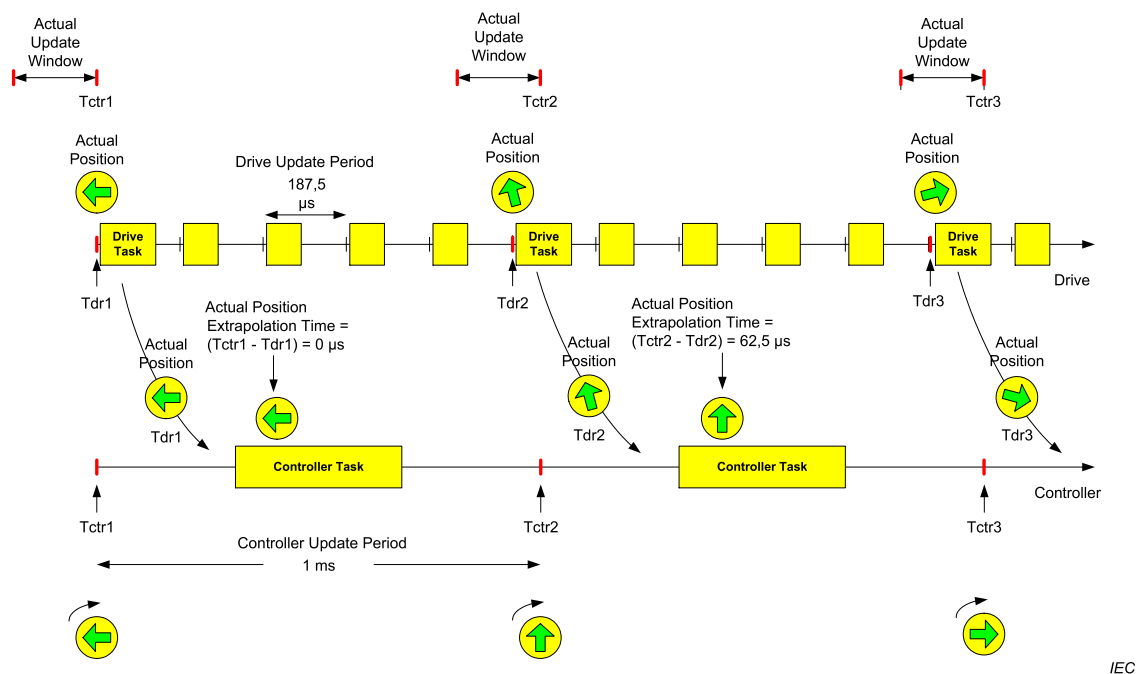
NOTE 2 Dans l'exemple ci-dessus, la période de mise à jour du contrôleur ne contient pas beaucoup de périodes de mise à jour du dispositif. Lorsque tel est le cas, l'interpolation fine n'est pas essentielle pour les performances du dispositif d'entraînement, et les données de consigne peuvent être appliquées plus directement à la structure de commande du dispositif d'entraînement par extrapolation, sans le délai supplémentaire exigé pour la prise en charge de l'interpolation fine. Toutefois, l'extrapolation présente l'inconvénient de révéler plus directement les erreurs d'extrapolation aux données de consigne générant un mouvement plus grossier que dans le cas d'une interpolation fine.

Toutes les données cycliques associées au paquet de connexion Contrôleur-Dispositif doivent être appliquées dans la mise à jour de la commande Tâche d'entraînement pour utiliser le plus tôt possible les dernières données de consigne, en calculant de nouveaux coefficients de polynôme d'interpolation/extrapolation.

6.4.6.3 Synchronisation de connexion Dispositif-Contrôleur

L'autre branche du cycle d'échange de données de connexion E/S CIP Motion est initiée par le dispositif d'entraînement via le paquet de connexion Dispositif-Contrôleur. En mode synchrone, la connexion Dispositif-Contrôleur CIP Motion inclut une datation du dispositif (et un décalage temporel du dispositif) avec les données réelles, pour permettre au contrôleur de calculer la position d'axe d'entraînement au moment de la mise à jour de la tâche du contrôleur. La datation permet au dispositif d'entraînement d'échantillonner la réaction et de calculer les valeurs de données réelles sur la base de ses propres mises à jour de dispositif. Contrairement à d'autres protocoles de réseau de commande de mouvement, il n'est pas nécessaire que le dispositif soit lié de manière stricte avec les mises à jour du contrôleur. La Figure 46 représente comment les données réelles et les datations fournies par la connexion Dispositif-Contrôleur servent à ajuster la position réelle d'axe d'entraînement, par exemple, à la base de temps du contrôleur.

Une nouvelle fois, pour des raisons de simplicité, les tâches de transmission et de réception de dispositif d'entraînement décrites dans la section de modèle de synchronisation ci-dessus ont été combinées avec le service d'interruption de dispositif d'entraînement dans une tâche du dispositif d'entraînement périodique singulière exécutée au cours de la période de mise à jour du dispositif.



IEC

Anglais	Français
Actual Update Window	Fenêtre de mise à jour réelle
Actual Position	Position réelle
Drive Update Period	Période de mise à jour du dispositif d'entraînement
Drive	Dispositif d'entraînement
Drive Task	Tâche du dispositif d'entraînement
Actual Position Extrapolation Time	Temps d'extrapolation de la position réelle
Controller Task	Tâche du contrôleur
Controller Update Period	Période de mise à jour du contrôleur
Controller	Contrôleur

Figure 46 – Utilisation de la datation pour ajuster la position réelle à la base de temps du contrôleur

Les étapes suivantes décrivent en détail la manière dont les données de connexion sont transférées du dispositif d'entraînement au contrôleur lors d'un cycle de connexion classique, dans le cas général d'une période de mise à jour du contrôleur (CUP) qui n'est pas un entier multiple de la période de mise à jour du dispositif. Bien entendu, cette séquence s'applique également au cas particulier où la période de mise à jour du contrôleur (CUP) est un entier multiple de la période de mise à jour du dispositif d'entraînement.

- 1) Vérification de mise à jour réelle: Si l'axe est synchronisé, le dispositif d'entraînement compare la datation de la tâche d'entraînement en cours à la Fenêtre de mise à jour réelle établie dans le cadre du dernier cycle de mise à jour. La durée de la Fenêtre de mise à jour réelle équivaut à 1 période de mise à jour du dispositif et se termine le plus tard possible pour permettre au dispositif d'entraînement d'assembler entièrement le paquet de connexion Dispositif-Contrôleur et de pouvoir le transmettre avant le début du cycle de mise à jour suivant. Si la datation de la tâche d'entraînement se trouve dans la fenêtre temporelle, il s'agit d'un cycle de mise à jour des données réel. Si la datation précède la fenêtre, le moment d'envoyer les données au contrôleur n'est pas venu (la Fenêtre de mise à jour réelle a pour objet de planifier la transmission de données réelles le plus près possible du début du cycle de mise à jour de connexion, tout en réduisant également au minimum la durée entre la saisie de réaction et le début du cycle de mise à jour de connexion). Les tâches de transmission entraînées par interruption de l'événement de synchronisation n'imposent pas de vérifier la mise à jour réelle dans la mesure où l'exécution au moment opportun du service d'événement de synchronisation de la tâche de transmission est déjà planifiée. Lorsque l'axe n'est pas synchronisé et que l'on vient de recevoir une mise à jour de consigne via la connexion Contrôleur-Dispositif, ou lorsque la durée écoulée depuis la dernière mise à jour de consigne a dépassé 1 période de mise à jour du contrôleur, il s'agit alors également d'un cycle de mise à jour réel, et il y a lieu de passer à l'étape de transmission d'entraînement.
- 2) Transmission d'entraînement: Lorsqu'il s'agit d'un cycle de mise à jour réel, la tâche du dispositif d'entraînement assemble et transmet alors le paquet de connexion Dispositif-Contrôleur au contrôleur avec les données réelles les plus récentes, l'ID de mise à jour incrémenté et, pour les dispositifs synchrones, la datation de dispositif associée (ainsi que le décalage temporel du dispositif en cas de changement depuis la dernière mise à jour). Pour de meilleures performances, il est recommandé que la transmission du paquet soit le plus proche possible du début du cycle Mise à jour de connexion. La seule contrainte de synchronisation réside dans le fait que, en fonctionnement normal, le dispositif d'entraînement doit lancer la transmission au plus tard au début du cycle de mise à jour réel. Il s'agit de la contrainte de synchronisation la plus importante associée au modèle de synchronisation CIP Motion. De fait, le traitement du paquet de connexion Dispositif-Contrôleur doit être prioritaire sur le traitement du paquet de connexion Contrôleur-Dispositif. Le délai supplémentaire que cela peut introduire au traitement du paquet de connexion Contrôleur-Dispositif n'a pas un impact significatif sur la qualité de commande de mouvement tant que la période de mise à jour du dispositif est bien plus courte que la période de mise à jour du contrôleur.
- 3) Vérification de l'ID de mise à jour: Dans la tâche du contrôleur suivante, le contrôleur examine les nouvelles données provenant du dispositif d'entraînement par une vérification d'un ID de mise à jour modifié. Si l'ID de mise à jour n'a pas changé depuis la dernière mise à jour, le contrôleur incrémente la valeur Perte de mises à jour associée à la connexion Dispositif-Contrôleur du dispositif d'entraînement. Les étapes suivantes sont réalisées indépendamment du changement ou non de l'ID de mise à jour.
- 4) Vérification du mode de synchronisation: Le dispositif d'entraînement vérifie le bit Mode sync de l'octet État de nœud de dispositif d'entraînement, afin de déterminer si l'axe d'entraînement est synchronisé. Si ce n'est pas le cas, passer à l'étape Appliquer les données réelles pour éviter la vérification de mise à jour retardée et la correction de datation. Éviter les étapes suivantes permet au dispositif d'entraînement de fonctionner lors du démarrage ou même si le dispositif d'entraînement ne dispose d'aucun service de synchronisation temporelle.
- 5) Vérification de décalage du temps système: Un fonctionnement synchrone exige une mise en correspondance du temps système du dispositif d'entraînement et du contrôleur, de manière à rendre significatives les datations échangées entre les dispositifs. Étant donné que le temps système peut être ajusté à tout instant par l'horloge maîtresse et que les

ajustements peuvent se propager vers les différents dispositifs du réseau à différents moments, il est possible que le temps système entre le contrôleur et le dispositif d'entraînement soit étalé. Cet étalement doit être détecté, auquel cas, le dispositif d'entraînement doit ajuster la datation du dispositif afin de le compenser. Le contrôleur effectue cette compensation de décalage uniquement si les décalages de temps système du contrôleur ou du dispositif ont varié depuis la dernière mise à jour. Voir 6.4.6.7 (compensation de décalage du temps système) pour plus de détails sur cet algorithme.

- 6) Vérification de mise à jour retardée: Le contrôleur calcule la différence entre la datation de la tâche du contrôleur en cours et la datation du dispositif dans le paquet de connexion Dispositif-Contrôleur. Si la différence est supérieure à (Limite inférieure du délai de mise à jour du contrôleur x Période de mise à jour du contrôleur), le contrôleur émet une alarme de synchronisation de commande. Si la différence est supérieure à (Limite supérieure du délai de mise à jour du contrôleur x Période de mise à jour du contrôleur), le contrôleur émet une alarme de synchronisation de commande.
- 7) Correction de datation: Si la différence de datation préalablement calculée est non nulle, extrapoler alors la valeur de données réelles en fonction de la précédente trajectoire réelle d'axe pour s'aligner sur la datation du contrôleur. Cette correction est nécessaire, le planificateur de mouvement supposant que les données d'entrée réelles sont horodatées de manière implicite au début de la période de mise à jour du contrôleur. En fonctionnement normal, la différence de datation calculée est faible par rapport à la période de mise à jour du contrôleur et la correction de position résultante est réduite, mais dans les cas où le paquet de connexion Dispositif-Contrôleur est retardé ou perdu, cette correction de position est importante et critique pour un fonctionnement permanent étant donné qu'elle permet de "parcourir" cette condition en fonction de la trajectoire précédente.
- 8) Appliquer les données réelles: Le contrôleur applique les données réelles en entrées au planificateur de mouvement, qui calcule les nouvelles données de référence de consigne pour la mise à jour Contrôleur-Dispositif suivante.

6.4.6.4 Planification de la mise à jour suivante

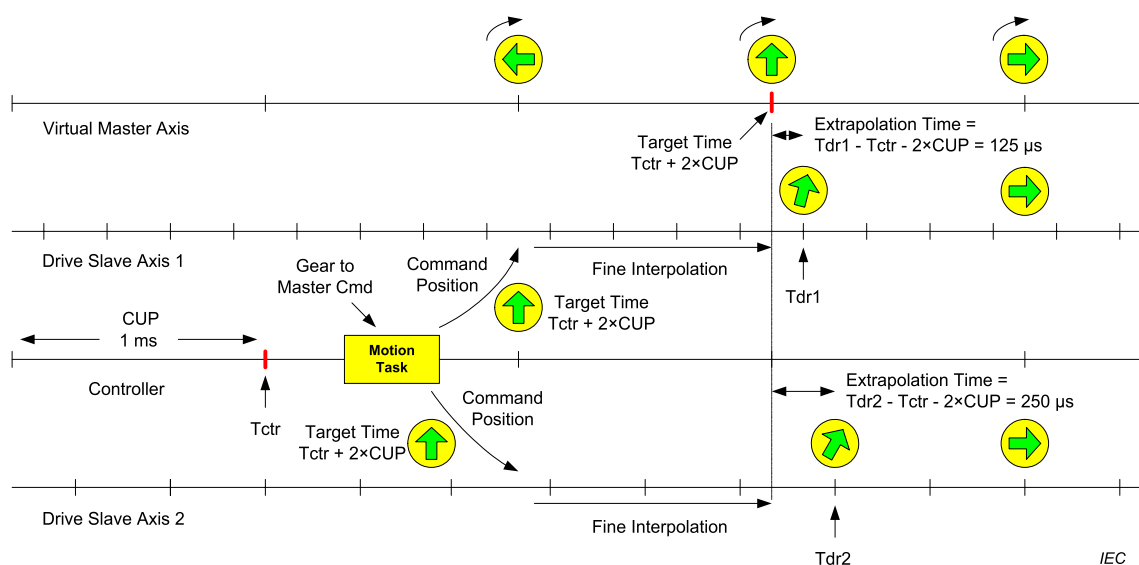
Afin de s'assurer qu'une mise à jour de la connexion Dispositif-Contrôleur a lieu indépendamment de la réception d'un paquet de connexion Contrôleur-Dispositif, le dispositif d'entraînement doit, au cours de chaque cycle, planifier le moment de la mise à jour suivante du cycle de connexion Dispositif-Contrôleur.

Pour un fonctionnement synchrone, le dispositif d'entraînement calcule la durée de la mise à jour suivante du cycle de connexion Dispositif-Contrôleur de sorte qu'elle soit suffisante avant la fin du cycle de mise à jour suivant, afin de prévoir une durée appropriée pour le rassemblement des données de connexion Dispositif-Contrôleur et leur transmission avant la fin du cycle. Pour les tâches de transmission entraînées par interruption de l'événement de synchronisation, cette nouvelle planification se produit automatiquement lors de la recharge du temporisateur périodique local. Pour les mises en œuvre où la connexion Dispositif-Contrôleur est traitée par la tâche du dispositif d'entraînement, la nouvelle planification doit être effectuée explicitement par le dispositif d'entraînement à la fin de chaque mise à jour de la connexion Dispositif-Contrôleur.

Pour un fonctionnement asynchrone, le dispositif d'entraînement déclenche une tâche d'assemblage et de transmission du paquet de connexion Dispositif-Contrôleur immédiatement après réception (et analyse) d'un paquet Contrôleur-Dispositif. Le dispositif d'entraînement planifie alors la transmission du paquet Dispositif-Contrôleur suivant pour qu'elle corresponde à 1 période de mise à jour du contrôleur à partir du moment de réception du paquet Contrôleur-Dispositif. Ainsi, la transmission de ce paquet Dispositif-Contrôleur se produit même si le paquet Contrôleur-Dispositif suivant est perdu ou retardé. L'ID de mise à jour pour ce paquet Dispositif-Contrôleur doit correspondre à l'ID de mise à jour + 1 du paquet Contrôleur-Dispositif de sorte que les ID de mise à jour pour les paquets Dispositif-Contrôleur et Contrôleur-Dispositif correspondent ensuite à un cycle de mise à jour donné.

6.4.6.5 Indépendance de la période de mise à jour du dispositif

Le diagramme de synchronisation de la Figure 47 représente comment les deux axes d'entraînement peuvent être très coordonnés malgré différentes périodes de mise à jour du dispositif, et malgré une période de mise à jour du contrôleur associée qui n'est pas un entier multiple de la période de mise à jour du dispositif.



Anglais	Français
Virtual Master Axis	Axe maître virtuel
Target Time	Temps cible
Extrapolation Time	Temps d'extrapolation
Drive Slave Axis 1	Axe esclave 1 du dispositif d'entraînement
Gear to Master Cmd	Engrenage vers la consigne principale
Command Position	Position de consigne
Fine Interpolation	Interpolation fine
Controller	Contrôleur
Motion Task	Tâche de mouvement
Drive Slave Axis 2	Axe esclave 2 du dispositif d'entraînement

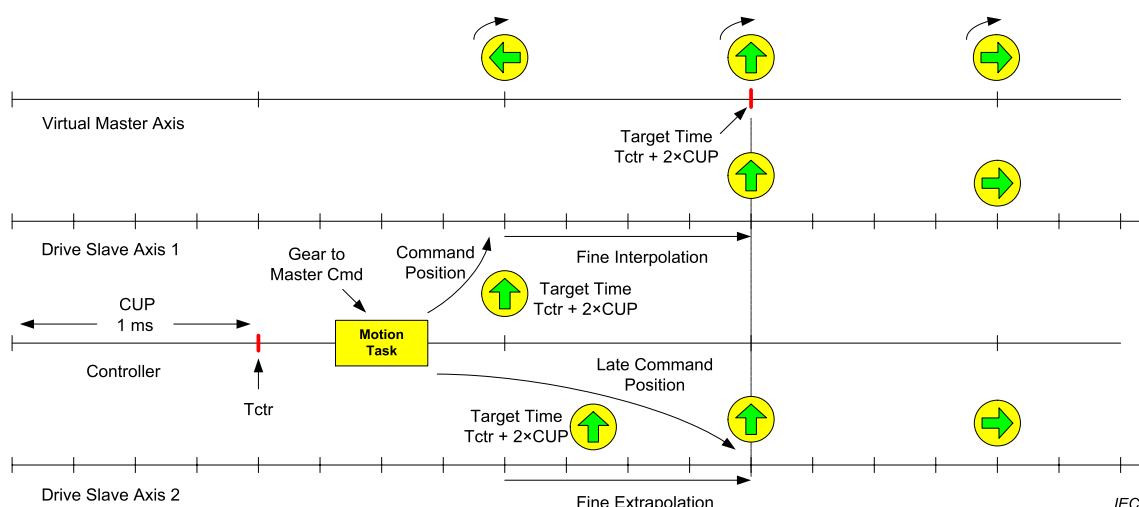
Figure 47 – Coordination des deux dispositifs d'entraînement avec différentes périodes de mise à jour

Dans le diagramme de synchronisation ci-dessus, la tâche du planificateur de mouvement du contrôleur envoie des positions de consigne et des datations identiques aux deux axes d'entraînement esclaves qui, lorsqu'ils sont synchronisés avec le temps système, fonctionnent à différentes vitesses de mise à jour de dispositif d'entraînement. Lorsque les données de position de consigne arrivent au niveau des deux dispositifs d'entraînement, elles utilisent la datation du contrôleur, le temps de consigne cible et la période de mise à jour du contrôleur pour calculer de nouveaux coefficients par rapport au polynôme d'interpolation, en s'appuyant sur la contrainte selon laquelle la valeur polynomiale égale en temps à (Datation du contrôleur + Temps de consigne cible x Période de mise à jour du contrôleur) est la valeur Position de consigne spécifiée. Compte tenu de l'absence de dépendance quant à la vitesse de mise à jour du dispositif d'entraînement, les coefficients polynomiaux calculés par chaque dispositif d'entraînement sont identiques. Aucune mise à jour des dispositifs d'entraînement ne coïncidant avec ce temps cible, les dispositifs d'entraînement utilisent le polynôme d'interpolation fine pour calculer la référence de position de consigne de chaque de mise à jour de dispositif d'entraînement tant qu'une nouvelle position de consigne n'a pas été reçue du contrôleur. Si une nouvelle position de consigne n'arrive pas bien après le temps cible, le

dispositif d'entraînement continue à utiliser la même équation polynomiale pour "extrapoler" la position de consigne pour les mises à jour de dispositif d'entraînement suivantes comme indiqué à la Figure 47. Cette extrapolation se poursuit jusqu'à ce que de nouvelles données arrivent et que de nouveaux coefficients puissent être calculés. De cette manière, qu'il s'agisse d'interpolation ou d'extrapolation, chaque axe esclave fonctionne de manière harmonieuse et les deux axes restent verrouillés en phase avec l'axe maître.

6.4.6.6 Indépendance de latence de transmission

La coordination précise de plusieurs axes de dispositif d'entraînement CIP Motion peut être maintenue même si les paquets de connexion Contrôleur-Dispositif impliquent des délais importants lors du passage dans le réseau CIP. Dans la Figure 48, le paquet propre à l'axe de dispositif d'entraînement esclave 2 a subi un retard important lors de la transmission. En conséquence, la position de consigne de cet axe doit être extrapolée à partir du dernier polynôme d'interpolation fine. Cela permet à l'axe de se déplacer de manière harmonieuse à travers une perturbation de latence de transmission. Lorsque les nouvelles données de consigne arrivent, la nouvelle valeur de consigne peut ne pas accepter la valeur extrapolée en raison d'une erreur d'extrapolation. Cette erreur peut perturber le profil de mouvement. L'amplitude de l'erreur d'extrapolation dépend des éléments dynamiques du profil de mouvement et de la vitesse de mise à jour du contrôleur. Dans la plupart des applications réelles, des latences de transmission durant plusieurs périodes de mise à jour peuvent se produire sans perturbation notable du profil de mouvement associé.



Anglais	Français
Virtual Master Axis	Axe maître virtuel
Target Time	Temps cible
Drive Slave Axis 1	Axe esclave 1 du dispositif d'entraînement
Gear to Master Cmd	Engrenage vers la consigne principale
Command Position	Position de consigne
Fine Interpolation	Interpolation fine
Controller	Contrôleur
Motion Task	Tâche de mouvement
Late Command Position	Position de consigne retardée
Drive Slave Axis 2	Axe esclave 2 du dispositif d'entraînement
Fine Extrapolation	Extrapolation fine

Figure 48 – Coordination de plusieurs axes de dispositif d'entraînement en cas de retard des paquets de connexion Contrôleur-Dispositif

6.4.6.7 Compensation de décalage du temps système

6.4.6.7.1 Généralités

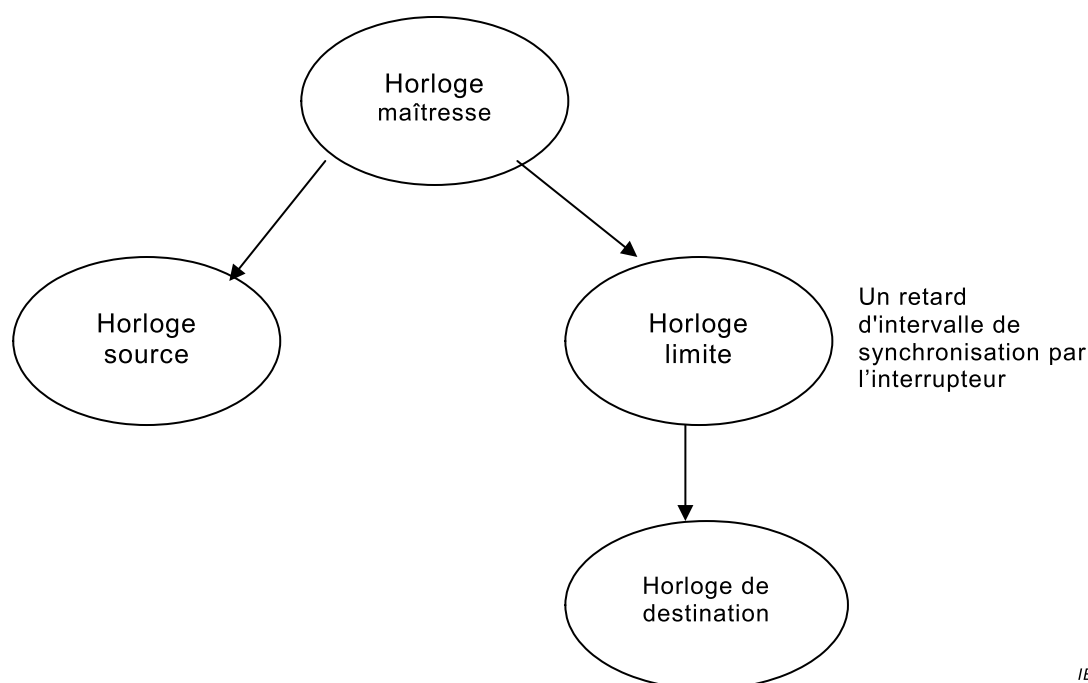
CIP Motion repose sur CIP Sync et son protocole de synchronisation temporelle IEC 61588:2009 sous-jacent, qui définit un mécanisme de distribution et de synchronisation du temps des dispositifs présents sur le réseau.

En utilisant le protocole CIP Sync défini dans l'IEC 61588:2009, chaque dispositif présent sur le réseau a une représentation du temps système étroitement synchronisée avec l'horloge maîtresse principale du réseau. Le temps système propre à chaque dispositif est défini comme la somme du temps d'horloge locale du dispositif et du décalage du temps système de ce même dispositif. Dans le cas du protocole CIP Sync, il est important de noter que même si l'horloge locale fonctionne à la même fréquence que l'horloge maîtresse principale, elle n'est pas strictement verrouillée en phase avec cette dernière. Seule la valeur Temps système du dispositif est verrouillée en phase avec l'horloge maîtresse principale, et est appropriée à la synchronisation de l'échange de données entre le contrôleur et le dispositif d'entraînement.

Selon le protocole CIP Sync, le temps système peut engendrer de larges réponses à un échelon, à savoir de quelques millisecondes à plusieurs années, ces réponses à un échelon étant propagées par le réseau à chaque dispositif. Ces réponses à un échelon dans la durée peuvent être provoquées, par exemple, par un utilisateur qui ajuste manuellement le temps absolu de l'horloge maîtresse pour corriger la dérive de l'horloge ou par une modification de la maîtrise du temps. L'IEC 61588:2009 ne traite pas des réponses à un échelon dans la durée comme partie intégrante de son protocole et, par conséquent, ne comporte pas de disposition de traitement de cette condition lors d'une opération de commande en temps réel. La norme CIP Sync définit un concept appelé Compensation de pas de temps qui, lorsqu'il est mis en œuvre dans l'Objet Axe de dispositif de mouvement, traite effectivement de cette condition. Le paragraphe suivant décrit cet algorithme en détail.

6.4.6.7.2 Compensation de pas de temps

Selon le protocole CIP Sync, les réponses à un échelon de l'heure de l'horloge maîtresse doivent se propager dans le système de temps réparti, la durée de propagation de ces réponses à un échelon dans le système entre l'horloge maîtresse et l'horloge esclave la plus éloignée étant fonction de l'intervalle de paquet de synchronisation (2 s par défaut) et du nombre de sauts (1 saut par horloge limite IEC 61588:2009, c'est-à-dire un interrupteur mettant en œuvre le protocole IEC 61588:2009, par exemple). Ceci est représenté à la Figure 49.



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Figure 49 – Propagation d'une réponse à un échelon dans le temps

Dans la Figure 49, l'horloge source est sur le point de recevoir un ajustement du temps système sur un intervalle de synchronisation avant l'horloge de destination. Par conséquent, le temps système de l'horloge source va être étalé par rapport à celui de l'horloge de destination pour un intervalle de paquet de synchronisation complet.

La variabilité de la propagation de cet ajustement chronologique dans le système crée un problème de commande de mouvement lié à la datation, en ce sens que la datation du dispositif source (le contrôleur, par exemple) peut ne pas s'appliquer lorsqu'il est reçu par le dispositif de destination (un dispositif d'entraînement, par exemple), car le temps système des dispositifs source et de destination est étalé. Cet effet est représenté en détail dans le Tableau 24.

Tableau 24 – Propagation d'une réponse à un échelon dans le temps (exemple 1)

Intervalle de synchronisation temporelle	Temps de l'horloge maîtresse	Temps source (contrôleur)	Temps de destination (dispositif d'entraînement)
1	100	100	100
2	200 + 1 000	200	200
3	1 300	300 + 1 000	300
4	1 400	1 400	400 + 1 000
5	1 500	1 500	1 500

Aux intervalles de synchronisation temporelle 1 et 2, l'horloge source de datation du contrôleur et l'horloge de destination de datation du dispositif d'entraînement présentent toutes deux la même notion du temps (c'est-à-dire le temps système). A l'intervalle de synchronisation 2, une réponse à un échelon du temps se produit au niveau de l'horloge maîtresse et se propage vers le contrôleur dans l'intervalle de synchronisation 3, affectant la datation de l'horloge source envoyée au dispositif d'entraînement. Ce n'est qu'à l'intervalle de synchronisation 4 que le dispositif d'entraînement détecte la réponse à un échelon. Toutes les données de datation envoyées du contrôleur au dispositif d'entraînement entre les intervalles

de synchronisation 3 et 4 ne sont pas corrélées avec la notion de temps du dispositif d'entraînement.

CIP Motion et CIP Sync étendent le protocole IEC 61588:2009 pour traiter cette condition. Pour ce faire, un algorithme d'exécution est défini dans le dispositif d'entraînement de destination qui peut détecter cette condition de pas de temps et ajuster la valeur de la datation reçue de manière précise. Deux conditions sont possibles:

- 1) le dispositif source a détecté une réponse à un échelon dans le temps, à l'inverse du dispositif de destination;
- 2) le dispositif de destination a détecté une réponse à un échelon dans le temps, à l'inverse du dispositif source.

La composante essentielle de l'algorithme est la valeur Décalage du temps système définie comme faisant partie intégrante de l'Objet Synchronisation temporelle de CIP Sync. Cette valeur de décalage est ajoutée à l'heure de l'horloge locale afin de générer le temps système, c'est-à-dire,

$$\text{Temps système} = \text{Horloge locale} + \text{Décalage du temps système (dispositif d'entraînement)}$$

Lorsque le système est synchronisé, à un moment donné, les valeurs Horloge locale et Décalage du temps système peuvent différer d'un dispositif à l'autre, mais la valeur Temps système doit être identique dans la précision de la mise en œuvre du protocole IEC 61588:2009.

NOTE La valeur Décalage du temps système associée au dispositif d'entraînement est fournie par l'Objet Synchronisation temporelle et est préfiltrée afin de limiter l'effet du bruit provoqué par le trafic sur le réseau.

Une réponse à un échelon dans le temps est indiquée par une modification de la valeur Décalage du temps système des dispositifs source ou de destination. Le décalage du temps système de la source est envoyé au dispositif de destination en tant que décalage, le temps système de la source étant envoyé en tant que datation de données. Le dispositif de destination compare le décalage à celui déjà reçu afin de déterminer si une réponse à un échelon s'est produite, puis ajuste la valeur de datation reçue en conséquence.

L'algorithme de compensation de décalage, dans la mesure où il s'applique à la datation reçue du contrôleur via la connexion C-D, est indiqué comme suit:

$$\begin{aligned} \text{Datation}_{\text{comp}} &= \text{Datation}_{\text{rec}} + \text{Compensation de décalage} \\ \text{Compensation de décalage} &= \text{Décalage}_{\text{dest}} - \text{Décalage}_{\text{dest}(\text{dernier})} - (\text{Décalage}_{\text{src}} - \text{Décalage}_{\text{src}(\text{dernier})}) \end{aligned}$$

où,

$\text{Datation}_{\text{rec}}$	= datation reçue du contrôleur.
$\text{Datation}_{\text{comp}}$	= datation compensée du contrôleur.
$\text{Décalage}_{\text{dest}}$	= la valeur actuelle du décalage du temps système à la destination.
$\text{Décalage}_{\text{dest}(\text{dernier})}$	= la valeur précédente du décalage du temps système à la destination.
$\text{Décalage}_{\text{src}}$	= la valeur reçue du décalage du temps système de la source.
$\text{Décalage}_{\text{src}(\text{dernier})}$	= la valeur précédente du décalage du temps système de la source.

Lorsque le dispositif d'entraînement détermine que le décalage du temps système source ou celui de destination a changé, le calcul suivant est effectué pour vérifier si une réponse à un échelon dans le temps à l'échelle du système a été apportée.

Si $|\text{Compensation de décalage}| \leq \text{Seuil d'échelon}$

$$\text{Décalage}_{\text{dest}(\text{dernier})} = \text{Décalage}_{\text{dest}}$$

$$\text{Décalage}_{\text{src}(\text{dernier})} = \text{Décalage}_{\text{src}}$$

où Seuil d'échelon est un nombre qui définit le plus petit palier de temps qui peut se produire dans le système. Pour les systèmes CIP Motion, ce nombre est généralement établi au RPI de la Connexion CIP Motion par l'intermédiaire de l'attribut de classe d'Objet Axe de dispositif de mouvement facultatif, Seuil d'échelon. Si l'attribut de Seuil d'échelon n'est pas pris en charge dans le dispositif, la valeur par défaut de Seuil d'échelon de 1 ms est utilisée.

Si la compensation de décalage est supérieure au seuil d'échelon, indiquant qu'une réponse à un échelon dans le temps à l'échelle du système a été apportée, l'algorithme de compensation de décalage comporte (et non réinitialise) les dernières valeurs de décalage temporel de "réponse préalable à un échelon" applicables au dispositif d'entraînement et au contrôleur jusqu'à ce que le changement de décalage temporel mis en correspondance soit détecté et abaisse la compensation de décalage en dessous du seuil d'échelon. Une fois apportée la réponse à un échelon dans le temps, la valeur Compensation de décalage est une nouvelle fois faible et le temps système du contrôleur correspond étroitement au temps système du dispositif d'entraînement. Selon l'emplacement du réseau de l'horloge maîtresse principale, le dispositif d'entraînement pourrait constater le changement de décalage temporel avant le contrôleur ou inversement. L'algorithme traite des deux cas.

Le Tableau 25 donne un exemple de fonctionnement de la compensation de décalage du temps système. Pour la clarté de l'illustration, un paquet est supposé être envoyé à l'intervalle de synchronisation sans retard de propagation.

Tableau 25 – Propagation d'une réponse à un échelon dans le temps (exemple 2)

Intervalle de synchronisation temporelle	Temps de l'horloge maîtresse	Source de datation (contrôleur)					Destination de datation (dispositif d'entraînement)					
		Horloge locale	Décalage	Dernier décalage	Temps local	Datation	Horloge locale	Décalage	Dernier décalage	Temps local	Comp. de décalage	Datation adj.
1	100	0	100	100	100	100	0	100	100	100	0	100
2	200 + 1 000	100	100	100	200	200	100	100	100	200	0	200
3	1 300	200	100 + 1 000	100	1 300	1 300	200	100	100	300	-1 000	300
4	1 400	300	1 100	100	1 400	1 400	300	100 + 1 000	100	1 400	0	1 400
5	1 500	400	1 100	1 100	1 500	1 500	400	1 100	1 100	1 500	0	1 500

À l'intervalle de synchronisation 2, une réponse à un échelon de 1 000 se produit dans l'horloge maîtresse et est propagée à l'horloge source de datation, par exemple, le contrôleur, dans l'intervalle de synchronisation 3. La datation reçue à l'horloge de destination, le dispositif d'entraînement dans cet exemple, est compensée par la valeur Compensation de décalage de 1 000 selon l'algorithme ci-dessus, produisant une valeur Datation de contrôleur ajustée conforme à l'heure locale du dispositif d'entraînement même si elle n'a pas encore constaté la réponse à un échelon. Dans l'intervalle de synchronisation 4, le dispositif d'entraînement observe la réponse à un échelon et la valeur Compensation de décalage calculée est une nouvelle fois égale à 0, étant donné que les termes de décalage delta de source et de destination s'annulent dans l'algorithme de compensation de décalage. Dans la mesure où la valeur Compensation de décalage est à présent inférieure au seuil d'échelon, les valeurs Dernier décalage sont initialisées par rapport aux valeurs Décalage actuelles comme indiqué dans l'intervalle de synchronisation 5, et aucune autre compensation de décalage n'est exigée.

6.4.6.7.3 Compensation d'échelon de fréquence

Tandis que l'algorithme ci-dessus est parfaitement adapté aux cas où une réponse à un échelon importante dans le cadre du temps système est propagée dans le système, il ne traite pas du cas où une réponse à un échelon est apportée par rapport à la fréquence de l'horloge maîtresse, comme cela peut se produire avec une nouvelle horloge maîtresse principale. Lorsque cela se produit, un changement important par rapport au décalage du temps système a lieu à chaque fois que le dispositif reçoit un message de synchronisation IEC 61588:2009 pour autant de cycles de synchronisation temporelle qui sont nécessaires pour une synchronisation complète de l'horloge du dispositif sur la nouvelle horloge maîtresse principale. Ces décalages temporels peuvent ou peuvent ne pas se produire pour le contrôleur, notamment si le contrôleur endosse le rôle de la nouvelle horloge maîtresse principale, auquel cas la valeur Décalage temporel du contrôleur ne change jamais. Lorsque le contrôleur ne fait jamais état d'un changement de décalage temporel correspondant, les changements de décalage temporels s'accumulent alors dans le dispositif et introduisent un décalage de phase entre ce dispositif et le contrôleur qui peut être assez important, c'est-à-dire supérieur à 1 ms. Ceci est généralement inacceptable lors de l'exécution d'une application de commande de mouvement à haute performance.

Afin de minimiser ce cas de changement de fréquence par échelon, l'algorithme de compensation de décalage ci-dessus doit être modifié pour compenser le décalage de phase associé. Le changement de fréquence maximum qui peut se produire au cours d'un changement de l'horloge maîtresse principale est limité à 0,02 % par la tolérance du quartz spécifiée dans l'IEC 61588:2009. Ceci limite l'ampleur des changements de décalage temporel au produit du différentiel de fréquence maximum, de l'intervalle de synchronisation temporelle et du nombre de sauts d'horloge limite entre l'horloge maîtresse principale et le dispositif. Une limite pratique applicable aux changements de décalage temporel dus au changement de l'horloge maîtresse principale est fixée à 1 ms environ.

Le niveau de décalage de phase qui peut être introduit entre les mises à jour de messages de synchronisation dues à l'échelon de fréquence peut être important, par exemple de 100 µs environ. Tenter d'ajuster la valeur Compensation de décalage selon ce niveau générerait immédiatement une perturbation de mouvement importante si le dispositif fonctionnait conjointement. Ainsi, pour éviter de perturber le mouvement, l'algorithme de compensation de décalage temporel doit être modifié afin de réduire progressivement la valeur Compensation de décalage à zéro pour éliminer le décalage de phase. Cette correction progressive est effectuée à une vitesse de 1 µs de correction de phase par milliseconde de temps d'exécution, assurant ainsi qu'une correction de phase de 1 ms peut être effectuée dans un intervalle de synchronisation temporelle de 1 s.

Lorsque le dispositif est à mi-parcours de réalisation de cette correction de phase, et lorsqu'un nouveau décalage temporel est transmis au dispositif par le contrôleur, il est possible que le processus de correction de phase soit sur le point de s'interrompre de manière prématurée, entraînant un décalage de phase persistant entre les horloges de temps système du contrôleur et du dispositif. Dans la mesure où cela n'est pas acceptable, l'algorithme modifié garantit que tout décalage de phase non corrigé est dirigé vers l'intervalle de synchronisation temporel suivant.

L'algorithme de compensation de décalage temporel modifié est le suivant:

$$\text{Datation}_{\text{comp}} = \text{Datation}_{\text{rec}} + \text{Compensation de décalage.}$$

où,

$$\text{Compensation de décalage} = \text{Variance} - \text{Correction de phase.}$$

Les valeurs applicables aux nouveaux termes Variance et Correction de phase sont calculées sur la base du changement ou non des valeurs Décalage temporel. Les deux cas de figure sont les suivants.

Si $\text{décalage}_{\text{dest}}$ ou $\text{Décalage}_{\text{src}}$ a changé par rapport à sa dernière valeur:

$$\text{Variance} = ((\text{Décalage}_{\text{dest}} - \text{Décalage}_{\text{dest}(\text{dernier})}) - (\text{Décalage}_{\text{src}} - \text{Décalage}_{\text{src}(\text{dernier})}))$$

$$\text{Variance} = \text{Variance} + \text{Phase Restante}$$

$$\text{Correction de phase} = 0$$

$$\text{Phase restante} = 0$$

Si $|\text{Variance}| \leq \text{Seuil d'échelon}$

Cas = Synchronisation normale

$$\text{Décalage}_{\text{dest}(\text{dernier})} = \text{Décalage}_{\text{dest}}$$

$$\text{Décalage}_{\text{src}(\text{dernier})} = \text{Décalage}_{\text{src}}$$

Sinon $|\text{Variance}| > \text{Seuil d'échelon}$

Cas = Réponse à un échelon de temps

Sinon, si $\text{Décalage}_{\text{dest}}$ et $\text{Décalage}_{\text{src}}$ n'ont pas changé:

Cas = Synchronisation normale

Si $\text{Variance} = \text{Correction de phase}$

Pas de fonctionnement

Sinon si $\text{Variance} > \text{Correction de phase}$

$$\text{Correction de phase} = \text{Correction de phase} + \text{RPI}/1000$$

Si $\text{Correction de phase} > \text{Variance}$

$$\text{Correction de phase} = \text{Variance}$$

$$\text{Phase Restante} = \text{Variance} - \text{Correction de phase}$$

Sinon si $\text{Variance} < \text{Correction de phase}$

$$\text{Correction de phase} = \text{Correction de phase} - \text{RPI}/1000$$

$\text{Correction de phase} < \text{Variance}$

$$\text{Correction de phase} = \text{Variance}$$

$$\text{Phase restante} = \text{Variance} - \text{Correction de phase.}$$

Cas = Réponse à un échelon de temps

Pas de fonctionnement

Cet algorithme de compensation de décalage temporel à version modifiée est très robuste, et couvre ainsi les réponses à un échelon dans le cadre du temps système, et les réponses à un échelon dans le cadre de la fréquence de l'horloge maîtresse principale. L'algorithme élimine les décalages de phase persistants entre le contrôleur et le dispositif dans ces circonstances sans perturber le mouvement en fonctionnement en temps réel.

6.5 Procédure de démarrage du dispositif

6.5.1 Généralités

Ce paragraphe présente le processus de démarrage d'un dispositif CIP Motion. Par souci de concision, les messages exacts et leurs formats ne sont pas présentés. Il s'agit de présenter les étapes et les échanges mis en place entre le contrôleur et le dispositif afin de configurer correctement le dispositif CIP Motion.

Le démarrage du dispositif implique trois processus distincts initiés par le contrôleur CIP Motion et auxquels tous les dispositifs CIP Motion présents sur le réseau répondent. Il s'agit de:

- 1) la création d'une connexion E/S de mouvement;

- 2) la configuration d'un objet de mouvement;
- 3) la synchronisation temporelle.

Ces processus peuvent se dérouler en parallèle ou consécutivement, selon la mise en œuvre particulière. La configuration du dispositif peut se produire via la Connexion E/S CIP Motion ou une messagerie explicite connectée distincte. Le processus de démarrage est présenté dans les paragraphes ci-dessous.

6.5.2 Création d'une connexion E/S de mouvement

Les connexions E/S de mouvement sont créées sur EtherNet/IP à l'aide des services CIP communs `Forward_Open` ou `Large_Forward_Open` (voir IEC 61158-5-2 et IEC 61158-6-2). Ces services spécifient la taille maximale des structures de données de connexion Contrôleur-Dispositif et Dispositif-Contrôleur et ciblent de manière explicite un point de connexion d'entrée et de sortie pour la classe d'Objet Axe de dispositif de mouvement.

Afin d'ouvrir cette connexion à la classe d'Objet Axe de dispositif de mouvement, l'expéditeur établit un service `Forward_Open` à l'aide du segment Clé électronique et deux chemins d'application qui représentent le bloc de données de configuration et l'ensemble des blocs de données de consommation/production à utiliser avec cette connexion. Se reporter à l'objet Gestionnaire de connexion dans l'IEC 61158-5-2 et l'IEC 61158-6-2 pour les détails complets du service `Forward_Open` service.

Le chemin de configuration est codé comme point de connexion 81 (0x51) de l'Objet Axe de dispositif de mouvement (0x42). Le chemin de données de connexion E/S est codé comme point de connexion 2 de l'Objet Axe de dispositif de mouvement, et doit toujours correspondre à la révision de format des structures de données des connexions C-D et D-C. Ensuite, le segment de données (0x80) contient 4 mots (8 octets) de données de configuration comme défini par le bloc de configuration de segments de données ci-dessous. Ainsi, pour cette version de la spécification, il convient que le chemin d'application et le segment de données du service `Forward_Open` apparaissent comme suit:

20 42 2C 51 2C 02 80 04 ...suivis du bloc de configuration à 4 mots (Rév 1)

Outre le chemin d'application, les informations suivantes sont également incluses dans le service `Forward_Open`:

Taille de connexion T-O: 220 octets (exemple uniquement)

Type de taille de connexion T-O: Variable

Type de connexion T-O: Point à Point

Taille de connexion O-T: 220 octets (exemple uniquement)

Type de taille de connexion O-T: Variable

Type de connexion O-T: Point à Point

Les informations de connexion O-T correspondent à la connexion Contrôleur-Dispositif tandis que les informations de connexion T-O correspondent à la connexion Dispositif-Contrôleur. Dans l'exemple ci-dessus, la taille de connexion T-O et O-T peut varier selon la mise en œuvre du contrôleur.

Comme indiqué ci-dessus, outre les informations concernant les points de connexion et de mise à la clé des dispositifs, le service `Forward_Open` contient également un bloc de configuration dans le segment de données de la demande de service. Le bloc de configuration sert à vérifier que la structure de puissance associée au dispositif correspond à celle du dispositif sélectionné par le contrôleur.

Le bloc de configuration applicable à la révision de format 1 est une structure de données à 8 octets définie comme présenté à la Figure 50.

← Mot à 32 bits (octet 1 octet 2 octet 3 octet 4) →		
Bloc de configuration de segments de données		
Révision de configuration	Bits de configuration	(réservé)
ID de classe de structure de puissance d'entraînement		

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Figure 50 – Révision de format 1 de bloc de configuration (Point de connexion 81)

L'élément Révision de format représente la révision de la structure de données du bloc de configuration. Si le format du bloc de connexion change dans les révisions futures de cette spécification, le numéro de révision de format doit être incrémenté comme doit l'être le point de connexion associé. La révision de format pour le bloc de configuration comme défini ci-dessus est de 1, ce qui correspond au point de connexion 81 (0x51).

Le champ Bits de configuration pour un dispositif CIP Motion ne comporte qu'un seul bit actuellement défini, à savoir le bit Vérifier les caractéristiques assignées de puissance en tant que bit 0. Lorsque ce bit est défini, le dispositif doit vérifier que l'ID de classe de structure de puissance d'entraînement du service Forward_Open correspond à la valeur d'attribut de l'ID de classe de structure de puissance d'entraînement. Lorsque les valeurs ne correspondent pas, la réponse du service Forward_Open doit indiquer une erreur de segment de données. Plus particulièrement, le dispositif d'entraînement doit renvoyer un état général de 0x09 (Erreur de segment de données) avec un état étendu de 0x04.

La structure du bloc de configuration pour la révision de format 2, à savoir le point de connexion 82 (0x52), est une structure de données de longueur variable qui étend la structure de révision de format 1 afin de contenir les informations concernant l'ID d'axe de structure de puissance d'entraînement et nécessaires pour vérifier les structures de puissance associées à chaque instance d'axe de l'entraînement multiaxe. La taille du segment de données, en mots, détermine le nombre d'éléments ID d'axe de structure de puissance d'entraînement dans la matrice.

Le bloc de configuration applicable à la révision de format 2 est une structure de données à 8 octets définie comme présenté à la Figure 51.

← Mot à 32 bits (octet 1 octet 2 octet 3 octet 4) →		
Bloc de configuration de segments de données		
Révision de format	Bits de configuration	(réservé)
ID de classe de structure de puissance d'entraînement		
ID d'axe de structure de puissance d'entraînement		

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Figure 51 – Révision de format 2 de bloc de configuration (Point de connexion 82)

Dans ce cas, lorsque le bit Vérifier caractéristiques assignées de puissance est défini, le dispositif doit vérifier soit que l'ID de classe de structure de puissance d'entraînement du service Forward_Open correspond à la valeur d'attribut de l'ID de classe de structure de puissance d'entraînement du dispositif de même nature, ou que les ID d'axe de structure de puissance d'entraînement équivalent aux valeurs d'attribut de l'ID d'axe de structure de puissance d'entraînement correspondantes associées à chaque instance d'axe d'entraînement. Si, dans l'un des deux cas, les valeurs ne correspondent pas, la réponse du service Forward_Open doit indiquer une erreur de segment de données. Plus particulièrement, le dispositif d'entraînement doit renvoyer un état général de 0x09 (Erreur de segment de données) avec un état étendu de 0x04 si les valeurs de l'ID de classe de structure de puissance d'entraînement ne correspondent pas. Si les valeurs de l'ID d'axe de structure de puissance d'entraînement ne correspondent pas, le service Forward_Open renvoie un état étendu de 0x05 + n, où n est l'indice du premier élément ID d'axe de structure de puissance d'entraînement dans le bloc de configuration défaillant. Il est nécessaire que les entraînements monoaxe ou multiaxe avec des structures de puissance identiques utilisent uniquement l'ID de classe de structure de puissance d'entraînement pour vérifier les

caractéristiques assignées de puissance, auquel cas le bloc de configuration de révision 2 se décline en structure de bloc de configuration de révision 1. Toutefois, dans le cas d'un entraînement multiaxe avec des structures de puissance indépendantes, l'ID de classe de structure de puissance d'entraînement est mis à 0 et la matrice des ID d'axe de structure de puissance d'entraînement contient tous les ID qu'il faut vérifier par rapport aux valeurs d'attribut de l'ID d'axe de structure de puissance associées à chaque instance d'axe d'entraînement. Voir les descriptions relatives à l'attribut d'ID de classe de structure de puissance d'entraînement et à l'attribut d'instance d'ID d'axe de structure de puissance d'entraînement pour plus de détails.

La prise en charge de la révision de format 2 de la structure du bloc de configuration est exigée lors de la mise en œuvre d'un dispositif d'entraînement CIP Motion multiaxe modulaire avec des structures de puissance mixtes. Il est exigé que tous les autres types de dispositifs CIP Motion (par exemple, entraînements monoaxe et entraînements multiaxe avec caractéristiques assignées de structure de puissance identiques, codeurs, convertisseurs, etc.) prennent en charge la révision de format 1 ou 2. Les dispositifs CIP Motion sont tenus de prendre en charge uniquement une seule révision de format. Il n'est pas exigé que les dispositifs CIP Motion qui utilisent la révision de format 1 prennent en charge la révision de format 2. De façon similaire, il n'est pas exigé que les dispositifs CIP Motion qui utilisent la révision de format 2 prennent en charge la révision de format 1. Les contrôleurs sont généralement capables d'envoyer au dispositif l'un ou l'autre des formats de bloc de configuration, et doivent utiliser toute révision de format prise en charge par le profil de dispositif spécifique.

Le RPI (Intervalle de paquets demandé) du service Forward_Open pour cette connexion E/S de classe 1 est défini sur la période de mise à jour du contrôleur. Cette valeur est appliquée à l'attribut de classe de période de mise à jour du contrôleur de l'Objet Axe de dispositif de mouvement. La synchronisation CIP Motion exige que l'API de connexion (Intervalle de paquets réel) corresponde au RPI de la demande de service Forward_Open du contrôleur. Ainsi, dans le contexte d'une connexion E/S CIP Motion, RPI et API sont synonymes.

Une fois les connexions C-D et D-C établies avec succès via le service Forward_Open, le contrôleur et le dispositif sont tous deux chargés de la transmission des paquets de données à l'intervalle RPI.

Le premier paquet C-D transmis par le contrôleur doit contenir un format de connexion et une révision de format valides. Les éléments ID de mise à jour, ID de dernière mise à jour, Compte d'instances et Ensemble de données temporelles sont tous mis à zéro. L'élément Contrôle de nœud est mis à 0x01 dans la mesure où l'objectif est la commande à distance du dispositif. Noter que le bit Données du contrôleur valides dans le champ Contrôle de nœud n'est pas défini, indiquant que le contrôleur n'envoie aucune donnée d'instance valide dans cette mise à jour. Aucune donnée supplémentaire n'est exigée pour la connexion C-D. La Figure 52 présente un bloc de données de connexion C-D initial type.

Bloc de données de connexion C-D initial			
Format de connexion = 6	Révision de format = 2	ID de mise à jour = 0	Contrôle de nœud = 1
Compte d'instances = 0	0	Dernier ID reçu = 0	Ensemble de données temporelles = 0

IEC

Figure 52 – Bloc de données de connexion C-D initial type

De façon similaire, le premier paquet D-C transmis par le dispositif, en réponse au paquet C-D initial, doit contenir un format de connexion et une révision de format valides. Les éléments ID de mise à jour, Compte d'instances et Ensemble de données temporelles sont tous mis à zéro. L'élément État de nœud est mis à 0x01 dans la mesure où l'objectif est la commande à distance du dispositif. Dans la mesure où il existe une faible connexion CIP Motion, voire aucune connexion, de traitement de fonctionnalité en cours d'exécution dans cette phase précoce de la séquence de démarrage, les éléments Alarmes de nœud et Défaut de nœud sont tous également mis à 0. Noter que le bit Données du dispositif valides dans le champ État de nœud n'est pas défini, indiquant que le dispositif n'envoie aucune donnée d'instance

valide à cet instant. Jusqu'à ce que le contrôleur définisse le bit Données du contrôleur valides dans le champ Contrôle de nœud, le dispositif continue simplement à n'envoyer que l'en-tête de connexion D-C avec tous les éléments ID de mise à jour, Compte d'instances et Ensemble de données temporelles mis à zéro. Aucune donnée supplémentaire n'est exigée pour la connexion D-C dans cette phase. La Figure 53 présente un bloc de données de connexion D-C initial type.

Bloc de données de connexion D-C initial			
Format de connexion = 7	Révision de format = 2	ID de mise à jour = 0	État de nœud = 0
Compte d'instances = 0	Défaut/Alarme de nœud = 0	ID de dernière mise à jour = 0	Ensemble de données temporelles = 0

IEC

Figure 53 – Bloc de données de connexion D-C initial type

Une fois les premiers paquets C-D et D-C transmis, chaque dispositif maintient sa partie de la Connexion E/S CIP Motion sous tension par l'envoi de paquets de rafraîchissement au RPI. Les paquets de rafraîchissement sont définis comme des paquets CIP qui ont le même compte de séquences CIP que le paquet CIP transmis précédemment. La transmission et le traitement de ces paquets de rafraîchissement sont généralement traités par des programmeurs d'application CIP à bas niveau et sont transparents pour le gestionnaire de Connexion CIP Motion de niveau supérieur.

Une fois que la tâche du contrôleur est en cours d'exécution et que le contrôleur est prêt à gérer la connexion CIP, ce même contrôleur est également prêt à envoyer les données de configuration au dispositif, marquant ainsi la fin de la phase de connexion de la procédure de démarrage et le début de la phase de configuration.

6.5.3 Configuration de l'Objet Axe de dispositif de mouvement

La configuration de l'Objet Axe de dispositif de mouvement est assurée à l'aide du service Set_Attributes_List commun ou du service Set_Axis_Attributes_List spécifique à l'objet défini par l'Objet Axe de dispositif de mouvement et pris en charge par la Connexion E/S CIP Motion. Le présent paragraphe suppose que le contrôleur utilise la connexion E/S CIP pour configurer le dispositif, dans la mesure où ceci serait plus typique, et où la méthode qui consiste à appliquer le service Définir liste d'attributs est déjà parfaitement comprise.

La première étape du processus de configuration consiste à établir les attributs de classe de l'Objet Axe de dispositif de mouvement. Pour ce faire, le contrôleur envoie un service Set_Axis_Attributes_List à la classe (c'est-à-dire instance 0) de l'objet. La Figure 54 représente le contenu type du premier paquet de configuration d'attribut de classe C-D.

Premier bloc de données de connexion C-D de configuration d'instance de classe			
Format de connexion = 6	Révision de format = 2	ID de mise à jour = 1	Contrôle de nœud = 5
Compte d'instances = 1	0	Dernier ID reçu = 0	Ensemble de données temporelles = 0
Numéro d'instance = 0	-	Taille de bloc d'instance = 8	Taille de bloc cyclique = 0
Taille de bloc de consigne = 0	Taille de bloc d'écriture = 0	Taille de bloc d'événement = 0	Taille de bloc de service = 6
ID de transaction = 1	Code de service = 76	0	0

Données de demande spécifiques à la Set_Axis_Attributes_List (2 attributs)

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Figure 54 – Contenu type du premier paquet de configuration d'attributs de classe C-D

Noter que l'ID de mise à jour augmente à présent avec chaque paquet envoyé par le contrôleur et que l'élément Contrôle de nœud est à présent mis à 5, indiquant que le contrôleur envoie à présent des données d'instance valides. Le compte d'instances est égal à 1 étant donné que dans ce cas il est envoyé des données uniquement à la classe, c'est-à-dire numéro d'instance 0. Noter également que l'ensemble de données temporelles est égal à 1 et que le contrôleur envoie la période de mise à jour correspondante au dispositif. Seules les

tailles des blocs d'instance et de service ont des valeurs non nulles. Une valeur de 0 relative aux tailles des blocs cyclique et d'événement indique que ces blocs ne sont pas inclus dans ce bloc de données de connexion C-D.

Le dispositif traite ce paquet C-D dès que possible afin d'assurer une transmission D-C au moment opportun. L'ID de mise à jour propre au paquet D-C augmente également à présent avec chaque paquet envoyé par le dispositif qui indique que les données d'instance sont désormais valides. Ce paquet peut ou peut ne pas inclure la demande de service `Set_Axis_Attributes_List` selon la priorité accordée au traitement de la demande de service. La Figure 55 représente une réponse type au premier paquet de configuration de classe C-D décrit ci-dessus, en supposant l'inclusion des données de réponse de service.

Premier bloc de données de connexion D-C de configuration d'instance de classe			
Format de connexion = 7	Révision de format = 2	ID de mise à jour = 1	État de nœud = 5
Compte d'instances = 1	Défaut/Alarme de nœud = 0	Dernier ID reçu = 1	Ensemble de données temporelles = 0
Numéro d'instance = 0	-	Taille de bloc d'instance = 7	Taille de bloc cyclique = 0
Taille de bloc réelle = 0	Taille de bloc de lecture = 0	Taille de bloc d'événement = 0	Taille de bloc de service = 5
ID de transaction = 1	Code de service = 76	État général = 0	État étendu = 0
Données de réponse spécifiques à la <code>Set_Axis_Attributes_List</code> (2 attributs)			

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Figure 55 – Réponse type au premier paquet de configuration de classe C-D

Outre l'envoi de cette réponse, le dispositif planifie également la transmission D-C suivante de sorte qu'elle corresponde à 1 période de mise à jour du contrôleur à partir du moment de réception du paquet C-D. Si le paquet C-D suivant envoyé au dispositif est retardé ou perdu, le dispositif envoie le paquet D-C suivant comme planifié précédemment, et planifie la transmission de paquet D-C suivante. Toutefois, le paquet C-D suivant arrive normalement avant la transmission D-C planifiée précédemment, et le dispositif traite alors le paquet, envoie immédiatement la réponse D-C et planifie la transmission suivante pour qu'elle ait lieu une période de mise à jour de contrôleur plus tard. Cette procédure synchronise les transmissions de paquets D-C de dispositif avec la planification de la tâche du contrôleur sans exiger la synchronisation réciproque des horloges du contrôleur et du dispositif.

NOTE En général, les ID de mise à jour associés aux paquets C-D et D-C pour un cycle donné correspondent.

Une fois les attributs de classes configurés, la configuration de l'Objet Axe de dispositif de mouvement se poursuit avec la configuration des instances d'objet individuelles jusqu'à ce que toutes les instances soient configurées. Cette configuration s'effectue généralement une instance après l'autre, mais pourrait utiliser plusieurs instances pour une configuration en parallèle de tous les axes. En fait, la configuration d'instance d'axe initiale pourrait commencer dans le même paquet que la configuration d'instance de classe. Le dispositif est tenu de traiter tous ces cas. La Figure 56 représente le contenu type du premier paquet de configuration d'instance d'axe C-D.

Premier bloc de données de connexion C-D de configuration d'instance d'axe			
Format de connexion = 6	Révision de format = 2	ID de mise à jour = 4	Contrôle de nœud = 5
Compte d'instances = 1	0	Dernier ID reçu = 3	Ensemble de données temporelles = 0
Numéro d'instance = 1	-	Taille de bloc d'instance = 84	Taille de bloc cyclique = 0
Taille de bloc de consigne = 0	Taille de bloc d'écriture = 0	Taille de bloc d'événement = 0	Taille de bloc de service = 82
ID de transaction = 1	Code de service = 76	0	0
Données de demande spécifiques à la <code>Set_Axis_Attributes_List</code> (40 attributs)			

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Figure 56 – Contenu type du premier paquet de configuration d'instance d'axe C-D

En réponse à ce paquet, le dispositif envoie le paquet de données comme présenté à la Figure 57.

Premier bloc de données de connexion D-C de configuration d'instance d'axe			
Format de connexion = 7	Révision de format = 2	ID de mise à jour = 4	État de nœud = 5
Compte d'instances = 1	Défaut/Alarme de nœud = 0	Dernier ID reçu = 4	Ensemble de données temporelles = 0
Numéro d'instance = 0	-	Taille de bloc d'instance = 45	Taille de bloc cyclique = 0
Taille de bloc réelle = 0	Taille de bloc de lecture = 0	Taille de bloc d'événement = 0	Taille de bloc de service = 43
ID de transaction = 1	Code de service = 76	État général = 0	État étendu = 0
Données de réponse spécifiques à la Set_Axis_Attributes_List (40 attributs)			

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Figure 57 – Réponse type au premier paquet de configuration d'axe C-D

Une fois toutes les instances d'objet configurées, le contrôleur marque la fin de la phase de configuration du processus de démarrage en définissant le bit Configuration achevée dans le champ État de commande du paquet C-D suivant envoyé au dispositif.

6.5.4 Synchronisation temporelle

Tous les nœuds de dispositif CIP Motion qui prennent en charge les services de synchronisation sont tenus d'être synchronisés avec un contrôleur CIP Motion dans le cadre d'un groupe, communément appelé Groupe Motion. Tous les dispositifs de synchronisation temporelle dans ce groupe reçoivent un service Group_Sync qui contient l'ID de l'horloge maîtresse principale associé au contrôleur. S'agissant d'un service de classe, la structure du paquet ressemble à celle présentée à la Figure 58. Bien que cet exemple montre seulement le Bloc de données d'instances de classe, les Blocs de données d'instance de classe et d'instance d'axe peuvent coexister dans le même paquet. Ceci permet l'exécution simultanée du processus de synchronisation temporelle et du processus de configuration de l'objet Dispositif de mouvement. Une exigence du contrôleur stipule cependant que tous les attributs de classe associés à l'Objet Axe de dispositif de mouvement de ce dispositif doivent être configurés avant que le contrôleur n'envoie la première demande de service Group_Sync.

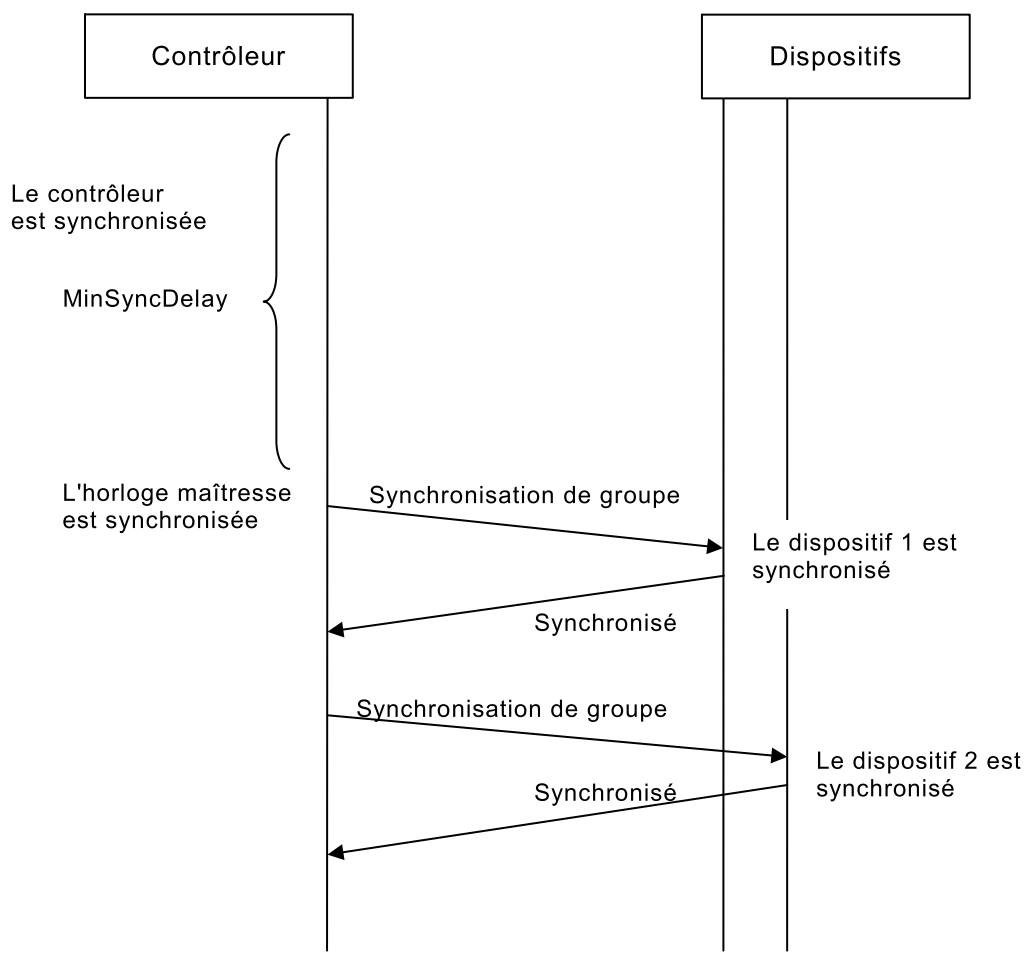
Bloc de données de connexion C-D de la demande de service de synchronisation temporelle			
Format de connexion = 6	Révision de format = 2	ID de mise à jour = 12	Contrôle de nœud = 5
Compte d'instances = 1	0	Dernier ID reçu = 11	Ensemble de données temporelles = 0
Numéro d'instance = 0	-	Taille de bloc d'instance = 5	Taille de bloc cyclique = 0
Taille de bloc de consigne = 0	Taille de bloc d'écriture = 0	Taille de bloc d'événement = 0	Taille de bloc de service = 3
ID de transaction = 3	Code de service = 28	0	0
Données de demande spécifiques à Group_Sync (ID d'horloge maîtresse principale)			

IEC

Figure 58 – Contenu type du paquet de demande de service de synchronisation temporelle C-D

Cette procédure a pour objet de garantir la synchronisation réciproque de tous les dispositifs dans le Groupe Motion ou, en d'autres termes, que leur notion du temps est identique en fonction d'une Horloge maîtresse principale commune. Outre la garantie que tous les dispositifs partagent la même Horloge maîtresse principale que le contrôleur, la procédure garantit également qu'aucune réponse à un échelon du Temps système ne s'est produite au moment de la synchronisation qui pourrait étaler de façon permanente une horloge de dispositif par rapport à l'horloge du contrôleur.

La Figure 59 représente la façon dont le contrôleur attend d'abord la synchronisation de son horloge locale à l'horloge de temps système maîtresse. L'Objet Synchronisation temporelle du contrôleur est chargé de déterminer si ce dernier est synchronisé. Le contrôleur attend ensuite une période minimale, MinSyncDelay, pour permettre aux autres composants dans le système de se synchroniser à une Horloge maîtresse principale commune. Une valeur type pour ce délai est de 10 s.



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Figure 59 – Synchronisation de groupe de dispositifs CIP Motion

A la fin de cette période, le contrôleur lance une série de services de synchronisation de groupe vers chacun des dispositifs CIP Motion. Chaque dispositif renvoie une réponse indiquant s'il est également synchronisé avec le contrôleur selon le critère décrit pour le service Group_Sync de l'Objet Axe de dispositif de mouvement. Si la réponse de Group_Sync indique que le dispositif est synchronisé avec le contrôleur, ce dernier effectue une dernière vérification d'étalement et en l'absence de tout étalement temporel, le dispositif et le contrôleur sont considérés comme totalement synchronisés. Lorsque tous les dispositifs sont totalement synchronisés avec le contrôleur, le Groupe Motion est considéré comme étant synchronisé et le contrôleur est libre de commencer à envoyer des données cycliques à tous les dispositifs CIP Motion associés au Groupe Motion.

Lorsque le contrôleur rencontre un dispositif qui ne renvoie pas une réponse "dispositif est synchronisé" au service Group_Sync après des essais répétés pendant au moins 1 minute, il doit indiquer qu'un défaut de connexion s'est produit pour ce dispositif.

Lorsque le contrôleur et les dispositifs CIP Motion du Groupe Motion sont synchronisés au groupe, le contrôleur peut alors définir le bit Commande synchrone de l'en-tête Connexion Contrôleur-Dispositif. Définir le bit Commande synchrone dans le paquet cyclique suivant envoyé aux dispositifs permet au dispositif de planifier sa mise à jour de connexion Dispositif-Contrôleur suivante, en fonction de la datation du contrôleur contenue dans les données de connexion. Plus particulièrement, le dispositif calcule la durée de mise à jour de connexion Dispositif-Contrôleur synchrone initiale comme suit:

Datation du contrôleur + 2 × Période de mise à jour du contrôleur (RPI de Connexion CIP Motion) – Temps de traitement de transmission.

Le temps de traitement de transmission est le temps nécessaire pour assembler et préparer la connexion Dispositif-Contrôleur en vue de sa transmission. Lorsque la transmission planifiée a lieu, le dispositif indique qu'il envoie à présent les données de connexion Dispositif-Contrôleur synchronisées en définissant le bit Mode synchrone dans l'en-tête du paquet Dispositif-Contrôleur envoyé au contrôleur.

L'ID de mise à jour de la connexion Dispositif-Contrôleur doit être initialisé pendant ce premier cycle de mise à jour synchrone afin de correspondre à l'ID de mise à jour de la Connexion Contrôleur-Dispositif pour le cycle concerné et doit ensuite être incrémenté de un à chaque période de mise à jour de connexion. Par conséquent, le maintien des ID de mise à jour correspondants par cycle de mise à jour ne nécessite pas de devoir traiter, voire recevoir, le paquet Contrôleur-Dispositif avant d'assembler le paquet Dispositif-Contrôleur.

L'échange de données entre le contrôleur et le dispositif se poursuit donc conformément au modèle de synchronisation CIP Motion. Ceci marque la fin de la phase de synchronisation du processus de démarrage.

Si, au point de réception de la demande de Commande synchrone dans l'en-tête de connexion Contrôleur-Dispositif, la durée (datation du contrôleur + période de mise à jour du contrôleur) diffère du temps de système de dispositif en cours de plus d'une période de mise à jour du contrôleur, les horloges de temps système du contrôleur et du dispositif sont étalées.

C'est-à-dire,

Si $|(Datation\ du\ contrôleur + CUP) - Temps\ système\ en\ cours| > 1\ CUP$

Les horloges du contrôleur et du dispositif sont étalées

sinon

Les horloges du contrôleur et du dispositif sont OK

Dans ce cas peu probable, le dispositif continue simplement de fonctionner de manière asynchrone et le bit Mode synchrone dans l'en-tête du paquet Dispositif-Contrôleur reste annulé tant que la durée de mise à jour calculée reste dans les limites de tolérance acceptables ou jusqu'à ce qu'une limite de temporisation de 1 minute soit dépassée. Au cours de l'intervalle de temporisation de 1 minute dans lequel les horloges sont étalées, le code d'alarme de nœud du dispositif indique une condition Alarme d'étalement d'horloge. Tout dépassement de la période de temporisation d'étalement d'horloge d'une durée de 1 minute entraînerait la défaillance du dispositif qui définirait le code de défaut de nœud pour indiquer une condition Défaut d'étalement d'horloge.

6.6 Visualisation du dispositif

Le dispositif CIP Motion exige des composants de visualisation permettant de diagnostiquer rapidement son état. Ces composants vont des LED bicolores aux affichages alphanumériques à plusieurs caractères. Le Tableau 26 définit les exigences concernant ces composants de visualisation, ainsi que les libellés appropriés à ces composants lorsque le dispositif les utilise.

Tableau 26 – Composants de visualisation CIP Motion

Composant de visualisation	Besoin dans la mise en œuvre	Libellé associé
LED de statut du réseau	Required (Obligatoire ou O)	“Statut du réseau” “NET”
LED de statut du module	Required (Obligatoire ou O)	“Statut du module” “MOD”
LED de statut d'axe	Conditionnel ^a	“Statut d'axe” “XS” “Statut de # d'axe” “XS#” ^b
Affichage à sept segments	Optional (Facultatif ou F)	N/A
Affichage alphanumérique à plusieurs caractères	Optional (Facultatif ou F)	N/A
^a Il peut être dérogé à cette exigence en utilisant un affichage alphanumérique à plusieurs caractères.		
^b Des dispositifs multiaxe ont besoin de plusieurs LED de statut d'axe (une par axe de dispositif).		

S'ils sont appliqués à des réseaux CIP différents, les comportements LED de statut du réseau et de statut du module doivent être conformes au comportement LED du réseau CIP associé auquel le dispositif est connecté. La mise en correspondance des états de la LED de statut du module du dispositif avec les états de l'Objet Axe de dispositif de mouvement est définie de manière exhaustive dans la section Objet Axe de dispositif de mouvement dédiée à la Visualisation.

6.7 Qualité de Service (Quality of Service ou QoS) Ethernet/IP

Lorsqu'un dispositif CIP Motion prenant en charge les services de synchronisation temporelle est ciblé pour fonctionner sur EtherNet/IP, le protocole Ethernet QoS (ou Qualité de Service) standard doit être utilisé pour garantir que les paquets IEC 61588:2009 CIP Sync et les paquets de Connexion E/S CIP Motion sont délivrés de manière opportune par l'intermédiaire des composants du réseau. QoS pour EtherNet/IP est définie dans l'IEC 61158-4-2 (comportement QoS et Objet QoS facultatif). Alors que les mécanismes QoS définis dans l'IEC 61158-4-2 sont spécifiés comme facultatifs pour les dispositifs EtherNet/IP, les dispositifs CIP Motion doivent au moins mettre en œuvre le marquage DSCP pour les paquets CIP Motion. Il est exigé de transmettre les paquets CIP Sync avec le marquage DSCP. Une priorité urgente CIP doit être définie par défaut par le service Forward_Open qui établit la Connexion CIP Motion. Il est prévu que la pile EtherNet/IP de dispositif CIP Motion accorde une priorité élevée au traitement des paquets CIP Motion et IEC 61588:2009 par rapport aux paquets de priorité moins élevée.

7 Objet Axe de dispositif de mouvement

7.1 Considérations d'ordre général

7.1.1 Généralités

Ce paragraphe définit le comportement d'un objet CIP appelé Objet Axe de dispositif de mouvement. Cet objet est le principal composant fonctionnel du profil de dispositif CIP Motion défini dans le cadre des extensions CIP Motion vers CIP Common. Les instances de cet objet sont obligatoires pour prendre en charge la commande de mouvement en cas de connexion à un contrôleur conforme à CIP Motion par l'intermédiaire d'un réseau CIP. EtherNet/IP est le réseau privilégié pour les commandes multiaxe synchronisées haute performance. Les autres réseaux CIP tels que ControlNet et DeviceNet pourraient être réservés aux applications de dispositif non synchronisé à faible performance telles que les dispositifs d'entraînement à fréquence variable simples, les dispositifs d'entraînement à boucle de vitesse et les dispositifs d'entraînement indexés.

7.1.2 Historique des révisions

L'Article 7 fournit une description complète des attributs d'objets individuels, de leur comportement et des services d'objets associés. Depuis l'édition initiale, des modifications

ont été apportées à la définition de cette classe d'objet qui nécessite donc de faire l'objet d'une révision. Le Tableau 27 présente l'historique des révisions.

Tableau 27 – Historique des révisions de l'Objet Axe de dispositif de mouvement

Révision	Raison de la mise à jour de la définition de l'objet
1	Révision initiale (IEC 61800-7 Edition 1, obsolète)
2	Modifications majeures des interfaces (déconseillé)
3	Modification du comportement du service Group_Sync concernant la détection de l'étalement de l'horloge. Recommandé pour les nouveaux dispositifs CIP Motion prenant en charge la synchronisation temporelle. Modification du service Get_Motor_Test_Data pour ajouter une structure facultative de demande de paramètre.

7.1.3 Présentation de l'objet

L'Objet Axe de dispositif de mouvement est l'un des deux principaux composants d'objet nécessaires à la prise en charge de la commande de mouvement dans l'architecture de commande CIP Motion. Ces composants d'objet sont représentés à la Figure 60.

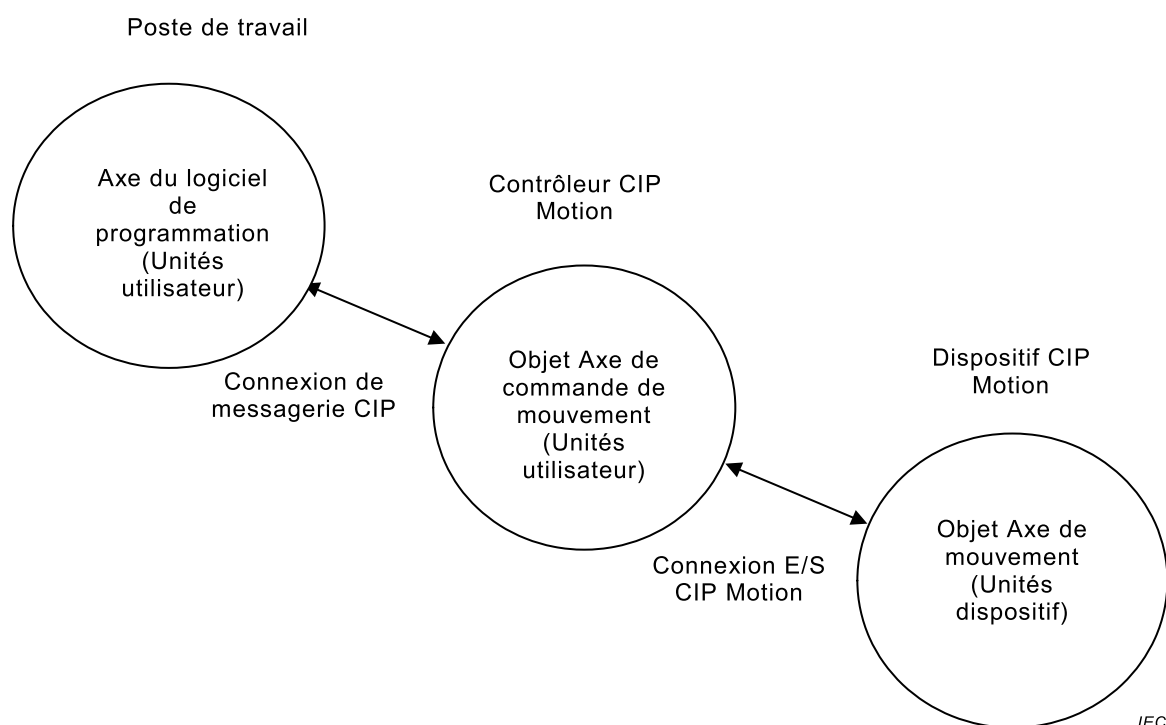


Figure 60 – Composants d'objet de l'architecture de commande CIP Motion

7.1.4 Abstraction de l'Objet Axe de dispositif de mouvement

L'Objet Axe de dispositif de mouvement est une abstraction qui peut être appliquée à n'importe quelle fonction de dispositif de commande de mouvement associée à un composant mobile d'un système de commande de mouvement. Un dispositif de commande de mouvement donné peut donc être représenté par une instance Objet Axe de dispositif de mouvement, et dans la mesure où la plupart de ces fonctions de dispositifs sont associées au mouvement d'un composant de machine, une instance Objet Axe de dispositif de mouvement est parfois simplement appelée "Axe". En règle générale, une instance Objet Axe de dispositif de mouvement est composée d'un ou de plusieurs des éléments intégraux suivants :

- actionneur de mouvement (moteurs rotatifs ou linéaires, actionneurs hydrauliques et pneumatiques, par exemple);
- capteur de mouvement (codeurs, résolveurs, transducteurs ultrasoniques de débattement, par exemple);
- commande d'actionneur (commande de moteur vectorielle, à fréquence et pas à pas, commande de robinet hydraulique, par exemple);
- l'amplificateur qui fournit la puissance de mouvement de l'actionneur (structure de puissance d'excitation des moteurs, par exemple);
- convertisseur: convertit l'alimentation de la ligne à courant alternatif en alimentation du bus à courant continu pour un ou plusieurs amplificateurs associés;
- E/S de mouvement: capteurs d'entrée d'enregistrement, actionneurs par came de sortie.

Les attributs d'instance sont inclus dans l'Objet Axe de dispositif de mouvement pour chacun de ces éléments intégraux afin de décrire le comportement d'axe. Lorsqu'un Objet Axe de dispositif de mouvement est doté d'un élément de commande d'actionneur, les attributs de commande sont inclus dans l'objet afin de contrôler les éléments dynamiques de l'actionneur. L'objet contient également des attributs qui décrivent les éléments dynamiques de l'actionneur, qu'ils soient estimés ou réels.

Le mode d'application le plus fréquent d'un Objet Axe de dispositif de mouvement étant un dispositif "d'entraînement", l'Objet Axe de dispositif de mouvement peut également être appliqué à d'autres dispositifs de mouvement (un codeur CIP Motion, un Convertisseur de puissance CIP Motion, un module d'entrée d'enregistrement CIP Motion, etc. par exemple). Pour ces dispositifs, seul un sous-ensemble de l'attribut Objet Axe de dispositif de mouvement défini est réellement mis en œuvre, le sous-ensemble étant constitué de tous les attributs applicables à la fonction du dispositif.

Dans le cas d'un dispositif de réaction à plusieurs ports, chaque port de réaction est considéré comme un axe et, à ce titre, une instance de l'Objet Axe de dispositif de mouvement lui est affectée. Dans le cas d'un dispositif d'entraînement à plusieurs amplificateurs, chaque amplificateur est associé à un axe, et une instance de l'Objet Axe de dispositif de mouvement lui est affectée. Pour les instances d'axe d'entraînement, l'Objet Axe de dispositif de mouvement peut représenter un entraînement asservi à commande vectorielle en boucle fermée complexe ou un simple dispositif d'entraînement Volts/Hertz à boucle ouverte (ou VFD). Les dispositifs d'entraînement sont souvent dotés de plusieurs ports de réaction qui peuvent être dédiés à des instances d'amplificateur de dispositif d'entraînement ou utilisés comme sources de réaction principales indépendantes. Dans ce dernier cas, le canal de réaction principal est considéré comme un axe, et une instance de l'Objet Axe de dispositif de mouvement lui est affectée.

7.1.5 Objet Axe de commande de mouvement

A chaque instance Objet Axe de dispositif de mouvement sur le dispositif dans le système de commande de mouvement correspond un Objet Axe de commande de mouvement sur le contrôleur. Les valeurs d'attribut contenues dans l'Objet Axe de dispositif de mouvement comportent des attributs de correspondance dans l'Objet Axe de commande de mouvement, la seule différence étant les unités dans lesquelles sont exprimés ces attributs. Dans l'Objet Axe de dispositif de mouvement, les attributs sont exprimés en unités de dispositif fixes (nombres de réactions ou r/min, par exemple) telles que définies dans l'Objet Axe de dispositif de mouvement. Dans l'Objet Axe de commande de mouvement, les attributs sont exprimés en unités définies par l'utilisateur (degrés, pouces ou centimètres, par exemple). Étant donné que le logiciel de programmation interagit avec le dispositif d'entraînement par l'intermédiaire de l'Objet Axe de commande de mouvement, l'ensemble de la programmation de la configuration, du diagnostic et du mouvement est assuré avec les unités préférentielles de l'utilisateur. Par conséquent, plutôt que de programmer l'application de commande de mouvement en nombres de réactions ou en r/min de l'arbre de moteur, par exemple, l'utilisateur peut programmer en centimètres, en pieds, en produits par minute ou en toute autre unité adaptée à l'application spécifique.

Lors du transfert des valeurs d'attribut de configuration ou de commande vers l'Objet Axe de dispositif de mouvement via la Connexion E/S CIP Motion, ces valeurs d'attribut sont ajustées par l'Objet Axe de commande de mouvement dans les valeurs de l'unité de dispositif fixe exigées par l'Objet Axe de dispositif de mouvement. De la même manière, lorsque des attributs de signal sont transférés vers l'Objet Axe de commande de mouvement via la Connexion E/S CIP Motion, ces valeurs d'unité de réaction sont ajustées par l'Objet Axe de commande de mouvement et archivées dans l'objet des unités utilisateur.

7.1.6 Classification de commande de dispositif

En fonction des variations en Mode de commande et Méthode de commande figurant en 4.2, il peut être défini un ensemble de Codes de fonction de dispositif de base autour desquels la plupart des attributs d'Objet Axe de dispositif de mouvement peuvent être organisés. Les codes de fonction de dispositif sont désignés dans les tableaux de l'Article 7 par un identifiant à une seule lettre qui peut être utilisé pour déterminer les attributs d'objet obligatoires (ou facultatifs) pour la mise en œuvre d'un dispositif CIP Motion donné. La liste des codes de fonction de dispositif est présentée ci-dessous:

Codes de fonction:

B – Convertisseurs de puissance de bus (Pas de mode de commande, pas de méthode de commande)

E – Réaction de codeur uniquement (Pas de mode de commande, pas de méthode de commande)

P – Boucle de position (Mode d'asservissement de position, Méthode de commande vectorielle en boucle fermée)

V – Boucle de vitesse (Mode de commande de vitesse, Méthode de commande vectorielle en boucle fermée)

T – Boucle de couple (Mode d'asservissement de couple, Méthode de commande vectorielle en boucle fermée)

F – Contrôle de fréquence (Mode de commande de vitesse, Méthode de contrôle de fréquence)

Combinaisons des codes de fonction de dispositif:

Les combinaisons de ces lettres peuvent être utilisées pour désigner une classe particulière de dispositifs CIP Motion afin d'identifier les attributs applicables.

EXEMPLE "FV" se rapporte à la classe de tous les entraînements à commande de vitesse, que ce soit à commande vectorielle ou à contrôle de fréquence.

Les combinaisons ci-dessous sont si fréquentes que des codes de fonction spéciaux ont été définis pour des raisons de commodité:

N = BE = Toutes les Fonctions de dispositif qui n'utilisent pas de Méthode de commande

O = F = Toutes les Fonctions de dispositif qui utilisent des Méthodes de commande en boucle ouverte (par exemple, Contrôle de fréquence)

C = PVT = Toutes les Fonctions de dispositif qui utilisent des Méthodes de commande en boucle fermée (par exemple, Commande vectorielle PI)

D = OC = Toutes les Fonctions de dispositif qui utilisent des Méthodes de commande (Méthode de commande != Pas de commande)

Outre ces combinaisons, de nombreux attributs sont applicables ou non à un fonctionnement des dispositifs sans capteur ou sans codeur, c'est-à-dire des instances d'axe de dispositif fonctionnant sans un dispositif de réaction. Les codes de fonction de dispositif peuvent être utilisés pour spécifier les règles de mise en œuvre conditionnelle suivantes pour les attributs. Dans ce cas, les règles de mise en œuvre conditionnelle suivantes sont spécifiées:

E = Fonctionnement des dispositifs par codeur.

IE = Fonctionnement des dispositifs sans codeur ou sans capteur.

7.1.7 Obligatoire par rapport à Facultatif dans une mise en œuvre

Dans les paragraphes qui suivent, les attributs et services sont définis comme étant obligatoires ou facultatifs dans la mise en œuvre de l'Objet Axe de dispositif de mouvement. Les attributs et services obligatoires doivent être pris en charge dans la mise en œuvre de l'objet. Les attributs et services facultatifs peuvent être pris en charge ou non dans la mise en œuvre, et sont laissés à la discrétion du fournisseur.

Pour les attributs d'instances, la détermination du caractère obligatoire ou facultatif d'un attribut ou service donné dépend souvent du code de fonction du dispositif associé défini ci-dessus. Si un attribut ou service est marqué comme étant obligatoire pour un code de fonction de dispositif donné, la mise en œuvre d'un dispositif doit alors le prendre en charge s'il est censé fonctionner dans ce mode. Par exemple, un attribut marqué comme étant obligatoire pour le code de fonction de dispositif "V" doit être mis en œuvre par un dispositif CIP Motion dont il est prévu qu'il prenne en charge le mode de boucle de Vitesse.

Dans certains cas, un attribut d'instance ou un service peut ne pas être applicable à un code de fonction de dispositif donné. Cette situation se produit lorsque l'attribut est défini comme n'étant ni obligatoire ni facultatif.

Le Tableau 29 fournit une liste pratique de tous les attributs d'instances de cet objet et identifie si l'attribut est obligatoire ou facultatif, conditionnel ou non applicable dans la mise en œuvre sur la base du code de fonction de dispositif ci-dessus. Les définitions d'énumération des attributs et des bits sont également désignées comme Obligatoires, Facultatifs ou Conditionnelles, avec un code de fonction de dispositif approprié, le cas échéant. Si aucune désignation n'est associée à la définition des énumérations ou des bits, il est alors supposé que l'énumération est obligatoire dans la mise en œuvre.

Pour certains attributs, il existe des règles de mise en œuvre conditionnelle qui dépassent le code de fonction de dispositif. Ces règles sont spécifiées dans la colonne de mise en œuvre conditionnelle du Tableau 29.

EXEMPLE Le Tableau 28 donne un exemple où l'attribut Résistance de moteur à aimants permanents est obligatoire (R) dans la mise en œuvre si le dispositif prend en charge le contrôle de fréquence, l'asservissement de position, la commande de vitesse ou l'asservissement de couple ET le dispositif prend en charge les moteurs à aimant permanent, c'est-à-dire: "Moteur à aimants permanents uniquement". L'attribut N'est Pas Applicable (-) pour un Convertisseur de puissance de bus ou un dispositif à réaction uniquement ou un dispositif d'entraînement qui ne prend pas en charge un moteur à aimant permanent.

Tableau 28 – Exemple de mise en œuvre d'attributs d'instance par rapport au Code de fonction de dispositif

Attribut d'instance			Mise en œuvre par code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
1327	Set*	Résistance du moteur à aimant permanent	-		O	O	O	O	Moteur à aimant permanent uniquement

Légende du Tableau 28 et du Tableau 29: (voir 7.1.6 et les alinéas ci-dessus pour plus de détails)

(O) Attribut obligatoire – doit être pris en charge dans la mise en œuvre.

(F) Attribut facultatif – la prise en charge est laissée à la discrétion du fournisseur.

(C) Attribut conditionnel.

(-) Attribut non applicable.

Set* Indique que l'attribut est en principe défini par le bloc de données de Connexion CIP Motion et non pas par un service Set.

Légende de mise en œuvre conditionnelle:

Moteur rotatif	Moteur à aimant permanent rotatif ou moteur à induction rotatif (type de moteur)
Moteur linéaire	Moteur à aimant permanent linéaire ou moteur à induction linéaire (type de moteur)
PM	Moteur à aimant permanent rotatif ou linéaire (SPM et IPM) (type de moteur)
IM	Moteur à induction rotatif ou linéaire (type de moteur)
SPM	Moteur à aimant permanent de surface rotatif ou linéaire (type de moteur)
IPM	Moteur à aimant permanent d'intérieur rotatif ou linéaire (type de moteur)
TT	AqB numérique (type de réaction)
TP	Parallèle numérique (type de réaction)
SC	Sinus/Cosinus (type de réaction)
HI	Hiperface® (type de réaction)
DSL	Hiperface DSL® (type de réaction)
E21	EnDat 2.1® (type de réaction)
E22	EnDat 2.2® (type de réaction)
SS	SSI (type de réaction)
LT	LDT ou Transducteur ultrasonique de débattement (type de réaction)
RS	Résolveur (type de réaction)
Rotative absolue	Unité de réaction – rév; Méthode de réaction et de démarrage- absolue
Linéaire absolue	Unité de réaction – mètre; Méthode de réaction et de démarrage- absolue
Mise à l'échelle du dispositif d'entraînement	Le dispositif d'entraînement prend en charge la fonctionnalité de mise à l'échelle de l'entraînement.

**Tableau 29 – Mise en œuvre d'attributs d'instances
par rapport au Code de fonction de dispositif**

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
40	Set*	Mode de commande	O	O	O	O	O	O	
41	Set	Méthode de commande	O	O	O	O	O	O	
45	Set	Configuration d'étalonnage du mouvement	-	O	O	O	O	O	Mise à l'échelle de l'entraînement uniquement
77	Set	Unité de mouvement	-	O	O	O	O	O	Mise à l'échelle de l'entraînement uniquement
78	Set	Résolution de mouvement	-	O	O	O	O	O	Mise à l'échelle de l'entraînement uniquement
79	Set	Polarité de mouvement	-	O	O	O	O	O	Mise à l'échelle de l'entraînement uniquement
83	Set	Commande de déroulement cyclique	-	O	O	O	O	O	Mise à l'échelle de l'entraînement uniquement
84	Set	Position de déroulement cyclique	-	O	-	O	O	O	Mise à l'échelle de l'entraînement uniquement; E
89	Set*	État de commande	F	F	F	F	F	F	
90	Set*	Ensemble de données réelles	-	O	O	O	O	O	
91	Set*	Ensemble de données de consigne	-	-	O	O	O	O	
92	Set*	Commande de consigne	-	-	F	O	O	O	
93	Get	Temps de consigne cible	-	-	F	O	F	F	
94	Set*	Ensemble de données d'état	O	O	O	O	O	O	
95	Set*	Ensemble de données d'enregistrement	-	O	O	O	O	O	Mise à l'échelle de l'entraînement uniquement; E
96	Set*	Ensemble de données d'origine	-	O	O	O	O	O	Mise à l'échelle de l'entraînement uniquement; E
97	Set*	Ensemble de données d'observation	-	O	O	O	O	O	Mise à l'échelle de l'entraînement uniquement; E
1310	Get	Numéro de lot du moteur	-	-	F	F	F	F	
1311	Get	Numéro de série du moteur	-	-	F	F	F	F	
1312	Get	Code de date du moteur	-	-	F	F	F	F	
1313	Set	Source de données du moteur	-	-	O	O	O	O	
1314	Set	Code de dispositif du moteur	-	-	O	O	O	O	
1315	Set	Type de moteur	-	-	O	O	O	O	
1316	Set	Unité moteur	-	-	O	O	O	O	
1317	Set	Polarité du moteur	-	-	F	F	F	F	
1318	Set	Tension assignée du moteur	-	-	O	O	O	O	
1319	Set	Courant continu assigné du moteur	-	-	O	O	O	O	
1320	Set	Courant de crête assigné du moteur	-	-	C	C	C	C	O – PM; F – IM

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
1321	Set	Puissance de sortie assignée du moteur	-	-	C	C	C	C	O – PM; F – IM
1322	Set	Limite de surcharge du moteur	-	-	F	F	F	F	
1323	Set	Thermocontacteur interne du moteur	-	-	F	F	F	F	
1324	Set	Temp. d'enroulement max du moteur	-	-	F	F	F	F	
1325	Set	Cap. Therm. Enroul-Amb. du moteur	-	-	F	F	F	F	
1326	Set	Rés. Therm. Enroul-Amb. du moteur	-	-	F	F	F	F	
1327	Set	Résistance du moteur à aimant permanent	-	-	O	O	O	O	Moteur à aimant permanent uniquement
1328	Set	Inductance du moteur à aimant permanent	-	-	O	O	O	O	Moteur à aimant permanent uniquement
1353	Set	Inductance Lq du moteur à aimant permanent	-	-	O	O	O	O	Moteur IPM uniquement
1354	Set	Inductance Ld du moteur à aimant permanent	-	-	O	O	O	O	Moteur IPM uniquement
1329	Set	Pôles du moteur rotatif	-	-	O	O	O	O	Moteur rotatif uniquement
1330	Set	Inertie du moteur rotatif	-	-	F	F	F	F	Moteur rotatif uniquement
1331	Set	Vitesse assignée du moteur rotatif	-	-	O	O	O	O	Moteur rotatif uniquement
1332	Set	Vitesse maximale du moteur rotatif	-	-	F	F	F	F	Moteur rotatif uniquement
1333	Set	Coefficient d'amortisseur du moteur rotatif	-	-	F	F	F	F	Moteur rotatif uniquement
1334	Set	Pas polaire du moteur linéaire	-	-	O	O	O	O	Moteur linéaire uniquement
1335	Set	Vitesse assignée du moteur linéaire	-	-	O	O	O	O	Moteur linéaire uniquement
1336	Set	Masse du moteur linéaire	-	-	F	F	F	F	Moteur linéaire uniquement
1337	Set	Vitesse maximale du moteur linéaire	-	-	F	F	F	F	Moteur linéaire uniquement
1338	Set	Coefficient d'amortisseur du moteur linéaire	-	-	F	F	F	F	Moteur linéaire uniquement
1339	Set	Couple assigné du moteur à aimant permanent	-	-	F	F	F	F	Moteur à aimant permanent rotatif uniquement
1340	Set	Constante de couple du moteur à aimant permanent	-	-	F	F	F	F	Moteur à aimant permanent rotatif uniquement
1341	Set	Constante de tension du moteur à aimant permanent rotatif	-	-	O	O	O	O	Moteur à aimant permanent rotatif uniquement
1342	Set	Force assignée du moteur à aimant permanent	-	-	F	F	F	F	Moteur à aimant permanent linéaire uniquement

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
1343	Set	Constante de force du moteur à aimant permanent	-	-	F	F	F	F	Moteur à aimant permanent linéaire uniquement
1344	Set	Constante de tension linéaire du moteur à aimant permanent	-	-	O	O	O	O	Moteur à aimant permanent linéaire uniquement
1345	Set	Fréquence assignée du moteur à induction	-	-	O	O	O	O	Moteur à induction uniquement
1346	Set	Courant de flux du moteur à induction	-	-	O	O	O	O	Moteur à induction uniquement
1347	Set	Résistance statorique du moteur à induction	-	-	O	O	O	O	Moteur à induction uniquement
1348	Set	Réactance de fuite statorique du moteur à induction	-	-	F	F	F	F	Moteur à induction uniquement
1349	Set	Réactance d'aimantation du moteur à induction	-	-	F	F	F	F	Moteur à induction uniquement
1350	Set	Résistance rotorique du moteur à induction	-	-	F	F	F	F	Moteur à induction uniquement
1351	Set	Réactance de fuite rotorique du moteur à induction	-	-	F	F	F	F	Moteur à induction uniquement
1352	Set	Vitesse de glissement assignée du moteur à induction	-	-	F	F	F	F	Moteur à induction uniquement
1370	Set	Type de charge	-	F	F	F	F	F	Mise à l'échelle de l'entraînement uniquement
1371	Set	Entrée de rapport de transmission	-	F	F	F	F	F	Mise à l'échelle de l'entraînement uniquement
1372	Set	Sortie de rapport de transmission	-	F	F	F	F	F	Mise à l'échelle de l'entraînement uniquement
1373	Set	Type d'actionneur	-	F	F	F	F	F	Mise à l'échelle de l'entraînement uniquement
1374	Set	Câble d'actionneur	-	F	F	F	F	F	Mise à l'échelle de l'entraînement uniquement
1375	Set	Unité de câble d'actionneur	-	F	F	F	F	F	Mise à l'échelle de l'entraînement uniquement
1376	Set	Diamètre de l'actionneur	-	F	F	F	F	F	Mise à l'échelle de l'entraînement uniquement
1377	Set	Unité de diamètre de l'actionneur	-	F	F	F	F	F	Mise à l'échelle de l'entraînement uniquement
1400 + o	Get	Référence de réaction n	-	F	-	F	F	F	E
1401 + o	Get	Numéro de série de réaction n	-	F	-	F	F	F	E
1402 + o	Get	Position de réaction n	-	O	-	O	O	O	E
1403 + o	Get	Vitesse de réaction n	-	O	-	O	O	O	E
1404 + o	Get	Accélération de réaction n	-	O	-	O	O	O	E
42	Set*	Mode de réaction	-	O	O	O	O	O	E
43	Set	Sélection de réaction principale	-	F	-	-	-	-	
44	Set	Rapport d'unité de réaction	-	-	-	F	F	-	E
1410 + o	Set	Unité de résolution de réaction n	-	F	-	F	F	F	E

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
1411 + o	Set	Unité de réaction n	-	O	-	O	O	O	E
1412 + o	Set	Sélection de port de réaction n	-	F	-	F	F	F	E
1413 + o	Set	Type de réaction n	-	O	-	O	O	O	E
1414 + o	Set	Polarité de réaction n	-	F	-	F	F	F	E
1415 + o	Set	Méthode de démarrage de réaction n	-	O	-	O	O	O	E
1416 + o	Set	Résolution de cycle de réaction n	-	O	-	O	O	O	E; pas LT
1417 + o	Set	Interpolation de cycle de réaction n	-	O	-	O	O	O	E; pas LT
1418 + o	Set	Rotations de réaction n	-	O	-	O	O	O	E; Rotative absolue uniquement
1419 + o	Set	Durée de réaction n	-	O	-	O	O	O	E; Linéaire absolue uniquement
1420 + o	Set	Longueur de données de réaction n	-	F	-	F	F	F	E; TP, SS
1421 + o	Set	Code de données de réaction n	-	F	-	F	F	F	E; TP, SS
1422 + o	Set	Rapport trans. du résolveur de réaction n	-	F	-	F	F	F	E; RS
1423 + o	Set	Tension d'excitation du résolveur de réaction n	-	F	-	F	F	F	E; RS
1424 + o	Set	Fréquence d'excitation du résolveur de réaction n	-	F	-	F	F	F	E; RS
1425 + o	Set	Équilibre du câble du résolveur de réaction n	-	F	-	F	F	F	E; RS
1426 + o	Set	Type LDT de réaction n	-	O	-	O	O	O	E; LT
1427 + o	Set	Recirculations LDT de réaction n	-	O	-	O	O	O	E; LT
1434 + o	Set	Bande passante de filtre de vitesse de réaction n	-	F	-	F	F	F	E
1435 + o	Set	Bande passante de filtre d'accélération de réaction n	-	F	-	F	F	F	E
60	Set*	Commande de vérification d'événement	-	F	-	F	F	F	E
61	Get	État de vérification d'événement	-	F	-	F	F	F	E
62	Get	Position du front positif de l'enregistrement 1	-	F	-	F	F	F	E
63	Get	Position du front négatif de l'enregistrement 1	-	F	-	F	F	F	E
64	Get	Position du front positif de l'enregistrement 2	-	F	-	F	F	F	E
65	Get	Position du front négatif de l'enregistrement 2	-	F	-	F	F	F	E
66	Get	Durée du front positif de l'enregistrement 1	-	F	-	F	F	F	E
67	Get	Durée du front négatif de l'enregistrement 1	-	F	-	F	F	F	E

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
68	Get	Durée du front positif de l'enregistrement 2	-	F	-	F	F	F	E
69	Get	Durée du front négatif de l'enregistrement 2	-	F	-	F	F	F	E
70	Get	Position d'événement d'origine	-	O	-	O	F	F	E
71	Get	Heure d'événement d'origine	-	F	-	F	F	F	E
72	Get	Heure d'événement 1 d'observation	-	O	-	O	F	F	E; Mise à l'échelle de l'entraînement uniquement
73	Get	Heure d'événement 2 d'observation	-	F	-	F	F	F	E; Mise à l'échelle de l'entraînement uniquement
360	Set*	Commande de position du contrôleur – Entier	-	-	-	F	-	-	
361	Set*	Commande de position du contrôleur – Virgule flottante	-	-	-	F	-	-	
362	Set*	Consigne de vitesse du contrôleur	-	-	-	F	O	-	
363	Set*	Consigne d'accélération du contrôleur	-	-	-	F	F	F	Mode de commande d'accélération – R
364	Set*	Consigne de couple du contrôleur	-	-	-	-	-	O	
365	Get	Position de consigne fine	-	-	-	F	-	-	
366	Get	Vitesse de consigne fine	-	-	-	F	F	-	
367	Get	Accélération de consigne fine	-	-	-	F	F	F	
370	Set	Vitesse de saut 1	-	-	F	-	-	-	
371	Set	Vitesse de saut 2	-	-	F	-	-	-	
372	Set	Vitesse de saut 3	-	-	F	-	-	-	
373	Set	Bande de vitesse de saut	-	-	F	-	-	-	
374	Set	Vitesse de rampe – positive	-	-	F	-	F	-	
375	Set	Vitesse de rampe – négative	-	-	F	-	F	-	
376	Set	Accélération de rampe	-	-	F	-	F	-	
377	Set	Décélération de rampe	-	-	F	-	F	-	
378	Set	Commande de rampe d'à-coups	-	-	F	-	F	-	
380	Set	Activation de l'amorçage instantané	-	-	F	-	F	-	
430	Get	Commande de position	-	-	-	O	-	-	
431	Set*	Compensation de position	-	-	-	O	-	-	
432	Get	Référence de position			-	O	-	-	
433	Get	Consigne d'action anticipatrice de vitesse	-	-	-	O	-	-	
434	Get	Retour en position	-	-	-	O	O	O	E
436	Get	Erreur de position	-	-	-	O	-	-	
437	Get	Sortie de l'intégrateur de position	-	-	-	O	-	-	
438	Get	Sortie de la boucle de position	-	-	-	O	-	-	
440	Set	Kvff	-	-	-	O	-	-	

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
441	Set	Kpp	-	-	-	O	-	-	
442	Set	Kpi	-	-	-	O	-	-	
443	Set	Tolérance de verrouillage de position	-	-	-	O	-	-	
444	Set	Tolérance d'erreur de position	-	-	-	O	-	-	
445	Set	Temps de tolérance d'erreur de position	-	-	-	F	-	-	
446	Set	Commande d'intégrateur de position	-	-	-	O	-	-	
447	Set	Préchargement d'intégrateur de position	-	-	-	F	-	-	
448	Set	Limite de position – Positive	-	F	-	F	F	F	Mise à l'échelle de l'entraînement uniquement; E
449	Set	Limite de position – négative	-	F	-	F	F	F	Mise à l'échelle de l'entraînement uniquement; E
450	Get	Consigne de vitesse	-	-	O	O	O	-	
451	Set*	Compensation de vitesse	-	-	O	O	O	-	
452	Get	Consigne d'action anticipatrice d'accélération	-	-	-	O	O	-	
453	Get	Référence de vitesse	-	-	O	O	O	-	
454	Get	Retour en vitesse	-	O	O	O	O	O	
455	Get	Erreur de vitesse	-	-	-	O	O	-	
456	Get	Sortie d'intégrateur de vitesse	-	-	-	O	O	-	
457	Get	Sortie de boucle de vitesse	-	-	-	O	O	-	
460	Set	Kaff	-	-	-	O	O	-	
461	Set	Kvp	-	-	-	O	O	-	
462	Set	Kvi	-	-	-	O	O	-	
464	Set	Kdr	-	-	F	F	F	-	
465	Set	Tolérance d'erreur de vitesse	-	-	-	F	F	-	
466	Set	Temps de tolérance d'erreur de vitesse	-	-	-	F	F	-	
467	Set	Commande d'intégrateur de vitesse	-	-	-	O	O	-	
468	Set	Préchargement d'intégrateur de vitesse	-	-	-	F	F	-	
469	Set	Bande passante du filtre passe-bas de vitesse	-	-	-	F	F	-	
470	Set	Seuil de vitesse	-	F	F	F	F	F	
471	Set	Tolérance de verrouillage de vitesse	-	-	F	F	F	-	
472	Set	Fenêtre de gel de vitesse	-	O	O	O	O	O	
473	Set	Limite de vitesse – Positive	-	-	F	F	F	-	
474	Set	Limite de vitesse – Négative	-	-	F	F	F	-	
480	Get	Consigne d'accélération	-	-	-	F	F	F	

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
481	Set*	Compensation d'accélération	-	-	-	F	F	F	
482	Get	Référence d'accélération	-	-	-	F	F	F	
483	Get	Retour en accélération	-	O	-	O	O	O	
485	Set	Limite d'accélération	-	-	-	F	F	F	
486	Set	Limite de décélération	-	-	-	F	F	F	
490	Get	Consigne de couple	-	-	-	O	O	O	
491	Set*	Compensation de couple	-	-	-	O	O	O	
492	Get	Référence de couple	-	-	-	O	O	O	
493	Get	Référence de couple filtrée	-	-	-	O	O	O	
494	Get	Référence de couple limitée	-	-	-	O	O	O	
496	Set	Kj	-	-	-	O	O	F	
498	Set	Compensation de frottement – Glissant	-	-	-	F	F	F	
499	Set	Compensation de frottement – Statique	-	-	-	F	F	F	
500	Set	Compensation de frottement – Visqueux	-	-	-	F	F	F	
502	Set	Bande passante du filtre passe-bas de couple	-	-	-	F	F	F	
503	Set	Fréquence de filtre coupe-bande de couple	-	-	-	F	F	F	
504	Set	Limite de couple positive	-	-	-	O	O	O	
505	Set	Limite de couple négative	-	-	-	O	O	O	
506	Set	Limite de débit de couple	-	-	-	F	F	F	
507	Set	Seuil de couple	-	-	-	F	F	F	
508	Set	Limite de surcouple	-	-	F	F	F	F	
509	Set	Temps de limite de surcouple	-	-	F	F	F	F	
510	Set	Limite de sous-couple	-	-	F	F	F	F	
511	Set	Temps de limite de sous-couple	-	-	F	F	F	F	
520	Get	Consigne de courant Iq	-	-	-	O	O	O	
521	Get	Limite de courant opératoire	-	-	-	F	F	F	
522	Get	Source de limite de courant	-	-	-	F	F	F	
523	Get	Angle électrique du moteur	-	-	-	O	O	O	Moteurs à aimant permanent uniquement
524	Get	Référence de courant Iq	-	-	-	F	F	F	
525	Get	Référence de courant Id	-	-	-	F	F	F	
527	Get	Erreur de courant Iq	-	-	-	F	F	F	
528	Get	Erreur de courant Id	-	-	-	F	F	F	
529	Get	Retour de courant Iq	-	-	-	F	F	F	
530	Get	Retour de courant Id	-	-	-	F	F	F	
531	Get	Découplage Vq	-	-	-	F	F	F	
532	Get	Découplage Vd	-	-	-	F	F	F	

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
533	Get	Sortie de tension Vq	-	-	-	F	F	F	
534	Get	Sortie de tension Vd	-	-	-	F	F	F	
535	Get	Sortie de tension U	-	-	-	F	F	F	
536	Get	Sortie de tension V	-	-	-	F	F	F	
537	Get	Sortie de tension W	-	-	-	F	F	F	
538	Get	Retour de courant U	-	-	-	F	F	F	
539	Get	Retour de courant V	-	-	-	F	F	F	
540	Get	Retour de courant W	-	-	-	F	F	F	
541	Get	Décalage de courant U	-	-	-	F	F	F	
542	Get	Décalage de courant V	-	-	-	F	F	F	
543	Get	Décalage de courant W	-	-	-	F	F	F	
553	Set	Limite de vecteur de courant	-	-	F	F	F	F	
554	Set	Kqp	-	-	-	F	F	F	
555	Set	Kqi	-	-	-	F	F	F	
556	Set	Kdp	-	-	-	F	F	F	
557	Set	Kdi	-	-	-	F	F	F	
558	Set	Commande de flux ascendant			F	F	F	F	Moteur à induction uniquement
559	Set	Temps de flux ascendant	-	-	F	F	F	F	Moteur à induction uniquement
560	Set	Méthode de démarrage par commutation	-	-	-	F	F	F	Moteurs à aimant permanent uniquement
561	Set	Décalage de commutation	-	-	-	O	O	O	Moteurs à aimant permanent uniquement
562	Set	Autodétection de courant par commutation	-	-	-	F	F	F	Moteurs à aimant permanent uniquement
563	Set	Polarité de commutation	-	-	-	F	F	F	Moteurs à aimant permanent uniquement
564	Set	Alignement de commutation	-	-	-	F	F	F	Moteurs à aimant permanent uniquement
565	Get	Compensation de glissement	-	-	O	-	-	-	
570	Set	Méthode de contrôle de fréquence	-	-	O	-	-	-	
572	Set	Tension maximale	-	-	O	-	-	-	
573	Set	Fréquence maximale	-	-	O	-	-	-	V/Hz de référence uniquement
575	Set	Tension de rupture	-	-	O	-	-	-	V/Hz de référence uniquement
576	Set	Fréquence de rupture	-	-	O	-	-	-	V/Hz de référence uniquement
577	Set	Suralimentation au démarrage	-	-	O	-	-	-	V/Hz de référence uniquement
578	Set	Suralimentation à l'exécution	-	-	O	-	-	-	V/Hz de référence uniquement
600	Get	Fréquence de sortie	-	-	O	F	F	F	
601	Get	Courant de sortie	-	-	O	O	O	O	

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
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			B	E	F	P	V	T	
602	Get	Tension de sortie	-	-	O	O	O	O	
603	Get	Puissance de sortie	-	-	O	O	O	O	
604	Set	Fréquence MID	-	-	F	F	F	F	
608	Set	Vitesse nulle	-	-	F	F	F	F	
609	Set	Durée de vitesse nulle	-	-	F	F	F	F	
610	Set	Action d'arrêt	-	-	O	O	O	O	
611	Set	Couple d'arrêt	-	-	-	O	O	O	
612	Set	Limite de temps d'arrêt	-	-	-	F	F	F	
613	Set	Délai de contact de frein résistif	-	-	F	F	F	F	Moteurs à aimant permanent uniquement
614	Set	Commande de frein mécanique	-	-	F	F	F	F	
615	Set	Délai de libération de frein mécanique	-	-	F	F	F	F	
616	Set	Délai d'engagement de frein mécanique	-	-	F	F	F	F	
590	Set	Configuration d'épreuve	-	-	F	F	F	F	
591	Set	Courant d'épreuve du couple	-	-	F	F	F	F	
592	Set	Couple d'essai de freinage	-	-	F	F	F	F	E
593	Set	Durée de rampe d'épreuve du frein	-	-	F	F	F	F	E
594	Set	Tolérance de glissement du frein	-	-	F	F	F	F	E
620	Get	Tension de bus à courant continu	O		O	O	O	O	
621	Get	Tension de bus à courant continu nominale	O	-	O	O	O	O	
622	Set	Configuration de bus	F	-	F	F	F	F	
623	Set	Sélection de tension de bus	-	-	O	O	O	O	
624	Set	Action du régulateur de bus	O	-	O	O	O	O	Convertisseur intégral uniquement
625	Set	Limite de force régénératrice	-	-	F	F	F	F	Convertisseur intégral uniquement
627	Set	Action de perte de puissance	-	-	F	F	F	F	
628	Set	Seuil de perte de puissance	F	-	F	F	F	F	
629	Set	Action d'interruption	F	-	F	F	F	F	
630	Set	Temps de perte de puissance	F	-	F	F	F	F	
634	Get	Convertisseur intégral	F	-	F	F	F	F	
635	Get	Capacité du moteur	-	-	O	O	O	O	
636	Get	Capacité de l'onduleur	-	-	O	O	O	O	
637	Get	Capacité du convertisseur	F	-	F	F	F	F	
638	Get	Capacité du régulateur de bus	F	-	F	F	F	F	
639	Get	Température ambiante	F	-	F	F	F	F	
640	Get	Température du dissipateur thermique de l'onduleur	-	-	F	F	F	F	

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
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			B	E	F	P	V	T	
641	Get	Température de l'onduleur	-	-	F	F	F	F	
642	Get	Température du moteur	-	-	F	F	F	F	
643	Get	Température de réaction 1	-	-	F	F	F	F	E
644	Get	Température de réaction 2	-	-	F	F	F	F	E
645	Get	Limite de surcharge de l'onduleur	-	-	F	F	F	F	
646	Set	Action de surcharge du moteur	-	-	F	F	F	F	
647	Set	Action de surcharge de l'onduleur	-	-	F	F	F	F	
648	Set	Choix du service	-	-	F	F	F	F	
650	Get	État d'axe	O	O	O	O	O	O	
651	Get	Statut d'axe	O	O	O	O	O	O	
652	Get	Statut d'axe – Mfg	O	O	O	O	O	O	
653	Get	Statut d'E/S d'axe	O	O	O	O	O	O	
654	Get	Statut d'E/S d'axe – Mfg	O	O	O	O	O	O	
655	Get	Exceptions d'axe	O	O	O	O	O	O	
656	Get	Exceptions d'axe Mfg	O	O	O	O	O	O	
657	Get	Défauts d'axe	O	O	O	O	O	O	
658	Get	Défauts d'axe Mfg	O	O	O	O	O	O	
659	Get	Alarmes d'axe	F	F	F	F	F	F	
660	Get	Alarmes d'axe Mfg	F	F	F	F	F	F	
661	Get	Code de défaut d'axe	O	O	O	O	O	O	
662	Get	Type de défaut d'axe	O	O	O	O	O	O	
663	Get	Datation de défaut d'axe	O	O	O	O	O	O	
664	Get	Code d'alarme d'axe	F	F	F	F	F	F	
665	Get	Type d'alarme d'axe	F	F	F	F	F	F	
666	Get	Sous-code de défaut d'axe	F	F	F	F	F	F	
667	Get	Sous-code d'alarme d'axe	F	F	F	F	F	F	
668	Get	Action de défaut d'axe	O	O	O	O	O	O	
669	Get	État d'alarme d'axe	F	F	F	F	F	F	
670	Get	Datation d'alarme d'axe	F	F	F	F	F	F	
678	Get	Limite définie en usine de survitesse de moteur rotatif	-	-	F	F	F	F	Moteur rotatif uniquement
679	Get	Limite définie en usine de survitesse de moteur linéaire	-	-	F	F	F	F	Moteur linéaire uniquement
680	Get	Limite définie en usine de surchauffe du moteur	-	-	F	F	F	F	
681	Get	Surcharge thermique du moteur Limite définie en usine	-	-	F	F	F	F	
682	Get	Limite définie en usine de surchauffe de l'onduleur	-	-	F	F	F	F	
683	Get	Limite définie en usine de surcharge thermique de l'onduleur	-	-	F	F	F	F	

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
684	Get	Limite définie en usine de surchauffe du convertisseur	F	-	F	F	F	F	
685	Get	Limite définie en usine de surcharge thermique du convertisseur	F	-	F	F	F	F	
693	Get	Limite définie en usine de courant de terre du convertisseur	F	-	F	F	F	F	
686	Get	Limite définie en usine de surchauffe du régulateur de bus	F	-	F	F	F	F	
687	Get	Limite définie en usine de surcharge thermique du régulateur de bus	F	-	F	F	F	F	
688	Get	Limite définie en usine de surtension du bus	F	-	F	F	F	F	
689	Get	Limite définie en usine de sous-tension du bus	F	-	F	F	F	F	
690	Get	Limite définie en usine du bruit de réaction	-	F	F	F	F	F	E
691	Get	Limite définie en usine de perte de signal de réaction	-	F	F	F	F	F	E
692	Get	Limite définie en usine de perte de données de réaction	-	F	F	F	F	F	E
694	Set	Limite de perte de phase du moteur	-	-	F	F	F	F	
695	Set	Limite utilisateur de survitesse du moteur	-	-	F	F	F	F	
696	Set	Limite utilisateur de surchauffe du moteur	-	-	F	F	F	F	
697	Set	Limite utilisateur de surcharge thermique du moteur	-	-	F	F	F	F	
698	Set	Limite utilisateur de surchauffe de l'onduleur	-	-	F	F	F	F	
699	Set	Limite utilisateur de surcharge thermique de l'onduleur	-	-	F	F	F	F	
700	Set	Limite utilisateur de surchauffe du convertisseur	F	-	F	F	F	F	Convertisseur intégral uniquement
701	Set	Limite utilisateur de surcharge thermique du convertisseur	F	-	F	F	F	F	Convertisseur intégral uniquement
709	Set	Limite utilisateur de courant de terre du convertisseur	F	-	F	F	F	F	Convertisseur intégral uniquement
702	Set	Limite utilisateur de surchauffe du régulateur de bus	F	-	F	F	F	F	Convertisseur intégral uniquement
703	Set	Limite utilisateur de surcharge thermique du régulateur de bus	F	-	F	F	F	F	Convertisseur intégral uniquement
704	Set	Limite utilisateur de surtension du bus	F	-	F	F	F	F	
705	Set	Limite utilisateur de sous-tension du bus	F	-	F	F	F	F	
706	Get	Limite utilisateur du bruit de réaction	-	F	F	F	F	F	E

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
707	Get	Limite utilisateur de perte de signal de réaction	-	F	F	F	F	F	E
708	Get	Limite utilisateur de perte de données de réaction	-	F	F	F	F	F	E
672	Set	Action d'exception d'axe	O	O	O	O	O	O	
673	Set	Action d'exception d'axe Mfg	O	O	O	O	O	O	
674	Get	Défauts d'initialisation	O	O	O	O	O	O	
675	Get	Défauts d'initialisation Mfg	O	O	O	O	O	O	
676	Get	Inhibitions de démarrage	O	O	O	O	O	O	
677	Get	Inhibitions de démarrage Mfg	O	O	O	O	O	O	
756	Get	Défauts APR	-	F	-	F	F	F	E; Mise à l'échelle de l'entraînement uniquement
757	Get	Défauts APR Mfg	-	F	-	F	F	F	E; Mise à l'échelle de l'entraînement uniquement
710	Get	Durée de mise sous tension de commande	F	-	F	F	F	F	
711	Get	Temps d'exécution cumulé	F	-	F	F	F	F	
712	Get	Utilisation d'énergie cumulée	F	-	F	F	F	F	
713	Get	Rotations de moteur cumulées	-	-	F	F	F	F	
714	Get	Cycles de puissance principale cumulés	F	-	F	F	F	F	Convertisseur intégral uniquement
715	Get	Cycles de puissance de commande cumulés	F	-	F	F	F	F	Convertisseur intégral uniquement
720	Get	Tension de sortie assignée de l'onduleur	-	-	O	O	O	O	
721	Get	Courant de sortie assigné de l'onduleur	-	-	O	O	O	O	
722	Get	Puissance de sortie assignée de l'onduleur	-	-	O	O	O	O	
723	Get	Courant de sortie assigné du convertisseur	F	-	F	F	F	F	
724	Get	Puissance de sortie assignée du convertisseur	F	-	F	F	F	F	
725	Get	ID d'axe de structure de puissance d'entraînement	-	-	F	F	F	F	
730	Get	Entrées numériques	F	-	F	F	F	F	
731	Set	Sorties numériques	F	-	F	F	F	F	
732	Get	Entrée analogique 1	F	-	F	F	F	F	
733	Get	Entrée analogique 2	F	-	F	F	F	F	
734	Set	Sortie analogique 1	F	-	F	F	F	F	
735	Set	Sortie analogique 2	F	-	F	F	F	F	
736	Set	Vérification de l'entrée d'activation du dispositif d'entraînement	F	-	F	F	F	F	
737	Set	Vérification de l'entrée de surcourse du matériel	-	-	F	F	F	F	
750	Set	Commande locale	F	F	F	F	F	F	

Attribut d'instance			Mise en œuvre par Code de fonction de dispositif						
ID Attribut	Règle d'accès	Nom de l'attribut	N		O	Commande en boucle fermée C			Mise en œuvre conditionnelle
			B	E	F	P	V	T	
760	Get	État de sécurité d'axe	-	F	F	F	F	F	Sécurité uniquement
761	Get	Statut de sécurité d'axe	-	F	F	F	F	F	Sécurité uniquement
762	Get	Statut de sécurité d'axe – Mfg	-	F	F	F	F	F	Sécurité uniquement
763	Get	Défauts de sécurité d'axe	-	F	F	F	F	F	Sécurité uniquement
764	Get	Défauts de sécurité d'axe – Mfg	-	F	F	F	F	F	Sécurité uniquement

7.2 Attributs de classe

7.2.1 Généralités

Le tableau des attributs du Tableau 30 s'applique à la classe Objet Axe de dispositif de mouvement, qui est référencée en instance 0. Ces attributs existent avant la création des instances Objet Axe de dispositif de mouvement. Étant donné que les attributs de classe ne sont pas liés à une instance d'axe particulière du dispositif CIP Motion, ils sont en général utilisés pour adresser le comportement paramétrique qui s'applique à toutes les instances d'axe (le comportement du nœud de communication, par exemple). Au fur et à mesure de leur création, des numéros d'instance consécutifs sont donnés aux instances de cette classe, en commençant par 1.

NOTE 1 Aucun attribut associé à cet objet n'exige un archivage non volatile. La valeur "No" est donc implicite pour tous les attributs de la colonne NV.

NOTE 2 Les bits spécifiques au fournisseur et les énumérations permettent aux fournisseurs de dispositif d'offrir des fonctions de produit supplémentaires.

Légende du Tableau 30: (voir 7.1.6 et 7.1.7 pour plus de détails)

(O) – Bit ou énumération obligatoire – doit être pris en charge dans la mise en œuvre.

(F) – Bit ou énumération facultatif – la prise en charge est laissée à la discrétion du fournisseur.

Set* – Indique que l'attribut est en principe défini par le bloc de données de Connexion CIP Motion et non pas par un service Set.

Tableau 30 – Attributs de classe de l'Objet Axe de dispositif de mouvement

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1	Obligatoire	Get		Révision	UINT	Révision de cet objet.	La valeur actuelle affectée à cet attribut pour cette révision de l'objet est trois (3). La révision 1 de cet objet est désignée comme obsolète. La révision 2 n'est plus recommandée pour les nouveaux dispositifs prenant en charge la synchronisation temporelle.
2 à 7	Ces attributs de classe sont facultatifs et sont décrits dans l'IEC 61158-5-2 et l'IEC 61158-6-2.						
14	Obligatoire	Set*		Contrôle de nœud	BYTE (OCTET)	Contient les bits utilisés pour contrôler le comportement du nœud de communication du dispositif d'entraînement associé.	Voir la sémantique en 7.2.2.1.
15	Obligatoire	Get		État de nœud	BYTE (OCTET)	Contient les bits utilisés pour indiquer l'état du nœud de communication du dispositif d'entraînement associé.	Voir la sémantique en 7.2.2.2.
16	Obligatoire	Get		Code de défaut de nœud	USINT	Contient le code numérique de la condition de défaut de nœud actif. Le code de défaut de nœud est utilisé pour donner des détails de diagnostic afin d'identifier la source de la condition de défaut.	Voir la sémantique en 7.2.2.3.
17	Facultatif	Get		Code d'alarme de nœud	USINT	Contient le code numérique de la condition d'alarme active en cours. Le code d'alarme de nœud est utilisé pour donner des détails de diagnostic afin d'identifier la source de la condition d'alarme.	Voir la sémantique en 7.2.2.4.
18	Obligatoire	Set*		Période de mise à jour du contrôleur	UDINT	Représente la période entre les mises à jour du contrôleur, qui est également la période entre les mises à jour de connexion C-D CIP Motion.	Nanosecondes
19	Obligatoire	Set*		Décalage temporel du contrôleur	LINT	Cet élément représente la valeur à 64 bits au début de la période de mise à jour du contrôleur associée à la valeur Datation du contrôleur. Le décalage temporel du contrôleur est la valeur ajoutée à l'horloge locale du contrôleur pour générer le temps système.	Nanosecondes

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
20	Obligatoire	Set*		Datation du contrôleur	LINT	Cet élément représente la valeur Temps système à 64 bits au début de la période de mise à jour du contrôleur lorsqu'un événement de synchronisation de la mise à jour du contrôleur se produit. Calculé par le contrôleur, il s'agit de la somme de la valeur d'horloge locale du contrôleur en cas d'événement de synchronisation de mise à jour et de la valeur Décalage du temps système du contrôleur donnée par le décalage temporel de ce même contrôleur. La datation du contrôleur est donc directement liée aux données de consigne contenues dans la connexion. Le format de datation est absolu et suit la norme CIP Sync, 0 correspondant au 1er janvier 1970.	Nanosecondes (CIP Sync absolu)
21	Facultatif	Set		Limite supérieure du délai de mise à jour du contrôleur	USINT	Représente la limite de délai supérieure d'une mise à jour de connexion Contrôleur-Dispositif (C-D). Ce délai est spécifié en unités de périodes de mise à jour du contrôleur. Le dépassement de cette limite donne lieu à un défaut de mise à jour de connexion de commande.	Nombre de périodes de mise à jour du contrôleur
22	Facultatif	Set		Limite inférieure du délai de mise à jour du contrôleur	USINT	Représente la limite de délai inférieure admise pour une mise à jour de connexion Contrôleur-Dispositif (C-D). Ce délai est spécifié en unités de périodes de mise à jour du contrôleur. Le dépassement de cette limite donne lieu à une alarme de mise à jour de connexion de commande.	Nombre de périodes de mise à jour du contrôleur
28	Facultatif	Get		Variation de synchronisation	UDINT	Représente la variation statistique en cours de la valeur Décalage du temps système par rapport à la valeur Décalage du temps système moyenne.	Nanosecondes
29	Facultatif	Set		Sync Threshold	UDINT	Détermine le seuil de la variation du temps système utilisé pour déterminer si le dispositif CIP Motion est considéré comme étant synchronisé à une horloge maîtresse PTP. Cette valeur est transmise au sous-système PTP du dispositif afin d'établir les critères de synchronisation.	Nanosecondes Défaut: valeur minimale dépendant du dispositif.
30	Facultatif	Get		Sync Update Delay	REAL	Cette valeur est le produit du nombre de sauts de réseaux qui sépare le dispositif de l'horloge maîtresse principale, déterminé par l'attribut Objet de synchronisation temporelle, Pas supprimés et l'intervalle de synchronisation temporelle, déterminé par l'attribut Objet de synchronisation temporelle, Intervalle de synchronisation.	Secondes

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
31	Facultatif	Set		Ensemble de données temporelles	BYTE (OCTET)	Cet octet de matriciel contient des indicateurs qui déterminent l'utilisation et le format des informations de synchronisation du contrôleur et du dispositif. Cette valeur est déduite de l'élément Configuration temporelle envoyé par le contrôleur dans le cadre d'une connexion Contrôleur-Dispositif (C-D) et devient la valeur Configuration temporelle envoyée par le dispositif dans le cadre d'une connexion Dispositif-Contrôleur (D-C). Bit de datation: Datation incluse dans la connexion D-C. Bit de décalage temporel: Décalage temporel inclus dans la connexion D-C. Bit de diagnostic de mise à jour: Données de diagnostic de mise à jour de dispositif incluses dans la connexion D-C. Inclut le nombre de pertes de mises à jour C-D et de mises à jour C-D retardées. Bit de diagnostic de synchronisation: Données de diagnostic de synchronisation de dispositif incluses dans la connexion D-C. Inclut la datation de réception des données C-D et la datation de transmission des données D-C. Il est nécessaire de définir les bits de décalage temporel et de diagnostic de mise à jour uniquement lorsque la valeur des données associées a changé depuis la dernière mise à jour D-C.	Champ de bit: 0 = Période de mise à jour 1 = Datations 2 = Diagnostic de mise à jour 3 = Diagnostic de synchronisation 4 à 7 = (Réservé)
32	Facultatif	Set		Perte de mises à jour du contrôleur	BYTE (OCTET)	Cette valeur représente le nombre cumulé de paquets perdus de connexion Contrôleur-Dispositif détectés depuis l'ouverture et la synchronisation de la connexion. Les paquets perdus sont détectés par examen de l'ID de mise à jour des paquets reçus provenant du contrôleur. Un paquet est déterminé comme perdu lorsque son ID de mise à jour prévu est omis. L'attribut Perte de mises à jour du contrôleur est incrémenté pour chaque ID de mise à jour omis.	Nombre de paquets perdus
33	Facultatif	Set		Mises à jour du contrôleur retardées	BYTE (OCTET)	Cette valeur d'octet représente le nombre de paquets de connexion Contrôleur-Dispositif retardés détectés depuis l'ouverture et la synchronisation de la connexion. Les paquets retardés sont détectés par examen de la datation du paquet reçu provenant du contrôleur. Un paquet est déterminé comme retardé lorsque la différence entre la datation de son contrôleur et la datation reçue des données du dispositif dépasse la période de mise à jour du contrôleur.	Nombre de paquets retardés

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
34	Obligatoire	Get		ID de classe de structure de puissance d'entraînement	UDINT	Cette valeur représente un ID affecté par le fournisseur du dispositif d'entraînement qui identifie de manière unique la structure de puissance associée à l'instance de classe. En tant qu'attribut d'instance de classe, l'ID reflète le matériel de la structure de puissance qui est commun à toutes les instances d'axe ou identique pour toutes les instances d'axe (en excluant les axes de réaction principale). Le matériel de la structure puissance unique à l'instance d'axe doit être identifié au moyen de l'attribut ID d'axe de structure de puissance d'entraînement pour cette instance. Dans ce cas, l'ID de classe de structure de puissance d'entraînement est simplement mis à 0. Un dispositif d'entraînement monoaxe doit toutefois utiliser l'ID de classe de structure de puissance d'entraînement pour identifier la structure de puissance de l'instance monoaxe. L'ID de classe de structure de puissance d'entraînement est inclus comme partie intégrante du service Forward_Open à la classe d'Objet Axe de dispositif de mouvement afin de confirmer que la structure de puissance d'entraînement correspond à la configuration du contrôleur.	Numéro d'ID
35	Conditionnel ^a	Set		Seuil d'échelon	UDINT	Détermine le seuil pour les réponses à un échelon dans le Temps système tel qu'il s'applique à l'algorithme de compensation d'échelon. Cette valeur est généralement définie par le contrôleur au RPI de la Connexion E/S CIP Motion. Pour plus de détail, voir la section Compensation d'échelon de la spécification de Profil de dispositif d'entraînement. La valeur de l'attribut Seuil d'échelon est également utilisée par le service Group_Sync pour identifier l'étalement de l'horloge. Pour plus de détail sur la détection de l'étalement de l'horloge au cours de l'initialisation du service, se reporter au service Group_Sync. Le Seuil d'échelon est exigé pour les dispositifs qui mettent en œuvre le service Group_Sync avec détection de l'étalement de l'horloge introduite dans la Révision 3 de cet objet. Le contrôleur peut appliquer l'accès réussi à cet attribut pour déterminer si le dispositif prend en charge l'algorithme Group_Sync avec détection de l'étalement de l'horloge.	Nanosecondes Par défaut: 1 000 000
^a L'attribut Seuil d'échelon est exigé lorsque le dispositif met en œuvre le service Group_Sync, sinon il n'est pas autorisé.							

7.2.2 Sémantique

7.2.2.1 Attribut N°14 – Contrôle de nœud

Contient les bits utilisés pour contrôler le comportement du nœud de communication du dispositif de mouvement associé, comme spécifié dans le Tableau 31.

Tableau 31 – Définitions de bit Contrôle de nœud

Bit	Obligatoire/ Facultatif	Nom	Description
0	O	Commande à distance	Ce bit est utilisé pour demander que le nœud du dispositif transmette le contrôle du dispositif de mouvement au contrôleur. Si ce bit est supprimé, le dispositif est sous le contrôle exclusif de l'interface locale. Le contrôleur CIP Motion ne peut pas agir sur le comportement du dispositif sans le rebasculer en mode Commande à distance. En cas de perte de la Connexion CIP Motion alors que le dispositif est sous commande à distance, ce dernier peut alors être contrôlé par l'interface locale.
1	F	Commande de synchronisation	Ce bit est utilisé par le contrôleur pour demander le fonctionnement synchrone du dispositif. Le fonctionnement synchrone se caractérise par le fait que le temporisateur local du nœud du dispositif est synchronisé avec le temps système, et que le nœud du dispositif commence à utiliser la datation du contrôleur et le décalage temporel pour traiter les données de connexion et planifier des mises à jour de connexion Dispositif-Contrôleur ultérieures. Le bit Commande synchrone implique que les valeurs Datation du contrôleur et Décalage temporel sont valides. Le bit doit rester défini pendant la durée du fonctionnement synchrone. En mode asynchrone, il n'existe aucune exigence de synchronisation ni de datation du temporisateur local, ni de données de temps associées valides. Généralement, ce bit est défini par le contrôleur lorsque tous les dispositifs CIP Motion associés au contrôleur ont donné une réponse Group_Sync positive.
2	O	Données de contrôleur valides	Ce bit doit être défini pour que le nœud de dispositif traite les Blocs de données d'instance. La séquence d'initialisation de la connexion peut présenter une période de temps dans laquelle les données de connexion des Blocs de données d'instance ne sont pas encore synchronisées. Cette condition peut être reconnue par le dispositif par une vérification préalable de ce bit et, s'il est supprimé, il n'est pas nécessaire que le dispositif traite les données de connexion éventuelles au-delà de l'en-tête de connexion.
3	O	Réinitialisation suite à un défaut de nœud	Ce bit sert à demander au dispositif de procéder à une réinitialisation du nœud de communication et de tenter d'effacer le code de défaut de nœud. Lorsque le contrôleur détecte que le code de défaut de nœud est effacé, le bit Réinitialisation suite à un défaut de nœud est annulé. Lorsque le défaut de communication persiste, le code de défaut de nœud le peut également, auquel cas un cycle de puissance ou un processus de reconfiguration complet doit être réalisé.
4 à 5			(Réservé)
6 à 7			(Spécifique au fournisseur)

7.2.2.2 Attribut N°15 – État du nœud

Contient les bits utilisés pour indiquer l'état du nœud de communication du dispositif CIP Motion associé, comme spécifié dans le Tableau 32.

Tableau 32 – Définitions du bit État du nœud

Bit	Obligatoire/ Facultatif	Nom	Description
0	O	Mode distant	Le bit Mode distant est utilisé pour indiquer que le nœud du dispositif a transmis le contrôle total du dispositif au contrôleur. Si ce bit est supprimé, le dispositif n'agit pas sur des données contenues dans les canaux cycliques ou les canaux de service Contrôleur-Dispositif.
1	F	Mode sync	Le bit Mode sync indique si le nœud du dispositif est synchronisé ou non. Le fonctionnement synchrone se caractérise par le fait que le temporisateur local du nœud du dispositif est synchronisé avec le temps système, et que le nœud du dispositif utilise les datations de connexion pour traiter les données de connexion. La définition du bit Mode sync implique également que la datation du dispositif est valide. Le bit doit rester défini pendant la durée du fonctionnement synchrone. Si le bit Mode sync est supprimé, le dispositif est supposé être en mode asynchrone. En mode asynchrone, il n'existe aucune exigence de synchronisation ni de datation nécessaire, voire valide, du temporisateur local. Généralement, ce bit est défini en réponse à la définition du bit Commande sync par le contrôleur.
2	O	Données de dispositif valides	Le bit Données de dispositif valides doit être défini pour que le contrôleur traite les Blocs de données d'instance provenant du nœud du dispositif. La séquence d'initialisation de la connexion peut présenter une période de temps dans laquelle les données de connexion des Blocs de données d'instance ne sont pas encore synchronisées. Cette condition peut être reconnue par le contrôleur par une vérification préalable de ce bit et, s'il est défini, il n'est pas nécessaire que le contrôleur traite les données de connexion éventuelles au-delà de l'en-tête de connexion.
3	O	Défaut de nœud du dispositif	Le bit Défaut de nœud du dispositif est utilisé pour indiquer que le dispositif a détecté une ou plusieurs conditions de défaut liées au nœud de communication. Les conditions de défaut spécifiques peuvent être déterminées par l'attribut Code de défaut de nœud. Si ce bit est supprimé, il n'existe aucune condition de défaut de nœud. Le bit Défaut de nœud du dispositif peut être annulé en définissant le bit Réinitialisation suite à un défaut de nœud du mot Contrôle de nœud. Tous les axes associés sont désactivés en présence d'une condition Défaut de nœud.
4 à 5			(Réservé)
6 à 7			(Spécifique au fournisseur)

7.2.2.3 Attribut N°16 – Code de défauts de nœud

Cet attribut est un code de défaut à 4 bits utilisé pour signaler la présence d'une condition de défaut particulière liée au nœud de communication du dispositif CIP Motion, comme spécifié dans le Tableau 33.

Tableau 33 – Définitions du code de défauts de nœud

Code	Nom	Description
0	Absence de défaut	Indique l'absence de condition de défaut au niveau du nœud de communication du dispositif.
1	Défaut de mise à jour de connexion de commande	Le code Défaut de mise à jour de connexion de commande est utilisé pour indiquer que les mises à jour provenant du contrôleur sur la connexion C-D ont beaucoup de retard ou ont été perdues de manière consécutive, comme déterminé par la valeur d'attribut Limite supérieure de délai de mise à jour du contrôleur.
2	Défaut de chien de garde du processeur	Le code Défaut de chien de garde du processeur indique que le processeur associé au nœud du dispositif a fait l'objet d'une condition de surcharge excessive qui a déclenché le mécanisme de chien de garde du processeur associé.
3	Défaut matériel	Le code Défaut matériel indique que le matériel de soutien critique (FPGA, ASIC, etc.) associé au nœud du dispositif a fait l'objet d'une condition de défaut.
4	Erreur de format de données	Ce code de défaut indique qu'une erreur s'est produite dans le format de données entre le contrôleur et le dispositif, par exemple une désadaptation de la révision de format.
5	Défaut d'étalement de l'horloge	Le code défaut d'étalement de l'horloge indique le dispositif d'entraînement a détecté une différence importante entre le temps système du dispositif d'entraînement et le temps système du contrôleur, ce qui a empêché le dispositif de basculer sur un fonctionnement synchrone après une période de temporisation.
6	Défaut de perte de connexion de commande	Le code de défaut de perte de connexion de commande indique la temporisation de la connexion C-D CIP Motion issue du contrôleur.
7	Défaut de synchronisation de l'horloge	La condition de défaut est une indication de la perte de synchronisation de l'horloge locale IEC 61588 avec l'horloge maîtresse et de l'incapacité de cette même horloge locale à se resynchroniser dans le délai de temporisation alloué (par exemple, 40 s à 60 s).
8	Chien de garde logique	Le code Défaut de chien de garde logique indique qu'un composant logique auxiliaire (par exemple, FPGA ou ASIC) associé au nœud du dispositif a fait l'objet d'une condition de surcharge excessive qui a déclenché le mécanisme de chien de garde logique associé.
9	Adresse dupliquée	Le code Défaut d'adresse dupliquée indique qu'un nœud de dispositif a été détecté sur le réseau qui utilise la même adresse de nœud que ce nœud de dispositif. Pour un réseau Ethernet, cette adresse serait l'adresse IP du dispositif.
10 à 15	(Réservé)	

7.2.2.4 Attribut N°17 – Code Alarme de nœud

Cet attribut est un code d'alarme à 4 bits utilisé pour indiquer des conditions d'alarme particulières du nœud de communication du dispositif CIP Motion associé, comme spécifié dans le Tableau 34. Les alarmes ne donnent pas lieu à une action directe. Les alarmes de nœud individuelles sont automatiquement interrompues par le dispositif d'entraînement après 10 s. Si la condition d'alarme persiste, l'alarme de nœud associée est relancée.

Tableau 34 – Définitions du code Alarme de nœud

Code	Nom	Description
0	Pas d'alarme	Indique l'absence de condition d'alarme au niveau du nœud de communication du dispositif.
1	Alarme de mise à jour de connexion de commande	Le code Alarme de mise à jour de connexion de commande est utilisé pour indiquer que les mises à jour provenant du contrôleur sur la connexion C-D ont beaucoup de retard ou ont été perdues de manière consécutive, comme déterminé par la valeur d'attribut Limite inférieure de délai de mise à jour du contrôleur.
2	Alarme de surcharge du processeur	Le code Alarme de surcharge du processeur indique que le processeur associé au dispositif fait l'objet de conditions de surcharge qui peuvent éventuellement donner lieu à un défaut.
3	Alarme d'instabilité de l'horloge	Le code Alarme d'instabilité de l'horloge indique que la variation de synchronisation a dépassé le seuil de synchronisation alors que le dispositif fonctionne en Mode sync.
4	Alarme d'étalement de l'horloge	Le code Alarme d'étalement de l'horloge indique le dispositif d'entraînement a détecté une différence importante entre le temps système du dispositif d'entraînement et le temps système du contrôleur, ce qui empêche le dispositif de basculer sur un fonctionnement synchrone.
5	Alarme de synchronisation de l'horloge	Le code Alarme de synchronisation de l'horloge indique que l'horloge locale du dispositif n'est plus synchronisée avec l'horloge maîtresse pendant une courte période (par exemple, 10 s à 20 s) en fonctionnement synchrone. Cette condition d'alarme peut également se produire en cas de détection d'un changement de la source de l'horloge maîtresse. L'Alarme de synchronisation de l'horloge indique que l'horloge locale IEC 61588 est revenue à son mode de démarrage afin de se synchroniser rapidement avec l'horloge maîtresse.
6	Alarme d'adresse de nœud	Le code Alarme d'adresse de nœud indique que le réglage Adresse de nœud du dispositif a été modifié lors de son fonctionnement et peut ne plus être valide.
7 à 15	(Réservé)	

7.3 Attributs d'instances

7.3.1 Généralités

Ce paragraphe 7.3 présente tous les attributs pris en charge d'une instance Objet Axe de dispositif de mouvement. Compte tenu de leur nombre important, les attributs énumérés en 7.3 ont été organisés en catégorie fonctionnelle. Chaque groupe fonctionnel peut être organisé davantage en indiquant d'abord les attributs d'état et de signal de l'objet, suivis des attributs de configuration de l'objet.

NOTE 1 En raison du nombre important d'attributs d'instance pris en charge par cet objet, des ID d'attribut à 16 bits sont utilisés.

NOTE 2 Aucun attribut associé à cet objet n'exige un archivage non volatile. La valeur "No" est donc implicite pour tous les attributs de la colonne NV. La fonction d'archivage non volatile est fournie par le contrôleur de mouvement et plus particulièrement, l'Objet Axe de commande de mouvement.

Saut indication contraire, la valeur par défaut de tous les attributs d'objet est 0 à la mise sous tension du dispositif. Étant donné que le contrôleur de mouvement, qui agit conjointement avec le logiciel de configuration adapté à l'utilisateur, définit les valeurs d'attributs de dispositif, la détermination de valeurs par défaut appropriées lui incombe également. Pour cette raison, la caractérisation des valeurs d'attributs par défaut est traitée dans l'objet Axe de commande de mouvement et ne relève pas du domaine d'application de la présente spécification.

Les limites de marge associées aux attributs de configuration sont spécifiées par le constructeur du dispositif. Tenter de définir un attribut pour une valeur hors des limites de marge doit générer un code d'erreur d'état général 9, pour une "Valeur d'attribut non valide"

('Invalid Attribute Value'). Dans la mesure où les outils de configuration d'axe sont en interface directe avec le contrôleur de mouvement, et non avec le dispositif CIP Motion, ce même contrôleur est chargé d'appliquer les limites de marge d'attributs du dispositif. Pour cette raison, la caractérisation des limites de marge d'attributs est traitée dans l'objet Axe de commande de mouvement et ne relève pas du domaine d'application de la présente spécification.

La seule exception à un "détournement" d'une valeur d'attribut de configuration hors des limites de marge est le cas où ladite valeur n'influe pas sur le comportement du dispositif. Un exemple de cette condition serait l'attribut de largeur de bande passante de filtre passe-bas de couple. Lorsque la largeur de bande passante passe-bas de couple maximale du dispositif est de 2 000 Hz sur la base de limites de débit d'échantillon et le contrôleur tente de régler l'attribut sur 5 000 Hz, le dispositif peut simplement appliquer une valeur de 2 000 Hz avec le même comportement produit. Dans ce cas, le fournisseur peut choisir de fixer la valeur sur la limite applicable au dispositif ou simplement de désactiver le filtre.

Tous les bis réservés et non utilisés, ainsi que les énumérations, sont mis à 0.

Légende des tableaux: (voir 7.1.6 et 7.1.7 pour plus de détails)

(O) – Bit ou énumération obligatoire – doit être pris en charge dans la mise en œuvre

(F) – Bit ou énumération facultatif – la prise en charge est laissée à la discrétion du fournisseur

Codes de fonction de dispositif:

B – Convertisseur de puissance de bus

E – Codeur, réaction uniquement

F – Contrôle de fréquence (V/Hz ou VFD)

P – Boucle d'asservissement de position

V – Boucle de commande de vitesse

T – Boucle d'asservissement de couple

N – Toutes les fonctions de dispositif n'utilisent pas de Méthode de commande

O – Toutes les fonctions de dispositif utilisent des Méthodes de commande en boucle ouverte (Contrôle de fréquence)

C – Toutes les fonctions de dispositif utilisent des Méthodes de commande vectorielle en boucle fermée (Méthode de commande vectorielle PI)

D – Toutes les fonctions de dispositif utilisent (Méthode de commande != Pas de commande)

All – Toutes les fonctions de dispositif

Règles de mise en œuvre conditionnelle:

E = Canal de réaction par codeur présent.

IE = Fonctionnement des dispositifs sans codeur ou sans capteur, absence de canal de réaction.

All – Tous les modes de commande

Set* – Indique que l'attribut est en principe défini par l'en-tête de Connexion CIP Motion et pas par un service Set.

% unités assignées du dispositif – défini comme étant le pourcentage des caractéristiques assignées continues du dispositif avec un fonctionnement de 100 % à la spécification assignée continue du dispositif. Cette unité peut être appliquée aux attributs liés à la vitesse, au couple, à la force, au courant, à la tension et à la puissance. Les "dispositifs" concernés

peuvent être le moteur, l'onduleur, le convertisseur ou le régulateur de bus. Cette unité peut être utilisée indépendamment du fait que la valeur d'attribut représente un niveau instantané ou moyenné dans le temps. L'unité appropriée à la puissance assignée du dispositif est implicite. Comme pour tous les attributs dont l'unité est exprimée en %, une valeur d'attribut de 100 signifie 100 %.

Unités dynamiques: En règle générale, les attributs relatifs à la dynamique du mouvement expriment le déplacement en termes d'unités de dispositif de réaction ou moteur sélectionné. Étant donné que le dispositif CIP Motion peut utiliser différents canaux de réaction pour différentes boucles d'asservissement en fonction du mode de réaction, la détermination du dispositif de réaction qui s'applique dépend, lorsqu'il existe, du mode de réaction.

Le Tableau 35 est un tableau croisé permettant de déterminer les nombres de réactions appropriées ou les unités à appliquer à un attribut en fonction de son type d'unité dynamique (position, vitesse ou accélération), ainsi que du mode de réaction. Si le dispositif CIP Motion prend en charge la fonctionnalité de mise à l'échelle, l'unité de commande de position peut alors être exprimée en nombres et en unités de mouvement, indépendamment du mode de réaction. La fonction de mise à l'échelle, lorsqu'elle est activée, convertit les nombres de mouvements en/à partir des nombres de réactions, et les unités de mouvement en/à partir des unités de réaction. Si une instance d'axe est configurée pour l'absence de réaction, c'est-à-dire un fonctionnement sans capteur/codeur (V/Hz, commande de vitesse sans capteur, etc.), les nombres de réactions ne s'appliquent pas, et la dynamique de mouvement doit donc être exprimée en termes de déplacement du moteur.

Tableau 35 – Unités dynamiques par rapport au mode de réaction

Unités dynamiques	Mise à l'échelle activée	Mode de réaction				
		Pas de réaction	Réaction principale	Réaction du moteur	Réaction de charge	Réaction double
Unités d'asservissement de position	Nombres de mouvements	Nombre de moteurs	Nombres de réactions n ^a	Nombre de réactions 1	Nombre de réactions 2	Nombre de réactions 2
Unités de commande de vitesse	Unités de mouvement	Unités moteur	Unités de réaction n ^a	Unités de réaction 1	Unités de réaction 2	Unités de réaction 1
Unités de commande d'accélération	Unités de mouvement	Unités moteur	Unités de réaction n ^a	Unités de réaction 1	Unités de réaction 2	Unités de réaction 1
^a Canal de réaction déterminé par l'attribut Sélection de réaction principale						

7.3.2 Attributs de configuration de commande de mouvement

7.3.2.1 Généralités

Le tableau d'attributs ci-dessous du Tableau 36 contient les attributs de configuration de commande de mouvement de base associés à une instance Objet Axe de dispositif de mouvement. Ces attributs régissent les aspects du comportement général de l'Objet Axe de dispositif de mouvement.

Set* → Ces attributs sont en général mis à jour par l'intermédiaire de l'Ensemble de données de consigne cycliques de la connexion C-D CIP Motion. S'ils sont inclus sous la forme de données de consigne cycliques, ces attributs ne peuvent pas être mis à jour via un service Set.

Tableau 36 – Attributs de configuration de dispositif de commande de mouvement

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
40	Obligatoire – Tous	Set*		Mode de commande	USINT	Attribut dont l'énumération à 4 bits inférieure détermine le comportement de commande dynamique général de l'instance d'axe de mouvement	Bits 0 à 3: Énumération: Bits 4 à 7: (Réservé) Voir la sémantique en 7.3.2.2.1.
41	Obligatoire – Tous	Set		Méthode de commande	USINT	Code énuméré qui détermine l'algorithme de base de commande de moteur appliqué par le dispositif pour commander le comportement dynamique du moteur.	Voir la sémantique en 7.3.2.2.2.

7.3.2.2 Sémantique

7.3.2.2.1 Attribut N° 40 – Mode de commande

L'attribut Mode de commande est une énumération à 4 bits qui détermine le comportement dynamique spécifique du moteur que le dispositif est tenu de commander pour cette instance d'axe. La vue système de ces modes de commande est décrite en détail en 4.2. Le Tableau 37 présente un bref récapitulatif des énumérations Mode de commande.

Tableau 37 – Définitions de l'énumération du mode de commande

Énum.	Obligatoire/Facultatif	Nom	Description
0	O/N	Pas d'asservissement	Aucune commande de moteur n'est fournie dans ce mode, mais l'interface vers un dispositif de réaction spécifique, en tant que source de réaction principale, est possible par l'intermédiaire de l'attribut Mode de réaction.
1	O/P	Asservissement de position	Le dispositif cherche à commander la position (ou l'orientation) du moteur.
2	O/PV	Commande de vitesse	Le dispositif cherche à commander la vitesse du moteur.
3	F	Commande d'accélération	Le dispositif cherche à commander l'accélération du moteur.
4	O/C	Asservissement de couple	Le dispositif cherche à commander la sortie de couple du moteur.
5 à 15		(Réservé)	-

7.3.2.2.2 Attribut N° 41 – Méthode de commande

L'attribut Méthode de commande est un code énuméré à 8 bits qui détermine l'algorithme de commande de base appliqué par le dispositif pour commander le comportement dynamique du moteur associé à une instance d'axe, comme spécifié dans le Tableau 38.

Tableau 38 – Définitions de l'énumération de la méthode de commande

Énum.	Obligatoire/ Facultatif	Nom	Description
0	O/N	Pas d'asservissement	Aucun asservissement n'est associé à un mode de commande en l'absence de commande de moteur explicite fournie par le dispositif pour cette instance d'axe.
1	O/F	Contrôle de fréquence	Le contrôle de fréquence est une méthode de commande en "boucle ouverte" qui applique la tension au moteur, en général proportionnellement à la fréquence ou la vitesse commandée. Cette méthode de commande est associée à des dispositifs d'entraînement à fréquence variable (VFD) ou dispositifs d'entraînement Volts/Hertz.
2	O/C	Commande vectorielle PI	La commande vectorielle PI est une méthode de commande en "boucle fermée" qui utilise la réaction réelle ou estimée de la commande PI en boucle fermée des éléments dynamiques du moteur, c'est-à-dire la position, la vitesse, l'accélération et le couple, incluant toujours le contrôle PI indépendant en boucle fermée des composants I_q et I_d du vecteur de courant du moteur.
3 à 127		(Réservé)	-
128 à 255		(Spécifique au fournisseur)	-

7.3.3 Attributs d'étalonnage du mouvement

7.3.3.1 Généralités

Le tableau d'attributs ci-dessous du Tableau 39 contient les attributs de configuration de mise à l'échelle du mouvement de base associés à l'instance Objet Axe de dispositif de mouvement. La fonction de mise à l'échelle, lorsqu'elle est activée, convertit les nombres de mouvements en/à partir des nombres de réactions, et les unités de mouvement en/à partir des unités de réaction. La fonctionnalité de mise à l'échelle englobe également le fonctionnement de déroulement cyclique et la polarité de mouvement.

Tableau 39 – Attributs de mise à l'échelle du mouvement

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
45	Obligatoire – Mise à l'échelle	Set		Configuration de mise à l'échelle du mouvement	USINT	Attribut énuméré utilisé pour déterminer si la fonction de mise à l'échelle est exécutée par le contrôleur ou le dispositif d'entraînement. La sélection de configuration de la mise à l'échelle de commande configure le système de commande pour effectuer les calculs de mise à l'échelle dans le contrôleur. Dans ce mode, le contrôleur interagit avec le dispositif d'entraînement en termes de nombres de réactions ou d'unités moteur, et de ce fait le dispositif d'entraînement n'exige aucune opération de mise à l'échelle. De même, en mode Mise à l'échelle de commande, le contrôleur est chargé des opérations de déroulement cycliques associées au mode de débatement cyclique. La sélection de de la mise à l'échelle du dispositif d'entraînement configure le système pour effectuer les calculs de mise à l'échelle dans ce même dispositif. Dans ce mode, le contrôleur interagit avec le dispositif d'entraînement en termes de nombres de réactions ou d'unités moteur, et le dispositif d'entraînement est chargé de la conversion en nombres de réactions et d'unités moteur équivalents. De même, en mode Mise à l'échelle du dispositif d'entraînement, ce dernier est chargé des opérations de déroulement cycliques associées au mode de débatement cyclique.	Énumération: 0 = Mise à l'échelle de commande (O) 1 = Mise à l'échelle du dispositif d'entraînement (F) 2 à 255 = réservé

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
77	Obligatoire – Mise à l'échelle	Set		Unité de mouvement	USINT	Cette valeur énumérée détermine l'unité de mesure utilisée pour exprimer la Résolution de mouvement utilisée par les fonctions du planificateur de mouvement. Voir la sémantique en 7.3.3.2.1.	Énumération: 0 = Rév de moteur 1 = Rév de charge 2 = Rév de réaction 3 = Moteur mm 4 = Charge mm 5 = Réaction mm 6 = Moteur pouce 7 = Charge pouce 8 = Réaction pouce 9 = Moteur r/s 10 = Charge r/s 11 = Moteur m/s 12 = Charge m/s 13 = Moteur pouce/s 14 = Charge pouce/s 15 à 255 = (réservé)
78	Obligatoire – Mise à l'échelle	Set		Résolution de mouvement	DINT	Valeur entière qui détermine le nombre de mouvements par unité de mouvement utilisé par la fonction de mise à l'échelle pour la conversion entre les nombres de mouvements et les nombres de réactions. Voir la sémantique en 7.3.3.2.1.	Nombres de mouvements / Unité de mouvement
79	Obligatoire – Mise à l'échelle	Set		Polarité de mouvement	USINT	Lorsque la configuration de mise à l'échelle du mouvement est définie pour la mise à l'échelle du dispositif d'entraînement, la polarité de mouvement peut être utilisée pour basculer l'orientation du système de commande de mouvement. (voir la sémantique en 7.3.3.2.2 pour plus de détails)	Énumération: 0 = Polarité normale 1 = Polarité inversée 2 à 255 = (Réservé)

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
83	Obligatoire – Mise à l'échelle	Set		Commande de déroulement cyclique	BOOL	Lorsque cet attribut booléen est vrai (1), il active la capacité de déroulement cyclique de la fonction de mise à l'échelle. Cette fonction assure une gamme de positionnement infinie par un "déroulement" de la position d'axe à chaque déplacement d'axe sur un cycle machine complet. Le nombre de mouvements par cycle machine d'axe est spécifié par l'attribut Position de déroulement cyclique. Ainsi, si l'axe représente un fonctionnement cyclique configuré, la mise en œuvre de l'attribut Position de déroulement cyclique est exigée. Lorsque l'attribut Déroulement cyclique est faux (0), indiquant un fonctionnement non cyclique, aucune opération de "déroulement" est effectuée, et un mouvement unidirectionnel poursuit le cumul de position et peut finalement dépasser la gamme de représentation de la position, en produisant un renversement. Pour cette raison, le débattement non cyclique est généralement limité pour les applications d'asservissement de position.	0 = Fonctionnement non cyclique (O) 1 = Fonctionnement cyclique (E)
84	Obligatoire – Mise à l'échelle E	Set		Position de déroulement cyclique	DINT	Cette valeur entière est utilisée pour "dérouler" les valeurs de consigne et de position réelle au cours de chaque cycle machine. La fonctionnalité de déroulement cyclique prévoit une gamme de position infinie pour les axes cycliques, en soustrayant la valeur Position de déroulement cyclique des positions réelle et de consigne chaque fois que l'axe atteint ou croise la position de déroulement cyclique. De façon similaire, lors d'un déplacement dans le sens inverse, la valeur Position de déroulement cyclique est ajoutée aux positions réelle et de consigne chaque fois que l'axe atteint ou croise la position zéro. Ainsi, les valeurs de consigne et réelle ne doivent jamais être en dehors de la gamme de 0 ou s'écarter de la position de déroulement cyclique. Pour éviter toute erreur cumulée due à l'arrondi avec des constantes de conversion irrationnelles, la valeur de déroulement est représentée comme un nombre entier de nombres de mouvements par cycle.	Nombres de mouvements / Cycle de déroulement

7.3.3.2 Sémantique

7.3.3.2.1 Attributs N° 77 et 78 – Unité/Résolution de mouvement

L'attribut Résolution de mouvement détermine la proportion de nombres de mouvements contenus dans une unité de mouvement. Un nombre de mouvements est l'unité de déplacement essentielle utilisée par le planificateur de mouvement et une unité de mouvement est l'unité technique standard de mesure du déplacement par mouvement. Les unités de mouvement peuvent être configurées en tant que Rotations (ou r/min), pouces ou millimètres selon l'application spécifique. En général, la valeur Résolution de mouvement, peut être configurée en nombres de mouvements par unité de mouvement, indépendamment de la résolution du ou des dispositifs de réaction utilisés. La fonction de mise à l'échelle du dispositif d'entraînement veille à la mise à l'échelle entre les nombres de réactions et les nombres de mouvements. Prévoir une valeur Résolution de mouvement configurable est particulièrement utile pour le traitement des applications de déroulement fractionné lorsqu'il est nécessaire de disposer d'un nombre entier de nombres de mouvements par cycle de déroulement.

Les sélections d'attribut Unité de mouvement valide sont déterminées par les valeurs Mode de réaction, Type de charge et Élément actionneur linéaire (Unité principale ou Unité de diamètre) selon le Tableau 40.

Tableau 40 – Règles de sélection de l'unité de mouvement

Mode de réaction	Type de charge	Élément actionneur linéaire	Unité de mouvement
Pas de réaction	Rotatif direct	-	Moteur r/s
Pas de réaction	Transmission rotative	-	Charge r/s
Pas de réaction	Actionneur linéaire	mm/r ou mm	Charge m/s
Pas de réaction	Actionneur linéaire	pouce/r ou pouce	Charge pouce/s
Réaction principale	Rotatif direct	-	Réaction Rév
Réaction principale	Linéaire direct	-	Réaction mm ou Réaction pouce
Réaction principale	Transmission rotative	-	Charge Rév
Réaction principale	Actionneur linéaire	mm/r ou mm	Charge mm
Réaction principale	Actionneur linéaire	pouce/r ou pouce	Charge pouce
Réaction du moteur	Rotatif direct	-	Moteur Rév
Réaction du moteur	Linéaire direct	-	Moteur mm ou Moteur pouce
Réaction du moteur	Transmission rotative	-	Charge Rév
Réaction du moteur	Actionneur linéaire	mm/r ou mm	Charge mm
Réaction du moteur	Actionneur linéaire	pouce/r ou pouce	Charge pouce
Charge ou Réaction double	Rotatif direct	-	Charge Rév
Charge ou Réaction double	Linéaire direct	-	Charge mm ou Moteur pouce
Charge ou Réaction double	Transmission rotative	-	Charge Rév
Charge ou Réaction double	Actionneur linéaire	mm/r ou mm	Charge mm
Charge ou Réaction double	Actionneur linéaire	pouce/r ou pouce	Charge pouce

7.3.3.2.2 Attribut N° 79 – Polarité de mouvement

Lorsque la configuration de mise à l'échelle du mouvement est définie pour la mise à l'échelle du dispositif d'entraînement, la polarité de mouvement peut être utilisée pour basculer l'orientation du système de commande de mouvement. Un réglage normal conserve le même signe des valeurs de consigne de commande de mouvement et des valeurs de signal réelles, que celui des valeurs correspondantes dans la structure de commande du dispositif d'entraînement. Un réglage inversé bascule le signe des valeurs de signal de consigne sur la structure de commande du dispositif d'entraînement et bascule le signe des valeurs de signal réelles provenant de cette même structure. La polarité de mouvement peut par conséquent être utilisée pour ajuster l'orientation de la direction positive du système de commande de mouvement afin qu'elle soit conforme à la direction positive sur la machine. Lorsque la configuration de mise à l'échelle du mouvement est définie sur la Mise à l'échelle du dispositif d'entraînement, l'inversion de la polarité de mouvement se produit entre l'interface de Connexion CIP Motion et la structure de commande du dispositif d'entraînement. Lorsque la configuration de mise à l'échelle du mouvement est définie sur la Mise à l'échelle du contrôleur, l'inversion de la polarité de mouvement est effectuée exclusivement par le contrôleur. Pour assurer une cohérence de direction, les signes de toutes les valeurs Attribut de signal lues de la structure de commande du dispositif d'entraînement ou saisies dans cette même structure sont déterminés par la polarité de mouvement. Une liste exhaustive de ces attributs de signal est définie dans le Tableau 41.

Tableau 41 – Attributs de signal affectés par la polarité de mouvement

ID Attribut	Règle d'accès	Nom de l'attribut de signal
1402 + o	Get	Position de réaction n
1403 + o	Get	Vitesse de réaction n
1404 + o	Get	Accélération de réaction n
62	Get	Position du front positif de l'enregistrement 1
63	Get	Position du front négatif de l'enregistrement 1
64	Get	Position du front positif de l'enregistrement 2
65	Get	Position du front négatif de l'enregistrement 2
70	Get	Position d'événement d'origine
360	Set*	Commande de position du contrôleur – Entier
361	Set*	Commande de position du contrôleur – Virgule flottante
362	Set*	Consigne de vitesse du contrôleur
363	Set*	Consigne d'accélération du contrôleur
364	Set*	Consigne de couple du contrôleur
365	Get	Position de consigne fine
366	Get	Vitesse de consigne fine
367	Get	Accélération de consigne fine
370	Set	Vitesse de saut 1
371	Set	Vitesse de saut 2
372	Set	Vitesse de saut 3
430	Get	Commande de position
431	Set*	Compensation de position
432	Get	Référence de position
433	Get	Consigne d'action anticipatrice de vitesse
434	Get	Retour en position
436	Get	Erreur de position
437	Get	Sortie de l'intégrateur de position
438	Get	Sortie de la boucle de position
450	Get	Consigne de vitesse
451	Set*	Compensation de vitesse
452	Get	Consigne d'action anticipatrice d'accélération

ID Attribut	Règle d'accès	Nom de l'attribut de signal
453	Get	Référence de vitesse
454	Get	Retour en vitesse
455	Get	Erreur de vitesse
456	Get	Sortie d'intégrateur de vitesse
457	Get	Sortie de boucle de vitesse
480	Get	Consigne d'accélération
481	Set*	Compensation d'accélération
482	Get	Référence d'accélération
483	Get	Retour en accélération
490	Get	Consigne de couple
491	Set*	Compensation de couple
492	Get	Référence de couple
493	Get	Référence de couple filtrée
494	Get	Référence de couple limitée
520	Get	Consigne de courant Iq
521	Get	Limite de courant opératoire
523	Get	Angle électrique du moteur
524	Get	Référence de courant Iq
525	Get	Référence de courant Id
527	Get	Erreur de courant Iq
528	Get	Erreur de courant Id
529	Get	Retour de courant Iq
530	Get	Retour de courant Id
531	Get	Découplage Vq
532	Get	Découplage Vd
533	Get	Sortie de tension Vq
534	Get	Sortie de tension Vd
535	Get	Sortie de tension U
536	Get	Sortie de tension V
537	Get	Sortie de tension W
538	Get	Retour de courant U
539	Get	Retour de courant V
540	Get	Retour de courant W
541	Get	Décalage de courant U
542	Get	Décalage de courant V
543	Get	Décalage de courant W
565	Get	Compensation de glissement
600	Get	Fréquence de sortie
601	Get	Courant de sortie
602	Get	Tension de sortie
603	Get	Puissance de sortie

La polarité de mouvement peut également influencer sur les attributs de limite de position directionnelle, de vitesse, d'accélération et de couple. Lorsque la configuration de mise à l'échelle du mouvement est définie sur Mise à l'échelle du dispositif d'entraînement, l'inversion de la polarité de mouvement exige que les valeurs positives et négatives de limite de position, de vitesse, d'accélération et de couple soient permutées entre l'interface de Connexion CIP Motion et la structure de commande interne du dispositif d'entraînement. Lorsque la configuration de mise à l'échelle du mouvement est définie sur Mise à l'échelle du convertisseur, l'inversion de la polarité de mouvement exige que les valeurs positives et négatives de limite de position, de vitesse, d'accélération et de couple dans l'Objet Axe de

commande de mouvement soient permutées avec les attributs correspondants dans l'Objet Axe de dispositif de mouvement. Par exemple, la valeur positive – limite de vitesse dans le contrôleur serait mise en correspondance avec la valeur négative – limite de vitesse dans le dispositif d'entraînement. Une liste exhaustive de ces attributs de limite de direction est définie dans le Tableau 42.

Tableau 42 – Attributs de limite de direction affectés par la polarité de mouvement

ID Attribut	Règle d'accès	Nom de l'attribut de signal
374	Set	Vitesse de rampe – positive
375	Set	Vitesse de rampe – négative
376	Set	Accélération de rampe
377	Set	Décélération de rampe
448	Set	Limite de position – Positive
449	Set	Limite de position – négative
473	Set	Limite de vitesse – Positive
474	Set	Limite de vitesse – Négative
485	Set	Limite d'accélération
486	Set	Limite de décélération
504	Set	Limite de couple positive
505	Set	Limite de couple négative

7.3.4 Attributs de données de connexion

7.3.4.1 Généralités

Le Tableau 43 contient les attributs de données de connexion associés à une instance Objet Axe de dispositif de mouvement. Ces attributs sont des éléments contenus dans l'en-tête de la structure des données de Connexion CIP Motion qui régissent le format et l'interprétation des données de connexion.

Tableau 43 – Attributs de données de connexion

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
89	Facultatif – Tous	Set*		État de commande	BYTE (OCTET)	Champ matriciel qui contient les bits d'état du contrôleur. Voir la sémantique en 7.3.4.2.2.	Champ de bit: 0 = Configuration complète 1 = (réservé) 2 = Bus de convertisseur activé 3 = Bus de convertisseur sans charge 4 = Perte de puissance en courant alternatif du convertisseur 5 = (réservé) 6 = (réservé) 7 = (réservé)
90	Obligatoire – Tous	Set*		Ensemble de données réelles	BYTE (OCTET)	Champ matriciel qui détermine les valeurs de données de réaction réelles transmises au contrôleur dans le cadre d'une connexion Dispositif-Contrôleur. Voir la sémantique en 7.3.4.2.2.	Champ de bit: 0 = Retour en position (O/C) 1 = Retour en vitesse (F/C) 2 = Retour en accélération (F) 3 = (réservé) 4 = (réservé) 5 = (réservé) 6 = Nombre de cycles de déroulement ^a 7 = Déplacement de position ^a
91	Obligatoire – D	Set*		Ensemble de données de consigne	BYTE (OCTET)	Champ matriciel qui détermine les valeurs de données de consigne mises à jour par le contrôleur Voir la sémantique en 7.3.4.2.4.	Champ de bit: 0 = Commande de position (O/P) 1 = Consigne de vitesse (O/V) 2 = Consigne d'accélération (F) 3 = (Consigne de couple) 4 = (réservé) 5 = (réservé) 6 = Nombre de cycles de déroulement ^a 7 = Déplacement de position ^a

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
92	Obligatoire – PVT Facultatif – F	Set*		Commande de consigne	BYTE (OCTET)	Contrôle le comportement des données de consigne cycliques, y compris le format des données de position de consigne et la méthode d'interpolation ou d'extrapolation appliquée aux données de consigne de position, vitesse et accélération provenant du planificateur de mouvement en fonction de la datation associée.	Voir la sémantique en 7.3.4.2.5.
93	Obligatoire – P Facultatif – FVT	Get		Temps de consigne cible	LINT	Datation associée aux données de consigne provenant du planificateur de mouvement et utilisée par les interpolateurs fins de commande. Le temps de consigne cible est la somme de la Datation du contrôleur, de la valeur Datation du dispositif et de la mise à jour de consigne cible associée à cette instance d'axe. Le format de datation est absolu et suit la norme CIP Sync, 0 correspondant au 1er janvier 1970.	Nanosecondes (CIP Sync absolu)
94	Obligatoire – Tous	Set*		Ensemble de données d'état	BYTE (OCTET)	Champ matriciel qui détermine les valeurs de données d'état à transmettre au contrôleur dans le cadre d'une connexion Dispositif-Contrôleur.	Voir la sémantique en 7.3.4.2.6.
95	Obligatoire – Mise à l'échelle E	Set*		Ensemble de données d'enregistrement	BYTE (OCTET)	Champ matriciel qui détermine les valeurs de données d'événement d'enregistrement actuellement mises à jour par le contrôleur dans le cadre de la connexion Contrôleur-Dispositif. La fonction Enregistrement fenêtré est applicable uniquement lorsque la configuration de mise à l'échelle est définie sur Mise à l'échelle du dispositif d'entraînement. Le format des données d'événement d'enregistrement est défini en 7.3.4.2.6	Champ de bit: 0 = Position du front positif de la fenêtre d'enregistrement 1 (F) 1 = Position du front négatif de la fenêtre d'enregistrement 1 (F) 2 = Position du front positif de la fenêtre d'enregistrement 1 (F) 3 = Position du front négatif de la fenêtre d'enregistrement 1 (F) 4 = (réservé) 5 = (réservé) 6 = (réservé) 7 = (réservé)

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
96	Obligatoire – Mise à l'échelle E	Set*		Ensemble de données d'origine	BYTE (OCTET)	Champ matriciel qui détermine les valeurs de données d'événement d'origine actuellement mises à jour par le contrôleur dans le cadre de la connexion Contrôleur-Dispositif. Le format des données d'événement d'origine est défini en 7.3.4.2.6	Champ de bit: 0 = Seuil de couple d'origine (F) 1 = Temps de couple d'origine (F) 2 = (réservé) 3 = Réservé 4 = (réservé) 5 = (réservé) 6 = (réservé) 7 = (réservé)
97	Obligatoire – Mise à l'échelle E	Set*		Ensemble de données d'observation	BYTE (OCTET)	Champ matriciel qui détermine les valeurs de données d'événement d'observation actuellement mises à jour par le contrôleur dans le cadre de la connexion Contrôleur-Dispositif. La fonction Position d'observation est applicable uniquement lorsque la configuration de mise à l'échelle est définie sur Mise à l'échelle du dispositif d'entraînement. Le format des données d'événement d'observation est défini en 7.3.4.2.6	Champ de bit: 0 = Position d'observation 1(O) 1 = Position d'observation 2 (F) 2 = (réservé) 3 = Réservé 4 = (réservé) 5 = (réservé) 6 = (réservé) 7 = (réservé)
^a Exigé en E avec mise à l'échelle du dispositif d'entraînement.							

7.3.4.2 Sémantique

7.3.4.2.1 Attribut N° 89 – État de commande

Les bits individuels dans le champ de bit État de commande sont utilisés comme indiqué ci-dessous.

Configuration complète: Ce bit est défini lorsque le contrôleur a achevé la configuration de tous les attributs d'instances d'axe pendant la phase d'initialisation.

Bus de convertisseur activé: Ce bit, le cas échéant, est défini par le contrôleur lorsque tous les convertisseurs associés ou les dispositifs d'entraînement CIP Motion à convertisseurs intégraux ou les dispositifs d'entraînement CIP Motion à convertisseurs externes non CIP, fournissant l'alimentation de bus à courant continu à ce dispositif, ont indiqué au contrôleur que la tension de bus à courant continu a atteint un niveau opérationnel tel qu'indiqué par le ou les convertisseurs définissant le bit Statut d'axe, bus activé. Ce bit n'est applicable qu'aux dispositifs d'entraînement qui prennent en charge la fonctionnalité Partage de bus à courant continu et qui sont en mesure de qualifier le statut Bus à courant continu activé du dispositif d'entraînement en fonction de ce bit.

Bus de convertisseur sans charge: Ce bit, le cas échéant, est défini par le contrôleur lorsqu'un convertisseur associé ou un dispositif d'entraînement CIP Motion à convertisseur intégral ou un dispositif d'entraînement CIP Motion connecté à un convertisseur externe non CIP, fournissant l'alimentation de bus à courant continu à ce dispositif, a demandé que ce dispositif arrête de s'alimenter par le bus à courant continu. Lorsque le bit Bus de convertisseur sans charge est défini, le dispositif doit générer une exception de Partage d'alimentation de bus si la structure de puissance du dispositif est activée et alimentée par le bus à courant continu. Ce bit n'est applicable qu'aux dispositifs d'entraînement qui prennent en charge la fonctionnalité Partage de bus à courant continu et qui sont en mesure de générer une exception de Partage d'alimentation de bus en fonction de ce bit.

Perte de puissance en courant alternatif du convertisseur: Ce bit, le cas échéant, est défini par le contrôleur lorsqu'un convertisseur associé ou un dispositif d'entraînement CIP Motion à convertisseur intégral ou un dispositif d'entraînement CIP Motion connecté à un convertisseur externe non CIP, a détecté une perte de puissance d'entrée en courant alternatif. Lorsque le bit Perte de puissance en courant alternatif du convertisseur est défini, le dispositif doit générer une exception de Perte de puissance en courant alternatif du convertisseur si la structure de puissance du dispositif est activée. Ce bit est supprimé lorsque le contrôleur a déterminé que tous les convertisseurs associés fournissant l'alimentation de bus à courant continu à ce dispositif ont la puissance d'entrée en courant alternatif suffisante pour les opérations de conversion. Ce bit n'est applicable qu'aux dispositifs d'entraînement qui prennent en charge la fonctionnalité Partage de bus à courant continu et qui sont en mesure de générer une exception de Perte de puissance en courant alternatif du convertisseur en fonction de ce bit.

7.3.4.2.2 Attribut N° 90 – Ensemble de données réelles

Généralement, la valeur Ensemble de données réelles est déterminée par le mode de commande opérationnel comme défini dans le Tableau 44.

Tableau 44 – Détermination de la valeur Ensemble de données réelles

Bit	Données réelles	N	F	P	V	T
0	Position	X		X	() ^a	() ^a
1	Vitesse		X		() ^a	() ^a
2	Accélération					
3	Couple					X
^a Le retour en position est sélectionné lorsque le dispositif de réaction est présent, et le retour en vitesse est sélectionné lorsqu'il est configuré pour un fonctionnement sans capteur ou sans codeur.						

7.3.4.2.3 Attribut N° 91 – Ensemble de données de consigne

Généralement, la valeur Ensemble de données de consigne est déterminée par le mode de commande opérationnel comme défini dans le Tableau 45.

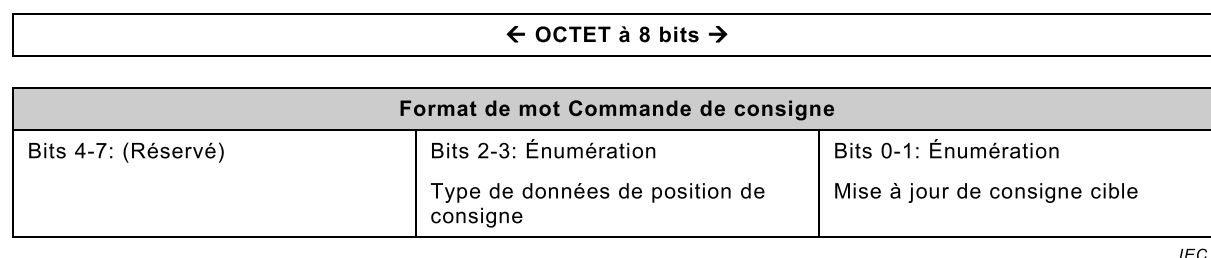
Tableau 45 – Détermination de la valeur Ensemble de données de consigne

Bit	Données de consigne	N	F	P	V	T
0	Position			() ^a		
1	Vitesse		X		X	
2	Accélération					
3	Couple					X

^a Le contrôleur envoie des données de position de consigne avec valeur à virgule flottante ou valeur entière, mais jamais les deux. L'entier est transmis uniquement lorsque le dispositif d'entraînement ne prend pas en charge les données de consigne de position à virgule flottante en double précision.

7.3.4.2.4 Attribut N° 92 – Commande de consigne

L'attribut Commande de consigne régit la méthode d'interpolation/extrapolation appliquée aux données de consigne de position, de vitesse et d'accélération provenant du planificateur de mouvement en fonction de la datation associée, comme spécifié à la Figure 61.



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Figure 61 – Champ de mot Commande de consigne

Les deux premiers bits de cet attribut à 8 bits représentent la mise à jour de consigne cible, qui détermine le nombre de périodes de mise à jour ajouté à la datation des données de consigne afin de déterminer le Temps absolu auquel est destinée la valeur de données de consigne, c'est-à-dire le temps de consigne cible (voir Tableau 46). Pour plus de détails sur la commande d'interpolation et d'extrapolation, voir 7.6.6.2.

Tableau 46 – Définition de l'énumération de mise à jour de consigne cible

Énum.	Obligatoire/ Facultatif	Nom	Description
0	O/P	Immédiate	Une mise à jour de consigne cible de 0 implique que les données de consigne sont à appliquer immédiatement à la structure de commande du dispositif.
1	O/P	Extrapoler (+1 UP)	Une mise à jour de consigne cible de 1 implique que l'extrapolation est à utiliser pour appliquer les données de consigne à la structure de commande du dispositif en fonction de l'ajout d'une période de mise à jour à la datation de consigne.
2	O/P	Interpoler (+2 UP)	Une mise à jour de consigne cible de 2 implique que l'interpolation fine est à utiliser pour appliquer les données de consigne en fonction de l'ajout de deux périodes de mise à jour à la datation de consigne.
3		(Réservé)	-

Le Tableau 47 spécifie les valeurs applicables au champ de type de données de position de consigne.

Tableau 47 – Définition de l'énumération de type de données de position de consigne

Énum.	Obligatoire/ Facultatif	Nom	Description
0	F/P ^a	LREAL (Virgule flottante à 64 bits)	Un type de données de position de consigne de 0 correspond au type LREAL, ou à un type de données à virgule flottante en double précision. Il est préférable que la mise en œuvre prenne en charge ce type de données dans la mesure où il fournit au dispositif d'entraînement une information de commande fractionnée, ce qui produit un mouvement régulier.
1	F/P ^a	DINT (entier à 32 bits)	Un type de données de position de consigne de 1 correspond au type DNIT, ou à un type de données à entier signé à 32 bits. Ce type de données est applicable aux dispositifs d'entraînement simples qui n'exigent pas la précision d'une valeur de position de consigne à virgule flottante ou ne comportent pas de prise en charge matérielle suffisante pour une valeur mathématique à virgule flottante en double précision.
3-4		(Réservé)	-
^a Lorsque l'axe est configuré pour le fonctionnement en mode d'asservissement de position, au moins un des types de données de position de consigne définis doit être pris en charge par le dispositif.			

7.3.4.2.5 Attribut N° 94 – Ensemble de données d'état

L'attribut Ensemble de données d'état est un ensemble de bits à 8 bits indiquant quels attributs État sont à transmettre au contrôleur dans le cadre d'une connexion Dispositif-Contrôleur (voir Tableau 48). Les données d'état apparaissent dans le même ordre que les numéros de bit. Ainsi, les données de type/code de défaut d'axe apparaissent avant les données de datation de défaut d'axe dans la structure Ensemble de données d'état. Plusieurs attributs comprenant un élément de données d'état sélectionné sont transmis dans l'ordre de haut en bas. Ainsi, le type de défaut d'axe est transmis avant le Code de défaut d'axe.

Tableau 48 – Définitions du bit Ensemble de données d'état

Bit	Élément de données d'état produit	Type de données
0	Type de défaut d'axe	USINT
	Code de défaut d'axe	USINT
	Sous-code de défaut d'axe	USINT
	Action de défaut d'axe	USINT
	Datation de défaut d'axe	LINT
1	Type d'alarme d'axe	USINT
	Code d'alarme d'axe	USINT
	Sous-code d'alarme d'axe	USINT
	État d'alarme d'axe	USINT
	Datation d'alarme d'axe	LINT
2	Statut d'axe	DWORD
	Statut d'axe – Mfg	DWORD
3	Statut d'E/S d'axe	DWORD
	Statut d'E/S d'axe Mfg	DWORD

Bit	Élément de données d'état produit	Type de données
4	Statut de sécurité d'axe	DWORD
	Statut de sécurité d'axe Mfg	DWORD
	État de sécurité d'axe	USINT
	Remplissage[3]	USINT[3]
5	(Réservé)	(Réservé)
6	(Spécifique au fournisseur)	(Spécifique au fournisseur)
7	(Spécifique au fournisseur)	(Spécifique au fournisseur)

7.3.4.2.6 Attributs N° 95 et 97 – Format de données d'événement d'enregistrement, d'origine et d'observation

Le format des données d'événement d'enregistrement est défini dans le Tableau 49

Tableau 49 – Format de données d'événement d'enregistrement

Bit	Données d'enregistrement	Format
0	Fenêtre Pos d'enreg 1	DINT (max) DINT (min)
1	Fenêtre Neg d'enreg 1	DINT (max) DINT (min)
2	Fenêtre Pos d'enreg 2	DINT (max) DINT (min)
3	Fenêtre Neg d'enreg 2	DINT (max) DINT (min)

Le format des données d'événement d'origine est défini dans le Tableau 50

Tableau 50 – Format de données d'événement d'origine

Bit	Données d'origine	Format
0	Seuil de couple d'origine	REAL
1	Temps de couple d'origine	REAL

Le format des données d'événement d'observation est défini dans le Tableau 51

Tableau 51 – Format de données d'événement d'observation

Bit	Données d'origine	Format
0	Position d'observation 1	DINT
1	Position d'observation 2	DINT

7.3.5 Attributs du moteur

7.3.5.1 Généralités

Les tableaux d'attribut ci-dessous contiennent les attributs de configuration du moteur associés à l'instance Objet Axe de dispositif de mouvement qui s'appliquent aux différentes technologies de motorisation. Ces technologies incluent les moteurs rotatifs, linéaires, à

aimant permanent et à induction triphasés. Les attributs du moteur sont donc organisés en fonction des différents types de moteur. La catégorie Besoin dans la mise en œuvre d'un attribut est fonction du contexte du type de moteur et, par conséquent, du contexte du tableau dans lequel ces attributs apparaissent. Le cas échéant, Standard peut être distingué de Facultatif par les abréviations PM (Aimant permanent – Permanent Magnet) et IM (Moteurs à induction – Induction Motors). Dans la famille des moteurs PM, il existe une autre différenciation entre moteur SPM (Aimant permanent de surface – Surface PM) et moteur IPM (Aimant permanent d'intérieur – Interior PM). Il est sous-entendu que ces attributs de moteur sont applicables à tous les modes d'entraînement, F, P, V et T, mais ne sont pas applicables pour les configurations d'axe N ou Pas de commande en l'absence de fonction de commande de moteur active.

Ce paragraphe 7.3.5 a pour objet de définir l'ensemble minimal d'attributs obligatoires à la prise en charge de l'interchangeabilité du dispositif CIP Motion. Il s'agit de s'assurer que le contrôleur a fourni le nombre suffisant de données paramétriques pour un dispositif conforme à CIP Motion, c'est-à-dire un dispositif d'entraînement, et qu'un moteur donné est commandé de manière efficace.

Pour les moteurs à induction, l'Objet Axe de dispositif de mouvement s'appuie sur le modèle de motorisation du circuit équivalent phase-neutre recommandé IEEE Std 112 reposant sur la configuration "Wye" (voir Figure 62). Les valeurs de réactance, X , sont associées à leurs valeurs d'inductance correspondantes, L , par l'équation $X = \omega \times L$, où ω est la fréquence assignée du moteur. La notation primaire (X_2' , R_2' , par exemple) indique que les valeurs de composant rotorique réelles X_2 et R_2 font référence au côté stator du rapport d'enroulement stator-rotor.

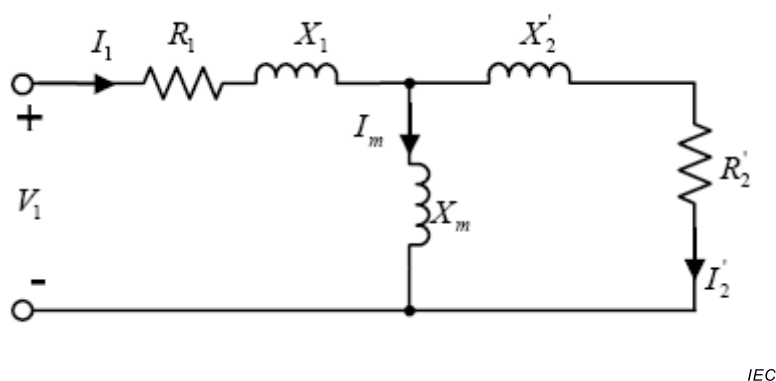


Figure 62 – IEEE Std 112 par modèle de moteur phasé

Pour les moteurs à aimant permanent, l'Objet Axe de dispositif de mouvement suppose de définir tous les paramètres moteur dans le contexte d'un modèle de moteur entre phases.

7.3.5.2 Attributs généraux du moteur

Le Tableau 52 et le Tableau 53 contiennent les attributs généraux du moteur qui s'appliquent à toutes les technologies de motorisation.

Tableau 52 – Attributs généraux d'informations sur le moteur

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1310	Facultatif	Get		Numéro de lot du moteur	SHORT STRING (CHAÎNE COURTE)	Chaîne de 32 caractères qui précise le numéro de lot du moteur. Lorsque le numéro de lot du moteur n'est pas disponible, le dispositif d'entraînement règle cet attribut sur une chaîne nulle.	Par exemple MPL-B310F
1311	Facultatif	Get		Numéro de série du moteur	SHORT STRING (CHAÎNE COURTE)	Chaîne de 16 caractères qui précise le numéro de série du moteur. Lorsque le numéro de série du moteur n'est pas disponible, le dispositif d'entraînement règle cet attribut sur une chaîne nulle.	Par exemple 0012003400560078
1312	Facultatif (Moteur NV)	Get		Code de date du moteur	SHORT STRING (CHAÎNE COURTE)	Chaîne de 16 caractères qui précise la date de fabrication du moteur. Lorsque le code de date du moteur n'est pas disponible, le dispositif d'entraînement règle cet attribut sur une chaîne nulle.	Par exemple Jan-01-2005

Tableau 53 – Attributs généraux de configuration du moteur

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1313	Obligatoire	Set		Source de données du moteur	USINT	Énumération qui précise la source des données du moteur pour le dispositif d'entraînement. NV du contrôleur implique que les attributs du moteur sont déduits de la mémoire non volatile du contrôleur lors du processus de configuration du dispositif d'entraînement. NV du dispositif d'entraînement implique que les attributs du moteur sont déduits directement de la mémoire non volatile du dispositif d'entraînement. Dans ce mode, seul un ensemble minimal de moteurs et de réaction du moteur (réaction 1) est exigé pour configurer le dispositif d'entraînement. NV moteur implique que les attributs du moteur sont déduits de la mémoire non volatile d'un dispositif de réaction intelligent monté sur le moteur équipé d'une interface série. De même, dans ce mode, seul un ensemble minimal de moteurs et de réaction du moteur (réaction 1) est exigé pour configurer le dispositif d'entraînement. Les attributs de moteur et de réaction du moteur transmis au dispositif d'entraînement en mode NV Dispositif d'entraînement ou Moteur sont tenus simplement de confirmer que le contrôleur et le dispositif d'entraînement conviennent des valeurs des attributs critiques pour une opération de mise à l'échelle. Lorsque les valeurs d'attribut NV dans le dispositif d'entraînement diffèrent des valeurs définies par le contrôleur, le dispositif d'entraînement doit rejeter le service Définir, l'État général indiquant une valeur Attribut non valide. La liste actuelle des attributs de moteur et de réaction de moteur transmis au dispositif d'entraînement dans les modes NV est la suivante: 1. Unité moteur 2. Unité de réaction 1 3. Type de réaction 1 4. Méthode de démarrage de réaction 1 5. Résolution de cycle de réaction 1 6. Interpolation de cycle de réaction 1 7. Rotations de réaction 1 8. Durée de réaction 1	Bits 0 à 3: Enum 0 = NV contrôleur (O) 1 = NV dispositif d'entraînement (F) 2 = NV moteur (F) 3 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1314	Obligatoire	Set		Code de dispositif du moteur	UDINT	Numéro unique affecté à un numéro de lot du moteur. Cette valeur permet de s'assurer que les données du moteur et les données de configuration du dispositif de réaction intégral monté sur le moteur provenant du contrôleur correspondent aux données réelles du moteur et de réaction associées au dispositif d'entraînement. Cette comparaison est uniquement valide si la source de données du moteur est NV contrôleur, le moteur étant équipé d'un dispositif de réaction intelligent. qui identifie le moteur de façon positive. Si les codes ne correspondent pas, le dispositif d'entraînement génère un acquittement négatif. Les codes de dispositif du moteur sont affectés par le constructeur du moteur. Une valeur de 0 applicable au code de dispositif du moteur doit être acceptée par le dispositif d'entraînement sans comparaison.	
1315	Obligatoire	Set		Type de moteur	USINT	Énumération qui précise la technologie du moteur. Lorsque la source de données du moteur est NV moteur ou NV dispositif d'entraînement, le type de moteur peut ne pas être connu du contrôleur, mais en revanche du dispositif d'entraînement, de sorte que ce dernier peut fonctionner dans ce cas sans que le contrôleur spécifie le type de moteur. Dans ce mode, le type de moteur ne doit pas être transmis par le contrôleur. S'il est reçu, il doit être rejeté, l'État général indiquant une valeur Attribut non valide. Lorsque la source de données du moteur est une fiche technique ou une base de données, un type de moteur non spécifié, lorsqu'il est reçu par le dispositif d'entraînement lors de la configuration, indique que la configuration du moteur n'a pas été définie et doit par conséquent être rejetée, l'État général indiquant une valeur Attribut non valide.	Énumération: 0 = non spécifié (O) 1 = à aimant permanent rotatif (F) 2 = à induction rotatif (F) 3 = à aimant permanent linéaire (F) 4 = à induction linéaire(F) 5 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)
1316	Obligatoire	Set		Unité moteur	USINT	Unité de mesure de déplacement du moteur. Cet attribut est également utilisé pour le fonctionnement sans codeur et sans capteur. dans la mesure où l'unité de réaction dans ce cas n'est pas applicable.	Énumération: 0 = Rév (O pour les moteurs rotatifs) 1 = Mètre (O pour les moteurs linéaires) 2 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1317	Facultatif	Set		Polarité du moteur	USINT	Valeur énumérée utilisée pour établir le sens du mouvement du moteur lorsque la phase des enroulements est conforme à la spécification définie en usine. La polarité normale est définie comme étant le sens de débattement du moteur lorsque les cosses du moteur ABC sont connectées conformément aux spécifications publiées du dispositif d'entraînement. La polarité inverse bascule le phasage ABC vers ACB de sorte que le moteur se déplace dans le sens opposé en réponse à une sortie de dispositif d'entraînement positive. Cet attribut peut être utilisé pour accorder le sens de débattement à la définition de débattement positif de l'utilisateur. Il peut être utilisé conjointement avec le bit Polarité de réaction afin de fournir une réaction négative, si un contrôle de boucle fermée est exigé. Lors de la commutation d'un moteur à aimant permanent, il est impératif que le séquençement de phase de commutation corresponde au séquençement de phase du moteur afin de contrôler correctement ce dernier.	Énumération: 0 = Polarité normale 1 = Polarité inverse 2 à 255 = (réservé)
1318	Obligatoire	Set		Tension assignée du moteur	REAL	Valeur flottante qui spécifie les caractéristiques assignées de tension à courant alternatif du moteur. Ceci représente la tension entre phases appliquée au moteur pour atteindre la vitesse assignée en pleine charge.	Volts (eff.)
1319	Obligatoire	Set		Courant continu assigné du moteur	REAL	Valeur flottante qui spécifie les caractéristiques assignées de courant continu du moteur. Ceci représente le courant appliqué au moteur dans des conditions de pleine charge à la vitesse et à la tension assignées.	Ampères (eff.)
1320	Obligatoire – PM Facultatif – IM	Set		Courant de crête assigné du moteur	REAL	Valeur flottante qui spécifie les caractéristiques assignées de courant de crête ou intermittent du moteur. Le courant assigné de crête du moteur est souvent déterminé par les contraintes thermiques de l'enroulement du stator ou par les limites de saturation du matériau magnétique du moteur à aimant permanent.	Ampères (eff.)
1321	Obligatoire – IM Facultatif – PM	Set		Puissance de sortie assignée du moteur	REAL	Valeur flottante qui spécifie les caractéristiques assignées de puissance de sortie du moteur. Ceci représente la puissance de sortie du moteur dans des conditions de pleine charge au courant, à la vitesse et à la tension assignés.	Kilowatts

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1322	Facultatif	Set		Limite de surcharge du moteur	REAL	Valeur flottante qui spécifie la limite de surcharge thermique maximale du moteur. Cette valeur est en général de 100 %, correspondant à la puissance dissipée dans le cadre d'un fonctionnement à courant assigné continu du moteur. Elle peut être plus élevée si des options de refroidissement sont appliquées. Pour les moteurs à induction, cet attribut est lié au facteur de service du moteur. Si la valeur de l'attribut Capacité du moteur associée au modèle thermique du moteur dépasse la limite de surcharge du moteur, le dispositif d'entraînement peut éventuellement déclencher une action Surcharge du moteur prédéterminée. La limite de surcharge du moteur peut également être utilisée par le dispositif d'entraînement pour déterminer la limite de capacité thermique absolue du moteur, c'est-à-dire la tolérance de fabrication de surcharge thermique du moteur qui, si elle est dépassée, génère une exception Tolérance de fabrication de surcharge thermique du moteur.	% de la valeur assignée du moteur
1323	Facultatif	Set		Thermo-contacteur interne du moteur	BOOL	Valeur booléenne qui indique si le moteur est doté d'un thermocontacteur intégral.	0 = Non 1 = Oui
1324	Facultatif	Set		Temp. d'enroulement max du moteur	REAL	Valeur flottante qui spécifie la température d'enroulement maximale du moteur.	°C
1325	Facultatif	Set		Capacité Enroulement-Ambiant du moteur	REAL	Valeur flottante qui spécifie la capacité thermique enroulement-ambient.	J/°C
1326	Facultatif	Set		Résistance Enroulement-ambient du moteur	REAL	Valeur flottante qui spécifie la résistance thermique enroulement-ambient.	°C/W

7.3.5.3 Attributs généraux du moteur à aimant permanent

Le Tableau 54 contient les attributs de configuration du moteur qui s'appliquent aux types de moteur à aimant permanent en général.

Tableau 54 – Attributs généraux de configuration du moteur à aimant permanent

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1327	Obligatoire	Set		Résistance du moteur à aimant permanent	REAL	Valeur flottante qui spécifie la résistance entre phases d'un moteur à aimant permanent.	Ohms
1328	Obligatoire	Set		Inductance du moteur à aimant permanent	REAL	Valeur flottante qui spécifie l'inductance entre phases d'un moteur à aimant permanent.	Henries
1353	Obligatoire (IPM uniquement)	Set		Inductance L_q du moteur à aimant permanent	REAL	Valeur flottante qui spécifie l'inductance entre phases, axe q, d'un moteur à aimant permanent d'intérieur.	Henries
1354	Obligatoire (IPM uniquement)	Set		Inductance L_d du moteur à aimant permanent	REAL	Valeur flottante qui spécifie l'inductance entre phases, axe d, d'un moteur à aimant permanent d'intérieur.	Henries

7.3.5.4 Attributs généraux du moteur rotatif

Le Tableau 55 contient les attributs de configuration du moteur qui s'appliquent spécifiquement aux types de moteur rotatif.

Tableau 55 – Attributs généraux de configuration du moteur rotatif

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1329	Obligatoire	Set		Pôles du moteur rotatif	UINT	Entier qui spécifie le nombre de pôles par révolution des moteurs rotatifs. Cette valeur est toujours un nombre pair, les pôles évoluant toujours par paires.	
1330	Facultatif	Set		Inertie du moteur rotatif	REAL	Valeur flottante qui spécifie l'inertie sans charge d'un moteur rotatif.	kg·m ²
1331	Obligatoire	Set		Vitesse assignée du moteur rotatif	REAL	Valeur flottante qui spécifie les caractéristiques assignées de vitesse d'un moteur rotatif. Pour les moteurs à induction, cette valeur est généralement spécifiée à la tension assignée en fonction du courant, du couple ou de la puissance assignés. Pour les moteurs à induction, cette valeur est la valeur du moteur entraîné à la fréquence assignée sous une charge de couple assignée. Cette valeur est synonyme du terme "vitesse de base".	r/min
1332	Facultatif	Set		Vitesse maximale du moteur rotatif	REAL	Valeur flottante qui spécifie la vitesse de fonctionnement maximale absolue d'un moteur rotatif en unités de r/min. Cette vitesse peut être déterminée par les limites du moteur ou par celles du système mécanique. De manière spécifique, elle peut représenter la vitesse maximale de fonctionnement sûr, la vitesse maximale continue "sans charge", la vitesse continue maximale du codeur ou la vitesse continue maximale de défilement du moteur. Cette valeur peut être utilisée par le dispositif d'entraînement pour déterminer la limite définie en usine de survitesse du moteur rotatif.	r/min
1333	Facultatif	Set		Coefficient d'amortisseur du moteur rotatif	REAL	Valeur flottante qui spécifie l'amortissement ou le frottement visqueux avec un moteur rotatif.	N·m/(rad/s)

7.3.5.5 Attributs généraux du moteur linéaire

Le Tableau 56 contient les attributs de configuration du moteur qui s'appliquent spécifiquement aux types de moteur linéaire.

Tableau 56 – Attributs généraux de configuration du moteur linéaire

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1334	Obligatoire	Set		Pas polaire du moteur linéaire	REAL	Valeur flottante qui spécifie le pas polaire d'un moteur linéaire en mètres, équivalent à la longueur de cycle électrique.	m
1335	Obligatoire	Set		Vitesse assignée du moteur linéaire	REAL	Valeur flottante qui spécifie les caractéristiques assignées de vitesse d'un moteur linéaire. Pour les moteurs à induction, cette valeur est généralement spécifiée à la tension assignée en fonction du courant, de la force ou de la puissance assignés. Pour les moteurs à induction, cette valeur est la valeur du moteur entraîné à la fréquence assignée sous une charge de force assignée. Cette valeur est synonyme du terme "vitesse de base".	m/s
1336	Facultatif	Set		Masse du moteur linéaire	REAL	Valeur flottante qui spécifie la masse en mouvement sans charge d'un moteur linéaire.	kg
1337	Facultatif	Set		Vitesse maximale du moteur linéaire	REAL	Valeur flottante qui spécifie la vitesse de fonctionnement maximale absolue d'un moteur linéaire en unités de m/s. Cette vitesse peut être déterminée par les limites du moteur ou par celles du système mécanique. De manière spécifique, elle peut représenter la vitesse maximale de fonctionnement sûr, la vitesse maximale continue "sans charge", la vitesse continue maximale du codeur ou la vitesse continue maximale de défilement du moteur. Cette valeur peut être utilisée par le dispositif d'entraînement pour déterminer la limite définie en usine de survitesse du moteur linéaire.	m/s
1338	Facultatif	Set		Coefficient d'amortissement du moteur linéaire	REAL	Valeur flottante qui spécifie l'amortissement ou le frottement visqueux avec un moteur linéaire.	N/(m/s)

7.3.5.6 Attributs du moteur à aimant permanent rotatif

Le Tableau 57 contient les attributs de configuration du moteur qui s'appliquent spécifiquement aux types de moteur rotatif.

Tableau 57 – Attributs de configuration du moteur à aimant permanent rotatif

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1339	Facultatif	Set		Couple assigné du moteur à aimant permanent	REAL	Valeur flottante qui spécifie les caractéristiques assignées de couple continu d'un moteur à aimant permanent rotatif.	N•m
1340	Facultatif	Set		Constante de couple du moteur à aimant permanent	REAL	Valeur flottante qui spécifie la constante de couple d'un moteur à aimant permanent rotatif, en newtons-mètres par ampère eff.	N•m/ A (eff.)
1341	Obligatoire	Set		Constante de tension du moteur à aimant permanent rotatif	REAL	Valeur flottante qui spécifie la constante de tension (ou force contre-électromotrice) d'un moteur à aimant permanent rotatif, en Volts eff entre phases par kr/min. Lorsque la constante facultative de couple de moteur à aimant permanent, K_t n'est pas explicitement prise en charge dans la mise en œuvre, la valeur peut être calculée à partir de la constante de tension de moteur à aimant permanent rotatif, K_e, selon l'équation suivante: $K_t \text{ (N•m/A)} = 0,016\ 54 \times K_e \text{ (V/kr/min)}$	Volts (eff.)/ Kr/min

7.3.5.7 Attributs du moteur à aimant linéaire

Le Tableau 58 contient les attributs de configuration du moteur qui s'appliquent spécifiquement aux types de moteur à aimant permanent linéaire.

Tableau 58 – Attributs de configuration du moteur à aimant permanent linéaire

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1342	Facultatif	Set		Force assignée du moteur à aimant permanent	REAL	Valeur flottante qui spécifie les caractéristiques assignées de la force continue d'un moteur à aimant permanent linéaire.	N
1343	Facultatif	Set		Constante de force du moteur à aimant permanent	REAL	Valeur flottante qui spécifie la constante de force d'un moteur à aimant permanent linéaire, en newtons par ampère eff.	N/A (eff.)
1344	Obligatoire	Set		Constante de tension linéaire du moteur à aimant permanent	REAL	Valeur flottante qui spécifie la constante de tension (ou force contre-électromotrice) d'un moteur à aimant permanent linéaire en Volts eff entre phases par m/s Lorsque la constante facultative de force de moteur à aimant permanent, K_f, n'est pas explicitement prise en charge dans la mise en œuvre, la valeur peut être calculée à partir de la constante de tension de moteur à aimant permanent linéaire, K_e, selon l'équation suivante: $K_f \text{ (N/A)} = 1,732 \times K_e \text{ (V/(m/s))}$	Volts (eff.) / (m/s)

7.3.5.8 Attributs du moteur à induction

Le Tableau 59 contient les attributs de configuration du moteur qui s'appliquent spécifiquement aux types de moteur à induction.

Tableau 59 – Attributs de configuration du moteur à induction

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1345	Obligatoire	Set		Fréquence assignée du moteur à induction	REAL	Valeur flottante qui spécifie les caractéristiques assignées de fréquence d'un moteur à induction.	Hertz
1346	Obligatoire	Set		Courant de flux du moteur à induction	REAL	Référence de courant I_d qui nécessite de générer un flux de moteur complet. Cette valeur est estimée par le courant assigné du moteur sans charge communément trouvé dans les fiches techniques du moteur à induction.	Ampères (eff.)
1347	Obligatoire	Set		Résistance statorique du moteur à induction	REAL	Valeur flottante qui spécifie la résistance d'enroulement phase-neutre du circuit Y du stator comme indiqué dans R_1 dans le modèle de moteur IEEE Std 112.	Ohms
1348	Obligatoire	Set		Réactance de fuite statorique du moteur à induction	REAL	Valeur flottante qui spécifie la réactance de fuite phase-neutre du circuit Y de l'enroulement statorique, à la fréquence assignée, comme indiqué dans X_1 dans le modèle de moteur IEEE Std 112.	Ohms
1349	Facultatif ^a	Set		Réactance d'aimantation du moteur à induction	REAL	Valeur flottante qui spécifie la réactance d'aimantation phase-neutre du circuit Y du moteur, à la fréquence assignée, comme indiqué dans X_m dans le modèle de moteur IEEE Std 112.	Ohms
1350	Facultatif ^a	Set		Résistance rotorique du moteur à induction	REAL	Valeur flottante qui spécifie la résistance d'enroulement référencée statorique équivalente phase-neutre du rotor comme indiqué dans R_2' dans le modèle de moteur IEEE Std 112.	Ohms
1351	Obligatoire	Set		Réactance de fuite rotorique du moteur à induction	REAL	Valeur flottante qui spécifie l'inductance de fuite équivalente statorique phase-neutre du circuit Y de l'enroulement rotorique, à la fréquence assignée, comme indiqué dans X_2' dans le modèle de moteur IEEE.	Ohms
1352	Facultatif – D	Set		Vitesse de glissement assignée du moteur à induction	REAL	Représente la valeur de glissement au courant assigné (pleine charge) et à la fréquence assignée du moteur.	r/min (moteur rotatif) m/s (moteur linéaire)
^a Ces paramètres comportent une composante de température du moteur importante que certains dispositifs d'entraînement pallient par le biais de diverses techniques d'adaptation ou de compensation.							

7.3.5.9 Attributs de transmission de charge et de configuration d'actionneur

Le Tableau 60 contient les attributs de configuration du moteur qui s'appliquent spécifiquement aux mécanismes de transmission rotative et d'actionneurs linéaires associés à l'axe.

Tableau 60 – Attributs de transmission de charge et de configuration d'actionneur

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1370	Facultatif – Mise à l'échelle	Set		Type de charge	USINT	Attribut énuméré utilisé pour déterminer comment la charge est associée mécaniquement au moteur. Les énumérations "Direct" indiquent que le moteur est couplé directement à la charge. Les énumérations "Rotatif" indiquent que la charge est rotative et que les éléments dynamiques de charge sont mesurés au moyen d'un système d'unités tournant. Les énumérations "Linéaire" indiquent que la charge se déplace de manière linéaire et que les éléments dynamiques de charge sont mesurés au moyen d'un système d'unités linéaire.	Énumération: 0 = Rotatif direct 1 = Linéaire direct 2 = Transmission rotative 3 = Actionneur linéaire 4 à 255 = réservé
1371	Facultatif – Mise à l'échelle	Set		Entrée de rapport de transmission	DINT	Nombre entier de rotations d'arbre d'entrée par cycle de transmission associé à la transmission rotative.	Rotations d'arbre d'entrée
1372	Facultatif – Mise à l'échelle	Set		Sortie de rapport de transmission	DINT	Nombre entier de rotations d'arbre de sortie par cycle de transmission associé à la transmission rotative.	Rotations d'arbre de sortie
1373	Facultatif – Mise à l'échelle	Set		Type d'actionneur	USINT	Cette valeur énumérée indique le type de mécanisme utilisé pour l'activation linéaire.	Énumération: 0 = Aucune (O) 1 = Vis (F) 2 = Courroie et poulie (F) 3 = Pignon d'entraînement (F) 4 = Crémaillère (F) 5 à 255 = réservé
1374	Facultatif – Mise à l'échelle NP uniquement	Set		Câble d'actionneur	REAL	Cette valeur à virgule flottante représente le guide ou le pas d'un actionneur à vis qui constitue une mesure du mouvement linéaire du mécanisme à vis par rotation de l'arbre.	Unités de câble d'actionneur
1375	Facultatif – Mise à l'échelle NP uniquement	Set		Unité de câble d'actionneur	USINT	Cette valeur énumérée indique les unités de l'attribut Câble d'actionneur.	Énumération: 0 = mm/r 1 = pouce/r 2 à 255 = réservé

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1376	Facultatif – Mise à l'échelle	Set		Diamètre de l'actionneur	REAL	Cette valeur à virgule flottante représente le diamètre de la poulie, du galet d'entraînement ou du pignon utilisé pour convertir le mouvement rotatif en déplacement linéaire tangentiel de la charge. Le diamètre d'actionneur est converti en interne en circonférence de la poulie, du galet d'entraînement ou du pignon pour déterminer le niveau de déplacement tangentiel par rotation.	Unités de diamètre de l'actionneur
1377	Facultatif- Mise à l'échelle	Set		Unité de diamètre de l'actionneur	USINT	Cette valeur énumérée indique les unités de l'attribut Diamètre de l'actionneur.	Énumération: 0 = mm 1 = pouce 2 à 255 = réservé

7.3.6 Attributs de réaction

7.3.6.1 Généralités

Le Tableau 64, le Tableau 65 et le Tableau 66 contiennent tous les attributs relatifs au retour en position associés à une instance Objet Axe de dispositif de mouvement qui s'appliquent à différentes techniques de dispositif de réaction et technologies d'interface de réaction.

NOTE Ces technologies d'interface de réaction incluent Digital AqB (signaux A quad B numériques), Sinus/Cosinus (signaux A quad B analogiques), Parallèle numérique (interface de bit numérique parallèle), SSI (Interface série synchrone), LDT (Transducteur ultrasonique de débattement) et Résolveur. D'autres interfaces de réaction modernes prises en charge sont Hiperface® (Stegmann) et EnDat 2.1® & EnDat 2.2® (Heidenhain).

La spécification "Besoin dans la mise en œuvre" d'un attribut de réaction repose souvent sur le contexte du Type de réaction Pour faciliter cette disposition, les abréviations concernant les différents types de réaction sont définies dans le Tableau 61.

Tableau 61 – Abréviations des types de réaction

Abréviation	Type de réaction
TT	AqB numérique
TP	Parallèle numérique
SC	Sinus/Cosinus
HI	Hiperface®
E21	EnDat 2.1®
E22	EnDat 2.2®
RS	Résolveur
SS	SSI
LT	LDT

Ce paragraphe 7.3.6 a pour objet de définir l'ensemble minimal d'attributs obligatoires à la prise en charge de l'interchangeabilité du dispositif CIP Motion. Il s'agit de s'assurer que le

contrôleur a fourni le nombre suffisant de données paramétriques pour un dispositif conforme à CIP Motion, permettant d'assurer une interface efficace avec un large éventail de types de dispositif de réaction.

Plusieurs interfaces de dispositif de réaction sont actuellement définies par l'Objet Axe de dispositif de mouvement par instance d'axe afin d'exécuter des fonctions de réaction de commande ou de maîtrise spécifiques. L'accès à ces dispositifs de réaction s'effectue via leurs canaux logiques affectés, par exemple, Réaction 1 et Réaction 2. Chaque canal de réaction logique est mis en correspondance avec un port d'interface de réaction physique du dispositif, par exemple, Port 1 et Port 2.

Le Tableau 62 énumère les fonctions de commande de canaux de réaction logiques.

Tableau 62 – Fonctions de commande de canaux de réaction logiques

Canal de réaction logique	Fonction de commande de mouvement	Fonction de réaction principale
Réaction 1	Réaction & Commutation du moteur	Réaction principale 1
Réaction 2	Réaction en aval	Réaction principale 2

Lorsque le Mode de commande est défini sur une position autre que Pas d'asservissement, Réaction 1 est en général associée au dispositif de réaction monté sur le moteur, tandis que Réaction 2 est associée au dispositif de réaction en aval ou monté sur la machine. Réaction 1 est toujours exigée pour la commutation du moteur à aimant permanent.

Lorsque le Mode de commande est défini sur Pas d'asservissement pour une instance Objet Axe de dispositif de mouvement, différents canaux de réaction logiques peuvent être utilisés comme source de réaction principale, par exemple, Réaction 1, Réaction 2, etc. Réaction 1 est généralement utilisée.

Pour limiter la longueur des tableaux d'attribut de réaction, la lettre "n" dans le nom d'attribut "Réaction n" générique permet de spécifier le numéro de canal de réaction associé. Les numéros de canal valides des attributs de réaction standard ouverts de l'Objet Axe de dispositif de mouvement sont 1, 2, 3 et 4. Les ID d'attribut sont affectés en fonction du numéro de canal. La prise en charge des canaux d'interface de réaction 1, 2, 3 et 4 est facultative dans la mise en œuvre du dispositif. Lorsque le dispositif ne comporte aucun canal d'interface de réaction, l'ensemble associé d'attributs de canaux de réaction n'est pas applicable. Toutefois, si le support matériel de l'un de ces canaux de réaction est disponible dans un dispositif donné, ces attributs sont clairement applicables dans la mise en œuvre et doivent respecter les règles propres au Besoin dans la mise en œuvre. Une règle de mise en œuvre de "Obligatoire – E" ou "Facultatif – E" indique que l'attribut est généralement applicable à tous les codes de fonction de dispositif où le canal de réaction lui-même est applicable, d'où l'utilisation de "E" pour "Codeur". Lorsqu'un canal de réaction logique spécifique n'est pas applicable en fonction du mode de réaction actuel, les attributs relatifs à la réaction n ne sont alors pas applicables; aucun attribut de configuration de réaction relatif à ce canal n'est défini par un logiciel de configuration, de même qu'aucun attribut de ce type n'est transmis au dispositif d'entraînement.

Le Tableau 63 présente ces règles.

Tableau 63 – Règles applicables aux canaux de réaction logiques

Mode de réaction	Réaction 1	Réaction 2
Pas de réaction	Non	Non
Réaction principale	Oui	Non ^a
Réaction du moteur	Oui	Non
Réaction de charge	Oui ^b	Oui
Réaction double	Oui	Oui
^a Canal de réaction 2 nécessaire uniquement si Sélection de réaction principale prend en charge l'option canal de réaction 2. ^b Canal de réaction 1 nécessaire pour la commutation des moteurs à aimant permanent.		

Les Tableau 64, Tableau 65 et Tableau 66 spécifient les attributs de réaction.

Tableau 64 – Attributs généraux d'informations sur la réaction

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1400 + (n-1)x50	Facultatif – E	Get		Référence de réaction n	SHORT STRING (CHAÎNE COURTE)	Chaîne de 32 caractères qui spécifie la référence du dispositif associé à la Réaction n. Si la référence de réaction n'est pas disponible, le dispositif d'entraînement règle cet attribut sur une chaîne vide.	par exemple SRM-50
1401 + (n-1)x50	Facultatif – E	Get		Numéro de série de réaction n	SHORT STRING (CHAÎNE COURTE)	Chaîne de 16 caractères qui spécifie le numéro de série du dispositif associé à la Réaction n. Si le numéro de série de réaction n'est pas disponible, le dispositif d'entraînement règle cet attribut sur une chaîne vide.	par exemple 001200340 0560078

Tableau 65 – Attributs généraux de signaux de réaction

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1402 + (n-1)x50	Obligatoire – E	Get		Position de réaction n	DINT	Position réelle d'axe en fonction de la Réaction n.	Nombres de réactions n
1403 + (n-1)x50	Obligatoire – E	Get		Vitesse de réaction n	REAL	Vitesse filtrée réelle d'axe en fonction de la Réaction n.	Unités/s de réaction n
1404 + (n-1)x50	Obligatoire – E	Get		Accélération de réaction n	REAL	Accélération filtrée réelle d'axe en fonction de la Réaction n.	Unités/s ² de réaction n

Tableau 66 – Attributs de Configuration de réaction

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
42	Obligatoire – Tous	Set*		Mode de réaction	BYTE (OCTET)	Cet attribut détermine la manière dont les différents canaux de réaction disponibles sont utilisés pour mettre en œuvre le Mode de commande sélectionné. Actuellement, seuls les bits 0 à 3 sont utilisés pour énumérer la configuration Mode de réaction. Les bits 4 à 7 sont réservés pour une utilisation ultérieure.	Bits 0 à 3: Mode de réaction Énumération: 0 = Pas de réaction 1 = Réaction principale 2 = Réaction du moteur 3 = Réaction de charge 4 = Réaction double 5 à 7 = (Réservé) 8 à 15 = (Spécifique au fournisseur) Bits 4 à 7: Réservé Voir la sémantique en 7.3.6.2.
43	Facultatif – N	Set		Sélection de réaction principale	USINT	Cet attribut détermine le canal logique affecté à cette instance d'axe lorsque le Mode de réaction est défini sur Réaction principale. Réaction 1 est la valeur par défaut.	Énumération: 0 = (Réservé) 1 = Réaction 1 2 = Réaction 2 2 à 255 = (réservé)
44	Facultatif – PV	Set		Rapport d'unité de réaction	REAL	Nombre d'unités de réaction 1 par unités de réaction 2. Cette valeur permet de convertir les unités ou les nombres de réactions 2 en unités ou nombres de réactions 1 lorsqu'ils sont configurés pour un fonctionnement de réaction en boucle double ou en aval. Le bloc de rapport de réaction qui s'applique qui s'applique le facteur de mise à l'échelle du rapport d'unité de réaction apparaît dans les schémas fonctionnels de référence de boucle et de couple de position.	Unités de réaction 1 par unités de réaction 2.

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1410 + (n-1)×50	Facultatif – E	Set		Unité de résolution de réaction n	USINT	Unité de mesure de la résolution de réaction utilisée par l'attribut Résolution de cycle de réaction n. Cycles/Unité représente la sélection par défaut où la résolution est exprimée en cycles de réaction par rotation pour les dispositifs de réaction rotatifs ou par mètre pour les dispositifs de réaction linéaires. Lorsque Unités/Cycle est sélectionné, Résolution de cycle de réaction n doit alors être exprimé en Nanomètres/Cycle pour les dispositifs de réaction linéaires. Cette sélection n'est pas applicable aux dispositifs rotatifs. Lorsque Bits/Unité est sélectionné, Résolution de cycle de réaction n est alors exprimée en 2n Cycles par rotation d'un dispositif de réaction rotatif, où n est le nombre de bits dans la représentation de position binaire du dispositif. Cette sélection n'est pas applicable aux dispositifs linéaires.	Énumération: 0 = Cycles/Unité (O) 1 = Unités/Cycle (F) (linéaires uniquement) 2 = Bits/Unité (F) (rotatifs uniquement) 3 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)
1411 + (n-1)×50	Obligatoire -E	Set		Unité de réaction n	USINT	Unité de mesure pour le dispositif de réaction désigné. L'unité de réaction pour la réaction 1 doit être identique à l'unité moteur configurée; si l'unité moteur est définie sur Rév, l'unité de réaction 1 doit être définie sur Rév; si l'unité moteur est définie sur Mètre, l'unité de réaction 1 doit être définie sur Mètre. Les dispositifs de réaction avec une unité de réaction Rév sont considérés comme des dispositifs rotatifs”, tandis que les dispositifs de réaction avec une unité de réaction Mètre sont considérés comme des dispositifs “linéaires”	Énumération: 0 = Rév 1 = Mètre 2 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)
1412 + (n-1)×50	Facultatif – E	Set		Sélection de port de réaction n	USINT	Met en correspondance le canal de réaction logique “n” avec un ID de port de réaction physique. Un ID de port de réaction est affecté à chaque port d'interface de réaction du dispositif par le fournisseur du dispositif d'entraînement. Lorsque cet attribut n'est pas pris en charge par le dispositif d'entraînement, le fournisseur doit coder en dur la mise en correspondance des ports de réaction avec les canaux de réaction logiques pour chaque instance d'axe. La prise en charge de l'attribut Sélection de port de réaction n permet de mettre en correspondance les canaux de réaction logiques avec différents ports de réaction. Sélection de port de réaction n par défaut = 0. Une valeur de 0 indique l'absence de mise en correspondance du canal de réaction n, et donc sa non-utilisation.	Énumération: 0 = Non utilisé (O) 1 à 255 = ID de port de réaction (F)

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1413 + (n-1)×50	Obligatoire – E	Set		Type de réaction n	USINT	Identifie le type d'interface de réaction du canal de dispositif de réaction associé. La prise en charge de tout type de réaction individuel est laissée à la discrétion du constructeur du dispositif. Toutefois, si un type de réaction particulier est pris en charge, les attributs associés à ce type sont en général obligatoires dans la mise en œuvre. Le type de réaction "Intégré" est spécifié pour les dispositifs conformes à CIP Motion dotés d'une fonction de transducteur de réaction intégral, par exemple, un codeur CIP Motion. Dans le cas d'un dispositif de réaction monté sur le moteur, lorsque la source de données du moteur est NV moteur ou NV dispositif d'entraînement, le type de réaction 1 peut ne pas être connu du contrôleur, mais en revanche du dispositif d'entraînement, de sorte que ce dernier peut fonctionner dans ce cas sans spécifier le type de réaction 1. Lorsque l'attribut facultatif Méthode de démarrage par commutation n'est pas pris en charge par le dispositif, ou lorsque cette méthode est définie sur A partir du type de réaction, le type de réaction 1 peut être utilisé pour spécifier de manière implicite la méthode de démarrage par commutation. Par exemple, en sélectionnant le type de réaction 1 avec ou sans signaux de commutation UVW, le dispositif applique la méthode de démarrage par commutation UVW ou la méthode de démarrage autonome, respectivement. Dans ce cas, les signaux de commutation UVW peuvent être déduits des pistes UVW internes au dispositif de réaction ou par l'intermédiaire de capteurs à effet Hall distincts internes au moteur. Toutes les autres sélections de type de réaction 1 appliqueraient la méthode de démarrage par commutation numérique. Dans le cas d'un dispositif de réaction monté sur le moteur, lorsque la source de données du moteur est une fiche technique ou une base de données, un type de réaction 1 non spécifié, lorsqu'il est reçu par le dispositif d'entraînement lors de la configuration, indique que la configuration de réaction du moteur n'a pas été définie et génère par conséquent une erreur d'état général ou une valeur Attribut non valide.	Énumération: 0 = Non spécifié (O) 1 = AqB numérique (F) 2 = AqB numérique avec UVW (F) 3 = Parallèle numérique (F) 4 = Sinus/Cosinus (F) 5 = Sinus/Cosinus avec UVW (F) 6 = Hiperface (F) 7 = EnDat 2.1 (F) 8 = EnDat 2.2 (F) 9 = Résolveur (F) 10 = SSI (F) 11 = LDT (F) 12 = Hiperface DSL (F) 13 = BiSS (F) 14 = Intégré (F) 15 à 127 = (Réservé) 128 à 255 = (spécifique au fournisseur)

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1414 + (n-1)×50	Facultatif – E	Set		Polarité de réaction n	USINT	Valeur énumérée permettant d'établir le sens de changement du compteur de réactions en réponse à un mouvement positif du dispositif de réaction associé. La polarité normale est définie comme étant celle donnant lieu à une augmentation des nombres de réactions lorsque le dispositif de réaction est connecté et déplacé dans le sens positif conformément aux spécifications publiées du dispositif. La polarité inverse bascule en interne la polarité de l'accumulateur de réactions de manière à diminuer le nombre lorsque le dispositif de réaction se déplace dans le sens positif. Cet attribut peut être utilisé pour accorder le sens de débattement à la définition de débattement positif de l'utilisateur. Il peut être utilisé conjointement avec le bit Polarité du moteur afin de fournir une réaction négative, lorsque ce canal de réaction est utilisé pour un contrôle de boucle fermée	Énumération: 0 = Polarité normale 1 = Polarité inverse 2 à 255 = (réservé)
1415 + (n-1)×50	Obligatoire -E	Set		Méthode de démarrage de réaction n	USINT	Détermine la manière dont le dispositif applique la valeur de nombre de réactions lors du démarrage du dispositif d'entraînement. S'il est configuré pour le mode incrémentiel, le dispositif annule l'accumulateur de nombre de réactions à la mise sous tension. La première valeur Position réelle envoyée au contrôleur dans le Bloc de données cycliques de la connexion Dispositif-Contrôleur à la mise sous tension doit être nulle. Il s'agit d'une indication informant le contrôleur que le dispositif d'entraînement a été soumis à un cycle d'alimentation et que l'axe du dispositif d'entraînement nécessite d'être remis à la position d'origine afin d'établir une position de référence machine. S'il est configuré pour le mode Absolu, le dispositif initialise l'accumulateur de nombre de réactions à la mise sous tension et applique la valeur de réaction absolue lue dans le dispositif de réaction. Lorsque la plage de position absolue du dispositif de réaction est inférieure à la représentation d'un entier signé à 32 bits de l'accumulateur de nombre de réactions, la position absolue est étendue à une valeur signée à 32 bits. Alors que de nombreux types de réaction prennent en charge le démarrage absolu, quelques types strictement incrémentiels ne le prennent pas, tels que AqB numérique et Sinus/Cosinus. Certains fournisseurs de dispositifs lient la méthode de démarrage par réaction à la sélection du type de réaction. Dans ces cas, une tentative du contrôleur de configurer de manière incorrecte la méthode de démarrage par réaction doit générer une erreur d'état général ou une valeur Attribut non valide.	Énumération: 0 = Incrémentiel (O) 1 = Absolu (F) 2 à 255 = (réservé)

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1416 + (n-1)×50	Obligatoire – E pas LT	Set		Résolution de cycle de réaction n	UDINT	Détermine la capacité de résolution du dispositif de réaction associé. Les unités applicables à cet attribut sont déterminées par l'unité de résolution de réaction n et l'unité rotative ou linéaire de réaction n comme indiqué dans la colonne "Sémantique". Pour les dispositifs de réaction rotatifs, cette valeur est exprimée comme nombre de cycles de réaction par rotation du dispositif, ou par le nombre de bits dans la représentation de position binaire du dispositif par rotation. Pour les dispositifs de réaction linéaires, cette valeur représente soit le nombre de cycles de réaction par mètre (m), soit le nombre de nanomètres (nm) par cycle de réaction. Les cycles de réaction applicables à un dispositif AqB numérique représentent la résolution en "ligne" du codeur. Les cycles applicables à un dispositif Sinus/Cosinus représentent la résolution de "cycle" sinusoïdal du codeur. Les cycles applicables à un résolveur représentent le nombre de "pôles" du dispositif. Pour les dispositifs de réaction en série numérique (par exemple, SSI) ou parallèles absolus, la valeur Cycle représente la résolution "par pas" ou "nombre" du dispositif.	Cycles/Unité (Rotatifs): Cycles de réaction / Rév (Rotatifs) Cycles/Unité (Linéaires): Cycles de réaction / m Unité/Cycle (Linéaires): nm / Cycle de réaction (Linéaires) Bits/Unité (Rotatifs): 2ⁿ Cycles / Rév
1417 + (n-1)×50	Obligatoire – E pas LT	Set		Interpolation de cycle de réaction n	UDINT	Nombre de réactions interpolées par cycle de réaction. Pour un dispositif AqB numérique, le matériel d'interface de réaction peut en général prendre en charge les valeurs d'interpolation 1, 2 ou 4. Pour un dispositif de réaction Sin/Cos, Hiperface, Endat 2.1 ou Résolveur, le nombre est en général plus important et est déterminé par la capacité d'interpolation du matériel d'interface du dispositif de réaction. La valeur 1 024 est en général utilisée dans ce cas. Pour les interfaces de dispositif de réaction en série numérique (par exemple, SSI) ou parallèles absolues, cette valeur est toujours de 1 étant donné qu'il n'existe aucune chance d'interpolation basée sur le dispositif. La résolution effective du dispositif de réaction dans le nombre de réactions par unité de réaction est déterminée par la combinaison des valeurs d'attribut de résolution de cycles de réaction et d'interpolation du cycle de réaction.	Nombres de réactions / Cycle de réaction

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1418 + (n-1)×50	Obligatoire – E (Rotative absolue)	Set		Rotations de réaction n	UDINT	Nombre maximum de rotations d'arbre spécifié pour un dispositif de réaction rotative absolue afin de maintenir sa référence de position absolue. Les dispositifs de réaction rotative absolue types spécifient un nombre absolu de rotations généralement comprises entre 1 et 4 096 en puissances de 2. Cet attribut peut être utilisé par le système de commande pour déterminer la gamme maximale du nombre de réactions du dispositif de réaction absolue, celle-ci étant le produit de la résolution, de l'interpolation et des rotations du cycle de réaction.	Unités de réaction (Rév)
1419 + (n-1)×50	Obligatoire – E (Linéaire absolue)	Set		Durée de réaction n	REAL	Longueur spécifiée d'un dispositif de réaction absolue linéaire. Les dispositifs de réaction absolue linéaire types spécifient la longueur en mètres. Cet attribut peut être utilisé par le système de commande pour déterminer la capacité de débattement maximale du dispositif de réaction absolue exprimée en nombre de réactions, cette capacité résultant de la combinaison de la résolution, de l'interpolation et de la durée du cycle de réaction.	Unités de réaction (Mètres)
1420 + (n-1)×50	Facultatif – E TP, SS	Set		Longueur de données de réaction n	USINT	Nombre de bits de données de réaction transférés sur le canal d'interface de données en série ou parallèle numérique d'un dispositif de réaction.	Nombre de bits
1421 + (n-1)×50	Facultatif – E TP, SS	Set		Code de données de réaction n	USINT	Type de codage de bits de données de réaction utilisé par le canal d'interface de données en série ou parallèle numérique désigné d'un dispositif de réaction.	Énumération: 0 = Binaire 1 = Gray 2 à 255 = (réservé)
1422 + (n-1)×50	Facultatif – E RS	Set		Rapport de transformateur du résolveur de réaction n	REAL	Spécification du rapport de transformateur du dispositif de réaction de résolveur désigné.	
1423 + (n-1)×50	Facultatif – E RS	Set		Tension d'excitation du résolveur de réaction n	REAL	Définit la tension d'excitation sinusoïdale appliquée au rotor du dispositif de réaction de résolveur désigné.	Volts (eff.)
1424 + (n-1)×50	Facultatif – E RS	Set		Fréquence d'excitation du résolveur de réaction n	REAL	Fréquence du signal d'excitation sinusoïdal appliqué au dispositif de réaction de résolveur désigné. La gamme ou les valeurs de fréquences valides de cet attribut dépendent de l'interface matérielle du dispositif spécifique.	Hertz

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
1425 + (n-1)×50	Facultatif – E RS	Set		Équilibre du câble du résolveur de réaction n	REAL	Ajuste l'amplitude relative des signaux Sinus et Cosinus provenant du résolveur afin de compenser l'impact du câble du résolveur.	%
1426 + (n-1)×50	Obligatoire – E LT	Set		Type LDT de réaction n	USINT	Détermine le type LDT. Les options sont Start/Stop et MID. Les transducteurs Start/Stop acceptent un signal d'entrée (interrogation) pour lancer le cycle de mesure et répondre par deux impulsions sur la ligne de retour. La synchronisation peut reposer sur le front de montée ou de descente. La durée entre les impulsions est proportionnelle à la position. Les transducteurs MID répondent au signal d'interrogation par une seule impulsion longue sur la ligne de retour. La largeur d'impulsion est proportionnelle à la position.	Énumération: 0 = MID 1 = Start/Stop Montée 2 = Start/Stop Descente 3 à 255 = (réservé)
1427 + (n-1)×50	Obligatoire – E LT	Set		Recirculations LDT de réaction n	USINT	Détermine le nombre de recirculations associées à un transducteur LDT de type MID. Plusieurs recirculations peuvent être utilisées pour augmenter la résolution du LDT au détriment de l'augmentation de la période d'échantillonnage.	Nombre de recirculations
1434 + (n-1)×50	Facultatif – E	Set		Bande passante de filtre de vitesse de réaction n.	REAL	Contrôle la bande passante du filtre passe-bas appliqué au signal de vitesse brut provenant de Réaction n. Une valeur nulle pour cet attribut désactive cette fonction.	rad/s
1435 + (n-1)×50	Facultatif – E	Set		Bande passante de filtre d'accélération de réaction n.	REAL	Contrôle la bande passante du filtre passe-bas appliqué au signal d'accélération brut provenant de Réaction n. Une valeur nulle pour cet attribut désactive cette fonction.	rad/s

7.3.6.2 Sémantique: Attribut N° 42 – Mode de réaction

Cet attribut contient une valeur énumérée à 4 bits qui détermine la manière dont sont utilisés les différents canaux de réaction logiques afin de mettre en œuvre le Mode de commande sélectionné pour cette instance d'axe.

L'énumération Mode de réaction (voir Tableau 67) fournit un support pour la fonctionnalité de commande de dispositif multiréaction pour les différents Modes de commande de dispositif actifs, c'est-à-dire lorsque le dispositif commande activement le moteur en fonction d'une réaction. Dans ces Modes de commande de dispositif actifs, le canal logique Réaction 1 est supposé être directement lié au moteur, Réaction 2 étant lié en aval de la transmission mécanique. Les signaux de commutation d'un moteur à aimant permanent sont toujours déduits de Réaction 1.

Tableau 67 – Définitions de l'énumération Mode de réaction

Énum.	Obligatoire/ Facultatif	Nom	Description
0	O/S	Pas de réaction	Pas de réaction est sélectionné si un contrôle de boucle fermée ou de boucle ouverte sans capteur/codeur est souhaité. Lors de l'exécution d'un contrôle de boucle ouverte, aucun signal de réaction n'est exigé. Dans le contrôle de boucle fermée, le signal de réaction exigé est estimé par un algorithme de commande sans capteur reposant sur les signaux de tension et de courant de phase du moteur.
1	O/N	Réaction principale	La Réaction principale attribue un canal de réaction non engagé, tel que spécifié par l'attribut Sélection de réaction principale, à cette instance d'axe de dispositif, pour faire office de source de "réaction principale" lorsque le dispositif est configuré pour le mode Pas d'asservissement.
2	O/C	Réaction du moteur	Si Réaction du moteur est sélectionné, les signaux de retour en accélération, en vitesse et en position de commutation sont alors tous déduits de Réaction 1 monté sur le moteur.
3	F/C	Réaction de charge	Si Réaction de charge est sélectionné, Réaction 1 montée sur le moteur est alors uniquement utilisée pour la commutation du moteur à aimant permanent, Réaction 2 en aval étant utilisée pour la position, la vitesse et l'accélération.
4	F/P	Réaction double	Si Réaction double est sélectionné, Réaction 1 montée sur le moteur est alors utilisée pour la commutation, l'accélération et la vitesse, Réaction 2 en aval étant utilisée exclusivement pour la position.
5 à 7	-	Réservé	
8 à 15	-	Spécifique au fournisseur	

7.3.7 Attributs Capture d'événement

7.3.7.1 Généralités

Le Tableau 68 contient tous les attributs d'événement associés à une instance Objet Axe de dispositif de mouvement. Il contient les événements d'enregistrement, de marquage et de retour à la position de référence. Les attributs Capture d'événement sont conçus pour prendre en charge jusqu'à 16 événements actifs éventuels par période de mise à jour du contrôleur. Le format de tous les attributs Datation est le Temps système absolu et suit la spécification CIP Sync, la valeur 0 correspondant au 1er janvier 1970.

L'Objet Axe de dispositif de mouvement prend actuellement en charge deux canaux d'entrée d'enregistrement indépendants par instance d'axe, qui peuvent être déclenchés sur les fronts de montée ou de descente du signal. En d'autres termes, Enregistrement 1 pour l'instance d'axe 1 n'est généralement pas le même signal qu'Enregistrement 1 pour l'instance d'axe 2. Si la mise en œuvre matérielle du dispositif le permet, les données d'heure et de position

d'événement peuvent être capturées simultanément pour quatre conditions d'événement. Les attributs Capture d'événement prennent également en charge le réarmement automatique pour les événements d'enregistrement. Cela permet au contrôleur de mettre en œuvre des fonctions importantes telles que l'Enregistrement fenêtré et la Reconnaissance de modèle d'enregistrement.

L'Objet Axe de dispositif de mouvement prend également en charge les événements Interrupteur d'origine, Marqueur et Marqueur d'interrupteur de la fonctionnalité de retour à la position de référence axe par axe. Les événements Marqueur sont typiquement générés par le dispositif de retour en position configuré pour l'instance d'axe associée.

Lorsque la Configuration de mise à l'échelle du mouvement est définie sur Mise à l'échelle du dispositif d'entraînement, la fonctionnalité Capture d'événement s'étend afin de prendre en charge les événements d'Enregistrement fenêtré et d'Observation.

Set* → Ces attributs sont en général mis à jour par l'intermédiaire de l'Ensemble de données de consigne cycliques de la connexion C-D CIP Motion. S'ils sont inclus sous la forme de données de consigne cycliques, ces attributs ne peuvent pas être mis à jour via un service Set.

Tableau 68 – Attributs d'événement

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
60	Facultatif – E	Set*		Commande de vérification d'événement	DWORD	Cet attribut est transmis au dispositif par le contrôleur dans le cadre d'une connexion Dispositif d'entraînement-Contrôleur afin d'armer les différentes entrées de dispositif, par exemple, les entrées de marqueur, d'interrupteur d'origine et d'enregistrement, afin de générer des événements pour le contrôleur. Lorsque ces événements sont activés, le dispositif capture l'heure et la position exacte d'axe lors du déroulement de l'événement. Cet attribut gère également le format et le contenu du bloc de données d'événement C-D.	Voir la sémantique en 7.3.7.2.1.

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
61	Facultatif – E	Get		État de vérification d'événement	DWORD	Cet attribut est transmis par le dispositif au contrôleur dans le cadre d'une connexion Dispositif d'entraînement-Contrôleur afin d'indiquer si le dispositif vérifie réellement les événements en fonction des entrées de dispositif (les entrées de marqueur, d'interrupteur d'origine et d'enregistrement, par exemple). La vérification d'événement est initiée lorsque le bit Commande de vérification d'événement correspondant est défini dans la connexion Contrôleur-Dispositif. Cet attribut gère également le format et le contenu du bloc de données d'événement D-C.	Voir la sémantique en 7.3.7.2.2.
62	Facultatif – E	Get		Position du front positif de l'enregistrement 1	DINT	Position de retour verrouillée sur le front de montée de l'Entrée d'enregistrement 1.	Unités d'asservissement de position
63	Facultatif – E	Get		Position du front négatif de l'enregistrement 1	DINT	Position de retour verrouillée sur le front de descente de l'Entrée d'enregistrement 1.	Unités d'asservissement de position
64	Facultatif – E	Get		Position du front positif de l'enregistrement 2	DINT	Position de retour verrouillée sur le front de montée de l'Entrée d'enregistrement 2.	Unités d'asservissement de position
65	Facultatif – E	Get		Position du front négatif de l'enregistrement 2	DINT	Position de retour verrouillée sur le front de descente de l'Entrée d'enregistrement 2.	Unités d'asservissement de position
66	Facultatif – E	Get		Durée du front positif de l'enregistrement 1	LINT	Datation système sur le front de montée de l'Entrée d'enregistrement 1.	Nano-secondes (CIP Sync absolu)
67	Facultatif – E	Get		Durée du front négatif de l'enregistrement 1	LINT	Datation système sur le front de descente de l'Entrée d'enregistrement 1.	Nano-secondes (CIP Sync absolu)
68	Facultatif – E	Get		Durée du front positif de l'enregistrement 2	LINT	Datation système sur le front de montée de l'Entrée d'enregistrement 2.	Nano-secondes (CIP Sync absolu)
69	Facultatif – E	Get		Durée du front négatif de l'enregistrement 2	LINT	Datation système sur le front de descente de l'Entrée d'enregistrement 2.	Nano-secondes (CIP Sync absolu)
70	Obligatoire – NP Facultatif – VT E	Get		Position d'événement d'origine	DINT	Retour en position verrouillé sur l'événement d'origine spécifié.	Unités d'asservissement de position
71	Facultatif – NP Facultatif – VT E	Get		Heure d'événement d'origine	LINT	Datation système verrouillée sur l'événement d'origine spécifié.	Nano-secondes (CIP Sync absolu)

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
72	Obligatoire – NP Facultatif – VT E (Mise à l'échelle du dispositif d'entraînement)	Get		Heure d'événement 1 d'observation	LINT	Datation système verrouillée sur l'événement d'observation spécifié.	Nano-secondes (CIP Sync absolu)
73	Obligatoire – NP Facultatif – VT E (Mise à l'échelle du dispositif d'entraînement)	Get		Heure d'événement 2 d'observation	LINT	Datation système verrouillée sur l'événement d'observation spécifié.	Nano-secondes (CIP Sync absolu)

7.3.7.2 Sémantique

7.3.7.2.1 Attribut N° 60 – Commande de vérification d'événement

Les 28 premiers bits de cet attribut à 32 bits (voir Figure 63) sont utilisés pour activer les différentes entrées du dispositif (les entrées de marqueur, d'interrupteur d'origine et d'enregistrement, par exemple) afin de générer des événements pour le contrôleur. Lorsque ces événements sont activés, le dispositif capture l'heure et la position exacte de l'instance d'axe de mouvement associée. Trois bits, parmi les 4 bits de poids fort, représentent le nombre de messages Acquiescement d'événement dans le bloc de données d'événement dans le cadre de la mise à jour de cette connexion Contrôleur-Dispositif, le dernier bit de poids fort indiquant si le format du bloc de données d'événement est étendu afin de prendre en charge les événements supplémentaires associés à la fonctionnalité Mise à l'échelle du dispositif d'entraînement (voir Configuration de mise à l'échelle du mouvement). La prise en charge du format étendu est facultative dans la mise en œuvre du dispositif. Pour une description détaillée du protocole de gestion des événements entre le dispositif et le contrôleur, voir la section Connexion du profil de dispositif CIP Motion approprié.

← DWORD à 32 bits →

Format de commande de vérification d'événement		
Bit 31: Booléen Format étendu	Bits 28-30: Entier Compte de blocs d'événement	Bits 0-27: Champ de bit Bits de commande de vérification d'événement

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Figure 63 – Champ de mot Commande de vérification d'événement

Pour les événements Marqueur – Interrupteur d'origine spécifiés dans le Tableau 69, le dispositif recherche en premier lieu le niveau de l'entrée d'interrupteur d'origine par rapport à la transition en fonction du symbole + ou –, puis recherche immédiatement la transition de l'entrée de marqueur en fonction du deuxième symbole + ou –.

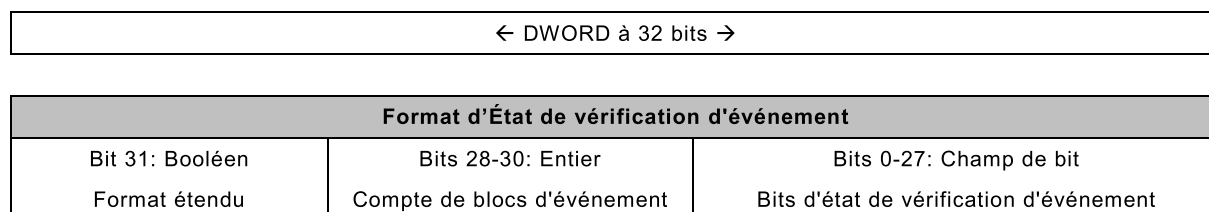
Tableau 69 – Définitions de bit Commande de vérification d'événement

Bit	Format étendu uniquement	Nom	Description
0		Front Pos d'enreg 1	Activer la vérification pour la transition de front positif de l'entrée Enregistrement 1
1		Front Neg d'enreg 1	Activer la vérification pour la transition de front négatif de l'entrée Enregistrement 1
2		Front Pos d'enreg 2	Activer la vérification pour la transition de front positif de l'entrée Enregistrement 2
3		Front Neg d'enreg 2	Activer la vérification pour la transition de front négatif de l'entrée Enregistrement 2
4	*	Fenêtre de front Pos d'enreg 1	Activer la vérification de fenêtre après la transition de front positif de l'entrée Enregistrement 1
5	*	Fenêtre de front Neg d'enreg 1	Activer la vérification de fenêtre après la transition de front négatif de l'entrée Enregistrement 1
6	*	Fenêtre de front Pos d'enreg 2	Activer la vérification de fenêtre après la transition de front positif de l'entrée Enregistrement 2
7	*	Fenêtre de front Neg d'enreg 2	Activer la vérification de fenêtre après la transition de front négatif de l'entrée Enregistrement 2
8		Réarm auto Pos d'enreg 1	Activer le réarmement automatique après la transition de front positif de l'entrée Enregistrement 1
9		Réarm auto Neg d'enreg 1	Activer le réarmement automatique après la transition de front négatif de l'entrée Enregistrement 1
10		Réarm auto Pos d'enreg 2	Activer le réarmement automatique après la transition de front positif de l'entrée Enregistrement 2
11		Réarm auto Neg d'enreg 2	Activer le réarmement automatique après la transition de front négatif de l'entrée Enregistrement 2
12	*	Position d'observation 1 Avt	Activer la vérification de la position d'observation 1 pour le croisement des positions d'axe dans le sens avant (positif).
13	*	Position d'observation 1 Inv	Activer la vérification de la position d'observation 1 pour le croisement des positions d'axe dans le sens inverse (négatif).
14	*	Position d'observation 2 Avt	Activer la vérification de la position d'observation 2 pour le croisement des positions d'axe dans le sens avant (positif).
15	*	Position d'observation 2 Inv	Activer la vérification de la position d'observation 2 pour le croisement des positions d'axe dans le sens inverse (négatif).
16		Front positif du marqueur	Activer la vérification du front positif de l'entrée de marqueur du canal de retour en position associé.
17		Front négatif du marqueur	Activer la vérification du front négatif de l'entrée de marqueur du canal de retour en position associé.
18		Front positif de l'interrupteur d'origine	Activer la vérification du front positif de l'entrée d'origine associée à cette instance d'axe.
19		Front négatif de l'interrupteur d'origine	Activer la vérification du front positif de l'entrée d'origine associée à cette instance d'axe.
20		Marqueur – Interrupteur d'origine ++	Activer la vérification de la séquence d'événements spécifiée en tant que transition de front positif de l'entrée Interrupteur d'origine suivie d'un front positif de l'entrée Marqueur.
21		Marqueur – Interrupteur d'origine +-	Activer la vérification de la séquence d'événements spécifiée en tant que transition de front négatif de l'entrée Interrupteur d'origine suivie d'un front positif de l'entrée Marqueur.
22		Marqueur – Interrupteur d'origine -+	Activer la vérification de la séquence d'événements spécifiée en tant que transition de front négatif de l'entrée Interrupteur d'origine suivie d'un front positif de l'entrée Marqueur.

Bit	Format étendu uniquement	Nom	Description
23		Marqueur – Interrupteur d'origine --	Activer la vérification de la séquence d'événements spécifiée en tant que transition de front négatif de l'entrée Interrupteur d'origine suivie d'un front négatif de l'entrée Marqueur.
24	*	Seuil de couple d'origine	Activer la vérification de l'événement de couple d'origine spécifié comme niveau de couple qui dépasse le seuil de couple d'origine pendant une période donnée par le temps de couple d'origine.
25 à 27		(Réservé)	

7.3.7.2.2 Attribut N° 61 – État de vérification d'événement

Les 28 premiers bits de cet attribut à 32 bits (voir Figure 64) sont utilisés pour indiquer si le dispositif vérifie réellement les événements en fonction des différentes entrées du dispositif, telles que les entrées de marqueur et d'enregistrement, par exemple (voir Tableau 70). La vérification d'événement est initiée lorsque le bit de Commande de vérification d'événement correspondant est défini dans la connexion Contrôleur-Dispositif. Lorsqu'un événement se produit, le dispositif capture l'heure et la position exacte d'axe, puis transmet ces informations au contrôleur dans des blocs de données de notification d'événement. Mais pour que le contrôleur puisse traiter les données d'événement, le bit d'État de vérification d'événement correspondant doit être défini. Trois bits, parmi les 4 bits de poids fort de cet attribut, représentent le nombre de messages Notification d'événement dans le bloc de données d'événement dans le cadre de la mise à jour de cette connexion Dispositif-Contrôleur, le dernier bit de poids fort indiquant si le format du bloc de données d'événement est étendu afin de prendre en charge les événements supplémentaires associés à la fonctionnalité Mise à l'échelle du dispositif d'entraînement (voir Configuration de mise à l'échelle du mouvement). La prise en charge du format étendu est facultative dans la mise en œuvre du dispositif. Pour une description détaillée du protocole de gestion des événements entre le dispositif et le contrôleur, voir la section Connexion.



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Figure 64 – Champ de mot État de vérification d'événement

Tableau 70 – Définitions de bit État de vérification d'événement

Bit	Format étendu uniquement	Nom	Description
0		Front Pos d'enreg 1	Vérifier la transition de front positif de l'entrée Enregistrement 1
1		Front Neg d'enreg 1	Vérifier la transition de front négatif de l'entrée Enregistrement 1
2		Front Pos d'enreg 2	Vérifier la transition de front positif de l'entrée Enregistrement 2
3		Front Neg d'enreg 2	Vérifier la transition de front négatif de l'entrée Enregistrement 2
4	*	Fenêtre de front Pos d'enreg 1	Appliquer la vérification de fenêtre après la transition de front positif de l'entrée Enregistrement 1
5	*	Fenêtre de front Neg d'enreg 1	Appliquer la vérification de fenêtre après la transition de front négatif de l'entrée Enregistrement 1
6	*	Fenêtre de front Pos d'enreg 2	Appliquer la vérification de fenêtre après la transition de front positif de l'entrée Enregistrement 2
7	*	Fenêtre de front Neg d'enreg 2	Appliquer la vérification de fenêtre après la transition de front négatif de l'entrée Enregistrement 2
8		Réarm auto Pos d'enreg 1	Activer le mécanisme de réarmement automatique après la transition de front positif de l'entrée Enregistrement 1
9		Réarm auto Neg d'enreg 1	Activer le mécanisme de réarmement automatique après la transition de front négatif de l'entrée Enregistrement 1
10		Réarm auto Pos d'enreg 2	Activer le mécanisme de réarmement automatique après la transition de front positif de l'entrée Enregistrement 2
11		Réarm auto Neg d'enreg 2	Activer le mécanisme de réarmement automatique après la transition de front négatif de l'entrée Enregistrement 2
12	*	Position d'observation 1 Avt	Vérification de l'événement d'observation avec position d'observation 1 pour le croisement des positions réel dans le sens avant (positif).
13	*	Position d'observation 1 Inv	Vérification de l'événement d'observation avec position d'observation 1 pour le croisement des positions réel dans le sens inverse (négatif).
14	*	Position d'observation 2 Avt	Vérification de l'événement d'observation avec position d'observation 2 pour le croisement des positions réel dans le sens avant (positif).
15	*	Position d'observation 2 Inv	Vérification de l'événement d'observation avec position d'observation 2 pour le croisement des positions réel dans le sens inverse (négatif).
16		Front positif du marqueur	Vérification du front positif de l'entrée de marqueur du canal de retour en position associé
17		Front négatif du marqueur	Vérification du front négatif de l'entrée de marqueur du canal de retour en position associé
18		Front positif de l'interrupteur d'origine	Vérification du front positif de l'entrée d'origine associée à cette instance d'axe
19		Front négatif de l'interrupteur d'origine	Vérification du front positif de l'entrée d'origine associée à cette instance d'axe
20		Marqueur – Interrupteur d'origine ++	Vérification de la séquence d'événements spécifiée en tant que transition de front positif de l'entrée Interrupteur d'origine suivie d'un front positif de l'entrée Marqueur
21		Marqueur – Interrupteur d'origine +-	Vérification de la séquence d'événements spécifiée en tant que transition de front positif de l'entrée Interrupteur d'origine suivie d'un front négatif de l'entrée Marqueur
22		Marqueur – Interrupteur d'origine -+	Vérification de la séquence d'événements spécifiée en tant que transition de front négatif de l'entrée Interrupteur d'origine suivie d'un front positif de l'entrée Marqueur
23		Marqueur – Interrupteur d'origine --	Vérification de la séquence d'événements spécifiée en tant que transition de front négatif de l'entrée Interrupteur d'origine suivie d'un front négatif de l'entrée Marqueur
24	*	Seuil de couple d'origine	Vérification de l'événement de couple d'origine spécifié comme niveau de couple qui dépasse le seuil de couple d'origine pendant une période donnée par le temps de couple d'origine.
25 à 27		(Réservé)	

7.3.8 Attributs de génération de référence de consigne

Les listes d'attributs dans le Tableau 71 et le Tableau 72 s'appliquent à la fonctionnalité de génération de référence de consigne du dispositif qui convertit la position, la vitesse et l'accélération de consigne, ainsi que les données de couple provenant d'un planificateur basé sur contrôleur, en signaux de références de consigne correspondants adressés aux structures de commande de moteur du dispositif. La fonctionnalité de génération de référence de consigne inclut des interpolateurs fins, des interrupteurs de signal et des limiteurs dynamiques.

Set* → Ces attributs sont en général mis à jour par l'intermédiaire de l'Ensemble de données de consigne cycliques de la connexion C-D CIP Motion. S'ils sont inclus sous la forme de données de consigne cycliques, ces attributs ne doivent pas être mis à jour via un service Set.

Tableau 71 – Attributs Signal du générateur de consigne

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
360	Facultatif – P (Si non pris en charge, une valeur à virgule flottante de consigne de position du contrôleur est alors exigée.)	Set*		Consigne de position du contrôleur – Entier	DINT	Valeur entière des données de position de consigne provenant du contrôleur, mise à jour au cours de la période de mise à jour du contrôleur, et alimentant le générateur fin de position de consigne. Cette représentation par un entier de la position de consigne est adaptée aux dispositifs de positionnement simples où le traitement en temps réel des valeurs à virgule flottante en double précision est trop chronophage et où le lissage du mouvement n'est pas un facteur d'entraînement.	Unités d'asservissement de position
361	Facultatif – P (Si non pris en charge, une valeur entière de consigne de position du contrôleur est alors exigée.)	Set*		Commande de position du contrôleur – Virgule flottante	LREAL	Valeur à virgule flottante des données de position de consigne provenant du contrôleur, mise à jour au cours de la période de mise à jour du contrôleur, et alimentant le générateur fin de position de consigne. La plage de cette valeur est limitée à celle d'un DINT par le planificateur de mouvement, mais contrairement à un DINT, elle comporte une composante fractionnée qui se traduit en un mouvement plus lisse, notamment pour un fonctionnement par action anticipatrice.	Unités d'asservissement de position

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
362	Obligatoire – VF Facultatif – P	Set*		Commande de vitesse du contrôleur	REAL	Valeur des données de vitesse de consigne provenant du contrôleur, mise à jour au cours de la période de mise à jour du contrôleur, et alimentant le générateur fin de vitesse de consigne.	Unités de commande de vitesse / s
363	Facultatif – PVT	Set*		Consigne d'accélération du contrôleur	REAL	Valeur des données d'accélération de consigne provenant du contrôleur, mise à jour au cours de la période de mise à jour du contrôleur, et alimentant le générateur fin d'accélération de consigne.	Unités de commande d'accélération / s ²
364	Obligatoire – T	Set*		Consigne de couple du contrôleur	REAL	Valeur des données de couple de consigne provenant du contrôleur utilisée en mode d'asservissement de couple, et mise à jour au cours de la période de mise à jour du contrôleur, et alimentant le générateur fin de couple de consigne.	% de la valeur assignée du moteur
365	Facultatif – P	Get		Position de consigne fine	DINT	Valeur de sortie provenant du générateur fin de position de consigne.	Unités d'asservissement de position
366	Facultatif – PV	Get		Vitesse de consigne fine	REAL	Valeur de sortie provenant du générateur fin de vitesse de Commande de consigne. En l'absence de signal Vitesse de consigne lors de l'exécution d'un asservissement de position, ce signal peut être déduit en ajustant la valeur de sortie Position différentielle du générateur fin de position de consigne.	Unités de commande de vitesse / s
367	Facultatif – C	Get		Accélération de consigne fine	REAL	Valeur de sortie provenant du générateur fin d'accélération de consigne. En l'absence de signal Accélération de consigne lors de l'exécution d'un asservissement de position ou d'une commande de vitesse, ce signal peut être déduit en ajustant la valeur de sortie Vitesse différentielle du générateur fin de vitesse de consigne. En l'absence de signal Vitesse de consigne, le signal d'accélération de consigne fine peut être déduit en ajustant la seconde valeur de sortie Position différentielle du générateur fin de position de consigne.	Unités de commande d'accélération / s ²

Tableau 72 – Attributs Configuration du générateur de consigne

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
370	Facultatif – F	Set		Vitesse de saut 1	REAL	Définit la vitesse centrale d'une bande de vitesse de saut dans laquelle le dispositif ne fonctionne pas. La valeur de vitesse de saut est signée.	Unités de commande de vitesse / s
371	Facultatif – F	Set		Vitesse de saut 2	REAL	Définit la vitesse centrale d'une bande de vitesse de saut dans laquelle le dispositif ne fonctionne pas. La valeur de vitesse de saut est signée.	Unités de commande de vitesse / s
372	Facultatif – F	Set		Vitesse de saut 3	REAL	Définit la vitesse centrale d'une bande de vitesse de saut dans laquelle le dispositif ne fonctionne pas. La valeur de vitesse de saut est signée.	Unités de commande de vitesse / s
373	Facultatif – F	Set		Bande de vitesse de saut	REAL	Détermine la fenêtre de vitesse voisine d'une vitesse de saut qui ne peut pas être commandée. Tout point de consigne de commande dans cette fenêtre est ajusté par le bloc Vitesse de saut de manière à atteindre la valeur limite Bande de vitesse de saut supérieure ou inférieure. Le dispositif peut accélérer ou décélérer doucement sur la bande de vitesse de saut en fonction de l'interrupteur de signal, mais il peut ne pas fonctionner à une vitesse définie dans la bande. La Bande de vitesse de saut est répartie à moitié au-dessus et à moitié au-dessous de la vitesse de saut. L'attribut Bande de vitesse de saut s'applique à toutes les vitesses de saut prises en charge dans le dispositif. Une valeur nulle pour cet attribut désactive cette fonction.	Unités de commande de vitesse / s
374	Facultatif – FV	Set		Vitesse de rampe – positive	REAL	Cette valeur positive définit la sortie de consigne de vitesse positive maximale de l'interrupteur de signal.	Unités de commande de vitesse / s
375	Facultatif – FV	Set		Vitesse de rampe – négative	REAL	Cette valeur négative définit la sortie de consigne de vitesse négative maximale de l'interrupteur de signal.	Unités de commande de vitesse / s
376	Facultatif – FV	Set		Accélération de rampe	REAL	Cette valeur positive définit l'accélération maximale (vitesse en augmentation) de la sortie de consigne de vitesse par l'interrupteur de signal.	Unités de commande de vitesse / s ²
377	Facultatif – FV	Set		Décélération de rampe	REAL	Cette valeur positive définit la décélération maximale (vitesse en diminution) de la sortie de consigne de vitesse par l'interrupteur de signal.	Unités de commande de vitesse / s ²

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
378	Facultatif – FV	Set		Commande de rampe d'à-coups	REAL	Cette valeur définit le pourcentage de temps d'accélération ou de décélération appliqué au ralentisseur de vitesse sous la forme d'une courbe S d'à-coups limités en fonction d'une réponse à un échelon de vitesse. Le temps de Courbe S est ajouté à la première et à la deuxième moitié de la rampe. Une valeur nulle ne donne lieu à aucune Courbe en S, c'est-à-dire une rampe d'accélération ou de décélération linéaire. Une valeur de 100 % donne lieu à un profil d'accélération triangulaire dont la crête est l'accélération / décélération de rampe configurée. La valeur d'accélération d'à-coups diminue proportionnellement à l'augmentation de la valeur de commande d'à-coups sur la base de $(0,5 \times 0,01 \times \text{Commande d'à-coups} \times \text{Vitesse de rampe Positive} / \text{Accél de rampe})$, et la valeur limite de décélération d'à-coups diminue également en fonction de $(0,5 \times 0,01 \times \text{Commande d'à-coups} \times \text{Vitesse de rampe Négative} / \text{Décél de rampe})$.	%
380	Facultatif – FV	Set		Activation de l'amorçage instantané	BOOL	Cette valeur booléenne permet d'activer ou de désactiver la fonction Amorçage instantané de l'interrupteur de signal du dispositif. Si la valeur "true" est affectée à Amorçage instantané activé, et que l'axe de mouvement est activé, le dispositif détermine la vitesse en cours du moteur dans le cadre des activités d'initialisation État du démarrage. Juste avant de passer à l'état Exécution, le dispositif prédéfinit la sortie de l'interrupteur de signal sur la vitesse en cours. De cette manière, le moteur passe en continu de sa vitesse en cours à la vitesse commandée. Si la valeur "false" est affectée à Amorçage instantané activé, la vitesse du moteur n'est pas déterminée et une valeur prédéfinie nulle est affectée à la sortie de l'interrupteur de signal.	0 = Amorçage instantané désactivé 1 = Amorçage instantané activé

7.3.9 Attributs du mode de commande

7.3.9.1 Attributs Boucle de position

Le Tableau 73 et le Tableau 74 contiennent tous les attributs d'asservissement de position associés à une instance Objet Axe de dispositif de mouvement.

Set* → cet attribut est en général mis à jour par l'intermédiaire de l'Ensemble de données de consigne cycliques de la connexion C-D CIP Motion. S'il est inclus sous la forme de données de consigne cycliques, il ne doit pas être mis à jour via un service Set.

Tableau 73 – Attributs Signal de boucle de position

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
430	Obligatoire – P	Get		Consigne de position	DINT	Sortie de position de consigne provenant du générateur fin de consigne (s'il est actif) dans la boucle de position s'il est configuré pour le contrôle de la boucle de position.	Unités d'asservissement de position
431	Obligatoire – P	Set*		Compensation de position	DINT	Consigne de position supplémentaire ajoutée à la Consigne de position afin de générer le signal Référence de position dans la jonction de somme de boucle de position.	Unités d'asservissement de position
432	Obligatoire – P	Get		Référence de position	DINT	Signal de référence de position de consigne dans la jonction de somme de boucle de position à comparer à un signal de retour en position.	Unités d'asservissement de position
433	Obligatoire – P	Get		Consigne d'action anticipatrice de vitesse	REAL	Signal de consigne d'action anticipatrice de vitesse qui représente une version ajustée du profil de vitesse de consigne. Ce signal est le signal fin de vitesse de consigne ajusté par Kvff et appliqué à la sortie de la boucle de position.	Unités de commande de vitesse/s
434	Obligatoire – E	Get		Retour en position	DINT	Valeur de retour en position canalisé dans la jonction de somme d'asservissement de position. Dans la plupart des cas, le signal de retour en position est déduit directement du dispositif de réaction spécifié par la sélection Mode de réaction. Lorsque l'axe est configuré pour le mode "Pas d'asservissement", le retour en position représente la position réelle du dispositif de réaction spécifié par la sélection de réaction principale.	Unités d'asservissement de position
436	Obligatoire – P	Get		Erreur de position	REAL	Erreur entre la position commandée et la position réelle qui est la sortie de la jonction de somme de boucle de position.	Unités d'asservissement de position
437	Obligatoire – P	Get		Sortie de l'intégrateur de position	REAL	Sortie de l'intégrateur de position représentant la contribution de l'intégrateur de position à la Sortie de la boucle de position.	Unités de commande de vitesse/s
438	Obligatoire – P	Get		Sortie de la boucle de position	REAL	Sortie de la chaîne d'action de boucle de position représentant la mesure de contrôle totale de la boucle de position.	Unités de commande de vitesse/s

Tableau 74 – Attributs Configuration de boucle de position

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
440	Obligatoire – P	Set		Kvff	REAL	Valeur de Gain d'action anticipatrice de vitesse qui multiplie le signal Consigne d'action anticipatrice de vitesse afin de former cette même consigne qui est appliquée à la sortie de la boucle d'asservissement de position. Une action anticipatrice de vitesse à 100 % applique le signal fin de vitesse maximale de consigne à la boucle de vitesse.	%
441	Obligatoire – P	Set		Kpp	REAL	Valeur de Gain proportionnel de position qui multiplie le signal Erreur de position afin de former le signal de commande proportionnel qui s'ajoute au signal de commande intégral pour générer la sortie de la boucle d'asservissement de position. Cette valeur détermine directement la bande passante de la boucle de position.	rad/s
442	Obligatoire – P	Set		Kpi	REAL	Valeur de Gain intégral de position qui, associée à Kpp, multiplie le signal Erreur d'intégrateur de position afin de former le signal de commande intégral qui s'ajoute au signal de commande proportionnel pour générer la sortie de la boucle d'asservissement de position. La valeur inverse de cette valeur, $1/Kpi$, représente la constante de temps de l'intégrateur pour la boucle de position. Une valeur nulle pour cet attribut désactive l'intégrateur.	rad/s
443	Obligatoire – P	Set		Tolérance de verrouillage de position	REAL	Cette valeur établit une fenêtre autour de la position de consigne en cours. Si la position réelle se trouve dans cette fenêtre, le bit d'état Verrouillage de position est défini. Si la position réelle ne se trouve pas dans cette fenêtre, le bit d'état Verrouillage de position est annulé.	Unités d'asservissement de position
444	Obligatoire – P	Set		Tolérance d'erreur de position	REAL	Détermine la valeur Erreur de position maximale absolue qui peut être tolérée sans générer une exception Erreur de position excessive.	Unités d'asservissement de position
445	Facultatif – P	Set		Temps de tolérance d'erreur de position	REAL	Détermine la durée maximale de dépassement de la Tolérance d'erreur de position sans générer une exception.	Secondes

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
446	Obligatoire – P	Set		Commande d'intégrateur de position	BYTE (OCTET)	Ce champ de bit contrôle le comportement de l'intégrateur de boucle de position tout en commandant le mouvement par l'intermédiaire du contrôleur. Si le bit d'activation du maintien de l'intégrateur est défini, l'intégrateur est maintenu pendant que le mouvement est commandé avec une vitesse non nulle. S'il est annulé, l'intégrateur fonctionne sans qualification. Si le bit de pré réglage automatique est défini, la valeur de préchargement de l'intégrateur est automatiquement chargée avec la consigne de vitesse en cours en présence d'une variation de mode de commande entre la commande de vitesse et l'asservissement de position. S'il n'est pas défini, l'intégrateur est chargé avec la valeur de préchargement d'intégrateur de position configuré.	Champ de bit: 0: Activation du maintien de l'intégrateur (O) 1: Pré réglage automatique (F) 2-7: (Réservé)
447	Facultatif – P	Set		Précharge ment d'intégrateur de position	REAL	Valeur affectée à l'intégrateur de position lorsque la boucle d'asservissement de position est activée.	Unités de commande de vitesse/s
448	Facultatif – E (Mise à l'échelle du dispositif d'entraînement)	Set		Limite de position – Positive	REAL	Cette valeur définit la valeur de retour en position la plus positive qui, lorsqu'elle est dépassée dans la direction positive, génère une exception Surcourse de position positive.	Unités d'asservissement de position
449	Facultatif – E (Mise à l'échelle du dispositif d'entraînement)	Set		Limite de position – négative	REAL	Cette valeur définit la valeur de retour en position la plus négative qui, lorsqu'elle est dépassée dans la direction négative, génère une exception Surcourse de position négative.	Unités d'asservissement de position

7.3.9.2 Attributs Boucle de vitesse

Le Tableau 75 et le Tableau 76 contiennent tous les attributs de commande de vitesse associés à une instance Objet Axe de dispositif de mouvement.

Set* ➔ cet attribut est en général mis à jour par l'intermédiaire de l'Ensemble de données de consigne cycliques de la connexion C-D CIP Motion. S'il est inclus sous la forme de données de consigne cycliques, il ne doit pas être mis à jour via un service Set.

Tableau 75 – Attributs Signal de boucle de vitesse

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
450	Obligatoire – FPV	Get		Consigne de vitesse	REAL	Sortie de vitesse de consigne provenant du générateur fin de consigne (s'il est actif) dans le cadre du contrôle de boucle de vitesse ou du contrôleur de fréquence s'il est configuré pour un fonctionnement de contrôle de fréquence.	Unités de commande de vitesse/s
451	Obligatoire – FPV	Set*		Compensation de vitesse	REAL	Consigne de vitesse supplémentaire ajoutée à la jonction de somme de boucle de vitesse ou de contrôle de fréquence.	Unités de commande de vitesse/s
452	Obligatoire – PV	Get		Consigne d'action anticipatrice d'accélération	REAL	Signal de consigne d'action anticipatrice d'accélération qui représente une version ajustée du profil d'accélération de consigne. Ce signal est le signal fin d'accélération de consigne ajusté par Kaff et appliqué à la sortie de la boucle de vitesse.	Unités de commande de vitesse /s ²
453	Obligatoire – FPV	Get		Référence de vitesse	REAL	Référence de vitesse de consigne dans une jonction de somme de boucle de vitesse, ou dans le cas du contrôle de fréquence, le signal ajusté pour devenir la référence de fréquence.	Unités de commande de vitesse/s
454	Obligatoire – Tous	Get		Réaction de vitesse	REAL	Vitesse réelle d'axe appliquée à la jonction de somme de vitesse, le cas échéant, en fonction de la sélection Mode de commande. Dans la plupart des cas, le signal de retour en vitesse est déduit directement du dispositif de réaction spécifié par la sélection Mode de réaction. Lorsque l'axe est configuré pour le mode "Pas d'asservissement", le retour en vitesse représente la vitesse réelle du dispositif de réaction spécifié par la sélection de réaction principale. Lorsque l'axe est configuré pour le contrôle de fréquence, le signal Retour en vitesse est déduit du signal Consigne de vitesse. S'il est configuré pour un fonctionnement de boucle de vitesse sans capteur, le retour en vitesse est estimé par l'algorithme de commande sans capteur.	Unités de commande de vitesse/s
455	Obligatoire – PV	Get		Erreur de vitesse	REAL	Erreur entre la référence de vitesse et la valeur de retour en vitesse qui est la sortie de la jonction de somme de boucle de vitesse.	Unités de commande de vitesse/s
456	Obligatoire – PV	Get		Sortie d'intégrateur de vitesse	REAL	Sortie de l'intégrateur de vitesse représentant la contribution de l'intégrateur de vitesse à la Sortie de la boucle de vitesse.	Unités de commande d'accélération/s ²
457	Obligatoire – PV	Get		Sortie de boucle de vitesse	REAL	Sortie de la chaîne d'action de vitesse représentant la mesure de contrôle totale de la boucle de vitesse.	Unités de commande d'accélération/s ²

Tableau 76 – Attributs Configuration de boucle de vitesse

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
460	Obligatoire – PV	Set		Kaff	REAL	Valeur de Gain d'action anticipatrice d'accélération qui multiplie le signal fin de consigne d'action anticipatrice d'accélération afin de former cette même consigne avant de l'appliquer à la sortie de la boucle de commande de vitesse. Un signal fin de consigne d'accélération à 100% applique le signal fin de consigne d'action anticipatrice d'accélération maximale à la sortie de la boucle de vitesse.	%
461	Obligatoire – PV	Set		Kvp	REAL	Valeur de Gain proportionnel de vitesse qui multiplie le signal Erreur de vitesse afin de former le signal de commande proportionnel qui s'ajoute au signal de commande intégral pour générer la sortie de la boucle de commande de vitesse. Cette valeur détermine directement la bande passante de la boucle de vitesse.	rad/s
462	Obligatoire – PV	Set		Kvi	REAL	Valeur de Gain intégral de vitesse qui, associée à Kvp, multiplie le signal Erreur d'intégrateur de vitesse afin de former le signal de commande intégral qui s'ajoute au signal de commande proportionnel pour générer la sortie de la boucle de commande de vitesse. La valeur inverse de cette valeur, $1/Kvi$, représente la constante de temps de l'intégrateur pour la boucle de vitesse. Une valeur nulle pour cet attribut désactive l'intégrateur.	rad/s
464	Facultatif – FPV	Set		Kdr	REAL	Valeur Statisme de vitesse qui assure la conformité avec l'intégrateur de vitesse en soustrayant une partie de l'effort de boucle de vitesse de l'entrée d'erreur de vitesse à l'intégrateur de vitesse. La présence d'un gain d'échelle Couple/Force, Kj, dans le parcours du signal de statisme permet de spécifier le Statisme de vitesse dans les unités de vitesse par % de sortie de couple assigné. Ce paramètre est également valide pour les dispositifs d'entraînement V/Hz, et son comportement est pratiquement identique, mais au lieu d'associer le % assigné au couple, le % assigné est associé au courant.	Unités de commande de vitesse/s/ % assigné
465	Facultatif – PV	Set		Tolérance d'erreur de vitesse	REAL	Détermine la valeur Erreur de vitesse maximale absolue qui peut être tolérée sans générer une exception Erreur de vitesse excessive.	Unités de commande de vitesse/s
466	Facultatif – PV	Set		Temps de tolérance d'erreur de vitesse	REAL	Détermine la durée maximale de dépassement de la Tolérance d'erreur de vitesse sans générer une exception.	secondes

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
467	Obligatoire – PV	Set		Commande d'intégrateur de vitesse	BYTE (OCTET)	Ce champ de bit contrôle le comportement de l'intégrateur de boucle de vitesse tout en commandant le mouvement par l'intermédiaire du contrôleur. Si le bit d'activation du maintien de l'intégrateur est défini, l'intégrateur est maintenu pendant que le mouvement est commandé avec une vitesse non nulle. S'il est annulé, l'intégrateur fonctionne sans qualification. Si le bit de prééclage automatique est défini, la valeur de préchargement de l'intégrateur est automatiquement chargée avec la consigne de couple en cours en présence d'une variation de mode de commande entre l'asservissement de couple et la commande de vitesse. S'il n'est pas défini, l'intégrateur est chargé avec la valeur de préchargement d'intégrateur de vitesse configuré.	Champ de bit: 0: Activation du maintien de l'intégrateur (O) 1: Prééclage automatique (F) 2-7: (Réservé)
468	Facultatif – PV	Set		Préchargement d'intégrateur de vitesse	REAL	Valeur affectée à l'intégrateur de vitesse lorsque la boucle de commande de vitesse est activée.	Unités de commande d'accélération/s ²
469	Facultatif – PV	Set		Bande passante du filtre passe-bas de vitesse	REAL	Contrôle la bande passante du filtre passe-bas appliqué au signal d'erreur de vitesse. Un filtre IIR à deux pôles constitue une mise en œuvre recommandée. Une valeur nulle pour cet attribut désactive cette fonction.	rad/s
470	Facultatif – ED	Set		Seuil de vitesse	REAL	Définit une vitesse absolue minimale. Lorsque l'amplitude du signal de retour en vitesse est inférieure à cette valeur, le bit d'état Seuil de vitesse est défini.	Unités de commande de vitesse/s
471	Facultatif – FPV	Set		Tolérance de verrouillage de vitesse	REAL	Cette valeur établit une fenêtre autour du signal de référence de vitesse non limitée. Si le signal de retour en vitesse se trouve dans cette fenêtre, le bit d'état Verrouillage de vitesse est défini. Si le signal de retour en vitesse ne se trouve pas dans cette fenêtre, le bit d'état Verrouillage de vitesse est annulé.	Unités de commande de vitesse/s
472	Obligatoire – ED	Set		Fenêtre de gel de vitesse	REAL	Cette valeur établit une fenêtre autour de la vitesse nulle. Si le signal de retour en vitesse se trouve dans cette fenêtre, le bit d'état Gel de vitesse est défini. Si le signal de retour en vitesse ne se trouve pas dans cette fenêtre, le bit d'état Gel de vitesse est annulé.	Unités de commande de vitesse/s
473	Facultatif – FPV	Set		Limite de vitesse – Positive	REAL	Cette valeur positive définit la valeur de référence de vitesse la plus positive dans la jonction de somme de vitesse. Lorsque cette limite de vitesse est dépassée, le dispositif répond en fixant la référence de vitesse à cette limite et en définissant le bit d'état Limite de vitesse.	Unités de commande de vitesse/s
474	Facultatif – FPV	Set		Limite de vitesse – Négative	REAL	Cette valeur négative définit la valeur de référence de vitesse la plus négative admise dans la jonction de somme de vitesse. Lorsque cette limite de vitesse est dépassée, le dispositif répond en fixant la référence de vitesse à cette limite et en définissant le bit d'état Limite de vitesse.	Unités de commande de vitesse/s

7.3.9.3 Attributs Commande d'accélération

Le Tableau 77 et le Tableau 78 contiennent tous les attributs d'accélération associés à une instance Objet Axe de dispositif de mouvement.

Set* → cet attribut est en général mis à jour par l'intermédiaire de l'Ensemble de données de consigne cycliques de la connexion C-D CIP Motion. S'il est inclus sous la forme de données de consigne cycliques, il ne doit pas être mis à jour via un service Set.

Tableau 77 – Attributs Signal d'accélération

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
480	Facultatif – C	Get		Consigne d'accélération	REAL	Sortie d'accélération de consigne provenant du générateur fin de consigne (s'il est actif) dans la boucle d'accélération s'il est configuré pour la commande d'accélération.	Unités de commande d'accélération/s ²
481	Facultatif – C	Set*		Compensation d'accélération	REAL	Consigne d'accélération supplémentaire ajoutée à la jonction de somme de boucle d'accélération.	Unités de commande d'accélération/s ²
482	Facultatif – C	Get		Référence d'accélération	REAL	Référence d'accélération de consigne dans une jonction de somme de boucle d'accélération.	Unités de commande d'accélération/s ²
483	Obligatoire – E	Get		Retour en accélération	REAL	Accélération réelle d'axe en fonction du dispositif de réaction sélectionné. Dans la plupart des cas, le signal de retour en accélération est déduit directement du dispositif de réaction spécifié par la sélection Mode de réaction. Lorsque l'axe est configuré pour le mode "Pas d'asservissement", le retour en accélération représente l'accélération réelle du dispositif de réaction spécifié par la sélection de réaction principale.	Unités de commande d'accélération/s ²

Tableau 78 – Attributs Configuration d'accélération

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
485	Facultatif – D	Set		Limite d'accélération	REAL	Cette valeur définit la valeur d'accélération maximale (augmentation de vitesse) dans la jonction de somme d'accélération. Lorsque cette limite d'accélération est dépassée, le dispositif répond en fixant la référence d'accélération à cette limite et en définissant le bit d'état Limite d'accélération.	Unités de commande d'accélération/s ²
486	Facultatif – D	Set		Limite de décélération	REAL	Cette valeur définit la valeur de décélération maximale (diminution de vitesse) dans la jonction de somme d'accélération. Lorsque cette limite de décélération est dépassée, le dispositif répond en fixant la référence d'accélération à cette limite et en définissant le bit d'état Limite de décélération.	Unités de commande d'accélération/s ²

7.3.9.4 Attributs Asservissement de couple/force

Le Tableau 79 et le Tableau 80 contiennent tous les attributs de couple/force associés à une instance Objet Axe de dispositif de mouvement.

Set* → cet attribut est en général mis à jour par l'intermédiaire de l'Ensemble de données de consigne cycliques de la connexion C-D CIP Motion. S'il est inclus sous la forme de données de consigne cycliques, il ne doit pas être mis à jour via un service Set.

Tableau 79 – Attributs Signal d'asservissement de couple/force

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
490	Obligatoire – C	Get		Consigne de couple	REAL	Sortie de couple de consigne provenant du générateur fin de consigne (s'il est actif) dans la jonction de somme d'entrée de couple s'il est configuré pour l'asservissement de couple.	% de la valeur assignée du moteur
491	Obligatoire – C	Set*		Compensation de couple	REAL	Consigne de couple supplémentaire ajoutée à la jonction de somme d'entrée de couple.	% de la valeur assignée du moteur
492	Obligatoire – C	Get		Référence de couple	REAL	Signal d'entrée de référence de couple commandé avant la section de filtre de couple, représentant la somme des entrées de signal Consigne de couple et Compensation de couple.	% de la valeur assignée du moteur
493	Obligatoire – C	Get		Référence de couple filtrée	REAL	Signal d'entrée de référence de couple commandé après la section de filtre de couple.	% de la valeur assignée du moteur
494	Obligatoire – C	Get		Référence de couple limitée	REAL	Signal d'entrée de référence de couple commandé après la section de limiteur de couple.	% de la valeur assignée du moteur

Tableau 80 – Attributs Configuration d'asservissement de couple/force

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
496	Obligatoire – PV Facultatif – T	Set		Kj	REAL	Valeur Gain d'échelle couple/force qui convertit l'accélération commandée en couple/force assignés équivalents. Correctement définie, cette valeur représente l'inertie ou la masse totale du système.	% de la valeur assignée du moteur/ (Unités moteur/ s ²)
498	Facultatif – C	Set		Compensation de frottement – Glissant	REAL	Valeur ajoutée à la consigne de courant/couple afin de décaler les effets du frottement coulomb.	% de la valeur assignée du moteur
499	Facultatif – C	Set		Compensation de frottement – Statique	REAL	Valeur ajoutée à la consigne de courant/couple afin de décaler les effets du frottement statique, parfois appelé "sticktion" en anglais.	% de la valeur assignée du moteur
500	Facultatif – C	Set		Compensation de frottement – Visqueux	REAL	Valeur ajoutée à la consigne de courant/couple afin de décaler les effets du frottement visqueux, c'est-à-dire le frottement proportionnel à la vitesse.	% de la valeur assignée du moteur/ (Unités moteur/ s)
502	Facultatif – C	Set		Bande passante du filtre passe-bas de couple	REAL	Fréquence de rupture du filtre passe-bas appliqué au signal de référence de couple.	rad/s
503	Facultatif – C	Set		Fréquence de filtre coupe-bande de couple	REAL	Fréquence centrale du filtre coupe-bande appliqué au signal de référence du couple. Une valeur nulle pour cet attribut désactive cette fonction.	rad/s
504	Obligatoire – C	Set		Limite de couple positive	REAL	Cette valeur positive détermine le couple positif maximum qui peut être appliqué au moteur. Si le dispositif tente de dépasser cette valeur, la consigne de couple est fixée à cette valeur.	% de la valeur assignée du moteur
505	Obligatoire – C	Set		Limite de couple négative	REAL	Cette valeur négative détermine le couple négatif maximum qui peut être appliqué au moteur. Si le dispositif tente de dépasser cette valeur, la consigne de couple est fixée à cette valeur.	% de la valeur assignée du moteur
506	Facultatif – C	Set		Limite de débit de couple	REAL	Limite le taux de variation du signal de référence de couple.	% de la valeur assignée du moteur/s
507	Facultatif – C	Set		Seuil de couple	REAL	Spécifie le seuil de l'amplitude du signal de référence de couple filtré qui, lorsqu'il est dépassé, entraîne la définition du bit d'état Seuil de couple.	% de la valeur assignée du moteur

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
508	Facultatif – D	Set		Limite de surcouple	REAL	Seuil maximum du couple qui produit l'amplitude de signal Retour de courant Iq. Lorsque le signal Retour de courant Iq est supérieur à cette valeur pour la durée spécifiée par l'attribut Temps de limite de surcouple, cela donne lieu à une exception de limite de surcouple. Cette fonction permet au dispositif de générer une exception en cas d'augmentation subite de l'amplitude du couple de charge en fonctionnement. Cette condition pourrait se produire en cas de défaillance du défilement ou d'une autre défaillance mécanique.	% de la valeur assignée du moteur
509	Facultatif – D	Set		Temps de limite de surcouple	REAL	Spécifie la durée admise d'une condition Limite de surcouple avant de générer une exception Limite de surcouple. Une valeur nulle pour cet attribut désactive la fonction de surcouple.	secondes
510	Facultatif – D	Set		Limite de sous-couple	REAL	Seuil minimum du couple qui produit l'amplitude de signal Retour de courant Iq. Lorsque le signal Retour de courant Iq est inférieur à cette valeur pour la durée spécifiée par l'attribut Temps de limite de sous-couple, cela donne lieu à une exception de limite de sous-couple. Cette fonction permet au dispositif de générer une exception en cas de diminution soudaine de l'amplitude du couple de charge en fonctionnement. Cette condition peut se produire en cas de rupture de couple de charge, de rupture de matériaux bridés, etc.	% de la valeur assignée du moteur
511	Facultatif – D	Set		Temps de limite de sous-couple	REAL	Spécifie la durée admise d'une condition Limite de sous-couple avant de générer une exception Limite de sous-couple. Une valeur nulle pour cet attribut désactive la fonction de sous-couple.	secondes

7.3.9.5 Attributs Asservissement de courant

Le Tableau 81 et le Tableau 82 contiennent tous les attributs d'asservissement de courant associés à une instance Objet Axe de dispositif de mouvement.

Tableau 81 – Attributs Signal d'asservissement de courant

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
520	Obligatoire – C	Get		Consigne de courant Iq	REAL	Représente la valeur instantanée du couple commandé qui génère le signal de courant, Iq, avant de traverser le limiteur de vecteur de courant. Elle est directement liée à la sortie du trajet de référence de couple après la mise à l'échelle 1/Kt qui représente l'effort de couple à appliquer au couple du dispositif d'entraînement qui génère la boucle de courant Iq. La valeur nominale de 1/Kt est basée sur un couple assigné de 100 % produit par le courant assigné de 100 %.	% de la valeur assignée du moteur
521	Facultatif – C	Get		Limite de courant opératoire	REAL	Représente la limite de courant opératoire en fonction de plusieurs sources de limite.	% de la valeur assignée du moteur
522	Facultatif – C	Get		Source de limite de courant	USINT	Représente la source opératoire d'une limite de courant lorsqu'une condition de limite de courant doit se produire.	Énumération: 0 = Non limité 1 = Limite de courant de crête de l'onduleur 2 = Limite de courant de crête du moteur 3 = Limite de courant thermique de l'onduleur 4 = Limite de courant thermique du moteur 5 = Limite de régulateur de shunt 6 = Limite de vecteur de courant 7 = Limite d'essai du frein 8 à 127 = réservé 128 à 255 = spécifique au fournisseur
523	Obligatoire – C (Moteur à aimant permanent)	Get		Angle électrique du moteur	REAL	Angle électrique calculé du moteur en fonction du nombre de pôles du moteur, du décalage de commutation et du dispositif d'entraînement de réaction sélectionné.	Degrés
524	Facultatif – C	Get		Référence de courant Iq	REAL	Signal de référence du courant, Iq, dans la jonction de somme de boucle de courant de couple.	% de la valeur assignée du moteur

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
525	Facultatif – C	Get		Référence de courant Id	REAL	Référence de courant de consigne, Id, dans la jonction de somme de boucle de courant de flux.	% de la valeur assignée du moteur
527	Facultatif – C	Get		Erreur de courant Iq	REAL	Erreur entre le courant commandé et le courant réel qui est la sortie de la jonction de somme de boucle de courant, q-axe, générant le couple.	% de la valeur assignée du moteur
528	Facultatif – C	Get		Erreur de courant Id	REAL	Erreur entre le courant commandé et le courant réel qui est la sortie de la jonction de somme de boucle de courant, d-axe, générant le flux.	% de la valeur assignée du moteur
529	Facultatif – C	Get		Retour de courant Iq	REAL	Courant de couple réel appliqué à l'axe en fonction du retour du capteur de courant.	% de la valeur assignée du moteur
530	Facultatif – C	Get		Retour de courant Id	REAL	Courant de flux réel appliqué à l'axe en fonction du retour du capteur de courant.	% de la valeur assignée du moteur
531	Facultatif – C	Get		Découplage Vq	REAL	Signal de tension instantané ajouté à la sortie de la boucle d'asservissement Iq pour compenser les effets de Id.	Volts
532	Facultatif – C	Get		Découplage Vd	REAL	Signal de tension instantané ajouté à la sortie de la boucle d'asservissement Id pour compenser les effets de Iq.	Volts
533	Facultatif – C	Get		Sortie de tension Vq	REAL	Couple/force instantanés générant la tension de sortie.	Volts
534	Facultatif – C	Get		Sortie de tension Vd	REAL	Flux instantané générant la tension de sortie.	Volts
535	Facultatif – C	Get		Sortie de tension U	REAL	Tension instantanée appliquée à la phase U.	Volts
536	Facultatif – C	Get		Sortie de tension V	REAL	Tension instantanée appliquée à la phase V.	Volts
537	Facultatif – C	Get		Sortie de tension W	REAL	Tension instantanée appliquée à la phase W.	Volts
538	Facultatif – C	Get		Retour de courant U	REAL	Courant instantané mesuré sur la phase U.	Ampères
539	Facultatif – C	Get		Retour de courant V	REAL	Courant instantané mesuré sur la phase V.	Ampères
540	Facultatif – C	Get		Retour de courant W	REAL	Courant instantané mesuré sur la phase W.	Ampères
541	Facultatif – C	Get		Décalage de courant U	REAL	Décalage du transducteur de courant de phase U.	Ampères
542	Facultatif – C	Get		Décalage de courant V	REAL	Décalage du transducteur de courant de phase V.	Ampères
543	Facultatif – C	Get		Décalage de courant W	REAL	Décalage du transducteur de courant de phase W.	Ampères

Tableau 82 – Attributs Configuration de contrôle de courant

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
553	Facultatif – D	Set		Limite de vecteur de courant	REAL	Valeur de limite de vecteur de courant appliquée au limiteur de vecteur de courant afin de fournir une limite configurable de l'amplitude du vecteur de courant.	% de la valeur assignée du moteur
554	Facultatif – C	Set		Kqp	REAL	Valeur Gain proportionnel Iq qui multiplie le signal Erreur de courant Iq . Cette valeur détermine directement la bande passante de la boucle de courant qui génère le couple.	rad/s
555	Facultatif – C	Set		Kqi	REAL	Valeur Gain intégral Iq qui, associée à Kdp, multiplie le signal Erreur de courant Iq avant de l'appliquer à l'accumulateur Erreur d'intégrateur Iq. La valeur inverse de cette valeur, 1/Kvdi, représente la constante de temps de l'intégrateur pour la boucle de courant de flux. Une valeur nulle pour cet attribut désactive l'intégrateur.	rad/s
556	Facultatif – C	Set		Kdp	REAL	Valeur Gain proportionnel Id qui multiplie le signal Erreur de courant Id. Cette valeur détermine directement la bande passante de la boucle de courant qui génère le flux.	rad/s
557	Facultatif – C	Set		Kdi	REAL	Valeur Gain intégral Id qui, associée à Kdp, multiplie le signal Erreur de courant Id avant de l'appliquer à l'accumulateur Erreur d'intégrateur Id. La valeur inverse de cette valeur, 1/Kvdi, représente la constante de temps de l'intégrateur pour la boucle de courant de flux. Une valeur nulle pour cet attribut désactive l'intégrateur.	rad/s
558	Facultatif – D (Moteur à induction)	Set		Commande de flux ascendant	USINT	Si l'axe de mouvement est activé, le courant en courant continu est appliqué à un moteur à induction afin de générer un flux de stator avant de passer à l'état Exécution. Cet attribut permet de contrôler l'application d'un flux à un moteur à induction à l'état Démarrage avant de passer à l'état Exécution. Si Pas de délai est sélectionné, l'axe passe immédiatement à l'état Exécution lors de la génération du flux du moteur. Avec Délai manuel, l'axe reste à l'état Démarrage pendant le Temps de flux ascendant afin de laisser le temps au moteur de recevoir la totalité du flux. Avec Délai automatique, le dispositif d'entraînement détermine le délai de génération du flux du moteur en fonction des données d'attribut de configuration du moteur ou en fonction des mesures.	Énumération: 0 = Pas de délai (O) 1 = Délai manuel (F) 2 = Délai automatique (F) 3 à 255 = (réservé)
559	Facultatif – D (Moteur à induction)	Set		Temps de flux ascendant	REAL	Définit le délai de génération d'un flux de moteur complet admis par le dispositif d'entraînement avant de passer à l'état Exécution.	secondes

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
560	Facultatif – CE (Moteur à aimant permanent)	Set		Méthode de démarrage par commutation	USINT	Spécifie la méthode utilisée par le dispositif d'entraînement pour établir un alignement absolu du rotor (ou de la bobine mobile) par rapport aux enroulements statoriques à des fins de commutation du moteur à aimant permanent lors du démarrage du dispositif d'entraînement. Si cet attribut n'est pas pris en charge, la méthode de démarrage par commutation est déterminée de manière implicite par la sélection du type de réaction 1. De même, si cet attribut est pris en charge et défini sur A partir du type de réaction, la méthode de démarrage par commutation est également déterminée de manière implicite par la sélection du type de réaction 1. La méthode de démarrage UVW utilise les signaux de même nature émis par les pistes du codeur monté sur le moteur ou par les capteurs à effet Hall, associés au décalage de commutation, afin d'aligner correctement le rotor sur les enroulements statoriques ou, dans le cas d'un moteur linéaire, la bobine mobile avec la piste magnétique. Une fois l'alignement établi, la commutation est maintenue via les signaux de position émis par le dispositif de réaction monté sur le moteur, c'est-à-dire la réaction 1. La méthode de démarrage numérique utilise les signaux de même nature émis par un dispositif de réaction absolue monté sur le moteur, associés au décalage de commutation, afin d'aligner correctement le rotor sur les enroulements statoriques ou, dans le cas d'un moteur linéaire, la bobine mobile avec la piste magnétique. La méthode de démarrage par autodétection applique le courant au stator du moteur lors de l'état Démarrage initial afin de forcer le rotor sur la position Nulle, et de ce fait, d'établir un alignement de commutation correct. Une fois l'alignement établi, la commutation est maintenue via les signaux de position émis par le dispositif de réaction monté sur le moteur, c'est-à-dire la réaction 1. Cette méthode est utilisée lorsqu'aucune réaction absolue ne permet d'aligner le moteur, par exemple un moteur comportant un codeur incrémentiel.	Énumération: 0 = A partir du type de réaction (O) 1 = UVW (F) 2 = Numérique (F) 3 = A autodétection (F) 4 à 255 = (Réservé)

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
561	Obligatoire – CE (Moteur à aimant permanent)	Set		Décalage de commutation	REAL	Valeur qui spécifie le décalage de commutation du dispositif de réaction monté sur un moteur à aimant permanent, en unités de degrés électriques. Cet attribut spécifie le décalage par rapport à une position de référence de commutation définie en appliquant le courant en courant continu à la borne A et hors des bornes B et C court-circuitées du moteur, et en permettant au rotor de se déplacer vers sa position nulle magnétique par rapport au stator. Sur un codeur ou résolveur absolu, le décalage est la différence entre la position absolue nulle du dispositif et la position de référence de commutation. Sur un codeur incrémentiel ou un capteur à effet Hall avec émission de signaux UVW, le décalage est la différence entre la position correspondant à une transition du canal du dispositif de commutation W (S3) (les canaux U (S1) et V (S2) étant respectivement haut et bas) et la position de référence de commutation. Le décalage de commutation s'applique uniquement dispositif Réaction 1 monté sur le moteur. Lorsque l'attribut Alignement de commutation facultatif est pris en charge et défini sur Décalage du contrôleur, le dispositif d'entraînement doit appliquer la valeur Décalage de commutation du contrôleur afin de déterminer l'angle électrique du moteur. Dans ce cas, une valeur valide Décalage de commutation doit être établie par le contrôleur. Dans le cas inhabituel où le décalage de commutation est également archivé dans le moteur et diffère de manière significative de la valeur Décalage de commutation du contrôleur, le dispositif d'entraînement doit passer à l'état Démarrage inhibé. Si l'attribut Alignement de commutation n'est pas défini sur Décalage du contrôleur, la valeur Décalage de commutation du contrôleur est ignorée par le dispositif d'entraînement et ce dernier doit déterminer sa valeur interne de décalage de commutation par d'autres moyens. En l'absence d'un décalage de commutation valide, le dispositif d'entraînement doit être à l'état Démarrage inhibé.	Degrés électriques
562	Facultatif – CE (Moteur à aimant permanent)	Set		Courant autodétecteur par commutation	REAL	Lorsqu'un dispositif d'entraînement de réaction monté sur moteur à aimant permanent est un codeur incrémentiel sans pistes UVW de commutation, un algorithme à autodétection doit être exécuté au cours de l'état Démarrage, qui détermine le décalage de commutation afin d'appliquer le retour en position. Cet algorithme applique un courant au stator du moteur afin d'orienter le rotor de manière à établir la mise en phase de commutation du moteur.	% de la valeur assignée du moteur

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
563	Facultatif – CE (Moteur à aimant permanent)	Set		Polarité de commutation	USINT	Lorsqu'un moteur à aimant permanent utilise les signaux UVW pour un démarrage par commutation, il est essentiel que les phases UVW du dispositif de commutation suivent la mise en phase du moteur. Une polarité normale implique une mise en phase UVW conforme à la spécification définie en usine lorsque le dispositif de commutation se déplace dans la direction positive également définie en usine. La polarité inverse bascule effectivement la mise en phase UVW sur la mise en phase UWV, inversant ainsi la direction du dispositif de commutation. Lorsqu'un essai de commutation permet de déterminer que la mise en phase du moteur et celle du dispositif de commutation ont une polarité opposée, cet attribut peut servir à compenser le défaut d'adaptation.	Énumération: 0 = Polarité normale 1 = Polarité inverse 2 à 255 = (réservé)
564	Facultatif – CE (Moteur à aimant permanent)	Set		Alignement de commutation	USINT	Ce paramètre énuméré est défini sur Décalage du contrôleur (1) lorsque le dispositif de réaction absolue monté sur le moteur est tenu d'être aligné sur les enroulements statoriques du moteur à aimant permanent selon la valeur Décalage de commutation. Dans certains cas, le décalage de commutation peut être prédéfini sur une valeur établie par un alignement défini en usine du dispositif de réaction du moteur, par rapport aux enroulements statoriques de ce dernier. Un paramètre de réglage Non Aligné (0) indique que le moteur n'est pas aligné, et que la valeur Décalage de commutation n'est pas valide. Lorsque le décalage de commutation n'est pas valide, il ne peut pas être utilisé par le dispositif d'entraînement pour déterminer l'angle de commutation. Toute tentative d'activation du dispositif d'entraînement avec un angle de commutation non valide doit générer une condition Inhibition de démarrage. L'alignement peut être réalisé au moyen d'un essai de commutation qui mesure et définit le décalage de commutation du moteur, ou par une entrée utilisateur directe. Si cet attribut est défini sur Décalage du moteur (2), le dispositif d'entraînement déduit le décalage de commutation directement du moteur. Lorsqu'il est défini sur Autodétection (4), le dispositif d'entraînement mesure automatiquement le décalage de commutation lorsqu'il passe à l'état Démarrage pour la première fois après un cycle d'alimentation. Ceci s'applique généralement à un moteur à aimant permanent comportant un dispositif de réaction incrémentielle simple.	Énumération: 0 = Non aligné (O) 1 = Décalage du contrôleur (O) 2 = Décalage du moteur (F) 3 = A autodétection (F) 4 à 255 = Réservé.

7.3.9.6 Attributs Contrôle de fréquence

Le Tableau 83 et le Tableau 84 contiennent tous les attributs associés à la méthode de fonctionnement Contrôle de fréquence d'une instance Objet Axe de dispositif de mouvement.

Tableau 83 – Attributs Signal de contrôle de fréquence

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
565	Obligatoire – F	Get		Compensation de glissement	REAL	Indique la valeur réelle de compensation de glissement actuellement appliquée.	r/min

Tableau 84 – Attributs Configuration de contrôle de fréquence

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
570	Obligatoire – F	Set		Méthode de contrôle de fréquence	SINT	La méthode de commande Volts/Hertz de base applique la tension au moteur, en général en relation directe avec la fréquence ou la vitesse commandée.	Énumération 0 = Volts/Hertz de base (O) 1 à 127 = réservé 128 à 255 = spécifique au fournisseur
572	Obligatoire – F	Set		Tension maximale	REAL	Définit la tension la plus élevée entre phases que le dispositif d'entraînement peut sortir.	Volts (eff.)
573	Obligatoire – F	Set		Fréquence maximale	REAL	Définit la fréquence la plus élevée que le dispositif d'entraînement peut sortir.	Hertz
575	Obligatoire – F	Set		Tension de rupture	REAL	Définit la tension de sortie entre phases du dispositif d'entraînement à la fréquence de rupture à laquelle se termine la suralimentation.	Volts (eff.)
576	Obligatoire – F	Set		Fréquence de rupture	REAL	Définit la fréquence de sortie du dispositif d'entraînement à la tension de rupture à laquelle se termine la suralimentation.	Hertz
577	Obligatoire – F	Set		Suralimentation au démarrage	REAL	Définit le niveau de suralimentation de tension entre phases pour le départ et l'accélération.	Volts (eff.)
578	Obligatoire – F	Set		Suralimentation à l'exécution	REAL	Définit le niveau de suralimentation de tension entre phases à la vitesse ou décélération en régime permanent.	Volts (eff.)

7.3.9.7 Attributs Sortie de dispositif d'entraînement

Le Tableau 85 contient tous les attributs de sortie de l'onduleur associés à une instance Objet Axe de dispositif de mouvement.

Tableau 85 – Attributs Sortie de dispositif d'entraînement

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
600	Obligatoire – F Facultatif – C	Get		Fréquence de sortie	REAL	Fréquence de sortie appliquée au moteur. La valeur de fréquence est exprimée en termes de cycles électriques.	Hertz
601	Obligatoire – D	Get		Courant de sortie	REAL	Courant de sortie appliqué au moteur.	Ampères (eff.)
602	Obligatoire – D	Get		Tension de sortie	REAL	Tension de sortie entre phases appliquée au moteur.	Volts (eff.)
603	Obligatoire – D	Get		Puissance de sortie	REAL	Puissance de sortie du moteur. Cette valeur repose sur le produit du signal Référence de couple et Retour en vitesse.	kW
604	Facultatif – D	Set		Fréquence MID	UINT	Définit la fréquence de porteuse de la sortie Modulation d'impulsions en durée appliquée au moteur. Le déclassement du dispositif d'entraînement est exigé à des fréquences MID plus élevées en raison de pertes de commutation. Le temps de mise à jour de la boucle de courant est directement lié à la fréquence MID. Par conséquent, les performances de la boucle augmentent en général au fur et à mesure de l'augmentation du taux MID.	Hertz

7.3.10 Attributs Arrêt et freinage

7.3.10.1 Généralités

Le Tableau 86 contient tous les attributs d'arrêt et de freinage associés à une instance Objet Axe de dispositif de mouvement.

Tableau 86 – Attributs Arrêt/Freinage

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
610	Obligatoire – D	Set		Action d'arrêt	USINT	Lors de la désactivation ou de l'abandon d'un axe, via une demande Désactivation ou une demande Abandon, cette valeur détermine la méthode d'arrêt à appliquer au moteur. L'état final appliqué à l'issue de la méthode d'arrêt est Arrêté. À l'état Arrêté, la structure de puissance de l'onduleur du dispositif doit être Désactivée (sélection Désactivé) et sans couple ou maintenue active (Sélection Maintien) dans une condition statique. Cet attribut ne détermine, en aucun cas, les actions d'arrêt appliquées en réponse aux conditions de défaut.	Voir la sémantique en 7.3.10.2.1.
611	Obligatoire – C	Set		Couple d'arrêt	REAL	Lors de la désactivation ou de l'abandon d'un axe, cette valeur détermine le couple maximal générant le courant disponible pour arrêter le moteur lorsque l'action d'arrêt est définie sur Décel. de courant.	% de la valeur assignée du moteur
612	Facultatif – C	Set		Limite de temps d'arrêt	REAL	Lors de la désactivation ou de l'abandon d'un axe, ce paramètre détermine la durée maximale dont a besoin le dispositif d'entraînement pour atteindre une vitesse nulle.	secondes
613	Facultatif – D (Moteur à aimant permanent)	Set		Délai de contact de frein résistif	REAL	Lorsqu'un frein résistif externe est utilisé, le Délai de contact de frein résistif peut être défini pour retarder l'activation de la structure de puissance d'entraînement pour que le frein résistif ait le temps de connecter le moteur au dispositif d'entraînement.	secondes Voir la sémantique en 7.3.10.2.2 pour plus de détails.
614	Facultatif – D	Set		Commande de frein mécanique	USINT	Cet attribut régit le fonctionnement de la sortie de frein mécanique du dispositif d'entraînement qui commande le mécanisme de freinage mécanique. Lorsqu'il est défini sur Automatique, le frein mécanique est sous le contrôle du diagramme d'états d'axe. Le séquençement du freinage est décrit en détail par les attributs Délai d'engagement de frein mécanique et Délai de libération de frein mécanique. Lorsqu'il est défini sur Libération de frein, le frein est libéré sans condition, et n'est plus sous le contrôle du diagramme d'états d'axe.	Énumération: 0 = Automatique 1 = Libération de frein 2 à 255 = (réservé)
615	Facultatif – D	Set		Délai de libération de frein mécanique	REAL	Lors de l'activation d'axe, la valeur Délai de libération de frein mécanique détermine la durée à laquelle le dispositif doit retarder le passage de l'état Démarrage à l'état Exécution ou Essais. Ce délai empêche tout mouvement commandé d'axe de mouvement pour laisser le temps au frein mécanique externe de se désengager.	secondes Voir la sémantique en 7.3.10.2.3 pour plus de détails.
616	Facultatif – D	Set		Délai d'engagement de frein mécanique	REAL	Lors de la désactivation d'axe de mouvement, la valeur Délai d'engagement de frein mécanique détermine pendant combien de temps la structure de puissance du dispositif doit rester activée après la décélération d'axe en vue de son arrêt. Cela laisse le temps à un frein mécanique externe de s'engager. L'action d'arrêt configurée détermine le type de séquence d'arrêt appliqué.	secondes Voir la sémantique en 7.3.10.2.4 pour plus de détails.

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
608	Facultatif – D	Set		Vitesse nulle	REAL	Cet attribut définit le seuil de vitesse associé aux critères de vitesse nulle de la séquence d'arrêt. La Vitesse nulle est spécifiée en pourcentage de la vitesse assignée du moteur. Lorsque l'attribut Durée de vitesse nulle est pris en charge, il définit le seuil de vitesse auquel le temporisateur de vitesse nulle démarre. Lorsque la vitesse d'axe est inférieure au seuil de Vitesse nulle pendant la durée de vitesse nulle, l'axe satisfait aux critères de vitesse nulle. Dans tous les cas d'arrêts, sauf ceux de Catégorie 2, ceci donne lieu à l'engagement du frein mécanique. Si cet attribut n'est pas pris en charge, le seuil de vitesse nulle est laissé à la discrétion du fournisseur, il est généralement défini à 1 % de la vitesse assignée du moteur. La vitesse d'axe décrite ci-dessus est fonction du signal de Réaction de vitesse, ou dans le cas d'un dispositif d'entraînement à Contrôle de fréquence, la vitesse d'axe est fonction du signal de Référence de vitesse.	% de la valeur assignée du moteur
609	Facultatif – D	Set		Durée de vitesse nulle	REAL	Cet attribut définit la durée pendant laquelle la vitesse d'axe doit être inférieure au seuil de vitesse nulle, défini par l'attribut Vitesse nulle ou établi par le fournisseur du dispositif d'entraînement, avant de pouvoir satisfaire aux critères de vitesse nulle. Dans tous les cas d'arrêts, sauf ceux de Catégorie 2, ceci donne lieu à l'engagement du frein mécanique. Si cet attribut n'est pas pris en charge, la durée nécessaire pour satisfaire aux critères de vitesse nulle est laissée à la discrétion du fournisseur, elle est généralement immédiate (0). La vitesse d'axe décrite ci-dessus est fonction du signal de Réaction de vitesse, ou dans le cas d'un dispositif d'entraînement à Contrôle de fréquence, la vitesse d'axe est fonction du signal de Référence de vitesse.	secondes

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
590	Facultatif – D	Set		Configuration de contrôle	USINT	Cet attribut active les fonctions Contrôle du couple et Contrôle du frein du dispositif d'entraînement fonctionnant conjointement avec la commande de frein mécanique. Lorsque le Contrôle est activé, le frein mécanique doit être établi dès que le dispositif d'entraînement est désactivé. Lorsque le frein est sous le contrôle du diagramme d'états d'axe, il est appliqué automatiquement. Lorsque le frein est contrôlé de manière externe, l'échec à l'établissement du frein lorsque le dispositif d'entraînement est désactivé peut provoquer une condition de chute libre sur une application verticale. Lorsqu'il est activé, le dispositif d'entraînement effectue un essai de Contrôle du couple du courant du moteur à l'état Démarrage afin de « contrôler » que le courant s'écoule correctement dans chacune des phases du moteur avant la libération du frein. En cas d'échec à l'essai de Contrôle du couple, une exception Perte de phase du moteur est générée. Lorsque la fonctionnalité Contrôle du couple est applicable aux Modes de commande du dispositif d'entraînement qui ne peuvent pas générer de couple de maintien fiable en fonction d'un dispositif de réaction, par exemple, Contrôle de fréquence et Commande de vitesse sans capteur, il convient de ne pas l'utiliser dans des applications pour lesquelles le couple de maintien est critique en fonctionnement de sécurité comme par exemple dans une application type de levage ou de grue. Si l'attribut Couple d'essai du frein facultatif est pris en charge, l'essai de Contrôle du couple comprend également un Essai du frein proactif permettant de s'assurer que le frein mécanique fonctionne correctement. Si l'Essai du frein détecte un glissement du frein, une exception Glissement du frein est générée. Lorsque le Contrôle est activé, le dispositif d'entraînement effectue également un essai de Contrôle du frein à l'état Arrêt ou Abandon afin de « contrôler » le bon fonctionnement du frein mécanique avant de désactiver la structure de puissance du dispositif d'entraînement. Si l'Essai de Contrôle du frein détecte un glissement du frein, une exception Glissement du frein est générée. À moins qu'une autre méthode spécifique au fournisseur ne soit utilisée pour traiter une condition de Glissement du frein à l'état Arrêt ou Abandon, l'Action Défaut appropriée pour l'exception Glissement du frein est Arrêt et Maintien de couple limité. Cette Action Défaut applique le couple de maintien pour arrêter le glissement du frein et fait passer l'axe à l'état Défaut principal. En général, la fonctionnalité Contrôle du frein n'est applicable qu'aux Modes de commande du dispositif d'entraînement qui peuvent générer le couple de maintien en fonction d'un dispositif de réaction. Le Contrôle du frein n'est par conséquent pas applicable au Contrôle de fréquence ou aux modes de Commande de vitesse sans capteur. Le séquençement des essais de « contrôle » de couple et de frein est décrit en détail dans les sections ci-après relatives à la sémantique des attributs Délai d'engagement de frein mécanique et Délai de libération de frein mécanique.	Énumération: 0 = Désactivé 1 = Activé 2 à 255 = (réservé)

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
591	Facultatif – D	Set		Courant de Contrôle du couple	REAL	Cet attribut définit le pourcentage de courant assigné du moteur appliqué au moteur par l'essai de Contrôle du couple dans le cadre de la fonction Contrôle du couple exécutée à l'état Démarrage. L'essai de Contrôle du couple applique le courant au moteur afin de « contrôler » qu'il s'écoule correctement dans chacune des phases du moteur avant la libération du frein.	% de la valeur assignée du moteur
592	Facultatif – D	Set		Couple d'essai du frein	REAL	Cet attribut définit le pourcentage de couple assigné du moteur appliqué au moteur par l'Essai du frein dans le cadre de la fonction Contrôle du couple exécutée à l'état Démarrage. L'Essai du frein vérifie de façon proactive l'aptitude du frein mécanique à maintenir la charge anticipée maximale avant de libérer le frein et permettre le fonctionnement. Si l'Essai du frein détecte un glissement du frein, une exception Glissement du frein est générée. Si la valeur de l'attribut Couple d'essai du frein est 0, l'Essai du frein ne doit pas être effectué à l'état Démarrage.	% de la valeur assignée du moteur
593	Facultatif – D E	Set		Durée de rampe de contrôle du frein	REAL	Cet attribut détermine la durée dont le dispositif d'entraînement doit disposer pour diminuer le couple appliqué du moteur jusqu'à zéro pendant l'essai de Contrôle du frein à l'état Arrêt ou Abandon. La Durée de rampe de contrôle du frein détermine la décélération de rampe de la sortie du couple appliqué en divisant la Limite de couple par la Durée de rampe de contrôle du frein. Dans ce cas, la Limite de couple correspond à la valeur maximale des valeurs configurées de Limite de couple positive et de Limite de couple négative. L'Essai de Contrôle du frein est effectué afin de vérifier le glissement du frein avant de désactiver la structure de puissance.	secondes
594	Facultatif – D E	Set		Tolérance de glissement du frein	REAL	Cet attribut détermine la limite admise de glissement du frein après l'engagement du frein. Si cette tolérance est dépassée lorsque le frein est engagé, une exception Glissement du frein est générée. Le Glissement du frein peut par conséquent être contrôlé dans tout état d'axe lorsque le frein est engagé.	Unités d'asservissement de position

7.3.10.2 Sémantique

7.3.10.2.1 Attribut N° 610 – Action d'arrêt

Lors de la désactivation ou de l'abandon d'un axe, cet attribut détermine la méthode d'arrêt à appliquer au moteur (voir Tableau 87). L'état final appliqué à l'issue de la méthode d'arrêt est Arrêté. À l'état Arrêté, la structure de puissance de l'onduleur du dispositif doit être Désactivée (sélection Désactivé) et sans couple ou maintenue active (Sélection Maintien) dans une condition statique. Cet attribut n'a aucun impact sur les profils d'accélération et de décélération générés par le planificateur ni ne leur est lié.

Tableau 87 – Définitions de l'énumération Action d'arrêt

Énum.	Obligatoire/ Facultatif	Nom	Description
0	O/D	Désactivé & arrêt progressif	Désactivé & arrêt progressif désactive immédiatement la structure de puissance du dispositif et les boucles d'asservissement actives, provoquant la mise en arrêt progressif du moteur, sauf si une certaine forme de freinage externe est appliquée. Ceci équivaut à l'arrêt de catégorie 0 de l'IEC 60204-1.
1	O/C F/F	Décél. de courant & Désactiver	Décél. de courant & Désactiver maintient activées la structure de puissance et les boucles d'asservissement actives en cours d'arrêt. Lorsqu'il est configuré pour le mode d'asservissement de position, le dispositif d'entraînement force la référence de position afin de maintenir sa valeur actuelle jusqu'à ce que la vitesse d'axe soit nulle. Une fois la vitesse nulle atteinte, la référence de position est immédiatement définie comme égale à la position réelle afin de maintenir l'axe à l'arrêt. En mode de commande de vitesse, le dispositif d'entraînement force la référence de vitesse à une valeur nulle. Dans chaque cas, forcer les signaux de référence de position ou de vitesse à une valeur fixe entraîne une augmentation rapide de la valeur de référence de couple d'axe mobile, qui sature le couple de production de courant du dispositif d'entraînement sur le couple d'arrêt configuré qui conduit à l'arrêt du moteur. Une fois arrêté, ou à l'expiration de la limite de temps d'arrêt configurée, le dispositif d'entraînement désactive la structure de puissance et les boucles d'asservissement. Ce mode d'arrêt est conforme à la catégorie 1 Arrêt de l'IEC 60204-1. En mode de contrôle de fréquence, l'attribut Limite de vecteur de courant, en non l'attribut Couple d'arrêt, sert à réguler le courant d'arrêt.
2	F/FV	Décél. rampée & Désactiver	Décél. rampée & Désactiver maintient également activées la structure de puissance du dispositif et les boucles d'asservissement actives au cours d'arrêt, mais utilise l'interrupteur de signal associé au bloc Générateur fin de consigne de vitesse pour décélérer le moteur jusqu'à son arrêt. Lors du déclenchement d'un arrêt Décél. rampée & Désactiver, l'interrupteur de signal est activé immédiatement et le dispositif d'entraînement ne suit plus la consigne du contrôleur. L'entrée de l'interrupteur de signal est mise à zéro et la sortie est initialisée sur la vitesse actuelle du moteur, conduisant ainsi la sortie de l'interrupteur à réduire la vitesse actuelle du moteur à une vitesse nulle selon les paramètres de commande de l'interrupteur. Une fois arrêté, ou à l'expiration de la limite de temps d'arrêt ou de temporisation définie en usine configurée, le dispositif désactive la structure de puissance et les boucles d'asservissement. Ce mode d'arrêt est également conforme à la catégorie 1 Arrêt de l'IEC 60204-1.
3	F/PV	Décél. de courant & Maintenir	Décél. de courant & Maintenir se comporte comme Décél. de courant & Désactiver, mais laisse la structure de puissance active avec le couple de maintien afin de maintenir la condition d'arrêt. La méthode de génération du couple de maintien est laissée à la discrétion du fournisseur du dispositif d'entraînement. Ce mode d'arrêt est conforme à la catégorie 2 Arrêt de l'IEC 60204-1.

Énum.	Obligatoire/ Facultatif	Nom	Description
4	F/PV	Décél. rampée & Maintenir	Décél. rampée & Maintenir se comporte comme Décél. rampée & Désactiver, mais laisse la structure de puissance active avec le couple de maintien afin de maintenir la condition d'arrêt. La méthode de génération du couple de maintien est laissée à la discrétion du fournisseur du dispositif d'entraînement. Ce mode d'arrêt est également conforme à la catégorie 2 Arrêt de l'IEC 60204-1.
5-127		(Réservé)	-
128-255		(Spécifique au fournisseur)	-

7.3.10.2.2 Attribut N° 613 – Délai de contact de frein résistif

Lorsqu'un frein résistif externe est utilisé, un contacteur externe commute les cosses du moteur ABC de la structure de puissance de l'onduleur vers une résistance de dissipation d'énergie pour arrêter le moteur. Cette commutation ne se produit pas instantanément, et l'activation prématurée de la structure de puissance peut donc provoquer un arc électrique dans le contacteur. Pour éviter cette condition, le Délai de contact de frein résistif peut être défini comme étant la durée maximale de fermeture totale du contacteur dans les lignes de moteur ABC de sorte que, lors de l'activation d'axe, la structure de puissance de l'onduleur n'est pas activée tant que le Délai de contact de frein résistif n'est pas activé. L'opération de freinage résistif s'applique uniquement aux types de moteur à aimant permanent. La séquence ci-dessous définit plus en détail les facteurs Délai de contact de frein résistif dans la séquence d'activation globale qui peut également inclure le fonctionnement d'un frein mécanique.

Séquence d'activation:

- 1) Passer à l'état Démarrage.
- 2) Activer le contacteur de frein résistif pour connecter le moteur à la structure de puissance de l'onduleur.
- 3) Attendre le "Délai de contact de frein résistif" pendant la fermeture des contacts de frein résistif.
- 4) Activer la structure de puissance de l'onduleur.
- 5) Activer la sortie Frein mécanique pour libérer le frein.
- 6) Attendre le "Délai de libération de frein mécanique" lors de la détente du frein.
- 7) Passer à l'état Exécution.

7.3.10.2.3 Attribut N° 615 – Délai de libération de frein mécanique

Lors de l'activation de l'axe avec un frein mécanique engagé, la valeur Délai de libération de frein mécanique détermine la durée à laquelle le dispositif d'entraînement doit retarder le passage de l'état Démarrage à l'état Exécution ou Essais. Ce délai empêche tout mouvement commandé d'axe de mouvement pour laisser le temps au frein mécanique externe de se désengager. S'il est pris en charge, une opération de Contrôle du couple est incluse dans cette séquence avant la libération du frein.

Séquence d'activation:

- 1) Passer à l'état Démarrage.
- 2) Activer le contacteur de frein résistif pour connecter le moteur à la structure de puissance de l'onduleur.
- 3) Attendre le "Délai de contact de frein résistif" pendant la fermeture des contacts de frein résistif.
- 4) Activer la structure de puissance de l'onduleur.

- 5) Effectuer (facultatif) l'opération de Contrôle du couple pour vérifier la commande de charge du moteur.
- 6) Activer la sortie Frein mécanique pour libérer le frein.
- 7) Attendre le "Délai de libération de frein mécanique" lors de la détente du frein.
- 8) Passer à l'état Exécution (ou Essais).

7.3.10.2.4 Attribut N° 616 – Délai d'engagement de frein mécanique

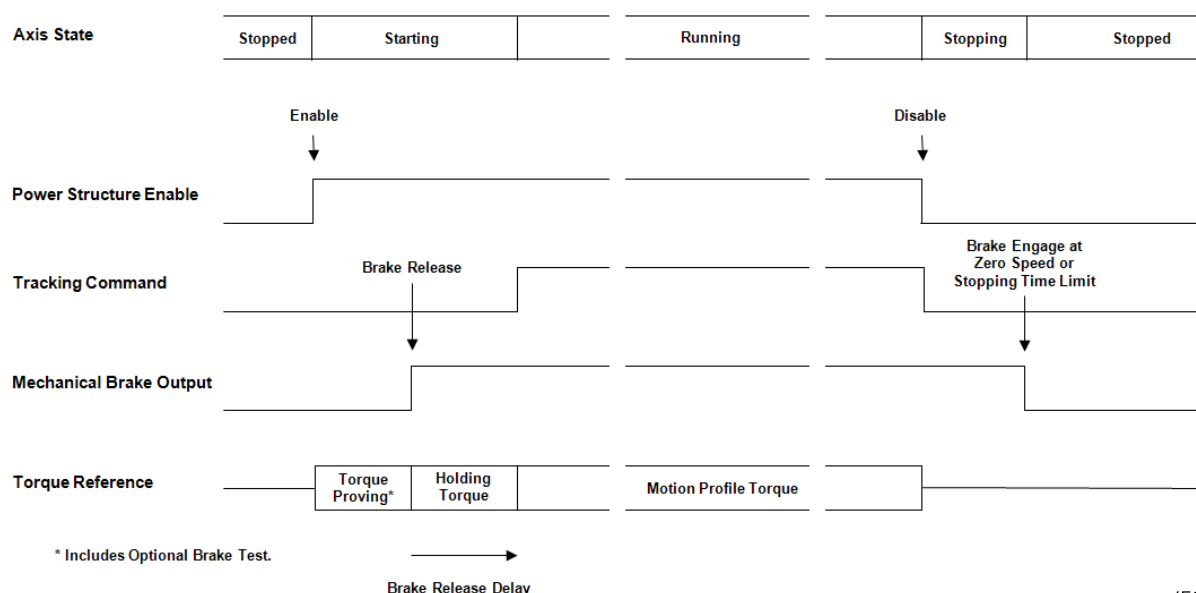
7.3.10.2.4.1 Généralités

Lors de la désactivation de l'axe de mouvement au moyen d'une Action d'arrêt de catégorie 1, la valeur Délai d'engagement de frein mécanique détermine pendant combien de temps la structure de puissance de l'onduleur doit rester activée après la décélération d'axe jusqu'à son arrêt. Cela laisse le temps à un frein mécanique externe de s'engager. L'action d'arrêt configurée détermine le type de séquence d'arrêt appliqué. S'il est pris en charge, une opération de Contrôle du frein est incluse dans la séquence d'arrêt de Catégorie 1 avant la désactivation de la structure de puissance.

7.3.10.2.4.2 Séquence de désactivation (Arrêt de catégorie 0):

L'onduleur est immédiatement désactivé. Le Contrôle du frein n'est pas applicable. La séquence correspondante est spécifiée ci-après et représentée à la Figure 65.

- 1) Passer à l'état Arrêt.
- 2) Désactiver la structure de puissance de l'onduleur.
- 3) Attendre que la vitesse soit nulle (voir 7.3.10.2.4.5), la "Limite de temps d'arrêt" ou une temporisation définie en usine, selon l'option qui se produit en premier.
- 4) Passer à l'état Arrêté.
- 5) Désactiver la sortie Frein mécanique pour engager le frein.
- 6) Désactiver le contacteur de frein résistif pour déconnecter le moteur de la structure de puissance de l'onduleur.



IEC

Anglais	Français
Axis State	État d'axe

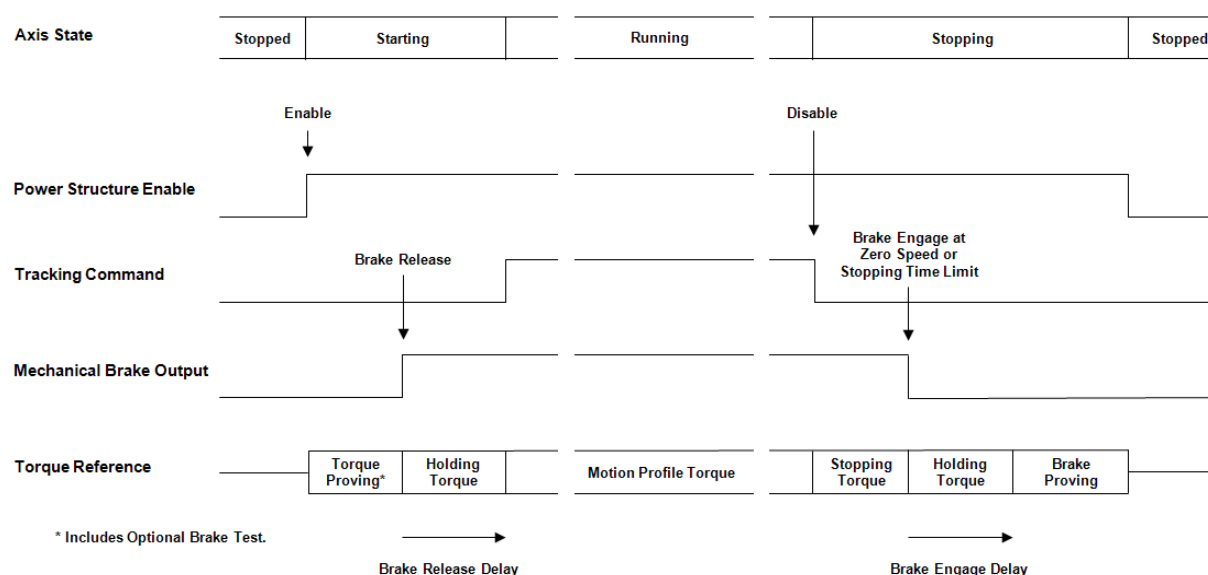
Anglais	Français
Stopped	Arrêté
Starting	Démarrage
Running	Exécution
Stopping	Arrêt
Enable	Activer
Disable	Désactiver
Power Structure Enable	Activer la structure de puissance
Tracking Command	Commande de suivi
Brake release	Libération du frein
Brake engage at Zero Speed or Stopping Time Limit	Engagement du frein à vitesse nulle ou limite de temps d'arrêt
Mechanical Brake Output	Sortie Frein mécanique
Torque Reference	Référence de couple
Torque Proving	Contrôle du couple
Holding Torque	Couple de maintien
Motion Profile Torque	Couple de profil de mouvement
Includes Optional Brake Test	Comprend l'essai du frein facultatif
Brake Release Delay	Délai de libération du frein

Figure 65 – Séquence de commande de frein (Arrêt de catégorie 0)

7.3.10.2.4.3 Séquence de désactivation (Arrêt de catégorie 1):

Le couple est appliqué pour arrêter le moteur. Le Contrôle du frein est applicable. La séquence correspondante est spécifiée ci-après et représentée à la Figure 66.

- 1) Passer à l'état Arrêt.
- 2) Appliquer la méthode "Décél. de courant" ou "Décél rampée" pour arrêter le moteur.
- 3) Attendre que la vitesse soit nulle (voir 7.3.10.2.4.5), la "Limite de temps d'arrêt" ou une temporisation définie en usine, selon l'option qui se produit en premier.
- 4) Désactiver la sortie Frein mécanique pour engager le frein.
- 5) Attendre le "Délai d'engagement de frein mécanique" lors de l'engagement du frein.
- 6) Effectuer (facultatif) l'opération de Contrôle du frein pour vérifier la commande de charge du frein.
- 7) Désactiver la structure de puissance de l'onduleur.
- 8) Passer à l'état Arrêté.
- 9) Désactiver le contacteur de frein résistif pour déconnecter le moteur de la structure de puissance de l'onduleur.



IEC

Anglais	français
Axis State	État d'axe
Stopped	Arrêté
Starting	Démarrage
Running	Exécution
stopping	Arrêt
Enable	Activer
Disable	Désactiver
Power Structure Enable	Activer la structure de puissance
Tracking Command	Commande de suivi
Brake Release	Libération du frein
Brake Engage at Zero Speed or Stopping Time Limit	Engagement du frein à vitesse nulle ou limite de temps d'arrêt
Mechanical Brake Output	Sortie Frein mécanique
Torque Reference	Référence de couple
Torque Proving	Contrôle du couple
Holding Torque	Couple de maintien
Motion Profile Torque	Couple de profil de mouvement
Stopping Torque	Couple d'arrêt
Brake Proving	Contrôle du frein
Includes Optional Brake Test	Comprend l'essai du frein facultatif
Brake Release Delay	Délai de libération du frein
Brake Engage Delay	Délai d'engagement du frein

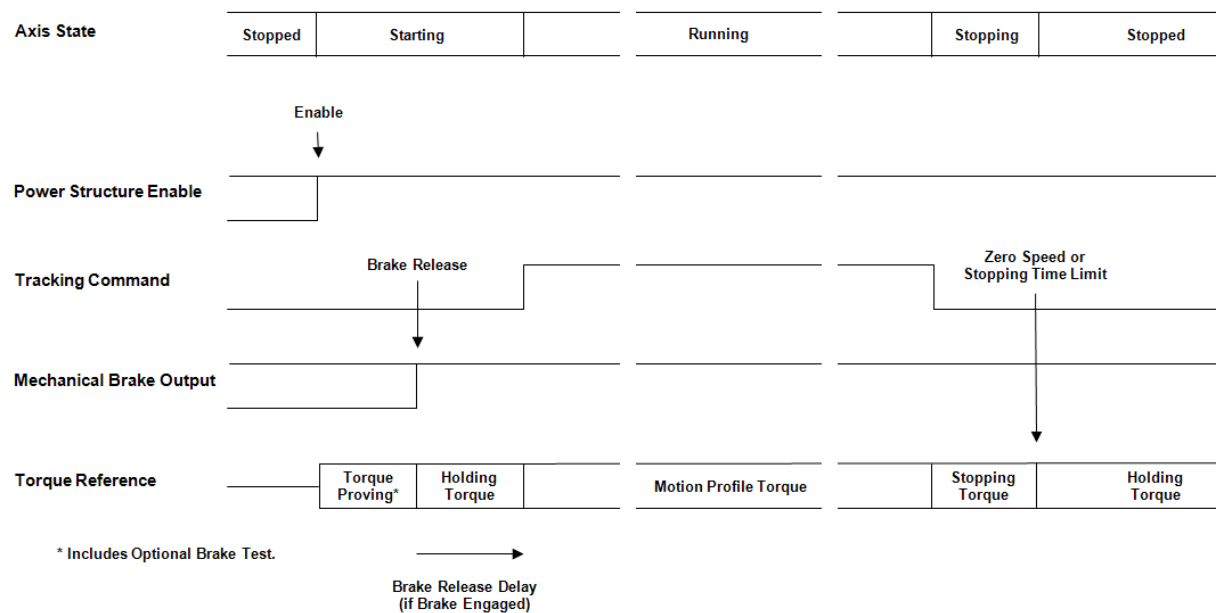
Figure 66 – Séquence de commande de frein (Arrêt de catégorie 1)

7.3.10.2.4.4 Séquence de désactivation (Arrêt de catégorie 2):

Le frein mécanique n'est pas utilisé. Le Contrôle du frein n'est pas applicable. La séquence correspondante est spécifiée ci-après et représentée à la Figure 67.

- 1) Passer à l'état Arrêt.
- 2) Appliquer la méthode "Décél. de courant" ou "Décél rampée" pour arrêter le moteur.

- 3) Attendre que la vitesse soit nulle (voir 7.3.10.2.4.5), la "Limite de temps d'arrêt" ou une temporisation définie en usine, selon l'option qui se produit en premier.
- 4) Passer à l'état Arrêté.



IEC

Anglais	Français
Axis State	État d'axe
Stopped	Arrêté
Starting	Démarrage
Running	Exécution
stopping	Arrêt
Enable	Activer
Power Structure Enable	Activer la structure de puissance
Tracking Command	Commande de suivi
Brake Release	Libération du frein
Zero Speed or Stopping Time Limit	Vitesse nulle ou limite de temps d'arrêt
Mechanical Brake Output	Sortie Frein mécanique
Torque Reference	Référence de couple
Torque Proving	Contrôle du couple
Holding Torque	Couple de maintien
Motion Profile Torque	Couple de profil de mouvement
Stopping Torque	Couple d'arrêt
Includes Optional Brake Test	Comprend l'essai du frein facultatif
Brake Release Delay (if Brake Engaged)	Délai de libération du frein (si Frein engagé)

Figure 67 – Séquence de commande de frein (Arrêt de catégorie 2)

7.3.10.2.4.5 Critères de vitesse nulle

Les critères recommandés pour l'obtention d'une vitesse nulle sont basés sur le retour en vitesse, ou dans le cas d'un dispositif d'entraînement à contrôle de fréquence, sont basés sur

la Référence de vitesse. Les critères de vitesse nulle peuvent être établis de façon explicite par l'intermédiaire des attributs facultatifs Vitesse nulle et Durée de vitesse nulle, ou de façon implicite à 1 % de la vitesse assignée du moteur ou encore laissés à la discrétion du fournisseur du dispositif d'entraînement.

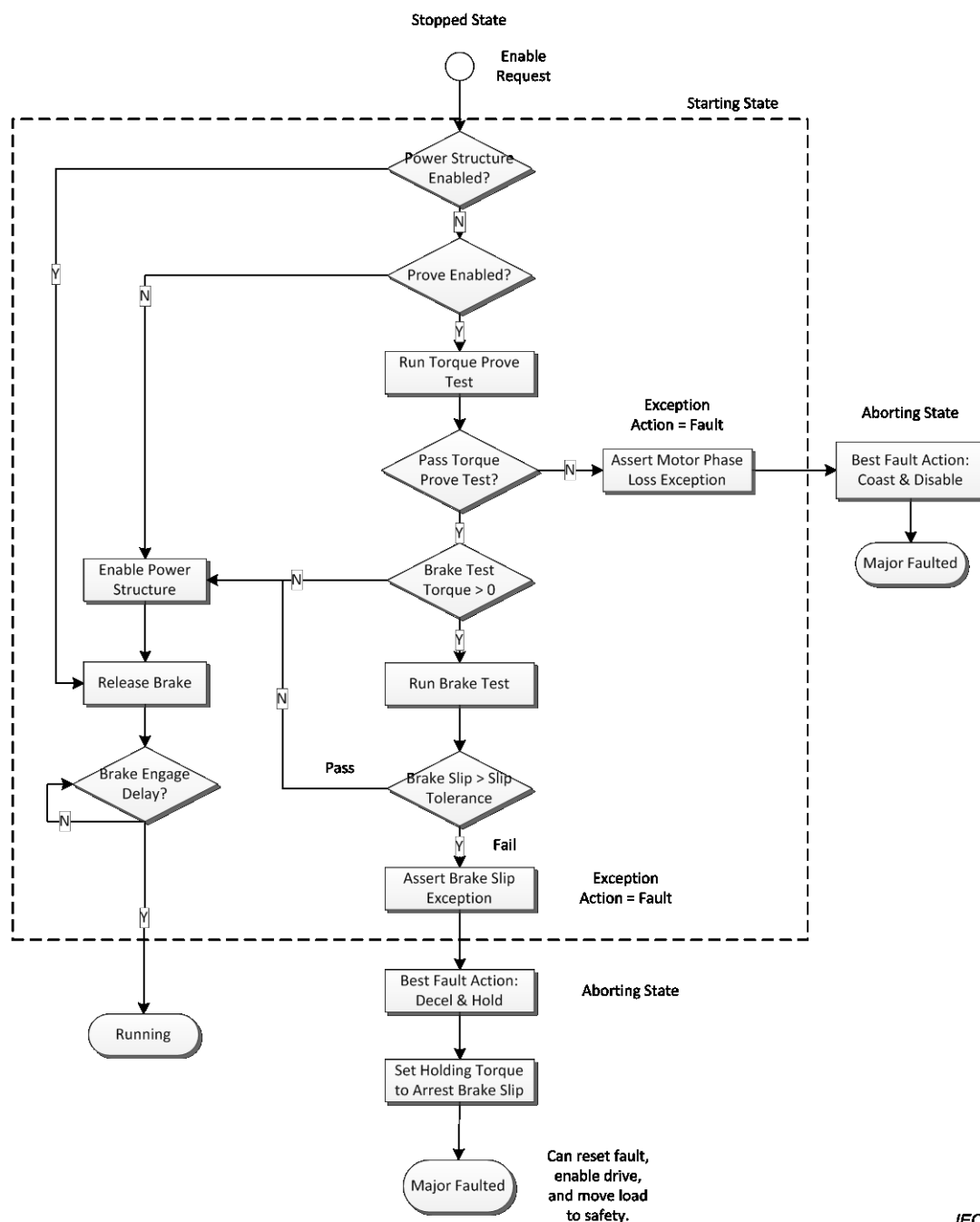
7.3.10.2.5 Attribut N° 590 – Configuration de contrôle

La propriété de Contrôle comprend un certain nombre de sous-propriétés facultatives dont la plupart dépendent de la prise en charge d'autres attributs de propriété de Contrôle. Le Tableau 88 définit ces dépendances d'attributs.

Tableau 88 – Dépendances d'attributs de sous-propriété de contrôle

Sous-propriété de contrôle	Attributs de contrôle	Conditions préalables d'attribut
Contrôle du couple		Configuration de contrôle
Essai du frein	Couple d'essai du frein Tolérance de glissement du frein	Configuration de contrôle
Contrôle du frein	Durée de rampe de contrôle du frein Tolérance de glissement du frein	Configuration de contrôle

Les essais de contrôle sont effectués à l'activation ou à la désactivation de l'axe du dispositif d'entraînement. Au cours de ces passages d'états, le dispositif d'entraînement effectue une série d'opérations afin de s'assurer du bon fonctionnement du moteur (Contrôle du couple) et du frein (Contrôle du frein). Les organigrammes de la Figure 68 et de la Figure 69 définissent ces séquences d'opérations dans le contexte d'une activation du dispositif d'entraînement et d'une désactivation du dispositif d'entraînement ou d'un abandon.

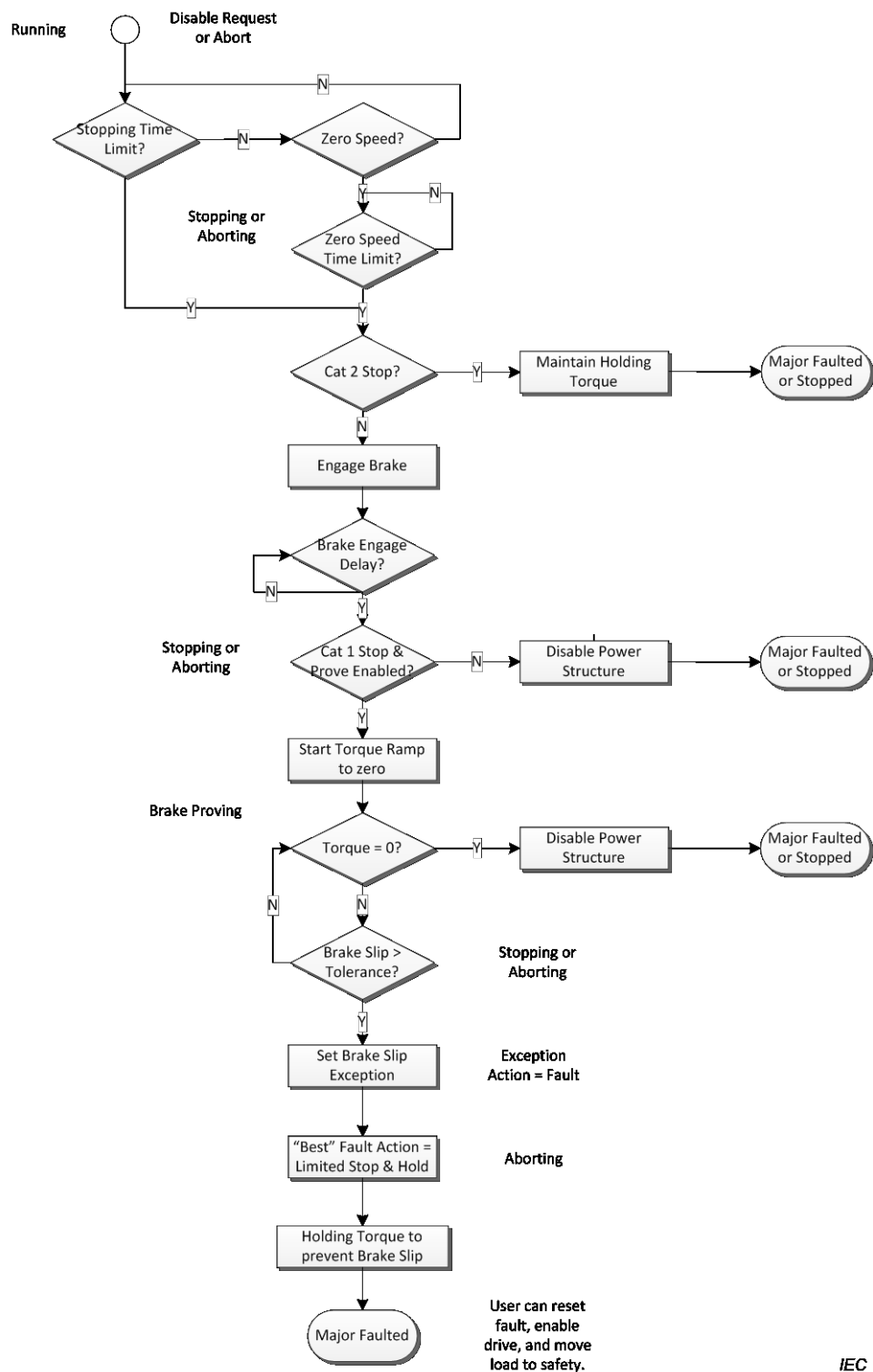


IEC

Anglais	Français
Stopped State	État Arrêté
Enable Request	Activer la demande
Starting State	État Démarrage
Power Structure Enabled?	Structure de puissance activée?
N	Non
Y	Oui
Prove Enabled?	Contrôle activé?
Run Torque Prove Test	Exécuter l'essai de contrôle du couple
Exception Action = fault	Action d'exception = défaut
Aborting State	État Abandon

Anglais	Français
Pass Torque Prove Test?	Essai de contrôle du couple réussi?
Assert Motor Phase Loss Exception	Déterminer l'exception Perte de phase du moteur
Best Fault Action: Const & Disable	Meilleure action de défaut: Const & Désactiver
Enable Power Structure	Activer la structure de puissance
Brake Test Torque > 0	Couple d'essai du frein > 0
Major Faulted	Défaut principal
Release Brake	Libérer le frein
Run Brake Test	Exécuter l'essai du frein
Brake Engage Delay?	Délai d'engagement du frein?
Brake Slip > Slip Tolerance	Glissement du frein > tolérance de glissement
Assert Brake Slip Exception	Déterminer l'exception Glissement du frein
Best Fault Action: Decel & Hold	Meilleure action de défaut: Décél & Maintenir
Aborting State	État Abandon
Running	Exécution
Set Holding Torque to Arrest Brake Slip	Définir le couple de maintien pour arrêter le glissement du frein
Can reset fault, enable drive, and move load to safety	Peut réinitialiser le défaut, activer le dispositif d'entraînement, et déplacer la charge en sécurité

Figure 68 – Séquence d'activation du dispositif d'entraînement avec propriété de Contrôle



IEC

Anglais	Français
Running	Exécution
Disable Request or Abort	Désactiver la demande ou abandonner
Stopping Time Limit?	Limite de temps d'arrêt?
Zero Speed	Vitesse nulle
N	Non
Y	Oui

Anglais	Français
Stopping or Aborting	Arrêt ou abandon
Zero Speed Time Limit?	Limite de durée de vitesse nulle
Cat 2 Stop?	Arrêt de catégorie 2?
Maintain Holding Torque	Maintenir le couple de maintien
Major Faulted or Stopped	Défaut principal ou arrêté
Engage Brake	Engager le frein
Brake Engage Delay?	Délai d'engagement du frein?
Cat 1 Stop & Prove Enabled?	Arrêt de catégorie 1 & contrôle activés?
Disable Power Structure	Désactiver la structure de puissance
Start Torque Ramp to zero	Démarrer la rampe de couple jusqu'à zéro
Brake Proving	Contrôle du frein
Torque = 0?	Couple = 0?
Brake Slip > Tolerance?	Glissement du frein > tolérance?
Set Brake Slip Exception	Définir l'exception Glissement du frein
Exception Action = Fault	Action d'exception = Défaut
Best fault action: Limited Stop & Hold	Meilleure action de défaut: Arrêt limité & Maintenir
Aborting	Abandon
Holding Torque to prevent Brake Slip	Couple de maintien pour empêcher le glissement du frein
User can reset fault, enable drive, and move load to safety	L'utilisateur peut réinitialiser le défaut, activer le dispositif d'entraînement, et déplacer la charge en sécurité

Figure 69 – Séquence de désactivation du dispositif d'entraînement avec propriété de Contrôle

7.3.11 Attributs Commande de bus à courant continu

Le Tableau 89 contient les attributs d'instance Objet Axe de dispositif de mouvement associés à la commande de bus à courant continu, comprenant la fonctionnalité visant à adresser les conditions de sous-tension et de surtension.

Tableau 89 – Attributs Commande de bus à courant continu

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
620	Obligatoire – BD	Get		Tension de bus à courant continu	REAL	Tension de bus à courant continu mesurée.	Volts
621	Obligatoire – BD	Get		Tension de bus à courant continu nominale	REAL	Tension de bus à courant continu normale en fonctionnement, déterminée par la moyenne de tension de bus à courant continu sur un intervalle de temps spécifique du dispositif. Cette valeur est la base des limites de surtension et de sous-tension du bus. NOTE Si le dispositif ne prend pas en charge ce concept de moyenne de tension de bus, une autre possibilité consiste à coder cette valeur.	Volts
622	Facultatif – BD	Set		Configuration de bus	USINT	Cette sélection énumérée indique comment le bus à courant continu va être utilisé. Autonome indique que l'alimentation de bus à courant continu fournie par l'onduleur du dispositif d'entraînement est appliquée uniquement à cette structure de puissance du dispositif. AC/DC partagé indique que l'onduleur associé à ce dispositif CIP Motion est tenu de fournir et de partager l'alimentation de bus à courant continu avec les autres dispositifs d'entraînement. Ceci entraînerait généralement un déclassement des caractéristiques assignées du courant continu pour le dispositif d'entraînement AC/DC partagé. DC partagé indique que ce dispositif d'entraînement partage l'alimentation de bus à courant continu générée par un autre dispositif d'entraînement AC/DC partagé. DC partagé – Convertisseur Non CIP indique que ce dispositif d'entraînement reçoit l'alimentation de bus à courant continu générée par un convertisseur AC/DC externe qui n'est pas conforme à CIP Motion et distribue son alimentation de bus à courant continu à d'autres dispositifs d'entraînement CIP Motion. Un dispositif d'entraînement configuré pour DC partagé – Convertisseur Non CIP est chargé de communiquer le statut du convertisseur externe au système de commande comme si le convertisseur externe était intégré au dispositif d'entraînement. Cette communication comprend tout particulièrement les bits d'état Bus activé et Décharger le bus à courant continu représentant l'état en cours du convertisseur externe associé.	Énumération: 0 = Autonome 1 = AC/DC partagé 2 = DC partagé 3 = DC partagé – Convertisseur Non CIP 4 à 255 = (réservé)

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
623	Facultatif – D	Set		Sélection de tension de bus	USINT	Cette valeur indique le niveau de tension de bus prévu de l'application du dispositif d'entraînement. La sélection de tension de bus élevée est habituellement associée avec un dispositif d'entraînement fonctionnant sur le réseau d'alimentation nord-américain, tandis qu'une sélection de tension de bus faible serait appropriée pour un fonctionnement sur le réseau européen. Ce paramètre peut être utilisé pour compenser ces différents niveaux de tension de bus dans la boucle de courant.	Énumération: 0 = Élevée (115 V, 230 V, 460 V) 1 = Faible (100 V, 200 V, 400 V) 2 à 255 = (réservé)
624	Obligatoire – BD (Convertisseur intégral)	Set		Action du régulateur de bus	USINT	Contrôle la méthode de fonctionnement du régulateur de bus à courant continu qui adresse les conditions de surtension régénératives qui peuvent se produire lors de la décélération d'un moteur. Si Désactivé est sélectionné, aucune régulation n'est appliquée au niveau du bus à courant continu pour contrôler l'énergie régénérative émise par le moteur. Cette sélection est valable uniquement lorsqu'elle est appliquée à un moteur à induction où le moteur peut être autorisé à s'arrêter progressivement. Si Régulateur de shunt est sélectionné, le matériel de régulation de shunt associé est appliqué au bus à courant continu afin de dissiper l'énergie régénérative par l'intermédiaire d'une résistance interne ou externe.	Énumération: 0 = Désactivé (O) 1 = Régulateur de shunt (O) 3 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)
625	Facultatif – D (Convertisseur intégral)	Set		Limite de force régénératrice	REAL	Limite la quantité d'alimentation admise pour le transfert entre le moteur et le bus à courant continu au cours d'un freinage par récupération de la charge du moteur. En cas d'utilisation d'une résistance de shunt externe, attribuer la valeur maximale. Dans la mesure où il s'agit d'une force régénératrice, la valeur de la limite est négative.	% de la valeur assignée du moteur

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
627	Facultatif – D	Set		Action de perte d'alimentation	USINT	Définit la réaction à une condition de sous-tension du bus à courant continu lorsque ce dernier chute en dessous d'un seuil incorporé au programme du dispositif ou sous le Seuil de perte de puissance configuré. Ceci fournit une réponse (configurée) spécifique à une perte de puissance à l'arrivée avec le dispositif d'entraînement/moteur en fonctionnement. Une sélection de l'action continuer configure le dispositif d'entraînement de manière à ignorer la condition de perte de puissance et à ce qu'il continue à fonctionner le plus longtemps possible. Une exception de sous-tension de bus peut se produire si la tension de bus à courant continu chute en dessous des limites définies en usine ou des limites de l'utilisateur. Dans le cas contraire, le fonctionnement se poursuit jusqu'à l'épuisement des alimentations basse tension. Le fonctionnement de la structure de puissance en dessous du point où les entraînements à coulisse commencent à libérer de l'énergie peut présenter des problèmes lorsque cette opération peut éventuellement endommager les dispositifs d'entraînement. Les actions d'exception de sous-tension de bus doivent être définies en conséquence. Une sélection de l'action jusqu'à l'arrêt progressif configure le dispositif d'entraînement de manière à que sa sortie PWM ait une valeur nulle, avec l'axe dans l'état Exécution. Lorsque l'alimentation à l'arrivée revient avant l'expiration de la période de temporisation, donnée par le temps de perte d'alimentation, le dispositif d'entraînement commence automatiquement à contrôler de nouveau le moteur. Si, toutefois, l'alimentation ne revient pas avant l'expiration de la période de temporisation de perte d'alimentation, une Exception de perte d'alimentation du bus est générée. La sélection de l'action de régénération de décélération configure le dispositif d'entraînement de manière à charger par régénération le bus à courant continu par une décélération du moteur à l'aide du régulateur de bus afin de réguler la tension de ce dernier à un niveau prédéterminé. L'alimentation à l'arrivée restaurée, le dispositif d'entraînement retrouve son fonctionnement normal. Si, toutefois, le dispositif d'entraînement atteint une vitesse nulle ou si la période de temporisation de perte d'alimentation expire avant la restauration de l'alimentation à l'arrivée, la structure de puissance du dispositif est désactivée et une exception de perte d'alimentation du bus est générée.	Énumération: 0 = Continuer (Ignorer) (O) 1 = Arrêt progressif (O) 2 = Régén de décél (F) 3 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)
628	Facultatif – BD	Set		Seuil de perte de puissance	REAL	Définit le niveau de perte de puissance en pourcentage de la tension de bus à courant continu nominale. Si cette valeur est nulle, le seuil incorporé au programme est utilisé.	%

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
629	Facultatif – BD	Set		Action d'interruption	USINT	Sélectionne l'action du dispositif lors du passage à l'état Interruption suite à une demande d'interruption ou en réponse à une action d'exception où la meilleure action d'arrêt disponible est l'interruption. L'action par défaut consiste à immédiatement désactiver la structure de puissance du dispositif. Lorsque Diminution de la tension en courant continu du bus est sélectionnée, une action peut être entreprise pour diminuer également la tension de bus à courant continu.. Ceci s'effectue généralement par l'ouverture d'une sortie Activer contacteur à courant alternatif du dispositif d'entraînement qui régule l'alimentation du contrôleur.	Énumération: 0 = Désactiver (O) 1 = Diminution de la tension de bus à courant continu (F) 2 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)
630	Facultatif – BD	Set		Temps de perte de puissance	REAL	Définit la valeur de temporisation avant que le dispositif d'entraînement ne génère une exception de perte d'alimentation du bus en réponse à une condition de perte de puissance. Voir Action de perte d'alimentation pour plus de détails.	secondes

7.3.12 Attributs de gestion de l'alimentation et de gestion thermique

Le Tableau 90 et le Tableau 91 contiennent tous les attributs d'alimentation et attributs thermiques associés à une instance Objet Axe de dispositif de mouvement.

Tableau 90 – Attributs État de gestion de l'alimentation et de gestion thermique

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
634	Facultatif – BD	Get		Convertisseur intégral	BOOL	Cette valeur booléenne indique si l'axe est associé à un convertisseur intégré (alimentation) qui fournit l'alimentation du bus à courant continu et la régulation du bus.	0 = Non 1 = Oui
635	Obligatoire – D	Get		Capacité du moteur	REAL	Estimation en temps réel de la capacité thermique assignée continue du moteur utilisée pendant le fonctionnement selon le modèle thermique du moteur. Une valeur de 100 % indique que le moteur est utilisé à 100 % de sa capacité thermique assignée, déterminée par le courant assigné continu du moteur.	% de la valeur assignée du moteur
636	Obligatoire – D	Get		Capacité de l'onduleur	REAL	Estimation en temps réel de la capacité thermique assignée continue de l'onduleur utilisée pendant le fonctionnement selon le modèle thermique de l'onduleur. Une valeur de 100 % indique que l'onduleur est utilisé à 100 % de sa capacité thermique assignée, déterminée par le courant assigné continu de l'onduleur.	% de la valeur assignée de l'onduleur
637	Facultatif – BD (Convertisseur intégral)	Get		Capacité du convertisseur	REAL	Estimation en temps réel de la capacité thermique assignée continue du convertisseur utilisée pendant le fonctionnement selon le modèle thermique du convertisseur. Une valeur de 100 % indique que le convertisseur est utilisé à 100 % de sa capacité thermique assignée, déterminée par le courant assigné continu du convertisseur.	% de la valeur assignée du convertisseur
638	Facultatif – BD (Convertisseur intégral)	Get		Capacité du régulateur de bus	REAL	Estimation en temps réel de la capacité thermique assignée continue du régulateur de bus utilisée pendant le fonctionnement selon le modèle thermique du régulateur de bus. Une valeur de 100 % indique que le régulateur de bus est utilisé à 100 % de sa capacité thermique assignée, déterminée par le courant assigné continu du régulateur de bus.	% de la valeur assignée du régulateur
639	Facultatif – BD	Get		Température ambiante	REAL	Température ambiante interne actuelle du boîtier du dispositif.	°C

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
640	Facultatif – D	Get		Température du dissipateur thermique de l'onduleur	REAL	Température actuelle du dissipateur thermique de l'onduleur du dispositif, mesurée en général par un capteur de température intégré.	°C
641	Facultatif – D	Get		Température de l'onduleur	REAL	Température actuelle du bloc d'alimentation utilisé dans la structure de puissance de l'onduleur, parfois appelée température de jonction du semi-conducteur.	°C
642	Facultatif – D	Get		Température du moteur	REAL	Température actuelle du stator du moteur, mesurée en général par un capteur de température intégré.	°C
643	Facultatif – E	Get		Température de réaction 1	REAL	Température actuelle du dispositif de réaction 1.	°C
644	Facultatif – E	Get		Température de réaction 2	REAL	Température actuelle du dispositif de réaction 2.	°C
645	Facultatif – D	Get		Limite de surcharge de l'onduleur	REAL	Limite maximale définie en usine de la Capacité de l'onduleur qui, si elle est dépassée, déclenche l'action Surcharge de l'onduleur sélectionnée.	% de la valeur assignée de l'onduleur

Tableau 91 – Attributs Configuration de gestion de l'alimentation et de gestion thermique

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
646	Facultatif – D	Set		Action de surcharge du moteur	USINT	Sélectionne la réponse du dispositif à une condition de surcharge du moteur selon un modèle thermique de moteur I2t. La condition de surcharge du moteur se produit lorsque le modèle thermique du moteur indique que la Capacité du moteur a dépassé la Limite de surcharge du moteur. Cette fonctionnalité Action de surcharge du moteur est indépendante de la fonctionnalité d'action d'exception de surcharge du moteur. Aucune action explicite n'est réalisée par le dispositif dans la condition de surcharge si l'option Aucune est l'action de surcharge sélectionnée. Toutefois, en sélectionnant l'action Réaction en cours, le courant de référence est diminué par l'intermédiaire de l'attribut Limite de courant thermique du moteur, dont la valeur est réduite proportionnellement au pourcentage de la différence entre la Capacité du moteur et la Limite de surcharge du moteur.	Énumération: 0 = Aucune (O) 1 = Réaction en cours (F) 2 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)

ID attribut	Besoin dans mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
647	Facultatif – D	Set		Action de surcharge de l'onduleur	USINT	Sélectionne la réponse du dispositif à une condition d'alarme de surcharge de l'onduleur selon un modèle thermique d'onduleur I2t. La condition d'alarme de surcharge de l'onduleur se produit lorsque le modèle thermique de l'onduleur indique que sa Capacité a dépassé la Limite de surcharge de l'onduleur. L'Action de surcharge de l'onduleur offre des possibilités de limiter la condition de surcharge sans arrêter le système. Cette fonctionnalité Action de surcharge de l'onduleur est indépendante de la fonctionnalité d'action d'exception de surcharge du moteur. Aucune action explicite n'est réalisée par le dispositif dans la condition de surcharge si l'option Aucune est l'action de surcharge sélectionnée. Toutefois, en sélectionnant l'action Réaction en cours, le courant de consigne est diminué par l'intermédiaire de l'attribut Limite de courant thermique de l'onduleur, dont la valeur est réduite proportionnellement au pourcentage de la différence entre la Capacité de l'onduleur et la Limite de surcharge de l'onduleur.	Énumération: 0 = Aucune (O) 1 = Réaction en cours (F) 2 à 127 = (réservé) 128 à 255 = (spécifique au fournisseur)

ID attribut	Besoin dans mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
648	Facultatif – D	Set		Choix du service	USINT	Cette valeur indique le niveau de service de l'application du dispositif d'entraînement et équilibre la capacité de surcharge continue et intermittente du dispositif d'entraînement et du moteur en conséquence. Service normal assure des caractéristiques assignées continues nominales et une capacité de surcharge nominale. Grande capacité assure une capacité de surcharge maximale aux dépens d'une puissance assignée continue réduite. Faible capacité assure une puissance assignée continue maximale aux dépens d'une capacité de surcharge réduite. La spécification des caractéristiques assignées continues et de surcharge dans des conditions de capacité normale, de grande capacité et de faible capacité est laissée à la discrétion du fournisseur du dispositif d'entraînement. Choix du service est utilisé pour déterminer le niveau de protection thermique du moteur et de l'onduleur lors du fonctionnement du dispositif d'entraînement.	Énumération: 0 = Normal (O) 1 = Grande capacité (F) 2 = Faible capacité (F) 3 à 255 = (réservé)

7.3.13 Attributs Statut d'axe

7.3.13.1 Généralités

Le Tableau 92 contient tous les attributs de statut associés à une instance Objet Axe de dispositif de mouvement.

Tableau 92 – Attributs Statut d'axe

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
650	Obligatoire – Tous	Get		État d'axe	USINT	Valeur énumérée indiquant l'état d'axe de mouvement.	Énumération: 0 = Initialisation 1 = Précharge 2 = Arrêté 3 = Démarrage 4 = Exécution 5 = Essais 6 = Arrêt 7 = Abandon 8 = Défaut principal 9 = Démarrage inhibé 10 = Interruption 11 à 255: Réservé
651	Obligatoire – Tous	Get		Statut d'axe	DWORD	Ensemble de bits standard indiquant le statut interne d'axe de mouvement.	Voir la sémantique en 7.3.13.2.1.
652	Obligatoire – Tous	Get		Statut d'axe – Mfg	DWORD	Ensemble de bits spécifiques au fournisseur indiquant le statut interne d'axe de mouvement.	Matriciel: 0 à 31: Spécifique au fournisseur (Publié dans le manuel du produit)
653	Obligatoire – Tous	Get		Statut d'E/S d'axe	DWORD	L'attribut Statut d'E/S d'axe est un ensemble de bits de 32 bits indiquant l'état des entrées et sorties numériques normalisées associées au fonctionnement de cet axe de mouvement. Une valeur nulle affectée à un bit d'entrée donné indique une valeur logique nulle, la valeur 1 indiquant une valeur logique 1.	Voir la sémantique en 7.3.13.2.2.
654	Obligatoire – Tous	Get		Statut d'E/S d'axe – Mfg	DWORD	Ensemble de bits indiquant l'état des entrées numériques spécifiques au fournisseur associées au fonctionnement de cet axe de mouvement. Une valeur nulle affectée à un bit d'entrée donné indique une valeur logique nulle, la valeur 1 indiquant une valeur logique 1.	Matriciel: 0 à 31: Spécifique au fournisseur (Publié dans le manuel du produit)

7.3.13.2 Sémantique

7.3.13.2.1 Attribut N° 651 – Statut d'axe

L'attribut Statut d'axe est un ensemble de bits de 32 bits indiquant différentes conditions de statut interne de l'instance d'axe, comme spécifié dans le Tableau 93. Les bits de statut qui ne sont pas applicables doivent être mis à 0.

Tableau 93 – Définitions du bit Statut d'axe

Bit	Obligatoire/ Facultatif	Nom	Description
0	O	Commande locale	Ce bit est défini si l'axe prend une référence de consigne et des services dans l'interface locale au lieu de les prendre dans l'interface (CIP Motion) distante. Ce bit est fondé sur l'état en cours du bit Mode distant de l'attribut Statut de nœud (voir 7.2.2.2).
1	O	Alarme	Ce bit est défini si l'axe a détecté une ou plusieurs conditions d'exception configurées pour générer une alarme. Ce bit est supprimé en l'absence de condition d'alarme d'axe de courant (voir 7.3.13).
2	O/BD	Bus à courant continu activé	Ce bit est défini si le bus à courant continu est chargé jusqu'à un niveau de tension opérationnel fondé sur le mesurage direct et, le cas échéant, le bit Statut de bus de convertisseur activé associé au(x) convertisseur(s) externe(s) fournissant l'alimentation de bus à courant continu à ce dispositif. Lorsqu'un axe est à l'état Précharge, le passage du bit Statut de bus à courant continu activé de 0 à 1 initie le passage à l'état Arrêté (axe de dispositif d'entraînement) ou à l'état Exécution (axe de convertisseur). Une fois défini, le bit Bus à courant continu activé est supprimé lorsque la tension du bus à courant continu a chuté en dessous d'un niveau de tension opérationnel, indépendamment de l'état du bit Statut de bus de convertisseur activé (voir 7.6.2.5).
3	O/D	Structure de puissance activée	Ce bit est défini si l'amplificateur d'axe est sous tension et en mesure de générer un flux et un couple de moteur. La valeur du bit Structure de puissance activée est déterminée par l'État d'axe (voir 7.6.1) et la valeur configurée de l'attribut Action d'arrêt (voir 7.3.10.2.1).
4	O/D	Flux de moteur activé	Ce bit est défini si le flux d'un moteur à induction a atteint un niveau opérationnel. Le passage du bit Flux de moteur activé est initié dans l'État Démarrage (voir 7.6.2.7) selon la valeur configurée de l'attribut Commande de flux activée (voir 7.3.9.5). Ce bit n'est applicable qu'aux types de moteurs à induction.
5	O/D	Commande de suivi	Ce bit est défini si la structure de commande d'axe assure le suivi actif de la référence de consigne à partir d'un planificateur de mouvement. Le bit Commande de suivi est directement associé à l'état Exécution du Modèle d'état d'axe (voir 7.6.2.8).
6	O/P	Verrouillage de position	Ce bit est défini si la position réelle se trouve dans la Tolérance de verrouillage de position de la position de consigne (voir 7.6.9.3)..
7	O/PV	Verrouillage de vitesse	Ce bit est défini si le signal de retour en vitesse se trouve dans la tolérance de verrouillage de vitesse de la référence de vitesse non limitée (voir 7.6.9.5).
8	O/ED	Gel de vitesse	Ce bit est défini si le signal de retour en vitesse se trouve dans la fenêtre de gel de vitesse de 0. Pour un dispositif d'entraînement à contrôle de fréquence, ce bit est défini si le signal de vitesse de référence se trouve dans la fenêtre de gel de vitesse de 0 (voir 7.6.9.5).
9	O/ED	Seuil de vitesse	Ce bit est défini si le signal de retour en vitesse absolue est inférieur au seuil de vitesse. Pour un dispositif d'entraînement à contrôle de fréquence, ce bit est défini si le signal de vitesse absolue de référence est inférieur au seuil de vitesse (voir 7.6.9.5).
10	O/FPV	Limite de vitesse	Ce bit est défini si le signal de référence de vitesse est actuellement limité par le limiteur de vitesse (voir 7.6.9.6.2).
11	O/C	Limite d'accélération	Ce bit est défini si le signal de référence d'accélération est actuellement limité par le limiteur d'accélération (voir 7.6.9.8.2).
12	O/C	Limite de décélération	Ce bit est défini si le signal de référence d'accélération est actuellement limité par le limiteur de décélération (voir 7.6.9.8.2).
13	O/C	Seuil de couple	Ce bit est défini si la référence de couple filtrée absolue est supérieure au Seuil de couple (voir 7.6.9.9).

Bit	Obligatoire/ Facultatif	Nom	Description
14	O/C	Limite de couple	Ce bit est défini si la référence de couple filtrée est actuellement limitée par le limiteur de couple (voir 7.6.9.9.7).
15	O/C	Limite de courant	Ce bit est défini si le courant de consigne, I _q , est actuellement limité par le Limiteur de vecteur de courant (voir 7.6.9.10.2).
16	O/C	Limite thermique	Ce bit est défini si la condition Limite de vecteur de courant d'axe est actuellement limitée par l'un des Modèles thermiques d'axe (voir 7.6.9.10.2).
17	O/E	Intégrité de réaction	Ce bit, lorsqu'il est défini, indique que le dispositif de réaction reflète avec précision les changements de position d'axe, et qu'aucune condition n'a été détectée qui serait susceptible de compromettre la qualité de la valeur de position de réaction. Le bit est défini à la mise sous tension en supposant que le dispositif de réaction satisfait à tout essai autonome de mise sous tension exigé. Si une exception de réaction se produit en cours de fonctionnement qui peut influencer sur la fidélité de position d'axe, le bit est annulé immédiatement. Le bit reste supprimé jusqu'à l'exécution d'une réinitialisation suite à un défaut par le dispositif d'entraînement ou jusqu'à ce que ce dernier soit soumis à un cycle d'alimentation. Noter que le comportement du bit Intégrité de réaction s'applique à un fonctionnement absolu et incrémentiel du dispositif de réaction.
18	O/BD	Interruption	Ce bit est défini lorsque l'axe est à l'état Interruption ou Défaut, mais passe à l'état d'interruption lorsque les défauts sont supprimés. Par conséquent, le bit Interruption est étroitement associé à l'État d'interruption du Modèle d'état d'axe (voir 7.6.2.14).
19	O	En cours de processus	Ce bit est défini pour toute la durée d'un processus actif. Un exemple de processus actif est une opération lancée par une demande de service telle que Run_Motor_Test, Run_Hookup_Test, ou Run_Inertia_Test. Il convient de définir le bit En cours de processus dès le lancement du processus et il convient que ce bit reste défini jusqu'à l'achèvement du processus. Un processus actif qui nécessite l'activation de la structure de puissance de l'axe donne lieu au passage à l'État Essais du Modèle d'état d'axe (voir 7.6.2.9).

Bit	Obligatoire/ Facultatif	Nom	Description
20	F/BD	Décharger le bus à courant continu	Ce bit est défini par un convertisseur CIP Motion ou un dispositif d'entraînement CIP Motion comportant un convertisseur intégral, ou un dispositif d'entraînement CIP Motion connecté à un convertisseur externe Non CIP, afin d'indiquer que le convertisseur ne peut plus fournir une alimentation de bus à courant continu à d'autres dispositifs d'entraînement sur un bus à courant continu partagé. Ceci est généralement le résultat d'une action de défaut d'interruption initiée par le dispositif d'entraînement ou le convertisseur, ou d'une demande d'interruption de la part du contrôleur. Ainsi, lorsque le bit Décharger le bus à courant continu est défini, il convient également de définir le bit Interruption (bit 18). En l'absence de sortie Activer le contacteur à courant alternatif destinée à diminuer la tension de bus à courant continu, une méthode se révèle nécessaire pour décharger le convertisseur de tous les autres dispositifs d'entraînement qui partagent le bus à courant continu. Par un contrôle du bit de statut Décharger le bus à courant continu, le système de commande peut diffuser une exception de partage de l'alimentation du bus à tous les dispositifs d'entraînement sur le Bus à courant continu partagé qui sont configurés pour un fonctionnement AC/DC ou DC partagé. Cette exception Partage d'alimentation du bus appelle l'action d'exception configurée qui, par défaut, désactive la structure de puissance du dispositif d'entraînement, et de ce fait la décharge. Le défaut de partage de l'alimentation du bus sur ces dispositifs d'entraînement partagés est un défaut permanent qui ne peut être supprimé tant que le bit Décharger le bus à courant continu est défini sur ce dispositif d'entraînement ou convertisseur d'origine. Le système de commande doit simplement régénérer les défauts de partage de l'alimentation du bus en fonction du bit de statut Décharger le bus à courant continu toujours défini. Noter que seul le dispositif d'entraînement ou convertisseur d'origine avec la condition Décharger le bus à courant continu peut provoquer des défauts de partage de l'alimentation du bus sur d'autres dispositifs d'entraînement partagés. En d'autres termes, aucun dispositif avec un Défaut de partage de l'alimentation du bus ne peut provoquer une exception Partage de l'alimentation du bus sur d'autres dispositifs d'entraînement partagés en définissant son bit Décharger le bus à courant continu. Cette qualification empêche tout blocage de rétablissement de bus à courant continu. Pour retrouver un fonctionnement optimal de bus à courant continu, le dispositif d'entraînement d'origine avec la condition Décharger le bus à courant continu doit tout d'abord être réinitialisé via une demande de réinitialisation suite à une interruption. Une fois les bits annulés, les défauts de partage de l'alimentation du bus sur les entraînements partagés peuvent alors être supprimés avec succès par l'intermédiaire d'une demande Réinitialisation suite à un défaut ou à une demande Réinitialisation suite à une interruption, permettant à ces entraînements de devenir opérationnels.

Bit	Obligatoire/ Facultatif	Nom	Description
21	F/BD	Perte de puissance en courant alternatif	Ce bit est défini lorsqu'un convertisseur CIP Motion, ou un dispositif d'entraînement CIP Motion comprenant un convertisseur intégral, ou un dispositif d'entraînement CIP Motion connecté à un convertisseur externe Non CIP, a détecté une perte de puissance d'entrée en courant alternatif. Ce bit est supprimé lorsque la puissance d'entrée en courant alternatif est déterminée comme suffisante pour le fonctionnement du convertisseur. Lorsqu'une condition de Perte de puissance en courant alternatif est détectée par un convertisseur fournissant l'alimentation à d'autres dispositifs d'entraînement sur le bus à courant continu partagé, une méthode est nécessaire pour générer une exception Perte de puissance en courant alternatif du convertisseur sur tout dispositif d'entraînement dont la structure de puissance est activée. A cet effet, le système de commande contrôle les bits de statut Perte de puissance en courant alternatif des convertisseurs fournissant l'alimentation de bus à courant continu et diffuse le statut Perte de puissance en courant alternatif à tous les dispositifs d'entraînement sur le Bus à courant continu partagé, c'est-à-dire les dispositifs d'entraînement qui sont configurés pour un fonctionnement AC/DC partagé ou DC partagé. Dès notification de la Perte de puissance en courant alternatif, les dispositifs d'entraînement ayant activé les structures de puissance doivent déterminer une exception Perte de puissance en courant alternatif du convertisseur et appeler l'action d'exception programmée. Les dispositifs d'entraînement désactivés ne doivent pas générer une action d'exception sur la Perte de puissance en courant alternatif. Ainsi, aucun défaut d'entraînement ne doit se produire en cas de suppression de puissance en courant alternatif du convertisseur tant qu'aucune structure de puissance de dispositif d'entraînement alimentée par le convertisseur concerné n'est activée.
22 à 31		Réservé	

Un grand nombre des bits Statut d'axe définis dans le Tableau 93 sont associés à l'État d'axe de courant comme représenté au Tableau 94.

Tableau 94 – Bit Statut d'axe par rapport à l'État d'axe

Bit	Nom du bit Statut d'axe	Initialisation	Précharge	Interruption	Démarrage inhibé	Arrêté	Démarrage	Exécution	Essais	Arrêt	Abandon	Défaut principal
0	Commande locale	x	x	x	x	x	x	x	x	x	x	x
1	Alarme	x	x	x	x	x	x	x	x	x	x	x
2	Bus à courant continu activé	x	0	x	1	1	1	1	1	1	1	1
3	Structure de puissance activée	0	0	0	0	0/1	1	1	1	1	1	0/1
4	Flux de moteur activé (IM uniquement)	0	0	0	0	0/1	1	1	1	1	1	0/1
5	Commande de suivi	0	0	0	0	0	0	1	0	0	0	0
6	Verrouillage de position	0	0	0	0	0/x	x	x	0	x	x	0/x
7	Verrouillage de vitesse	0	0	0	0	0/x	x	x	0	x	x	0/x
8	Gel de vitesse	0	0	0	0	0/x	x	x	0	x	x	0/x
9	Seuil de vitesse	0	0	0	0	0/x	x	x	0	x	x	0/x
10	Limite de vitesse	0	0	0	0	0/x	x	x	0	x	x	0/x
11	Limite d'accélération	0	0	0	0	0/x	x	x	0	x	x	0/x
12	Limite de décélération	0	0	0	0	0/x	x	x	0	x	x	0/x
13	Seuil de couple	0	0	0	0	0/x	x	x	0	x	x	0/x
14	Limite de couple	0	0	0	0	0/x	x	x	0	x	x	0/x
15	Limite de courant	0	0	0	0	0/x	x	x	x	x	x	0/x
16	Limite thermique	x	x	x	x	x	x	x	x	x	x	x
17	Intégrité de réaction	x	x	x	x	x	x	x	x	x	x	x
18	Interruption	0	0	1	0	0	0	0	0	0	x	x
19	En cours de processus	0	x	x	x	x	0	0	1	0	0	0
20	Décharger le bus à courant continu	0	0	x	0	0	0	0	0	0	x	x
Légende: x = 0 ou 1 0/1 = 0 pour Arrêt de Catégorie 0 ou 1, 1 pour Arrêt de Catégorie 2 0/x = 0 pour Arrêt de Catégorie 0 ou 1, x pour Arrêt de Catégorie 2												

La correspondance entre l'Action d'arrêt et la Catégorie d'arrêt est donnée au Tableau 95.

Tableau 95 – Action d'arrêt par rapport à la Catégorie d'arrêt

Enum. Action d'arrêt	Nom de l'Action d'arrêt	IEC-60204-1
0	Désactiver & Arrêt progressif	Arrêt de Catégorie 0
1	Décél courant & Désactiver	Arrêt de Catégorie 1
2	Décél rampée & Désactiver	Arrêt de Catégorie 1
3	Décél courant & Maintenir	Arrêt de Catégorie 2
4	Décél rampée & Maintenir	Arrêt de Catégorie y 2

7.3.13.2.2 Attribut N° 653 – Statut d'E/S d'axe

L'attribut Statut d'E/S d'axe est un ensemble de bits de 32 bits indiquant l'état des entrées et sorties numériques normalisées associées au fonctionnement de cet axe de mouvement, comme spécifié dans le Tableau 96.

Une valeur nulle affectée à un bit d'entrée donné indique une valeur logique nulle, la valeur 1 indiquant une valeur logique 1. Les bits de statut qui ne sont pas applicables doivent être mis à 0.

Tableau 96 – Définitions du bit Statut d'E/S d'axe

Bit	Obligatoire/ Facultatif	Nom
0	O	Activer l'entrée
1	O/E	Entrée Origine
2	O/E	Entrée Enregistrement 1
3	F/E	Entrée Enregistrement 2
4	O/P	Entrée de surcourse positive
5	O/P	Entrée de surcourse négative
6	F/E	Thermostat Réaction 1
7	F/D	Sortie de frein résistif
8	F/D	Sortie de frein mécanique
9 à 31		Réservé

7.3.14 Attributs d'exception, de défaut et d'alarme

7.3.14.1 Généralités

Le Tableau 97 contient tous les attributs d'exception, de défaut et d'alarme associés à une instance Objet Axe de dispositif de mouvement. Les exceptions sont des conditions qui peuvent se produire pendant le fonctionnement d'axe de mouvement présentant un potentiel de génération de défauts ou d'alarmes.

Tableau 97 – Attributs Exception, défaut et alarme

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
655	Obligatoire – Tous	Get		Exceptions d'axe	LWORD	Matriciel qui représente l'état actuel de toutes les conditions d'exception standard. Voir la définition matricielle du Tableau Exceptions standard en 7.3.14.2 pour plus de détails sur les positions du bit. Une Action d'exception d'axe est associée à chaque exception. Les exceptions configurées pour être ignorées sont uniquement visibles dans cet attribut.	Voir la sémantique en 7.3.14.2.
656	Obligatoire – Tous	Get		Exceptions d'axe Mfg	LWORD	Matriciel qui représente l'état actuel de toutes les conditions d'exception spécifiques au constructeur. Voir le Tableau Exception Mfg publié dans le manuel du produit d'entraînement. Une Action d'exception d'axe est associée à chaque exception. Les exceptions configurées pour être ignorées sont uniquement visibles dans cet attribut.	Voir le Tableau Exception Mfg (Publié dans le manuel du produit)
657	Obligatoire – Tous	Get		Défauts d'axe	LWORD	Matriciel qui représente l'état de tous les défauts d'exécution standard. Ce matriciel est identique à celui de l'attribut Exception d'axe. Les bits de défaut lorsqu'ils sont définis sont verrouillés jusqu'à la réinitialisation suite à un défaut. Une réinitialisation suite à un défaut annule les bits de défaut d'exécution, mais les bits sont de nouveau définis immédiatement si la condition d'exception sous-jacente est toujours présente. Toutes les exceptions pour lesquelles l'Action d'exception d'axe est configurée pour ignorer ou consigner sous forme d'alarmes n'apparaissent pas dans cet attribut.	Voir la sémantique en 7.3.14.2.
658	Obligatoire – Tous	Get		Défauts d'axe Mfg	LWORD	Matriciel qui représente l'état de tous les défauts d'exécution spécifiques au constructeur. Ce matriciel est identique à celui de l'attribut Exception d'axe Mfg. Les bits de défaut lorsqu'ils sont définis sont verrouillés jusqu'à la réinitialisation suite à un défaut. Une réinitialisation suite à un défaut annule les bits de défaut d'exécution, mais les bits sont de nouveau définis immédiatement si la condition d'exception sous-jacente est toujours présente. Toutes les exceptions pour lesquelles l'Action d'exception d'axe est configurée pour ignorer ou consigner sous forme d'alarmes n'apparaissent pas dans cet attribut.	Voir le Tableau Exception Mfg (Publié dans le manuel du produit)

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
659	Obligatoire – Tous	Get		Alarmes d'axe	LWORD	Matriciel qui représente l'état actuel de toutes les situations d'alarme standard. Ce matriciel est identique à celui de l'attribut Exception d'axe standard. Seules les conditions d'exception dont l'Action Exception d'axe est configurée pour consigner sous forme d'alarme apparaissent dans cet attribut, mais ne sont pas consignées dans l'attribut Défauts d'axe. Les bits d'alarme, lorsqu'ils sont définis, ne sont pas verrouillés et sont annulés dès que la condition d'exception sous-jacente est corrigée.	Voir la sémantique en 7.3.14.2.
660	Obligatoire – Tous	Get		Alarmes d'axe Mfg	LWORD	Matriciel qui représente l'état actuel de toutes les situations d'alarme spécifiques au constructeur. Ce matriciel est identique à celui de l'attribut Exception d'axe Mfg. Seules les conditions d'exception dont l'Action Exception d'axe est configurée pour consigner sous forme d'alarme apparaissent dans cet attribut, mais ne sont pas consignées dans l'attribut Défauts d'axe. Les bits d'alarme, lorsqu'ils sont définis, ne sont pas verrouillés et sont annulés dès que la condition d'exception sous-jacente est corrigée.	Voir le Tableau Exception Mfg. (Publié dans le manuel du produit)

7.3.14.2 Sémantique: Tableau Exceptions standard

Le Tableau 98 définit une liste des conditions d'exception standard associées aux attributs Exceptions d'axe, Défauts d'axe et Alarmes d'axe. Alors que les attributs Exceptions d'axe, Défauts d'axe et Alarmes d'axe sont tous Obligatoires dans la mise en œuvre du dispositif CIP Motion, la prise en charge de chacune des conditions d'exception individuelles décrites ici est Facultative. La colonne Règle dans le Tableau 98 indique les codes de fonction de dispositif où l'exception associée est applicable.

Tableau 98 – Tableau Exceptions standard

Bit	Règle	Exception	Description
0	-	-- Réservé --	Ce bit ne peut pas être utilisé car les Codes d'alarme et le Code de défaut sont définis par le numéro de bit d'exception associé, un Code d'alarme ou un Code de défaut nul indiquant l'absence d'une condition d'alarme ou de défaut.
1	D	Surintensité du moteur	Le courant du moteur a dépassé sa valeur de crête assignée ou sa limite de courant instantané.
2	D	Commutation du moteur	Problème de commutation du moteur à aimant permanent détecté. Par exemple: état interdit "111" ou "000" d'un dispositif de commutation UVW.
3	D	Limite de fabrication de survitesse du moteur	La vitesse du moteur a dépassé sa limite maximale donnée par l'attribut Limite de fabrication de survitesse du moteur associé au type de moteur.
4	D	Limite utilisateur de survitesse du moteur	La vitesse du moteur a dépassé la limite de vitesse définie par l'utilisateur donnée par la Limite utilisateur de survitesse du moteur.
5	D	Limite de fabrication de surchauffe du moteur	La température du moteur a dépassé la limite de température définie en usine donnée par la Limite de fabrication de surchauffe du moteur, ou le thermocontacteur intégral du moteur a été déclenché.
6	D	Limite utilisateur de surchauffe du moteur	La température du moteur a dépassé la limite de température définie par l'utilisateur donnée par la Limite utilisateur de surchauffe du moteur.
7	D	Limite de fabrication de surcharge thermique du moteur	Le modèle thermique du moteur a dépassé sa limite de capacité thermique définie en usine donnée par la Limite de fabrication de surcharge thermique du moteur.
8	D	Limite utilisateur de surcharge thermique du moteur	Le modèle thermique du moteur a dépassé sa capacité thermique définie par l'utilisateur donnée par la Limite utilisateur de surcharge thermique du moteur.
9	D	Perte de phase du moteur	Le courant dans une ou plusieurs phases du moteur a été perdu ou a été inférieur au seuil défini en usine ou, si prise en charge, à la Limite configurée de perte de phase du moteur. Cette exception est également associée à la fonction facultative de Contrôle du couple qui vérifie par essai le courant du moteur en fonction d'un frein mécanique engagé. En fonctionnement normal à l'état Exécution, l'essai de perte de phase du moteur effectue des cycles sur les courants des moteurs triphasés pour vérifier que le courant dans chaque phase du moteur dépasse le niveau de seuil. Lorsque la phase vérifiée dépasse le niveau, la vérification passe à la phase suivante. Si une quelconque phase ne parvient pas à dépasser le niveau dans les limites d'un délai spécifié par le fournisseur, par exemple une seconde, cette exception est émise. L'essai de perte de phase du moteur n'est effectué que lorsque le moteur tourne à une vitesse supérieure à celle spécifiée par le fournisseur. Lorsque le Contrôle du couple est activé, le courant de phase du moteur est vérifié à l'état Démarrage. Le courant est appliqué au moteur à un angle fixe réputé fournir le courant dans les trois phases du moteur. Ainsi, la durée d'exécution de cet essai est très courte. La Limite de perte de phase du moteur est utilisée pour déterminer si le dispositif d'entraînement peut produire un couple. Il est nécessaire que le courant mesuré dans les trois phases dépasse ce niveau pour qu'un passage se produise.
10	D	Surintensité de l'onduleur	Le courant de l'onduleur a dépassé la limite maximale définie en usine ou la limite de courant instantané.
11	D	Limite de fabrication de surchauffe de l'onduleur	La température de l'onduleur a dépassé sa limite de température définie en usine donnée par la Limite de fabrication de surchauffe de l'onduleur.
12	D	Limite utilisateur de surchauffe de l'onduleur	La température de l'onduleur a dépassé la limite de température définie par l'utilisateur donnée par la Limite utilisateur de surchauffe de l'onduleur.
13	D	Limite de fabrication de surcharge thermique de l'onduleur	Le modèle thermique de l'onduleur a dépassé sa limite de capacité thermique définie en usine donnée par la Limite de fabrication de surcharge thermique de l'onduleur.

Bit	Règle	Exception	Description
14	D	Limite utilisateur de surcharge thermique de l'onduleur	Le modèle thermique de l'onduleur a dépassé sa capacité thermique définie par l'utilisateur donnée par la Limite utilisateur de surcharge thermique de l'onduleur.
15	BD	Surintensité du convertisseur	Le courant du convertisseur a dépassé la limite maximale définie en usine ou la limite de courant instantané.
16	BD	Limite de fabrication du courant de terre du convertisseur	Le courant de terre a dépassé sa limite de courant définie en usine donnée par la limite de fabrication du courant de terre du convertisseur.
17	BD	Limite utilisateur du courant de terre du convertisseur	Le courant de terre a dépassé la limite définie par l'utilisateur donnée par la limite utilisateur du courant de terre du convertisseur.
18	BD	Limite de fabrication de surchauffe du convertisseur	La température du convertisseur a dépassé sa limite de température définie en usine donnée par la Limite de fabrication de surchauffe du convertisseur.
19	BD	Limite utilisateur de surchauffe du convertisseur	La température du convertisseur a dépassé la limite de température définie par l'utilisateur donnée par la Limite utilisateur de surchauffe du convertisseur.
20	BD	Limite de fabrication de surcharge thermique du convertisseur	Le modèle thermique du convertisseur a dépassé sa limite de capacité thermique définie en usine donnée par la Limite de fabrication de surcharge thermique du convertisseur.
21	BD	Limite utilisateur de surcharge thermique du convertisseur	Le modèle thermique du convertisseur a dépassé sa capacité thermique définie par l'utilisateur donnée par la Limite utilisateur de surcharge thermique du convertisseur
22	BD	Perte de puissance en courant alternatif du convertisseur	Plusieurs phases en courant alternatif ont été perdues sur la ligne à courant alternatif vers le convertisseur, généralement suite à l'ouverture d'un contacteur de ligne en courant alternatif. Lorsqu'elle est associée à un convertisseur externe dans une configuration AC/DC partagé ou Bus à courant continu partagé, la condition Perte de puissance en courant alternatif détectée par le convertisseur peut être acheminée via l'élément Statut de commande de la Connexion CIP Motion. Généralement, cette exception n'est pas déterminée sauf si la structure de puissance du dispositif est activée.
23	BD	Perte monophasée en courant alternatif du convertisseur	Une phase en courant alternatif a été perdue sur la ligne à courant alternatif vers le convertisseur.
24	BD	Court-circuit de phase en courant alternatif du convertisseur	Un court-circuit a été détecté entre une phase en courant alternatif et une autre phase en courant alternatif ou la terre.
25	BD	Défaillance de précharge du convertisseur	Un problème a été détecté dans les circuits de précharge du convertisseur, ce qui empêche de charger le bus à courant continu à un niveau de tension acceptable.
26	-	-- Réservé --	
27	BD	Limite de fabrication de surchauffe du régulateur de bus	La température du régulateur de bus a dépassé sa limite de température définie en usine donnée par la Limite de fabrication de surchauffe du régulateur de bus.
28	BD	Limite utilisateur de surchauffe du régulateur de bus	La température du régulateur de bus a dépassé la limite de température définie par l'utilisateur donnée par la Limite utilisateur de surchauffe du régulateur de bus.
29	BD	Limite de fabrication de surcharge thermique du régulateur de bus	Le modèle thermique du régulateur de bus a dépassé sa limite de capacité thermique définie en usine donnée par la Limite de fabrication de surcharge thermique du régulateur de bus.
30	BD	Limite utilisateur de surcharge thermique du régulateur de bus	Le modèle thermique du régulateur de bus a dépassé sa capacité thermique définie par l'utilisateur donnée par la Limite utilisateur de surcharge thermique du régulateur de bus.
31	D	Défaillance du régulateur de bus	Le module régulateur de bus (shunt) dans un système multiaxe est défaillant.
32	-	-- Réservé --	
33	BD	Limite de fabrication de sous-tension du bus	Le niveau de tension du bus à courant continu est inférieur à la limite définie en usine donnée par la limite de fabrication de sous-tension du bus.

Bit	Règle	Exception	Description
34	BD	Limite utilisateur de sous-tension du bus	Le niveau de tension du bus à courant continu est inférieur à la limite définie par l'utilisateur donnée par la limite utilisateur de sous-tension du bus.
35	BD	Limite de fabrication de surtension du bus	Le niveau de tension du bus à courant continu est supérieur à la limite définie en usine donnée par la limite de fabrication de surtension du bus.
36	BD	Limite utilisateur de surtension du bus	Le niveau de tension du bus à courant continu est supérieur à la limite définie par l'utilisateur donnée par la limite utilisateur de surtension du bus.
37	BD	Perte d'alimentation du bus	Le niveau de tension du bus à courant continu est inférieur au seuil de perte d'alimentation du bus pour une valeur supérieure à la valeur de temps de perte d'alimentation spécifiée par la période de temporisation.
38	BD	Alimentation du bus avec fusible grillé	Perte d'alimentation du bus à courant continu due à un fusible grillé.
39	D	Fuite d'alimentation du bus	Une fuite d'alimentation du bus c.c. a été détectée lorsqu'il est configuré pour un fonctionnement autonome. Ceci peut se produire lorsque le dispositif d'entraînement, configuré pour un fonctionnement autonome, n'est pas correctement câblé pour partager l'alimentation du bus à courant continu
40	D	Partage d'alimentation du bus	Un convertisseur externe qui partage l'alimentation du bus à courant continu avec cet entraînement dans une configuration en courant alternatif/en courant continu ou en courant continu partagée a demandé audit entraînement de ne plus s'alimenter à partir du bus à courant continu partagé. Ceci peut exiger la désactivation de l'entraînement afin que ce dernier supprime sa charge d'alimentation de bus à courant continu du convertisseur défaillant. En l'absence de liaison de communication entre ce dispositif d'entraînement et le convertisseur externe, le contrôleur peut surveiller le bit Décharger le bus à courant continu d'axe du convertisseur et, s'il est défini, il peut également initier les exceptions de partage d'alimentation sur tous les entraînements associés au convertisseur. Voir la définition du bit d'état Décharger le bus à courant continu associée à l'attribut État d'axe pour une description détaillée de ce comportement.
41	E	Limite de fabrication du bruit de signal de réaction	Des changements d'état du canal A/B induits par le bruit (états illégaux) d'un dispositif de réaction ont été détectés par le dispositif d'entraînement. De manière spécifique, le nombre de ces événements de bruit effectifs sur ce canal a dépassé la limite de fabrication du bruit de réaction. Le numéro de canal de réaction en cause est codé dans le sous-code Défaut/Alarme associé.
42	E	Limite utilisateur du bruit de signal de réaction	Des changements d'état du canal A/B induits par le bruit (états illégaux) d'un dispositif de réaction ont été détectés sur un canal de réaction. De manière spécifique, le nombre de ces événements de bruit effectifs sur ce canal a dépassé la limite utilisateur du bruit de réaction. Le canal de réaction en cause est codé dans le sous-code Défaut/Alarme associé.
43	E	Limite de fabrication de perte de signal de réaction	Une ou plusieurs connexions de signal d'un dispositif de réaction du canal A/B sont ouvertes, court-circuitées, manquantes ou gravement atténuées. De manière spécifique, les niveaux de tension détectés des signaux sont inférieurs à la limite de fabrication de perte de signal de réaction. Le canal de réaction en cause est codé dans le sous-code Défaut/Alarme associé.
44	E	Limite utilisateur de perte de signal de réaction	Une ou plusieurs connexions de signal d'un dispositif de réaction du canal A/B sont ouvertes, court-circuitées, manquantes ou gravement atténuées. De manière spécifique, les niveaux de tension détectés des signaux sont inférieurs à la limite utilisateur de perte de signal de réaction. Le canal de réaction en cause est codé dans le sous-code Défaut/Alarme associé.
45	E	Limite de fabrication de perte de données de réaction	Le nombre de paquets de données en série consécutifs manqués ou corrompus sur le canal de données série d'un dispositif de réaction a dépassé la limite de fabrication de perte de données de réaction. Le canal de réaction en cause est codé dans le sous-code Défaut/Alarme associé.

Bit	Règle	Exception	Description
46	E	Limite utilisateur de perte de données de réaction	Le nombre de paquets de données en série consécutifs manqués ou corrompus sur le canal de données série d'un dispositif de réaction a dépassé la limite utilisateur de perte de données de réaction. Le canal de réaction en cause est codé dans le sous-code Défaut/Alarme associé.
47	E	Défaillance du dispositif de réaction	Le dispositif de réaction a détecté une erreur interne.
48	-	-- Réservé --	
49	D	Glissement du frein	Le déplacement du moteur a dépassé la Tolérance de glissement du frein pendant l'engagement du frein mécanique.
50	D	Surcourse positive du matériel	L'axe s'est déplacé au-delà des limites de course physiques dans le sens positif et a activé l'interrupteur de fin de course Surcourse positive.
51	D	Surcourse négative du matériel	L'axe s'est déplacé au-delà des limites de course physiques dans le sens négatif et a activé l'interrupteur de fin de course Surcourse négative.
52	E	Surcourse positive de position	La position réelle d'axe a dépassé la limite de position configurée – valeur d'attribut Positive dans le sens positif. (Mise à l'échelle de l'entraînement uniquement)
53	E	Surcourse négative de position	La position réelle d'axe a dépassé la limite de position configurée – valeur d'attribut Négative dans le sens négatif. (Mise à l'échelle de l'entraînement uniquement)
54	D	Erreur de position excessive	La valeur Erreur de position de la boucle d'asservissement de position a dépassé la valeur configurée de Tolérance d'erreur de position.
55	D	Erreur de vitesse excessive	La valeur Erreur de vitesse de la boucle de commande de vitesse a dépassé la valeur configurée de Tolérance d'erreur de vitesse.
56	D	Limite de surcouple	Le couple du moteur a dépassé le niveau de couple maximal défini par l'utilisateur donné par Limite de surcouple.
57	D	Limite de sous-couple	Le couple du moteur est passé au-dessous du niveau de couple minimal défini par l'utilisateur donné par Limite de sous-couple.
58	-	-- Réservé --	
59	-	-- Réservé --	
60	Tous	Mode Commande interdite	Le contrôleur a spécifié un mode de commande ou de réaction non pris en charge
61	BD	Activer l'entrée désactivée	L'activation de l'entrée a été désactivée alors que la structure de puissance de l'axe était activée et fournissait le courant au Bus à courant continu ou au moteur.
62	Tous	Exception initiée par le contrôleur	Exception générée spécifiquement par le contrôleur.
63	Tous	Entrée Exception externe	Exception générée par l'entrée externe au dispositif.

7.3.15 Attributs Journaux de défauts et d'alarmes

Les attributs du Tableau 99 sont utilisés avec les journaux de défauts et d'alarmes en fonction du dispositif. Ces journaux se présentent fondamentalement sous la forme d'une liste de fiches, chaque fiche correspondant à une condition de défaut ou d'alarme détectée par le dispositif. La spécification Objet Axe de dispositif de mouvement ne définit pas de manière particulière les structures du journal de défauts ou du journal d'alarmes; la structure du journal est laissée à la discrétion du fournisseur du dispositif. La spécification exige toutefois que les journaux de défauts et d'alarmes contiennent certaines informations.

Pour les besoins de l'examen suivant, une structure représentative d'un journal de défauts peut se présenter comme suit:

```
Struct
{
```

```

Index;          // Index to the latest fault record element
Fault Log Record Struct  // Array of fault records
{
Fault Type;    // Type of fault
Fault Code;    // Specific fault identifier
Fault Sub Code;    // Extended fault diagnostic info
Fault Action;  // Drives response to the fault condition
Time Stamp;    // Time when this fault occurred
} [25]         // Store last 25 fault records in this circular buffer.
}

```

De façon similaire, un journal d'alarmes peut se présenter comme suit:

```

Struct
{
Index;          // Index to the latest alarm record element
Alarm Log Record Struct  // Array of alarm records
{
Alarm Type;    // Type of alarm
Alarm Code;    // Specific alarm identifier
Alarm Sub Code;    // Extended alarm diagnostic info
Alarm State;   // Alarm turned on (1) or turned off (0)
Time Stamp;    // Time stamp when this alarm transition occurred.
} [25]         // Store last 25 alarm records in circular buffer
}

```

Les journaux de défauts et d'alarmes ont plusieurs objectifs dans le cadre du dispositif CIP Motion. Tout d'abord, ils offrent un mécanisme de mise en file d'attente de plusieurs événements de défaut et d'alarme avec leurs datations associées pour leur transfert ultérieur au contrôleur afin de permettre une reproduction des journaux de défauts et d'alarmes dans le contrôleur, qui soit accessible à l'utilisateur. Ensuite, les éléments constitutifs du journal offrent eux-mêmes un moyen efficace de communication des informations de défaut et d'alarme au contrôleur. Le dispositif, au lieu d'envoyer 24 octets d'attributs de défaut au contrôleur lors de chaque mise à jour, a besoin d'envoyer uniquement un code de défaut d'1 octet correspondant à un défaut qui s'est effectivement produit.

Afin de faciliter le transfert des journaux de défauts et d'alarmes au contrôleur, des attributs sont définis pour chacun des éléments de journal associés mis directement en correspondance avec les éléments de l'Ensemble de données d'état de la connexion Dispositif-Contrôleur CIP Motion, et avec les attributs correspondants de l'Objet Axe de commande de mouvement.

Un journal de défaut et un journal d'alarme peuvent être transférés au contrôleur par mise à jour de connexion. Le dernier ID reçu est utilisé par le dispositif d'entraînement pour déterminer si les données envoyées ont été traitées avec succès, c'est-à-dire que les codes de défauts et d'alarmes transmis comme partie intégrante du bloc de données d'état de la connexion Dispositif-Contrôleur ont été saisis dans les journaux de défauts et d'alarmes du contrôleur. Cette condition permet au dispositif d'envoyer les codes de défaut et d'alarme suivants dans la séquence. Lorsque le dispositif a transmis avec succès l'événement de défaut ou d'alarme le plus récent, il envoie alors l'indication Pas de défaut pour l'absence de nouveaux défauts ou Pas d'alarme pour l'absence de nouvelles alarmes à consigner. En variante, le dispositif peut annuler les bits Ensemble de données d'état associés, en indiquant au contrôleur l'absence de défaut ou d'alarme à consigner.

Tableau 99 – Attributs Journaux de défauts et d'alarmes

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
661	Obligatoire – Tous	Get		Code de défaut d'axe	USINT	Énumération à 8 bits qui, lorsque la valeur du type de défaut d'axe n'est pas nulle, spécifie le numéro de bit d'une condition de défaut dans un journal de défaut d'axe. Les sources de défauts possibles incluent les défauts d'initialisation standards et spécifiques au constructeur, les défauts d'axe et les défauts de sécurité d'axe. Dans le cas d'un type de défaut = 0, cette valeur indique la source d'un événement d'annulation de défaut. Les événements d'annulation de défaut possibles sont la réinitialisation d'un dispositif, une demande de réinitialisation suite à un défaut, une demande de réinitialisation suite à une interruption ou une réinitialisation suite à l'ouverture d'une Connexion CIP Motion. Une réinitialisation doit être consignée indépendamment d'un défaut actuel ou non d'axe. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire les attributs et le journal de défauts du dispositif. Une valeur de 0 indique que tous les défauts ont été supprimés et qu'il n'y a pas de nouveau journal de défauts à consigner.	Si le Type de défaut d'axe != 0 Énumération: 0 à 255: Numéro de bit de défaut Si le Type de défaut d'axe = 0 Énumération: 0 = Pas de défaut 1 = Réinitialisation du dispositif (Cycle d'alimentation) 2 = Réinitialisation suite à un défaut 3 = Réinitialisation suite à un arrêt 4 = Réinitialisation suite à une connexion 5 à 255 = (réservé)

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
662	Obligatoire – Tous	Get		Type de défaut d'axe	USINT	Énumération à 8 bits qui spécifie la source d'une condition de défaut ou d'un événement de suppression de défaut dans un journal de défaut d'axe. Les sources de défauts possibles incluent les défauts d'initialisation standards et spécifiques au constructeur, les défauts d'axe et les défauts de sécurité d'axe. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire les attributs et le journal de défauts du dispositif. Une valeur de 0 indique que les défauts ont été supprimés. Dans ce cas, le code de défaut d'axe indique quel événement de suppression de défaut s'est effectivement produit.	Énumération: 0 = Défauts supprimés 1 = Défaut d'initialisation 2 = Défaut d'initialisation – Mfg 3 = Défaut d'axe 4 = Défaut d'axe – Mfg 5 = Défaut APR 6 = Défaut APR – Mfg 7 = Défaut de sécurité d'axe 8 = Défaut de sécurité d'axe – Mfg 9 à 127: Réservé 128 à 255: Spécifique au fournisseur
663	Obligatoire – Tous	Get		Datation de défaut d'axe	LINT	Datation à 64 bits de la condition de défaut d'axe ou d'un événement de suppression de défaut dans un journal de défaut d'axe. Le format de datation suit la norme CIP Sync, 0 correspondant au 1er janvier 1970. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire le journal de défauts du dispositif. Lorsque les attributs Type de défaut d'axe du journal de défauts et Code de défaut d'axe sont 0, la datation de défaut d'axe indique l'heure de dernière suppression des défauts d'axe.	Nanosecondes (CIP Sync)
664	Facultatif – Tous	Get		Code d'alarme d'axe	USINT	Énumération à 8 bits qui spécifie le numéro de bit d'une condition d'alarme dans un journal d'alarme d'axe. Les sources d'alarmes possibles incluent les alarmes d'axe standard et spécifiques au constructeur. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire l'état et le journal d'alarmes du dispositif. Une valeur de 0 indique qu'il n'existe actuellement aucune condition d'alarme pour cette instance d'axe.	Énumération: 0 à 255: Numéro de bit d'alarme

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
665	Facultatif – Tous	Get		Type d'alarme d'axe	USINT	Énumération à 8 bits qui spécifie la source d'une condition d'alarme dans un journal d'alarme d'axe. Les sources d'alarmes possibles incluent les alarmes d'axe standard et spécifiques au constructeur. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire les attributs et le journal d'alarmes du dispositif. Une valeur de 0 indique qu'il n'existe actuellement aucune condition d'alarme pour cette instance d'axe.	Énumération: 0 = Pas d'alarmes 1 = Inhibition de démarrage 2 = Inhibition de démarrage – Mfg 3 = Alarme d'axe 4 = Alarme d'axe – Mfg 5 à 127: Réservé 128 à 255: Spécifique au fournisseur
666	Facultatif – Tous	Get		Sous-code de défaut d'axe	USINT	Énumération à 8 bits qui spécifie le sous-code de défaut d'une condition de même nature dans un journal de défaut d'axe. Cet attribut permet de fournir des détails supplémentaires dans le cas de certains défauts, et a une interprétation différente pour chaque défaut. Des valeurs non nulles indiquent la présence d'un détail supplémentaire. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire le journal de défauts du dispositif.	Énumération: spécifique à la condition de défaut.
667	Facultatif – Tous	Get		Sous-code d'alarme d'axe	USINT	Énumération à 8 bits qui spécifie le sous-code d'alarme d'une condition de même nature dans un journal d'alarme d'axe. Cet attribut permet de fournir des détails supplémentaires dans le cas de certaines alarmes, et a une interprétation différente pour chaque alarme. Des valeurs non nulles indiquent la présence d'un détail supplémentaire. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire le journal d'alarmes du dispositif.	Énumération: spécifique à la condition d'alarme.

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
668	Obligatoire – Tous	Get		Action de défaut d'axe	USINT	Valeur à 8 bits composée de deux quartets dont les énumérations indiquent l'action entreprise par le dispositif en réponse à la condition de défaut énumérée dans un journal de défaut d'axe. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire les attributs et le journal de défauts du dispositif. L'action entreprise par le dispositif en réponse au défaut peut également être utilisée par le contrôleur pour modifier le fonctionnement du planificateur de mouvement et les instructions de mouvement. L'Action de défaut appliquée par le dispositif est définie par une Action d'arrêt suivie d'une Action de changement d'état.	Bits 0 à 3: Action d'arrêt Énumération: 0 = Défaut secondaire (Pas d'action) 1 = (réservé) 2 = Arrêt rampé 3 = Arrêt de limite de couple 4 = Arrêt progressif 5 à 15 = réservé. Bits 4 à 7: Changement d'état Énumération: 0 = Défaut secondaire (Pas d'action) 1 = Maintenir 2 = Désactiver 3 = Interruption 4 à 15 = réservé.
669	Facultatif – Tous	Get		État d'alarme d'axe	USINT	Énumération à 8 bits qui indique l'état actuel d'une condition d'alarme associé à un journal d'alarme d'axe. Les registres d'alarme sont journalisés dans le cas d'une condition d'alarme, c'est-à-dire lors de l'activation d'un bit d'alarme et lors de l'annulation d'une condition d'alarme, c'est-à-dire lors de la désactivation de ce même bit. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire les attributs et le journal d'alarmes du dispositif. Lorsque le code d'alarme d'axe du journal d'alarme est 0, c'est-à-dire lorsque le bit Pas d'alarme est actif, la valeur de cet attribut est également 0.	Énumération: 0 = Bit d'alarme Désactivé – 0 1 = Bit d'alarme Activé – 1 2 à 255 = réservé.

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
670	Facultatif – Tous	Get		Datation d'alarme d'axe	LINT	Datation à 64 bits de la condition d'alarme d'axe associée à un journal d'alarme d'axe. Le format de datation suit la norme CIP Sync, 0 correspondant au 1er janvier 1970. Cet attribut est conçu pour être transmis au contrôleur via le Bloc de données cycliques de Connexion CIP Motion afin de reproduire le journal d'alarmes du dispositif. Lorsque les attributs Type d'alarme d'axe et Code d'alarme d'axe du journal d'alarme ont une valeur de 0, la valeur de datation d'alarme d'axe est inappropriée et peut également être mise à 0.	Nanosecondes (CIP Sync)

7.3.16 Attributs de limite d'exception

Le Tableau 100 et le Tableau 101 contiennent tous les attributs de limite d'exception associés à une instance Objet Axe de dispositif de mouvement. Les attributs de limite d'exception définissent les conditions de génération d'une exception correspondante pendant le fonctionnement d'axe de mouvement ayant la possibilité de générer un défaut ou une alarme. Ils sont en général associés aux conditions de température, de courant et de tension du dispositif, par nature continues. Les Tolérances de fabrication (TF) associées aux exceptions sont en général intégrées au programme du dispositif et donnent souvent lieu à une condition de défaut principal. Les Limites utilisateur (LU) associées aux exceptions peuvent être configurées et sont en général utilisées pour générer une condition de défaut secondaire ou une condition d'alarme. C'est la raison pour laquelle les Limites utilisateur sont en général définies dans les Tolérances de fabrication correspondantes. Le déclenchement d'une exception Limite utilisateur n'interdit pas le déclenchement de l'exception Tolérance de fabrication correspondante, les deux conditions de déclenchement d'exception étant totalement indépendantes l'une de l'autre.

Tableau 100 – Attributs Informations de tolérance de fabrication d'exception

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
678	Facultatif – D	Get		Tolérance de fabrication de survitesse de moteur rotatif	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de survitesse du moteur en fonction d'une valeur définie en usine déterminée par les valeurs d'attributs Vitesse assignée de moteur rotatif et Vitesse maximale de moteur rotatif.	r/min
679	Facultatif – D	Get		Tolérance de fabrication de survitesse de moteur linéaire	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de survitesse du moteur en fonction d'une valeur définie en usine déterminée par les valeurs d'attributs Vitesse assignée de moteur linéaire et Vitesse maximale de moteur linéaire.	m/s
680	Facultatif – D	Get		Tolérance de fabrication de surchauffe du moteur	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surchauffe du moteur en fonction d'une valeur définie en usine relative à la température d'enroulement maximale du moteur.	°C
681	Facultatif – D	Get		Tolérance de fabrication de surcharge thermique du moteur	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surcharge thermique du moteur en fonction d'une valeur définie en usine relative aux caractéristiques assignées de surcharge thermique du moteur.	% de la valeur assignée du moteur
682	Facultatif – D	Get		Tolérance de fabrication de surchauffe de l'onduleur	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surchauffe de l'onduleur.	°C
683	Facultatif – D	Get		Tolérance de fabrication de surcharge thermique de l'onduleur	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surcharge thermique de l'onduleur.	% de la valeur assignée de l'onduleur

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
684	Facultatif – BD	Get		Tolérance de fabrication de surchauffe du convertisseur	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surchauffe du convertisseur.	°C
685	Facultatif – BD	Get		Tolérance de fabrication de surcharge thermique du convertisseur	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surcharge thermique du convertisseur.	% de la valeur assignée du convertisseur
693	Facultatif – BD	Get		Tolérance de fabrication de courant de terre du convertisseur	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de courant de terre du convertisseur.	Ampères
686	Facultatif – BD	Get		Tolérance de fabrication de surchauffe du régulateur de bus	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surchauffe du régulateur de bus.	°C
687	Facultatif – BD	Get		Tolérance de fabrication de surcharge thermique du régulateur de bus	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surcharge thermique du régulateur de bus.	% de la valeur assignée du régulateur
688	Facultatif – BD	Get		Tolérance de fabrication de surtension du bus	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de surtension du bus.	Volts
689	Facultatif – BD	Get		Tolérance de fabrication de sous-tension du bus	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de sous-tension du bus.	Volts
690	Facultatif – E	Get		Tolérance de fabrication du bruit de réaction	UDINT	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication du bruit de réaction.	Comptes de bruit
691	Facultatif – E	Get		Tolérance de fabrication de perte de signal de réaction	REAL	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de perte de signal de réaction.	% de la tension nominale
692	Facultatif – E	Get		Tolérance de fabrication de perte de données de réaction	UDINT	Renvoie la tolérance de fabrication relative à l'exception de tolérance de fabrication de perte de données de réaction.	Paquets de données perdus consécutifs

Tableau 101 – Attributs Configuration de limite utilisateur d'exception

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
694	Facultatif – D	Set		Limite de perte de phase du moteur	REAL	Définit la valeur minimale du courant de phase du moteur pour l'exception Perte de phase du moteur. Le courant dans chaque phase du moteur doit dépasser cette valeur pendant l'essai de perte de phase du moteur ou une exception Perte de phase du moteur se produit. La réduction de cette valeur d'attribut diminue la sensibilité aux conditions de perte de phase. Une valeur de 0 doit désactiver de manière efficace l'essai de perte de phase du moteur.	% de la valeur assignée du moteur
695	Facultatif – D	Set		Limite utilisateur de survitesse du moteur	REAL	Définit la limite utilisateur de survitesse par rapport à la vitesse assignée du moteur rotatif ou à la vitesse maximale du moteur linéaire admissible avant d'initier une exception de limite utilisateur de survitesse du moteur.	% de la valeur assignée du moteur
696	Facultatif – D	Set		Limite utilisateur de surchauffe du moteur	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surchauffe du moteur.	°C
697	Facultatif – D	Set		Tolérance de fabrication de surcharge thermique du moteur	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surcharge thermique du moteur.	% de la valeur assignée du moteur
698	Facultatif – D	Set		Limite utilisateur de surchauffe de l'onduleur	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surchauffe de l'onduleur.	°C
699	Facultatif – D	Set		Limite utilisateur de surcharge thermique de l'onduleur	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surcharge thermique de l'onduleur.	% de la valeur assignée de l'onduleur
700	Facultatif – BD (Convertisseur intégral uniquement)	Set		Limite utilisateur de surchauffe du convertisseur	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surchauffe du convertisseur.	°C
701	Facultatif – BD (Convertisseur intégral uniquement)	Set		Limite utilisateur de surcharge thermique du convertisseur	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surcharge thermique du convertisseur.	% de la valeur assignée du convertisseur
709	Facultatif – BD (Convertisseur intégral uniquement)	Set		Limite utilisateur de courant de terre du convertisseur	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de courant de terre du convertisseur	% tolérance de fabrication

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
702	Facultatif – BD (Convertisseur intégral uniquement)	Set		Limite utilisateur de surchauffe du régulateur de bus	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surchauffe du régulateur de bus.	°C
703	Facultatif – BD (Convertisseur intégral uniquement)	Set		Limite utilisateur de surcharge thermique du régulateur de bus	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surcharge thermique du régulateur de bus.	% de la valeur assignée du régulateur
704	Facultatif – BD	Set		Limite utilisateur de surtension du bus	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de surtension du bus. A l'inverse de la tolérance de fabrication correspondante, exprimée en Volts, la limite utilisateur repose sur le pourcentage de tension de bus nominale en fonctionnement.	% de tension de bus à courant continu
705	Facultatif – BD	Set		Limite utilisateur de sous-tension du bus	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de sous-tension du bus. A l'inverse de la tolérance de fabrication correspondante, exprimée en Volts, la limite utilisateur repose sur le pourcentage de tension de bus nominale en fonctionnement.	% de tension de bus à courant continu
706	Facultatif – E	Set		Limite utilisateur du bruit de réaction	UDINT	Définit la limite utilisateur relative à l'exception de limite utilisateur de surcharge du bruit de réaction. Un exemple de comptes de bruit se présente sous la forme de transitions simultanées des canaux A et B d'un dispositif de réaction à codeur à quadrature.	Comptes de bruit
707	Facultatif – E	Set		Limite utilisateur de perte de signal de réaction	REAL	Définit la limite utilisateur relative à l'exception de limite utilisateur de perte de signal de réaction. Le matériel d'interface de réaction contrôle généralement les niveaux de tension moyens des signaux entrants. Les conditions de perte de signal de réaction existent lorsque les niveaux de tension moyens baissent en dessous d'un pourcentage de la chute de tension admise par la tolérance de fabrication de perte de signal de réaction.	% de chute de tension avec tolérance de fabrication

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
708	Facultatif – E	Set		Limite utilisateur de perte de données de réaction	UDINT	Définit la limite utilisateur relative à l'exception de tolérance de fabrication de perte de données de réaction. Pour les dispositifs de réaction numériques, le matériel d'interface de réaction contrôle l'intégrité des données transférées au dispositif de réaction via la connexion en série. Les conditions de perte de données de réaction existent lorsque deux paquets de données ou plus consécutifs sont perdus ou corrompus.	Paquets de données perdus consécutifs

7.3.17 Attribut de configuration d'action d'exception d'axe

7.3.17.1 Généralités

L'attribut de configuration du Tableau 102 commande l'action réalisée par le dispositif suite à une condition d'exception. Une seule action d'exception est définie pour chaque exception prise en charge.

Tableau 102 – Attribut Configuration d'action d'exception d'axe

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
672	Obligatoire – Tous	Set		Action d'exception d'axe	USINT [64]	L'attribut Action d'exception d'axe est une matrice de 64 éléments d'octets énumérés qui spécifie l'action dédiée à l'exception standard associée. Voir la sémantique en 7.3.17.2 pour plus de détails.	Énumération: 0 = Ignorer (F) 1 = Alarme (F) 2 = Défaut secondaire (F) 3 = Défaut principal (O) 4 à 254 = réservé 255 = Non pris en charge
673	Obligatoire – Tous	Set		Action d'exception d'axe Mfg	USINT [64]	L'attribut Action d'exception d'axe est une matrice de 64 éléments d'octets énumérés qui spécifie l'action dédiée à l'exception spécifique au fournisseur associée. Voir la sémantique en 7.3.17.2 pour plus de détails	Énumération: 0 = Ignorer (F) 1 = Alarme (F) 2 = Défaut secondaire (F) 3 = Défaut principal (O) 4 à 254 = réservé 255 = Non pris en charge

7.3.17.2 Sémantique: Attributs N° 672 et 673 – Action d'exception d'axe

Les attributs Action d'exception d'axe et Action d'exception d'axe – Mfg. sont des matrices de 64 éléments d'octets énumérés qui spécifient l'action correspondant respectivement à l'exception standard ou spécifique au constructeur associée (voir Tableau 103). Pour une exception donnée, certaines actions d'exception peuvent ne pas être prises en charge. Tenter une prise en charge doit générer une erreur. Chaque produit du dispositif doit spécifier les actions disponibles pour chaque exception prise en charge. Si une exception spécifique n'est pas prise en charge par le dispositif, la seule énumération d'action d'exception valide est

“Non pris en charge”. Tenter de saisir toute autre valeur dans l'élément associé à une exception non prise en charge génère une erreur.

Tableau 103 – Définitions de l'action d'exception d'axe

Énum.	Besoin dans la mise en œuvre	Nom	Description
0	Facultatif	Ignorer	Demande au dispositif d'ignorer complètement la condition d'exception. Pour certaines exceptions essentielles au fonctionnement d'axe, il peut ne pas être possible d'ignorer la condition.
1	Facultatif	Alarme	L'action Alarme définit simplement le bit associé dans le mot Alarmes d'axe, mais n'a aucun impact sur le comportement d'axe. Pour certaines exceptions essentielles au fonctionnement du dispositif, il peut ne pas être possible de sélectionner cette action ou toute autre action qui n'a aucun impact sur le fonctionnement du dispositif.
2	Facultatif	Défaut secondaire	L'action Défaut secondaire signale l'exception en définissant le bit associé dans le mot Défauts d'axe, mais n'a aucun impact sur le comportement d'axe. Les défauts secondaires permettent au contrôleur d'engager une action spécifique à l'application en réponse à la condition d'exception. Pour certaines exceptions essentielles au fonctionnement du dispositif, il peut ne pas être possible de simplement indiquer la condition comme étant un défaut secondaire et de n'avoir aucun impact sur le fonctionnement du dispositif. Les convertisseurs (B) exécutant une action d'exception Défaut secondaire continuent à fournir l'alimentation du bus à courant continu et ne définissent pas le bit Décharger le bus à courant continu dans l'attribut Statut d'axe.

Énum.	Besoin dans la mise en œuvre	Nom	Description
3	Obligatoire	Défaut principal	L'action Défaut principal se traduit par la définition du bit associé dans le mot Défauts et par l'arrêt d'axe selon la "meilleure" méthode d'arrêt disponible définie en usine. Cette "meilleure" méthode d'arrêt inclut la méthode de décélération du moteur jusqu'à l'arrêt, et l'état final d'axe compte tenu du niveau prévu de commande encore disponible. Le niveau de commande d'axe disponible dépend de la condition d'exception spécifique et du mode de commande configuré. Les méthodes de décélération disponibles sont définies par l'attribut Action d'arrêt. Les actions d'arrêt standard, classées en niveaux de contrôle de décélération décroissants, sont les suivantes: 1. Décel rampée 2. Décel de limite de courant 3. Arrêt progressif En général, la "meilleure" action d'arrêt est la méthode de décélération la mieux contrôlée toujours disponible compte tenu de la condition d'exception. Alors que l'état d'axe final après un défaut principal est toujours l'état de défaut principal, l'état final de la structure de puissance en réponse à l'action d'exception de défaut principal peut être l'un des états suivants classés en niveaux de fonctionnalité de commande décroissants: 1. Maintenir (arrêté avec couple de maintien) 2. Désactiver (arrêté avec structure de puissance désactivée) 3. Interruption (arrêté avec action d'arrêt) Le "meilleur" état final de la structure de puissance est celui présentant le plus de fonctionnalités de commande encore disponibles compte tenu de la condition d'exception. L'action d'arrêt spécifique et l'état final associé à une action d'exception Défaut principal donnée sont capturés dans l'attribut Action Défaut d'axe qui est inclus dans l'enregistrement de journal de défauts. Les énumérations Action Défaut d'axe sont les suivantes: Énumérations Action d'arrêt: 0 = Pas d'action 1 = (réservé) 2 = Arrêt rampé 3 = Arrêt de limite de courant 4 = Arrêt progressif Énumérations Changement d'état: 0 = Pas d'action 1 = Maintenir 2 = Désactiver 3 = Interruption Si l'application exige une action d'exception qui est une action d'arrêt plus sévère que la "meilleure" méthode définie en usine, le contrôleur doit initier cette action. Si l'application exige une action d'exception moins sévère que la "meilleure" méthode définie en usine, le contrôleur doit configurer l'instance d'axe de dispositif pour une action d'exception Défaut secondaire et traiter directement le défaut. Cela peut mettre en danger le dispositif et les composants du moteur, et le dispositif doit l'admettre uniquement s'il peut rester actif, quoique temporel. Cela est particulièrement important dans les applications dans lesquelles la valeur du produit est supérieure à celle du moteur ou du dispositif.

Énum.	Besoin dans la mise en œuvre	Nom	Description
			Lorsque l'action d'exception Défaut principal est appliquée à un dispositif de convertisseur (B), l'action d'arrêt n'est pas applicable (0 = Pas d'action). Les états finaux de Désactiver ou Interruption pour le convertisseur sont applicables avec cependant Interruption exécutant l'Action d'interruption configurée. Dans tous les cas, le bit Décharger le bus à courant continu de l'attribut Statut d'axe du convertisseur est défini permettant au système de commande de générer une exception Partage du bus sur tous les dispositifs d'entraînement qui consomment l'alimentation du bus à courant continu provenant de ce convertisseur. Lorsque plusieurs défauts principaux se produisent avec différentes actions d'arrêt, l'action d'arrêt associée la plus sévère est appliquée, c'est-à-dire l'action d'arrêt qui exige le niveau de fonctionnalité de commande le plus faible. Cette règle s'applique également à l'action d'arrêt associée à l'Action d'arrêt, elle-même associée à une demande Désactivation.
4 à 254		Réservé	-
255	Obligatoire	Non pris en charge	L'Action d'exception non prise en charge est la valeur affectée aux Exceptions qui ne sont pas prises en charge par le dispositif. Tenter d'affecter une action d'exception, autre que l'action Non pris en charge, à une exception non prise en charge par le dispositif génère une erreur.

7.3.18 Attributs de défaut d'initialisation

7.3.18.1 Généralités

Le Tableau 104 contient tous les attributs de défaut d'initialisation associés à une instance Objet Axe de dispositif de mouvement. Les Défauts d'initialisation sont des conditions qui se produisent lors du processus d'initialisation du dispositif, empêchant son fonctionnement normal.

Tableau 104 – Attributs Défaut d'initialisation

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
674	Obligatoire – Tous	Get		Défauts d'initialisation	DWORD	Matriciel qui représente l'état de tous les défauts d'initialisation standard. Ces défauts empêchent tout mouvement et ne présentent aucune action de défaut configurable. Il s'agit, par exemple, de données de mémoire corrompues, d'erreurs d'étalonnage, de problèmes de démarrage de micrologiciel ou d'une valeur d'attribut de configuration non valide. Les défauts d'initialisation ne peuvent pas être supprimés avec une demande Réinitialisation suite à un défaut, bien qu'un cycle d'alimentation offre une nouvelle opportunité à l'initialisation.	Voir la sémantique en 7.3.18.2.
675	Obligatoire – Tous	Get		Défauts d'initialisation Mfg	DWORD	Matriciel qui représente l'état de tous les défauts d'initialisation spécifiques au constructeur. Ces défauts empêchent tout mouvement et ne présentent aucune action de défaut configurable. Il s'agit, par exemple, de données de mémoire corrompues, d'erreurs d'étalonnage, de problèmes de démarrage de micrologiciel ou d'une valeur d'attribut de configuration non valide. Les défauts d'initialisation ne peuvent pas être supprimés avec une demande Réinitialisation suite à un défaut, bien qu'un cycle d'alimentation offre une nouvelle opportunité à l'initialisation.	Voir le Tableau de Défaut d'initialisation Mfg (Publié dans le manuel du produit)

7.3.18.2 Sémantique: Tableau Défaut d'initialisation standard

Le Tableau 105 définit une liste de défauts standard associés à l'attribut Défauts d'initialisation et également applicable à l'attribut Code de défaut d'initialisation.

Tableau 105 – Tableau Défaut d'initialisation standard

Bit	Exception	Description
0	-- Réservé --	Ce bit ne peut pas être utilisé car le Code de défaut est défini par le numéro de bit d'exception associé, un Code de défaut nul indiquant l'absence d'une condition de défaut.
1	Défaut de somme de contrôle de bloc d'amorçage	Erreur de somme de contrôle ou erreur CRC pour le bloc d'amorçage du dispositif CIP Motion détectée comme partie intégrante d'un autotest
2	Défaut de somme de contrôle de bloc principal	Erreur de somme de contrôle ou erreur CRC pour le bloc principal du dispositif CIP Motion détectée comme partie intégrante d'un autotest.
3	Défaut de somme de contrôle de mémoire non volatile	Erreur de somme de contrôle ou erreur CRC pour la mémoire non volatile du dispositif CIP Motion détectée comme partie intégrante d'un autotest.
4 à 31	-- Réservé --	

7.3.19 Attributs Inhibition de démarrage

7.3.19.1 Généralités

Le Tableau 106 contient tous les attributs d'inhibition de démarrage associés à une instance Objet Axe de dispositif de mouvement. Les Inhibitions de démarrage sont des conditions qui empêchent l'axe de mouvement de passer de l'état Arrêté à l'un des états opérationnels.

Tableau 106 – Attributs d'inhibition de démarrage

ID attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
676	Obligatoire – D	Get		Inhibitions de démarrage	WORD	Matriciel qui spécifie l'état en cours de toutes les conditions standard qui bloquent le démarrage d'axe.	Voir la sémantique en 7.3.19.2.
677	Obligatoire – D	Get		Inhibitions de démarrage Mfg	WORD	Matriciel qui spécifie l'état en cours de toutes les conditions spécifiques au constructeur qui bloquent le démarrage d'axe.	Voir le Tableau d'Inhibition de démarrage Mfg. (Publié dans le manuel du produit)

7.3.19.2 Sémantique: Tableau Inhibition de démarrage standard

Le Tableau 107 définit une liste d'inhibitions de démarrage standard associées à l'attribut Inhibitions de démarrage.

Tableau 107 – Tableau Inhibition de démarrage standard

Bit	Exception	Description
0	-- Réservé --	Ce bit ne peut pas être utilisé car le Code d'inhibition de démarrage est défini par le numéro de bit associé, un Code d'inhibition de démarrage nul indiquant l'absence d'une condition de défaut.
1	Activer l'entrée d'axe	Code non actif.
2	Moteur non configuré	Le moteur associé n'a pas été configuré pour être utilisé.
3	Réaction non configurée	Le dispositif de réaction associé n'a pas été configuré pour être utilisé.
4	Commutation non configurée	La fonction de commutation du moteur à aimant permanent associée n'a pas été configurée pour être utilisée.
5	Couple de sécurité désarmé actif	La fonction intégrée de sécurité Couple de sécurité désarmé est active en fonction du bit Couple de sécurité désactivé actif (bit 3) du Statut Sécurité d'axe (Attribut N° 761) défini
6 à 15	-- Réservé --	

7.3.20 Attributs Défauts APR

7.3.20.1 Généralités

Le Tableau 108 contient tous les attributs de défaut APR (Rétablissement de position absolue) associés à une instance Objet Axe de dispositif de mouvement. Les défauts APR sont des conditions qui peuvent se produire lors du processus d'initialisation du dispositif lors de toute tentative de rétablissement de la position absolue d'un axe. Contrairement aux défauts d'initialisation, ces défauts sont réparables et peuvent être supprimés via une demande Réinitialisation suite à un défaut.

Tableau 108 – Attributs Défauts APR

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
756	Facultatif – Mise à l'échelle du dispositif d'entraînement E	Get		Défauts APR	WORD	Matriciel qui représente l'état de tous les défauts APR (Rétablissement de position absolue) standard. Un défaut APR est généré lorsque le système ne retrouve pas la position absolue d'axe après un cycle d'alimentation, une réinitialisation ou une reconnexion. Les défauts APR sont détectés au cours de la configuration ou de l'initialisation initiale d'axe d'entraînement. Dans le cas d'un défaut APR, la position réelle d'axe n'est plus corrélée à la position d'axe préalablement à un cycle d'alimentation, une réinitialisation ou une reconnexion. Exemples de défauts APR standards: désadaptation du numéro de série de réaction, désadaptation de l'ID d'axe, changement de configuration de mise à l'échelle, etc. Les défauts APR sont réparables et peuvent être supprimés via une demande Réinitialisation suite à un défaut.	Voir la sémantique en 7.3.20.2.
757	Facultatif – Mise à l'échelle du dispositif d'entraînement E	Get		Défauts APR Mfg	WORD	Matriciel qui représente l'état de tous les défauts APR (Rétablissement de position absolue) spécifiques au constructeur. Un défaut APR est généré lorsque le système ne retrouve pas la position absolue d'axe après un cycle d'alimentation, une réinitialisation ou une reconnexion. Les défauts APR sont détectés au cours de la configuration ou de l'initialisation initiale d'axe d'entraînement. Dans le cas d'un défaut APR, la position réelle d'axe n'est plus corrélée à la position d'axe préalablement à un cycle d'alimentation, une réinitialisation ou une reconnexion. Les défauts APR sont réparables et peuvent être supprimés via une demande Réinitialisation suite à un défaut.	Voir le Tableau Défauts APR Mfg (Publié dans le manuel du produit)

7.3.20.2 Sémantique: Tableau Défauts APR standard

Le Tableau 109 définit une liste de défauts standard associés à l'attribut Défauts APR et également applicable à l'attribut Code de défaut APR.

Tableau 109 – Tableau Défauts APR standard

Bit	Exception	Description
0	-- Réservé --	Ce bit ne peut pas être utilisé car le Code de défaut est défini par le numéro de bit d'exception associé, un Code de défaut nul indiquant l'absence d'une condition de défaut.
1	Erreur d'écriture mémoire	Erreur de sauvegarde des données de position absolue dans la mémoire NV.
2	Erreur de lecture mémoire	Erreur de lecture des données de position absolue contenues dans la mémoire NV.
3	Désadaptation du numéro de série de réaction	Le numéro de série de réaction de position ne correspond pas au numéro de série de réaction sauvegardé
4	Défaut d'affectation de mémoire tampon	Défaut dû à l'absence de mémoire RAM suffisante pour la sauvegarde des données APR.
5	Changement de la configuration de mise à l'échelle	La configuration de l'attribut de mise à l'échelle pour cet axe ne correspond pas à la configuration de mise à l'échelle sauvegardée.
6	Changement du mode de réaction	Le mode Réaction a changé et ne correspond pas à la configuration du mode réaction sauvegardée.
7 à 15	-- Réservé --	

7.3.21 Attributs de statistiques d'axe

Le Tableau 110 contient les attributs donnant des statistiques utiles sur le fonctionnement d'axe de mouvement.

Tableau 110 – Attributs Statistiques d'axe

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
710	Facultatif – BD	Get		Durée de mise sous tension de commande	REAL	Temps écoulé depuis la dernière application de la puissance de Commande d'axe.	Secondes
711	Facultatif – BD	Get		Temps d'exécution cumulé	REAL	Temps cumulé de mise sous tension d'axe à l'état Exécution.	Heures
712	Facultatif – BD	Get		Utilisation d'énergie cumulée	REAL	Énergie de sortie cumulée d'axe.	kW•h
713	Facultatif – D	Get		Rotations de moteur cumulées	LINT	Nombre de tours cumulés d'axe de moteur (Moteurs rotatifs uniquement)	
714	Facultatif – BD	Get		Cycles d'alimentation principale cumulés	DINT	Nombre de tours cumulés de cycles du secteur alternatif.	
715	Facultatif – BD	Get		Cycles de puissance de commande cumulés	DINT	Nombre de tours cumulés de cycles de puissance de commande.	

7.3.22 Attributs d'informations sur l'axe

Le Tableau 111 contient les attributs donnant des informations relatives aux capacités matérielles associées de l'instance Objet Axe de dispositif de mouvement.

Tableau 111 – Attributs Informations sur l'axe

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
720	Obligatoire – D	Get		Tension de sortie assignée de l'onduleur	REAL	Valeur assignée de la tension de sortie de l'onduleur d'entraînement. Cette valeur est incorporée au programme du dispositif.	Volts (eff.)
721	Obligatoire – D	Get		Courant de sortie assigné de l'onduleur	REAL	Valeur assignée du courant de sortie de l'onduleur d'entraînement. Cette valeur est incorporée au programme du dispositif.	Ampères (eff.)
722	Obligatoire – D	Get		Puissance de sortie assignée de l'onduleur	REAL	Valeur assignée de la puissance de sortie de l'onduleur d'entraînement. Cette valeur est incorporée au programme du dispositif.	kW
723	Facultatif – BD (Convertisseur intégral uniquement)	Get		Courant de sortie assigné du convertisseur	REAL	Valeur assignée du courant de sortie du convertisseur d'entraînement. Cette valeur est déterminée par l'axe de mouvement à partir du convertisseur associé.	Ampères
724	Facultatif – BD (Convertisseur intégral uniquement)	Get		Puissance de sortie assignée du convertisseur	REAL	Valeur assignée de la puissance de sortie du convertisseur d'entraînement. Cette valeur est déterminée par l'axe de mouvement à partir du convertisseur associé.	kW

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
725	Facultatif – D	Get		ID d'axe de structure de puissance d'entraînement	UDINT	Cette valeur représente un ID affecté par le fournisseur du dispositif d'entraînement qui identifie de manière unique la structure de puissance associée à cette instance d'axe. En revanche, le matériel de structure de puissance commun ou identique à toutes les instances d'axe du dispositif d'entraînement (à l'exclusion des axes de réaction principale) peut être identifié à l'aide de l'attribut ID de classe de structure de puissance d'entraînement. L'ID Axe de structure de puissance d'entraînement peut être inclus comme partie intégrante du segment de données dans le service Forward_Open de la Connexion CIP Motion afin de confirmer que la structure de puissance propre à cette instance d'axe correspond à la configuration du contrôleur. Les instances d'axe avec des structures de puissance qui ne sont pas configurées sont mises à 0, indiquant au dispositif qu'il n'est pas nécessaire de vérifier l'ID Axe de structure de puissance d'entraînement pour ce type d'instance d'axe.	Numéro ID

7.3.23 Attributs E/S à usage général

Le Tableau 112 contient les attributs offrant des E/S analogiques et numériques d'usage général associés à l'instance Objet Axe de dispositif de mouvement.

Tableau 112 – Attributs E/S à usage général de dispositif d'entraînement

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
730	Facultatif – BD	Get		Entrées numériques	DWORD	Mot à 32 bits dont les bits peuvent être affectés par le fournisseur aux entrées numériques à usage général.	Matriciel spécifique au fournisseur
731	Facultatif – BD	Set		Sorties numériques	DWORD	Mot à 32 bits dont les bits peuvent être affectés par le fournisseur aux sorties numériques à usage général.	Matriciel spécifique au fournisseur
732	Facultatif – BD	Get		Entrée analogique 1	REAL	Entrée analogique de niveau 1 à usage général.	% pleine échelle
733	Facultatif – D	Get		Entrée analogique 2	REAL	Entrée analogique de niveau 2 à usage général.	% pleine échelle
734	Facultatif – BD	Set		Sortie analogique 1	REAL	Sortie analogique de niveau 2 à usage général.	% pleine échelle
735	Facultatif – BD	Set		Sortie analogique 2	REAL	Sortie analogique de niveau 2 à usage général.	% pleine échelle
736	Facultatif – BD	Set		Vérification de l'entrée d'activation du dispositif d'entraînement	BOOL	Valeur booléenne qui contrôle si le dispositif d'entraînement doit ou non vérifier régulièrement l'état de l'entrée d'activation du dispositif d'entraînement. Lorsque la vérification de l'entrée d'activation du dispositif d'entraînement est activée, une entrée d'activation du dispositif inactive génère une condition Inhibition de démarrage. Lorsque l'entrée d'activation du dispositif d'entraînement est désactivée, une exception Entrée d'activation du dispositif d'entraînement désactivée est générée. Lorsque la vérification de l'entrée d'activation du dispositif d'entraînement est désactivée, le dispositif d'entraînement ne doit pas vérifier l'état de l'entrée d'activation du dispositif d'entraînement.	0 = Désactivé 1 = Activé

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
737	Facultatif – D	Set		Vérification de l'entrée de surcourse du matériel	BOOL	Valeur booléenne qui contrôle si le dispositif d'entraînement doit ou non vérifier régulièrement l'état des entrées de surcourse du matériel positive et négative. Lorsque la Vérification de l'entrée de surcourse du matériel est activée, une entrée Surcourse du matériel inactive génère une exception Surcourse du matériel positive ou négative associée. Lorsque la Vérification de surcourse du matériel d'entraînement est désactivée, le dispositif d'entraînement ne doit pas vérifier l'état des entrées Surcourse du matériel.	0 = Désactivé 1 = Activé

7.3.24 Attributs Mode local

Le Tableau 113 contient tous les attributs de mode local associés à une instance Objet Axe de dispositif de mouvement. Ces attributs régissent le comportement de l'instance Objet Axe de dispositif de mouvement lorsqu'une interface de dispositif d'entraînement (c'est-à-dire non CIP Motion) locale demande le contrôle du comportement d'axe.

Tableau 113 – Attributs Configuration du mode local

ID Attribut	Besoin dans la mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
750	Facultatif	Set		Commande locale	USINT	Mécanisme permettant au contrôleur d'autoriser une interface locale à demander une commande locale. La commande implique, dans ce cas, la possibilité de modifier l'état ou le comportement d'axe. Commande locale Non admise configure le dispositif afin d'empêcher une interface locale de prendre le contrôle du dispositif d'entraînement. Commande locale Admise sous condition configure le dispositif afin d'empêcher une interface locale de prendre le contrôle d'axe, ce dernier étant à un état opérationnel (Exécution ou Essais, par exemple). Commande locale Admise configure le dispositif afin d'autoriser une interface locale à prendre le contrôle d'axe, même dans un État opérationnel. Certains dispositifs peuvent ne pas prendre en charge cet état.	Énumération: 0 = Non admis (O) 1 = Admis sous condition (F) 2 = Admis (F) 3 à 255 = réservé

7.3.25 Attributs Sécurité d'axe

Le tableau d'attributs suivant comprend les attributs d'axe associés à la fonctionnalité de sécurité « sur réseau » associée à une instance Objet Axe de dispositif de mouvement incluse dans un Dispositif d'entraînement de sécurité CIP Motion. Ces attributs représentent l'état en cours d'un Noyau de sécurité intégré dans le dispositif conçu pour interagir avec un Contrôleur de sécurité externe via une connexion CIP Safety™ 17.

Les attributs État de sécurité d'axe, Statut de sécurité d'axe et Défaut de sécurité d'axe définis dans le Tableau 114 sont basés sur les valeurs lues des attributs résidants dans des objets associés au Noyau de sécurité et sont utilisés par le système de commande de mouvement pour contrôler le comportement du Noyau de sécurité via la Connexion CIP Motion.

NOTE Le Dispositif d'entraînement de sécurité CIP Motion, le Noyau de sécurité, la Connexion CIP Safety™ ainsi que les objets relatifs à la sécurité associés et leurs attributs ne relèvent pas du domaine d'application de la présente norme. Leur spécification peut être consultée en [30].

Tableau 114 – Attributs Statut de sécurité d'axe

ID Attribut	Besoin dans mise en œuvre	Règle d'accès	NV	Nom de l'attribut	Type de données	Description d'attribut	Sémantique des valeurs
760	Facultatif – Sécurité DE uniquement	Get		État de sécurité d'axe	USINT	Énumération à 8 bits qui indique l'état de l'Objet Superviseur de sécurité associé du dispositif. Il n'y a qu'un seul Objet Superviseur de sécurité desservant le dispositif CIP Motion de sorte que son état concerne toutes les instances d'Axe applicables du dispositif. Ceci signifie que toutes les instances de cet objet ont le même état pour cet attribut	Énum
761	Facultatif – Sécurité DE uniquement	Get		Statut de sécurité d'axe	DWORD	Ensemble de bits indiquant le statut des fonctions de sécurité normalisées pour l'axe tel qu'indiqué par le Noyau de sécurité intégré	Matriciel
762	Facultatif – Sécurité DE uniquement	Get		Statut de sécurité d'axe – Mfg	DWORD	Ensemble de bits indiquant le statut des fonctions de sécurité spécifiques au constructeur pour l'axe tel qu'indiqué par le Noyau de sécurité intégré	Matriciel 0 à 31 Spécifique au constructeur
763	Facultatif – Sécurité DE uniquement	Get		Défauts de sécurité d'axe	DWORD	Ensemble de bits indiquant le statut de défaut de sécurité de l'axe associé à la fonctionnalité de sécurité normalisée tel qu'indiqué par le Noyau de sécurité intégré	Matriciel
764	Facultatif – Sécurité DE uniquement	Get		Défauts de sécurité d'axe – Mfg	DWORD	Ensemble de bits indiquant le statut de défaut de sécurité de l'axe associé à la fonctionnalité de sécurité spécifique au constructeur tel qu'indiqué par le Noyau de sécurité intégré	Matriciel 0 à 31 Spécifique au constructeur
NOTE Voir [30] pour une définition complète des énumérations, matriciels et comportement d'état correspondant.							

¹⁷ CIP Safety™ est une marque de ODVA, Inc. Cette information est fournie pour la commodité des utilisateurs de la présente norme internationale et ne constitue en aucun cas un entérinement par l'IEC du détenteur de la marque ou de l'un quelconques de ses produits. La conformité à ce profil n'implique pas l'utilisation de la marque CIP Safety™. L'utilisation de la marque CIP Safety™ nécessite l'autorisation de ODVA, Inc.

7.4 Services communs

7.4.1 Services pris en charge

L'Objet Axe de dispositif de mouvement fournit les services communs CIP spécifiés dans le Tableau 115.

Tableau 115 – Objet Axe de dispositif de mouvement – Services communs

Code de service (Hex)	Besoin dans la mise en œuvre		Nom de service	Description du service
	Classe	Instance		
03 _{hex}	Obligatoire	Obligatoire	Get_Attributes_List	Renvoie le contenu des attributs extractibles sélectionnés d'une instance donnée.
04 _{hex}	Obligatoire	Obligatoire	Set_Attributes_List	Définit le contenu des attributs définissables sélectionnés d'une instance donnée.
0E _{hex}	Obligatoire	Obligatoire	Get_Attribute_Single	Renvoie le contenu de l'attribut extractible sélectionné d'une instance donnée.
10 _{hex}	Obligatoire	Obligatoire	Set_Attribute_Single	Définit le contenu de l'attribut définissable sélectionné d'une instance donnée.
1C _{hex}	Conditionnel ^a	N/A	Group_Sync	Transmet l'ID de l'Horloge maîtresse principale du contrôleur au dispositif et signale si ce dispositif peut être synchronisé avec le contrôleur dans le cadre d'un groupe de dispositifs associés.
^a Le service Group_Sync est exigé pour les dispositifs CIP Motion qui prennent en charge le service Synchronisation temporelle et incluent l'Objet Synchronisation temporelle.				

7.4.2 Données spécifiques au service

7.4.2.1 Généralités

Les services communs pris en charge par l'Objet Axe de dispositif de mouvement comportent des données de demande et des données de réponse associées. Le format et la syntaxe des données spécifiques au service dépendent du code de service spécifié. Les descriptions des données spécifiques au service des services Set/Get_Attribute_List et Set/Get_Attribute_Single sont déjà définies dans CIP (voir IEC 61158-5-2 et IEC 61158-6-2). Une description des données spécifiques au service associée au service Group_Sync (partie des extensions CIP Sync aux spécifications CIP) est présentée en détail au 7.4.2.2.

7.4.2.2 Group_Sync

Le service Group_Sync permet de synchroniser l'Objet Axe de dispositif de mouvement associé, avec ce nœud de dispositif, au contrôleur à l'origine de la demande de service et, finalement, à tous les autres dispositifs composant le Groupe Motion du contrôleur. La réponse du service indique "dispositif synchronisé" si le dispositif est actuellement synchronisé sur la même horloge maîtresse principale IEC 61588:2009 que le contrôleur.

Le format et la syntaxe des données de demande Group_Sync sont spécifiés dans le Tableau 116.

Tableau 116 – Structure des données de demande Group_Sync

Nom	Type	Description du paramètre de demande	Sémantique des valeurs
ID de l'horloge maîtresse principale	SINT[8]	ID de l'horloge maîtresse principale associée au contrôleur.	Voir Objet Synchronisation temporelle dans l'IEC 61158-5-2 et l'IEC 61158-6-2

L'ID de l'horloge maîtresse principale est utilisé pour vérifier que l'horloge maîtresse principale PTP dédiée au contrôleur et définie dans l'IEC 61588:2009 est la même horloge maîtresse principale associée au dispositif d'entraînement.

Le format et la syntaxe des données de réponse Group_Sync sont spécifiés dans le Tableau 117.

Tableau 117 – Structure des données de réponse Group_Sync

Nom	Type	Description du paramètre de réponse	Sémantique des valeurs
État de synchronisation	SINT	Valeur énumérée qui indique si le dispositif fait ou non actuellement l'objet d'une synchronisation de groupe ou s'il comporte une horloge maîtresse différente du contrôleur.	0 = dispositif synchronisé 1 = dispositif non synchronisé 2 = Horloge maîtresse principale différente 3 = étalement de l'horloge détecté 4 à 255 = réservé

Lorsque la demande du service Group_Sync est exécutée, le dispositif vérifie tout d'abord si l'horloge maîtresse principale associée au contrôleur correspond à l'horloge maîtresse principale associée du dispositif. Ceci est condition préalable importante pour une synchronisation satisfaisante de l'horloge locale du dispositif avec le contrôleur. Si les ID d'horloge maîtresse principale ne correspondent pas, le service Group_Sync répond par un État de synchronisation de 2, en indiquant "horloge maîtresse principale différente". Si les ID d'horloge maîtresse principale correspondent, le service Group_Sync procède alors à la vérification de l'état de la synchronisation de l'horloge du dispositif.

Le dispositif doit ensuite interroger l'Objet Synchronisation temporelle pour déterminer si le dispositif est synchronisé. Ceci exige que les variations du Décalage de temps système aient été inférieures au Seuil de synchronisation temporelle (généralement 10 µs) au cours d'une période prédéterminée. Si l'horloge du dispositif est synchronisée avec une horloge Maîtresse principale commune, le service Group_Sync effectue alors une vérification quant à la présence éventuelle d'un étalement de l'horloge. Dans le cas contraire, le service renvoie un Statut de synchronisation "Dispositif non synchronisé" (1).

Si les conditions ci-dessus sont satisfaites, le dispositif vérifie la présence éventuelle d'un étalement du temps entre le Temps système du contrôleur et le Temps système du dispositif. Cette vérification est rendue possible par le processus de configuration de dispositif du contrôleur qui permet d'envoyer régulièrement au dispositif les valeurs de Datation et de Décalage temporel avant d'envoyer une demande de service Group_Sync, qui à son tour exige le retour régulier au contrôleur des valeurs de Datation et de Décalage temporel avant d'envoyer une réponse de service Group_Sync. Le dispositif effectue une vérification d'étalement de l'horloge en chargeant la Datation du contrôleur provenant du paquet Contrôleur-Dispositif le plus récent. Si la durée (Datation du contrôleur + Période de mise à jour du contrôleur (CUP)) diffère du Temps système du dispositif en cours de plus d'une Période de mise à jour du contrôleur (CUP), les horloges de Temps système du contrôleur et du dispositif sont étalées.

C'est-à-dire:

Si $|(Datation\ du\ contrôleur + CUP) - Temps\ système\ du\ dispositif\ en\ cours| > 1\ CUP$

Les horloges du contrôleur et du dispositif sont étalées

sinon

Les horloges du contrôleur et du dispositif sont OK

Si l'horloge du dispositif n'est pas étalée, les horloges de Temps système du contrôleur et du dispositif sont considérées comme totalement synchronisées à ce moment. Sur la base de cette hypothèse, le service Group_Sync initialise la valeur du Dernier Décalage de temps système du contrôleur avec la Datation du contrôleur. De la même façon, le service initialise la valeur du Dernier Décalage de temps système du dispositif avec la valeur associée à la Datation du dispositif en cours. Ces deux variables doivent être utilisées ensuite lors de l'exécution de l'algorithme de Compensation d'échelon de temps. Enfin, le Group_Sync renvoie le Statut de synchronisation « dispositif synchronisé » (0). Dans le cas contraire, si l'étalement de l'horloge est détecté, le service renvoie un Statut de synchronisation « étalement de l'horloge détecté » (3).

Lorsque le contrôleur reçoit une réponse « dispositif synchronisé » du dispositif, il effectue sa propre vérification d'étalement et initialisation du Décalage temporel. En l'absence de détection d'étalement de l'horloge, le dispositif est alors considéré comme Synchronisé au groupe. Si le contrôleur reçoit une réponse Statut de synchronisation non nulle du dispositif ou si son essai d'étalement n'est pas satisfaisant, le contrôleur doit envoyer une autre demande de service Group_Sync. Ce processus doit se poursuivre pendant au moins 1 min jusqu'à ce que le dispositif soit Synchronisé au groupe ou que la temporisation du processus entraîne un défaut de connexion initié par le contrôleur.

Lorsque le contrôleur et tous les dispositifs CIP Motion sont synchronisés au groupe, le contrôleur définit alors le bit Commande synchrone de l'en-tête Connexion Contrôleur-Dispositif du paquet cyclique suivant envoyé au dispositif, indiquant que le dispositif peut planifier sa mise à jour de connexion Dispositif-Contrôleur suivante en fonction de la Datation du contrôleur et de la Période de mise à jour du contrôleur contenues dans les données de connexion. Le dispositif indique alors qu'il envoie les données de connexion planifiées en fonction de la Période de mise à jour du contrôleur en définissant le bit Mode synchrone de l'en-tête connexion Dispositif-Contrôleur du paquet cyclique suivant envoyé au contrôleur. L'échange de données entre le contrôleur et le dispositif se poursuit donc conformément au modèle de synchronisation CIP Motion.

7.5 Services spécifiques à l'objet

7.5.1 Services pris en charge

L'Objet Axe de dispositif de mouvement fournit les services spécifiques à l'objet spécifiés dans le Tableau 118.

Tableau 118 – Objet Axe de dispositif de mouvement – Services spécifiques à l'objet

Code de service (Hex)	Besoin dans la mise en œuvre		Nom de service	Description du service
	Classe	Instance		
4B _{hex}	n/a	Obligatoire	Get_Axis_Attributes_List	Renvoie le contenu des attributs Objet Axe de dispositif de mouvement extractibles sélectionnés.
4C _{hex}	n/a	Obligatoire	Set_Axis_Attributes_List	Définit le contenu des attributs Objet Axe de dispositif de mouvement définissables sélectionnés.
4D _{hex}	n/a	Obligatoire	Set_Cyclic_Write_List	Définit le contenu des attributs du bloc Données d'écriture cycliques.
4E _{hex}	n/a	Obligatoire	Set_Cyclic_Read_List	Définit le contenu des attributs du bloc Données de lecture cycliques.
4F _{hex}	n/a	Facultatif – D	Run_Motor_Test	Initie l'essai sélectionné sur le moteur afin de mesurer les différents paramètres moteur.
50 _{hex}	n/a	Facultatif – D	Get_Motor_Test_Data	Renvoie les résultats du précédent service Run_Motor_Test.
51 _{hex}	n/a	Obligatoire – C	Run_Inertia_Test	Initie l'essai sélectionné sur le moteur afin de mesurer l'inertie.
52 _{hex}	n/a	Obligatoire – C	Get_Inertia_Test_Data	Renvoie les résultats du précédent service Run_Inertia_Test.

Code de service (Hex)	Besoin dans la mise en œuvre		Nom de service	Description du service
	Classe	Instance		
53 _{hex}	n/a	Obligatoire	Run_Hookup_Test	Initie l'essai sélectionné afin de déterminer la condition des connexions du moteur et du dispositif de réaction.
54 _{hex}	n/a	Obligatoire	Get_Hookup_Test_Data	Renvoie les résultats du précédent service Run_Hookup_Test.
55 à 63 _{hex}			Réservé	

7.5.2 Données spécifiques au service

7.5.2.1 Généralités

Les services spécifiques à l'objet pris en charge par l'Objet Axe de dispositif de mouvement comportent des données de demande et des données de réponse associées. Le format et la syntaxe des données spécifiques au service dépendent du code de service spécifié. Tous les paramètres contenus dans les Données spécifiques au service sont du texte aligné. Cela est vrai, que les données de demande spécifique au service soient transmises dans le cadre de la connexion Contrôleur-Dispositif ou d'une connexion de messagerie explicite. Une description des données spécifiques au service associées à chaque service est présentée de 7.5.2.2 à 7.5.2.11.

7.5.2.2 Get_Axis_Attributes_List

Le service de liste Get_Axis_Attributes offre un mécanisme de lecture de la valeur en cours d'un ou de plusieurs attributs Objet Axe de dispositif de mouvement, y compris ceux comportant un type de données matricielles multidimensionnelles. Le mécanisme de transfert mémoire tampon/matrice peut être utilisé pour transférer des matrices volumineuses de consignation de données à partir du dispositif, afin de faciliter les fonctions telles que le flux à grande vitesse.

Le format du Bloc de données de demande pour ce service est représenté à la Figure 70.

← Mot à 32 bits →		
Format de demande Get_Axis_Attributes_List		
Nombre d'attributs	-	
ID attr 1	Dimension attr 1	Dimension d'élément attr 1
Index de début attr 1 (matrice uniquement)	Éléments de données attr 1 (matrice uniquement)	
...	...	
ID attr 2	Dimension attr 2	Dimension d'élément attr 2
Index de début attr 2 (matrice uniquement)	Éléments de données attr 2 (matrice uniquement)	
...	...	
ID attr n	Dimension attr n	Dimension d'élément attr n
Index de début attr n (matrice uniquement)	Éléments de données attr n (matrice uniquement)	
...	...	

IEC

Figure 70 – Format de demande Get_Axis_Attributes_List

Les définitions des paramètres individuels de cette structure de données sont les suivantes:

- Nombre d'attributs: représente le nombre d'attributs que contient la demande de service Extraction d'attribut d'axe.
- ID attr: identifie l'attribut Objet Axe de dispositif de mouvement cible à extraire.
- Dimension attr: détermine la dimension de la matrice d'attribut. Une dimension nulle signifie que l'attribut est un élément de données singulier et, par conséquent, pas réellement une matrice. Les matrices multidimensionnelles (dimension > 1) sont prises en charge en ajoutant les valeurs Index de début attr et Éléments de données attr. En cas

d'erreur liée à une opération d'extraction d'attribut demandée spécifique, le dispositif doit le signaler en affectant la valeur 0xFF à Dimension attr, auquel cas le champ Dimension d'élément contient le code d'erreur spécifique. En présence d'un code d'erreur, ni les paramètres d'index de matrice ni les champs de données d'attribut ne sont renvoyés.

- Dimension d'élément attr: détermine la taille, en octets, de l'élément de données d'attribut. Les éléments de données doivent être du texte aligné. Les mots à 32 bits et à 16 bits sont respectivement alignés sur 32 et 16 bits. Le remplissage peut être ajouté pour conserver l'alignement des mots. En cas d'erreur liée à une opération d'extraction d'attribut demandée spécifique, c'est-à-dire Dimension attr = 0xFF, le champ Dimension d'élément contient alors le code d'erreur État général CIP Common, tel que défini dans l'IEC 61158-6. En présence d'un code d'erreur, ni les paramètres d'index de matrice ni les champs de données d'attribut ne sont renvoyés.
- Index de début attr: identifie l'index de début de la matrice de valeurs d'attribut dans la section Données attr. Ce champ est uniquement présent si le type de données d'attribut est une matrice (c'est-à-dire si la dimension > 0).
- Éléments de données attr: détermine le nombre de valeurs d'élément de données dans la section Données attr de l'index associé. Ce champ est uniquement présent si le type de données d'attribut est une matrice (c'est-à-dire si la dimension > 0).

La structure du Bloc de données de réponse pour ce service est représentée à la Figure 71.

← Mot à 32 bits →		
Format de réponse Get_Axis_Attributes_List		
Nombre d'attributs	-	
ID attr 1	Dimension attr 1	Dimension d'élément attr 1
Index de début attr 1 (matrice uniquement)	Éléments de données attr 1 (matrice uniquement)	
...	...	
Données attr 1		
...		
ID attr 2	Dimension attr 2	Dimension d'élément attr 2
Index de début attr 2 (matrice uniquement)	Éléments de données attr 2 (matrice uniquement)	
...	...	
Données attr 2		
...		
ID attr n	Dimension attr n	Dimension d'élément attr n
Index de début attr n (matrice uniquement)	Éléments de données attr n (matrice uniquement)	
...	...	
Données attr n		
...		

IEC

Figure 71 – Format de réponse Get_Axis_Attributes_List

Les définitions des paramètres individuels de cette structure de données sont identiques à celles de la structure de données de demande, à laquelle a été ajouté l'élément Données d'attribut, qui est défini comme suit.

- Données attr: contient la/les valeur(s) en cours de l'attribut Objet Axe de dispositif de mouvement ciblé indiqué par l'ID attr. Si l'attribut est une matrice, la/les valeur(s) sont répertoriées en fonction de l'Index de début attr et du nombre d'Éléments de données attr. Pour les matrices multidimensionnelles (dimension > 1), la séquence de données se déplace de manière séquentielle de droite à gauche par l'intermédiaire des index. Des exemples sont présentés ci-après.

EXEMPLE 1 La Figure 72 définit une réponse Get_Axis_Attributes_List pour un attribut scalaire DINT simple de 4 octets. Étant donné que Dim attr est nul, les champs Index de début attr et Éléments de données attr sont ignorés.

Réponse Get_Axis_Attributes_List – Exemple de données sans dimension		
Nombre d'attributs = 1	-	
ID attr = 25	Dim attr = 0	Dim Élém attr = 4
Données attr = ?		

IEC

Figure 72 – Réponse Get_Axis_Attributes_List – Attribut de 4 octets simple

EXEMPLE 2 La Figure 73 définit une réponse Get_Axis_Attributes_List pour un attribut scalaire de 2 octets simples qui montre l'octet de remplissage ajouté pour conserver l'alignement des mots à 32 bits.

Réponse Get_Axis_Attributes_List – Exemple de données sans dimension		
Nombre d'attributs = 1	-	
ID attr = 20	Dim attr = 0	Dim Élém attr = 2
Données attr = ?	(Remplissage)	

IEC

Figure 73 – Réponse Get_Axis_Attributes_List – Attribut de 2 octets simple

EXEMPLE 3 La Figure 74 définit une réponse Get_Axis_Attributes_List pour les trois premiers éléments d'un attribut de matrice unidimensionnelle qui est un élément UINT. Étant donné que Dim Élém attr de cet attribut est une valeur de 2 octets, un octet de remplissage est ajouté pour conserver l'alignement des mots pour les données de connexion qui suivent.

Réponse Get_Axis_Attributes_List – Exemple de matrice unidimensionnelle		
Nombre d'attributs = 1	-	
ID attr = 43	Dim attr = 1	Dim Élém attr = 2
Index de début attr 1 = 0	Éléments de données attr 1 = 3	
Données attr (0) = ?	Données attr (1) = ?	
Données attr (2) = ?	(Remplissage)	

IEC

Figure 74 – Réponse Get_Axis_Attributes_List – Matrice d'attributs d'octets

EXEMPLE 4 La Figure 75 définit une réponse Get_Axis_Attributes_List pour six éléments d'une matrice bidimensionnelle de REAL. Les Données attr se déplacent de manière séquentielle par l'index de matrice le plus à gauche d'abord en commençant par Index de début attr 1 dont la valeur est nulle.

Réponse Get_Axis_Attributes_List – Exemple de matrice bidimensionnelle		
Nombre d'attributs = 1	-	
ID attr = 27	Dim attr = 2	Dim Élém attr = 4
Index de début attr 1 = 0	Éléments de données attr 1 = 2	
Index de début attr 2 = 2	Éléments de données attr 2 = 3	
Données attr (0, 2) = ?		
Données attr (0, 3) = ?		
Données attr (0, 4) = ?		
Données attr (1, 2) = ?		
Données attr (1, 3) = ?		
Données attr (1, 4) = ?		

IEC

Figure 75 – Réponse Get_Axis_Attributes_List – Matrice d'attributs bidimensionnelle

En cas d'erreur liée à une opération d'extraction d'attribut demandée spécifique, le dispositif doit le signaler en affectant la valeur 0xFF à Dimension attr, auquel cas le champ Dimension d'élément peut contenir le code d'erreur spécifique. En présence d'un code d'erreur, ni les paramètres d'index de matrice ni les champs de données d'attribut ne sont renvoyés.

EXEMPLE 5 La Figure 76 définit ce type de cas dans lequel Attribut 29 n'est pas pris en charge par l'axe de mouvement ciblé, tel qu'indiqué par le code État général de 0x14.

Réponse Get_Axis_Attributes_List – Exemple d'erreur d'attribut		
Nombre d'attributs = 3	-	
ID attr = 25	Dim attr = 0	Dim Élém attr = 4
Données attr = 43,5		
ID attr = 29	Dim attr = 0xFF	Code d'erreur = 0x14
ID attr = 43	Dim attr = 1	Dim Élém attr = 2
Index de début attr 1 = 0	Éléments de données attr 1 = 3	
Données attr [0] = 55	Données attr [1] = 45	
Données attr [2] = 35	(Remplissage)	

IEC

Figure 76 – Réponse Get_Axis_Attributes_List – Exemple d'erreur

7.5.2.3 Set_Axis_Attributes_List

Le service Set_Axis_Attributes_List offre un mécanisme d'écriture d'une valeur dans un ou plusieurs attributs Objet Axe de dispositif de mouvement définissables, y compris ceux comportant un type de données matricielles multidimensionnelles. Le mécanisme de transfert mémoire tampon/matrice peut être utilisé pour générer des tables de paramètres dans le dispositif pour des applications d'indexation ou de came.

Le format du Bloc de données de demande pour ce service est représenté à la Figure 77.

← Mot à 32 bits →		
Format de demande Set_Axis_Attributes_List		
Nombre d'attributs	-	
ID attr 1	Dimension attr 1	Dimension d'élément attr 1
Index de début attr 1 (matrice uniquement)	Éléments de données attr 1 (matrice uniquement)	
...	...	
Données attr 1		
...		
ID attr 2	Dimension attr 2	Dimension d'élément attr 2
Index de début attr 2 (matrice uniquement)	Éléments de données attr 2 (matrice uniquement)	
...	...	
Données attr 2		
...		
...		
ID attr 3	Dimension attr n	Dimension d'élément attr n
Index de début attr n (matrice uniquement)	Éléments de données attr n (matrice uniquement)	
...	...	
Données attr 3		
...		

IEC

Figure 77 – Format de demande Set_Axis_Attributes_List

Les définitions des paramètres individuels de cette structure de données sont les suivantes:

- Nombre d'attributs: représente le nombre d'attributs que contient la demande de service Définition d'attribut d'axe.
- ID attr: identifie l'attribut de configuration Objet Axe de dispositif de mouvement cible à définir.
- Dimension attr: détermine la dimension de la matrice d'attribut. Une dimension nulle signifie que l'attribut est un élément de données singulier et, par conséquent, pas réellement une matrice. Les matrices multidimensionnelles (dimension > 1) sont prises en charge en ajoutant les valeurs Index de début attr et Éléments de données attr avant la séquence Données attr.
- Dimension d'élément attr: détermine la taille, en octets, de l'élément de données d'attribut. Les éléments de données doivent être du texte aligné. Les mots à 32 bits et à 16 bits sont

respectivement alignés sur 32 et 16 bits. Le remplissage peut être ajouté pour conserver l'alignement des mots.

- Index de début attr: identifie l'index de début de la matrice de valeurs d'attribut dans la section Données attr. Ce champ est uniquement présent si le type de données d'attribut est une matrice (c'est-à-dire si la dimension > 0).
- Éléments de données attr: détermine le nombre de valeurs d'élément de données en fonction de l'index de début inclus dans la section Données attr. Ce champ est uniquement présent si le type de données d'attribut est une matrice (c'est-à-dire si la dimension > 0).
- Données attr: contient la/les nouvelle(s) valeur(s) à appliquer à l'attribut de configuration du dispositif ciblé indiqué par l'ID attr. Si l'attribut est une matrice, la/les nouvelle(s) valeur(s) sont appliquées en fonction de l'Index de début attr et du nombre d'Éléments de données attr. Pour les matrices multidimensionnelles (dimension > 1), la liste des éléments Données attr se déplace de manière séquentielle de gauche à droite par l'intermédiaire des index. Le type de données de Données attr doit correspondre à celui de l'attribut spécifié par l'ID attr.

EXEMPLE 1 La Figure 78 définit une demande Set_Axis_Attributes_List pour un attribut scalaire simple de 4 octets. Étant donné que Dim attr est nul, les champs Index de début attr et Éléments de données attr sont ignorés.

Demande Set_Axis_Attributes_List – Exemple de données sans dimension		
Nombre d'attributs = 1	-	
ID attr = 25	Dim attr = 0	Dim Élém attr = 4
Données attr = ?		

IEC

Figure 78 – Demande Set_Axis_Attributes_List – Attribut de 4 octets simple

EXEMPLE 2 La Figure 79 définit une demande Set_Axis_Attributes_List pour un attribut scalaire de 2 octets simples qui montre l'octet de remplissage ajouté pour conserver l'alignement des mots à 32 bits.

Demande Set_Axis_Attributes_List – Exemple de données sans dimension		
Nombre d'attributs = 1	-	
ID attr = 20	Dim attr = 0	Dim Élém attr = 2
Données attr = ?	(Remplissage)	

IEC

Figure 79 – Demande Set_Axis_Attributes_List – Attribut de 2 octets simple

EXEMPLE 3 La Figure 80 définit une demande Set_Axis_Attributes_List pour les trois premiers éléments d'une matrice unidimensionnelle de UINT. Étant donné que Dim Élém attr est une valeur de 2 octets, un octet de remplissage est ajouté pour conserver l'alignement des mots pour les données de connexion qui suivent.

Demande Set_Axis_Attributes_List – Exemple de matrice unidimensionnelle		
Nombre d'attributs = 1	-	
ID attr = 43	Dim attr = 1	Dim Élém attr = 2
Index de début attr 1 = 0	Éléments de données attr 1 = 3	
Données attr (0) = ?	Données attr (1) = ?	
Données attr (2) = ?	(Remplissage)	

IEC

Figure 80 – Demande Set_Axis_Attributes_List – Matrice d'attributs à 2 octets

EXEMPLE 4 La Figure 81 définit une demande Set_Axis_Attributes_List pour six éléments d'une matrice bidimensionnelle de REAL. Les Données attr se déplacent de manière séquentielle par l'index de matrice le plus à gauche d'abord en commençant par Index de début attr 1 dont la valeur est nulle.

Demande Set_Axis_Attributes_List – Exemple de matrice bidimensionnelle		
Nombre d'attributs = 1	-	
ID attr = 27	Dim attr = 2	Dim Élém attr = 4
Index de début attr 1 = 0	Éléments de données attr 1 = 2	
Index de début attr 2 = 2	Éléments de données attr 2 = 3	
Données attr (0, 2) = ?		
Données attr (0, 3) = ?		
Données attr (0, 4) = ?		
Données attr (1, 2) = ?		
Données attr (1, 3) = ?		
Données attr (1, 4) = ?		

IEC

Figure 81 – Demande Set_Axis_Attributes_List – Matrice d'attributs bidimensionnelle

La structure du Bloc de données de réponse pour ce service est représentée à la Figure 82.

← Mot à 32 bits →		
Format de réponse Set_Axis_Attributes_List		
Nombre d'attributs	-	
ID attr 1	État attr 1	Index attr 1 (matrice uniquement)
ID attr 2	État attr 2	Index attr 2 (matrice uniquement)
...		
ID attr n	État attr n	Index attr n (matrice uniquement)

IEC

Figure 82 – Format de réponse Set_Axis_Attributes_List

- ID attr: identifie l'attribut de configuration Objet Axe de dispositif de mouvement ciblé de la demande de définition de liste.
- État attr: indique si l'action de définition de liste pour l'attribut ciblé a abouti. La valeur d'un état d'attribut est nulle si l'écriture de l'attribut ciblé a abouti. Une valeur État attr différente de zéro indique qu'une erreur s'est produite et a empêché la mise à jour de l'attribut. Les codes d'erreur suivent les Codes d'état général CIP Common (voir IEC 61158-6-2). Une valeur État attr différente de zéro pour tout élément dans la liste doit également être représentée dans cet élément État général du service en indiquant une Erreur de liste d'attributs, 0x0A.
- Index attr: indique l'index de matrice de l'élément matriciel en charge de la valeur de code État attr différente de zéro. Cette valeur est nulle lorsque la valeur État attr est nulle (succès) ou lorsque le type de données d'attribut n'est pas une matrice.

7.5.2.4 Set_Cyclic_Write_List

Le service Set_Cyclic_Write_List offre un mécanisme permettant de déterminer la liste des attributs à transmettre dans le Bloc de données d'écriture cycliques de la connexion Contrôleur-Dispositif.

Le format du Bloc de données de demande pour ce service est représenté à la Figure 83.

← Mot à 32 bits →	
Format de demande Set_Cyclic_Write_List	
Nombre d'attributs	-
ID attr 1	ID attr 2
...	

IEC

Figure 83 – Format de demande Set_Cyclic_Write_List

- Nombre d'attributs: représente le nombre d'attributs que contient la Liste d'écriture cyclique.
- ID attr: identifie l'attribut de configuration de dispositif spécifique à mettre à jour grâce au Bloc de données d'écriture cycliques de la connexion Contrôleur-Dispositif. L'ID attr détermine le type de données et la sémantique implicite des données en fonction des spécifications de l'attribut associé.

L'ordre des données d'attribut dans le Bloc de données d'écriture cycliques est déterminé par celui des ID d'attribut de la demande Set_Cyclic_Write_List. Les éléments de données d'attribut doivent être du texte aligné. Les mots à 32 bits et à 16 bits sont respectivement alignés sur 32 et 16 bits. Le remplissage peut être ajouté pour conserver l'alignement des mots.

La structure du Bloc de données de réponse pour ce service est représentée à la Figure 84.

← Mot à 32 bits →		
Format de réponse Set_Cyclic_Write_List		
Nombre d'attributs	ID de bloc d'écriture cyclique	
ID attr 1	État attr 1	-
ID attr 2	État attr 2	-
...		
ID attr n	État attr n	-

IEC

Figure 84 – Format de réponse Set_Cyclic_Write_List

- Nombre d'attributs: représente le nombre d'attributs que contient la Liste d'écriture cyclique demandée.
- ID de bloc d'écriture cyclique: Si tous les attributs de la liste demandée peuvent être pris en charge dans le Bloc de données d'écriture cycliques, la réponse Set_Cyclic_Write_List fournit un nouvel ID de bloc d'écriture cyclique, qui est simplement l'incrément de l'ID de bloc d'écriture cyclique en cours. Ce nouvel ID peut être utilisé dans la mise à jour suivante de la connexion Contrôleur-Dispositif. Si la Set_Cyclic_Write_List n'a pas abouti, l'ID de bloc d'écriture cyclique conserve la valeur en cours.
- ID attr: identifie l'attribut de configuration d'axe spécifique dont la mise à jour a été demandée grâce au Bloc de données d'écriture cycliques de la connexion Contrôleur-Dispositif.
- État attr: La valeur de État attr est nulle si l'attribut associé indiqué par l'ID attr peut être pris en charge dans le Bloc de définition de données d'écriture cycliques. Un code État attr différent de zéro indique la raison spécifique pour laquelle l'attribut associé ne peut pas être pris en charge. Ces codes suivent les Codes d'état général CIP Common (voir IEC 61158-6-2). Si l'État attr d'un ou de plusieurs ID attr de la liste indique une erreur, la demande Set_Cyclic_Write_List est considérée comme n'ayant pas abouti.

7.5.2.5 Set_Cyclic_Read_List

Le service Set_Cyclic_Read_List offre un mécanisme permettant de déterminer la liste des attributs à transmettre dans le Bloc de données de lecture cycliques de la connexion Dispositif-Contrôleur.

Le format du Bloc de données de demande pour ce service est représenté à la Figure 85.

← Mot à 32 bits →	
Format de demande Set_Cyclic_Read_List	
Nombre d'attributs	-
ID attr 1	ID attr 2
...	

IEC

Figure 85 – Format de demande Set_Cyclic_Read_List

- Nombre d'attributs: représente le nombre d'attributs que contient la Liste de lecture cyclique.
- ID attr: identifie l'état de dispositif spécifique ou l'attribut de signal à mettre à jour grâce au Bloc de données de lecture cycliques de la connexion Dispositif-Contrôleur. L'ID attr détermine le type de données et la sémantique implicite des données en fonction des spécifications de l'attribut associé.

L'ordre des données d'attribut dans le Bloc de données de lecture cycliques est déterminé par celui des ID d'attribut de la demande Set_Cyclic_Read_List. Les éléments de données d'attribut doivent être du texte aligné. Les mots à 32 bits et à 16 bits sont respectivement alignés sur 32 et 16 bits. Le remplissage peut être ajouté pour conserver l'alignement des mots.

La structure du Bloc de données de réponse pour ce service est représentée à la Figure 86.

← Mot à 32 bits →		
Format de réponse Set_Cyclic_Read_List		
Nombre d'attributs	ID de bloc de lecture cyclique	
ID attr 1	État attr 1	-
ID attr 2	État attr 2	-
...		
ID attr n	État attr n	-

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Figure 86 – Format de réponse Set_Cyclic_Read_List

- Nombre d'attributs: représente le nombre d'attributs que contient la Liste de lecture cyclique demandée.
- ID de bloc de lecture cyclique: Si tous les attributs de la liste demandée peuvent être pris en charge dans le Bloc de données de lecture cycliques, la réponse Set_Cyclic_Read_List fournit un nouvel ID de bloc de lecture cyclique, qui est simplement l'incrément de l'ID de bloc de lecture cyclique en cours. Ce nouvel ID peut être utilisé dans la mise à jour suivante de la connexion Dispositif-Contrôleur. Si la Set_Cyclic_Read_List n'a pas abouti, l'ID de bloc de lecture cyclique conserve la valeur en cours.
- ID attr: identifie l'attribut de configuration d'axe spécifique dont la mise à jour a été demandée grâce au Bloc de données de lecture cycliques de la connexion Dispositif-Contrôleur.
- État attr: La valeur de État attr est nulle si l'attribut associé indiqué par l'ID attr peut être pris en charge dans le Bloc de définition de données de lecture cycliques. Un code État attr différent de zéro indique la raison spécifique pour laquelle l'attribut associé ne peut pas être pris en charge. Ces codes suivent les Codes d'état général CIP Common (voir IEC 61158-6-2). Si l'état attr d'un ou de plusieurs ID attr de la liste indique une erreur, la demande Set_Cyclic_Read_List est considérée comme n'ayant pas abouti.

7.5.2.6 Run_Motor_Test

Le service Run_Motor_Test déclenche diverses opérations d'essai sur le moteur afin de déterminer les paramètres de modèle de moteur importants. Pour les essais actifs qui activent la structure de puissance, l'état d'axe doit être Arrêté pour que la demande de ce service soit acceptée. Pour les essais passifs qui n'activent pas la structure de puissance, l'état d'axe peut être Arrêté ou dans les états d'Attente suivants: Interruption, Précharge ou Inhibition de

démarrage. Le Run_Motor_Test déclenche le processus d'essai spécifié et définit le bit En cours de processus dans l'attribut État d'axe. Pour les essais actifs, l'état d'axe passe à l'état Essai. Dans la mesure où la réalisation d'un processus d'essai actif peut durer relativement longtemps, une réponse de service est renvoyée au contrôleur dès le lancement du processus d'essai. Ceci permet de ne pas utiliser le canal de service de la Connexion CIP Motion pour d'autres demandes de service pendant toute la durée du processus. Une fois le processus d'essai actif achevé, l'état d'axe passe de l'état Essai à l'état Arrêté ou à l'état Défaut principal. Une fois cette transition d'état achevée, le bit En cours de processus est annulé.

Ce service prend en charge plusieurs types d'essai différents (voir Tableau 119). L'essai statique applique un flux statique directionnel au moteur, les essais dynamiques étant réalisés sur le moteur et générant un mouvement. Lorsque le Type d'essai est Calculé, le dispositif d'entraînement évalue les paramètres du modèle de moteur sur l'hypothèse selon laquelle le moteur correspond parfaitement à la puissance assignée du dispositif d'entraînement. Les paramètres mesurés par ces essais sont extraits par le contrôleur grâce au service Get_Motor_Test_Data.

Tableau 119 – Structure de demande Run_Motor_Test

Nom	Type	Description du paramètre de demande	Sémantique des valeurs
Type d'essai	USINT	Énumération qui spécifie le type d'essai du moteur à effectuer:	Énumération: 0 = essai statique 1 = essai dynamique 2 = calculé 3 à 255 = réservé

Il n'existe pas de paramètres de réponse spécifiques pour ce service.

7.5.2.7 Get_Motor_Test_Data

Le service Get Motor Test Data permet d'accéder aux mesures de paramètre moteur générées par la dernière commande Run_Motor_Test. Le contrôleur utilise ces données pour mettre à jour le dispositif avec les meilleures estimations des paramètres moteur critiques. Les paramètres renvoyés dépendent du Type de Moteur configuré au moment du dernier service Run_Motor_Test. Le service Get_Motor_Test_Data renvoie une valeur nulle pour les paramètres qui n'ont pas été mesurés ou calculés de manière explicite par la dernière procédure Run_Motor_Test. Le Tableau 120 répertorie les paramètres du moteur qui sont généralement mesurés ou calculés par le processus Run_Motor_Test selon le Type d'essai.

Tableau 120 – Get_Motor_Test_Data mesurées par le type d'essai

Paramètres moteur	Type d'essai		
	Statique	Dynamique	Calculé
Résistance statorique (tous)	oui	oui	oui
Inductance de fuite (IM, SPM)	oui	oui	oui
Courant de flux (IM)	non	oui	oui
Vitesse de glissement (IM)	non	oui	oui
Force contre-électromotrice (PM)	non	oui	non
Vitesse maximale (IPM)	non	oui	non
Inductance Lq (IPM)	non	oui	non
Inductance Ld (IPM)	non	oui	non

Pour différencier les données d'essai moteur normalisées de celles spécifiques au fournisseur, un paramètre Données d'essai moteur réinitialisées est inclus de manière facultative dans la demande de service Get_Motor_Test_Data. Si la valeur des Données

d'essai moteur réinitialisées n'est pas incluse dans la demande de service, la demande de service s'applique aux données d'essai moteur normalisées.

Les paramètres (facultatifs) de demande pour ce service sont définis dans le Tableau 121.

Tableau 121 – Structure de demande Get_Motor_Test_Data (facultatif)

Nom	Type	Description du paramètre de demande	Sémantique des valeurs
Données d'essai moteur réinitialisées	USINT	Énumération qui spécifie que les données d'essai moteur réinitialisées pour le service doivent être renvoyées en fonction du Type de moteur configuré	Énumération: 0 = données normalisées 1 = données spécifiques au fournisseur 2 à 255 = réservé

Les paramètres de réponse pour ce service sont définis dans le Tableau 122, le Tableau 123 et le Tableau 124.

Si l'Essai moteur venait à être abandonné, à être expiré ou à ne pas aboutir pour quelque raison que ce soit, le dispositif doit renvoyer les valeurs de paramètre qui ont été mesurées avec succès pendant l'essai et réinitialiser les paramètres qui n'ont pas été mesurés avec succès.

**Tableau 122 – Structure normalisée de réponse Get_Motor_Test_Data
(Type de moteur = à induction)**

Nom	Type	Description du paramètre de réponse	Sémantique des valeurs
État d'essai	USINT	Indique le résultat de la dernière commande Essai moteur.	Énumération: 0 = processus d'essai abouti 1 = essai en cours 2 = processus d'essai abandonné 3 = processus d'essai expiré 4 = processus d'essai non abouti
(Réservé)	USINT[3]	Octets de remplissage pour l'alignement de mots à 32 bits	
Résistance statorique	REAL	Résistance statorique entre phases mesurée du moteur.	Ohms
Inductance de fuite	REAL	Inductance de fuite entre phases mesurée du moteur.	Henries
Courant de flux	REAL	Courant mesuré permettant d'obtenir tout le flux du moteur à induction	Ampères
Vitesse de glissement	REAL	Glissement mesuré au courant assigné du moteur à induction	r/min (moteur rotatif) m/s (moteur linéaire)
(Réservé)	REAL	Remplissage pour rétrocompatibilité	

**Tableau 123 – Structure normalisée de réponse Get_Motor_Test_Data
(Type de moteur = SPM)**

Nom	Type	Description du paramètre de réponse	Sémantique des valeurs
État d'essai	USINT	Indique le résultat de la dernière commande Essai moteur.	Énumération: 0 = processus d'essai abouti 1 = essai en cours 2 = processus d'essai abandonné 3 = processus d'essai expiré 4 = processus d'essai non abouti
(Réservé)	USINT[3]	Octets de remplissage pour l'alignement de mots à 32 bits	
Résistance statorique	REAL	Résistance statorique entre phases mesurée du moteur.	Ohms
Inductance de fuite	REAL	Inductance de fuite entre phases mesurée du moteur.	Henries
(Réservé)	REAL	Remplissage pour rétrocompatibilité	
(Réservé)	REAL	Remplissage pour rétrocompatibilité	
Force contre-électromotrice	REAL	Chute de tension CEMF mesurée à la vitesse assignée	Volts

**Tableau 124 – Structure normalisée de réponse
Get_Motor_Test_Data (Type de moteur = IPM)**

Nom	Type	Description du paramètre de réponse	Sémantique des valeurs
État d'essai	USINT	Indique le résultat de la dernière commande Essai moteur.	Énumération: 0 = processus d'essai abouti 1 = essai en cours 2 = processus d'essai abandonné 3 = processus d'essai expiré 4 = processus d'essai non abouti
(Réservé)	USINT[3]	Octets de remplissage pour l'alignement de mots à 32 bits	
Résistance statorique	REAL	Résistance statorique entre phases mesurée du moteur.	Ohms
Inductance Lq	REAL	Inductance statorique d'axe q entre phases mesurée du moteur.	Henries
Inductance Ld	REAL	Inductance statorique d'axe d entre phases mesurée du moteur.	Henries
Force contre-électromotrice	REAL	Chute de tension CEMF mesurée à la vitesse assignée	Volts

7.5.2.8 Run_Inertia_Test

Le service Run_Inertia_Test réalise une rampe d'accélération et de décélération sur l'axe et procède à des mesures temporelles dans le processus (voir Tableau 125).

S'agissant d'un essai actif qui active la structure de puissance, l'état d'axe doit être Arrêté pour que la demande de ce service soit acceptée. Le Run_Inertia_Test déclenche le processus d'essai spécifié et définit le bit En cours de processus dans l'attribut État d'axe. Pour cet essai actif, l'état d'axe passe à l'état Essai. Dans la mesure où la réalisation d'un processus d'essai actif peut durer relativement longtemps, une réponse de service est renvoyée au contrôleur dès le lancement du processus d'essai. Ceci permet de ne pas utiliser le canal de service de la Connexion CIP Motion pour d'autres demandes de service pendant toute la durée du processus. Une fois le processus d'essai actif achevé, l'état d'axe passe de l'état Essai à l'état Arrêté ou à l'état Défaut principal. Une fois cette transition d'état achevée, le bit En cours de processus est annulé.

Le contrôleur peut accéder aux mesures temporelles résultantes grâce à la commande `Get_Inertia_Test_Data`, et ces mesures sont utilisées en dernier lieu pour calculer une valeur d'inertie précise pour le moteur et la charge.

Tableau 125 – Structure de demande `Run_Inertia_Test`

Nom	Type	Description du paramètre de demande	Sémantique des valeurs
Sens de l'essai	USINT	Énumération qui sélectionne le sens de l'essai.	Énumération: 0 = Vers l'avant 1 = inversé 2 à 255 = réservé
(Réservé)	USINT[3]	Octets de remplissage pour l'alignement de mots à 32 bits	
Vitesse d'essai	REAL	Détermine la vitesse maximale atteinte pendant le profil d'essai.	Moteur Unités / s Pour les moteurs linéaires, Unité moteur = Mètres, pour les moteurs rotatifs, Unité moteur = Rotations.
Couple d'essai	REAL	Détermine le couple maximal appliqué pendant le profil d'essai.	% de la valeur assignée du moteur
Limite de course d'essai	REAL	Établit l'excursion maximale de l'axe en unités moteur admise dans le sens de l'essai.	Unités moteur Pour les moteurs linéaires, Unité moteur = Mètres, pour les moteurs rotatifs, Unité moteur = Rotations.

Il n'existe pas de paramètres de réponse spécifiques pour ce service.

7.5.2.9 `Get_Inertia_Test_Data`

Le service `Get_Inertia_Test_Data` permet d'accéder aux mesures temporelles générées par le dernier service `Run_Inertia_Test`. Le contrôleur utilise ces données pour calculer en dernier lieu une valeur d'inertie précise pour le moteur et la charge. Le service `Get_Inertia_Test_Data` renvoie une valeur nulle pour les paramètres qui ne sont pas mesurés avec succès par la dernière procédure `Run_Inertia_Test`.

Il n'existe pas de paramètres de demande spécifiques pour ce service.

Les paramètres de réponse pour ce service sont définis dans le Tableau 126. Si l'Essai Inertie venait à être abandonné, à être expiré ou à ne pas aboutir pour quelque raison que ce soit, le dispositif doit renvoyer les valeurs de paramètre qui ont été mesurées avec succès pendant l'essai et réinitialiser les paramètres qui n'ont pas été mesurés avec succès.

Tableau 126 – Structure de réponse Get_Inertia_Test_Data

Nom	Type	Description du paramètre de réponse	Sémantique des valeurs
État d'essai	USINT	Énumération qui indique le résultat du dernier service Run_Inertia_Test:	Énumération: 0 = processus d'essai abouti 1 = essai en cours 2 = processus d'essai abandonné 3 = processus d'essai expiré 4 = processus d'essai non abouti 5 = l'essai a atteint la limite 6 = polarité d'essai incorrecte 7 = erreur de mesure d'essai 8 = erreur de configuration d'essai 9 à 255 = réservé
(Réservé)	USINT[3]	Octets de remplissage pour l'alignement de mots à 32 bits	
Durée d'accélération	REAL	Durée mesurée de l'accélération permettant d'atteindre la vitesse d'essai.	Secondes
Durée de décélération	REAL	Durée mesurée de la décélération permettant d'atteindre une vitesse nulle à partir de la vitesse d'essai.	Secondes

7.5.2.10 Run_Hookup_Test

Le service Run_Hookup_Test réalise un certain nombre d'essais différents visant à vérifier que l'interface avec le moteur et/ou le dispositif de réaction est correcte. Pour les essais actifs qui activent la structure de puissance du dispositif d'entraînement, l'état d'axe doit être Arrêté pour que la demande de ce service soit acceptée. Pour les essais passifs qui n'activent pas la structure de puissance, l'état d'axe peut être Arrêté ou dans les états d'Attente suivants: Interruption, Précharge ou Inhibition de démarrage. Le Run_Hookup_Test déclenche le processus d'essai spécifié et définit le bit En cours de processus dans l'attribut État d'axe. Pour les essais actifs, l'état d'axe passe à l'état Essai. Dans la mesure où la réalisation d'un processus d'essai actif peut durer relativement longtemps, une réponse de service est renvoyée au contrôleur dès le lancement du processus d'essai. Ceci permet de ne pas utiliser le canal de service de la Connexion CIP Motion pour d'autres demandes de service pendant toute la durée du processus. Une fois le processus d'essai actif achevé, l'état d'axe passe de l'état Essai à l'état Arrêté ou à l'état Défaut principal. Une fois cette transition d'état achevée, le bit En cours de processus est annulé.

Ce service prend en charge plusieurs types d'essai différents. Une description succincte de chacun de ces types d'essai est donnée ci-dessous.

- L'essai Moteur/Réaction est un essai actif qui tente de déplacer le moteur dans le sens avant pour une distance spécifiée donnée par la Distance d'essai, ou pendant une durée spécifiée donnée par la Durée d'essai. Le sens Vers l'avant est défini comme le sens de mouvement du moteur lorsqu'il suit le séquençement de phases UVW normal. Pendant l'essai, le dispositif contrôle le changement du nombre de réactions sur la durée de l'essai pour les canaux de réaction applicables. Lorsque le mode de réaction est défini pour une Réaction double ou une Réaction de charge, les deux canaux de réaction sont soumis à l'essai.
- L'essai Réaction est un essai passif qui permet de surveiller l'axe lorsqu'un agent externe le déplace et indique qu'il a abouti si le nombre de réactions d'axe dépasse la distance spécifiée par la distance d'essai. Le paramètre Canal de réaction détermine quel canal de réaction est tenu d'être contrôlé.
- L'essai Marqueur est un essai passif qui permet de surveiller le signal de marquage du canal de réaction de positionnement d'axe lorsqu'un agent externe le déplace le long de l'emplacement du marqueur et indique que l'essai a abouti lorsqu'une impulsion de marquage est détectée. Le paramètre Canal de réaction détermine quel signal de marquage est tenu d'être soumis à l'essai.

- L'essai de commutation, qui concerne uniquement les moteurs à aimant permanent, est un essai actif qui applique un courant au moteur afin d'aligner le rotor, vérifier la bonne mise en phase d'un codeur UVW ou d'un capteur à effet Hall, le cas échéant, et mesurer le décalage de commutation. L'essai de commutation cible toujours Réaction 1, qui est le canal de réaction du moteur.

Une limite de temporisation propre au service Run_Hookup_Test est appliquée par le fournisseur du dispositif afin de traiter du cas où les critères de réussite ne peuvent pas être satisfaits. Cette limite est généralement de 30 s environ.

Les données obtenues générées par ces essais sont accessibles par le service Get Test Data, et sont utilisées par le contrôleur pour définir automatiquement les polarités correctes de l'interface de réaction et de l'interface du moteur.

Les paramètres de demande pour ce service sont définis dans le Tableau 127.

Tableau 127 – Structure de demande Run_Hookup_Test

Nom	Type	Description du paramètre de demande	Sémantique des valeurs
Type d'essai	USINT	Détermine l'essai de connexion spécifique à exécuter:	Énumération: 0 = essai moteur/réaction 1 = essai de réaction 2 = essai de marquage 3 = essai de commutation (F) 4 à 255 = réservé
Canal de réaction	USINT	Détermine le numéro de canal logique ciblé pour l'essai de réaction. Lorsque cela n'est pas applicable, ce paramètre doit être mis à 0.	Numéro de canal
(Réservé)	USINT[2]	Octets de remplissage pour l'alignement de mots à 32 bits	
Distance d'essai	REAL	Établit la distance à parcourir par l'axe, en unités moteur ou unités de réaction n, afin d'indiquer qu'un essai a abouti. Le type d'essai détermine les unités. Un type d'essai "essai moteur & réaction" utilise les unités moteur. Un type d'essai "essai de réaction" utilise les unités de réaction où n est déterminé par le canal de réaction. La distance d'essai est valide si l'axe n'est pas configuré pour un fonctionnement sans codeur/capteur. Lorsque cela n'est pas applicable, ce paramètre doit être mis à 0.	Unités moteur ou unités de réaction n
Durée d'essai	REAL	Établit la durée de mouvement d'axe pendant l'essai. La durée d'essai est valide si l'axe est configuré pour un fonctionnement sans codeur/capteur. Lorsque cela n'est pas applicable, cette valeur doit être mise à 0.	secondes

Il n'existe pas de paramètres de réponse spécifiques pour cette demande de service.

7.5.2.11 Get_Hookup_Test_Data

Le service Get_Hookup_Test_Data permet d'accéder aux résultats d'essai de connexion générés par le dernier service Run_Hookup_Test (voir Tableau 128). Le contrôleur utilise ces données pour signaler un problème de câblage à l'utilisateur et pour calculer les paramètres de bit de configuration de polarité d'axe. Les résultats de cet essai sont indépendants des valeurs configurées pour la polarité du moteur, de réaction et de commutation. Le service Get_Hookup_Test_Data renvoie une valeur nulle pour les paramètres qui ne sont pas déterminés par la dernière procédure Run_Hookup_Test.

Tableau 128 – Get_Hookup_Test_Data mesurées par le type d'essai

Paramètres Get_Hookup_Test	Type d'essai			
	Moteur/Réaction	Réaction	Marqueur	Commutation
Décalage de commutation	non	non	non	oui
Polarité de commutation	non	non	non	oui
Direction de réaction 1	Oui ^a	oui ^b	non	non
Direction de réaction 2	oui ^a	oui ^b	non	non
^a Lors de l'exécution d'un essai de connexion Moteur/Réaction, la direction de réaction 1 est toujours mesurée à moins que le mode Réaction ne soit défini sur Pas de réaction (fonctionnement sans codeur/capteur). La direction de réaction 2 est mesurée lorsque le mode Réaction est défini sur Réaction double ou Réaction de charge. ^b Lors de l'exécution d'un essai de connexion de réaction, les directions de réaction 1 et 2 sont toutes deux mesurées en fonction du canal de réaction spécifié par le dernier Run_Hookup_Test.				

Il n'existe pas de paramètres de demande spécifiques pour ce service.

Les paramètres de réponse pour ce service sont définis dans le Tableau 129. Si l'Essai Connexion venait à être abandonné, à être expiré ou à ne pas aboutir pour quelque raison que ce soit, le dispositif doit renvoyer les valeurs de paramètre qui ont été mesurées avec succès pendant l'essai et réinitialiser les paramètres qui n'ont pas été mesurés avec succès.

Tableau 129 – Structure de réponse Get_Hookup_Test_Data

Nom	Type	Description du paramètre de réponse	Sémantique des valeurs
État d'essai	USINT	Énumération qui indique le résultat de la dernière commande de l'essai de connexion:	Énumération: 0 = processus d'essai abouti 1 = essai en cours 2 = processus d'essai abandonné 3 = processus d'essai expiré 4 = processus d'essai non abouti 5 = échec essai pas de réaction 1 6 = échec essai pas de réaction 2 7 à 255 = réservé
Direction de réaction 1	USINT	Énumération qui indique le sens de mouvement d'axe associé à la dernière commande de l'essai d'exécution au cours d'un essai de réaction 1 ou de moteur/réaction, le cas échéant. Sens positif indique que l'essai a produit des nombres de réactions 1 à incrémentation positive.	Énumération: 0 = Positive 1 = Négative 2 à 255 = réservé
Direction de réaction 2	USINT	Énumération qui indique le sens de mouvement d'axe associé à la dernière commande de l'essai d'exécution au cours d'un essai de réaction 2 ou de moteur/réaction, le cas échéant. Sens positif indique que l'essai a produit des nombres de réactions 2 à incrémentation positive.	Énumération: 0 = Positive 1 = Négative 2 à 255 = réservé
Polarité de commutation	USINT	Lors de la réalisation d'un essai de commutation sur un moteur à aimant permanent, l'essai vérifie si la mise en phase UVW du codeur ou du capteur à effet Hall correspond à la mise en phase du moteur. Si la mise en phase correspond, la polarité de commutation est Normale. Lorsqu'il est établi que la mise en phase du moteur et du dispositif de commutation ne correspond pas, ce paramètre indique que la mise en phase UVW est inversée. Voir l'attribut Polarité de commutation pour plus de détails.	Énumération: 0 = Normale 1 = Inversée 2 à 255 = réservé
Décalage de commutation	REAL	Décalage mesuré lors de l'essai de commutation qui doit être appliqué à l'accumulateur de position du moteur afin d'aligner le signal Angle électrique sur les enroulements statoriques du moteur. Voir l'attribut Décalage de commutation.	Degrés électriques

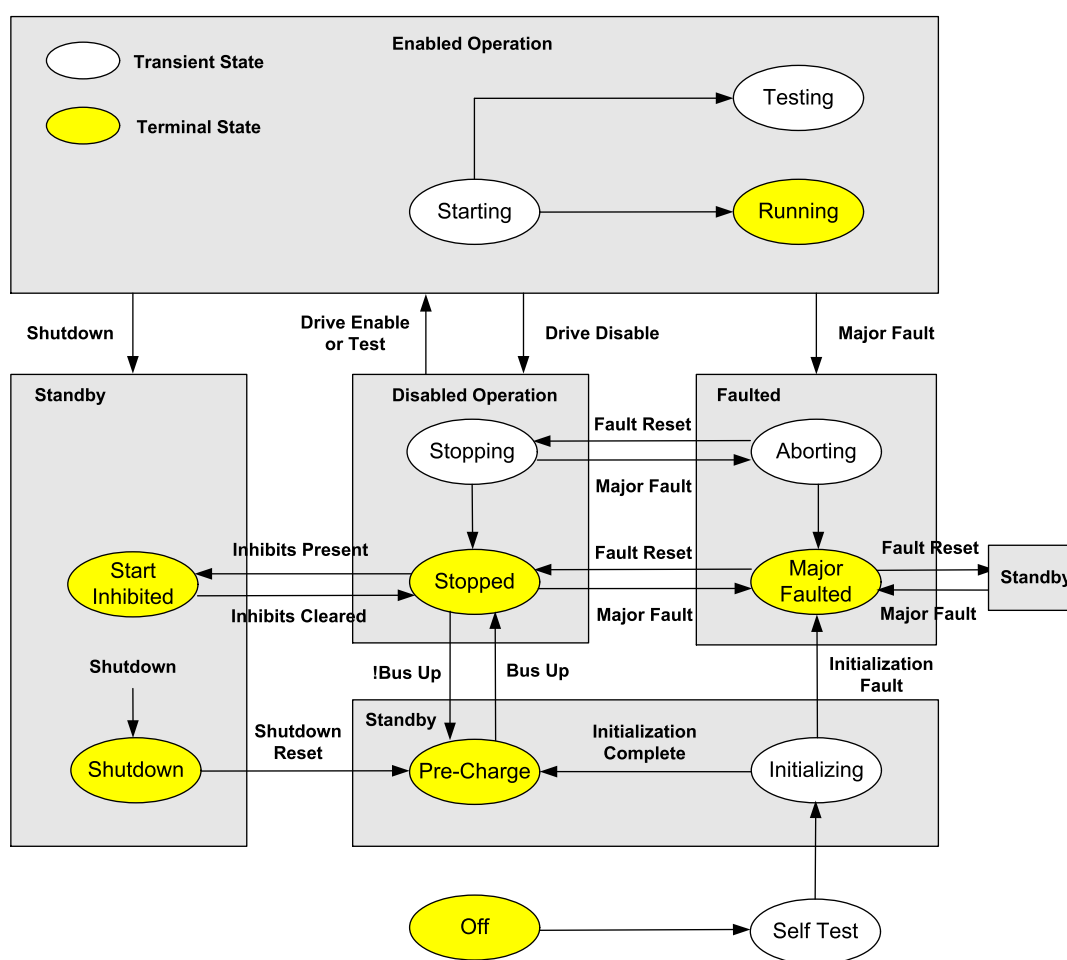
7.6 Comportement

7.6.1 Modèle d'état

7.6.1.1 Généralités

Le modèle d'état Objet Axe de dispositif de mouvement repose sur les éléments des modèles d'état standard S88 et Pack/ML. L'état en cours de l'instance Objet Axe de dispositif de mouvement est indiqué par l'attribut État d'axe (ID attr = 650). Les transitions d'état peuvent être initiées directement grâce au mécanisme de demande Commande d'axe ou par des conditions se produisant lors du fonctionnement du dispositif.

La Figure 87 présente les états de fonctionnement de base de l'Objet Axe de dispositif de mouvement lors du contrôle du mouvement de l'axe (Mode de commande != Pas d'asservissement). Les zones grisées représentent la mise en correspondance des États d'axe avec les états Objet Identité correspondants.



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Anglais	Français
Transient State	État transitoire
Terminal State	État final
Enabled Operation	Fonctionnement activé
Starting	Démarrage
Testing	Essais

Anglais	Français
Running	Exécution
Shutdown	Interruption
Drive Enable or Test	Activer le dispositif d'entraînement ou Essai
Drive Disable	Désactiver le dispositif d'entraînement
Major Fault	Défaut principal
Standby	Attente
Start Inhibited	Démarrage inhibé
Disabled Operation	Fonctionnement désactivé
Stopping	Arrêt
Stopped	Arrêté
Aborting	Abandon
Major Faulted	Défaut principal
Pre-Charge	Précharge
Initialization Complete	Initialisation achevée
Initializing	Initialisation
Off	Désactivé
Self Test	Autotest
Inhibits Present	Inhibitions présentes
Inhibits Cleared	Inhibitions supprimées
Major Fault	Défaut principal
Fault Reset	Réinitialisation suite à un défaut
Faulted	Défaut
Bus Up	Bus activé
Shutdown Reset	Réinitialisation suite à une interruption
Initialization fault	Défaut d'initialisation

Figure 87 – Modèle d'état de l'Objet Axe de dispositif de mouvement

Les transitions valides pour le Modèle d'état d'axe sont définies de manière explicite dans le Tableau 130.

Tableau 130 – Transitions de diagramme d'états d'axe

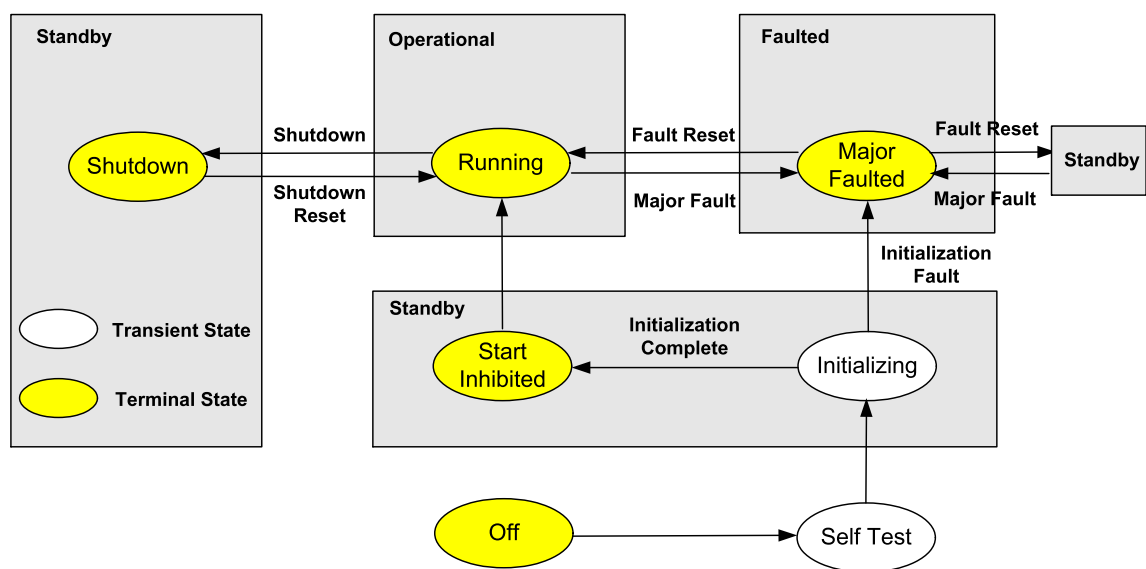
État en cours	Événement	Conditions	État suivant
Désactivé	Alimentation activée		Autotest
Autotest	Autotest achevé		Initialisation
Initialisation	Défaut d'initialisation		Défaut principal
Initialisation	Initialisation achevée		Précharge
Interruption	Défaut principal ou abandon		Défaut principal
Interruption	Réinitialisation suite à une interruption		Précharge
Précharge	Interruption		Interruption
Précharge	Défaut principal		Défaut principal
Précharge	Bus activé		Arrêté
Démarrage inhibé	Interruption		Interruption
Démarrage inhibé	Défaut principal		Défaut principal
Démarrage inhibé	Inhibitions supprimées		Arrêté
Défaut principal	Réinitialisation suite à un défaut	SD = 1	Interruption
Défaut principal	Réinitialisation suite à un défaut	SD = 0, BU = 0	Précharge
Défaut principal	Réinitialisation suite à un défaut	SD = 0, BU = 1, SI > 0	Démarrage inhibé
Défaut principal	Réinitialisation suite à un défaut	SD = 0, BU = 1, SI = 0	Arrêté
Défaut principal	Reconnexion		Initialisation
Arrêté	!Bus activé		Précharge
Arrêté	Interruption		Interruption
Arrêté	Défaut principal		Défaut principal
Arrêté	Inhibition de démarrage		Inhibition de démarrage
Arrêté	Activer		Démarrage
Arrêté	Essai		Démarrage
Démarrage	Interruption		Interruption
Démarrage	Défaut principal ou abandon		Abandon
Démarrage	Désactiver		Arrêt
Démarrage	Démarrage achevé	IP = 0	Exécution
Démarrage	Démarrage achevé	IP = 1	Essais
Arrêt	Interruption		Interruption
Arrêt	Arrêt achevé		Arrêté
Arrêt	Défaut principal		Abandon
Arrêt	Activer	Amorçage instantané activé	Démarrage
Abandon	Arrêt achevé		Défaut principal
Abandon	Réinitialisation suite à un défaut		Arrêt
Essais	Interruption		Interruption
Essais	Défaut principal ou abandon		Abandon
Essais	Désactiver		Arrêt
Exécution	Interruption		Interruption
Exécution	Défaut principal ou abandon		Abandon
Exécution	Désactiver		Arrêt
Tout état	Fermeture de connexion		Initialisation
Tout état	Perte de connexion		Défaut principal

Certaines des transitions de diagramme d'états d'axe du Tableau 130 présentent des dépendances aux conditions de statut en cours de l'axe définies dans le Tableau 131.

Tableau 131 – Conditions de diagramme d'états d'axe

Condition	Symbole	Description
Bus activé	BU	Le bit Statut d'axe est défini si le bus à courant continu est chargé dans les limites de la plage de fonctionnement.
Interruption	SD	Le bit Statut d'axe est défini en présence d'une interruption active. Ceci est différent de l'état d'interruption. Le bit Statut d'interruption peut être actif à l'état en défaut.
En cours de processus	IP	Le bit Statut d'axe indique qu'un processus d'essai associé à une demande de service est actif. Noter que le bit En cours de processus peut être actif sans que le diagramme d'états de l'axe soit à l'état Essais.
Inhibition de démarrage	SI	Ceci représente l'ensemble des Inhibitions de démarrage actives. Une valeur SI de 0 indique l'absence d'inhibition de démarrage active. Noter que les inhibitions de démarrage peuvent être présentes sans que le diagramme d'états de l'axe soit à l'état inhibition de démarrage.

Lorsque l'Objet Axe de dispositif de mouvement n'est pas en train de contrôler le mouvement de l'axe (Mode de commande = Pas d'asservissement), le diagramme d'état à la Figure 87 se réduit à celui présenté à la Figure 88 pour un axe Réaction uniquement ou un type de dispositif Codeur CIP Motion.



IEC

Anglais	Français
Standby	Attente
Operational	Opérationnel
Faulted	Défaut
Shutdown	interruption
Shutdown Reset	Réinitialisation suite à une interruption
Running	Exécution
Fault Reset	Réinitialisation suite à un défaut
Major Fault	Défaut principal
Initialization Complete	Initialisation achevée
Initialization Fault	Défaut d'initialisation
Initializing	Initialisation
Off	Désactivé
Self Test	Autotest
Transient State	État transitoire
Terminal State	État final

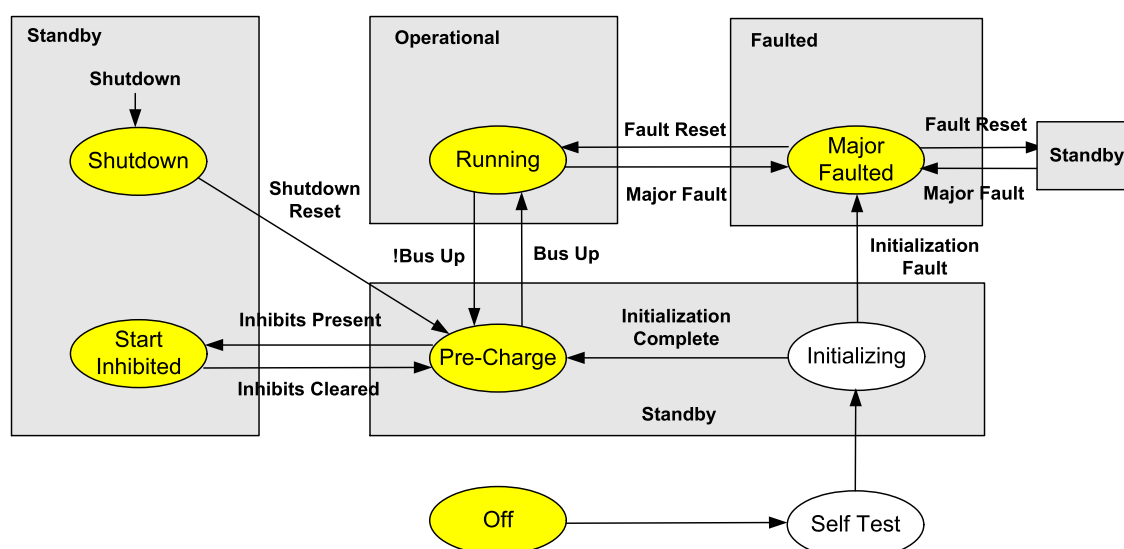
Figure 88 – Modèle d'état de l'Objet Axe de dispositif de mouvement pour le mode Réaction uniquement

Les transitions valides pour le Modèle d'état d'axe d'un axe Réaction uniquement ou CIP Motion sont définies de façon explicite dans le Tableau 132.

Tableau 132 – Transitions de diagramme d'états d'axe (Réaction uniquement)

État en cours	Événement	Conditions	État suivant
Désactivé	Alimentation activée		Autotest
Autotest	Autotest achevé		Initialisation
Initialisation	Défaut d'initialisation		Défaut principal
Initialisation	Initialisation achevée		Démarrage inhibé
Interruption	Défaut principal		Défaut principal
Interruption	Réinitialisation suite à une interruption		Exécution
Démarrage inhibé	Défaut principal		Défaut principal
Démarrage inhibé	Inhibitions supprimées		Exécution
Défaut principal	Réinitialisation suite à un défaut	SD = 1	Interruption
Défaut principal	Réinitialisation suite à un défaut	SD = 0, SI > 0	Démarrage inhibé
Défaut principal	Réinitialisation suite à un défaut	SD = 0, SI = 0	Exécution
Défaut principal	Reconnexion		Initialisation
Exécution	Interruption		Interruption
Exécution	Défaut principal		Défaut principal
Tout état	Fermeture de connexion		Initialisation
Tout état	Perte de connexion		Défaut principal

Lorsque l'Objet Axe de dispositif de mouvement est associé à un convertisseur CIP Motion, le diagramme d'états Commande active à la Figure 87 se réduit à celui présenté à la Figure 89.



IEC

Anglais	Français
Standby	Attente
Operational	Opérationnel
Faulted	Défaut
Shutdown	interruption
Shutdown Reset	Réinitialisation suite à une interruption
Running	Exécution
Fault Reset	Réinitialisation suite à un défaut
Major Fault	Défaut principal
Major Faulted	Défaut principal
Bus up	Bus activé
Initialization Complete	Initialisation achevée
Initialization Fault	Défaut d'initialisation
Initializing	Initialisation
Off	Désactivé
Self Test	Autotest
Start inhibited	Démarrage inhibé
Inhibits Present	Inhibitions présentes
Inhibits Cleared	Inhibitions supprimées
Pre-charge	précharge

Figure 89 – Modèle d'état de l'Objet Axe de dispositif de mouvement pour le Convertisseur

Les transitions valides pour le Modèle d'état d'axe d'un axe Convertisseur CIP Motion sont définies de façon explicite dans le Tableau 133.

Tableau 133 – Transitions de diagramme d'états d'axe (Convertisseur)

État en cours	Événement	Conditions	État suivant
Désactivé	Alimentation activée		Autotest
Autotest	Autotest achevé		Initialisation
Initialisation	Défaut d'initialisation		Défaut principal
Initialisation	Initialisation achevée		Précharge
Interruption	Défaut principal		Défaut principal
Interruption	Réinitialisation suite à une interruption		Précharge
Démarrage inhibé	Inhibitions supprimées		Précharge
Précharge	Inhibition de démarrage		Démarrage inhibé
Précharge	Interruption		Interruption
Précharge	Défaut principal		Défaut principal
Précharge	Bus activé		Exécution
Défaut principal	Réinitialisation suite à un défaut	SD = 1	Interruption
Défaut principal	Réinitialisation suite à un défaut	SD = 0	Précharge
Défaut principal	Reconnexion		Initialisation
Exécution	!Bus activé		Précharge
Exécution	Interruption		Interruption
Exécution	Défaut principal		Défaut principal
Tout état	Fermeture de connexion		Initialisation
Tout état	Perte de connexion		Défaut principal

Étapes de transition d'état Réinitialisation suite à un défaut

Dans l'état Défaut principal présenté dans les diagrammes d'états d'axe ci-dessus, l'état d'axe peut passer à l'un des différents états existants en réponse à un événement Réinitialisation suite à un défaut, ou l'état d'axe peut revenir directement à l'état Défaut principal en présence d'une condition de défaut persistante. L'état auquel passe l'axe dépend des autres conditions d'état/statut d'axe. En fait, il peut être possible que plusieurs conditions d'état soient présentes simultanément, par exemple, Interruption, Démarrage inhibé, etc. Étant donné que le modèle d'état d'axe peut représenter uniquement un seul axe à un instant donné, l'état d'axe est déterminé selon la condition préalable suivante:

- Défaut principal
- Interruption
- Précharge
- Démarrage inhibé
- Arrêté

Cet aspect et tout autre comportement de passage à un état sont examinés plus en détail en 7.6.1.2 et en 7.6.2.

7.6.1.2 Commande d'état

La méthode principale de modification de l'état d'une instance Objet Axe de dispositif de mouvement consiste à utiliser le mécanisme Commande d'axe/Réponse d'axe intégré à l'entête Connexion E/S CIP Motion. La modification de l'état d'axe de mouvement est simplement réalisée en plaçant le Code de demande approprié dans l'élément Commande d'axe de l'entête Connexion Contrôleur-Dispositif. Les codes de demande de commande sont définis dans

le profil de dispositif CIP Motion et reproduits dans le Tableau 134 pour des raisons de commodité.

Tableau 134 – Code de demande de Commande d'axe

Code de demande	Opération demandée
F	Pas de demande
1	Demande d'activation
2	Demande de désactivation
3	Demande d'interruption
4	Demande de réinitialisation suite à une interruption
5	Demande d'abandon
6	Demande de réinitialisation suite à un défaut
7	Demande d'arrêt de processus
8	Demande de changement de référence de position réelle
9	Demande de changement de référence de position de consigne
10 à 126	(Réservé)
127	Demande d'annulation
128 à 255	(Spécifique au fournisseur)

Si une transition d'état est demandée via l'élément Commande d'axe de la connexion Contrôleur-Dispositif, le dispositif initie la transition d'état, puis acquitte la demande de transition via l'attribut Réponse d'axe à l'issue de la transition d'état demandée ou si elle n'a pas abouti. Les Codes d'acquiescement de l'élément Réponse d'axe sont identiques aux Codes de demande de l'élément Commande d'axe comme défini dans le profil de dispositif CIP Motion (voir Tableau 135).

Tableau 135 – Codes d'acquiescement de réponse d'axe

Code d'acquiescement	Réponse d'axe
F	Pas d'acquiescement
1	Acquiescement d'activation
2	Acquiescement de désactivation
3	Acquiescement d'interruption
4	Acquiescement de réinitialisation suite à une interruption
5	Acquiescement d'abandon
6	Acquiescement de réinitialisation suite à un défaut
7	Acquiescement d'arrêt de processus
8	Acquiescement de changement de référence de position réelle
9	Acquiescement de changement de référence de position de consigne
127	Acquiescement d'annulation

Les critères d'exécution réussie du fonctionnement demandé dépendent du Code de demande spécifique présenté dans le Tableau 136.

Tableau 136 – Critères d'exécution réussie du fonctionnement demandé

Code de demande	Critères d'exécution réussie
1	Passage réussi d'axe à l'état Exécution avec le bit Commande de suivi défini dans l'attribut Statut d'axe.
2	Passage réussi d'axe de l'état Fonctionnement activé avec le bit Commande de suivi annulé dans l'attribut Statut d'axe.
3	Passage réussi d'axe à l'état Interruption ou l'état Défaut, avec le bit Interruption défini dans l'attribut Statut d'axe.
4	Passage réussi d'axe à l'état Préchargé Arrêté ou Inhibé avec le bit Interruption annulé dans l'attribut Statut d'axe, ainsi que l'attribut Défauts d'axe également annulé provisoirement. ^a Lorsque l'état d'axe n'est pas Défaut ou Interruption, l'opération Réinitialisation suite à une interruption n'a aucun impact sur l'axe.
5	Passage réussi d'axe à l'état Abandon ou Défaut avec le bit Exception initiée par le contrôleur défini dans l'attribut défauts d'axe.
6	Passage réussi d'axe à l'état Préchargé Arrêté, En cours d'arrêt, Inhibé ou Interruption avec l'attribut Défauts d'axe annulé provisoirement. ^a Lorsque l'état d'axe n'est pas Défaut, l'opération Réinitialisation suite à un défaut n'a aucun impact sur l'axe.
7	L'axe arrête avec succès un processus actif, annulant le bit En cours de processus dans l'attribut Statut d'axe.
8	L'axe a appliqué le mouvement à la Position réelle avec succès afin de passer à un nouveau système de référence de position.
9	L'axe a appliqué le mouvement à la Position de consigne avec succès afin de passer à un nouveau système de référence de position.
127	S'accomplit toujours.
^a Dans le cas d'une Réinitialisation suite à un défaut ou suite à une interruption, et lorsque des conditions d'exception produisant un défaut persistant, l'axe quitte uniquement provisoirement l'état Défaut dans la mesure où il doit ne plus fonctionner une nouvelle fois de manière immédiate et revenir à l'état Défaut.	

Si le dispositif détermine que la transition d'état demandée ne peut pas être accomplie avec succès, il acquitte la demande de commande avec un code d'erreur différent de zéro dans l'élément Statut de réponse. Les conditions d'erreur possibles sont présentées dans le Tableau 137.

Tableau 137 – Conditions d'erreur possibles du fonctionnement demandé

Code de demande	Conditions d'erreur possibles
1	L'axe passe à l'état Défaut ou Démarrage inhibé après réception de la demande avant d'atteindre l'état Exécution.
2	Aucune
3	Aucune
4	Aucune
5	Aucune
6	Aucune
7	Aucune
8	Aucune
127	Aucune.

Lorsque le contrôleur reçoit le Code d'acquiescement, et l'absence de demande de commande à lancer en attente, il annule alors le Code de demande correspondant. Lorsque le dispositif

détecte le Code de demande annulé, il efface le Code d'acquiescement et le Statut de réponse associé.

Si l'axe ne peut pas passer à l'état demandé, la condition d'erreur spécifique est indiquée par une valeur d'attribut Statut de réponse différente de zéro transmise avec la Réponse d'axe dans le cadre de la connexion Dispositif-Contrôleur.

Ce comportement est décrit plus en détail dans le Tableau 138 et le Tableau 139.

Tableau 138 – Cycle de demande de commande d'axe réussie

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur définit le Code de demande dans l'élément Commande d'axe pour demander un changement d'état.	→ Commande d'axe = x	Le dispositif détecte un Code de demande x différent de zéro dans l'élément Commande d'axe, et initie le changement d'état demandé.
	← Réponse d'axe = x ← Statut de réponse = 0	Si l'opération de changement d'état a abouti, le dispositif l'acquiesce en attribuant le Code de demande x d'origine au Code d'acquiescement de l'élément Réponse d'axe. L'élément Statut de réponse est nul, indiquant un succès.
Le contrôleur détecte le Code d'acquiescement différent de zéro dans l'élément Réponse d'axe et annule le Code de demande d'origine dans l'élément Commande d'axe. Le changement d'état demandé a abouti, comme le prouve le Statut de réponse nul.	→ Commande d'axe = 0	
La transaction de demande de changement d'état est à présent terminée. Une autre demande Commande d'axe peut être traitée à cet instant.	← Réponse d'axe = 0 ← Statut de réponse = 0	Le dispositif détecte un Code de demande nul dans l'élément Commande d'axe et annule le Code d'acquiescement associé dans l'élément Réponse d'axe.

Tableau 139 – Cycle de demande de commande d'axe non réussie

Action du contrôleur	Connexion	Action du dispositif
Le contrôleur définit le Code de demande dans l'élément Commande d'axe pour demander un changement d'état.	→ Commande d'axe = x	Le dispositif détecte un Code de demande x différent de zéro dans l'élément Commande d'axe, et initie le changement d'état demandé.
	← Réponse d'axe = x ← Statut de réponse = Code d'erreur	Si l'opération de changement d'état n'aboutit pas, le dispositif signale le problème par un Code d'erreur de statut de réponse non nul, avec un Code d'acquiescement correspondant au Code de demande d'origine, x.
Le contrôleur détecte le Statut de réponse différent de zéro associé au Code de demande, x, dans la Réponse d'axe, traite la condition d'erreur en fonction du Code d'erreur, puis annule le Code de demande d'origine dans l'élément Commande d'axe.	→ Commande d'axe = 0	
La transaction de demande de changement d'état est à présent terminée. Une autre demande Commande d'axe peut être traitée à cet instant.	← Réponse d'axe = 0 ← Statut de réponse = 0	Le dispositif détecte un Code de demande nul dans l'élément Commande d'axe et annule le Code d'acquiescement associé dans l'élément Réponse d'axe.

Dans le cas d'une demande de commande en attente lorsque le Code d'acquiescement propre au cycle de demande de commande précédent est traité par le contrôleur, ce dernier peut

lancer immédiatement la demande de commande en attente. Le délai d'exécution de la demande de commande en attente est ainsi réduit. Ce comportement est décrit plus en détail dans le Tableau 140.

Tableau 140 – Cycle de demande de commande d'axe en attente

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur définit le Code de demande dans l'élément Commande d'axe pour demander un changement d'état x.	→ Commande d'axe = x	Le dispositif détecte un Code de demande x différent de zéro dans l'élément Commande d'axe, et initie le changement d'état demandé.
Le contrôleur a une demande de changement d'état y en attente puisqu'il reçoit le Code d'acquiescement issu de la demande de changement d'état x	← Réponse d'axe = x ← Statut de réponse = 0	Si l'opération de changement d'état a abouti, le dispositif l'acquiesce en attribuant le Code de demande x d'origine au Code d'acquiescement x de l'élément Réponse d'axe. L'élément Statut de réponse est nul, indiquant un succès.
En fonction du Code d'acquiescement différent de zéro dans l'élément Réponse d'axe, le contrôleur définit immédiatement le Code de demande y dans l'élément Commande d'axe afin de demander le changement d'état en attente.	→ Commande d'axe = y	Le dispositif détecte un Code de demande y différent de zéro dans l'élément Commande d'axe, et initie le changement d'état demandé.
	← Réponse d'axe = y ← Statut de réponse = 0	Si l'opération de changement d'état a abouti, le dispositif l'acquiesce en attribuant le Code de demande x d'origine au Code d'acquiescement y de l'élément Réponse d'axe. L'élément Statut de réponse est nul, indiquant un succès.
Le contrôleur détecte le Code d'acquiescement différent de zéro dans l'élément Réponse d'axe et annule le Code de demande d'origine dans l'élément Commande d'axe. Le changement d'état demandé a abouti, comme le prouve le Statut de réponse nul.	→ Commande d'axe = 0	
La dernière transaction de demande de changement d'état est à présent terminée. Une autre demande Commande d'axe peut être traitée à cet instant.	← Réponse d'axe = 0 ← Statut de réponse = 0	Le dispositif détecte un Code de demande nul dans l'élément Commande d'axe et annule le Code d'acquiescement associé dans l'élément Réponse d'axe.

Les transitions d'état de l'objet peuvent également être initiées par des demandes de service (Essais, par exemple). En général, ces demandes de changement d'état utilisent des services car elles impliquent de transmettre un ou plusieurs paramètres au dispositif afin d'initier le changement d'état, ou ne sont pas essentielles pour les performances en temps réel.

Dans certains cas, l'opération Demande de commande peut durer relativement longtemps, par exemple une demande Activation. Pour empêcher que ce type de demande ne monopolise le mécanisme de demande de commande, un code Demande d'annulation spécial est défini. La demande d'annulation est acquiescée immédiatement par le dispositif afin d'exécuter le cycle d'établissement d'une liaison, autorisant ainsi une demande de commande ultérieure (voir Tableau 141).

Tableau 141 – Cycle de demande d'annulation

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur définit le Code de demande dans l'élément Commande d'axe pour demander un changement d'état, une demande Activation dans cet exemple.	→ Commande d'axe = Demande Activation	Le dispositif détecte le Code de demande d'activation dans l'élément Commande d'axe, et initie le changement d'état demandé. Ce changement d'état exige parfois une durée d'exécution significative.
Entretemps, le contrôleur détermine qu'il est nécessaire d'annuler la dernière demande de changement d'état Activer par une demande Désactivation. Pour libérer le mécanisme de demande de commande de la nécessité d'envoyer la demande Désactivation, l'élément Commande d'axe doit tout d'abord être défini sur Demande d'annulation	→ Commande d'axe = Demande d'annulation ← Réponse d'axe = Acquiescement Annulation ← Statut de réponse = 0	Le dispositif détecte le Code de demande d'annulation dans l'élément Commande d'axe et envoie immédiatement le Code d'acquiescement d'annulation dans l'élément Réponse d'axe. Le dispositif doit poursuivre l'opération de changement d'état d'origine jusqu'à réception d'une autre demande de commande.
Le contrôleur détecte le Code d'acquiescement différent de zéro dans l'élément Réponse d'axe et annule le Code de demande d'origine dans l'élément Commande d'axe. La transaction de demande d'annulation est à présent terminée. Une autre demande Commande d'axe peut à présent être traitée à cet instant.	→ Commande d'axe = 0 ← Réponse d'axe = 0 ← Statut de réponse = 0	Le dispositif détecte un Code de demande nul dans l'élément Commande d'axe et annule le Code d'acquiescement associé dans l'élément Réponse d'axe.
Le contrôleur définit le Code de demande dans l'élément Commande d'axe pour solliciter une demande Désactivation.	→ Commande d'axe = Demande Désactivation	Le dispositif détecte un Code de demande différent de zéro dans l'élément Commande d'axe, et initie le changement d'état Désactiver demandé. Ceci peut se substituer à toute opération de changement d'état annulée en cours d'exécution.
	← Réponse d'axe = Acquiescement Désactivation ← Statut de réponse = 0	Une fois le changement d'état Désactiver abouti, le dispositif l'acquiesce en attribuant le Code de demande d'origine au Code d'acquiescement de l'élément Réponse d'axe. L'élément Statut de réponse est nul, indiquant un succès.
Le contrôleur détecte le Code d'acquiescement différent de zéro dans l'élément Réponse d'axe et annule le Code de demande d'origine dans l'élément Commande d'axe. Le changement d'état demandé a abouti, comme le prouve le Statut de réponse nul.	→ Commande d'axe = 0	
La transaction de demande de changement d'état est à présent terminée. Une autre demande Commande d'axe peut être traitée à cet instant.	← Réponse d'axe = 0 ← Statut de réponse = 0	Le dispositif détecte un Code de demande nul dans l'élément Commande d'axe et annule le Code d'acquiescement associé dans l'élément Réponse d'axe.

7.6.1.3 Changement de système de référence de position

Lorsque la fonction de mise à l'échelle est activée dans le dispositif d'entraînement (voir définition du mode de commande), le mécanisme Commande d'axe/Réponse d'axe permet de coordonner la transition des positions réelle et de commande dans le contrôleur et le dispositif d'entraînement vers un nouveau système de référence de position absolue. La capacité de changer le système de référence de position absolue est essentielle pour les

opérations Retour à la position de référence et Redéfinir la position associées aux systèmes d'asservissement de position.

Les opérations Retour à la position de référence et Redéfinir la position déterminent tout d'abord une valeur de déplacement entre le système de référence de position actuel et le système de référence de position nouveau ou souhaité. La manœuvre consiste à appliquer la valeur de déplacement aux valeurs Position réelle et Position de consigne dans le dispositif d'entraînement et le contrôleur de manière à ne pas interrompre le mouvement en cours.

Le processus de coordination de la transition vers le nouveau système de référence de position commence par l'intégration du Code de demande de changement de la position réelle dans l'élément Commande d'axe de la connexion Contrôleur-Dispositif et d'un élément Déplacement de position dans les données cycliques. Lorsque le dispositif d'entraînement reçoit la demande de changement de la position réelle, il applique le déplacement de position à sa position réelle. La mise à jour Dispositif-Contrôleur suivante transmet la Position réelle en fonction du nouveau système de référence de position et le Déplacement de position au contrôleur, associés au Code d'acquiescement de changement de la position réelle dans l'élément Réponse d'axe. Le contrôleur reconnaît le Code d'acquiescement de changement de la position réelle, et de ce fait, lors du calcul du delta de position réelle (utilisé dans les opérations d'engrenage et de came), la valeur Déplacement de position est appliquée afin de compenser le changement de référence de position.

Après réception du Code d'acquiescement de changement de la position réelle par le contrôleur, ce dernier applique alors la valeur Déplacement de position à la Position de consigne et envoie cette dernière, en fonction du nouveau système de référence de position, au dispositif d'entraînement dans le cadre de la mise à jour Contrôleur-Dispositif suivante, accompagnée du Code de demande de changement de la position de consigne dans l'élément Commande d'axe. Une fois encore, le Déplacement de position est inclus comme élément des données cycliques. En fonction de la réception de la demande de changement de la position de consigne, le dispositif d'entraînement applique le Déplacement de position afin de compenser le changement de référence de position lors du calcul du delta de position de consigne.

L'adoption du nouveau système de référence de position est officielle avec la retransmission de l'acquiescement de changement de la position de consigne au contrôleur dans le cadre de la mise à jour Dispositif d'entraînement-Contrôleur suivante. Il n'est pas nécessaire de transmettre un Déplacement de position au contrôleur dans le cadre de cette mise à jour. Le Tableau 142 définit plus en détail le cycle complet de redéfinition de la référence de position.

Tableau 142 – Cycle de redéfinition de la référence de position

Action du contrôleur	Données de connexion	Action du dispositif
Le contrôleur définit le code de demande dans l'élément Commande d'axe pour solliciter une demande de changement de la position réelle, chg posréelle, et met à jour le dispositif d'entraînement avec Déplacement.	→ Commande d'axe = chg posréelle → Données cycliques = déplacement	Le dispositif détecte le Code de demande différent de zéro, à savoir une demande de changement de la position réelle, dans l'élément Commande d'axe, et applique Déplacement à la position réelle dans le dispositif d'entraînement.
Le contrôleur reçoit l'acquiescement de changement de la position réelle provenant du dispositif d'entraînement et applique Déplacement au calcul de la position réelle delta afin de corriger le changement de référence de position. Le contrôleur applique Déplacement pour déplacer la position de consigne vers le nouveau système de référence de position.	← Réponse d'axe = chg posréelle ← Statut de réponse = 0 ← Données cycliques = nouvelle posréelle ← Données cycliques = déplacement	Le dispositif d'entraînement acquiesce la demande en retransmettant le Code d'acquiescement de changement de la position réelle, chg posréelle, au contrôleur, accompagné de la nouvelle Position réelle dans le nouveau système de référence de position. L'élément Statut de réponse est nul, indiquant un succès.
Le contrôleur définit le Code de demande sur Demande de changement de la position de consigne, chg posconsig, et met à jour le dispositif d'entraînement avec la nouvelle Position de consigne en fonction du nouveau système de référence de position et du Déplacement.	→ Commande d'axe = chg posconsig → Données cycliques = nouvelle posconsig → Données cycliques = déplacement	Le dispositif d'entraînement reçoit l'acquiescement de changement de la position de consigne provenant du contrôleur et applique Déplacement au calcul de la position de consigne delta afin de corriger le changement de référence de position.
Le contrôleur indique que l'opération d'origine de retour à la position de référence ou de redéfinition de la position est achevée.	← Réponse d'axe = chg posconsig ← Statut de réponse = 0	Le dispositif d'entraînement acquiesce le changement de la demande de changement de la position de consigne en retransmettant le Code d'acquiescement, chg posconsig, au contrôleur dans l'élément Réponse d'axe. L'élément Statut de réponse est nul, indiquant un succès.
Le contrôleur détecte le Code d'acquiescement différent de zéro dans l'élément Réponse d'axe et annule le Code de demande d'origine dans l'élément Commande d'axe.	→ Commande d'axe = 0	
La dernière transaction de demande de changement d'état est à présent terminée. Une autre demande Commande d'axe peut être traitée à cet instant.	← Réponse d'axe = 0 ← Statut de réponse = 0	Le dispositif détecte un Code de demande nul dans l'élément Commande d'axe et annule le Code d'acquiescement associé dans l'élément Réponse d'axe.

7.6.2 Comportement d'état

7.6.2.1 Généralités

Les paragraphes 7.6.2.2 à 7.6.2.14 donnent une description détaillée de chacun des états et transitions d'état du modèle d'état Objet Axe de dispositif de mouvement.

7.6.2.2 État Hors tension

Il s'agit de l'état de mise hors tension de l'Objet Axe de dispositif de mouvement.

7.6.2.3 État Autotest

Lorsque le dispositif est mis sous tension ou réinitialisé, il procède généralement à une série de diagnostics d'autotest, et les paramètres internes du dispositif prennent leurs valeurs par

défaut à la mise sous tension. À l'issue de ces opérations, le dispositif et toutes ses instances d'axe associées passent à l'état Initialisation et peuvent être initialisés par le contrôleur associé. Dans le cas contraire, le dispositif et toutes ses instances d'axe associées passent immédiatement à l'état Défaut principal en déclarant un Défaut d'initialisation classé comme étant Irrécupérable selon la terminologie définie par l'Objet Identité. Ce défaut peut uniquement être annulé par un cycle d'alimentation et relève probablement plus d'un problème matériel du dispositif.

Lorsque le dispositif prend en charge une opération autonome sous commande locale avec des données de configuration locales, il peut passer librement de l'état Autotest à l'état Précharge, puis à l'état Arrêté. Lorsque le dispositif reçoit un service Forward_Open ultérieur pour ouvrir une Connexion CIP Motion, le dispositif d'entraînement doit désactiver tous les axes et revenir à l'état Initialisation, suivant la séquence d'états présentée ci-dessous.

Lorsque le dispositif ne prend pas en charge une opération autonome et dépend des données de configuration à distance à fournir via une connexion CIP, il doit passer à l'état Initialisation et attendre (Attente) que le service Forward_Open du contrôleur ouvre la Connexion CIP Motion.

7.6.2.4 État Initialisation

Au cours de l'état Initialisation, le dispositif attend que ses connexions CIP Motion soient établies par le contrôleur via un service Forward_Open. Une fois le service Forward_Open achevé avec succès, le dispositif initialise tous les attributs sur leurs valeurs par défaut définies en usine, réinitialise tous les défauts actifs et les conditions de statut d'axe applicables y compris le bit d'interruption, afin de préparer la configuration des attributs du dispositif.

Une fois établies les connexions, le contrôleur envoie des services Définir au dispositif afin d'affecter aux attributs de configuration de l'Objet Axe de dispositif de mouvement les valeurs archivées dans le contrôleur. Toute erreur de configuration qui apparaît au cours de ce processus, par exemple, en raison d'une valeur hors des limites établies ou d'une valeur non applicable, doit être traitée par une réponse erronée du service Définir, et non par la génération d'un défaut d'initialisation. Le contrôleur ne doit pas achever le processus de configuration tant que tous les attributs de configuration n'ont pas été acquittés avec succès.

Si le dispositif prend en charge le fonctionnement synchrone, le contrôleur se synchronise alors avec le dispositif à l'aide du service Group_Sync. A l'issue de ce processus, le dispositif et toutes ses instances d'axe associées passent à l'état Précharge. Si un problème est détecté lors de ce processus d'initialisation qui ne relève pas du champ d'application d'une erreur du service Définir, le dispositif génère un Défaut d'initialisation. Un Défaut d'initialisation étant perçu comme un défaut irrécupérable, il peut uniquement être annulé par un service de cycle d'alimentation ou de réinitialisation du dispositif dans l'Objet Identité associé.

Lorsque la Connexion CIP Motion est fermée délibérément en cours de fonctionnement pour quelque raison que ce soit par l'intermédiaire d'un service Forward_Close, le dispositif annule tous les défauts et alarmes actifs et revient à l'état Initialisation. En cas de perte de la Connexion CIP Motion en cours de fonctionnement pour quelque raison que ce soit, le dispositif génère un Défaut de nœud et passe à l'état Défaut principal. Dans tous les cas, le dispositif doit attendre que ses connexions CIP Motion soient rétablies par le contrôleur via un service Forward_Open.

L'état Initialisation est classé comme un état Attente de l'Objet Identité et, par conséquent, le dispositif doit s'assurer que toutes les structures de puissance associées sont désactivées.

7.6.2.5 État Précharge

À l'état Précharge, le cas échéant, le dispositif attend la charge complète du bus à courant continu (le bit d'état Bus à courant continu activé est supprimé). Lorsque le bus à courant continu atteint un niveau de tension opérationnel (le bit d'état Bus à courant continu activé est défini), l'axe passe soit à l'état Arrêté (axe d'entraînement) soit à l'état Exécution (axe du convertisseur). La structure de puissance de l'onduleur du dispositif d'entraînement est toujours désactivée dans cet état (bit d'état Structure de puissance activée supprimé). Toutes les tentatives du contrôleur pour activer un dispositif d'entraînement grâce au mécanisme Commande d'axe alors qu'il est à l'état Précharge sont signalées au contrôleur comme des erreurs dans le Statut de réponse, et l'axe reste à l'état Précharge.

L'état Précharge est classé comme un état Attente de l'Objet Identité et, par conséquent, exige que la structure de puissance d'entraînement associée soit désactivée.

7.6.2.6 État Arrêté

À l'état Arrêté, la structure de puissance de l'onduleur du dispositif d'entraînement doit être désactivée et sans couple (bit d'état Structure de puissance activée supprimé) ou maintenue dans une condition statique via une boucle d'asservissement active (bit d'état Structure de puissance activée défini). Aucun mouvement ne peut être initié par le dispositif à l'état Arrêté, et le dispositif ne peut pas non plus répondre à des références de consigne générées par un planificateur (bit d'état Commande de suivi supprimé). En règle générale, l'axe ne doit pas bouger, mais si une force ou un couple extérieur(e) est appliqué(e) à la charge, un frein peut s'avérer nécessaire pour maintenir la condition de repos. À l'état Arrêté, l'alimentation principale est appliquée au dispositif, et le bus à courant continu se trouve à un niveau de tension opérationnel. En présence de conditions Inhibition de démarrage dans cet état, l'axe passe à l'état Démarrage inhibé. Si une demande Activation ou l'une des demandes du service Exécution des essais est appliquée à un axe à l'état Arrêté, l'axe de mouvement passe à l'état Démarrage.

7.6.2.7 État Démarrage

Si une demande Activation est adressée à un axe à l'état Arrêté ou En cours d'arrêt lors de l'exécution d'un Amorçage instantané, l'axe passe immédiatement à l'état Démarrage. Dans cet état, le dispositif vérifie les différentes conditions avant de passer à l'état Exécution. Ces conditions peuvent inclure le Délai de libération de frein et le niveau de flux du moteur à induction. À l'état Démarrage, les structures de commande et de puissance du dispositif sont activées (bit d'état Puissance de structure activée défini), mais la référence de consigne est définie sur une valeur statique locale et ne suit pas la référence de consigne déduite du planificateur de mouvement (bit d'état Commande de suivi supprimé). Si toutes les conditions de démarrage sont satisfaites, l'axe passe à l'état Exécution ou Essais.

7.6.2.8 État Exécution

L'état Exécution est celui dans lequel le travail est réalisé. Dans cet état, la structure de puissance du dispositif est active (bit d'état Structure de puissance activée défini) et le Mode de commande sélectionné est activé et suit activement les données de consigne provenant du contrôleur en s'appuyant sur la sortie du planificateur de mouvement pour affecter le mouvement d'axe (bit d'état Commande de suivi défini). L'axe de mouvement reste à l'état Exécution tant qu'un défaut ne s'est pas produit ou qu'il n'est pas explicitement invité à s'arrêter via une demande Commande d'axe.

Dans le cas d'un axe sans fonction de commande active (Mode de commande = Pas d'asservissement), l'état Exécution indique simplement que le dispositif est totalement opérationnel (bit d'état Structure de puissance activée et bit Commande de suivi supprimés). L'axe de mouvement reste à l'état Exécution tant qu'un défaut ne s'est pas produit ou qu'il n'est pas explicitement invité à s'arrêter (interruption) via une demande Commande d'axe.

À l'état Exécution, le dispositif permet de modifier uniquement un sous-ensemble de tous les attributs de configuration d'instance d'axe. Le Tableau 143 énumère tous les attributs que le

dispositif est tenu de prendre en charge à l'état Exécution. L'accès à tous les autres attributs de configuration à l'état Exécution est laissé à la discrétion du fournisseur du dispositif.

Tableau 143 – État Exécution – Attributs configurables

Id Attribut	Nom de l'attribut	B	E	F	P	V	T
370	Vitesse de saut 1	-	-	F	-	-	-
371	Vitesse de saut 2	-	-	F	-	-	-
372	Vitesse de saut 3	-	-	F	-	-	-
373	Bande de vitesse de saut	-	-	F	-	-	-
374	Vitesse de rampe – positive	-	-	F	-	F	-
375	Vitesse de rampe – négative	-	-	F	-	F	-
376	Accélération de rampe	-	-	F	-	F	-
377	Décélération de rampe	-	-	F	-	F	-
378	Commande de rampe d'à-coups	-	-	F	-	F	-
440	Kvff	-	-	-	O	-	-
441	Kpp	-	-	-	O	-	-
442	Kpi	-	-	-	O	-	-
443	Tolérance de verrouillage de position	-	-	-	O	-	-
444	Tolérance d'erreur de position	-	-	-	O	-	-
445	Temps de tolérance d'erreur de position	-	-	-	F	-	-
446	Commande d'intégrateur de position	-	-	-	O	-	-
447	Préchargement d'intégrateur de position	-	-	-	F	-	-
460	Kaff	-	-	-	O	O	-
461	Kvp	-	-	-	O	O	-
462	Kvi	-	-	-	O	O	-
464	Kdr	-	-	F	F	F	-
465	Tolérance d'erreur de vitesse	-	-	-	F	F	-
466	Temps de tolérance d'erreur de vitesse	-	-	-	F	F	-
467	Commande d'intégrateur de vitesse	-	-	-	O	O	-
468	Préchargement d'intégrateur de vitesse	-	-	-	F	F	-
469	Bande passante du filtre passe-bas de vitesse	-	-	-	F	F	-
470	Seuil de vitesse	-	F	F	F	F	F
471	Tolérance de verrouillage de vitesse	-	-	F	F	F	-
472	Fenêtre de gel de vitesse	-	O	O	O	O	O
473	Limite de vitesse – Positive	-	-	F	F	F	-
474	Limite de vitesse – Négative	-	-	F	F	F	-
485	Limite d'accélération	-	-	-	F	F	F
486	Limite de décélération	-	-	-	F	F	F
496	Kj	-	-	-	O	O	F
498	Compensation de frottement – Glissant	-	-	-	F	F	F
499	Compensation de frottement – Statique	-	-	-	F	F	F
500	Compensation de frottement – Visqueux	-	-	-	F	F	F
502	Bande passante du filtre passe-bas de couple	-	-	-	F	F	F
503	Fréquence de filtre coupe-bande de couple	-	-	-	F	F	F
504	Limite de couple positive	-	-	-	O	O	O

Id Attribut	Nom de l'attribut	B	E	F	P	V	T
505	Limite de couple négative	-	-	-	O	O	O
506	Limite de débit de couple	-	-	-	F	F	F
507	Seuil de couple	-	-	-	F	F	F
508	Limite de surcouple	-	-	F	F	F	F
509	Temps de limite de surcouple	-	-	F	F	F	F
510	Limite de sous-couple	-	-	F	F	F	F
511	Temps de limite de sous-couple	-	-	F	F	F	F
553	Limite de vecteur de courant	-	-	F	F	F	F
554	Kqp	-	-	-	F	F	F
555	Kqi	-	-	-	F	F	F
556	Kdp	-	-	-	F	F	F
557	Kdi	-	-	-	F	F	F
731	Sorties numériques	-	-	F	F	F	F
734	Sortie analogique 1	-	-	F	F	F	F
735	Sortie analogique 2	-	-	F	F	F	F

7.6.2.9 État Essais

Lorsque l'un des services de demande Exécution des essais est envoyé à l'axe de mouvement qui est à l'état Arrêté, c'est-à-dire les services qui exigent l'exécution d'une structure de puissance active, l'axe passe immédiatement à l'état Démarrage (bit d'état Structure de puissance activée défini), puis, une fois les conditions de démarrage satisfaites, à l'état Essais. L'état Essais s'apparente à l'état Exécution en ce sens que la structure de puissance du dispositif est active (bit d'état Structure de puissance activée défini), mais qu'à l'état Essais, l'un des algorithmes d'essai intégrés du dispositif contrôle le moteur et non les données de consigne d'un planificateur de mouvement (bit d'état Commande de suivi supprimé). À l'état Essais, le dispositif excite le moteur de différentes manières tout en procédant à des mesures visant à déterminer les caractéristiques du moteur et de la charge. L'axe de mouvement reste dans cet état pendant la durée de la procédure d'essai demandée, puis retourne à l'état Arrêté. L'axe de mouvement peut également quitter l'état Essais par un défaut ou une demande Commande d'axe explicite.

7.6.2.10 État Démarrage inhibé

L'état Démarrage inhibé est identique à l'état Arrêté, sauf que l'axe comporte une ou plusieurs conditions "d'inhibition de démarrage" l'empêchant de passer à l'état Démarrage. Ces conditions peuvent être contenues dans les attributs Inhibition de démarrage. Une fois corrigé, l'axe repasse automatiquement à l'état Arrêté.

Pour un axe sans fonction de commande active (Mode de commande = Pas d'asservissement), l'état d'axe Démarrage inhibé empêche de passer à l'état Exécution tant que les conditions Inhibition de démarrage spécifiques ne sont pas résolues, telles que celles où le dispositif associé n'est pas totalement configuré pour fonctionner. Là encore, une fois ces conditions corrigées, l'axe repasse automatiquement à l'état Exécution.

L'état Démarrage Inhibé est classé comme un état Attente de l'Objet Identité et, par conséquent, exige que la structure de puissance associée, le cas échéant, soit désactivée.

7.6.2.11 État Arrêt

Lorsqu'une demande Désactivation est adressée à l'Objet Axe de dispositif de mouvement à l'état Démarrage, Exécution ou Essais, l'axe passe immédiatement à l'état Arrêt. Dans cet état, l'axe est en cours d'arrêt et ne suit plus les données de consigne provenant du

planificateur de mouvement (bit d'état Commande de suivi supprimé). Un certain nombre d'actions d'arrêt différentes sont prises en charge par l'Objet Axe de dispositif de mouvement. La plupart de ces actions d'arrêt permettent de décélérer activement l'axe jusqu'à son arrêt. La structure de puissance peut rester active (bit d'état Structure de puissance activée défini) pendant toute la durée d'exécution de la procédure d'action d'arrêt. A l'issue de la procédure d'action d'arrêt sélectionnée, l'axe passe à l'état Arrêté. Toutefois, lorsque l'action d'arrêt consiste en une désactivation et un arrêt progressif, la structure de puissance est désactivée immédiatement (bit d'état Structure de puissance activée supprimé) et l'axe s'arrête progressivement alors qu'il est à l'état d'arrêt. Dans tous les cas, le dispositif d'entraînement doit attendre que la vitesse d'axe soit nulle pour passer à l'état Arrêté. Dans certains cas, tels que lorsque l'axe est fixe, cette transition peut être immédiate. Les critères recommandés pour l'obtention d'une vitesse nulle sont basés sur le retour en vitesse, ou dans le cas d'un dispositif d'entraînement à contrôle de fréquence, ceci implique que la Référence de vitesse soit inférieure à 1 % de la vitesse assignée du moteur. Finalement, ces critères sont laissés à la discrétion du fournisseur.

Lorsqu'une Demande d'activation est adressée à un axe à l'état Arrêt avec Amorçage instantané activé, l'axe doit immédiatement passer à l'état Démarrage.

7.6.2.12 État Abandon

Si un Défaut principal se produit dans le dispositif de mouvement alors que l'axe est à l'état Démarrage, Exécution, Essais ou Arrêt, l'axe passe immédiatement à l'état Abandon. Dans cet état, l'axe est en cours d'arrêt et ne suit plus les données de consigne provenant du planificateur de mouvement (bit d'état Commande de suivi supprimé). L'état Abandon permet d'exécuter l'action d'arrêt appropriée spécifiée par le fournisseur du dispositif. Lorsqu'elle arrête de manière active l'axe à l'état Abandon, la structure de puissance reste active (bit d'état de Structure de puissance activée défini) pendant toute la durée d'exécution de l'action d'arrêt. Dans certains cas, la structure de puissance doit être désactivée immédiatement, de sorte que l'axe peut s'arrêter progressivement alors qu'il est à l'état Abandon. Dans tous les cas, le dispositif d'entraînement doit attendre que la vitesse d'axe soit nulle pour passer à l'état Défaut principal. À l'issue de la procédure d'arrêt et lorsque la vitesse d'axe est nulle, l'axe passe à l'état Défaut principal. Dans certains cas, tels que lorsque l'axe est fixe, cette transition peut être immédiate.

Lorsqu'une demande Abandon est transmise à l'Objet Axe de dispositif de mouvement, une Exception initiée par le contrôleur est générée. Lorsque l'action d'exception d'axe associée est définie pour générer un Défaut principal, le dispositif d'entraînement arrête l'axe conformément à l'action d'arrêt configurée avant de passer à l'état Défaut principal. Les critères recommandés pour l'obtention d'une vitesse nulle sont basés sur le retour en vitesse, ou dans le cas d'un dispositif d'entraînement à contrôle de fréquence, ceci implique que la Référence de vitesse soit inférieure à 1 % de la vitesse assignée du moteur. Finalement, ce critère est laissé à la discrétion du fournisseur et peut être spécifique à l'application.

7.6.2.13 État Défaut principal

L'état Défaut principal est identique à l'état Arrêté (ou, si une action de défaut Interruption a été initiée, à l'état Interruption) sauf qu'un ou plusieurs Défauts principaux sont actifs. En d'autres termes, un axe à l'état Défaut principal est un axe à l'état Arrêté (ou état Interruption) avec une condition Défaut principal. Les défauts étant des conditions verrouillées, une demande Réinitialisation suite à un défaut provenant du contrôleur est exigée pour supprimer le défaut et, en supposant que la condition de défaut d'origine a été retirée, l'axe passe à l'état Arrêté (ou état Interruption).

Il existe quatre différents Défauts principaux: Défauts de nœud, Défauts d'initialisation, Défauts d'axe et Défauts de sécurité d'axe.

Défauts d'initialisation

Ce type de défaut ne peut se produire qu'à l'état Initialisation. Un défaut d'initialisation ne peut pas être généré dans un autre état du dispositif, c'est-à-dire l'occurrence d'un défaut au

cours du fonctionnement du dispositif après avoir quitté l'état Initialisation. Les défauts d'initialisation peuvent s'appliquer à un axe spécifique ou au dispositif dans son ensemble, auquel cas toutes les instances d'axe indiquent le défaut d'initialisation. La structure de puissance du dispositif, le cas échéant, est désactivée en présence d'un défaut d'initialisation.

Défauts de nœud

Ces défauts s'appliquent toujours à l'ensemble du dispositif et affectent de la même manière tous les axes. Ces défauts peuvent survenir à tout moment au cours du fonctionnement du dispositif. Les défauts de nœud sont principalement des défauts de communication, mais peuvent comprendre les défauts de matériel généraux, lorsque ces conditions de défaut sont vérifiées en cours d'exécution. Un défaut de chien de garde CPU constitue un exemple de défaut de nœud de matériel. La structure de puissance du dispositif, le cas échéant, est désactivée en présence d'un défaut de nœud.

Défauts d'axe

Comme le nom l'indique, les défauts d'axe s'appliquent à des instances d'axe spécifiques. Les défauts d'axe sont le résultat direct des exceptions d'axe configurées pour générer une réponse par défaut. Ces conditions d'exception peuvent s'appliquer à des instances d'axe individuelles ou à toutes les instances d'axe. Dans tous les cas, les applications peuvent exiger que le dispositif soit configuré pour traiter ces exceptions différemment pour des axes différents. Les conditions d'exécution associées aux composants du moteur, de l'onduleur, du convertisseur, du régulateur de bus et de réaction, doivent généralement être traitées comme des exceptions d'axe. La structure de puissance du dispositif, le cas échéant, peut ou peut ne pas être désactivée en présence d'un défaut d'axe selon l'action d'arrêt spécifique appliquée par le dispositif en réponse à la condition de défaut.

Défauts de sécurité d'axe

Les défauts de sécurité d'axe s'appliquent également aux instances d'axe spécifiques. Les défauts de sécurité sont signalés par le Noyau de sécurité intégré du dispositif qui est chargé de contrôler la condition des diverses fonctions de sécurité critiques associées à l'axe.

NOTE Voir [30] pour de plus amples informations.

Les défauts qui se produisent après que l'axe du dispositif a quitté l'état Initialisation, doivent être définis comme des défauts de nœud, des Défauts d'axe ou des Défauts de sécurité d'axe.

7.6.2.14 État Interruption

Lorsque le dispositif exécute une demande Interruption, l'axe ciblé passe à l'état Interruption. Dans le cas d'une demande Interruption, l'axe passe immédiatement de n'importe quel état à l'état Interruption. L'état Interruption a les mêmes caractéristiques de base que l'état Arrêté, à l'exception du fait que la structure de puissance de l'onduleur du dispositif doit être désactivée et exempte de couple (bit d'état Structure de puissance activée supprimé), et l'attribut Action d'interruption peut être configuré pour réduire la puissance du bus à courant continu à la structure de puissance du dispositif (bit d'état Bus à courant continu activé supprimé).

NOTE Ceci s'effectue généralement par l'ouverture d'une sortie Activer contacteur à courant alternatif, le cas échéant, du dispositif qui régule l'alimentation du contrôleur.

Que le bus à courant continu soit déconnecté ou non, cet état exige une demande Réinitialisation suite à une interruption explicite de la part du contrôleur pour passer à l'état Précharge. Si le dispositif est configuré pour maintenir active l'alimentation du bus à courant continu pendant l'état Interruption, l'axe de mouvement passe alors à l'état Précharge, puis à l'état Arrêté. L'état Interruption offre un niveau supplémentaire de sécurité contre les mouvements intempestifs.

Dans le cas où une action de défaut d'interruption est initiée par le dispositif en réponse à une condition d'exception configurée pour être un Défaut principal, le dispositif exécute ladite action, mais l'axe passe à l'état défaut principal, et non à l'état Interruption. De la même façon, lorsque l'axe est à l'état Interruption et en présence d'une condition de défaut principal, l'axe passe à l'état Défaut principal. En d'autres termes, la condition de défaut principale est prioritaire sur la condition d'interruption et cette dernière peut être considérée comme un sous-état. Dans tous les cas, une demande Réinitialisation suite à un défaut provenant du contrôleur supprime le défaut et, en supposant que la condition de défaut d'origine a été retirée, l'axe passe alors à l'état Arrêté (ou état Interruption). Une demande Réinitialisation suite à une interruption provenant du contrôleur supprime toutefois le défaut et procède à cette réinitialisation, de sorte que, en supposant que la condition de défaut d'origine a été retirée, l'axe passe à l'état Précharge comme décrit ci-dessus.

Outre la fonction d'action d'interruption, l'état Interruption peut également servir au contrôleur à désactiver les fonctions d'applications d'engrenage ou de came esclaves du planificateur de mouvement, qui référencent cet axe de dispositif comme étant un axe maître. Pour cette raison, l'état Interruption est applicable à Réaction uniquement où l'instance d'axe est simplement associée à un dispositif de réaction sans fonction de commande active.

L'état Interruption est classé comme un état Attente de l'Objet Identité et, par conséquent, exige que la structure de puissance d'entraînement associée soit désactivée.

7.6.3 Comportement de défaut et d'alarme

7.6.3.1 Généralités

Les fonctions de traitement des défauts et alarmes de l'Objet Axe de dispositif de mouvement répondent aux besoins d'un nombre de plus en plus important de défauts et d'alarmes spécifiques, au besoin d'actions programmables et au besoin de signalement opportun de ces défauts et alarmes au contrôleur. De plus, aucun compromis n'est possible pour limiter la résolution des défauts et alarmes signalés, de sorte que le contrôleur a toujours accès à la condition d'axe unique et à un diagnostic significatif. De nombreux attributs de défaut et d'alarme peuvent être inclus dans la partie fixe de la connexion Dispositif-Contrôleur cyclique, de sorte que le contrôleur puisse surveiller la condition d'axe de mouvement en temps réel, sans scrutation interminable.

L'attribut Statut d'axe contient des bits indiquant si une condition d'alarme est présente. L'énumération Statut d'axe indique si l'axe présente un défaut principal, qui peut être un Défaut d'axe ou un Défaut d'initialisation récurrent lors de la phase d'exécution. Le Code de défaut d'axe et ses attributs associés permettent de signaler la condition de défaut spécifique, la datation et l'action de défaut au contrôleur afin d'établir un journal des défauts. Mais avant d'entrer dans les détails, il est nécessaire de clairement définir les termes utilisés pour décrire les fonctionnalités Défaut et Alarme de l'Objet Axe de dispositif de mouvement.

7.6.3.2 Exceptions

Les exceptions sont des conditions d'exécution que le dispositif vérifie en permanence, et susceptibles de signaler un comportement incorrect d'axe de mouvement ou un fonctionnement hors norme. Une exception peut donner lieu à une alarme, un défaut secondaire ou un défaut principal, selon la configuration de l'Action d'exception d'axe associée (une exception peut même être configurée pour être ignorée). Les exceptions sont automatiquement annulées par le dispositif lorsque la condition d'exception sous-jacente disparaît.

7.6.3.3 Actions d'exception

Pour chaque exception, l'axe de mouvement peut être programmé pour exécuter un large éventail d'actions grâce à l'attribut Action d'exception d'axe. Les Actions d'exception peuvent aller de la génération d'un défaut principal se traduisant par l'arrêt d'axe de mouvement, à l'absence d'action. L'attribut Défaut d'axe permet au contrôleur d'accéder immédiatement à

toutes les exceptions configurées pour générer un défaut principal ou secondaire. L'attribut Alarmes d'axe permet au contrôleur d'accéder immédiatement à toutes les exceptions configurées pour être signalées en tant qu'alarmes.

7.6.3.4 Alarmes

Les alarmes sont des conditions d'exception d'exécution pour lesquelles le dispositif n'exécute aucune autre action que de signaler une alarme. Les alarmes et avertissements sont donc synonymes. Sur un produit de dispositif donné, certaines conditions d'exception peuvent ne pas pouvoir être signalées en tant qu'alarme sans leur associer une action (un défaut IPM dans lequel le module d'alimentation s'arrête automatiquement sans intervention logicielle, par exemple). Les conditions d'alarme sont automatiquement annulées lorsque la condition d'exception sous-jacente disparaît.

7.6.3.5 Défauts principaux

Les défauts principaux peuvent être des défauts d'initialisation, des défauts de sécurité ou des conditions d'exception d'exécution que le dispositif, de par sa configuration, considère comme étant des défauts principaux. Si un défaut d'exécution se produit lors d'un état opérationnel (Exécution ou Essais, par exemple), le dispositif arrête (ou abandonne) le mouvement d'axe. Les défauts principaux font en dernier lieu passer l'axe à l'état Défaut principal. Un Défaut principal apparu suite à une condition d'exception est verrouillé et n'est pas supprimé lors de la suppression de la condition d'exception. Un défaut peut uniquement être supprimé par une demande de service Réinitialisation suite à un défaut adressée par le contrôleur. Si la condition de défaut est classée comme étant un "défaut irrécupérable", seul un cycle d'alimentation ou une réinitialisation du dispositif peut supprimer la condition de défaut.

7.6.3.6 Défauts secondaires

Les défauts secondaires sont des conditions d'exception que le dispositif, de par sa configuration, signale au contrôleur comme étant un défaut, mais sans exécuter d'action directe. Cela offre la possibilité au contrôleur de réaliser une action de défaut spécifique à l'application qui peut ne pas être prise en charge par le dispositif comme l'une des actions d'exception définies. Comme les alarmes, les Défauts secondaires ne sont pas à l'origine d'un changement d'état, même en présence d'un état Défaut secondaire. Ils permettent à l'axe de mouvement de continuer à fonctionner dans l'état dans lequel il se trouve. Mais à l'inverse des alarmes, un Défaut secondaire est verrouillé, c'est-à-dire que le défaut n'est pas supprimé lorsque la condition d'exception l'est. Les défauts principaux et secondaires peuvent uniquement être supprimés par une demande de service Réinitialisation suite à un défaut adressée par le contrôleur.

7.6.3.7 Défauts d'initialisation

Les Défauts d'initialisation sont générés lors de la procédure de mise sous tension ou de réinitialisation du dispositif, lorsque ce dernier détecte un problème qui empêche son fonctionnement normal. Il peut s'agir d'un problème matériel ou de micrologiciel détecté dans le cadre d'essais d'autodiagnostic ou d'un problème lié au processus de configuration d'attribut. Ces défauts ne sont pas la conséquence de conditions d'exception et ne font donc l'objet d'aucune action configurable. Il s'agit, par exemple, de données de mémoire corrompues, d'erreurs d'étalonnage ou de problèmes de démarrage de micrologiciel. Les Défauts d'initialisation se traduisant par l'état Défaut sont considérés comme des "défauts irrécupérables" et ne peuvent pas être supprimés avec une demande de service Réinitialisation suite à un défaut. Par conséquent, les mouvements sont impossibles dans cet état. Seul un cycle d'alimentation ou seule une réinitialisation du dispositif peuvent supprimer ce type de défaut.

7.6.3.8 Défauts de sécurité

Les défauts de sécurité sont les défauts signalés par le Noyau de sécurité intégré du dispositif d'entraînement qui fait partie intégrante d'un système de sécurité comprenant ce

dispositif, un Contrôleur de sécurité et une connexion réseau CIP Safety. Lorsque le système de sécurité détecte un problème qui empêche le fonctionnement normal en toute sécurité du dispositif d'entraînement, il génère un défaut de sécurité. Ces défauts ne sont pas à l'origine de conditions d'exception et n'ont donc pas d'actions configurables. À l'instar des défauts d'exécution, les défauts de sécurité qui donnent lieu à l'état Défaut sont considérés comme des « défauts récupérables » et ils peuvent être supprimés avec une demande de Réinitialisation suite à un défaut.

EXEMPLE Défauts de sécurité de réaction, défauts de sécurité d'arrêt ou défauts de sécurité de vitesse.

7.6.3.9 Défauts de nœud

Il s'agit de défauts générés par l'interface de communication du dispositif d'entraînement. Ces défauts ne sont pas spécifiques à l'axe, mais s'appliquent à la classe Objet Axe de dispositif de mouvement. Si un défaut de nœud se produit lors d'un état opérationnel (Exécution ou Essais, par exemple), le dispositif arrête (ou abandonne) tous les axes commandés qui lui sont associés. Les défauts de nœud font en dernier lieu passer l'axe à l'état Défaut principal. Un défaut principal issu d'un défaut de nœud est verrouillé. Un défaut peut uniquement être supprimé par une demande de service Réinitialisation suite à un défaut de nœud adressée par le contrôleur. Si la condition de défaut de nœud est classée comme étant un "défaut irrécupérable", seul un cycle d'alimentation ou seule une réinitialisation du dispositif peuvent supprimer la condition de défaut.

7.6.3.10 Codes de défaut

L'attribut Code de défaut est une énumération qui indique la condition d'exception d'exécution, la condition d'initialisation ou la condition de sécurité, ayant généré le défaut. La source de la condition est spécifiée par l'attribut Type de défaut. Comme les bits d'état de défaut, cette valeur d'attribut est verrouillée et ne change pas, sauf si 1) le défaut est "récupérable" et une demande Réinitialisation suite à un défaut est initiée, ou 2) une autre condition de défaut se produit. La valeur Code de défaut correspond à la position des bits de l'attribut de défaut associé. Une valeur nulle indique l'absence effective de condition de défaut pour toutes les sources de défaut potentielles. Les Codes de défaut sont utilisés principalement à des fins de visualisation et d'établissement d'un journal de défauts dans le contrôleur associé.

7.6.4 Comportement Inhibition de démarrage

Une Inhibition de démarrage est une condition qui inhibe le démarrage d'axe, c'est-à-dire le passage à l'état Démarrage pour un fonctionnement d'axe activé. Cette condition ne génère pas d'exception en cas de tentative de démarrage. Si les circonstances ayant mené à l'Inhibition de démarrage sont levées, la condition d'inhibition de démarrage est automatiquement supprimée par le dispositif, renvoyant l'axe à l'état Arrêté.

Si l'axe de mouvement est à l'état Inhibition de démarrage, cela indique qu'une ou plusieurs conditions empêchent l'axe de passer au fonctionnement activé. L'attribut Inhibition de démarrage signale la condition spécifique qui inhibe l'axe.

7.6.5 Comportement de visualisation

7.6.5.1 Généralités

Le comportement d'état de l'Objet Axe de dispositif de mouvement a un impact direct sur les composants de visualisation du dispositif de mouvement. Ces composants vont des LED bicolores ou des voyants de type LED aux affichages alphanumériques à plusieurs caractères. Cette section définit dans quelle mesure les états de l'Objet Axe de dispositif de mouvement ont un impact sur le comportement de ces composants de visualisation.

7.6.5.2 LED de statut du module

Les états de l'Objet Axe de dispositif de mouvement sont liés au comportement d'état de l'Objet Identité du dispositif CIP Motion et à sa LED de statut du module associée. Le Tableau 144 met en correspondance les états de l'instance Objet Axe de dispositif de mouvement principale avec les états appropriés de l'Objet Identité. Les dispositifs compatibles CIP Motion sont tenus de prendre en charge la LED de statut du module.

Pour plus d'informations relatives au modèle d'état de l'Objet Identité, voir la spécification Objet Identité (voir IEC 61158-5-2).

7.6.5.3 LED d'état d'axe

Pour accroître davantage les informations visuelles fournies par la LED de statut du module standard, le profil de dispositif CIP Motion définit également le comportement d'une deuxième LED. Cette LED, dite d'état d'axe, offre par exemple une indication visuelle de la mise sous tension ou non du bus à courant continu, de l'activation ou la désactivation d'axe, et même des conditions d'alarmes actives ou de défaut secondaire (voir Tableau 144). Cette LED est également exigée par les dispositifs compatibles CIP Motion, sauf s'ils sont dotés d'un affichage alphanumérique à plusieurs caractères.

La LED de statut d'axe utilise trois couleurs qui peuvent être générées par une LED Rouge/Verte bicolore standard, à savoir le rouge, le vert et l'orange. L'orange (ou le jaune) est la couleur générée lorsque les mélanges du vert et du rouge de la LED bicolore sont activés. La signification générale de ces trois couleurs est la suivante:

- vert – Indique une mise sous tension normale ou un état opérationnel;
- orange – Indique la présence d'une condition d'alarme ou d'inhibition de démarrage;
- rouge – Indique la présence d'une certaine forme de condition de défaut.

Lors de la mise sous tension normale ou de la réinitialisation du dispositif, les LED de statut du module et de statut d'axe doivent d'abord être rouges (sous contrôle du matériel) pendant l'amorçage du processeur, puis devenir vertes pendant environ 1 s lorsque le dispositif démarre son autotest. Dans l'idéal, les deux LED doivent être rouges pendant environ 1 s, puis vertes pendant environ 1 s avant de passer à une indication de l'état Attente. La séquence Rouge-Vert confirme le bon fonctionnement des voyants LED.

Tableau 144 – Mise en correspondance de l'état d'axe avec l'Objet Identité grâce au comportement de la LED

État de l'Objet Identité	LED de statut du module	État de l'Objet Axe de dispositif de mouvement	LED de statut d'axe
Inexistant – Hors tension	Désactivé	Désactivé	Désactivé
Autotest du dispositif	Rouge/vert clignotant	Autotest	Rouge/vert clignotant
Attente	Vert clignotant	Initialisation – Bus non activé	Désactivé
		Initialisation – Bus activé	Vert clignotant
		Interruption – Bus non activé	Désactivé
		Interruption – Bus activé	Orange clignotant ^a
		Précharge – Bus non activé	Désactivé
		Démarrage inhibé	Orange clignotant ^a
Opérationnel	Vert fixe	Arrêté	Vert clignotant ^{a,b}
		Arrêt	Vert fixe ^{a,b}
		Démarrage	Vert fixe ^{a,b}
		Exécution	Vert fixe ^{a,b}
		Essais	Vert fixe ^{a,b}
Défaut principal récupérable	Rouge clignotant	Abandon	Rouge clignotant
		Défaut principal	Rouge clignotant
Défaut principal irrécupérable	Rouge fixe	Abandon	Rouge fixe
		Défaut principal	Rouge fixe
^a L'Objet Axe de dispositif de mouvement et l'Objet Identité définissent des conditions de défaut secondaire. Alors qu'un défaut secondaire n'a aucun impact sur la LED de statut du module, il a un impact sur la LED de statut d'axe. Si une condition de défaut secondaire est détectée, une indication de LED Verte fixe devient Rouge-Vert-Rouge-Vert en alternance, une indication de LED Vert fixe normale devient Rouge-Désactivé-Vert-Désactivé en alternance, et des indications de LED Orange deviennent Rouge-Désactivé-Orange-Désactivé.			
^b L'Objet Axe de dispositif de mouvement définit également des conditions d'alarme. Si une condition d'alarme est détectée, une indication de LED Vert fixe devient Orange-Vert-Orange-Vert en alternance, et une indication de LED Vert clignotant devient Orange-Désactivé-Vert-Désactivé en alternance.			

7.6.5.4 Affichage alphanumérique

7.6.5.4.1 Généralités

Outre la visualisation requise assurée par la LED de statut du module et la LED de statut d'axe, l'Objet Axe de dispositif de mouvement définit également le comportement d'un affichage alphanumérique facultatif pour indiquer de manière plus explicite la condition du dispositif. Ce type d'affichage est particulièrement utile pour surveiller la progression du processus d'initialisation et fournir des informations de diagnostic détaillées relatives aux conditions de défaut, d'alarme et d'inhibition. Les affichages alphanumériques peuvent aller des simples affichages à sept segments aux affichages alphanumériques à plusieurs caractères.

7.6.5.4.2 Affichage à sept segments

Si le dispositif est doté d'un affichage à sept segments, ce dernier peut être utilisé pour indiquer la progression à l'état Initialisation et aux différentes conditions de défaut, d'alarme et d'inhibition. L'affichage doit au moins prendre en charge la mise en correspondance avec les différentes conditions du dispositif spécifiées dans le Tableau 145.

**Tableau 145 – Comportement de l'affichage
à sept segments du dispositif CIP Motion**

Chiffre d'affichage	Condition du dispositif
8	Exécution de l'autotest du dispositif
0	Attente de la connexion au contrôleur
1	Configuration des attributs du dispositif
2	Attente de la synchronisation de groupe
3	Attente de la charge du bus à courant continu
4	Dispositif opérationnel
I##	Code de défaut d'initialisation
Ic##	Code de défaut d'initialisation – Extensions du constructeur
F##	Code de défaut d'axe
Fc##	Code de défaut d'axe – Extensions du constructeur
A##	Code d'alarme
Ac##	Code d'alarme d'axe – Extensions du constructeur
S##	Code d'inhibition de démarrage
Sc##	Code d'inhibition de démarrage – Extensions du constructeur
SF##	Code de défaut de sécurité
SFc##	Code de défaut de sécurité – Extensions du constructeur
nF##	Code de défaut de nœud
nA##	Code d'alarme de nœud

7.6.5.4.3 Affichage alphanumérique à plusieurs caractères

Si le dispositif est doté d'un affichage alphanumérique à plusieurs caractères, des informations encore plus utiles relatives au dispositif peuvent être apportées à l'utilisateur, par défilement ou zones de caractères fixes. En règle générale, les capacités de l'affichage dictent la longueur des chaînes de caractères qui peuvent être effectivement affichées. Néanmoins, pour assurer la cohérence de comportement des produits de dispositif compatibles CIP Motion, l'Objet Axe de dispositif de mouvement dicte exactement ce qui s'affiche pour les conditions indiquées au Tableau 146.

**Tableau 146 – Comportement de l'affichage
alphanumérique à plusieurs caractères CIP Motion**

Chaîne d'affichage	Condition du dispositif
SELF-TEST	Exécution de l'autotest du dispositif
STANDBY	Attente du service Forward_Open
CONNECTING	Attente du premier service Définir les attributs. (Rafraîchissement)
CONFIGURING	Configuration des Attributs Instance de classe & Axe
SYNCING	Attente d'un service Group_Sync abouti.
PRE-CHARGE	Attente d'activation du bus à courant continu
SHUTDOWN	L'axe a été interrompu
STOPPED	L'axe a été arrêté
STARTING	L'axe est en cours de démarrage
RUNNING	L'axe est en cours d'exécution
TESTING	L'axe exécute une procédure d'essai
STOPPING	Décélération jusqu'à l'arrêt suite à une désactivation
ABORTING	Décélération jusqu'à l'arrêt suite à un défaut
INIT FLT S##	Défaut d'initialisation – Code de défaut standard
INIT FLT M##	Défaut d'initialisation – Code de défaut Mfg
FLT S##	Défaut d'axe – Code de défaut standard
FLT M##	Défaut d'axe – Code de défaut Mfg
ALARM S##	Alarme d'axe – Code d'alarme standard
ALARM M##	Alarme d'axe – Code d'alarme Mfg
INHIBIT S##	Inhibition de démarrage – Code d'inhibition standard
INHIBIT M##	Inhibition de démarrage – Code d'inhibition Mfg
SAFE FLT S##	Défaut de sécurité d'axe – Code de défaut de sécurité standard
SAFE FLT M##	Défaut de sécurité d'axe – Code de défaut de sécurité Mfg
NODE FLT ##	Code de défaut de nœud
NODE ALARM ##	Code d'alarme de nœud

Outre ces fonctionnalités de base, l'affichage alphanumérique peut également fournir des descriptions détaillées des conditions de défaut, d'alarme et d'inhibition. La définition de ces chaînes spécifiques va bien au-delà du domaine d'application de cet objet et est laissée à la discrétion du fournisseur du dispositif.

7.6.5.4.4 Visualisation du dispositif multiaxe

Si plusieurs instances d'axe de mouvement sont prises en charge par le dispositif, le comportement ci-dessus implique des explications plus approfondies.

En premier lieu, il existe une seule LED de statut du module par nœud de dispositif. Par conséquent, sa condition est un cumul des états de toutes les instances Objet Axe de dispositif de mouvement. À l'inverse, une LED de statut d'axe est associée à chaque instance Objet Axe de dispositif de mouvement dans le dispositif doté d'une structure de puissance. Un affichage alphanumérique à plusieurs caractères peut aisément gérer plusieurs instances d'axe de mouvement.

Dans le cas d'une instance d'axe de mouvement configurée sans fonction de commande active (Mode de commande = Pas d'asservissement), les informations d'état à visualiser sont très peu nombreuses. L'axe de mouvement est à l'état opérationnel (appelé état Exécution) et

y reste tant qu'un défaut ne l'a pas passé à l'état Défaut principal. Un axe Pas d'asservissement n'a donc aucun effet sur la LED de statut du module tant qu'un défaut ne s'est pas produit, auquel cas la LED de statut du module est Rouge clignotant (Défaut principal récupérable). Un axe Pas d'asservissement n'exige pas de LED de statut d'axe distincte.

En présence de plusieurs instances d'axe de mouvement, chacune dotée d'une structure de puissance distincte, le comportement de la LED de statut du module est un cumul des états des axes du dispositif. Si les axes du dispositif sont dans des états différents, la condition LED de statut du module repose sur la condition préalable suivante:

- 1) Défaut principal irrécupérable;
- 2) Défaut principal récupérable;
- 3) Attente;
- 4) Opérationnel.

En d'autres termes, en ce qui concerne la LED de statut du module, un Défaut principal récupérable sur l'Axe 1 remplace l'Axe 2 qui est à l'état Attente, Inhibition de démarrage.

NOTE La LED de statut du module ne reconnaît pas les défauts secondaires et les alarmes.

Les affichages à plusieurs caractères gèrent aisément les instances à plusieurs axes en ajoutant un préfixe "X#" à la chaîne d'affichage, afin de spécifier le numéro d'instance d'axe de mouvement associé. Pour les instances d'axe Réaction principale, aucune chaîne d'affichage spécifique n'apparaît tant que l'Axe maître présente une condition de défaut ou d'alarme.

Le Tableau 147 précise la mise en correspondance de l'affichage et de la condition du dispositif dans le cas de plusieurs instances d'axe de dispositif de mouvement.

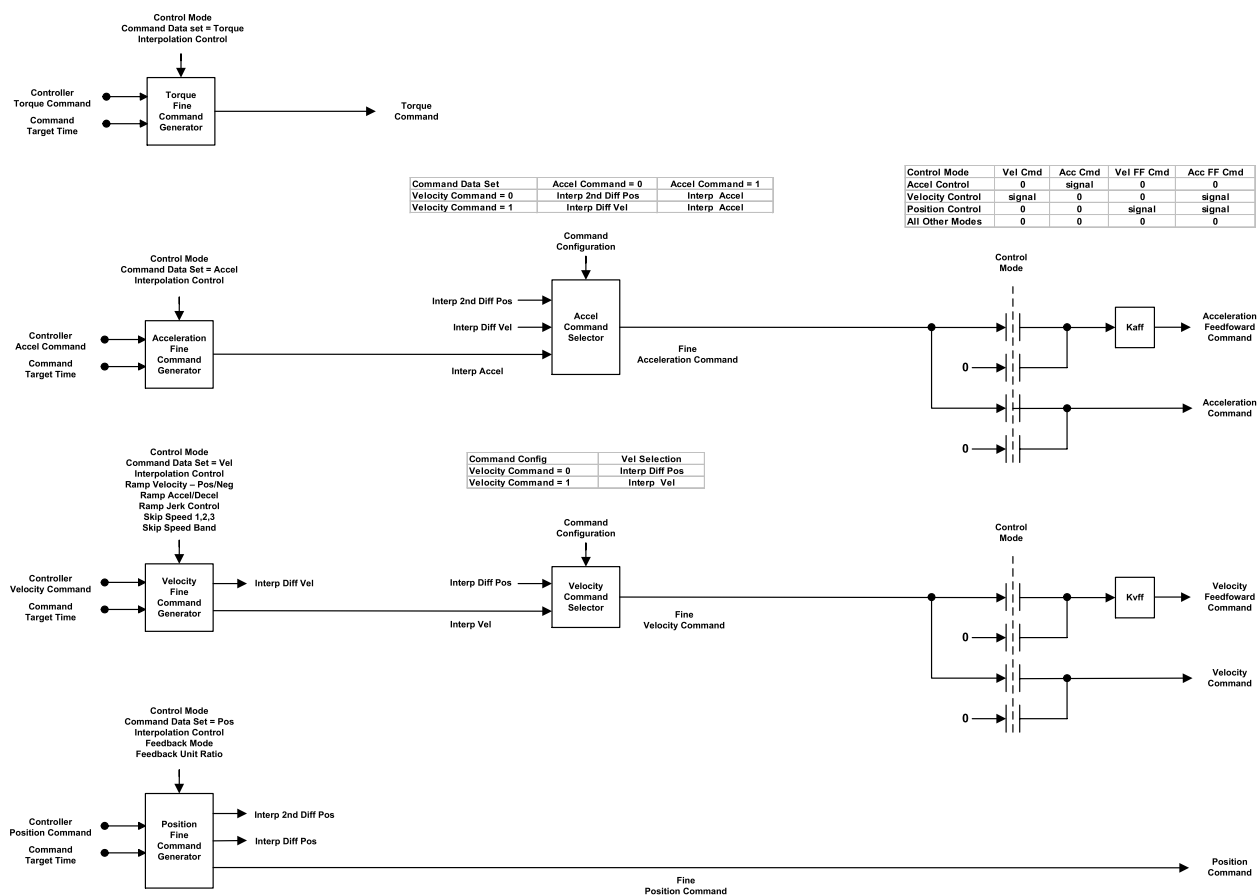
Tableau 147 – Comportement de l'affichage alphanumérique à plusieurs caractères multiaxe

Chaîne d'affichage ^a	Condition du dispositif
SELF-TEST	Exécution de l'autotest du dispositif
STANDBY	Attente du service Forward_Open
CONNECTING	Attente du premier service Définir les attributs. (Rafraîchissement)
CONFIGURING	Configuration des Attributs Instance de classe & Axe
SYNCING	Attente d'un service Group_Sync abouti.
PRE-CHARGE	Attente d'activation du bus à courant continu
X#:SHUTDOWN	L'axe a été interrompu
X#:STOPPED	L'axe a été arrêté
X#:INHIBITED	L'axe est à l'état Inhibition de démarrage
X#:STARTING	L'axe est en cours de démarrage
X#:RUNNING	L'axe est en cours d'exécution
X#:TESTING	L'axe exécute une procédure d'essai
X#:STOPPING	Décélération jusqu'à l'arrêt suite à une désactivation
X#:ABORTING	Décélération jusqu'à l'arrêt suite à un défaut
X#:INIT FLT S##	Défaut d'initialisation – Code de défaut standard
X#:INIT FLT M##	Défaut d'initialisation – Code de défaut Mfg
X#:FLT S##	Défaut d'axe – Code de défaut standard
X#:FLT M##	Défaut d'axe – Code de défaut Mfg
X#:ALARM S##	Alarme d'axe – Code d'alarme standard
X#:ALARM M##	Alarme d'axe – Code d'alarme Mfg
X#:INHIBIT S##	Inhibition de démarrage – Code d'inhibition standard
X#:INHIBIT M##	Inhibition de démarrage – Code d'inhibition Mfg
X#:SAFE FLT S##	Défaut de sécurité d'axe – Code d'inhibition standard
X#:SAFE FLT M##	Défaut de sécurité d'axe – Code d'inhibition Mfg
NODE FLT ##	Code de défaut de nœud
NODE ALARM ##	Code d'alarme de nœud
^a X# = Numéro d'instance d'axe (0, 1, 2, 3 ...) → 0 = instance de classe	

7.6.6 Comportement de génération de consigne

7.6.6.1 Sources de données de consigne

Les données de consigne qui affectent le mouvement d'axe peuvent provenir de sources bien différentes. La source de données de consigne la plus courante est celle provenant d'un planificateur de mouvement du contrôleur via la connexion C-D CIP Motion. Dans ce contexte, les données de consigne peuvent prendre la forme des commandes de position, vitesse, accélération et couple de contrôleur générées par le planificateur de mouvement (voir Figure 90). Les éléments de données de consigne fournis sont spécifiés par l'attribut Ensemble de données de consigne, qui repose sur le Mode de commande sélectionné. L'élément de données de consigne principal peut être augmenté par des éléments de consigne d'ordre supérieur afin de pouvoir générer des signaux d'action anticipatrice de meilleure qualité. En variante, ces éléments de consigne d'ordre supérieur peuvent être déduits par le dispositif à partir des données de consigne principales. Dans tous les cas, un interpolateur fin de consigne est généralement appliqué aux données de consigne afin de produire des signaux de référence de consigne destinés à la structure de commande du dispositif à sa vitesse de mise à jour.



IEC

Anglais	Français
Control Mode	Mode de commande
Command Data Set = Torque	Ensemble de données de consigne = Couple
Interpolation Control	Commande d'interpolation
Controller Torque Command	Consigne de couple du contrôleur
Command Target Time	Temps de consigne cible
Torque Fine command generator	Générateur de consigne fin de couple
Torque Command	Consigne de couple
Command Data Set	Ensemble de données de consigne
Velocity Command = 0	Consigne de vitesse = 0
Velocity Command = 1	Consigne de vitesse = 1
Accel Command = 0	Consigne d'accélération = 0
Interp 2nd Diff Pos	2ème pos diff de l'interpolateur
Interp Diff Vel	Vitesse diff interp
Accel Command = 1	Consigne d'accélération = 1
Interp Accel	Accélération d'interpolateur
Accel Control	Commande d'accélération
Velocity Control	Commande de vitesse
Position Control	Asservissement de position
All Other Modes	Tous les autres modes
Vel Cmd	Consigne de vitesse

Anglais	Français
Acc Cmd	Consigne d'accélération
Vel FF Cmd	Consigne FF vitesse
Acc FF Cmd	Consigne FF accél
Command Data Set = Accel	Ensemble de données de consigne = accél
Acceleration Fine command generator	Générateur de consigne fin d'accélération
Command Configuration	Configuration de consigne
Fine Acceleration Command	Consigne d'accélération fine
Accel Command Selector	Sélecteur de consigne d'accélération
Acceleration Feedforward Command	Consigne d'action anticipatrice d'accélération
Acceleration Command	Consigne d'accélération
Control Data Set = Vel	Ensemble de données de commande = vitesse
Ramp Velocity	Vitesse de rampe
Ramp Accel/decel	Accél/décél de rampe
Ramp Jerk Control	Commande d'à-coups de rampe
Skip Speed	Vitesse de saut
Skip Speed band	Largeur de vitesse de saut
Velocity Fine command generator	Générateur de consigne fin de vitesse
Velocity Command Selector	Sélecteur de consigne de vitesse
Velocity Feedforward Command	Consigne d'action anticipatrice de vitesse
Velocity Command	Consigne de vitesse
Fine Velocity Command	Consigne de vitesse fine
Command Data Set = Pos	Ensemble de données de consigne = Pos
Feedback Mode	Mode de réaction
Feedback Unit Ratio	Rapport d'unité de réaction
Position Fine command generator	Générateur de consigne fin de position
Fine Position Command	Consigne de position fine
Position Command	Consigne de position

Figure 90 – Générateur de consigne

7.6.6.2 Interpolation fine de consigne

Pour les applications synchronisées haute performance utilisant CIP Motion, les données de consigne sont reçues dans le cadre d'une connexion C-D CIP Motion et sont basées sur l'élément de mise à jour de consigne cible de la connexion défini sur "Interpolation", puis traitées par la fonction Interpolateur fin des blocs de génération de consigne. L'interpolateur fin est chargé de calculer les coefficients par rapport à un polynôme de trajectoire conçu pour atteindre les données de consigne à leur temps de consigne cible. En fonction de l'élément de données de consigne spécifique, il peut s'agir d'une trajectoire polynomiale de 1^{er}, 2^{ème} ou 3^{ème} ordre dont les conditions initiales reposent sur les éléments dynamiques d'axe en cours. Le polynôme étant fonction du temps, une nouvelle valeur de consigne fine peut être calculée à tout moment, à chaque fois que le dispositif CIP Motion a besoin de procéder à un calcul de commande. Il n'est donc pas nécessaire que la période de calcul de commande du dispositif soit totalement divisible dans la Période de mise à jour du contrôleur.

Afin d'améliorer l'interchangeabilité du dispositif, l'Objet Axe de dispositif de mouvement recommande un ordre minimal des interpolateurs fins. Les planificateurs de mouvement actuels générant souvent leurs trajectoires en fonction de polynômes de 3^{ème} ordre dans la position, il est important que les interpolateurs fins reproduisent ces trajectoires avec la plus grande fidélité. Par conséquent, l'interpolateur fin de position est défini en 3^{ème} ordre,

l'interpolateur de vitesse en 2^{ème} ordre et les interpolateurs d'accélération et de couple en 1^{er} ordre. Des interpolateurs fins d'ordre supérieur peuvent être utilisés et sont laissés à la discrétion du fournisseur.

Polynôme d'interpolation fine de position:

$$P(t) = a_0 + a_1 \times (t - t_0) + a_2 \times (t - t_0)^2 + a_3 \times (t - t_0)^3$$

Polynôme d'interpolation fine de vitesse:

$$V(t) = b_0 + b_1 \times (t - t_0) + b_2 \times (t - t_0)^2$$

Polynôme d'interpolation fine d'accélération:

$$A(t) = c_0 + c_1 \times (t - t_0)$$

Polynôme d'interpolation fine de couple:

$$T(t) = d_0 + d_1 \times (t - t_0)$$

Dans ces équations, t_0 représente le Temps de consigne cible de la mise à jour de planificateur de mouvement précédente, de sorte que si $t = t_0$ les valeurs de consigne de position, de vitesse, d'accélération et de couple sont égales aux valeurs envoyées lors de la mise à jour de planificateur de mouvement précédente (c'est-à-dire P_{-1} , V_{-1} , A_{-1} , et T_{-1}). Cela établit les coefficients d'ordre 0 des polynômes.

$$P(t_0) = P_{-1} = a_0$$

$$V(t_0) = V_{-1} = b_0$$

$$A(t_0) = A_{-1} = c_0$$

$$T(t_0) = T_{-1} = d_0$$

Les coefficients polynomiaux d'ordre supérieur sont calculés de telle sorte que lors de la mise à jour de planificateur de mouvement suivante, correspondant au Temps de consigne cible, t_1 , les valeurs de consigne de position, de vitesse, d'accélération et de couple soient celles envoyées dans la dernière mise à jour de planificateur de mouvement (c'est-à-dire P_0 , V_0 , A_0 , et T_0).

$$P(t_1) = P_0$$

$$V(t_1) = V_0$$

$$A(t_1) = A_0$$

$$T(t_1) = T_0$$

Grâce aux équations d'interpolation polynomiale ci-dessus, le dispositif CIP Motion peut à tout moment calculer les valeurs de consigne de position, de vitesse, d'accélération et de couple en insérant la valeur Temps système en cours du dispositif dans la variable t . Cela permet de procéder au calcul de commande du dispositif selon une planification indépendante de la planification de mise à jour du contrôleur.

Toutefois, si une chose doit être faite, c'est d'ajuster le Temps de consigne cible, t_0 , même en cas de Décalage de temps système du dispositif; t_0 et t doivent toujours reposer sur le même système de référence Temps système. Par exemple, soit Décalage_0 , le Décalage de temps système du dispositif lors de la réception de la datation de consigne de commande, t_0 . Si l'équation d'interpolation de consigne est à appliquer à $t = t_1$ et que le Décalage de temps système en cours est défini comme étant Décalage_1 alors t_0 doit être ajusté comme suit avant d'exécuter le polynôme:

$$\text{ajusté } t_0 = t_0 + (\text{Décalage}_1 - \text{Décalage}_0)$$

En variante, les valeurs de t , t_0 et t_1 peuvent être basées sur l'heure locale plutôt que sur le temps système en utilisant le décalage de temps système afin d'effectuer la conversion

Temps système – heure locale. Ceci peut être plus pratique pour la mise en œuvre de l'interpolateur et est laissé à la discrétion du fournisseur.

Les coefficients polynomiaux sont calculés selon des formules normalisées qui sont fonction de l'historique des valeurs de consigne au cours des toutes dernières mises à jour. Le nombre de valeurs de consigne historiques utilisées dans les formules dépend de l'ordre du polynôme. Par exemple, le polynôme de position de consigne de troisième ordre utilise les trois valeurs de position de consigne précédentes. Pour des raisons pratiques, les formules de coefficient polynomial de l'interpolateur sont les suivantes:

Coefficients polynomiaux d'interpolation fine de position:

$$\begin{aligned}a_0 &= P_{-1} \\a_1 &= 1/T \times (\Delta P_0 - 1/2 \times \Delta V_0 - 1/6 \times \Delta A_0) \\a_2 &= 1/T^2 \times (1/2 \times \Delta V_0) \\a_3 &= 1/T^3 \times (1/6 \times \Delta A_0)\end{aligned}$$

Coefficients polynomiaux d'interpolation fine de vitesse:

$$\begin{aligned}b_0 &= V_{-1} \\b_1 &= 1/T \times (\Delta V_0 - 1/2 \times \Delta A_0) \\b_2 &= 1/T^2 \times (1/2 \times \Delta A_0)\end{aligned}$$

Coefficients polynomiaux d'interpolation fine d'accélération (le couple se présente sous la même forme que l'accélération):

$$\begin{aligned}c_0 &= A_{-1} \\c_1 &= 1/T \times \Delta A_0\end{aligned}$$

Les équations ci-dessus reposent sur la nomenclature suivante:

$$\begin{aligned}T &= \text{Période de mise à jour du contrôleur} \\ \Delta P_0 &= (P_0 - P_{-1}) \\ \Delta V_0 &= (V_0 - V_{-1}) = (P_0 - 2P_{-1} + P_{-2}) \\ \Delta A_0 &= (A_0 - A_{-1}) = (V_0 - 2V_{-1} + V_{-2}) = (P_0 - 3P_{-1} + 3P_{-2} - P_{-3})\end{aligned}$$

Il convient d'appliquer les coefficients polynomiaux ci-dessus à l'interpolateur fin dès que possible après que t est supérieure ou égale à t_0 . Une application trop prématurée des nouveaux coefficients, c'est-à-dire avec nettement inférieure à t_0 , peut générer une erreur inutile dans la trajectoire de consigne lors de la connexion du dernier segment de l'interpolateur fin au nouveau segment de ce même interpolateur à la valeur t_0 .

Si $t > t_1$, le polynôme d'interpolation fine devient un polynôme d'extrapolation. En l'absence d'une nouvelle mise à jour provenant du planificateur de mouvement, le polynôme d'extrapolation peut être utilisé pour fournir des données de consigne estimées à la structure de commande du dispositif tant que les nouvelles données de consigne du planificateur de mouvement ne sont pas disponibles. Une fois les nouvelles données de consigne à disposition, de nouveaux coefficients polynomiaux doivent être calculés sans attendre. De cette manière, la commande de mouvement peut être maintenue ("maintien") malgré les paquets de données de connexion éventuellement retardés ou perdus dans une solution robuste de réseau de commande de mouvement répartie. Pour être clair, les données de connexion retardées sont toujours appliquées et ne sont jamais supprimées. Il s'agit toujours des dernières données disponibles provenant du contrôleur, le polynôme d'extrapolation assurant l'application des données de consigne de manière à maintenir une trajectoire de mouvement uniforme, en dépit des variations dans la livraison des données de consigne.

Si la période de mise à jour du planificateur de mouvement est suffisamment courte par rapport aux éléments dynamiques de la trajectoire de consigne ou qu'elle est comparable à la

période de calcul de commande du dispositif, l'interpolation fine peut s'avérer inutile. Le planificateur de mouvement peut procéder à cette détermination en comparant la période de mise à jour du planificateur à la période de calcul de commande du dispositif. Si l'interpolation fine est utilisée, le planificateur doit ajouter des périodes de mise à jour à sa datation. Il est donc avantageux d'éliminer ce délai de période de mise à jour du planificateur si l'interpolation est inutile.

Même si l'interpolation fine peut s'avérer inutile dans certains cas, cela ne signifie pas que les données de consigne sont à appliquer directement à la structure de commande du dispositif. Il peut toujours être nécessaire de calculer les polynômes ci-dessus de sorte que le dispositif puisse extrapoler la valeur de consigne en cas de mise à jour de commande du dispositif. En effet, il n'est en général pas utile que la datation de mise à jour de commande du dispositif corresponde à la datation des données de consigne.

Enfin, certaines applications et certains types de dispositif CIP Motion n'exigent pas la précision dynamique qu'offrent l'interpolation et l'extrapolation horodatées. Diverses applications de commande de vitesse et d'asservissement de couple peuvent par exemple entrer dans cette catégorie. En général, les données de consigne peuvent également être appliquées aux structures de commande des dispositifs d'entraînement à fréquence variable sans interpolation ni extrapolation.

7.6.6.3 Sélecteurs de consigne

Les sélecteurs de vitesse et d'accélération permettent de sélectionner la source des signaux de consigne de vitesse fine et de consigne d'accélération fine. La sélection repose sur la valeur Ensemble de données de consigne du contrôleur.

Par exemple, si le contrôleur fournit la consigne de vitesse, la consigne de vitesse fine provient de la sortie du générateur de consigne de vitesse fine. Lorsqu'il est configuré pour une interpolation ou une extrapolation fine, le polynôme utilisé pour calculer la consigne de vitesse fine est donné par:

$$V(t) = b_0 + b_1 \times (t - t_0) + b_2 \times (t - t_0)^2$$

Les coefficients polynomiaux sont les suivants:

$$b_0 = V_{-1}$$

$$b_1 = 1/T \times (\Delta V_0 - 1/2 \times \Delta A_0)$$

$$b_2 = 1/T^2 \times (1/2 \times \Delta A_0)$$

Dans ce cas, les valeurs de ΔV_0 et ΔA_0 sont calculées en termes de valeurs Consigne de vitesse, V_0 , V_{-1} , et V_{-2} :

$$\Delta V_0 = (V_0 - V_{-1})$$

$$\Delta A_0 = (V_0 - 2V_{-1} + V_{-2})$$

Lorsque le contrôleur ne fournit pas la consigne de vitesse, le signal de consigne de vitesse fine est alors le signal de position différentielle interpolée provenant du générateur de consigne de position fine qui est la dérivée temporelle du signal de consigne de position fine. Lorsqu'ils sont configurés pour une interpolation ou une extrapolation fine, les polynômes utilisés pour calculer la consigne de vitesse fine sont les mêmes que ceux décrits ci-dessus, mais les coefficients polynomiaux sont calculés en utilisant les valeurs ΔV_0 et ΔA_0 déduites des valeurs Consigne de position, P_0 , P_{-1} , P_{-2} , et P_{-3} :

$$\Delta V_0 = (P_0 - 2P_{-1} + P_{-2})$$

$$\Delta A_0 = (P_0 - 3P_{-1} + 3P_{-2} - P_{-3})$$

7.6.6.4 Générateur de rampe de consigne

La fonction Générateur de rampe du bloc Générateur de consigne est appliquée à la valeur Données de consigne envoyée par le contrôleur lorsque l'élément Mise à jour de consigne cible de la connexion est défini sur le mode "Immédiat". En mode Immédiat, les données de consigne sont appliquées immédiatement à la structure de commande du dispositif. Dans la mesure où aucun planificateur de mouvement ne génère habituellement les données de consigne dans ce mode, la valeur Données de consigne émise par le contrôleur peut changer considérablement d'une mise à jour à l'autre. Pour traiter cette condition, une fonction Générateur de rampe est nécessaire pour ajuster le moteur à la nouvelle valeur Données de consigne dans les limites dynamiques du système. Exemple: lorsque la valeur Consigne de vitesse du contrôleur passe subitement de 0 à 30 tours/secondes dans le mode Immédiat, le générateur de rampe produit un signal Consigne de vitesse fine qui contribue à l'accélération du moteur jusqu'à la valeur Consigne de vitesse du contrôleur en fonction des valeurs d'attribut configurées Accélération de rampe et Commande d'à-coups. L'attribut Commande d'à-coups de rampe détermine le pourcentage de rampes d'accélération ou de décélération de la Courbe en S, la partie restante de la rampe étant régie par les valeurs d'attribut Accélération ou Décélération de rampe fixes.

Alors qu'une fonction Générateur de rampe pourrait être intégrée dans chacun des blocs Générateur de consigne fin pour les consignes de position, de vitesse et d'accélération, cette version de la spécification de l'Objet Axe de dispositif de mouvement prend en charge uniquement un générateur de rampe dans le bloc Générateur de consigne de vitesse fine.

Le générateur de rampe applique les limites de vitesse directionnelles aux données de consigne, en s'assurant que la consigne de vitesse ne dépasse jamais les valeurs Pos/Nég de vitesse maximale configurées.

Le générateur de rampe prend également en charge la fonction Amorçage instantané. Lors de l'activation du dispositif d'entraînement alors que le moteur est toujours en mouvement, la sortie du générateur de rampe est initialisée sur la vitesse actuelle du moteur. À partir de cette vitesse, le générateur de rampe accélère ou décélère doucement le moteur par rapport à la consigne de vitesse du contrôleur en cours.

Enfin, le générateur de rampe prend en charge les largeurs de saut qui sont le plus souvent utilisées dans les applications de contrôle de fréquence lorsque certaines vitesses excitent la fréquence de résonance mécanique du moteur et de la charge. La fonction Largeur de saut permet de définir trois Vitesses de saut distinctes qui décalent le signal Consigne de vitesse afin d'éviter, ou d'ignorer, ces vitesses problématiques. La Largeur de vitesse de saut détermine la plage de vitesses centrée sur les trois Vitesses de saut que le dispositif évite. Si la baisse de la consigne de vitesse se situe dans la Largeur de saut, mais qu'elle est inférieure à la Vitesse de saut, la sortie Consigne de vitesse est définie sur la Vitesse de saut moins la moitié de la Largeur de vitesse de saut. Si la baisse de la consigne de vitesse se situe dans la Largeur de saut, mais qu'elle est supérieure à la Vitesse de saut, la sortie Consigne de vitesse est définie sur la Vitesse de saut plus la moitié de la Largeur de vitesse de saut.

7.6.6.5 Sélection du signal d'action anticipatrice

Les générateurs fins de consigne définis dans le cadre de l'Objet Axe de dispositif de mouvement peuvent générer des dérivées plus élevées de l'entrée de données de consigne pouvant faire office de signaux d'action anticipatrice. Les unités propres aux signaux d'action anticipatrice de vitesse et d'accélération diffèrent généralement des unités dérivées, et les signaux dérivés doivent ainsi être mis à l'échelle de manière appropriée. Toutefois, une qualité de signal supérieure peut être fournie par les générateurs de trajectoire du planificateur de mouvement. Les blocs de sélection d'action anticipatrice prélèvent le meilleur signal d'action anticipatrice en fonction des bits définis dans l'attribut Ensemble de données de consigne. Le meilleur signal est défini comme étant le signal déduit utilisant le moins d'opérations de différenciation. La position de consigne fine est appliquée directement à la boucle d'asservissement de position sans les déréférencements et décalages habituels. Ces

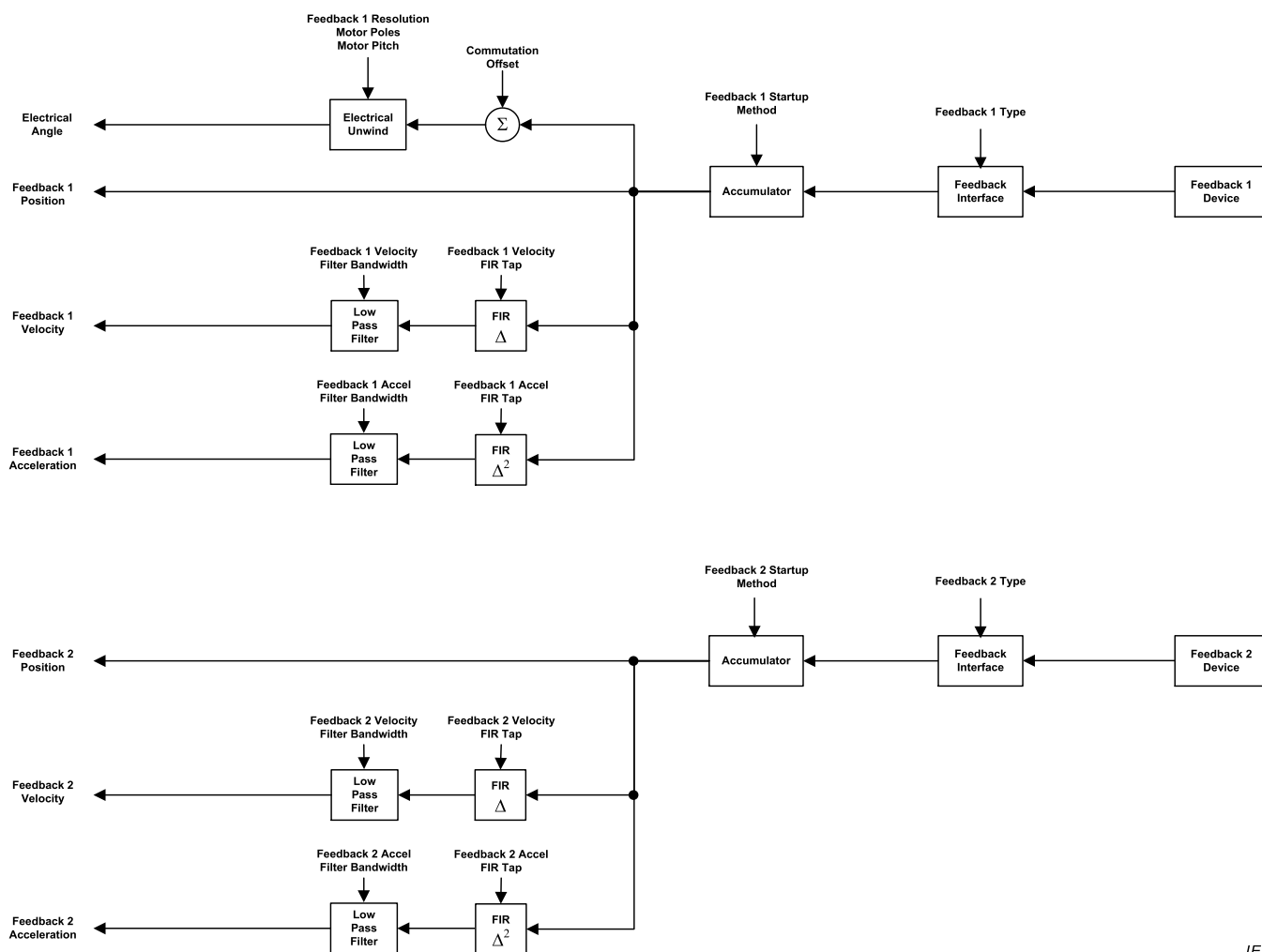
opérations sont supposées être réalisées par le planificateur de mouvement basé sur le contrôleur.

7.6.7 Comportement de l'interface de réaction

7.6.7.1 Sources de réaction

Les signaux de réaction définis par l'Objet Axe de dispositif de mouvement peuvent être déduits de l'un des 4 différents canaux d'interface de réaction (voir Figure 91). Les deux principaux canaux de réaction utilisés par les différents modes de contrôle de boucle fermée sont appelés Réaction 1 et Réaction 2. Ceci permet aux boucles d'asservissement de fonctionner avec un dispositif de réaction basé sur le moteur (généralement associé au canal Réaction 1) ou avec un dispositif de réaction en aval connecté au canal Réaction 2. Le choix de la source de réaction utilisée par la boucle est régi par l'attribut Mode de réaction.

Chaque interface de réaction est en mesure de prendre en charge un certain nombre de types de dispositif de réaction différents énumérés par l'attribut Type de réaction. La sortie de l'interface de réaction est le nombre de réactions que le dispositif de réaction a déplacées depuis son dernier échantillonnage. Si le dispositif de réaction est un dispositif absolu, l'interface de réaction détermine également la position absolue du dispositif de réaction lors de la mise sous tension, puis communique cette valeur à l'accumulateur de réaction pour le prédéfinir.



IEC

Anglais	Français
Electrical Angle	Angle électrique
Feedback 1 Position	Position de Réaction 1
Feedback 1 Velocity	Vitesse de Réaction 1
Feedback 1 Acceleration	Accélération de Réaction 1
Feedback 2 Position	Position de Réaction 2
Feedback 2 Velocity	Vitesse de Réaction 2
Feedback 2 Acceleration	Accélération de Réaction 2
Feedback 1 Resolution	Résolution de Réaction 1
Motor Poles	Pôles de moteur
Motor Pitch	Pas du moteur
Electrical Unwind	Déroutement électrique
Feedback 1 Velocity	Retour en vitesse 1
Filter Bandwidth	Bande passante du filtre
Low Pass Filter	Filtre passe bas
Feedback 1 Accel	Retour en accélération 1
Feedback 2 Velocity	Retour en vitesse 2
Feedback 2 Accel	Retour en accélération 2

Anglais	Français
Commutation Offset	Décalage de commutation
Feedback 1 Startup Method	Méthode de démarrage de Réaction 1
Feedback 1 Type	Type de Réaction 1
Feedback 2 Startup Method	Méthode de démarrage de Réaction 2
Feedback 2 Device	Dispositif de Réaction 2
Feedback 2 Type	Type de Réaction 2
Feedback 2 Mode	Mode de Réaction 2
Accumulator	Accumulateur
Feedback Interface	Interface de réaction
Feedback 1 Device	Dispositif de Réaction 1
Feedback 2 Device	Dispositif de Réaction 2

Figure 91 – Canaux de réaction 1 et 2

7.6.7.2 Accumulateur de réaction

Le rôle de l'accumulateur de réaction dépend de la méthode de démarrage de réaction n configurée, qui peut être Incrémentielle ou Absolue. Si le mode Incrémentiel est sélectionné, l'accumulateur accumule simplement les changements de valeurs de nombre de réactions (un entier signé à 32 bits) à chaque mise à jour du dispositif. Si le mode Absolu est déterminé, l'accumulateur de réaction fonctionne de la même manière qu'en mode Incrémentiel. La seule différence est l'initialisation de l'accumulateur à la mise sous tension du dispositif. En mode Incrémentiel, l'accumulateur de réaction est défini sur zéro, alors qu'en mode Absolu, il est initialisé à la position absolue du dispositif de réaction. Cela permet de récupérer la position absolue dans le cadre d'un cycle d'alimentation, tant que le mouvement de mise hors tension du dispositif de réaction absolu est limité à la moitié de la plage de réaction absolue du dispositif. Il n'existe pas d'exigence d'extension de la plage de position absolue du dispositif de réaction par stockage non volatil de l'accumulateur. Ce simple mécanisme de gestion de réaction absolue est dû au fait que CIP Motion place normalement la responsabilité de cette extension de la plage de position absolue d'axe, et l'établissement de la référence de position de machine absolue, sur le contrôleur.

7.6.7.3 Déroulement et décalage de commutation

Un bloc Déroulement électronique est également connecté à l'interface Réaction 1. Ce bloc est conçu pour dérouler ou moduler la sortie de l'accumulateur de position pour générer un signal proportionnel à l'angle électrique d'un moteur à aimant permanent, en fonction du nombre de pôles ou du pas polaire du moteur. Pour aligner ce signal aux enroulements ABC physiques du moteur plutôt que le zéro du dispositif de réaction, un Décalage de commutation configurable est ajouté avant le bloc Déroulement électrique.

7.6.7.4 Filtrage de réaction

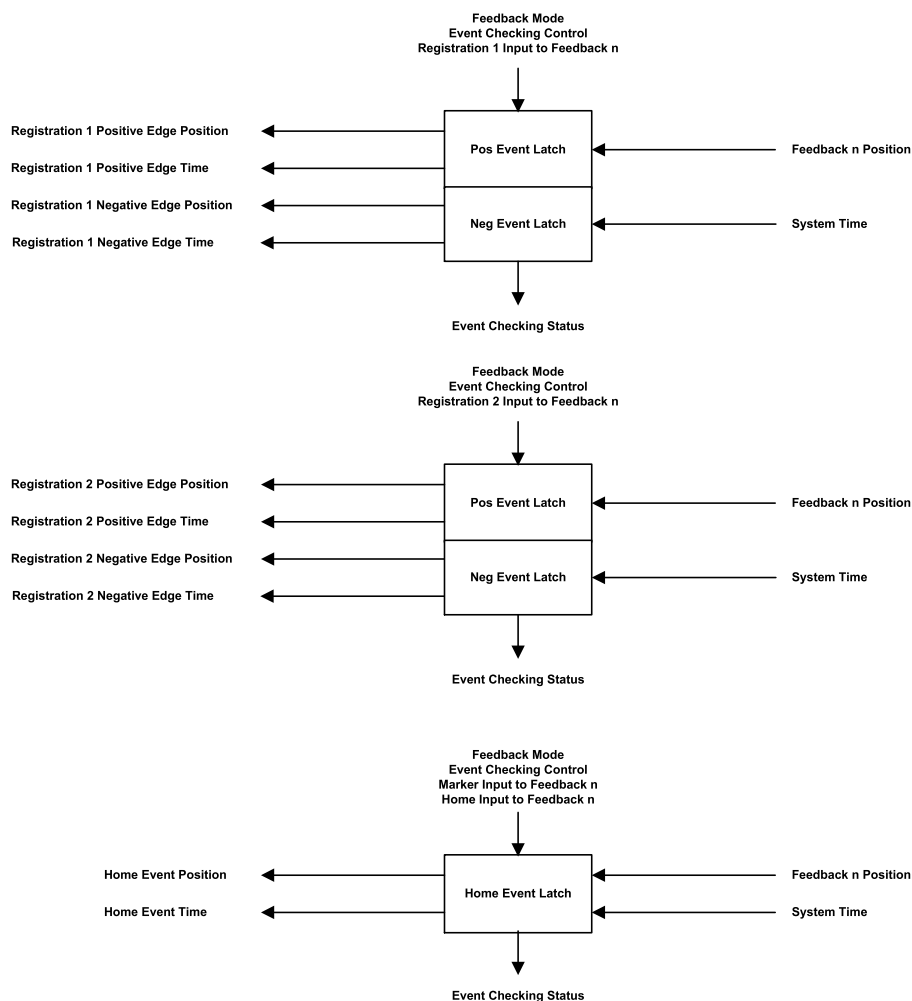
Un filtre IIR passe-bas configurable est défini par l'Objet Axe de dispositif de mouvement pour filtrer les estimations de vitesse et d'accélération pour chaque canal de réaction. Ces filtres peuvent être utilisés pour réduire le niveau de bruit de quantification associé aux signaux de réaction numérique de différenciation. La bande passante des filtres IIR de vitesse et d'accélération pour chaque canal de réaction peut être programmée individuellement.

7.6.8 Comportement de capture d'événement

7.6.8.1 Sources d'entrée d'événement

L'Objet Axe de dispositif de mouvement définit un mécanisme de capture de la position de réaction et de la datation associées aux transitions d'état spécifiques des sources d'entrée d'événement sélectionnées. Les sources d'entrée d'événement actuellement prises en charge

par l'objet sont Enregistrement 1, Enregistrement 2, Marqueur et Interrupteur d'origine. Ces 4 sources d'entrée d'événement s'appliquent à chaque canal de réaction pris en charge (voir Figure 92).



IEC

Anglais	Français
Feedback Mode	Mode de réaction
Event Checking Control	Commande de vérification d'événement
Registration 1 Input to Feedback n	Entrée d'enregistrement 1 dans Réaction n
Registration 1 Positive Edge Position	Position du front positif 1 de l'enregistrement
Registration 1 Positive Edge Time	Durée du front positif 1 de l'enregistrement
Registration 1 Negative Edge Position	Position du front négatif 1 de l'enregistrement
Registration 1 Negative Edge Time	Durée du front négatif 1 de l'enregistrement
Pos Event Latch	Verrouillage d'événement positif
Neg Event Latch	Verrouillage d'événement négatif
Feedback n Position	Position de Réaction n
System Time	Temps système
Event Checking Status	État de vérification d'événement
Registration 2 Input to Feedback n	Entrée d'enregistrement 2 dans Réaction n
Registration 2 Positive Edge Position	Position du front positif 2 de l'enregistrement
Registration 2 Positive Edge Time	Durée du front positif 2 de l'enregistrement

Anglais	Français
Registration 2 Negative Edge Position	Position du front négatif 2 de l'enregistrement
Registration 2 Negative Edge Time	Durée du front négatif 2 de l'enregistrement
Marker Input to Feedback n	Entrée de marqueur dans réaction n
Home Input to Feedback n	Entrée d'origine dans réaction n
Home Event Position	Position d'événement d'origine
Home Event Time	Durée d'événement d'origine
Home Event Latch	Verrouillage d'événement d'origine

Figure 92 – Fonctionnalité de capture d'événement

7.6.8.2 Verrouillages d'événement

Pour faciliter la capture précise de la position et de l'heure de réaction, des verrouillages d'événement matériel sont généralement mis en œuvre (voir le schéma ci-dessous). Deux verrouillages indépendants sont définis pour chaque entrée d'enregistrement, l'un pour capturer les événements de transition de front positif et l'autre pour capturer les événements de transition de front négatif. Cette conception permet de capturer les événements d'enregistrement dans des applications à impulsions d'enregistrement étroites dans lesquelles les fronts de montée et de descente se produisent presque en même temps. Outre les verrouillages d'enregistrement, un autre verrou est également défini pour la capture d'événement d'origine. L'événement d'entrée d'origine qui déclenche le Verrou d'événement d'origine peut être un nombre de combinaisons différentes d'événements d'entrée d'interrupteur d'origine ou de marqueur, c'est-à-dire des transitions de marqueur, des transitions d'interrupteur ou des transitions d'interrupteur suivies d'une transition de marqueur.

Avec les verrouillages d'événement basés sur le matériel, la précision de capture d'événement est en général uniquement limitée par la latence de l'entrée d'événement associée. Les entrées d'événement Enregistrement et Marqueur sont filtrées de manière à obtenir une précision de capture d'événement de l'ordre de 1 μ s. En termes de précision de capture de position, cette valeur est calculée comme étant le produit de la précision de capture d'événement et de la vitesse d'axe. Les entrées d'interrupteur d'origine sont en général très fortement filtrées, et sont donc limitées à une précision de capture d'événement de 1 ms à 10 ms. Par conséquent, pour obtenir une capture de position précise en fonction d'une transition d'entrée d'interrupteur d'origine, une séquence de retour à la position de référence à faible vitesse est exigée.

7.6.8.3 Datations d'événement

Étant donné que la datation d'enregistrement est transmise au contrôleur comme partie intégrante des données Notification d'événement, le contrôleur peut appliquer la datation de l'événement à l'historique des positions des autres axes du système afin d'interpoler leurs positions. Cette possibilité est particulièrement utile dans les applications dans lesquelles il est nécessaire de déterminer l'emplacement de plusieurs axes au moment d'un événement d'enregistrement unique. Plus la datation est précise, plus le contrôleur peut déterminer précisément ces positions.

On doit toutefois ajuster la datation de l'événement, t_0 , quand bien même un décalage du temps système pour le dispositif venait à se produire avant sa transmission au contrôleur; la datation de l'événement doit toujours être basée sur le même système de référence de temps système au moment de la transmission. Par exemple, si Décalage_0 est le décalage du temps système du dispositif lors de la datation effective de l'événement t_0 , et t_1 est le décalage du temps système au moment où l'événement est à transmettre au contrôleur, t_0 doit alors être ajusté pour être t_1 préalablement à la transmission au contrôleur avec le reste des données d'événement associées, c'est-à-dire: $t_1 = t_0 + (\text{Décalage}_1 - \text{Décalage}_0)$.

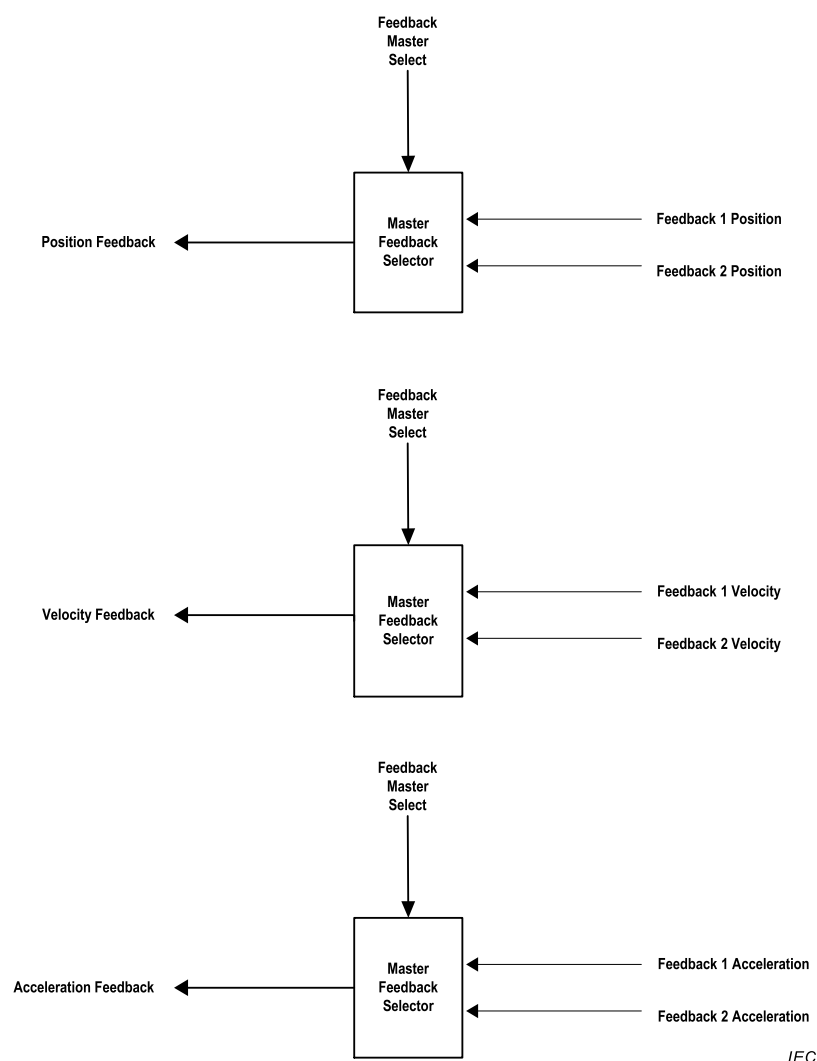
7.6.9 Comportement du mode de commande

7.6.9.1 Généralités

Les attributs d'instance définis en 7.3 ont un impact sur le comportement du dispositif dans le contexte du Mode de commande, de la Méthode de commande et du Mode de réaction. Comme indiqué en 4.2 concernant le domaine d'application de l'Objet Axe de dispositif de mouvement, il existe 4 Modes de commande de base communs aux dispositifs: l'asservissement de position, la commande de vitesse, l'asservissement de couple et pas d'asservissement. Cette liste a été complétée par un nouveau mode de commande, la commande d'accélération, pour compléter la progression de la commande de vitesse à l'asservissement de couple. Ce paragraphe 7.6.9 propose un schéma pour chaque mode de commande afin de mieux définir le comportement collectif des différents attributs Objet Axe de dispositif de mouvement.

7.6.9.2 Mode Pas d'asservissement (Réaction uniquement)

Une instance Objet Axe de dispositif de mouvement peut être configurée pour le mode Pas d'asservissement. Cette sélection de Mode de commande s'applique à plusieurs Fonctions de dispositif différentes qui comprennent les Convertisseurs d'alimentation de bus et les dispositifs Réaction uniquement. La fonctionnalité d'axe Réaction uniquement permet au contrôleur d'accéder à la position, la vitesse et l'accélération d'un des quatre canaux de réaction possibles par l'intermédiaire de la connexion Dispositif-Contrôleur. Ces signaux peuvent alors être répartis dans le système de commande de mouvement en tant qu'axe maître pour les opérations d'engrenage et de came. Dans ce mode, l'attribut Sélection de réaction principale, s'il est pris en charge, détermine le canal de réaction qui génère les signaux Retour en position, Retour en vitesse et Retour en accélération (voir Figure 93).



Anglais	Français
Feedback Master Select	Sélection de réaction principale
Position Feedback	Retour en position
Master Feedback Selector	Sélecteur de réaction principale
Feedback 1 Position	Retour en position 1
Feedback 2 Position	Retour en position 2
Velocity Feedback	Retour en vitesse
Feedback 1 Velocity	Retour en vitesse 1
Feedback 2 Velocity	Retour en vitesse 2
Acceleration Feedback	Retour en accélération
Feedback 1 Acceleration	Retour en accélération 1
Feedback 2 Acceleration	Retour en accélération 2

Figure 93 – Pas d'asservissement (Réaction uniquement)

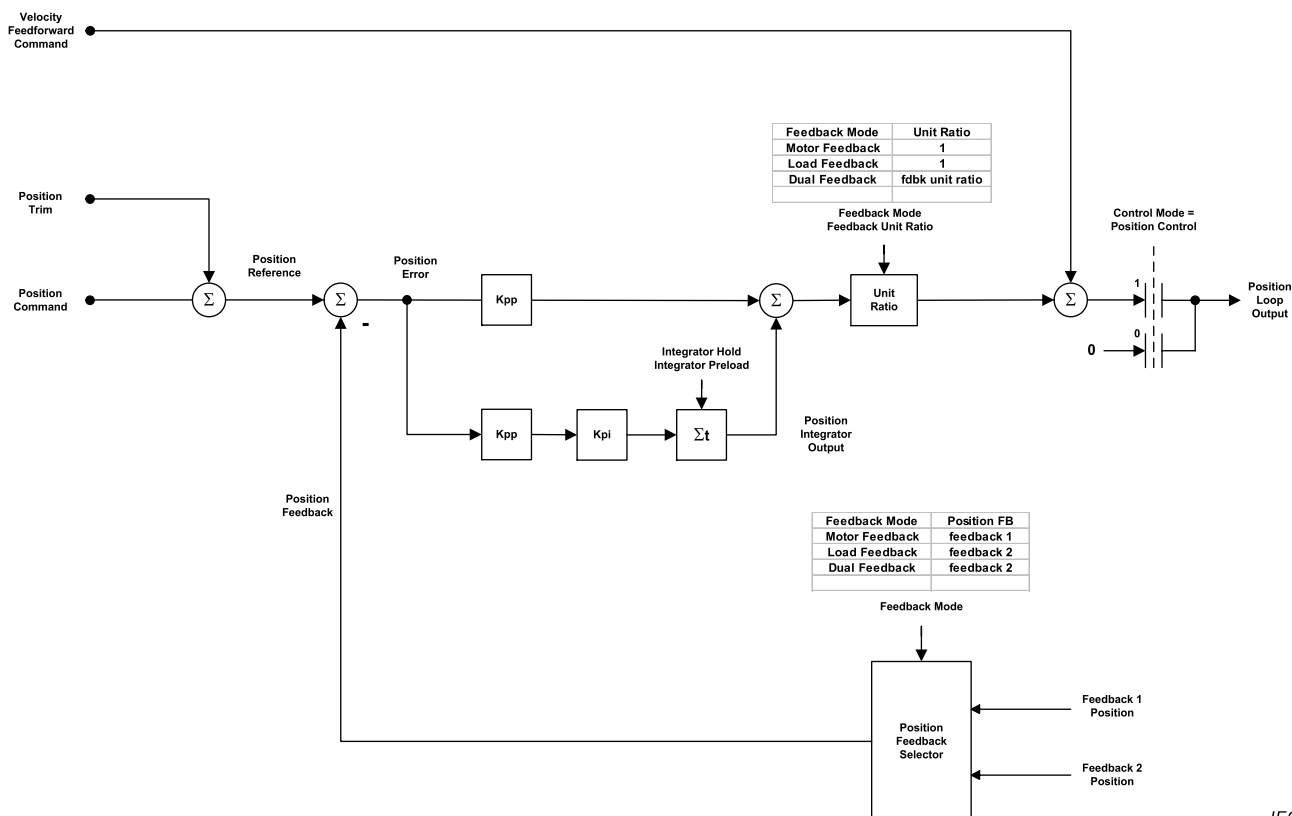
7.6.9.3 Asservissement de position

En mode Asservissement de position, la seule méthode de commande opérationnelle prise en charge par l'objet en cours est le contrôle asservi en boucle fermée. À une date ultérieure, l'objet peut être étendu afin d'inclure une méthode d'asservissement de position pas à pas.

7.6.9.4 Asservissement de position en boucle fermée

7.6.9.4.1 Généralités

Lors de l'exécution d'un asservissement de position en boucle fermée, le dispositif applique la sortie de signal Consigne de position du Générateur de consigne à la jonction de somme de boucle de position. Outre la Consigne de position, une entrée Compensation de position est fournie et peut être utilisée pour décaler la boucle de position. La boucle d'asservissement PI classique génère un signal Sortie de la boucle de position vers une boucle de vitesse interne (voir Figure 94).



IEC

Anglais	Français
Velocity Feedforward Command	Commande d'action anticipatrice de vitesse
Position Trim	Compensation de position
Position Command	Consigne de position
Position Reference	Référence de position
Position Error	Erreur de position
Feedback Mode	Mode de réaction
Motor Feedback	Réaction de moteur
Load Feedback	Réaction de charge
Dual Feedback	Réaction double
Unit Ratio	Rapport d'unité
Feedback Unit Ratio	Rapport d'unité de réaction
Control Mode = Position Control	Mode de commande = Asservissement de position
Position Loop Output	Sortie de boucle de position
Integrator Hold	Maintien de l'intégrateur

Anglais	Français
Integrator Preload	préchargement de l'intégrateur
Position Integrator Output	Sortie d'intégrateur de position
Position Feedback	Retour en position
Position FB	Retour en position
Feedback 1	Réaction 1
Feedback 2	Réaction 2
Position Feedback Selector	Sélecteur de retour en position
Feedback 1 Position	Retour en position 1
Feedback 2 Position	Retour en position 2

Figure 94 – Asservissement de position en boucle fermée

7.6.9.4.2 Sélection du retour en position

La réaction au régulateur PI peut être déduite à partir de deux canaux de réaction différents. Cette souplesse permet à la boucle de position de fonctionner avec un dispositif de réaction basé sur le moteur (généralement associé au canal Réaction 1) ou avec un dispositif de réaction en aval connecté au canal Réaction 2. Le choix de la source de réaction utilisée par la boucle est régi par l'attribut Mode de réaction.

Lorsque le Mode de réaction appelle une opération Réaction double, la boucle de position et la boucle de vitesse utilisent respectivement le canal Réaction 2 et le canal Réaction 1. Étant donné que les deux canaux de réaction peuvent ne pas présenter la même résolution de réaction, il est nécessaire de convertir la sortie de boucle de position du nombre de Réaction 1 en nombre de Réaction 2 avant d'appliquer la sortie à la jonction de somme de boucle de vitesse. Ceci est effectué en ajustant la sortie de boucle de position via le bloc Mise à l'échelle d'unités avec le Rapport d'unité de réaction.

7.6.9.4.3 Gains PI de position

Le Gain proportionnel du contrôleur PI classique définit la bande passante de gain unitaire de la boucle de position en rad/s, alors que le Gain intégral est utilisé pour attribuer une valeur nulle au signal Erreur de position afin de compenser les effets d'un couple ou de forces statiques et quasi statiques à la charge.

7.6.9.4.4 Action anticipatrice de vitesse

La boucle de vitesse interne implique une entrée de commande non nulle afin de générer une vitesse moteur d'axe en régime permanent. Pour fournir une sortie non nulle au moteur à partir du dispositif, une sortie de boucle de position non nulle est exigée, laquelle traduit en erreur de position différente de zéro. Cette erreur dynamique entre la position de consigne et la position réelle pendant le mouvement est appelée "erreur suivante". La plupart des applications de commande de mouvement en boucle fermée ne souhaitent aucune "erreur suivante", définitivement. Cela est possible dans une certaine mesure grâce à l'utilisation de la commande de gain intégral de position (voir ci-dessus). Mais, en règle générale, le temps de réponse de l'action d'intégrateur est trop lent pour être efficace dans des applications de contrôle de mouvement haute performance. Une autre approche présentant une réponse dynamique supérieure consiste à utiliser l'Action anticipatrice de vitesse.

La fonction Action anticipatrice de vitesse est utilisée en mode Asservissement de position afin de fournir l'essentiel des entrées Référence de vitesse pour générer la vitesse moteur souhaitée. Pour ce faire, la sortie de signal Consigne de vitesse fine du Générateur de consigne est ajustée par le Gain d'action anticipatrice de vitesse, puis le signal Consigne d'action anticipatrice de vitesse résultant est ajouté à la Sortie de la boucle de position générée par la boucle de position pour former le signal Référence de vitesse. Avec cette fonction, la boucle de position n'a pas besoin de générer trop d'effort pour produire le niveau

de consigne de vitesse exigé, la valeur Erreur de position étant réduite de manière significative. Le signal Consigne d'action anticipatrice de vitesse permet de réduire l'erreur suivante de la boucle d'asservissement de position à une valeur proche de zéro, dans le cadre d'une exécution à vitesse constante. Cela est important dans les applications telles que les applications d'engrenage et de synchronisation électroniques, dans lesquelles il est nécessaire que la position réelle d'axe ne soit pas décalée de manière trop significative derrière la position commandée à tout moment.

En théorie, la valeur optimale du Gain d'action anticipatrice de vitesse est de 100 %. Dans la réalité cependant, il peut s'avérer nécessaire de peaufiner la valeur afin d'adapter les boucles de vitesse au gain de boucle fini. Un aspect susceptible de forcer une valeur Action anticipatrice de vitesse inférieure se présente sous la forme d'actions anticipatrices toujours plus nombreuses qui ont tendance à exacerber le dépassement d'axe. C'est la raison pour laquelle l'action anticipatrice n'est pas recommandée pour les applications de positionnement point à point.

7.6.9.5 Commande de vitesse

En mode Commande de vitesse, deux méthodes de commande opérationnelles sont prises en charge par l'objet: la commande de vitesse en boucle fermée et le Contrôle de fréquence en boucle ouverte.

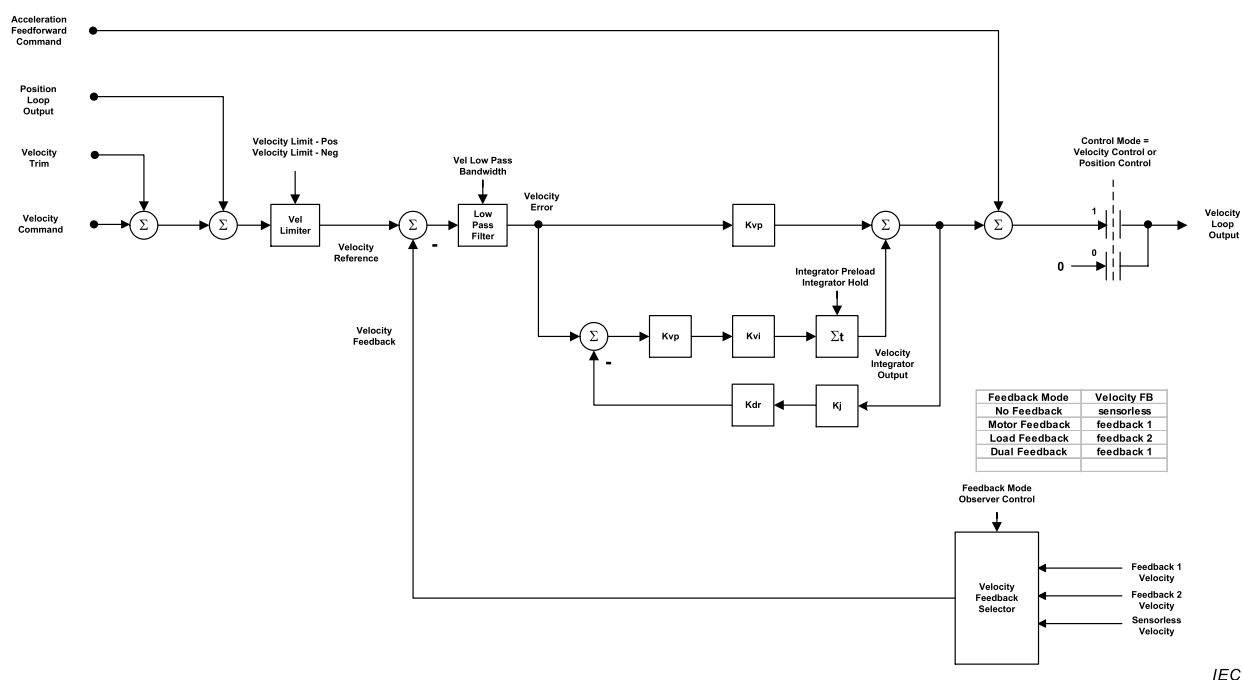
7.6.9.6 Commande de vitesse en boucle fermée

7.6.9.6.1 Généralités

La méthode Commande de vitesse en boucle fermée s'adresse aux applications nécessitant une régulation de vitesse importante. L'entrée de commande dans la boucle de vitesse peut être déduite directement de la Consigne de vitesse du Générateur de consigne en configuration Mode Commande de vitesse, ou de la sortie de boucle de position en configuration Mode Asservissement de position, comme indiqué en 7.6.9.3 (voir Figure 95).

S'il s'agit d'une boucle de vitesse externe en Mode Commande de vitesse, le dispositif applique l'entrée Consigne de vitesse à la jonction de somme de consigne de vitesse afin de générer le signal Référence de vitesse dans un régulateur PI classique. De même, l'entrée Compensation de vitesse contribue à la jonction de somme de consigne de vitesse, et peut être utilisée conjointement avec une boucle d'asservissement externe afin de légèrement ajuster la vitesse du moteur.

S'il s'agit d'une boucle de vitesse interne en Mode Asservissement de position, le dispositif applique le signal Sortie de boucle de position à l'entrée de la jonction de somme de consigne de vitesse. Une valeur nulle est en général affectée aux signaux d'entrée qui ne s'appliquent pas au mode de commande configuré.



IEC

Anglais	Français
Acceleration Feedforward Command	Commande d'action anticipatrice d'accélération
Position Loop Output	Sortie de la boucle de position
Velocity Trim	Compensation de vitesse
Velocity Command	Consigne de vitesse
Velocity limit	Limite de vitesse
Velocity limiter	Limiteur de vitesse
Velocity Reference	Référence de vitesse
Vel Low Pass Bandwidth	Bande passante passe-bas de vitesse
Low Pass Filter	Filtre passe-bas
Velocity Error	Erreur de vitesse
Integrator Preload	Préchargement d'intégrateur
Integrator Hold	Maintien de l'intégrateur
Velocity Integrator Output	Sortie d'intégrateur de vitesse
Control Mode = Velocity Control or Position Control	Mode de commande = Commande de vitesse ou Asservissement de position
Velocity Loop Output	Sortie de boucle de vitesse
Velocity Feedback	Retour en vitesse
Feedback Mode	Mode de réaction
No Feedback	Absence de réaction
Feedback 1	Réaction 1
Feedback 2	Réaction 2
Dual Feedback	Réaction double
Velocity FB	Retour en vitesse
Motor Feedback	Réaction de moteur
Load Feedback	Réaction de charge
Observer Control	Commande d'observation

Anglais	Français
Velocity Feedback Selector	Sélecteur de retour en vitesse
Feedback 1 Velocity	Retour en vitesse 1
Feedback 2 Velocity	Retour en vitesse 2
Sensorless velocity	Vitesse sans capteur

Figure 95 – Commande de vitesse en boucle fermée**7.6.9.6.2 Limiteur de vitesse**

La sortie du signal de jonction de somme de consigne de vitesse traverse un bloc de limite classique pour produire le signal Référence de vitesse dans la boucle de vitesse. Le bloc Limiteur de vitesse applique une limite de vitesse directionnelle – Limite de vitesse Pos ou Nég – à l'entrée du signal de consigne de vitesse établie sur le signe.

7.6.9.6.3 Sélection de réaction de vitesse

La réaction au régulateur PI peut être déduite de l'un des deux transducteurs de réaction disponibles, Réaction 1 ou Réaction 2. Le choix de la source de réaction utilisée par la boucle est régi par l'énumération Mode de réaction. Si le mode Réaction est Pas de réaction, indiquant un fonctionnement sans capteur, le signal Retour en vitesse est le signal Vitesse sans capteur généré par l'algorithme de commande sans capteur.

7.6.9.6.4 Filtre d'erreur de vitesse

Un filtre passe-bas peut être éventuellement appliqué au signal d'erreur de vitesse généré par la jonction de somme de boucle de vitesse. La sortie de ce filtre devient le signal Erreur de vitesse, qui est par la suite actionné par l'algorithme de commande PI de boucle de vitesse. Si le filtre est utilisé, il est défini sur 5 à 10 fois la bande passante de boucle de vitesse. Il est recommandé que ce filtre soit un filtre IIR bipolaire afin d'optimiser son efficacité au filtrage du bruit de quantification.

7.6.9.6.5 Gains PI de vitesse

La boucle de vitesse génère un signal Sortie de boucle de vitesse adressé à la boucle interne suivante par l'intermédiaire d'une structure de boucle d'asservissement PI classique. Le Gain proportionnel du contrôleur définit la bande passante de gain unitaire de la boucle de position en rad/s, alors que le Gain intégral est utilisé pour attribuer une valeur nulle au signal Erreur de vitesse afin de compenser le couple ou les forces statiques et quasi statiques appliqués à la charge. Le trajet de l'intégrateur inclut un gain proportionnel de sorte que les unités du gain intégral représentent la bande passante de l'intégrateur en rad/s.

La section intégrale du régulateur de vitesse comprend une fonction antitorsion. La fonction antitorsion maintient automatiquement le terme intégral du régulateur lorsqu'une condition limite est atteinte dans la chaîne d'action. Elle est conditionnée par le signe arithmétique de l'entrée de l'intégrateur. L'intégrateur est maintenu lorsque le signe de l'entrée est tel que la sortie de l'intégrateur se déplace dans la limite active. En d'autres termes, l'intégrateur est autorisé à fonctionner (non maintenu) si l'entrée a tendance à amener sa valeur de sortie à l'extérieur de la limite active.

L'intégrateur peut également être configuré pour un fonctionnement de maintien d'intégrateur. Si la valeur true (vrai) est affectée à l'attribut Maintien de l'intégrateur, le régulateur empêche l'intégrateur de procéder à l'accumulation lorsqu'il est demandé à l'axe de se déplacer. Ce comportement est utile dans les applications de positionnement point à point.

Une fonction de préréglaage automatique du terme intégral du régulateur de vitesse se déclenche en cas de transition du mode d'asservissement de couple au mode de commande de vitesse, à l'aide du paramètre de sélection Mode de commande. Lors de la transition en mode de vitesse, le terme intégral du régulateur de vitesse est prérégulé sur le paramètre de

référence de couple du moteur. Si l'erreur de vitesse est petite, cela offre une transition "sans secousse" entre la dernière valeur de référence de couple qui précède l'entrée en mode de vitesse.

7.6.9.6.6 Statisme de vitesse

Le statisme de vitesse est une autre fonction du régulateur de vitesse. L'entrée d'erreur de vitesse dans le terme intégral est réduite par une fraction de la sortie du régulateur de vitesse, contrôlée par le paramètre de gain de statisme K_{dr} . Au fur et à mesure de l'augmentation du couple de charge sur le moteur, la vitesse réelle du moteur diminue proportionnellement au gain de statisme. Cela est utile lorsqu'un certain niveau de conformité est exigé en raison d'un couplage mécanique rigide entre deux moteurs.

7.6.9.6.7 Action anticipatrice d'accélération

La boucle de vitesse implique une sortie de boucle de vitesse non nulle afin de générer une accélération moteur d'axe en régime permanent. Pour permettre au dispositif d'entraînement de fournir une sortie non nulle au moteur, une erreur de vitesse non nulle est en général exigée. Dans les applications d'asservissement de position, cette erreur de vitesse non nulle devient une erreur de boucle de position non nulle. Étant donné que la plupart des applications de commande de mouvement en boucle fermée exigent une erreur de boucle d'asservissement proche de zéro, ce comportement n'est pas souhaitable. De même, l'erreur de boucle de position et de vitesse peut être réduite en appliquant la commande de gain intégral de vitesse comme indiqué ci-dessus, mais l'intégrateur est toujours trop lent pour être très efficace. L'approche privilégiée avec la réponse dynamique supérieure consiste à utiliser l'Action anticipatrice d'accélération.

La fonction Action anticipatrice d'accélération permet de générer la plus grande partie de la référence d'accélération nécessaire pour produire l'accélération commandée. Pour ce faire, il suffit d'ajuster l'Action anticipatrice d'accélération fine produite par le générateur de consigne par le Gain d'action anticipatrice relative à l'accélération et d'ajouter le signal Consigne d'action anticipatrice relative à l'accélération résultant en tant que décalage à la sortie de la boucle de vitesse. Avec cette fonction, la boucle de vitesse n'a pas besoin de générer une mesure de contrôle supplémentaire, et donc de limiter l'erreur de boucle d'asservissement.

En théorie, la valeur optimale de l'Action anticipatrice d'accélération est de 100 %. Toutefois, dans la réalité, il peut s'avérer nécessaire de peaufiner la valeur afin d'adapter les variations d'inertie de charge et de constante de couple du moteur. À l'instar de l'Action anticipatrice de vitesse, l'Action anticipatrice d'accélération peut donner lieu à un comportement de dépassement. Elle ne doit donc pas être utilisée dans des applications de positionnement point à point.

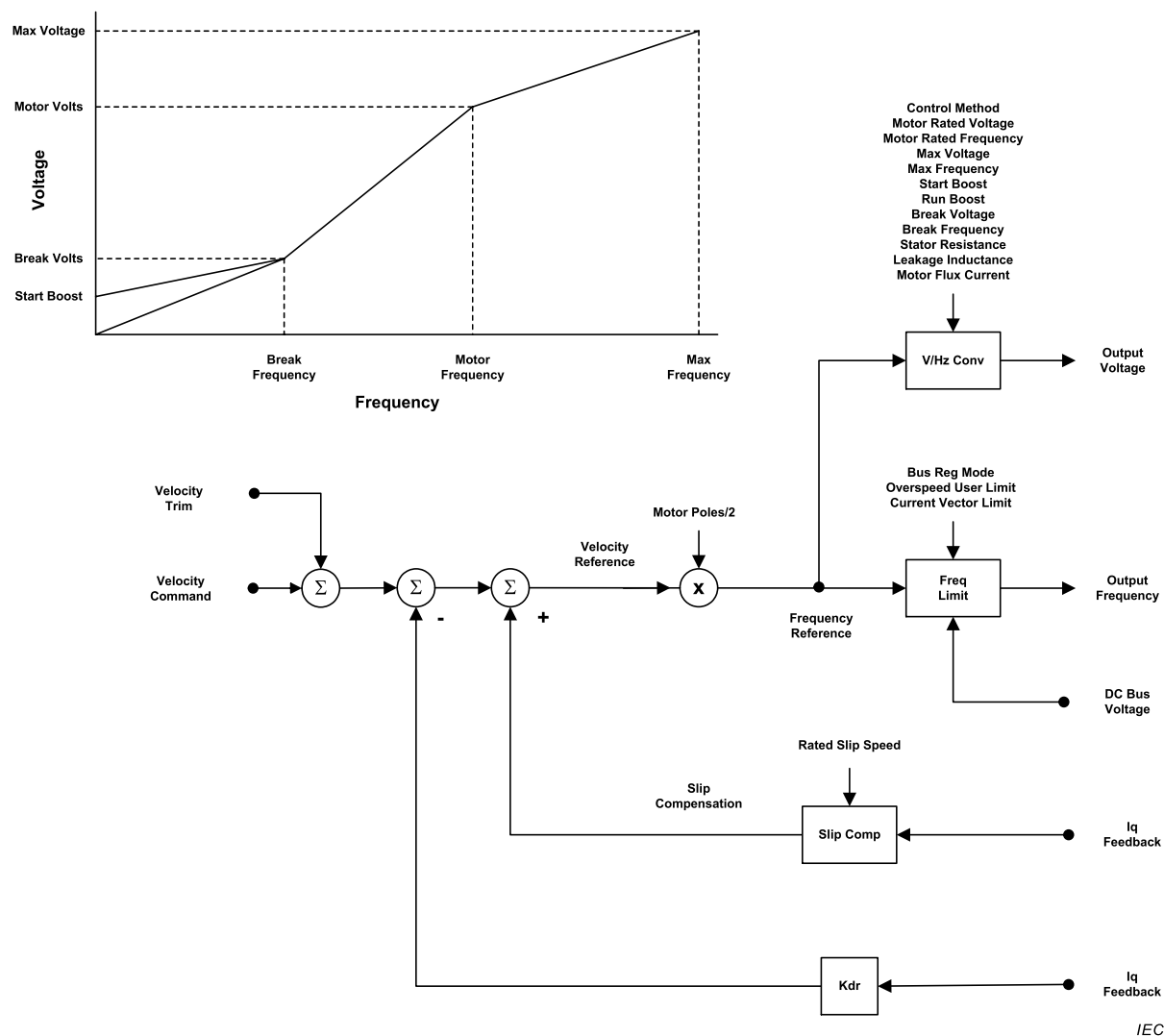
Une utilisation conjointe de l'Action anticipatrice d'accélération et de l'Action anticipatrice de vitesse permet de réduire l'erreur suivante de boucle d'asservissement de position ou de commande de vitesse à une valeur proche de zéro pendant les phases d'accélération et de décélération du mouvement. Cela est important dans les applications de suivi qui utilisent des opérations d'engrenage et de came électroniques pour synchroniser précisément un axe esclave aux mouvements d'un axe maître.

7.6.9.7 Contrôle de fréquence en boucle ouverte

7.6.9.7.1 Généralités

Une autre méthode de commande de vitesse est le Contrôle de fréquence en boucle ouverte associé à des dispositifs d'entraînement Volts/Hertz ou à des dispositifs d'entraînement à fréquence variable (VFD) dépourvus de boucle d'asservissement et qui entraînent en général un moteur à induction. Avec cette méthode, la commande de vitesse est obtenue en contrôlant la sortie de tension et de fréquence du dispositif d'entraînement, de manière à ce que la tension soit en général proportionnelle à la fréquence. Dans le cas d'un moteur à induction, la vitesse du moteur est déterminée par la Fréquence de sortie du dispositif

d'entraînement divisée par le nombre de Pôles du moteur. Cette méthode de commande est adaptée aux applications de commande de vitesse qui n'ont pas besoin d'une régulation de vitesse importante, et donc d'un dispositif de réaction. La Figure 96 représente cette méthode de commande de vitesse en boucle ouverte.



Anglais	Français
Max Voltage	Tension max
Motor Volts	Volts moteur
Voltage	Tension
Break Volts	Volts de rupture
Start Boost	Suralimentation au démarrage
Break Frequency	Fréquence de rupture
Motor Frequency	Fréquence du moteur
Max Frequency	Fréquence max
Frequency	Fréquence
Control Method	Méthode de commande
Motor Rated Voltage	Tension assignée du moteur
Motor Rated Frequency	Fréquence assignée du moteur

Anglais	Français
Run Boost	Suralimentation à l'exécution
Break Voltage	Tension de rupture
Stator Resistance	Résistance statorique
Leakage Inductance	Inductance de fuite
Motor Flux Current	Courant de flux du moteur
Output Voltage	Tension de sortie
V/Hz Conv	Conv. V/Hz
Velocity Trim	Compensation de vitesse
Velocity Command	Consigne de vitesse
Velocity Reference	Référence de vitesse
Motor Poles	Pôles du moteur
Frequency Reference	Référence de fréquence
Freq Limit	Limite de fréquence
Output Frequency	Fréquence de sortie
Bus Reg Mode	Mode de régulateur de bus
Overspeed User Limit	Limite utilisateur de survitesse
Current Vector Limit	Limite de vecteur de courant
DC Bus Voltage	Tension de bus à courant continu
Rated Slip Speed	Vitesse de glissement assignée
Slip Compensation	Compensation de glissement
Iq Feedback	Réaction Iq

Figure 96 – Contrôle de fréquence en boucle ouverte**7.6.9.7.2 Fonctionnement Volts/Hertz de base**

L'Objet Axe de dispositif de mouvement offre un certain nombre d'attributs utilisés pour spécifier la relation qu'utilise le dispositif d'entraînement entre la fréquence de sortie (vitesse) et la tension de sortie d'un moteur (à induction) donné. Les attributs Fréquence de rupture et Tension de rupture définissent le point de la courbe Volts/Hertz en dessous duquel la fonction Suralimentation au démarrage est appliquée. Comme son nom l'indique, Suralimentation au démarrage permet de fournir une tension de sortie non nulle au moteur lors du gel de vitesse pour faciliter le démarrage. La contribution de la Suralimentation au démarrage à la tension de sortie du dispositif d'entraînement tombe à zéro lorsque le moteur atteint la Fréquence de rupture. Au-dessus du point de rupture, la tension de sortie et la fréquence de sortie suivent une pente linéaire jusqu'au point défini par la Fréquence assignée du moteur et la Tension assignée du moteur. À partir de ce point, la courbe Volts/Hertz suit une autre pente linéaire jusqu'au point défini par les attributs Fréquence maximale et Tension maximale. Ce segment de la courbe Volts/Hertz permet un fonctionnement au-dessus de la fréquence assignée et de la tension assignée du moteur dans des applications qui l'exigent.

7.6.9.7.3 Fonctionnement avec vecteur sans capteur

Le vecteur sans capteur est une méthode alternative de commande de vitesse qui n'exige pas la configuration d'une courbe Volts/Hertz. En revanche, le dispositif d'entraînement, connaissant la résistance statorique et l'inductance de fuite du moteur, peut calculer la tension de sortie appropriée exigée pour une fréquence de sortie donnée. Cette méthode assure un meilleur comportement de commande de vitesse faible que la méthode Volts/Hertz de base.

7.6.9.7.4 Compensation de glissement

Lorsqu'un moteur à induction est entraîné à une fréquence spécifique, la vitesse réelle du moteur est en général inférieure à la vitesse de consigne, donnée par la fréquence de sortie divisée par le nombre de pôles du moteur, d'une valeur proportionnelle au couple de charge appliqué au moteur. Cette différence de vitesse est appelée "glissement", et est un attribut de configuration associé au moteur. L'Objet Axe de dispositif de mouvement prend en charge une fonction Compensation de glissement commune aux dispositifs d'entraînement à fréquence variable. La valeur Compensation de glissement appliquée à la Référence de vitesse est le produit du couple mesuré générant le courant, I_q , et de la Vitesse de glissement assignée configurée du moteur à induction.

7.6.9.7.5 Statisme de vitesse

Une autre fonction définie pour la méthode Contrôle de fréquence est la fonction de statisme. Cette fonction permet de réduire la référence de vitesse par une fraction proportionnée du couple générant le courant, I_q , contrôlé par le paramètre de gain de statisme, K_{dr} . Au fur et à mesure de l'augmentation du couple de charge sur le moteur, la vitesse réelle du moteur diminue proportionnellement au gain de statisme. Cela est utile lorsqu'un certain niveau de conformité est exigé en cas de partage de couple entre deux moteurs sur une charge commune.

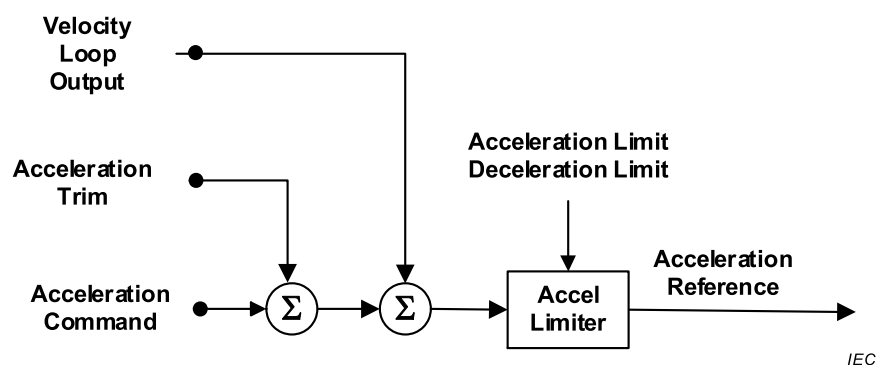
7.6.9.8 Commande d'accélération

7.6.9.8.1 Généralités

La commande de moteur dynamique par consigne d'accélération n'étant pas habituelle dans le secteur industriel, la commande d'accélération a été ajoutée à l'Objet Axe de dispositif de mouvement pour compléter la progression dynamique de la commande de vitesse à l'asservissement de couple. La sortie de la boucle de vitesse comporte également des unités d'accélération. Par conséquent, comme les autres modes de commande, les contributions de Consigne d'accélération, de Compensation d'accélération et de Sortie de boucle de vitesse sont ajoutées pour former le signal Référence d'accélération, lequel fait office de l'une des entrées principales de la section Asservissement de couple.

7.6.9.8.2 Limiteur d'accélération

La sortie du signal de jonction de somme de consigne d'accélération traverse un bloc de limite classique pour produire le signal Référence d'accélération. Le bloc Limiteur d'accélération applique une limite d'accélération directionnelle, la limite d'accélération et la limite de décélération, au signal de consigne d'entrée en fonction du signe du signal (voir Figure 97).



Anglais	Français
Velocity Loop Output	Sortie de boucle de vitesse
Acceleration Trim	Compensation d'accélération

Anglais	Français
Acceleration Command	Consigne d'accélération
Acceleration Reference	Référence d'accélération
Acceleration Limit	Limite d'accélération
Deceleration Limit	Limite de décélération
Accel limiter	Limiteur d'accélération

Figure 97 – Commande d'accélération

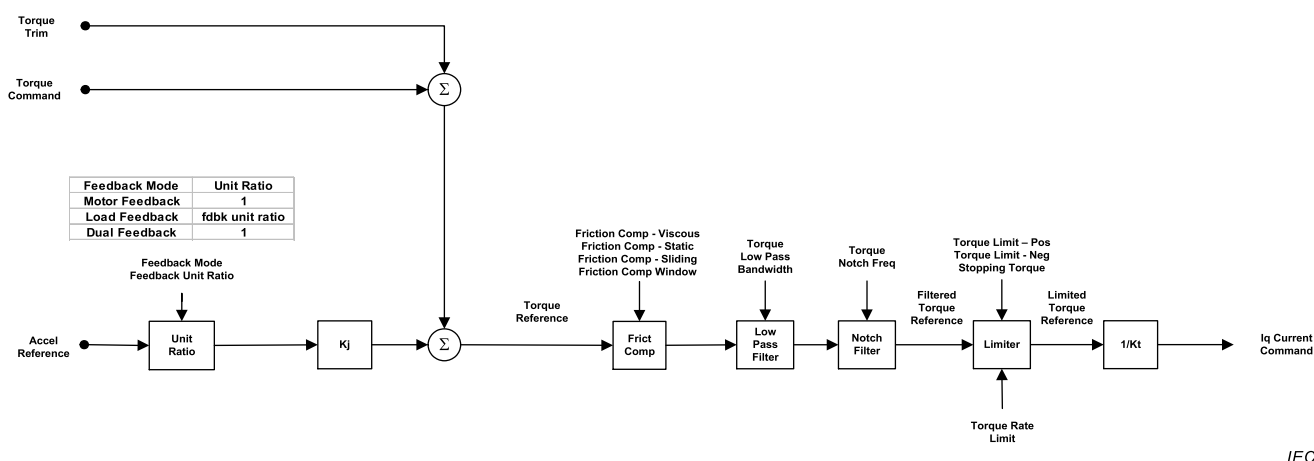
7.6.9.9 Asservissement de couple

7.6.9.9.1 Généralités

En règle générale, le couple est proportionnel à l'accélération et au couple générant le courant du moteur, I_q . La structure Asservissement de couple a pour objet de combiner les signaux d'entrée pour créer une Référence de couple à partir d'un certain nombre de sources différentes, en fonction du Mode de commande, et d'appliquer différents filtres et algorithmes de compensation à la Référence de couple pour créer une Référence de couple filtrée. Le signal Référence de couple filtrée est ajusté par la réciproque de la constante de couple K_t du moteur afin de devenir l'entrée Consigne de courant I_q dans la boucle de courant. Étant donné que le courant du moteur est également fonction du % de courant assigné du moteur, la constante de couple, K_t , est égale à 1. En d'autres termes, il est généralement supposé qu'un courant assigné à 100 % produit un couple assigné à 100 %.

7.6.9.9.2 Sources d'entrée de couple

La section Asservissement de couple peut intégrer des entrées provenant d'un large éventail de sources dépendantes du Mode de commande. L'entrée du trajet de Référence de couple peut passer par la Consigne de couple cyclique ou le signal Compensation de couple du mode Asservissement de couple. En mode d'asservissement de position ou de commande de vitesse, l'entrée de couple est déduite de la boucle de vitesse externe en plaçant les signaux d'accélération obtenus et en les ajustant dans le couple équivalent (voir Figure 98).



IEC

Anglais	Français
Torque Trim	Compensation de couple
Torque Command	Consigne de couple
Accel Reference	Référence d'accélération
Torque Reference	Référence de couple
Feedback Mode	Mode de réaction

Anglais	Français
Motor Feedback	Réaction de moteur
Load Feedback	Réaction de charge
Dual Feedback	Réaction double
Unit Ratio	Rapport d'unité
Feedback Unit ratio	Rapport d'unité de réaction
Friction Comp	Compensation de frottement
Viscous	visqueux
Static	statique
Sliding	glissant
Window	Fenêtre
Torque	Couple
Low Pass Bandwidth	Bande passante passe-bas
Low Pass Filter	Filtre passe-bas
Notch Freq	Fréq. de rejet
Notch filter	Filtre coupe-bande
Filtered Torque Reference	Référence de couple filtrée
Torque Limit – Pos	Limite de couple – Pos
Torque Limit – Neg	Limite de couple – Nég
Stopping Torque	Couple d'arrêt
Limiter	Limiteur
Torque Rate Limit	Limite de débit de couple
Iq Current Command	Consigne de courant Iq

Figure 98 – Asservissement de couple**7.6.9.9.3 Ajustement de l'accélération par rapport au couple**

Dans la mesure où les signaux d'entrée d'accélération dans la section d'asservissement du couple sont exprimés en unités d'accélération, un facteur de mise à l'échelle K_j est nécessaire pour convertir les unités d'accélération en unités de pourcentage de couple assigné. Ce facteur de mise à l'échelle, lorsqu'il est correctement configuré, représente l'inertie ou la masse totale du système qui inclut le moteur et la charge, et a la capacité d'annuler les effets de l'inertie/masse du système sur la réponse de la boucle d'asservissement et les paramètres de gain de boucle. Étant donné que les unités de couple sont exprimées en % de couple assigné du moteur, les unités propres à l'attribut Inertie du système sont exprimées en % assigné par unités moteur/s². Toutefois, les unités d'accélération peuvent être exprimées en unités de réaction 1 ou 2 en fonction du paramètre Mode de réaction. Par conséquent, dans le cas où la réaction 2 s'applique, il est nécessaire d'ajuster le signal d'accélération par le rapport d'unité de réaction comme le montre le bloc Rapport d'unité à la Figure 98.

7.6.9.9.4 Compensation de frottement**7.6.9.9.4.1 Généralités**

La compensation de frottement applique un couple ou une force directionnel(le) de compensation au moteur afin de surmonter les effets de frottement sur le système mécanique, réduisant ainsi le nombre de mesures de contrôle exigées. Des attributs individuels ont été définis afin de prendre en charge la compensation du frottement statique, du frottement par glissement (Coulomb) et du frottement visqueux. Un attribut de fenêtre de compensation est également fourni afin d'atténuer les vibrations du moteur, associé aux méthodes de compensation de frottement traditionnelles.

7.6.9.9.4.2 Compensation de frottement statique

Il n'est pas rare qu'un axe subisse un frottement statique conséquent, communément appelé "sticktion en anglais", dans les applications d'asservissement statique que le système mécanique refuse de traiter. Bien entendu, il est possible d'utiliser le gain intégral de position pour générer une sortie suffisante du dispositif d'entraînement afin de corriger l'erreur, mais cette approche peut ne pas être satisfaisante pour l'application. Une méthode alternative consiste à appliquer la compensation de frottement statique pour rompre le frottement statique avec une erreur de position non nulle. Ceci peut être réalisé par l'ajout ou le retrait d'un niveau de couple fixe, tel que déterminé par l'attribut Compensation de frottement statique, à la valeur de signal Référence de couple en fonction de son signe actuel. Cette forme de compensation de frottement doit être appliquée uniquement lorsque l'axe est statique, c'est-à-dire lorsque la consigne de position ne change pas.

La valeur Compensation de frottement statique doit être juste inférieure à la valeur qui supprimerait le frottement statique. Une valeur plus grande entraîne une "vibration" d'axe, à savoir un phénomène qui décrit un mouvement avant et arrière rapide d'axe centré sur la position commandée, du fait qu'il surcompense le frottement statique (sticktion).

Afin de traiter le problème de vibration lors de l'application de la Compensation de frottement statique, il est appliqué une fenêtre de compensation de frottement au voisinage de la position de consigne actuelle lorsque l'axe est au repos. Si la position réelle se situe dans la Fenêtre de compensation de frottement, la valeur Compensation de frottement statique est appliquée à la sortie asservie, mais est ajustée par le rapport du signal Erreur de position sur la Fenêtre de compensation de frottement. À l'intérieur de la fenêtre, les intégrateurs de boucle de position et de boucle de vitesse sont également désactivés afin d'éviter l'effet de pompage qui se produit lors du déroulement ascendant des intégrateurs. Ainsi, dès que l'erreur de position atteint ou dépasse la valeur de l'attribut Fenêtre de compensation de frottement, la valeur Compensation de frottement statique intégrale est appliquée. Bien entendu, si la fenêtre de compensation de frottement doit être initialisée, cette fonction est effectivement désactivée.

Une fenêtre de compensation de frottement non nulle a pour effet d'adoucir la compensation de frottement statique dans la mesure où elle l'applique à la référence de couple, et de réduire par ailleurs les effets de vibration et de pompage qu'elle peut générer. Cette fonction permet généralement d'appliquer des valeurs supérieures de compensation de frottement statique, permettant un meilleur positionnement point à point.

7.6.9.9.4.3 Compensation de frottement par glissement

Le frottement par glissement ou frottement Coulomb, par définition, est la composante de frottement indépendante de la vitesse tant que le système mécanique est en mouvement. Le frottement par glissement est toujours moins important que le frottement statique pour un système mécanique donné. La méthode de compensation du frottement par glissement est fondamentalement la même que celle de compensation du frottement statique, mais le niveau de couple ajouté à la référence de couple est toutefois inférieur à celui appliqué pour surmonter le frottement statique et est déterminé par l'attribut Compensation de frottement par glissement. La Compensation de frottement par glissement est appliquée uniquement lorsque l'axe fait l'objet d'une commande de mouvement.

7.6.9.9.4.4 Compensation de frottement visqueux

Le frottement visqueux, par définition, est la composante de frottement qui augmente linéairement avec la vitesse du système mécanique. La méthode de compensation du frottement visqueux consiste à multiplier la valeur configurée Compensation de frottement visqueux par la vitesse du moteur et à appliquer le résultat au signal de référence de couple. La Compensation de frottement visqueux est appliquée uniquement lorsque l'axe fait l'objet d'une commande de mouvement.

7.6.9.9.5 Filtre passe-bas

Le filtre passe-bas est efficace en matière de contrôle de résonance lorsque la fréquence de résonance naturelle est beaucoup plus élevée ($>5x$) que la bande passante de la boucle d'asservissement. Ce filtre fonctionne en réduisant la quantité d'énergie haute fréquence de sortie du dispositif qui excite la résonance naturelle. Le filtre passe-bas peut être conçu avec un seul pôle ou plusieurs pôles. Toutefois, il doit être veillé à limiter la quantité de retard de phase introduite par ce filtre dans la boucle d'asservissement afin d'éviter une éventuelle instabilité.

7.6.9.9.6 Filtre coupe-bande

Le filtre coupe-bande est efficace en matière de contrôle de résonance lorsque la fréquence de résonance naturelle est plus élevée que la bande passante de la boucle d'asservissement. À l'instar du filtre passe-bas, le filtre coupe-bande fonctionne en réduisant de manière significative la quantité d'énergie dans la sortie du dispositif qui peut exciter la résonance naturelle. Il peut être utilisé même lorsque la fréquence de résonance naturelle est relativement proche de la bande passante de la boucle d'asservissement. En effet, le retard de phase introduit par le filtre coupe-bande se trouve aux alentours de la fréquence de rejet. Pour que le filtre coupe-bande soit efficace, il faut que sa Fréquence soit très proche de la fréquence de résonance naturelle de la charge.

Une équation classique pour le filtre coupe-bande est la suivante:

$$G(s) = \frac{s^2 + \omega_n^2}{s^2 + s \times \omega_n/Q + \omega_n^2}$$

Dans cette équation, Q représente l'intensité du rejet. Dans la plupart des mises en œuvre, l'intensité, Q, est en général incorporée au programme du dispositif. La profondeur d'affaiblissement du filtre coupe-bande est infinie.

7.6.9.9.7 Limiteur de couple

Le signal Référence de couple filtrée passe par un bloc de limite afin de générer un signal Référence de couple limitée. Le bloc Limiteur de couple applique une limite de couple au signal en fonction du signe de l'entrée du signal de référence de couple et de l'état d'axe. En fonctionnement normal, il s'agit des attributs Limite de couple – Positive et Limite de couple – Négative définis par l'utilisateur et appliqués au signal de référence de couple. S'il est demandé à l'axe de s'arrêter suite à une condition de demande de désactivation ou de défaut principal, le dispositif applique la Limite de couple d'arrêt.

Un bloc de limite de couple est également intégré à la Limite de taux de variation de couple. Cette fonction limite le taux de variation de la sortie de référence de couple.

7.6.9.9.8 Ajustement du couple par rapport au courant

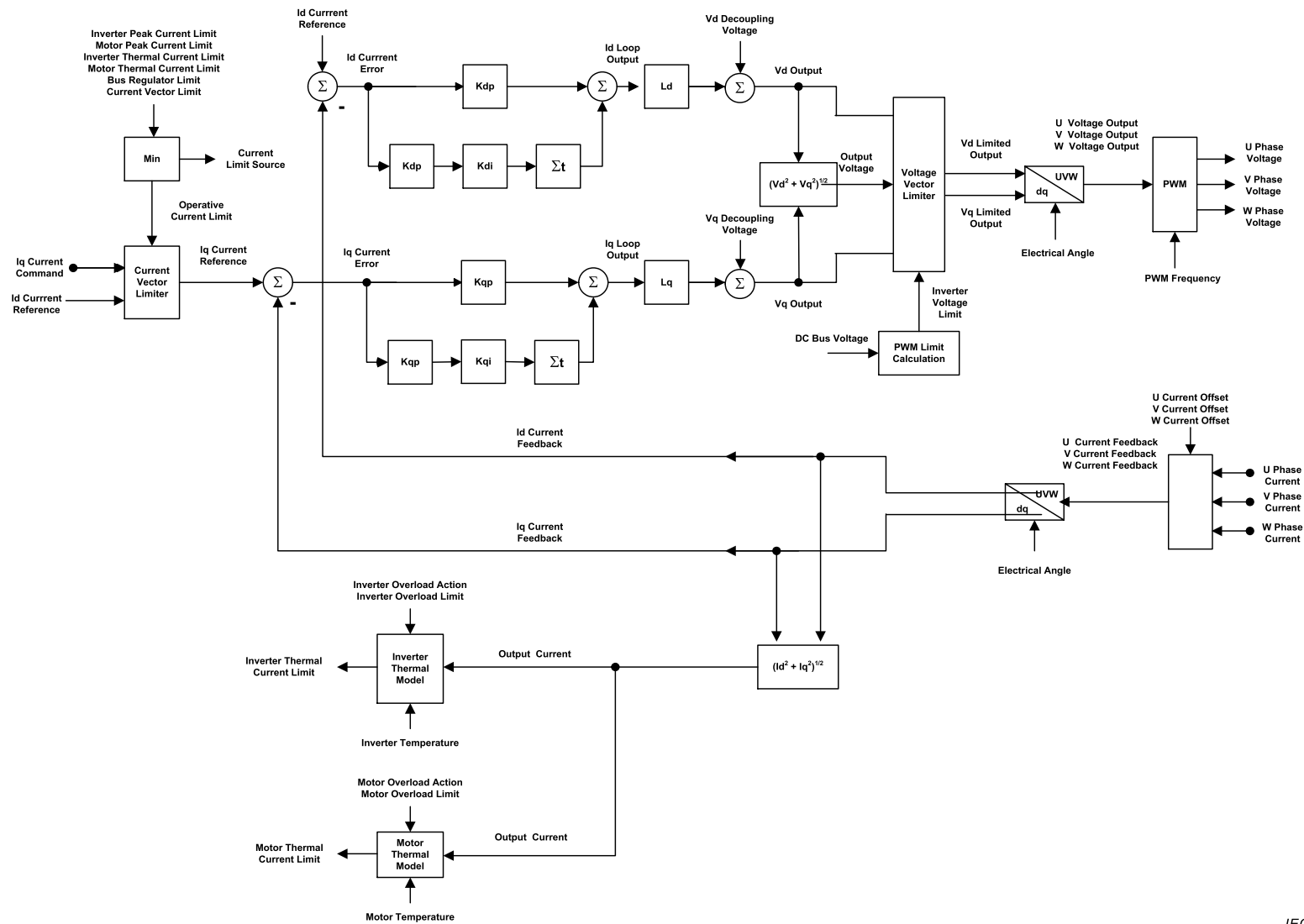
Le résultat final de l'ensemble de ces fonctions de filtrage de signal, de compensation et de limitation appliquées au couple se présente sous la forme du signal de référence de couple filtrée qui, lorsqu'il est ajusté par la réciproque de la constante de couple du moteur, $1/K_t$, devient le couple qui produit le signal de consigne de courant I_q pour la boucle de courant. En théorie, la relation entre le couple et le courant du moteur est indépendante de la position, de la durée, du courant et des conditions ambiantes. En d'autres termes, la mise à l'échelle $1/K_t$ à la Figure 98 a une valeur nominale de 1, c'est-à-dire qu'un couple assigné à 100 % donne un courant assigné à 100 %. Dans la pratique, ceci peut ne pas être le cas. La compensation peut être appliquée à la valeur $1/K_t$ pour traiter de ces questions à la discrétion du fournisseur du dispositif d'entraînement.

7.6.9.10 Asservissement de courant

7.6.9.10.1 Généralités

En général, le couple du moteur est contrôlé en fonction de l'orientation et de l'amplitude du vecteur de courant statorique du moteur par rapport au vecteur de flux magnétique du rotor. La boucle Asservissement de courant est chargée de fournir ce contrôle et est composée effectivement de deux boucles PI, l'une qui contrôle le couple générant le courant, I_q , et l'autre qui contrôle le flux générant le courant, I_d . La composante quadratique du courant, I_q , est utilisée pour l'asservissement de couple dynamique (voir Figure 99).

Dans le cas d'un moteur à induction, le flux générant le courant, I_d , est à lui seul chargé de générer le flux rotorique. Dans le cas d'un moteur à aimant permanent, le flux rotorique est généré par les aimants du moteur, I_d étant uniquement utilisé dans certains cas pour étendre la plage de vitesses du moteur en modifiant l'angle du champ statorique par rapport au champ rotorique. Dans ce cas, l'angle de I_q par rapport au champ rotorique reste le même, c'est-à-dire à quadrature. Mais étant donné que la combinaison vectorielle de I_q et I_d détermine l'angle de flux statorique par rapport au rotor, la valeur croissante de I_d peut décaler le flux statorique de la quadrature afin d'étendre la plage de vitesses du moteur au détriment du couple.



IEC

Anglais	Français
Inverter Peak Current Limit	Limite de courant de crête de l'onduleur
Motor Peak Current Limit	Limite de courant de crête du moteur
Inverter Thermal Current Limit	Limite de courant thermique de l'onduleur
Motor Thermal Current Limit	Limite de courant thermique du moteur
Bus Regulator Limit	Limite du régulateur de bus
Current Vector Limit	Limite de vecteur de courant
Id Current Reference	Référence de courant Id
Current Limit Source	Source de limite de courant
Operative Current Limit	Limite de courant opératoire
Id Current Command	Consigne de courant Id
Current Vector Limiter	Limiteur de vecteur de courant
Id Current Error	Erreur de courant Id
Id Loop Output	Sortie de boucle Id
Vd Decoupling Voltage	Tension de découplage Vd
Vd Output	Sortie Vd
Iq Loop Output	Sortie de boucle Iq
Vq Decoupling Voltage	Tension de découplage Vq
Vq Output	Sortie Vq
Output Voltage	Tension de sortie
Vd Limited Output	Sortie limitée Vd
Vq Limited Output	Sortie limitée Vq
U Voltage Output	Sortie de tension U
V Voltage Output	Sortie de tension V
W Voltage Output	Sortie de tension W
U Phase Voltage	Tension de phase U
V Phase Voltage	Tension de phase V
W Phase Voltage	Tension de phase W
PWM Frequency	Fréquence MID

Anglais	Français
Electrical Angle	Angle électrique
Inverter Voltage Limit	Limite de tension de l'onduleur
PWM Limit Calculation	Calcul de limite MID
DC Bus Voltage	Tension de bus à courant continu
Vq Output	Sortie Vq
U Current Offset	Décalage de courant U
V Current Offset	Décalage de courant V
W Current Offset	Décalage de courant W
U Phase Current	Courant de phase U
V Phase Current	Courant de phase V
W Phase Current	Courant de phase W
U Current Feedback	Retour de courant U
V Current Feedback	Retour de courant V
W Current Feedback	Retour de courant W
Id Current Feedback	Retour de courant Id
Iq Current Feedback	Retour de courant Iq
Output Current	Courant de sortie
Inverter Overload Action	Action de surcharge de l'onduleur
Inverter Overload Limit	Limite de surcharge de l'onduleur
Inverter Thermal Current Limit	Limite de courant thermique de l'onduleur
Inverter Thermal Model	Modèle thermique de l'onduleur
Inverter Temperature	Température de l'onduleur
Motor Overload Action	Action de surcharge du moteur
Motor Overload Limit	Limite de surcharge du moteur
Motor Thermal current limite	Limite de courant thermique du moteur
Motor Thermal Model	Modèle thermique du moteur
Motor Temperature	Température du moteur

Figure 99 – Commande vectorielle de courant en boucle fermée

7.6.9.10.2 Limiteur de vecteur de courant

La Consigne de courant I_q passe par un bloc Limiteur de vecteur de courant avant de devenir le signal Référence de courant I_q . Ce bloc de limite calcule l'amplitude vectorielle combinée de la Référence de courant I_q et des signaux Référence de courant I_d . L'amplitude vectorielle de courant obtenue est comparée à la Limite de courant opératoire qui représente la limite en cours minimale parmi un ensemble de limites en cours potentielles du dispositif d'entraînement et du moteur. Si l'amplitude vectorielle dépasse la Limite de courant opératoire, la Référence de courant I_q est réduite, de sorte que l'amplitude vectorielle soit égale à la Limite de courant opératoire. Les sources de limite de courant potentielles peuvent être les valeurs assignées Limite de courant de crête, ainsi que les Limites thermiques de l'onduleur du moteur et du dispositif d'entraînement. Une autre source de limite possible est l'attribut Limite de vecteur de courant configurable par l'utilisateur. Certaines de ces limites sont conditionnelles et dynamiques, comme les Limites de courant thermiques du moteur et de l'onduleur déduites des modèles thermiques de ces dispositifs. Ces limites sont uniquement actives si les attributs Action de surcharge du moteur et Action de surcharge de l'onduleur correspondants sont définis pour fournir un repli de courant. Dans ce cas, les limites de courant thermiques diminuent au fur et à mesure de l'augmentation de la température simulée des dispositifs modélisés. La Limite du régulateur de bus est uniquement appliquée lorsque le moteur génère à nouveau l'alimentation dans le bus à courant continu et repose sur la Limite de puissance régénératrice.

Avec toutes ces sources de limite de courant potentielles qui peuvent être opérationnelles, un attribut Source de limite de courant a été inclus dans l'Objet Axe de dispositif de mouvement pour identifier la source de la limite de courant active.

7.6.9.10.3 Sortie de tension

La sortie de chaque boucle de courant est ajustée par l'inductance du moteur afin de générer une commande de tension par rapport au bloc de transformation vectoriel. Le bloc de transformation vectoriel est chargé de transformer les signaux de commande de génération du couple, V_q , et de flux, V_d , de la trame de référence synchrone de rotation en trame de référence statorique stationnaire. Les valeurs Tension de sortie U, V et W obtenues sont alors appliquées au moteur par Modulation d'impulsions en durée (MID). La fréquence MID est également un attribut configurable de l'Objet Axe de dispositif de mouvement.

L'amplitude du vecteur V_q , V_d est calculée en temps réel en tant que signal Tension de sortie total. Le signal Tension de sortie maximal qui peut être appliqué au moteur est en dernier lieu limité par la Tension de bus à courant continu et mis en pratique par le Limiteur de vecteur de tension. Toutes les tentatives de dépassement de cette valeur donnent lieu à une condition Limite de tension de l'onduleur.

7.6.9.10.4 Retour de courant

Les signaux de retour de courant à la boucle de courant sont fournis par deux ou trois capteurs de courant. Les signaux provenant de ces capteurs sont conditionnés et corrigés des décalages spécifiques au dispositif pour devenir les signaux Retour de courant U, V et W associés à la carcasse stator du moteur stationnaire. Ces trois signaux sont transformés en trame de référence synchrone afin de générer les signaux Retour de courant I_q et I_d . L'amplitude du vecteur de courant I_q , I_d est calculée en temps réel et utilisée en entrée pour les modèles thermiques de l'onduleur et du moteur.

7.6.9.10.5 Commutation du moteur

La commutation du moteur est essentielle à la commande de moteur en boucle fermée. L'orientation du rotor du moteur peut être déterminée à partir d'une source de réaction montée sur le moteur. La source de commutation réelle est le dispositif de réaction du moteur affecté à Réaction 1. Une fois le dispositif de réaction étalonné en fonction de l'orientation absolue du rotor à l'aide de l'attribut Décalage de commutation, le bloc de commutation peut alors générer l'Angle électrique véritable du rotor. Ce signal est utilisé pour procéder aux

transformations vectorielles entre les carcasses de moteur stationnaire et rotatif, et peut également l'être pour d'autres algorithmes nécessitant de connaître la position du rotor.

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INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch